

The planar log gas in the disk

Infinite limits below 2, empty limits at and above 4,
conditional Gibbs laws, and hyperbolic geometry

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Abstract

We study the unweighted logarithmic gas in the open unit disk with density proportional to $\prod_{i < j} |z_i - z_j|^\beta$ at fixed $\beta > 0$. The unscaled point processes are tight for every positive β . For every $0 < \beta < 2$, every subsequential limit is almost surely infinite and satisfies $\sum_{z \in \Xi} (1 - |z|) = \infty$. We prove this by an exterior-only interpolation identity, weighted Hölder inequalities, and radial bounds uniform under holomorphic product weights. Every limiting intensity is bounded below by a positive multiple of the Bergman intensity, and we obtain quantitative lower-count tails and almost sure logarithmic growth of the accumulated radial depth. At $\beta = 2$ the full limit is the Bergman process. For $\beta \geq 4$ the full sequence converges to the empty configuration, locally in total variation, with an explicit rate above 4. For every subsequential limit we construct analytic canonical conditional densities, a meromorphic electric field, and consistent augmented DLR kernels. The corrected field is independent of the Gaussian harmonic modes. Exact Gaussian elimination gives a canonical hyperbolic Green specification with an explicit geometric one-body factor and a positive-kernel factorization of the unmarked density. At $\beta = 2$ we identify the complete count-resolved compensated DLR law. An independent perturbation argument identifies the first inverse-temperature response of every correlation order at 2, proves its hyperbolic covariance, and bounds the second temperature derivatives uniformly in particle number. We now evaluate the second intensity response at every interior point: its locally uniform limit is $1/[\pi(1 - |z|^2)^2]$, agreeing with the proposed density $(4/\beta - 1)/[\pi(1 - |z|^2)^2]$. For $0 < \beta < 2$, signed central-charge moments determine the radial-depth coefficient $2/\beta - 1/2$ in every positive moment along a density-one set of particle numbers. The same coefficient holds at every fixed hyperbolic center. These results do not prove the proposed local intensity: an explicit order-one mean-potential defect remains undetermined. Euclidean area coordinates give an exact screening equation, and a neutral hyperbolic plasma representation identifies the conjecture with local neutrality. We prove an energy obstruction at the moving finite-volume boundary and an exact connected-force correction to the screening identity. For bounded square-integrable one-body tilts of the Bergman process, we prove a strict nonlinear screening sign and uniqueness of the associated self-consistency equation; this does not identify a general-temperature gas limit. A Blaschke interpolation argument reduces nonemptiness above 2 to a stated, unproved boundary moment estimate. We construct infinitely many distinct infinite, rotationally invariant solutions of the same canonical Green equations for every $0 < \beta < 4$. For $\beta \leq 2$ these states are mutually singular with every actual disk-gas limit. Canonical Gibbs uniqueness therefore fails, while uniqueness and full hyperbolic invariance for $0 < \beta < 4$, $\beta \neq 2$, remain unproved here; nonemptiness remains unresolved only for $2 < \beta < 4$.

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1 Model, conventions, and scope

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$, let $B_r = \{z : |z| < r\}$, and let dA be ordinary, unnormalized planar Lebesgue measure. We also write $\mathbb{D}_r = B_r$ when using centered polydisks. For $\beta > 0$ and $N \geq 1$, put

$$\mathbb{P}_{N,\beta}(dz_1 \cdots dz_N) = \frac{1}{Z_{N,\beta}} |\Delta_N(z)|^\beta \prod_{i=1}^N \mathbf{1}_{\mathbb{D}}(z_i) dA(z_i), \quad \Delta_N(z) = \prod_{1 \leq i < j \leq N} (z_i - z_j). \quad (1.1)$$

Empty products equal 1. The partition function is finite because every pair distance is at most 2, and is positive because the integrand is positive off the collision diagonals. There is no external potential in the interior of the disk. This is the constant-potential disk model in [1, Section 3]; the boundary background in that formulation adds only a configuration-independent energy.

The associated counting measure and disk counts are

$$\Xi_N = \sum_{i=1}^N \delta_{z_i}, \quad M_N(r) = \Xi_N(B_r).$$

We work in $\mathcal{M}_p(\mathbb{D})$, the space of locally finite, nonnegative, integer-valued Radon measures, with the vague topology. Multiplicities are allowed in this state space until simplicity is proved. Vague convergence means convergence of $\xi(f) = \int f d\xi$ for every $f \in C_c(\mathbb{D})$. It allows particles to disappear through $\partial\mathbb{D}$. It does not allow infinitely many particles to accumulate in a compact subset of $\mathbb{D}$. The closed disk would be an inappropriate state space for the unnormalized measures, whose total mass is N .

We use the standard Polish-space, Prokhorov, and monotone-coupling facts for counting measures and stochastic order; see [3, 4]. All estimates specific to (1.1), including the product association inequality needed below, are proved here. No claim of bibliographic novelty is made for the resulting estimates.

Theorem 1.1 (Subsequential existence). *Fix $\beta > 0$. The laws of $(\Xi_N)_{N \geq 1}$ are tight in $\mathcal{M}_p(\mathbb{D})$. In particular, every sequence of particle numbers tending to infinity has a subsequence along which Ξ_N converges in distribution to a locally finite counting measure Ξ . Every such limit is simple. For $0 < r < 1$ and integers $m \geq 2$,*

$$\sup_{N \geq 1} \mathbb{P}_{N,\beta} \{M_N(r) \geq m\} \leq \min \left\{ 1, \exp \left(m - \frac{\beta}{4} m(m-1) \log \frac{1}{r} \right) \right\}. \quad (1.2)$$

The same bound holds with $M_N(r)$ replaced by $M(r) = \Xi(B_r)$. All positive moments and all real exponential moments of compact counts are finite, uniformly before taking the limit.

The proof of tightness is in Section 3; simplicity and the passage of count bounds to limits are justified without assuming simplicity in Section 4. The later sections establish stronger properties. Throughout, “a limit” means a subsequential limit in distribution unless full-sequence convergence is stated.

Theorem 1.2 (Full-sequence uniqueness for $\beta \geq 4$). *For every fixed $\beta \geq 4$, the laws of Ξ_N converge to the law concentrated on the empty configuration. For $\beta > 4$ and each $R < 1$,*

$$\sup_{|z| \leq R} \rho_{1,N}(z) + \mathbb{P}\{\Xi_N(\overline{B_R}) > 0\} \leq C_{\beta,R} N^{2-\beta/2} (\log(eN))^\beta.$$

Thus convergence on every relatively compact window holds in total variation. At $\beta = 2$, the full sequence instead converges to the Bergman determinantal process with kernel $K(z, w) = 1/[\pi(1 - z\overline{w})^2]$ relative to ordinary area.

The assertion for $\beta > 4$ is proved in Theorem 20.1. The critical case 4 is proved in Theorem 36.1, with an independent conformal proof in Theorem 37.1; no rate at 4 is obtained. The assertion at 2 is proved in Section 12. The next theorem establishes almost sure infiniteness throughout $0 < \beta < 2$. Uniqueness remains unresolved for $0 < \beta < 4$, $\beta \neq 2$, and nonemptiness for $2 < \beta < 4$. In particular, we do not identify 4 as the exact transition parameter.

Theorem 1.3 (Infinite subsequential limits below 2). *For every $0 < \beta < 2$, every subsequential limiting process satisfies*

$$\mathbb{P}\{\Xi(\mathbb{D}) = \infty\} = 1, \quad \rho_1(z) \geq \frac{c_\beta}{\pi(1 - |z|^2)^2}, \quad z \in \mathbb{D},$$

where $c_\beta > 0$ depends only on β . One may take $c_\beta = 1$ for $\beta \leq 1$. Every factorial correlation has a strictly positive lower bound after division by its Vandermonde factor. For each fixed $m \geq 0$,

$$\mathbb{P}\{\Xi(B_R) \leq m\} \leq C_{\beta,\tau,m} (1 - R^2)^{2\tau/(m\beta+2)},$$

where $\tau = 1$ for $\beta \leq 1$, and any $0 < \tau < 2 - \beta$ is allowed for $1 < \beta < 2$. These conclusions concern every subsequence; they do not assume uniqueness or hyperbolic invariance. Moreover, the stronger boundary-mass conclusion is

$$\sum_{z \in \Xi} (1 - |z|) = \infty \quad \text{almost surely.}$$

Writing $H(R) = \sum_{|z| < R} -\log |z|$ and $\kappa_\beta = \min(1, 2 - \beta)$, one has

$$\mathbb{E} e^{-\beta H(R)} \leq C_{\beta,\tau} (1 - R^2)^\tau, \quad \liminf_{R \uparrow 1} \frac{H(R)}{\log(1/(1 - R^2))} \geq \frac{\kappa_\beta}{\beta} \quad \text{almost surely.}$$

Proof. The intensity and factorial lower bounds are Theorems 48.2 and 48.4. Almost sure infiniteness and the lower-count estimate are Theorem 48.6 and Corollary 48.8. The stronger non-Blaschke and depth-growth conclusions are Theorem 49.2. $\square$

Remark 1.4 (New results concerning the intensity conjecture). The conjecture under investigation is

$$\rho_\beta(z) = \frac{4/\beta - 1}{\pi(1 - |z|^2)^2}, \quad 0 < \beta < 4.$$

It remains unproved except at $\beta = 2$. The new exact perturbative calculation, Theorem 62.1, establishes

$$\lim_N \partial_\beta^2 \log \rho_{N,\beta}(z) \Big|_2 = 0, \quad \lim_N \partial_\beta^2 \rho_{N,\beta}(z) \Big|_2 = \frac{1}{\pi(1 - |z|^2)^2}.$$

The convergence is locally uniform in z . These limits concern derivatives of the finite gases at 2, and do not interchange the particle-number and temperature limits. Theorem 63.2 proves, for $0 < \beta < 2$,

$$\frac{\sum_i -\log |z_i|}{\log N} \longrightarrow \frac{2}{\beta} - \frac{1}{2}$$

in every positive absolute moment along a set of particle numbers of natural density one. It also proves the signed-charge free energy for every charge $t > -2$. Corollary 63.6 extends the depth coefficient to every fixed hyperbolic center. The exact mean-potential defect in Proposition 64.8 vanishes if and only if the intensity conjecture holds. Theorem 64.6 gives a sufficient affine-field symmetry criterion, whose symmetry assumption has not been proved for the selected gas limit. Numerical checks in Section 65 retain substantial finite-particle corrections and are not used as proof.

Remark 1.5 (What the screening investigation proves). Sections 66–70 give a rigorous test of the proposed energy strategy. The map $T(z) = z/\sqrt{1 - |z|^2}$ sends hyperbolic area $dA/(1 - |z|^2)^2$ to ordinary planar area, making the conjectured intensity constant. Proposition 67.1 identifies the original gas, up to a dilation tending to the identity, with an exactly neutral hyperbolic plasma of background density $(4/\beta - 1)/\pi$. Global neutrality alone does not prove local neutrality. Even at $\beta = 2$, positive mean electrostatic energy of order N persists near the moving boundary (Propositions 66.6 and 67.5).

Theorem 68.3 identifies the precise connected-force work and cutoff boundary estimate that would force the conjecture. Both remain unproved for a general-temperature limit. In an explicit reference family, however, the sign is established: Theorem 70.3 gives strict nonlinear response monotonicity for every bounded L^2 one-body tilt of the Bergman process. Corollary 70.4 then proves uniqueness of its self-consistent screening potential, and Proposition 70.5 gives a uniform quantitative response bound. These are theorems about the stated reference family, not uniqueness or the intensity formula for the original gas at $\beta \neq 2$.

The quantitative proof for $\beta > 4$ has two inputs. Corollary 19.5 bounds compact insertion expectations uniformly in N , using transport and tightness under exponential tilts; it requires no central limit theorem. Harmonic boundary transport proves an adjacent-partition-ratio lower bound that diverges when $\beta > 4$. Their combination forces every compact intensity to zero. Corollaries 21.1 and 21.2 quantify all fixed correlations, count moments, and the probability of multiple interior points.

At every positive β , the arguments also establish analytic Gaussian fluctuations independent of the interior, a leading partition-function asymptotic (Theorem 14.1), and an exact criterion sufficient for uniqueness (Proposition 27.3). The covariance estimate in that criterion remains a hypothesis; the large- β uniqueness proof does not use it.

The critical proof uses a different mechanism. The independent Gaussian harmonic modes force a component of the electric-force variance. A hypothetical nonempty limit at

4 has a bounded renormalized intensity, forcing its remaining score fluctuations to vanish under boundary translation. The translated limiting Ward identity then contradicts retained two-point repulsion.

The other extensions supply further information about the model. Theorem 22.1 identifies adjacent-ratio exponents on a set of full natural density. Theorem 23.2 controls boundary-vanishing statistics and permits explicit limiting changes of potential. Theorem 24.1 proves the exact Gaussian mean shift produced by finitely many fixed charges, and Corollary 24.2 gives relative correlation invariance under fixed analytic tilts. The local Ward identity, Proposition 25.1, separates the proved $O(N^{-1})$ response from the stronger estimate needed for conformal invariance. These statements concern the original fixed temperature; no interchange of limits in temperature and particle number is made.

Conditional laws and hyperbolic geometry. For the DLR investigation, the main unconditional statements are in Sections 41 and 42. If $B \Subset \mathbb{D}$ is a disk and $m \geq 1$ is an allowed conditional count, the ordered conditional density has the form

$$p_{\eta,m}(\mathbf{z}) = |\Delta_m(\mathbf{z})|^\beta g_{\eta,m}(\mathbf{z}), \quad \left[\partial_{z_i} \partial_{\bar{z}_j} \log g_{\eta,m} \right]_{ij} \succeq \frac{\beta}{2} \left[(1 - z_i \bar{z}_j)^{-2} \right]_{ij},$$

with $g_{\eta,m} > 0$ real analytic. This is a canonical conditional-law theorem; it does not identify the relative probabilities of different counts. Section 43 constructs a simultaneous meromorphic force field and proves nested-set consistency of the canonical resampling kernels on that augmented space. The stronger separation theorem in Section 51 permits exact elimination of the independent Gaussian modes. The resulting canonical Green kernels, their explicit geometric factor, and their affine isometry covariance are proved in Section 52. Section 46 identifies those probabilities and the full compensated DLR equation at $\beta = 2$. The results of Section 44 distinguish covariance of the score and Gaussian field from invariance of the point-process law. Section 45 proves why an invariant Green specification needs compensation, then states precisely the additional hypotheses needed for general-temperature DLR and intensity conclusions.

Theorem 1.6 (Canonical Gibbs nonuniqueness with infinitely many particles). *For each $0 < \beta < 4$, infinitely many distinct infinite point-process laws admit augmentations satisfying the same canonical Green DLR specification as the actual disk-gas limits. They may all be chosen rotation and reflection invariant. None of these constructed states is fully hyperbolically invariant. For $0 < \beta \leq 2$, every one is mutually singular with every actual disk-gas limit: the constructed states have finite Blaschke mass, whereas actual limits have infinite Blaschke mass almost surely.*

Proof. The construction and all the distinctions in the statement are proved in Theorem 54.2. Proposition 54.3 gives an explicit determinantal family at $\beta = 2$. $\square$

This is a nonuniqueness theorem for the canonical equations. It does not assert that the original finite- N disk gas has two subsequential limits. Section 60 gives an exact oscillation criterion for that different question; no test satisfying its nonzero-oscillation condition has been exhibited.

2 Two elementary inequalities

Lemma 2.1 (Subharmonic powers and annular means). *If P is holomorphic on a planar domain and $\beta > 0$, then $|P|^\beta$ is nonnegative and subharmonic. If $u \geq 0$ is subharmonic on a*

neighborhood of $\overline{B(z, b)}$ and $0 \leq a < b$, then

$$u(z) \leq \frac{1}{\pi(b^2 - a^2)} \int_{a < |w-z| < b} u(w) dA(w).$$

The circular mean of u about z is nondecreasing in its radius.

Proof. Write $\alpha = \beta/2$. For $\varepsilon > 0$, direct differentiation gives

$$\Delta(|P|^2 + \varepsilon)^\alpha = 4\alpha(|P|^2 + \varepsilon)^{\alpha-2}(\varepsilon + \alpha|P|^2)|P'|^2 \geq 0.$$

Letting $\varepsilon \downarrow 0$ locally uniformly proves the first claim. For a smooth subharmonic function the derivative of its circular mean at radius s is $(2\pi s)^{-1} \int_{B(z,s)} \Delta u dA \geq 0$. Regularization gives the same conclusion for a general subharmonic function. The submean inequality on circles, integrated against $2\pi s ds$ for $a < s < b$, gives the annular statement. The same argument applies to separately subharmonic functions one variable at a time. $\square$

Lemma 2.2 (Association and increasing tilts). *Let μ be a product probability measure on a finite product of intervals. If f, g are bounded coordinatewise nondecreasing real functions, then $\text{Cov}_\mu(f, g) \geq 0$. Consequently, if $F \geq 0$ is bounded, nondecreasing, and $\mu(F) > 0$, and $d\nu = F d\mu / \mu(F)$, then*

$$\nu(A) \leq \mu(A) \tag{2.1}$$

for every coordinatewise decreasing Borel set A . More generally, $\nu(g) \leq \mu(g)$ for every bounded nondecreasing function of a coordinatewise decreasing vector-valued function of the coordinates.

Proof. For one coordinate, with independent identically distributed X, X' , twice the covariance is

$$\mathbb{E}[(f(X) - f(X'))(g(X) - g(X'))] \geq 0.$$

Induct on the number of coordinates. Condition on the last coordinate and apply the induction hypothesis to the conditional covariance. The two conditional expectations are nondecreasing functions of the last coordinate, so the covariance of these expectations is nonnegative by the one-dimensional case. The covariance decomposition proves association. Apply it to F and $-\mathbf{1}_A$ for (2.1), or to F and the negative of the indicated decreasing function for the final statement. Truncation extends the expectation inequality to nonnegative unbounded functions. This is the product association property in [2]. $\square$

3 Radial comparison and tightness

Configurations with a zero radius or repeated radii have probability zero, by absolute continuity. Sort radii and relabel the angles so that $0 < r_1 < \dots < r_N < 1$. Symmetry makes the density on this ordered region $N!$ times the original density. This constant will be absorbed in normalization. Set

$$t_k = \frac{r_k}{r_{k+1}}, \quad r_{N+1} = 1, \quad r_j = \prod_{k=j}^N t_k, \quad a_k = 2k + \frac{\beta}{2}k(k-1). \tag{3.1}$$

Thus $t \in (0, 1)^N$. Factoring each pair distance and computing the triangular Jacobian gives

$$\prod_{j=1}^N r_j dr_j = \prod_{k=1}^N t_k^{2k-1} dt_k, \quad \prod_{j=1}^N r_j^{\beta(j-1)} = \prod_{k=1}^N t_k^{\beta k(k-1)/2}.$$

For example, the Jacobian $|\det(\partial r/\partial t)|$ is $\prod_{k=1}^N t_k^{k-1}$, whereas $\prod_j r_j = \prod_k t_k^k$. After angular integration the ratio density is therefore proportional to

$$F_N(t) \prod_{k=1}^N t_k^{a_k-1}, \quad F_N(t) = \int_{[0,2\pi)^N} \prod_{i < j} \left| 1 - \left(\prod_{\ell=i}^{j-1} t_\ell \right) e^{i(\theta_i - \theta_j)} \right|^\beta \prod_{j=1}^N \frac{d\theta_j}{2\pi}. \quad (3.2)$$

This is a bounded continuous nonnegative function on $[0, 1]^N$, and its integral against the product beta density is positive.

Lemma 3.1 (Monotonicity of the angular factor). *The function F_N is nondecreasing in each coordinate t_k . It is independent of t_N .*

Proof. Fix $k < N$ and all other ratios. Only factors with $i \leq k < j$ depend on t_k ; write their radial ratios as $t_k q_{ij}$ with q_{ij} independent of t_k . Average additionally over the change $\theta_i \mapsto \theta_i + \phi$ for all $i \leq k$, keeping the outer angles fixed. Haar invariance makes this extra average leave F_N unchanged. The noncrossing factors are unchanged under this common rotation, and the crossing product is

$$\left| P_\theta(t_k e^{i\phi}) \right|^\beta, \quad P_\theta(w) = \prod_{i \leq k < j} (1 - q_{ij} e^{i(\theta_i - \theta_j)} w).$$

The circular mean in ϕ is nondecreasing in t_k by Lemma 2.1. All remaining factors are nonnegative and independent of t_k , so their integration preserves the inequality. No ratio in (3.2) involves t_N . $\square$

Let U_k be independent with distribution $\text{Beta}(a_k, 1)$, namely density $a_k u^{a_k-1} \mathbf{1}_{(0,1)}(u)$. Under their product law, the ratio law is the increasing tilt by F_N . Thus the actual ratios stochastically dominate $(U_1, \dots, U_N)$. Equivalently, all bounded coordinatewise increasing functions of the ordered depths

$$X_{N,j} = -\log r_j = \sum_{k=j}^N (-\log t_k)$$

have expectation at most the corresponding function of

$$T_j^{(N)} = \sum_{k=j}^N E_k, \quad E_k = -\log U_k \sim \text{Exp}(a_k) \quad \text{independently.} \quad (3.3)$$

The exponential convention is $\mathbb{P}(E_k > x) = e^{-a_k x}$ for $x \geq 0$. Indeed, each depth is decreasing in every ratio, so this is exactly Lemma 2.2. In particular, for $N \geq m$ and $s = \log(1/r)$,

$$\mathbb{P}\{M_N(r) \geq m\} \leq \mathbb{P}\{T_m^{(N)} > s\} \leq \mathbb{P}\{T_m \geq s\}, \quad T_m = \sum_{k=m}^\infty E_k. \quad (3.4)$$

The infinite sums are almost surely finite, since $\sum_k a_k^{-1} < \infty$.

For $m \geq 2$, take $\lambda = \beta m(m-1)/4$. Since $a_k \geq \beta k(k-1)/2$, for $k \geq m$ we have $\lambda/a_k \leq 1/2$ and

$$\begin{aligned} \log \mathbb{E} e^{\lambda T_m^{(N)}} &= \sum_{k=m}^N -\log(1 - \lambda/a_k) \leq 2\lambda \sum_{k=m}^N a_k^{-1} \\ &\leq \frac{4\lambda}{\beta} \sum_{k=m}^\infty \frac{1}{k(k-1)} = m. \end{aligned}$$

Exponential Markov inequality proves (1.2) at finite N . The case $N < m$ is automatic. It also follows that for every $r < 1$, $q > 0$, and $t \geq 0$,

$$\sup_N \mathbb{E} M_N(r)^q < \infty, \quad \sup_N \mathbb{E} e^{tM_N(r)} < \infty. \quad (3.5)$$

For instance, for $m \geq 2$ the summands in $\sum_m e^{tm} \mathbb{P}(M_N(r) = m)$ are bounded by $\exp((t+1)m - c_r m(m-1))$, with $c_r = (\beta/4) \log(1/r) > 0$. The same calculation shows $\sup_N \mathbb{E} e^{tM_N(r)} \leq C_r e^{C_r(t+1)^2}$, where constants may also depend on β .

Proof of tightness in Theorem 1.1. Choose an increasing sequence $r_\ell \uparrow 1$. Given $\varepsilon > 0$, choose integers b_ℓ such that $\sup_N \mathbb{P}(M_N(r_\ell) > b_\ell) < \varepsilon 2^{-\ell}$. The set

$$\mathcal{K} = \bigcap_{\ell \geq 1} \{ \xi \in \mathcal{M}_p(\mathbb{D}) : \xi(B_{r_\ell}) \leq b_\ell \}$$

is closed, since counting an open relatively compact set is lower semicontinuous under vague convergence. Its elements have uniformly bounded mass on every compact set. Such a set is relatively compact in the vague topology, and its vague limits are again counting measures. These standard facts can also be seen by diagonal extraction on compact exhaustions, where boundedly many atom positions have convergent subsequences; atoms escaping an exhaustion are treated at the next level. Thus $\mathcal{K}$ is compact. A union bound gives $\inf_N \mathbb{P}(\Xi_N \in \mathcal{K}) \geq 1 - \varepsilon$. The space $\mathcal{M}_p(\mathbb{D})$ is Polish, so Prokhorov's theorem proves tightness and subsequential existence. $\square$

4 Correlation densities and local regularity

We continue to fix $\beta > 0$. Let Ξ_{N_j} be any subsequence converging in law, for the vague topology on $\mathbb{D}$, to a locally finite integer-valued random measure Ξ . At this stage multiplicities are allowed in the definition of Ξ . The compact count estimates proved above imply, for every $R < 1$ and every $t > 0$,

$$\sup_N \mathbb{E} e^{tM_N(R)} < \infty, \quad \mathbb{E} e^{tM(R)} < \infty. \quad (4.1)$$

For the limiting bound, fix $R < S < 1$ and choose $\chi \in C_c(\mathbb{D})$ with $0 \leq \chi \leq 1$, $\chi = 1$ on $\overline{B_R}$, and support in B_S . The continuous mapping theorem and the lower-semicontinuous form of Portmanteau give

$$\mathbb{E} e^{tM(R)} \leq \mathbb{E} e^{t\Xi(\chi)} \leq \liminf_j \mathbb{E} e^{t\Xi_{N_j}(\chi)} \leq \sup_N \mathbb{E} e^{tM_N(S)} < \infty.$$

We use unnormalized planar area measure throughout this section, and write

$$(m)_p = m(m-1) \cdots (m-p+1), \quad \Delta_p(z) = \prod_{1 \leq i < j \leq p} (z_i - z_j).$$

Empty products have value one.

Theorem 4.1 (Regular correlation hierarchy). *The limit Ξ is simple. For each integer $p \geq 1$, it has a continuous factorial correlation density of the form*

$$\rho_p(z) = |\Delta_p(z)|^\beta h_p(z), \quad (4.2)$$

where h_p is nonnegative, plurisubharmonic, and locally Hölder continuous with exponent $\min\{1, \beta\}$. Along the given subsequence,

$$h_{p, N_j} \longrightarrow h_p, \quad \rho_{p, N_j} \longrightarrow \rho_p \quad \text{locally uniformly on } \mathbb{D}^p. \quad (4.3)$$

For every $r < 1$ there is a finite constant $C_{p,r,\beta}$, independent of N , such that

$$\rho_{p,N}(z) \leq C_{p,r,\beta} |\Delta_p(z)|^\beta, \quad \rho_p(z) \leq C_{p,r,\beta} |\Delta_p(z)|^\beta, \quad z \in \overline{\mathbb{D}_r^p}. \quad (4.4)$$

Here $\rho_{p,N} = h_{p,N} = 0$ when $p > N$.

We give all details, including the compactness argument needed to identify the limiting densities. Here plurisubharmonic means upper semicontinuous with subharmonic restriction to every complex affine line on each component of its intersection with the domain.

For $N \geq p$, direct integration of the joint density gives

$$\rho_{p,N}(z) = |\Delta_p(z)|^\beta h_{p,N}(z), \quad (4.5)$$

$$h_{p,N}(z) = \frac{(N)_p}{Z_{N,\beta}} \int_{\mathbb{D}^{N-p}} |\Delta_{N-p}(y)|^\beta \prod_{i=1}^p \prod_{\ell=1}^{N-p} |z_i - y_\ell|^\beta \, dA^{N-p}(y). \quad (4.6)$$

In particular $h_{p,N}$ is continuous. The integrand, as a function of z , is the β th power of the modulus of a holomorphic polynomial, and is therefore plurisubharmonic. Integration preserves this property. Only separate subharmonicity is needed for the following estimate.

Lemma 4.2 (Separated annular averages). *Fix $p \geq 1$, $\beta > 0$, and $0 < r < R < 1$. There is a constant $C = C(p, \beta, r, R) < \infty$ such that every nonnegative separately subharmonic continuous function u on $\mathbb{D}_R^p$ satisfies*

$$\sup_{\overline{\mathbb{D}_r^p}} u \leq C \int_{\mathbb{D}_R^p} |\Delta_p(w)|^\beta u(w) \, dA^p(w). \quad (4.7)$$

Proof. Fix a tuple $z^* \in \overline{\mathbb{D}_r^p}$. Partition its coordinates into groups with common values $c_1, \dots, c_q$. Choose $\eta > 0$ so small that

$$100p\eta \leq R - r, \quad 100p\eta \leq \min_{a \neq b} |c_a - c_b|,$$

omitting the second condition when $q = 1$. In each group assign its coordinates distinct labels $\ell = 1, \dots, m_a$.

Suppose that $\max_i |z_i - z_i^*| < \eta/4$. For the coordinate with label ℓ_i , average over the annulus

$$A_i(z_i) = \{w : (3\ell_i - 2)\eta \leq |w - z_i| \leq (3\ell_i - 1)\eta\}.$$

All these annuli lie in $\mathbb{D}_R$. For two coordinates in the same group, the radial intervals have a gap of at least 2η , whereas their centers are at distance less than $\eta/2$. Their annuli are thus at distance at least $3\eta/2$. For different groups, the lower bound on the distance between the c_a 's gives a still larger separation. Consequently

$$|w_i - w_j| \geq \eta \quad (i \neq j), \quad |A_i(z_i)| \geq 3\pi\eta^2.$$

The circular submean inequality, integrated in radius, is also an annular submean inequality. Applying it successively in all variables and then using the displayed separation gives

$$u(z) \leq (3\pi)^{-p} \eta^{-2p-\beta p(p-1)/2} \int_{\mathbb{D}_R^p} |\Delta_p(w)|^\beta u(w) \, dA^p(w).$$

The neighborhoods of the tuples z^* constructed in this way cover $\overline{\mathbb{D}_r^p}$. A finite subcover gives a single finite constant in (4.7). $\square$

Since

$$\int_{\mathbb{D}_R^p} \rho_{p,N}(w) \, dA^p(w) = \mathbb{E}(M_N(R))_p,$$

Lemma 4.2 and (4.1) show that $h_{p,N}$ are uniformly bounded on compact subsets of $\mathbb{D}^p$. This already proves the finite- N part of (4.4).

Lemma 4.3 (Uniform Hölder estimates). *For each $p \geq 1$ and $r < 1$, there is a constant $C < \infty$ such that*

$$|h_{p,N}(z) - h_{p,N}(w)| \leq C \|z - w\|^{\min\{1, \beta\}}, \quad z, w \in \overline{\mathbb{D}_r^p},$$

uniformly in N .

Proof. Write (4.6) as

$$h_{p,N}(z) = \int |F_y(z)|^\beta \, d\mu_N(y), \quad F_y(z) = \prod_{i,\ell} (z_i - y_\ell),$$

where μ_N is the finite positive measure containing the prefactor and the factor $|\Delta_{N-p}(y)|^\beta$. Fix $r < s < t < 1$. For every holomorphic function F on $\mathbb{D}_t^p$, iterated scalar submean inequalities, followed by Cauchy estimates on an intermediate polydisk, give

$$\sup_{\overline{\mathbb{D}_s^p}} |F|^\beta + \sum_{i=1}^p \sup_{\overline{\mathbb{D}_s^p}} |\partial_i F|^\beta \leq C \int_{\mathbb{D}_t^p} |F(z)|^\beta \, dA^p(z). \quad (4.8)$$

These estimates hold for every $\beta > 0$: one first applies the submean inequality to $|F|^\beta$, and only then uses the usual supremum form of Cauchy's derivative estimate. Integrating (4.8) with respect to μ_N and using the uniform local bound on $h_{p,N}$ yields

$$\int (A_y^\beta + D_y^\beta) \, d\mu_N(y) \leq C,$$

where

$$A_y = \sup_{\overline{\mathbb{D}_s^p}} |F_y|, \quad D_y = \sum_i \sup_{\overline{\mathbb{D}_s^p}} |\partial_i F_y|.$$

The polydisk is convex, so $|F_y(z) - F_y(w)| \leq C_p D_y \|z - w\|$ for z, w in the smaller polydisk. If $0 < \beta \leq 1$, the inequality

$$||a|^\beta - |b|^\beta| \leq |a - b|^\beta$$

proves the required estimate after integration. If $\beta > 1$, use

$$||a|^\beta - |b|^\beta| \leq \beta |a - b| (|a| + |b|)^{\beta-1}$$

and Hölder's inequality to obtain

$$|h_{p,N}(z) - h_{p,N}(w)| \leq C \|z - w\| \left(\int D_y^\beta \, d\mu_N \right)^{1/\beta} \left(\int A_y^\beta \, d\mu_N \right)^{(\beta-1)/\beta}.$$

The right side is uniformly bounded by $C \|z - w\|$. $\square$

For completeness, factorial moment measures also behave continuously in the present setting. If ξ is an integer-valued locally finite measure, its p th factorial measure $\xi^{[p]}$ counts ordered selections of distinct particle labels, even when some labels occupy the same location. For $f \in C_c(\mathbb{D}^p)$,

$$\langle f, \xi^{[p]} \rangle = \sum_{\pi \in \mathcal{P}_p} (-1)^{p-|\pi|} \prod_{B \in \pi} (|B| - 1)! \int f(x_{\pi(1)}, \dots, x_{\pi(p)}) \, d\xi^{\otimes |\pi|}(x). \quad (4.9)$$

Here $\mathcal{P}_p$ is the set of partitions of $\{1, \dots, p\}$; the blocks are ordered by their smallest elements, and $\pi(i)$ is the index of the block containing i . To verify the formula, expand each product measure as a sum over labels and group terms by their label-equality partition. The coefficient for a block of n equal labels is the sum of the signs of all permutations of n elements: a partition into cycles contributes $(-1)^{n-|\pi|} \prod_{B \in \pi} (|B| - 1)!$. This sum is 1 for $n = 1$ and 0 for $n \geq 2$, since multiplication by a transposition pairs permutations with opposite signs. Hence only distinct labels survive.

Every term on the right is continuous for vague convergence. Indeed, continuous functions on the relevant compact product can be uniformly approximated by sums of products of one-variable continuous functions, using compactly supported cutoffs. Vaguely convergent measures have uniformly bounded mass on those compact sets, so this approximation reduces the assertion to the defining one-variable convergence. The diagonal substitution in each term has compact support because every block variable occurs among the arguments of f . Thus $\xi \mapsto \langle f, \xi^{[p]} \rangle$ is continuous. Its absolute value is at most $\|f\|_\infty \xi(K)^p$ for a compact set K containing all coordinate projections of $\text{supp } f$. The count estimates give uniform integrability. Consequently,

$$\mathbb{E} \langle f, \Xi_{N_j}^{[p]} \rangle \longrightarrow \mathbb{E} \langle f, \Xi^{[p]} \rangle. \quad (4.10)$$

Proof of Theorem 4.1. The local bounds and Lemma 4.3 give local uniform precompactness of h_{p, N_j} by Arzelà–Ascoli. Let h be the limit along any further subsequence. Equations (4.5) and (4.10) show that the p th factorial moment measure of Ξ has density $|\Delta_p|^\beta h$. This identifies h uniquely away from the diagonals. Since that set is dense and h is continuous, it identifies h everywhere. Every locally uniformly convergent further subsequence therefore has the same limit, proving (4.3) along the original process-convergent subsequence. Local uniform limits preserve plurisubharmonicity and the Hölder bounds.

In particular the second factorial moment measure of Ξ is absolutely continuous with respect to planar area on $\mathbb{D}^2$. Its mass on the diagonal is zero. On every compact set K , this mass equals

$$\mathbb{E} \sum_{x \in K} \Xi(\{x\}) (\Xi(\{x\}) - 1).$$

It follows, by a countable compact exhaustion, that Ξ has no multiplicities almost surely. The remaining assertions follow from the estimates already proved. $\square$

Corollary 4.4 (Null sets and small clusters). *The process Ξ almost surely avoids every fixed area-null Borel subset of $\mathbb{D}$. Moreover, for each compact $K \subset \mathbb{D}$ and each $p \geq 1$, there are constants $C, \varepsilon_0 > 0$ such that*

$$\mathbb{P}\{\Xi(B(z, \varepsilon)) \geq p\} \leq C \varepsilon^{2p + \beta p(p-1)/2}, \quad z \in K, \quad 0 < \varepsilon \leq \varepsilon_0. \quad (4.11)$$

The same estimate holds uniformly for Ξ_N .

Proof. The first assertion follows by integrating ρ_1 over the null set and using a compact exhaustion. For the second, choose a compact disk containing all the balls in question. Markov’s inequality for the factorial count, followed by (4.4), gives

$$\mathbb{P}\{\Xi(B(z, \varepsilon)) \geq p\} \leq \frac{1}{p!} \int_{B(z, \varepsilon)^p} \rho_p(w) \, dA^p(w) \leq \frac{C}{p!} (2\varepsilon)^{\beta p(p-1)/2} (\pi \varepsilon^2)^p.$$

The finite- N proof is identical. $\square$

Corollary 4.5 (Convergence of disk counts). *For each $r < 1$, along the chosen subsequence, $M_{N_j}(r) \Rightarrow M(r)$ and all their positive moments converge. Their exponential moments converge for every real parameter. The Gaussian tail bound (1.2) holds for $M(r)$ with the same constants as at finite N .*

Proof. The fixed circle ∂B_r is an area-null set and therefore contains no point of Ξ almost surely. Counting B_r is consequently vaguely continuous at Ξ almost surely. The continuous mapping theorem gives convergence in distribution of the integer-valued counts. The uniform exponential bounds give uniform integrability of every fixed power and exponential, proving convergence of the expectations. Since the counts are integer valued, probabilities of $M \geq m$ converge by using the continuity threshold $m - 1/2$. This passes (1.2) to Ξ and completes all assertions of Theorem 1.1. $\square$

Corollary 4.6 (Leading small-cluster asymptotics). *Fix $a \in \mathbb{D}$ and $p \geq 1$, and put*

$$d_p = 2p + \frac{\beta}{2}p(p-1), \quad Z_{p,\beta} = \int_{\mathbb{D}^p} |\Delta_p(w)|^\beta dA^p(w).$$

As $\varepsilon \downarrow 0$,

$$\mathbb{E}(\Xi(B(a, \varepsilon)))_p = \varepsilon^{d_p} [Z_{p,\beta} h_p(a, \dots, a) + o(1)], \quad (4.12)$$

$$\mathbb{P}\{\Xi(B(a, \varepsilon)) \geq p\} = \varepsilon^{d_p} \left[\frac{Z_{p,\beta}}{p!} h_p(a, \dots, a) + o(1) \right], \quad (4.13)$$

and the same leading asymptotic holds for the probability of exactly p points. The coefficient is permitted to vanish.

Proof. In the integral of ρ_p over $B(a, \varepsilon)^p$, make the change of variables $z_i = a + \varepsilon w_i$. The area factor is ε^{2p} and the Vandermonde factor contributes $\varepsilon^{\beta p(p-1)/2}$. Continuity of h_p gives (4.12). If $L = \Xi(B(a, \varepsilon))$, then

$$0 \leq \mathbb{E} \binom{L}{p} - \mathbb{P}(L \geq p) \leq (p+1) \mathbb{E} \binom{L}{p+1} = O(\varepsilon^{d_{p+1}}).$$

Since $d_{p+1} > d_p$, this proves (4.13). The probability of at least $p+1$ points has this same smaller order, proving the assertion for exactly p points. $\square$

Remark 4.7 (Even inverse temperatures). If $\beta = 2q$ with $q \in \mathbb{N}$, the functions h_p and ρ_p are real analytic. Indeed, polarize the representation in the proof of Lemma 4.3:

$$H_N(z, \bar{w}) = \int F_y(z)^q \overline{F_y(w)^q} d\mu_N(y).$$

This is holomorphic in z and antiholomorphic in w , and

$$|H_N(z, \bar{w})| \leq h_{p,N}(z)^{1/2} h_{p,N}(w)^{1/2}.$$

The uniform local bounds and Montel's theorem give a polarized holomorphic limit along a further subsequence. Its diagonal is h_p , so h_p is real analytic. The factor $|\Delta_p|^{2q}$ is a polynomial in the real and imaginary coordinates, proving the same assertion for ρ_p .

5 Janossy densities and convergence in local total variation

Let $B \subset \mathbb{D}$ be any relatively compact Borel set. No regularity or null-boundary assumption is imposed on B . Write $M_B = \Xi(B)$. Its m th Janossy measure is

$$J_{B,m}(A) = \mathbb{E}[\mathbf{1}_{\{M_B=m\}} \Xi_B^{[m]}(A)], \quad A \subset B^m.$$

For $m = 0$, this means the scalar $\mathbb{P}(M_B = 0)$. The convention used here gives total mass $m! \mathbb{P}(M_B = m)$ for $m \geq 1$.

Proposition 5.1 (Janossy expansion). *For every $m \geq 1$, $J_{B,m}$ has a density $j_{B,m}$ satisfying $0 \leq j_{B,m} \leq \rho_m$ almost everywhere on B^m and*

$$j_{B,m}(z) = \sum_{\ell=0}^{\infty} \frac{(-1)^\ell}{\ell!} \int_{B^\ell} \rho_{m+\ell}(z, y) \, dA^\ell(y). \quad (5.1)$$

The series is absolutely convergent in $L^1(B^m)$. The same formula holds for $m = 0$, with $\rho_0 = 1$, and gives the hole probability. It also holds for every finite- N process.

Proof. The domination $J_{B,m} \leq \mathbb{E}\Xi_B^{[m]}$ gives absolute continuity and the density bound. For a finite configuration with n points in B , the inclusion–exclusion identity for each ordered selection of m labels is

$$\sum_{\ell=0}^{n-m} \frac{(-1)^\ell}{\ell!} (n-m)_\ell = (1-1)^{n-m} = \mathbf{1}_{\{n=m\}}.$$

For $n < m$ there are no such selections. Integrate against a bounded test function on B^m and take expectations. The sum of the L^1 norms of the terms in (5.1) is

$$\sum_{\ell \geq 0} \frac{1}{\ell!} \mathbb{E}(M_B)_{m+\ell} = \mathbb{E}[(M_B)_m 2^{M_B-m} \mathbf{1}_{\{M_B \geq m\}}] < \infty.$$

The exponential count moments therefore justify all interchanges and prove the claimed L^1 identity. The proof for $m = 0$ is the same binomial identity without the initial selection of labels. $\square$

Corollary 5.2 (Absolute continuity in a finite window). *For every relatively compact Borel set B , the law of $\Xi|_B$ is absolutely continuous with respect to a Poisson process of unit area intensity on B . On a configuration of m distinct points $z_1, \dots, z_m$, a version of the density is $e^{|B|} j_{B,m}(z_1, \dots, z_m)$, where $|B|$ is its area. If $\mathbb{P}(M_B = m) > 0$, a uniformly randomly labeled configuration conditional on $M_B = m$ has density $j_{B,m}/(m! \mathbb{P}(M_B = m))$ on B^m .*

Proof. The Poisson Janossy density on B^m is the constant $e^{-|B|}$, and the probability law is its symmetric pushforward with weight $1/m!$. Comparing it with $j_{B,m}$ proves the first formula. The second follows by dividing the Janossy measure by its mass $m! \mathbb{P}(M_B = m)$. If $|B| = 0$, both processes are empty on B almost surely, and the first assertion holds with the same convention. $\square$

Theorem 5.3 (Local total variation convergence). *Along every subsequence $\Xi_{N_j} \Rightarrow \Xi$ considered above, and for every relatively compact Borel set $B \subset \mathbb{D}$,*

$$\|\mathcal{L}(\Xi_{N_j}|_B) - \mathcal{L}(\Xi|_B)\|_{\text{TV}} \longrightarrow 0. \quad (5.2)$$

Here total variation is the supremum of the differences of probabilities over measurable events in the space of finite configurations on B .

Proof. For fixed m and ℓ , local uniform convergence of the correlations implies

$$\int_{B^\ell} \rho_{m+\ell, N_j}(\cdot, y) \, dA^\ell(y) \longrightarrow \int_{B^\ell} \rho_{m+\ell}(\cdot, y) \, dA^\ell(y) \quad \text{in } L^1(B^m).$$

This conclusion uses only the fact that $\overline{B}$ is compact in $\mathbb{D}$. For the inclusion–exclusion terms with $\ell \geq L$, the sum of their L^1 norms is at most

$$\mathbb{E}[(M_B)_m 2^{M_B-m} \mathbf{1}_{\{M_B \geq m+L\}}].$$

The analogous expression for Ξ_N tends to zero uniformly in N as $L \rightarrow \infty$, by (4.1) applied to a disk containing $\overline{B}$. It follows that, for every $m \geq 0$,

$$j_{B,m,N_j} \longrightarrow j_{B,m} \quad \text{in } L^1(B^m), \quad (5.3)$$

with the natural scalar interpretation when $m = 0$.

The law of a finite simple configuration is the pushforward, under $z \mapsto \sum_i \delta_{z_i}$, of its symmetric Janossy measures with weights $1/m!$. Hence its total variation distance is bounded by

$$\frac{1}{2} \sum_{m=0}^{\infty} \frac{1}{m!} \|j_{B,m,N_j} - j_{B,m}\|_{L^1(B^m)}.$$

For $m \leq L$ this tends to zero by (5.3). The remaining terms are at most one half the sum of the two probabilities of more than L points in B , which tend to zero uniformly as $L \rightarrow \infty$. This proves (5.2). $\square$

6 Entire transforms and determination by correlations

Proposition 6.1 (Generating functional). *For every bounded complex-valued Borel function g with relatively compact support in $\mathbb{D}$,*

$$\mathbb{E} \prod_{x \in \Xi} (1 + g(x)) = \sum_{p=0}^{\infty} \frac{1}{p!} \int_{\mathbb{D}^p} \prod_{i=1}^p g(z_i) \rho_p(z) \, dA^p(z). \quad (6.1)$$

The series is absolutely convergent. In particular, the full correlation hierarchy determines the law of Ξ .

Proof. Expand the finite product over the points in a relatively compact Borel set B containing the support of g . The absolute sum of the expected terms is at most

$$\sum_{p \geq 0} \frac{\|g\|_{\infty}^p}{p!} \mathbb{E}(M_B)_p = \mathbb{E}(1 + \|g\|_{\infty})^{M_B} < \infty.$$

This proves the identity and absolute convergence. Taking $g = e^{-f} - 1$ for nonnegative continuous compactly supported f recovers the Laplace functional, which determines the law of a locally finite random measure. $\square$

Corollary 6.2 (Entire moment and probability generating functions). *For every relatively compact Borel set B , the function*

$$G_B(s) = \mathbb{E} s^{M_B}, \quad s \in \mathbb{C},$$

is entire, and

$$G_B(s) = \sum_{p=0}^{\infty} \frac{(s-1)^p}{p!} \int_{B^p} \rho_p(z) \, dA^p(z). \quad (6.2)$$

For bounded compactly supported real-valued Borel functions $f_1, \dots, f_q$, the function

$$(z_1, \dots, z_q) \mapsto \mathbb{E} \exp \left(\sum_{a=1}^q z_a \langle f_a, \Xi \rangle \right)$$

is entire on $\mathbb{C}^q$. Its moments determine the joint distribution of the linear statistics. Along a process-convergent subsequence, the corresponding finite- N transforms converge locally uniformly in the complex parameters, even for these Borel test functions.

Proof. On each compact set of complex parameters, the integrands and all their parameter derivatives are dominated by a constant times e^{tM_B} for some finite t , where B contains all supports. The exponential moment bounds justify differentiation and the power-series expansions of every order. Formula (6.2) is also the special case $g = (s-1)\mathbf{1}_B$ of (6.1).

For convergence, first restrict to configurations with at most L points in B . The integrands are then bounded, so local total variation convergence applies. The contribution of the omitted configurations tends to zero uniformly in N and uniformly on compact parameter sets, by a slightly larger exponential moment. This proves the final assertion. Entire transforms are uniquely determined by their Taylor coefficients. Their values on imaginary parameters give the joint characteristic function, proving moment determination. $\square$

Remark 6.3 (Growth from the Gaussian count tail). The stronger compact count estimate obtained from the radial comparison has the form $\mathbb{P}(M_B \geq m) \leq Ce^{-cm^2}$, after adjustment of constants. Summing the tail and completing the square gives

$$\mathbb{E}e^{tM_B} \leq C \exp(Ct^2 + Ct), \quad t \geq 0.$$

Thus the entire moment generating function of each bounded compactly supported linear statistic has growth order at most two. Likewise,

$$\log \sup_{|s| \leq R} |G_B(s)| = O((\log(1+R))^2 + 1), \quad R \rightarrow \infty.$$

All these estimates may be chosen uniformly for the finite- N processes.

7 A simultaneous comparison for the ranked radii

For a locally finite counting measure ξ on $\mathbb{D}$, let $R_k(\xi)$ be the k th smallest modulus of its points, counting multiplicities. Set $R_k(\xi) = 1$ if $\xi(\mathbb{D}) < k$, and put

$$X_k(\xi) = -\log R_k(\xi), \quad -\log 0 = +\infty.$$

Thus the ranked depths form a nonincreasing sequence in $[0, +\infty]$; zeros are appended when there are only finitely many points. Recall that E_k , $k \geq 1$, are independent exponentials with rates

$$a_k = 2k + \frac{\beta}{2}k(k-1), \quad T_k = \sum_{j=k}^{\infty} E_j.$$

Since $\sum_j a_j^{-1} < \infty$, all T_k are finite almost surely and $T_k \downarrow 0$ almost surely.

Proposition 7.1 (Coupling of all ranked depths). *For each fixed N , there is a coupling of Ξ_N with $(T_k)_{k \geq 1}$ such that*

$$X_k(\Xi_N) \leq T_k \quad \text{for every } k \geq 1 \quad \text{almost surely.} \quad (7.1)$$

The same assertion holds, in a separate coupling, for every subsequential limit Ξ furnished by Theorem 1.1. In particular, if

$$L(t) = \#\{k \geq 1 : T_k > t\}, \quad t > 0,$$

then in these couplings

$$M_N(r) \leq L(\log(1/r)), \quad M(r) \leq L(\log(1/r)), \quad 0 < r < 1, \quad (7.2)$$

simultaneously in r .

Proof. The radial comparison used in Theorem 1.1 shows that the gap vector is stochastically dominated by $(E_1, \dots, E_N)$. Summing coordinates gives coordinatewise stochastic domination of the ranked depths by

$$\left(\sum_{j=k}^N E_j \right)_{1 \leq k \leq N},$$

with zeros appended. This latter vector is coordinatewise at most $(T_k)_{k \geq 1}$. Apply Strassen's theorem first to the finite depth vector and $(T_k^{(N)})_{k \leq N}$. Differences of the latter recover the independent $E_1, \dots, E_N$. Append independent E_j for $j > N$; this gives (7.1) for each fixed N .

For completeness, ranked radii are continuous under vague convergence of locally finite counting measures on $\mathbb{D}$, with values in $[0, 1]$. Indeed, to locate a ranked radius one brackets it by radii $u < v < 1$ whose circles contain no points of the limiting configuration; the counts in B_u and B_v then converge. Such brackets can approach the ranked radius from both sides. At the endpoints 0 and 1, only the nontrivial side of the bracket is needed. Consequently the map to $(X_k)_{k \geq 1}$ is continuous into the product space $[0, +\infty]^\mathbb{N}$, where the interval has its compact topology.

Along any convergent subsequence of Ξ_N , take the couplings above and extract a weakly convergent subsequence of their joint laws of ranked depths and (T_k) . These laws are tight, and their second marginal is fixed. The set

$$\{(x, t) : x_k \leq t_k \text{ for all } k \geq 1\}$$

is closed. The limiting joint law therefore couples the ranked depths of Ξ with (T_k) as claimed. If desired, a regular conditional distribution of Ξ given its ranked depths lifts this to a coupling of the full configuration with (T_k) ; the same observation applies to the finite- N couplings. Finally, $M(r) = \#\{k : X_k(\Xi) > \log(1/r)\}$, and the same identity holds for $M_N(r)$, proving (7.2). $\square$

8 Explicit occupancy and hole bounds

Proposition 8.1 (A gamma-function occupancy bound). *Let $c = 4/\beta$. For every $k \geq 1$, $0 < r < 1$, and every N ,*

$$\mathbb{P}\{M_N(r) \geq k\} \leq \frac{\Gamma(2k + c)}{\Gamma(k + 1)\Gamma(k + c)} r^{2k + \beta k(k-1)/2}. \quad (8.1)$$

The same bound holds with M_N replaced by M for every subsequential limit. In particular,

$$\mathbb{P}\{M(r) > 0\} \leq (1 + 4/\beta)r^2, \quad \mathbb{P}\{M_N(r) > 0\} \leq (1 + 4/\beta)r^2. \quad (8.2)$$

Proof. Write $t = \log(1/r)$. By Proposition 7.1, it suffices to bound $\mathbb{P}\{T_k > t\}$. Conditioning on T_{k+1} and using its independence of E_k gives

$$\mathbb{P}\{E_k + T_{k+1} > t \mid T_{k+1}\} \leq e^{-a_k t} e^{a_k T_{k+1}}.$$

We compute the exponential moment on the right. Since

$$a_j = \frac{\beta}{2}j(j + c - 1), \quad a_j - a_k = \frac{\beta}{2}(j - k)(j + k + c - 1),$$

for $J > k$ independence gives

$$\begin{aligned}\mathbb{E} \exp \left(a_k \sum_{j=k+1}^J E_j \right) &= \prod_{j=k+1}^J \frac{a_j}{a_j - a_k} \\ &= \frac{\Gamma(J+1)}{\Gamma(k+1)\Gamma(J-k+1)} \frac{\Gamma(J+c)}{\Gamma(k+c)} \frac{\Gamma(2k+c)}{\Gamma(J+k+c)}.\end{aligned}$$

The usual ratio asymptotic $\Gamma(x+b)/\Gamma(x+d) \sim x^{b-d}$ shows that the last expression tends to $\Gamma(2k+c)/(\Gamma(k+1)\Gamma(k+c))$. Monotone convergence proves that this limit is exactly $\mathbb{E} e^{a_k T_{k+1}}$. Thus

$$\mathbb{P}\{T_k > t\} \leq e^{-a_k t} \frac{\Gamma(2k+c)}{\Gamma(k+1)\Gamma(k+c)},$$

which proves the assertions. $\square$

Proposition 8.2 (Uniform positive hole probabilities). *Fix $0 < r < 1$ and put*

$$t = \log(1/r), \quad J = \max \left\{ 1, \left\lceil \frac{8}{\beta t} \right\rceil \right\}, \quad h_\beta(r) = \frac{1}{2} \prod_{j=1}^J \left(1 - e^{-a_j t/(2J)} \right) > 0.$$

Then

$$\mathbb{P}\{M_N(r) = 0\} \geq h_\beta(r), \quad \mathbb{P}\{M(r) = 0\} \geq h_\beta(r). \quad (8.3)$$

Consequently every compact subset of $\mathbb{D}$ has a positive hole probability, uniformly in N and in the choice of subsequential limit.

Proof. For $j \geq 2$, $a_j \geq \beta j(j-1)/2$, whence

$$\mathbb{E} T_{J+1} \leq \frac{2}{\beta J} \leq \frac{t}{4}.$$

Markov's inequality gives $\mathbb{P}\{T_{J+1} \leq t/2\} \geq 1/2$. Independently, the event $E_j < t/(2J)$ for every $1 \leq j \leq J$ has probability equal to the product in the definition of $h_\beta(r)$. Their intersection implies $T_1 < t$. By (7.2), this event forces both counts, in their respective couplings, to vanish. Any compact subset of $\mathbb{D}$ is contained in some B_r . $\square$

9 The pure-death comparison and boundary growth

Proposition 9.1 (A mean occupancy bound). *For every $0 < r < 1$,*

$$\mathbb{E} M_N(r) \leq \frac{1}{1 - r^{\beta/2}} \leq 1 + \frac{2}{\beta \log(1/r)}, \quad \mathbb{E} M(r) \leq \frac{1}{1 - r^{\beta/2}}. \quad (9.1)$$

Proof. For $n \geq 1$, define

$$L_n(t) = \# \left\{ 1 \leq k \leq n : \sum_{j=k}^n E_j > t \right\}.$$

This is a pure-death chain starting from n : its holding time at state k is E_k , and the transition from k to $k-1$ has rate a_k . Set $a_0 = 0$ and $m_n(t) = \mathbb{E} L_n(t)$. The elementary generator identity and Jensen's inequality give

$$m'_n(t) = -\mathbb{E} a_{L_n(t)} \leq -\frac{\beta}{2} \mathbb{E}[L_n(t)(L_n(t) - 1)] \leq -\frac{\beta}{2} m_n(t)(m_n(t) - 1).$$

Comparing with the scalar differential equation with initial value n yields

$$m_n(t) \leq \frac{1}{1 - (1 - 1/n)e^{-\beta t/2}}.$$

In the construction using the common sequence (E_j) , $L_n(t) \uparrow L(t)$ for each $t > 0$. Monotone convergence therefore gives

$$\mathbb{E}L(t) \leq \frac{1}{1 - e^{-\beta t/2}}.$$

Apply Proposition 7.1. The second inequality in (9.1) uses $1/(1 - e^{-x}) = 1 + 1/(e^x - 1) \leq 1 + 1/x$ for $x > 0$. $\square$

Lemma 9.2 (The reference process near the boundary). *Almost surely,*

$$kT_k \rightarrow \frac{2}{\beta} \quad (k \rightarrow \infty), \quad tL(t) \rightarrow \frac{2}{\beta} \quad (t \downarrow 0). \quad (9.2)$$

Proof. Since $a_j \sim \beta j^2/2$,

$$b_k := \mathbb{E}T_k = \sum_{j=k}^{\infty} a_j^{-1} \sim \frac{2}{\beta k}, \quad \text{Var}(T_k) = \sum_{j=k}^{\infty} a_j^{-2} = O(k^{-3}).$$

For every $\varepsilon > 0$, Chebyshev's inequality gives

$$\sum_{\ell=1}^{\infty} \mathbb{P}\{\ell^2 |T_{\ell^2} - b_{\ell^2}| > \varepsilon\} < \infty.$$

The Borel–Cantelli lemma, first for rational $\varepsilon > 0$, implies $\ell^2 T_{\ell^2} \rightarrow 2/\beta$ almost surely. If $\ell^2 \leq k \leq (\ell + 1)^2$, monotonicity of T_k gives

$$\ell^2 T_{(\ell+1)^2} \leq kT_k \leq (\ell + 1)^2 T_{\ell^2}.$$

Both outer quantities tend to $2/\beta$, proving the first assertion.

All E_j are strictly positive almost surely, so T_k is strictly decreasing and $L(t) \rightarrow \infty$ as $t \downarrow 0$. If $k = L(t)$, then $T_{k+1} \leq t < T_k$. Hence

$$kT_{k+1} \leq kt < kT_k.$$

The first assertion makes both outer quantities tend to $2/\beta$. This proves the second assertion. $\square$

Corollary 9.3 (Almost sure upper boundary growth). *Every subsequential limit satisfies*

$$\limsup_{r \uparrow 1} (1 - r)M(r) \leq \frac{2}{\beta} \quad \text{almost surely.} \quad (9.3)$$

This is an upper bound; no positive lower boundary growth or infinitude of the limiting configuration is asserted.

Proof. In the coupling of Proposition 7.1, $M(e^{-t}) \leq L(t)$ simultaneously for all $t > 0$. Apply Lemma 9.2 and $(1 - e^{-t})/t \rightarrow 1$. $\square$

10 Weighted summability of the limiting configuration

Proposition 10.1 (Exponential moments of depth norms). *For $p > 1$, define*

$$S_p(\xi) = \sum_{z \in \xi} (-\log |z|)^p,$$

where points are counted with multiplicity. For every $0 \leq \lambda < 2$,

$$\sup_{N \geq 1} \mathbb{E} \exp\{\lambda S_p(\Xi_N)^{1/p}\} \leq \prod_{j=1}^{\infty} \left(1 - \frac{\lambda j^{1/p}}{a_j}\right)^{-1} < \infty. \quad (10.1)$$

The same bound holds for $\mathbb{E} \exp\{\lambda S_p(\Xi)^{1/p}\}$ for any subsequential limit. In particular, these weighted sums have all positive moments, and

$$\sum_{z \in \Xi} (1 - |z|)^p < \infty \quad \text{almost surely for every } p > 1. \quad (10.2)$$

Proof. By the ranked coupling and Minkowski's inequality in ℓ^p ,

$$\begin{aligned} S_p(\Xi)^{1/p} &\leq \left(\sum_{k=1}^{\infty} T_k^p \right)^{1/p} \\ &= \left\| \sum_{j=1}^{\infty} E_j (\mathbf{1}_{\{k \leq j\}})_{k \geq 1} \right\|_{\ell^p} \leq \sum_{j=1}^{\infty} j^{1/p} E_j. \end{aligned}$$

The same inequalities hold for Ξ_N . One can first apply Minkowski's inequality to finite sums and then use monotone convergence. Since $p > 1$,

$$\sum_{j=1}^{\infty} \frac{j^{1/p}}{a_j} < \infty,$$

so the sum on the right is finite almost surely. Independence and monotone convergence give

$$\mathbb{E} \exp \left\{ \lambda \sum_{j=1}^{\infty} j^{1/p} E_j \right\} = \prod_{j=1}^{\infty} \left(1 - \frac{\lambda j^{1/p}}{a_j} \right)^{-1}.$$

Every factor is finite: $a_j \geq 2j$ implies $\lambda j^{1/p}/a_j \leq \lambda/2 < 1$. Moreover,

$$\sum_{j=1}^{\infty} -\log \left(1 - \frac{\lambda j^{1/p}}{a_j} \right) \leq \frac{\lambda}{1 - \lambda/2} \sum_{j=1}^{\infty} \frac{j^{1/p}}{a_j} < \infty.$$

This proves (10.1) and all asserted moment bounds.

The inequality $1 - r \leq -\log r$ proves (10.2) for each fixed $p > 1$. To obtain the statement simultaneously for all such p , first intersect the probability-one events for rational $p > 1$, and then, for any given $p > 1$, choose a rational q with $1 < q < p$ and use $(1 - r)^p \leq (1 - r)^q$. $\square$

Remark 10.2. The radial estimates provide upper occupancy bounds, positive local hole probabilities, and summability restrictions. They supply neither a lower bound on the probability of seeing a point in a compact set nor uniqueness of the subsequential law. In particular, they do not by themselves exclude the identically empty limit.

11 Symmetry and the radial intensity

Throughout this section, Ξ is a subsequential limit and ρ_p denotes its continuous p -point correlation density provided by Theorem 4.1. Along the subsequence defining Ξ , write $\rho_{p,N} \rightarrow \rho_p$ for the locally uniform convergence established there. All assertions concern this arbitrary subsequential limit.

Proposition 11.1 (Rotation and reflection symmetry). *The law of Ξ is invariant under every rotation about the origin and under complex conjugation. Consequently, for every $p \geq 1$, $\theta \in \mathbb{R}$, and $z_1, \dots, z_p \in \mathbb{D}$,*

$$\rho_p(e^{i\theta} z_1, \dots, e^{i\theta} z_p) = \rho_p(z_1, \dots, z_p), \quad \rho_p(\bar{z}_1, \dots, \bar{z}_p) = \rho_p(z_1, \dots, z_p).$$

In particular, the one-point intensity is radial.

Proof. Rotations and conjugation preserve planar area, the disk, and every pairwise distance, and therefore preserve each finite- N Gibbs law. If T is one of these maps, the induced map on point measures is continuous for the vague topology: for $f \in C_c(\mathbb{D})$, $(T_*\xi)(f) = \xi(f \circ T)$ and $f \circ T \in C_c(\mathbb{D})$. The continuous mapping theorem passes invariance to every subsequential limit. The corresponding correlation measures are invariant as well. Their continuous densities satisfy the displayed identities first almost everywhere, and then everywhere by continuity. $\square$

Lemma 11.2 (Log-subharmonicity of the finite- N intensity). *For every $N \geq 1$, the function $\rho_{1,N}$ is strictly positive on $\mathbb{D}$, and $\log \rho_{1,N}$ is subharmonic. The radial function*

$$g_N(t) := \log \rho_{1,N}(e^t), \quad -\infty < t < 0,$$

is convex and nondecreasing. In particular, $r \mapsto \rho_{1,N}(r)$ is nondecreasing on $[0, 1)$.

Proof. Integrating out $N - 1$ particles gives a representation

$$\rho_{1,N}(z) = \int |P_y(z)|^\beta \, d\nu_N(y), \quad P_y(z) = \prod_{j=1}^{N-1} (z - y_j),$$

where ν_N is the finite positive measure containing the normalization constant and the Vandermonde interaction among the y_j . For $N = 1$, the polynomial and the integral over the empty tuple have their usual empty-product interpretation. The integrand is strictly positive for almost every y , for each fixed z , so $\rho_{1,N}(z) > 0$.

For $\varepsilon > 0$, set

$$\phi_{y,\varepsilon}(z) = \frac{\beta}{2} \log(|P_y(z)|^2 + \varepsilon), \quad H_\varepsilon(z) = \log \int e^{\phi_{y,\varepsilon}(z)} \, d\nu_N(y).$$

The functions $\phi_{y,\varepsilon}$ are smooth and subharmonic; indeed,

$$\Delta \log(|P_y|^2 + \varepsilon) = \frac{4\varepsilon |P'_y|^2}{(|P_y|^2 + \varepsilon)^2} \geq 0,$$

where Δ is the ordinary real Laplacian. Polynomial bounds on compact sets justify differentiating the integral. Let $\mathbb{E}_{z,\varepsilon}$ denote expectation for the probability measure proportional to $e^{\phi_{y,\varepsilon}(z)} \, d\nu_N(y)$. Direct differentiation gives

$$\Delta H_\varepsilon = \mathbb{E}_{z,\varepsilon}[\Delta \phi_{y,\varepsilon}] + \mathbb{E}_{z,\varepsilon}[|\nabla \phi_{y,\varepsilon}|^2] - |\mathbb{E}_{z,\varepsilon}[\nabla \phi_{y,\varepsilon}]|^2 \geq 0.$$

As $\varepsilon \downarrow 0$, these subharmonic functions decrease to the finite function $\log \rho_{1,N}$. The decreasing-limit property for subharmonic functions proves the first assertion. Equivalently, one can pass directly to their submean inequalities, using the positive continuous function $\rho_{1,N}$ to give a local lower bound.

Rotation symmetry makes $\log \rho_{1,N}$ radial. On an annulus $e^a < |z| < e^b$, the harmonic function with its two constant boundary values is the affine interpolation of $g_N(a)$ and $g_N(b)$ in the variable $\log |z|$. The maximum principle therefore gives the convexity inequality for g_N . If g_N had a negative secant slope, convexity would imply $g_N(t) \rightarrow +\infty$ as $t \rightarrow -\infty$. This contradicts boundedness of $\log \rho_{1,N}$ near the origin. Thus g_N is nondecreasing. Continuity at the origin extends monotonicity to $r = 0$. $\square$

Proposition 11.3 (Intensity dichotomy and logarithmic convexity). *Exactly one of the following alternatives holds:*

1. Ξ is almost surely the empty configuration and $\rho_1 \equiv 0$;
2. Ξ is not almost surely empty and $\rho_1(r) > 0$ for every $0 < r < 1$.

In the second alternative,

$$t \mapsto \log \rho_1(e^t), \quad -\infty < t < 0,$$

is convex and nondecreasing. The density ρ_1 is continuous and radially nondecreasing on $\mathbb{D}$, and $\log \rho_1$, with $\log 0 = -\infty$, is subharmonic. Moreover, every fixed nonempty open set $U \subset \mathbb{D}$ satisfies

$$\mathbb{P}\{\Xi(U) > 0\} > 0.$$

The proposition does not assert $\rho_1(0) > 0$ or $\mathbb{P}\{\Xi(\mathbb{D}) > 0\} = 1$.

Proof. The identity $\mathbb{E}\Xi(A) = \int_A \rho_1 \, dA$, first for relatively compact sets and then by monotone convergence, shows that $\rho_1 \equiv 0$ is equivalent to Ξ being almost surely empty. Local uniform convergence and Lemma 11.2 show that ρ_1 is continuous, radial, and nondecreasing.

Suppose ρ_1 is not identically zero. Its intensity measure is nonzero, so there is $r_0 \in (0, 1)$ with $\rho_1(r_0) > 0$. Monotonicity already gives positivity for $r \geq r_0$. For $0 < s < r_0$, choose $r_0 < R < 1$ and set

$$\alpha = \frac{\log R - \log r_0}{\log R - \log s} \in (0, 1).$$

The finite- N convexity inequality is

$$\rho_{1,N}(r_0) \leq \rho_{1,N}(s)^\alpha \rho_{1,N}(R)^{1-\alpha}.$$

Passing to the locally uniform limit would contradict $\rho_1(r_0) > 0$ if $\rho_1(s) = 0$. Thus $\rho_1(s) > 0$ for every $s > 0$. Logs now converge uniformly on each compact radial interval in $(0, 1)$, and their limit is convex and nondecreasing in $\log r$.

A radial function that is convex in $\log r$ is subharmonic on the punctured disk. This follows, for example, by smoothing the convex function on compact logarithmic intervals and using $\Delta[g(\log r)] = g''(\log r)/r^2$ for the smooth approximants. The resulting function $\log \rho_1$ is locally bounded above near the origin. Its subharmonic extension there takes the value

$$\limsup_{z \rightarrow 0} \log \rho_1(z) = \log \rho_1(0),$$

by continuity and radial monotonicity of ρ_1 ; this value may be $-\infty$. The removable-singularity property for subharmonic functions gives the asserted subharmonicity on $\mathbb{D}$.

Finally, any nonempty open U contains a disk with compact closure in $\mathbb{D} \setminus \{0\}$. The integral of the positive continuous intensity over that disk is strictly positive. Its point count is a nonnegative integrable random variable with positive expectation and therefore has positive probability of being positive. This proves the open-set assertion. $\square$

Corollary 11.4 (An explicit local intensity bound). *For $0 \leq r < R < 1$, both the finite- N and limiting radial intensities satisfy*

$$\rho_{1,N}(r), \rho_1(r) \leq \frac{1}{\pi(R^2 - r^2)(1 - R^{\beta/2})}.$$

In particular, for $z \in \mathbb{D}$,

$$\rho_{1,N}(z), \rho_1(z) \leq \frac{8}{\pi \min\{\beta/2, 1\}(1 - |z|)^2}.$$

Proof. Radial monotonicity and Proposition 9.1 give

$$\pi(R^2 - r^2)\rho_1(r) \leq \int_{B_R \setminus B_r} \rho_1(z) dA(z) \leq \mathbb{E}M(R) \leq \frac{1}{1 - R^{\beta/2}},$$

and identically at finite N . Take $R = (1 + r)/2$ and use $R^2 - r^2 \geq (1 - r)/4$ and $1 - R^{\beta/2} \geq \min\{\beta/2, 1\}(1 - R)$. $\square$

Remark 11.5. The distinction between “not almost surely empty” and “almost surely nonempty” is essential. The positive-intensity alternative above does not exclude an atom at the empty configuration in the law of Ξ . That atom, and indeed every finite total count, will be excluded for $0 < \beta < 2$ by Theorem 48.6. Nor do the Euclidean symmetries proved here establish invariance under general conformal automorphisms of the disk.

12 The determinantal case as a check

This section verifies that the normalization and topology recover the Bergman process when $\beta = 2$. It is not used in the preceding estimates. The functions

$$p_j(z) = \sqrt{\frac{j+1}{\pi}} z^j, \quad j \geq 0,$$

are orthonormal in $L^2(\mathbb{D}, dA)$, by direct polar integration. The Vandermonde determinant and expansion of determinants give

$$\frac{|\Delta_N(z)|^2}{Z_{N,2}} = \frac{1}{N!} \det[K_N(z_i, z_j)]_{i,j=1}^N, \quad K_N(z, w) = \sum_{j=0}^{N-1} p_j(z) \overline{p_j(w)}, \quad Z_{N,2} = \pi^N.$$

Indeed, integrating the square of the determinant of the orthonormal functions leaves precisely the $N!$ matching permutations; the product of their squared leading coefficients is $N!/\pi^N$. Orthogonality also gives $\int K_N(z, u) K_N(u, w) dA(u) = K_N(z, w)$ and $\int K_N(u, u) dA(u) = N$. Expanding a determinant and using these two identities gives

$$\int_{\mathbb{D}} \det[K_N(z_i, z_j)]_{i,j=1}^k dA(z_k) = (N - k + 1) \det[K_N(z_i, z_j)]_{i,j=1}^{k-1}.$$

To check the coefficient, permutations fixing k give N times the smaller determinant, whereas inserting k into each of the $k - 1$ positions in the cycles of a smaller permutation gives the negative contribution $-(k - 1)$ times that determinant. Iterating this identity proves

$$\rho_{p,N}(z_1, \dots, z_p) = \det[K_N(z_i, z_j)]_{i,j=1}^p.$$

The geometric series and its derivative imply local uniform convergence

$$K_N(z, w) \longrightarrow K(z, w) = \frac{1}{\pi(1 - z\bar{w})^2}.$$

Every subsequential limit therefore has correlations $\det[K(z_i, z_j)]$. The determination-by-correlations result proved above makes their laws equal. Tightness then gives convergence of the entire sequence to this law, the Bergman determinantal process; see also [5, 6].

For this process,

$$\begin{aligned}\mathbb{E}M(r) &= \int_{B_r} \frac{dA(z)}{\pi(1-|z|^2)^2} = \frac{r^2}{1-r^2}, \\ \text{Var } M(r) &= \mathbb{E}M(r) - \int_{B_r^2} |K(z, w)|^2 dA(z) dA(w) \leq \mathbb{E}M(r).\end{aligned}$$

The variance identity follows by integrating $\rho_2 = \rho_1 \rho_1 - |K|^2$ and using $M^2 = (M)_2 + M$. Chebyshev's inequality now shows that the total number of points is infinite almost surely: for every fixed integer L , $\mathbb{P}(M(r) \leq L) \rightarrow 0$ as $r \uparrow 1$, and $\{\Xi(\mathbb{D}) \leq L\} \subset \{M(r) \leq L\}$. In fact

$$\lim_{r \uparrow 1} (1-r)M(r) = \frac{1}{2} \quad \text{almost surely.}$$

For a proof, use $r_n = 1 - n^{-2}$, $n \geq 2$. The means of $(1-r_n)M(r_n)$ tend to $1/2$, while their variances are $O(n^{-2})$. Chebyshev and Borel–Cantelli give almost sure convergence along this sequence. Monotonicity of $M(r)$ and $(1-r_n)/(1-r_{n+1}) \rightarrow 1$ extend it to all $r \uparrow 1$.

13 A positive radial expansion and its limitations for uniqueness

We give an exact refinement of the radial comparison. It applies to every real $\beta > 0$, without an integrality assumption. Although the resulting unnormalized angular factors are monotone in N , we then show that two natural ways to turn this observation into a convergence argument fail already at $\beta = 2$.

Order the radii as $0 < r_1 < \dots < r_N < 1$, which is possible almost surely, and put

$$t_k = r_k/r_{k+1}, \quad r_{N+1} = 1, \quad a_k = 2k + \frac{\beta}{2}k(k-1), \quad p = \frac{\beta}{2}.$$

Use normalized Haar measure on each angular circle, and define

$$F_N(t) = \int_{[0, 2\pi)^N} \prod_{i < j} \left| 1 - \left(\prod_{\ell=i}^{j-1} t_\ell \right) e^{i(\theta_i - \theta_j)} \right|^{2p} \prod_{i=1}^N \frac{d\theta_i}{2\pi}. \quad (13.1)$$

In particular, F_N does not depend on t_N . The ordered-ratio density, with respect to $dt_1 \dots dt_N$, is

$$\frac{1}{W_N} F_N(t) \prod_{k=1}^N t_k^{a_k-1}, \quad W_N = \frac{Z_{N,\beta}}{N!(2\pi)^N}. \quad (13.2)$$

For completeness, the change of variables gives $\prod_j r_j dr_j = \prod_k t_k^{2k-1} dt_k$. Factoring r_j^β from each pair $i < j$ contributes $\prod_k t_k^{\beta k(k-1)/2}$. Angular integration and the $N!$ possible radial orders give (13.2).

Proposition 13.1 (Positive Fourier expansion). *For $n \in \mathbb{N}_0$, let*

$$b_n = (-1)^n \binom{p}{n}, \quad \binom{p}{0} = 1, \quad \binom{p}{n} = \frac{p(p-1) \dots (p-n+1)}{n!} \quad (n \geq 1).$$

For $m = (m_1, \dots, m_{N-1}) \in \mathbb{N}_0^{N-1}$, define

$$B_{N,m} = \sum_{\substack{n_{ij} \in \mathbb{N}_0 \ (i < j) \\ \sum_{i \leq k < j} n_{ij} = m_k \ (1 \leq k < N)}} \prod_{i < j} b_{n_{ij}}. \quad (13.3)$$

Each sum in (13.3) is finite. For every $t \in (0, 1)^N$,

$$F_N(t) = \sum_{m \in \mathbb{N}_0^{N-1}} B_{N,m}^2 \prod_{k=1}^{N-1} t_k^{2m_k}. \quad (13.4)$$

The series has nonnegative terms and is finite at every such t .

Proof. The branch of $(1 - u)^p$ that equals 1 at $u = 0$ is holomorphic for $|u| < 1$ and has the absolutely convergent series $(1 - u)^p = \sum_{n \geq 0} b_n u^n$. Since p is real, its modulus is $|1 - u|^p$. Apply this expansion to the analytic product

$$A_N(t, \theta) = \prod_{i < j} \left(1 - \left(\prod_{\ell=i}^{j-1} t_\ell \right) e^{i(\theta_i - \theta_j)} \right)^p.$$

For fixed $t \in (0, 1)^N$, all the arguments have modulus strictly below 1. The product of the finitely many absolutely convergent series therefore converges absolutely and uniformly in θ .

A choice of edge multiplicities $(n_{ij})_{i < j}$ contributes the radial monomial $\prod_{k < N} t_k^{m_k}$, where $m_k = \sum_{i \leq k < j} n_{ij}$, and the angular character $e^{i \sum_i \alpha_i \theta_i}$, where

$$\alpha_i = \sum_{j > i} n_{ij} - \sum_{j < i} n_{ji}.$$

These quantities satisfy

$$m_k = \sum_{i=1}^k \alpha_i.$$

Conversely, with the conventions $m_0 = m_N = 0$, the angular charge is uniquely determined by m through $\alpha_i = m_i - m_{i-1}$. Thus different crossing vectors m yield different Fourier characters. For a fixed m , every contributing edge multiplicity obeys

$$0 \leq n_{ij} \leq \min_{i \leq k < j} m_k.$$

There are only finitely many such edge arrays. This proves the asserted finiteness of (13.3) and justifies regrouping the absolutely convergent series as

$$A_N(t, \theta) = \sum_m B_{N,m} \left(\prod_{k < N} t_k^{m_k} \right) \exp \left(i \sum_{i=1}^N (m_i - m_{i-1}) \theta_i \right).$$

Orthogonality of distinct characters in normalized Haar measure, or Parseval's identity, now gives (13.4), since $|A_N(t, \theta)|^2$ is the integrand in (13.1). The coefficients are real, so their squared moduli are $B_{N,m}^2$. $\square$

Corollary 13.2 (An exact mixture of independent gaps). *In every expression indexed by $m \in \mathbb{N}_0^{N-1}$, set $m_N = 0$. The numbers*

$$\pi_N(m) = \frac{B_{N,m}^2}{W_N} \prod_{k=1}^N \frac{1}{a_k + 2m_k} \quad (13.5)$$

form a probability mass function. The ordered radial gaps $X_k = -\log t_k$ have the following exact construction: first sample m with law π_N , and then, conditionally on m , sample independently

$$X_k \sim \text{Exp}(a_k + 2m_k), \quad 1 \leq k \leq N.$$

Here $\text{Exp}(\lambda)$ denotes the exponential distribution of rate λ . In particular, $X_N \sim \text{Exp}(a_N)$ is independent of $(X_1, \dots, X_{N-1})$.

Proof. Tonelli's theorem applied to the nonnegative series (13.4) gives

$$W_N = \int_{(0,1)^N} F_N(t) \prod_k t_k^{a_k-1} dt = \sum_m B_{N,m}^2 \prod_{k=1}^N \frac{1}{a_k + 2m_k}.$$

The normalization is finite by (13.2) and positive because $B_{N,0} = 1$. Substituting this identity into the ratio density expresses that density as the mixture, with weights (13.5), of the product densities

$$\prod_{k=1}^N (a_k + 2m_k) t_k^{a_k+2m_k-1}.$$

Taking negative logarithms gives the claimed exponential distributions. Since the conditional distribution of X_N is the same for every m , its conditional independence also gives its unconditional independence from all the other gaps. $\square$

Proposition 13.3 (Monotone unnormalized factors do not give a martingale). *The angular factors satisfy*

$$F_{N+1}(t_1, \dots, t_N) \geq F_N(t_1, \dots, t_{N-1}), \quad t_k \in (0, 1). \quad (13.6)$$

Nevertheless, on the common product reference space

$$\mu_\infty = \bigotimes_{k \geq 1} \text{Beta}(a_k, 1), \quad H_N = \frac{F_N(U_1, \dots, U_{N-1})}{\mathbb{E}_{\mu_\infty} F_N},$$

the normalized factors need not be a martingale, a submartingale, or a supermartingale for the filtration $\mathcal{F}_N = \sigma(U_1, \dots, U_{N-1})$.

Proof. If the last crossing coordinate for $N+1$ vertices is $m_N = 0$, then $\sum_{i \leq N} n_{i,N+1} = 0$. Nonnegativity forces all edges into vertex $N+1$ to have multiplicity zero. Hence

$$B_{N+1,(m,0)} = B_{N,m}.$$

The terms with last coordinate zero reproduce the whole series for F_N ; all remaining terms in F_{N+1} are nonnegative. This proves (13.6).

To test the normalized factors, take $\beta = 2$, so that $a_k = k(k+1)$. Put $x = U_1^2$ and $y = U_2^2$. Direct expansion of the Vandermonde, or (13.4), gives

$$F_2 = 1 + x, \quad F_3 = 1 + x + y + xy^2 + x^2y + x^2y^2.$$

The reference laws give $\mathbb{E}y = 6/(6+2) = 3/4$ and $\mathbb{E}y^2 = 6/(6+4) = 3/5$. Therefore

$$\mathbb{E}[F_3 \mid \mathcal{F}_2] = \frac{7}{4} + \frac{8}{5}x + \frac{27}{20}x^2.$$

Since $\mathbb{E}x = 1/2$ and $\mathbb{E}x^2 = 1/3$, we have $\mathbb{E}F_2 = 3/2$ and $\mathbb{E}F_3 = 3$. It follows that

$$\mathbb{E}[H_3 \mid \mathcal{F}_2] - H_2 = \frac{1}{3} \left(-\frac{1}{4} - \frac{2}{5}x + \frac{27}{20}x^2 \right). \quad (13.7)$$

The polynomial in parentheses is negative at $x = 0$ and positive at $x = 1$. Since x is uniform on $(0, 1)$, both signs occur with positive probability. This disproves each of the three asserted types of conditional ordering for this example. $\square$

Remark 13.4 (Positive coefficients do not imply log-supermodularity). There is a second obstruction to inferring a monotone coupling from the positive expansion. With $u = t_1^2$ and $v = t_2^2$, the same $\beta = 2$ example gives

$$F_3(u, v) = 1 + u + v + uv^2 + u^2v + u^2v^2.$$

At $(u, v) = (0, 0)$, its value and first derivatives equal 1, whereas $\partial_u \partial_v F_3(0, 0) = 0$. Consequently,

$$\partial_u \partial_v \log F_3(0, 0) = -1.$$

This mixed derivative remains negative for all sufficiently small positive u, v , by continuity. Thus F_3 is not log-supermodular even in the interior of its domain. Equivalently, in the original positive coordinates the mixed derivative is multiplied by $4t_1t_2 > 0$ and still has negative sign. Multiplication by the product reference density does not change a mixed derivative of its logarithm. Therefore positive Fourier coefficients alone do not supply the log-supermodularity hypothesis used in the corresponding FKG lattice criterion. Failure of that sufficient criterion is not a claim that the resulting law fails every possible association inequality.

Proposition 13.5 (Failure of uniform integrability at $\beta = 2$). *At $\beta = 2$, the normalized factors (H_N) in Proposition 13.3 are not uniformly integrable on $(\mu_\infty, \mathcal{F}_\infty)$. The reference ordered depths and the Bergman ordered depths have mutually singular laws: their almost-sure asymptotics are respectively*

$$k \sum_{j \geq k} (-\log U_j) \longrightarrow 1, \quad k D_k \longrightarrow \frac{1}{2}. \quad (13.8)$$

The only determinantal input in the proof is the $\beta = 2$ radial law, which we recall and derive below; no general- β convergence or uniqueness statement is used.

Proof. Under μ_∞ , the variables $X_j = -\log U_j$ are independent exponentials of rates $j(j+1)$. Their tail sums $S_k = \sum_{j \geq k} X_j$ are finite almost surely, and

$$\mathbb{E} S_k = \sum_{j \geq k} \frac{1}{j(j+1)} = \frac{1}{k}, \quad \text{Var } S_k = \sum_{j \geq k} \frac{1}{j^2(j+1)^2} \leq \frac{C}{k^3}.$$

For any $\varepsilon > 0$, Chebyshev's inequality gives $\mathbb{P}(|kS_k - 1| > \varepsilon) \leq C/(\varepsilon^2 k)$. Apply Borel–Cantelli at $k = n^2$, first for every positive rational ε , and then interpolate between successive squares using the monotonicity of S_k . This proves the first almost-sure limit in (13.8).

At $\beta = 2$, the unordered radii of the finite- N gas have the law of independent variables R_j with densities $2j r^{2j-1} dr$, $1 \leq j \leq N$, after forgetting their labels. Indeed, expand $\Delta_N(z) = \det[z_i^{j-1}]_{i,j=1}^N$. In its squared modulus, angular integration kills all terms with distinct permutations. The remaining sum of radial monomials, multiplied by $\prod_i r_i dr_i$, is exactly the symmetrization of these independent radial densities after normalization. Thus the corresponding unordered depths have the law of independent variables $Y_j = -\log R_j \sim \text{Exp}(2j)$. We may use one infinite independent sequence (Y_j) to couple these radial laws for every N .

For $x > 0$, let

$$M(x) = \sum_{j \geq 1} \mathbf{1}_{\{Y_j \geq x\}}.$$

This count is almost surely finite, since

$$\mathbb{E} M(x) = \frac{1}{e^{2x} - 1}, \quad \text{Var } M(x) \leq \mathbb{E} M(x).$$

Chebyshev and Borel–Cantelli at $x = n^{-2}$ show that $xM(x) \rightarrow 1/2$ almost surely along this sequence. Monotonicity of M and the fact that successive values of n^{-2} have ratio tending to 1 extend the limit to all $x \downarrow 0$. The sequence (Y_j) has no ties almost surely and contains infinitely many positive numbers, with only finitely many above each positive threshold. It therefore has a decreasing ordering $D_1 > D_2 > \dots > 0$, with $D_k \downarrow 0$ and $M(D_k) = k$. The full limit $xM(x) \rightarrow 1/2$, evaluated at $x = D_k$, gives $kD_k \rightarrow 1/2$ almost surely. This is the Bergman radial law and proves the second limit in (13.8). The two almost-sure limits define disjoint measurable sets of depth sequences, which proves the stated mutual singularity.

We give a direct argument showing why this singularity actually precludes uniform integrability of (H_N) ; uniform integrability alone would not imply L^1 convergence of an arbitrary sequence. Define Q_N on the common reference space by $dQ_N = H_N d\mu_\infty$. By (13.2), the first N coordinates under Q_N are exactly the finite- N gap variables. In particular,

$$D_{k,N} := \sum_{j=k}^N X_j$$

under Q_N has the law of the k th largest of $Y_1, \dots, Y_N$. For every fixed k , the latter increases almost surely to D_k as $N \rightarrow \infty$.

For $N \geq k$, set

$$A_{k,N} = \left\{ \left| k \sum_{j=k}^N X_j - \frac{1}{2} \right| < \frac{1}{8} \right\}.$$

For fixed k , as $N \rightarrow \infty$,

$$\begin{aligned} \mu_\infty(A_{k,N}) &\longrightarrow \mu_\infty \left(\left| kS_k - \frac{1}{2} \right| < \frac{1}{8} \right), \\ Q_N(A_{k,N}) &\longrightarrow \mathbb{P} \left(\left| kD_k - \frac{1}{2} \right| < \frac{1}{8} \right). \end{aligned}$$

There are no boundary atoms: S_k is the sum of an exponential variable and an independent variable, and D_k is one of a countable family of continuously distributed variables. The two right-hand sides tend respectively to 0 and 1 as $k \rightarrow \infty$, by (13.8). Choose an increasing sequence $N_k \geq k$ large enough that both preceding finite- N probabilities are within $1/k$ of their respective limits. Then

$$\mu_\infty(A_{k,N_k}) \longrightarrow 0, \quad \int_{A_{k,N_k}} H_{N_k} d\mu_\infty = Q_{N_k}(A_{k,N_k}) \longrightarrow 1.$$

This contradicts uniform absolute continuity of the integrals of a uniformly integrable family. Hence (H_N) is not uniformly integrable. $\square$

Remark 13.6. The positive expansion supplies an exact finite- N mixture and a stronger form of radial domination. It does not establish convergence of the normalized mixing probabilities π_N , nor does it identify the angular law of a possible limit. The preceding obstructions concern specific monotonicity, martingale, and uniform-integrability arguments; they neither prove nor disprove uniqueness for general β .

14 Partition functions and a vanishing central-density subsequence

Throughout this section, dA is ordinary planar Lebesgue measure, so $A(\mathbb{D}) = \pi$. Set

$$Z_{N,\beta} = \int_{\mathbb{D}^N} |\Delta_N(z)|^\beta dA^{\otimes N}(z), \quad Q_{N,\beta} = \frac{Z_{N,\beta}}{N!}, \quad Q_{0,\beta} = Z_{0,\beta} = 1.$$

Here $\rho_{N,\beta} = \rho_{1,N}$ denotes the one-point correlation density relative to dA .

Theorem 14.1 (Leading partition-function asymptotics). *For every fixed $\beta > 0$,*

$$\log Q_{N,\beta} = \left(\frac{\beta}{2} - 2\right) N \log N + O_\beta(N), \quad N \longrightarrow \infty. \quad (14.1)$$

More explicitly, for $\beta \geq 2$, $N \geq 2$, and $s_N = 1 - 1/N$,

$$\pi^N (1 - s_N^2)^N s_N^{\beta N(N-1)/2} (N!)^{\beta/2-1} \leq Q_{N,\beta} \leq \frac{\pi^N N^{(\beta/2-1)N}}{N!}. \quad (14.2)$$

For $0 < \beta \leq 2$ and $N \geq 1$,

$$\frac{\pi^N N^{(\beta/2-1)N}}{N!} \leq Q_{N,\beta} \leq \frac{(2\pi)^N (N!)^{\beta/2}}{\prod_{j=1}^N \left(2j + \frac{\beta}{2} j(j-1)\right)}. \quad (14.3)$$

Proof. First, Hadamard's determinant inequality, applied to the Vandermonde matrix whose i th row is $(1, z_i, \dots, z_i^{N-1})$, gives

$$|\Delta_N(z)| \leq N^{N/2}, \quad z \in \overline{\mathbb{D}}^N. \quad (14.4)$$

The exact identity $Z_{N,2} = \pi^N$ follows by expanding the two determinants in $|\Delta_N|^2$ and using

$$\int_{\mathbb{D}} z^j \bar{z}^k dA(z) = \mathbf{1}_{\{j=k\}} \frac{\pi}{j+1}.$$

Indeed, the only surviving pairs of permutations coincide, and their total contribution is

$$N! \prod_{j=0}^{N-1} \frac{\pi}{j+1} = \pi^N.$$

Suppose first that $\beta \geq 2$. Then (14.4) consequently yields

$$Z_{N,\beta} \leq N^{(\beta-2)N/2} Z_{N,2} = \pi^N N^{(\beta/2-1)N}.$$

This proves the upper bound in (14.2).

For the lower bound, let $dm(\theta) = d\theta/(2\pi)$ and define

$$A_{N,\beta}(r_1, \dots, r_N) = \int_{[0,2\pi)^N} |\Delta_N(r_1 e^{i\theta_1}, \dots, r_N e^{i\theta_N})|^\beta dm^{\otimes N}(\theta).$$

For fixed values of all coordinates except the i th, the integrand is the β th power of the modulus of a polynomial in the i th complex coordinate. It is therefore subharmonic in that coordinate. The circular mean of a subharmonic function on a disk is nondecreasing in its radius. Integrating over the other angles proves that $A_{N,\beta}$ is nondecreasing in each r_i . By homogeneity, for $r_i \in [s_N, 1]$,

$$A_{N,\beta}(r_1, \dots, r_N) \geq s_N^{\beta N(N-1)/2} A_{N,\beta}(1, \dots, 1).$$

On the torus, orthogonality of the monomials similarly gives

$$A_{N,2}(1, \dots, 1) = N!.$$

Jensen's inequality for the convex function $x \mapsto x^{\beta/2}$, applied to normalized angular measure, now implies

$$A_{N,\beta}(1, \dots, 1) \geq (N!)^{\beta/2}.$$

Integrate over the product of the annuli $s_N \leq |z_i| < 1$. Since $dA = 2\pi r dr dm(\theta)$ and each annulus has area $\pi(1 - s_N^2)$, we obtain

$$Z_{N,\beta} \geq \pi^N (1 - s_N^2)^N s_N^{\beta N(N-1)/2} (N!)^{\beta/2}.$$

Dividing by $N!$ proves the lower bound.

For these values of β ,

$$\log(1 - s_N^2) = -\log N + O(1), \quad \frac{\beta N(N-1)}{2} \log s_N = O_\beta(N),$$

and $\log(N!) = N \log N - N + O(\log N)$. Substitution into the two bounds proves (14.1) when $\beta \geq 2$.

Now suppose that $0 < \beta \leq 2$. Since $\beta - 2 \leq 0$, the direction of the pointwise estimate from Hadamard's inequality is reversed:

$$|\Delta_N(z)|^\beta \geq N^{(\beta/2-1)N} |\Delta_N(z)|^2.$$

This inequality first follows where $\Delta_N(z) \neq 0$ and extends to its zero set because both sides vanish there. Integration and $Z_{N,2} = \pi^N$ give the lower bound in (14.3).

To prove its upper bound, use the radial-ratio representation. With ordered radii $0 < r_1 < \dots < r_N < 1$, put $r_{N+1} = 1$ and $t_j = r_j/r_{j+1}$. In the notation of the radial comparison lemma,

$$Q_{N,\beta} = (2\pi)^N \int_{(0,1)^N} F_{N,\beta}(t) \prod_{j=1}^N t_j^{a_j-1} dt, \quad a_j = 2j + \frac{\beta}{2}j(j-1). \quad (14.5)$$

The factor $N!$ from ordering the radii has been canceled by the definition of $Q_{N,\beta}$; the factor $(2\pi)^N$ converts normalized angular measure to ordinary polar area measure. Recall that

$$F_{N,\beta}(t) = \int_{[0,2\pi)^N} \prod_{i < j} \left| 1 - \left(\prod_{h=i}^{j-1} t_h \right) e^{i(\theta_i - \theta_j)} \right|^\beta dm^{\otimes N}(\theta).$$

The radial comparison lemma proves that this factor is nondecreasing in each coordinate. It extends continuously to $[0, 1]^N$, as follows directly from the displayed integral. Therefore

$$F_{N,\beta}(t) \leq F_{N,\beta}(1, \dots, 1) = A_{N,\beta}(1, \dots, 1) \leq (N!)^{\beta/2}.$$

The last inequality is Jensen's inequality for the concave function $x \mapsto x^{\beta/2}$ on normalized angular measure, together with $A_{N,2}(1, \dots, 1) = N!$. Inserting this bound into (14.5) and integrating each power $t_j^{a_j-1}$ proves (14.3).

For clarity, the product in this upper bound can also be evaluated exactly. Writing $b = 4/\beta$,

$$\prod_{j=1}^N a_j = \left(\frac{\beta}{2}\right)^N N! \frac{\Gamma(N+b)}{\Gamma(b)}.$$

Consequently the upper bound equals

$$\left(\frac{4\pi}{\beta}\right)^N (N!)^{\beta/2-1} \frac{\Gamma(b)}{\Gamma(N+b)}.$$

Stirling's formula, with fixed $b > 0$, shows that its logarithm is $(\beta/2-2)N \log N + O_\beta(N)$. The logarithm of the lower bound in (14.3) has the same leading term and an $O_\beta(N)$ remainder. This completes the proof for all $\beta > 0$. $\square$

Proposition 14.2 (Exact central insertion and a telescoping lower bound). *Fix $\beta > 0$, and write $b = 4/\beta$. For every $N \geq 1$,*

$$\rho_{N,\beta}(0) = \frac{Q_{N-1,\beta}}{Q_{N,\beta}} \mathbb{E}_{N-1,\beta} \left[\prod_{j=1}^{N-1} |z_j|^\beta \right]. \quad (14.6)$$

In particular,

$$\frac{b(b+1)}{(N-1+b)(N+b)} \frac{Q_{N-1,\beta}}{Q_{N,\beta}} \leq \rho_{N,\beta}(0) \leq \frac{Q_{N-1,\beta}}{Q_{N,\beta}}. \quad (14.7)$$

The empty product in (14.6) equals one.

Proof. Fixing one particle at the origin in the definition of the one-point correlation density gives

$$\rho_{N,\beta}(0) = \frac{N}{Z_{N,\beta}} \int_{\mathbb{D}^{N-1}} |\Delta_{N-1}(w)|^\beta \prod_{j=1}^{N-1} |w_j|^\beta dA^{\otimes(N-1)}(w).$$

The identity $NZ_{N-1,\beta}/Z_{N,\beta} = Q_{N-1,\beta}/Q_{N,\beta}$ proves (14.6); its upper bound follows from $|w_j| \leq 1$.

For completeness, apply the increasing-tilt comparison for the radial ratios to an n -particle configuration. Write its ordered radii as $0 < r_1 < \dots < r_n < 1$, put $r_{n+1} = 1$ and $t_k = r_k/r_{k+1}$, and recall

$$a_k = 2k + \frac{\beta}{2}k(k-1).$$

That comparison says that the actual ratio law has the form of the product of the Beta($a_k, 1$) laws tilted by a coordinatewise nondecreasing function. Harris's inequality for product measures consequently gives $\mathbb{E}[G(t)] \geq \mathbb{E}[G(T)]$ for each bounded nondecreasing G , where the T_k are independent with those beta laws. Use the increasing function

$$G(t) = \prod_{j=1}^n r_j^\beta = \prod_{k=1}^n t_k^{\beta k}.$$

The beta moments and a telescoping product give

$$\begin{aligned} \mathbb{E}_{n,\beta} \left[\prod_{j=1}^n |z_j|^\beta \right] &\geq \prod_{k=1}^n \frac{a_k}{a_k + \beta k} = \prod_{k=1}^n \frac{k-1+b}{k+1+b} \\ &= \frac{b(b+1)}{(n+b)(n+b+1)}. \end{aligned}$$

For $n = 0$ the same formula equals one. Taking $n = N-1$ in (14.6) proves the lower bound. $\square$

Theorem 14.3 (A central-density subsequence for $\beta > 4$). *Fix $\beta > 4$ and set $c_\beta = \beta/2 - 2 > 0$. For every $\varepsilon \in (0, c_\beta)$ there are infinitely many integers N such that*

$$\rho_{N,\beta}(0) \leq N^{-c_\beta + \varepsilon}. \quad (14.8)$$

In particular, $\liminf_{N \rightarrow \infty} \rho_{N,\beta}(0) = 0$.

Proof. The lower bound from Theorem 14.1 gives a constant C_β such that

$$\log Q_{N,\beta} \geq c_\beta N \log N - C_\beta N.$$

Suppose, to obtain a contradiction, that only finitely many N satisfy

$$\frac{Q_{N,\beta}}{Q_{N-1,\beta}} \geq N^{c_\beta - \varepsilon}.$$

Then this inequality is reversed strictly for all $N \geq N_0$ for some N_0 . Summing logarithms of the ratios gives

$$\begin{aligned} \log Q_{N,\beta} &\leq \log Q_{N_0-1,\beta} + (c_\beta - \varepsilon) \sum_{n=N_0}^N \log n \\ &= (c_\beta - \varepsilon)N \log N + O_{\beta,\varepsilon,N_0}(N). \end{aligned}$$

This contradicts the preceding lower bound for large N . Thus the displayed ratio bound holds for infinitely many N , and the upper bound in (14.7) proves (14.8). $\square$

What this does and does not imply. If one uses the previously established compactness of the correlation densities in the topology of local uniform convergence, the indices in Theorem 14.3 admit a further subsequence whose limiting continuous one-point density satisfies $\rho_\infty(0) = 0$. This statement by itself does not imply that the limiting point process is empty: a nonnegative radial subharmonic function may vanish at the origin without vanishing elsewhere. The theorem also does not assert full-sequence convergence of the central densities.

In particular, the $O_\beta(N)$ remainder in (14.1) cannot be differenced as though it were a smooth asymptotic expansion. For example, for fixed real c and $a > 0$, the positive numerical sequence

$$B_N = \exp\{cN \log N + aN(-1)^N\}, \quad N \geq 1,$$

satisfies $\log B_N = cN \log N + O(N)$, whereas

$$\log \frac{B_N}{B_{N-1}} = c(N \log N - (N-1) \log(N-1)) + a(2N-1)(-1)^N$$

for $N \geq 2$. Its adjacent ratios therefore have arbitrarily large oscillations. This example concerns only what an $O(N)$ asymptotic allows; it is not a proposed behavior of the gas partition functions. The later harmonic boundary transport supplies the additional adjacent-ratio lower bound, while the uniform analytic exponential-moment bound controls compact insertions. Together they strengthen the central subsequence statement to a full empty-limit theorem for $\beta > 4$. The elementary partition asymptotic alone does not give this conclusion.

15 A global Gaussian mode independent of every interior limit

Write $S_{k,N} = \sum_{i=1}^N z_i^k$ and

$$c_N = 2 + \frac{\beta}{2}(N-1).$$

The following statement identifies a global fluctuation without assuming uniqueness of the interior point process.

Theorem 15.1 (Gaussian center of charge and independence). *For every $\beta > 0$,*

$$\mathbb{E} S_{1,N} = 0, \quad \mathbb{E} |S_{1,N}|^2 = \frac{N}{c_N}.$$

Moreover, $S_{1,N}$ converges in law to a centered circular complex Gaussian G_β with density

$$\frac{\beta}{2\pi} \exp\left(-\frac{\beta}{2}|g|^2\right) dA(g).$$

If $\Xi_{N_j} \Rightarrow \Xi$ vaguely on the open disk, then

$$(\Xi_{N_j}, S_{1,N_j}) \Rightarrow (\Xi, G_\beta),$$

where the two variables on the right are independent. For every fixed $t \in \mathbb{C}$ and every bounded continuous functional F on the space of locally finite counting measures with its vague topology,

$$\lim_{j \rightarrow \infty} \mathbb{E} \left[F(\Xi_{N_j}) e^{-2 \operatorname{Re}(t S_{1,N_j})} \right] = e^{2|t|^2/\beta} \mathbb{E} F(\Xi).$$

Proof. We first record an elementary macroscopic estimate. For every fixed integer $k \geq 1$ and every $\varepsilon > 0$,

$$\mathbb{P}(|S_{k,N}| \geq \varepsilon N) \leq \exp \left[(1 + \beta/2)N \log N + 2\beta N - \frac{\beta 4^{-k}}{2k} \varepsilon^2 N^2 \right], \quad N \geq 2. \quad (15.1)$$

To prove it, put $q = 1 - 1/N$. The annulus $q < |z| < 1$ has area at least π/N . Under normalized area measure on that annulus, independent Z, W satisfy

$$\mathbb{E} \log |Z - W| = \mathbb{E} \log \max(|Z|, |W|) \geq \log q \geq -2/N.$$

The equality follows by angular integration of the logarithmic kernel. Jensen's inequality applied to the partition function restricted to this annulus gives

$$Z_{N,\beta} \geq (\pi/N)^N e^{-\beta(N-1)}. \quad (15.2)$$

For arbitrary points in the disk the identity

$$|1 - q^2 z_i \bar{z}_j|^2 - q^2 |z_i - z_j|^2 = (1 - q^2 |z_i|^2)(1 - q^2 |z_j|^2) \geq 0$$

implies

$$\begin{aligned} \log |\Delta_N(z)| &\leq -\frac{N(N-1)}{2} \log q - \frac{1}{2} \sum_{\ell \geq 1} \frac{q^{2\ell}}{\ell} \left(|S_{\ell,N}|^2 - \sum_i |z_i|^{2\ell} \right) \\ &\leq N + \frac{N}{2} \log N - \frac{4^{-k}}{2k} |S_{k,N}|^2. \end{aligned}$$

The series is absolutely convergent since $q < 1$. In the second line we used $q \geq 1/2$, $-\log q \leq 2/N$, and $1 - q^2 \geq 1/N$. Integrating this upper bound over a subset of the disk product and using (15.2) proves (15.1). In particular, $S_{2,N}/N \rightarrow 0$ in probability.

For $a \in \mathbb{D}$, let

$$T_a(z) = \frac{z+a}{1+\bar{a}z}, \quad R_{N,a}(z) = (1-|a|^2)^{c_N N} \prod_{i=1}^N |1 + \bar{a}z_i|^{-2c_N}.$$

The distance and Jacobian formulas for this disk automorphism are

$$|T_a(z) - T_a(w)| = \frac{(1-|a|^2)|z-w|}{|1+\bar{a}z||1+\bar{a}w|}, \quad |T'_a(z)|^2 = \frac{(1-|a|^2)^2}{|1+\bar{a}z|^4}.$$

A change of variables therefore gives, for every bounded measurable F ,

$$\mathbb{E} [F((T_a)_* \Xi_N) R_{N,a}] = \mathbb{E} F(\Xi_N). \quad (15.3)$$

In particular, $\mathbb{E} R_{N,a} = 1$. Expansion at $a = 0$ is legitimate under the integral for each fixed N , since all derivatives are uniformly bounded on a sufficiently small closed disk in the a

variable. The coefficient of $|a|^2$ in $R_{N,a}$ is $c_N^2 |S_{1,N}|^2 - c_N N$. Its expectation vanishes, proving the variance identity. The mean vanishes by rotation symmetry.

Fix $t \in \mathbb{C}$ and set $a_N = t/c_N$ for all sufficiently large N . Taylor expansion, uniformly over all particle configurations, yields

$$\log R_{N,a_N} = -2 \operatorname{Re}(\bar{t} S_{1,N}) - \frac{N}{c_N} |t|^2 + \frac{1}{c_N} \operatorname{Re}(\bar{t}^2 S_{2,N}) + O_t(N/c_N^2). \quad (15.4)$$

Here $c_N \asymp_\beta N$, and the error bound holds deterministically. Thus, writing

$$Q_{N,t} = \exp \left[-2 \operatorname{Re}(\bar{t} S_{1,N}) - \frac{N}{c_N} |t|^2 \right],$$

we have $\log(R_{N,a_N}/Q_{N,t}) \rightarrow 0$ in probability. This logarithm is also bounded in absolute value by a constant depending only on t and β , uniformly for all sufficiently large N .

For clarity, uniform integrability does not need to be assumed here. The deterministic bounded-error estimate and $\mathbb{E} R_{N,a_N} = 1$ give

$$\sup_{N \geq N_t} \mathbb{E} e^{-2 \operatorname{Re}(\bar{t} S_{1,N})} < \infty.$$

The same argument with pt in place of t gives the bound for every fixed real $p > 1$. Consequently, both $(R_{N,a_N})_N$ and $(Q_{N,t})_N$ have uniformly bounded p th moments. Their bounded logarithmic ratio converges to zero in probability, so

$$\mathbb{E} |R_{N,a_N} - Q_{N,t}| \rightarrow 0. \quad (15.5)$$

For example, this follows from Hölder's inequality applied to $Q_{N,t} |\exp(\log(R_{N,a_N}/Q_{N,t})) - 1|$.

The tiny automorphisms do not change the vague limit. Indeed, for $f \in C_c(\mathbb{D})$, choose a compact $K' \subset \mathbb{D}$ containing both $\operatorname{supp} f$ and $T_{a_N}^{-1}(\operatorname{supp} f)$ for all sufficiently large N . Uniform continuity of the zero extension of f and uniform convergence $T_{a_N} \rightarrow \operatorname{id}$ on K' give

$$\left| \int f d((T_{a_N})_* \Xi_N) - \int f d\Xi_N \right| \leq \omega_f(C|a_N|) \Xi_N(K') \rightarrow 0$$

in probability, by the already proved tightness of compact-disk counts. Applying this to a countable family defining a metric for the vague topology shows that the distance between these two configurations tends to zero in probability. Since $(\Xi_N)_N$ is tight, every $F \in C_b$ therefore satisfies

$$F((T_{a_N})_* \Xi_N) - F(\Xi_N) \rightarrow 0$$

in probability. Boundedness of F and the uniform p th moment bound for R_{N,a_N} make the corresponding weighted difference tend to zero in L^1 .

Combining this observation with (15.3) and (15.5) gives

$$\mathbb{E} \left[F(\Xi_N) e^{-2 \operatorname{Re}(\bar{t} S_{1,N})} \right] - \exp \left(\frac{N}{c_N} |t|^2 \right) \mathbb{E} F(\Xi_N) \rightarrow 0.$$

Along N_j , this proves the asserted mixed Laplace-transform limit. The variance identity makes $S_{1,N}$ tight. Every joint subsequential limit of (Ξ_{N_j}, S_{1,N_j}) satisfies the same transform identity, because the exponential moment bounds justify passage to the limit. Taking $F = 1$, uniqueness of the moment-generating function in $\mathbb{R}^2$ identifies its second coordinate as G_β . For each nonnegative bounded continuous F with positive expectation, the probability measure on that second coordinate obtained by weighting by F has the same Gaussian moment-generating function. It is therefore the same Gaussian law. Since bounded continuous functionals determine probability measures on the configuration space, this proves independence. Every joint subsequential limit has consequently been identified, which proves the joint convergence along N_j . $\square$

Remark 15.2. The theorem does not identify the law of Ξ . It shows, in particular, that identifying the first global harmonic mode cannot by itself establish uniqueness of the interior limit: that Gaussian variable is asymptotically independent of the entire interior configuration.

16 A disk transport for each harmonic mode

Fix an integer $k \geq 1$. Throughout this section $D = \{z : |z| < 1\}$, $S_j = \sum_{i=1}^N z_i^j$ for every integer j for which this expression is defined, and $S_0 = N$. Negative powers are used only on configurations without a point at zero. All constants in the deterministic estimates may depend on k and on a fixed smooth cutoff, but not on N, r, a , or the configuration.

Lemma 16.1 (A smooth global extension of a branched Möbius map). *There are constants $c_k, C_k > 0$ with the following property. If $0 < r < 1/4$ and $|a|r^{-k} < c_k$, there is a smooth orientation-preserving diffeomorphism $T_{a,r} : \overline{D} \rightarrow \overline{D}$ which is the identity on $\{|z| \leq r\}$ and agrees on $\{|z| \geq 2r\}$ with*

$$f_a(z) = z(1 + az^{-k})^{1/k}(1 + \bar{a}z^k)^{-1/k}. \quad (16.1)$$

Here both fractional powers are defined by the logarithmic power series at 1. Moreover,

$$\|T_{a,r} - \text{id}\|_\infty \leq C_k |a| r^{1-k}, \quad \|DT_{a,r} - I\|_\infty \leq C_k |a| r^{-k}. \quad (16.2)$$

If $\eta = C_k |a| r^{-k} < 1$, then

$$(1 - \eta)(1 - |z|) \leq 1 - |T_{a,r}(z)| \leq (1 + \eta)(1 - |z|), \quad z \in \overline{D}. \quad (16.3)$$

After decreasing c_k if necessary, $T_{a,r}(\{|z| < 2r\}) \subset \{|z| < 3r\}$.

Proof. Choose a smooth function $\chi : [0, \infty) \rightarrow [0, 1]$ which is zero on $[0, 1]$ and one on $[2, \infty)$. On $|z| > r$ define

$$T_{a,r}(z) = z + \chi(|z|/r)(f_a(z) - z),$$

and set $T_{a,r}(z) = z$ on $|z| \leq r$. The flatness of χ at the end of its constant interval makes this a smooth map across $|z| = r$. On $r \leq |z| \leq 1$, the power series in (16.1) and their first derivatives give

$$|f_a(z) - z| \leq C_k |a| |z|^{1-k}, \quad |f'_a(z) - 1| \leq C_k |a| |z|^{-k}.$$

The product rule, including the derivative $r^{-1}\chi'(|z|/r)$, proves (16.2). The estimates remain true on the closed disk. Since the disk is convex, the derivative bound implies

$$(1 - \eta)|z - w| \leq |T_{a,r}(z) - T_{a,r}(w)| \leq (1 + \eta)|z - w|.$$

In particular the map is injective, has nonsingular derivative, and is orientation preserving when $\eta < 1$.

The identity

$$f_a(z)^k = \frac{z^k + a}{1 + \bar{a}z^k} \quad (16.4)$$

shows that $|f_a(z)| = 1$ on $|z| = 1$ and $|f_a(z)| < 1$ where $|z| < 1$ and the formula is used. Since the patched map is a convex combination of z and $f_a(z)$, it maps D into D and the boundary into the boundary. Its interior image is nonempty and open by the inverse function theorem. It is also relatively closed in D : if $T_{a,r}(z_n) \rightarrow w \in D$, choose a subsequence with $z_n \rightarrow z \in \overline{D}$. The point z cannot be on the boundary, since its image would then be on the boundary. Thus $w = T_{a,r}(z) \in T_{a,r}(D)$. Connectedness of D proves that the interior image is

D . Every boundary point is also in the image by approximating it from inside D and taking a convergent subsequence of preimages. The inverse function theorem, including an extension of the formula to a neighborhood of the boundary, proves the asserted diffeomorphism property. Taking the infimum of the two-sided Lipschitz estimate over $w \in \partial D$, whose images fill ∂D , proves (16.3). Finally, on $|z| < 2r$ the displacement is at most $C_k|a|r^{1-k} = C_k(|a|r^{-k})r$; decreasing c_k proves the last assertion. $\square$

Lemma 16.2 (Uniform divided-difference estimates). *For distinct z, w with $r \leq |z|, |w| \leq 1$, put*

$$D_m(z, w) = \frac{z^m - w^m}{z - w}, \quad m \in \mathbb{Z}.$$

If $|a|r^{-k}$ is sufficiently small, then the logarithm below is the branch near zero and

$$\begin{aligned} \log \frac{f_a(z) - f_a(w)}{z - w} &= \frac{a}{k} D_{1-k} - \frac{\bar{a}}{k} D_{1+k} \\ &\quad + \frac{a^2}{2k^2} \{(1-k)D_{1-2k} - D_{1-k}^2\} \\ &\quad + \frac{\bar{a}^2}{2k^2} \{(k+1)D_{1+2k} - D_{1+k}^2\} \\ &\quad + \frac{|a|^2}{k^2} \{-1 + D_{1-k}D_{1+k}\} + O_k(|a|^3 r^{-3k}). \end{aligned} \quad (16.5)$$

The error is uniform even when z and w approach one another. Furthermore,

$$\begin{aligned} \log f'_a(z) &= -\frac{k-1}{k} a z^{-k} - \frac{k+1}{k} \bar{a} z^k - |a|^2 \\ &\quad + \frac{k-1}{2k} a^2 z^{-2k} + \frac{k+1}{2k} \bar{a}^2 z^{2k} + O_k(|a|^3 r^{-3k}). \end{aligned} \quad (16.6)$$

Proof. For $m \geq 1$, the polynomial identity for a divided difference gives $|D_m(z, w)| \leq m$. For $m = -q \leq -1$, one has the exact identity

$$D_{-q}(z, w) = -\sum_{\ell=1}^q z^{-\ell} w^{-(q+1-\ell)},$$

and therefore $|D_{-q}(z, w)| \leq q r^{-(q+1)}$. Also $D_0 = 0$. Consequently, for $u, v \geq 0$ with $u+v = n$,

$$|D_{1+k(v-u)}(z, w)| \leq (1 + kn) r^{-kn}. \quad (16.7)$$

Expand the two binomial factors in (16.1). This gives an absolutely convergent series whose (u, v) term is $b_u c_v a^u \bar{a}^v z^{1+k(v-u)}$, where $b_u = \binom{1/k}{u}$ and $c_v = \binom{-1/k}{v}$. There is a constant C_k such that $|b_u| + |c_u| \leq C_k 2^u$ for all $u \geq 0$; for instance Cauchy's coefficient bound on the circle of radius $1/2$ gives this estimate. After taking divided differences, (16.7) implies that the sum of terms of total degree at least three is $O_k(|a|r^{-k})^3$, provided $|a|r^{-k}$ is smaller than a fixed constant. The terms of degree at most two give

$$\begin{aligned} \frac{f_a(z) - f_a(w)}{z - w} &= 1 + \frac{a}{k} D_{1-k} - \frac{\bar{a}}{k} D_{1+k} + \frac{1-k}{2k^2} a^2 D_{1-2k} \\ &\quad + \frac{k+1}{2k^2} \bar{a}^2 D_{1+2k} - \frac{|a|^2}{k^2} + O_k(|a|^3 r^{-3k}). \end{aligned}$$

The perturbation of 1 is $O_k(|a|r^{-k})$. Applying $\log(1+u) = u - u^2/2 + O(|u|^3)$ proves (16.5). This argument uses endpoint monomial identities and does not assume that the segment from z to w avoids the origin.

Differentiating (16.4) gives the exact formula

$$f'_a(z) = (1 - |a|^2)(f_a(z)/z)^{1-k}(1 + \bar{a}z^k)^{-2}.$$

Its logarithm, with the branches equal to zero when $a = 0$, is

$$\log f'_a(z) = \log(1 - |a|^2) - \frac{k-1}{k} \log(1 + az^{-k}) - \frac{k+1}{k} \log(1 + \bar{a}z^k).$$

Taylor expansion proves (16.6). $\square$

Proposition 16.3 (All aggregate quadratic coefficients). *Suppose the points $z_1, \dots, z_N$ are distinct and $r \leq |z_i| \leq 1$. With logarithms chosen near zero, set*

$$L_N = \sum_{i < j} \log \frac{f_a(z_i) - f_a(z_j)}{z_i - z_j}, \quad J_N = \sum_i \log f'_a(z_i).$$

Then

$$\begin{aligned} L_N = & \frac{a}{2k} \left\{ (k-1)S_{-k} - \sum_{p=1}^{k-1} S_{-p}S_{-(k-p)} \right\} \\ & - \frac{\bar{a}}{2k} \left\{ (2N - k - 1)S_k + \sum_{p=1}^{k-1} S_pS_{k-p} \right\} \\ & + \frac{a^2}{4k^2} \sum_{p=1}^{2k-1} |k-p| \{ S_{-p}S_{-(2k-p)} - S_{-2k} \} \\ & + \frac{\bar{a}^2}{4k^2} \sum_{p=0}^{2k} |k-p| \{ S_pS_{2k-p} - S_{2k} \} \\ & + \frac{|a|^2}{k^2} \left\{ -\frac{kN^2}{2} + \frac{k^2N}{2} - \sum_{p=1}^{k-1} (k-p)S_pS_{-p} \right\} + O_k(N^2|a|^3r^{-3k}), \end{aligned} \quad (16.8)$$

and

$$\begin{aligned} J_N = & -\frac{k-1}{k}aS_{-k} - \frac{k+1}{k}\bar{a}S_k - N|a|^2 \\ & + \frac{k-1}{2k}a^2S_{-2k} + \frac{k+1}{2k}\bar{a}^2S_{2k} + O_k(N|a|^3r^{-3k}). \end{aligned} \quad (16.9)$$

Empty sums are zero, so these formulas also cover $k = 1$.

Proof. The linear terms follow from

$$\begin{aligned} \sum_{i < j} D_{1+k}(z_i, z_j) &= \frac{1}{2} \left\{ (2N - k - 1)S_k + \sum_{p=1}^{k-1} S_pS_{k-p} \right\}, \\ \sum_{i < j} D_{1-k}(z_i, z_j) &= -\frac{1}{2} \left\{ \sum_{p=1}^{k-1} S_{-p}S_{-(k-p)} - (k-1)S_{-k} \right\}. \end{aligned}$$

To verify them, expand each divided difference into monomials and use

$$\sum_{i \neq j} z_i^u z_j^v = S_u S_v - S_{u+v}.$$

The coefficient arrays are symmetric in u, v , so an unordered pair sum is one half of the corresponding ordered pair sum.

For clarity the quadratic identities at a single pair are

$$\begin{aligned}(1-k)D_{1-2k} - D_{1-k}^2 &= \sum_{p=1}^{2k-1} |k-p|z^{-p}w^{-(2k-p)}, \\ (k+1)D_{1+2k} - D_{1+k}^2 &= \sum_{p=0}^{2k} |k-p|z^p w^{2k-p}, \\ D_{1-k}D_{1+k} &= -(k-1) - \sum_{p=1}^{k-1} (k-p)(z^p w^{-p} + z^{-p} w^p).\end{aligned}$$

The first two identities follow by counting the coefficients in the squares of finite geometric sums: the triangular convolution coefficients, subtracted from $k-1$ or $k+1$, are exactly $|k-p|$. For the last identity multiply the two geometric sums directly. It follows that

$$\begin{aligned}\sum_{i < j} \{-1 + D_{1-k}(z_i, z_j)D_{1+k}(z_i, z_j)\} &= -\frac{kN(N-1)}{2} - \sum_{p=1}^{k-1} (k-p)(S_p S_{-p} - N) \\ &= -\frac{kN^2}{2} + \frac{k^2 N}{2} - \sum_{p=1}^{k-1} (k-p)S_p S_{-p}.\end{aligned}$$

Substitution in (16.5) proves (16.8). Summing (16.6) proves (16.9). $\square$

Lemma 16.4 (Negative powers controlled by radial deficits). *Fix $0 < \rho < 1$ and an integer $j \geq 1$. If all $|z_i| \geq r$ with $0 < r < \rho$, and if $\mathcal{D}_N = \sum_i (1 - |z_i|)$ and $M_N(\rho) = \#\{i : |z_i| < \rho\}$, then*

$$|S_{-j} - \overline{S_j}| \leq C_{j,\rho} \mathcal{D}_N + 2r^{-j} M_N(\rho). \quad (16.10)$$

Proof. For $z = se^{i\theta}$, $|z^{-j} - \bar{z}^j| = |s^{-j} - s^j|$. On $\rho \leq s \leq 1$, the mean-value theorem bounds this by $C_{j,\rho}(1-s)$. On $r \leq s < \rho$, it is at most $2r^{-j}$. Sum these two bounds. $\square$

Proposition 16.5 (A probabilistic criterion for the transport expansion). *Let $k \geq 1$, fix $t \in \mathbb{C}$, and choose*

$$0 < \alpha < \frac{1}{3k}, \quad r_N = N^{-\alpha}, \quad a_N = \frac{2kt}{\beta N}, \quad T_N = T_{a_N, r_N}.$$

For finitely many small N the map may be defined to be the identity. Let $\mathcal{L}_N$ be any sequence of probability laws on configurations of N distinct points in D . Suppose, under these laws, that

1. $\mathcal{L}_N\{\min_i |z_i| < 2r_N\} \rightarrow 0$;
2. $M_N(\rho) = O_{\mathcal{L}_N}(1)$ for one fixed $0 < \rho < 1$;
3. $\mathcal{D}_N = O_{\mathcal{L}_N}(\log(N+1))$;
4. $S_j/N \rightarrow 0$ in probability for every $1 \leq j \leq 2k$;
5. $S_j = O_{\mathcal{L}_N}(1)$ for $1 \leq j < k$.

Define the deterministic likelihood function

$$R_N(z) = \prod_{i < j} \left| \frac{T_N(z_i) - T_N(z_j)}{z_i - z_j} \right|^\beta \prod_i \det DT_N(z_i). \quad (16.11)$$

Then

$$\log R_N = -2 \operatorname{Re}(\bar{t} S_k) - \frac{2k}{\beta} |t|^2 + o_{\mathcal{L}_N}(1). \quad (16.12)$$

No claim about exponential moments of the error is part of this statement.

Proof. Work on the event $G_N = \{\min_i |z_i| \geq 2r_N\}$, whose probability tends to one. The preceding negative-power bound implies

$$S_{-j} = \overline{S_j} + O_{\mathcal{L}_N}(\log(N+1) + N^{\alpha j}), \quad 1 \leq j \leq 2k.$$

In particular $S_{-j} = o_{\mathcal{L}_N}(N)$ for $j \leq 2k$, since $2k\alpha < 1$. For $j < k$, the positive modes are tight, so

$$S_{-j} = O_{\mathcal{L}_N}(\log(N+1) + N^{\alpha j}).$$

For $1 \leq p < k$ this gives

$$N^{-1} S_{-p} S_{-(k-p)} = o_{\mathcal{L}_N}(1), \quad N^{-1} S_p S_{k-p} = o_{\mathcal{L}_N}(1),$$

because $\alpha k < 1$. Consequently the first line of (16.8) is $o_{\mathcal{L}_N}(1)$, and its second line is $-\bar{a}_N N S_k / k + o_{\mathcal{L}_N}(1)$.

Every factor S_j or S_{-j} of nonzero degree appearing in the quadratic terms is $o_{\mathcal{L}_N}(N)$. The terms involving $S_0 = N$ are paired with $S_{2k} = o_{\mathcal{L}_N}(N)$. Thus the pure a_N^2 and $\bar{a}_N^2$ contributions are $o_{\mathcal{L}_N}(1)$. The mixed contribution is $-|a_N|^2 N^2 / (2k) + o_{\mathcal{L}_N}(1)$. All terms in (16.9) are $o_{\mathcal{L}_N}(1)$. Finally the largest deterministic remainder satisfies

$$N^2 |a_N|^3 r_N^{-3k} = O_{k,t,\beta}(N^{-1+3k\alpha}) \longrightarrow 0.$$

On G_N the map is holomorphic at every point, so its real Jacobian is $|f'_{a_N}(z_i)|^2$ and $\log R_N = \beta \operatorname{Re} L_N + 2 \operatorname{Re} J_N$. Substitution of $a_N = 2kt/(\beta N)$ gives (16.12). Discarding G_N^c is legitimate for convergence in probability. $\square$

Lemma 16.6 (Effect on power sums). *For every fixed integer $j \geq 1$, on G_N ,*

$$S_j(T_N z) - S_j(z) = \frac{j}{k} \{a_N S_{j-k} - \bar{a}_N S_{j+k}\} + O_{j,k}(N |a_N|^2 r_N^{-2k}). \quad (16.13)$$

In particular, under any laws satisfying the small-hole, radial-deficit, and local-count assumptions above and satisfying $S_\ell/N \rightarrow 0$ for the positive index $\ell = j+k$ and, when $j \neq k$, also $\ell = |j-k|$,

$$S_j(T_N z) - S_j(z) \longrightarrow \begin{cases} 2kt/\beta, & j = k, \\ 0, & j \neq k, \end{cases}$$

in probability. Regardless of macroscopic-mode convergence, on G_N there is the deterministic bound

$$|S_k(T_N z) - S_k(z)| \leq C_{k,t,\beta}. \quad (16.14)$$

Proof. Raise (16.1) to the j th power and expand both binomial factors to first order. The pointwise remainder is at most $C_{j,k} |a_N|^2 r_N^{-2k}$, because $|z|^j \leq 1$. This proves (16.13). Its total remainder is $O(N^{-1+2k\alpha}) = o(1)$. The negative modes with indices between 1 and $k-1$, when they occur, are $o(N)$ by (16.10); the other nonconstant modes are $o(N)$ by hypothesis. For $j = k$ the constant mode contributes $a_N N = 2kt/\beta$. Finally, (16.4) gives directly

$$|f_{a_N}(z)^k - z^k| = \left| \frac{a_N - \bar{a}_N z^{2k}}{1 + \bar{a}_N z^k} \right| \leq \frac{2|a_N|}{1 - |a_N|}.$$

Summing proves (16.14). $\square$

Lemma 16.7 (Stability under the transported law). *Let P_N be the disk-gas law and let Q_N have density R_N from (16.11) with respect to P_N . The change of variables gives $\mathbb{E}_{P_N} R_N = 1$ and makes $Y = T_N Z$ under Q_N have law P_N . Assume under P_N that*

$$P_N\{M_N(s) > 0\} \leq C s^2 \quad (0 < s < s_0), \quad \mathcal{D}_N = O_{P_N}(\log(N+1)),$$

that compact-disk counts are tight, that $S_j/N \rightarrow 0$ for all fixed positive j , and that S_j is tight for $1 \leq j < k$. Then all five hypotheses of Proposition 16.5 hold under Q_N , as well as under P_N .

Proof. Under Q_N , $Y = T_N Z$ has the original law. If some $|z_i| < 2r_N$, then $|Y_i| < 3r_N$. Consequently $Q_N(G_N^c) \leq P_N\{M_N(3r_N) > 0\} = O(r_N^2)$. The uniform displacement in (16.2) tends to zero. Thus, for any fixed $\rho < \rho' < 1$, and all sufficiently large N , $M_Z(\rho) \leq M_Y(\rho')$. This proves tightness of local counts under Q_N . Equation (16.3) gives $\mathcal{D}_N(Z) \leq (1 - \eta_N)^{-1} \mathcal{D}_N(Y)$ with $\eta_N \rightarrow 0$, which proves the radial-deficit bound under Q_N . For each fixed positive j ,

$$N^{-1}|S_j(Y) - S_j(Z)| \leq j\|T_N - \text{id}\|_\infty \rightarrow 0,$$

so macroscopic-mode convergence transfers too.

It remains to check tightness of the lower modes without assuming it under Q_N . On G_N , the first-order bound on the binomial factors gives, for $1 \leq j < k$,

$$|f_{a_N}(z)^j - z^j| \leq C_{j,k} |a_N| (|z|^{j-k} + |z|^{j+k}).$$

Splitting the sum at any fixed $0 < \rho < 1$ yields

$$|S_j(Y) - S_j(Z)| \leq C_{j,k,\rho} |a_N| \{N + r_N^{j-k} M_Z(\rho)\}.$$

The first term is bounded and the second is $o_{Q_N}(1)$ because $\alpha(k-j) < 1$ and the local count is tight. Since $S_j(Y)$ has the original tight law and $Q_N(G_N^c) \rightarrow 0$, this proves tightness of $S_j(Z)$ under Q_N , completing the proof. $\square$

17 All harmonic modes are Gaussian and independent of the interior

For integers j , set $S_{j,N} = \sum_{i=1}^N z_i^j$ whenever all $z_i \neq 0$; this holds almost surely. Set $S_{0,N} = N$. Let $\mathcal{D}_N = \sum_i (1 - |z_i|)$. Three estimates already proved for the disk gas will be used:

1. For every fixed $0 < r_0 < 1$, the counts $\Xi_N(B_{r_0})$ are tight, and $\sup_N \sup_{|z| \leq r_0} \rho_{1,N}(z) < \infty$.
2. $\mathcal{D}_N = O_{\mathbb{P}}(\log(N+1))$.
3. For every fixed positive integer j , $S_{j,N}/N \rightarrow 0$ in probability.

Here the second estimate follows directly from radial stochastic domination: if $a_j = 2j + \beta j(j-1)/2$ are the comparison exponential rates, then

$$\mathbb{E} \mathcal{D}_N \leq \mathbb{E} \sum_i (-\log |z_i|) \leq \sum_{j=1}^N \frac{j}{a_j} = O_\beta(\log(N+1)).$$

The last estimate follows from the elementary exponential concentration bound established above using the annular lower bound on $Z_{N,\beta}$.

Theorem 17.1 (Joint Gaussian harmonic modes). *For every $\beta > 0$ and every fixed integer $m \geq 1$,*

$$(S_{1,N}, \dots, S_{m,N}) \implies (G_1, \dots, G_m),$$

where the G_j are independent centered circular complex Gaussians with $\mathbb{E}|G_j|^2 = 2j/\beta$. If $\Xi_{N_\ell} \Rightarrow \Xi$ in the vague topology of the open disk, then

$$(\Xi_{N_\ell}, S_{1,N_\ell}, \dots, S_{m,N_\ell}) \implies (\Xi, G_1, \dots, G_m),$$

where the Gaussian vector is independent of Ξ .

Proof. The proof is by induction on the mode k . The already established Möbius argument supplies $k = 1$. We give the induction step in detail. Fix $k \geq 2$, suppose that all conclusions have been proved through $k - 1$, and put

$$r_N = N^{-\alpha}, \quad \alpha = \frac{1}{100k}, \quad a_N = \frac{2kt}{\beta N}, \quad t \in \mathbb{C}.$$

Use the smooth disk diffeomorphism $T_{N,t}$ from the preceding transport lemma. It agrees for $|z| \geq 2r_N$ with

$$f_{a_N}(z) = z(1 + a_N z^{-k})^{1/k} (1 + \bar{a}_N z^k)^{-1/k},$$

where the powers are defined by the analytic logarithms near 1. Its exact likelihood ratio under change of variables is

$$R_{N,t}(z) = \prod_i J_{T_{N,t}}(z_i) \prod_{i < j} \left| \frac{T_{N,t}(z_i) - T_{N,t}(z_j)}{z_i - z_j} \right|^\beta.$$

Define $Q_{N,t}$ by $dQ_{N,t} = R_{N,t} dP_{N,\beta}$. Then

$$\mathbb{E}_P R_{N,t} = 1, \quad (T_{N,t})_* Q_{N,t} = P_{N,\beta}, \quad (17.1)$$

where the second formula denotes coordinatewise transport of configurations.

Let A_N be the event that every particle has modulus at least $2r_N$. Uniform local intensity bounds give $P(A_N^c) = O(r_N^2)$. The construction also gives $T_{N,t}(B_{2r_N}) \subset B_{3r_N}$ for all sufficiently large N . Hence the exact pullback identity gives

$$Q_{N,t}(A_N^c) \leq P\{\Xi_N(B_{3r_N}) \geq 1\} = O(r_N^2). \quad (17.2)$$

In particular, expansions on A_N need not be integrated over its complement; that complement has vanishing probability under *both* measures.

We first verify all input bounds for the expansions under both P and $Q = Q_{N,t}$. For $p \geq 1$, on any event where $\min_i |z_i| \geq r_N$,

$$|S_{-p,N} - \overline{S_{p,N}}| \leq C_{p,r_0} \mathcal{D}_N + C_p r_N^{-p} \Xi_N(B_{r_0}), \quad 0 < r_0 < 1 \text{ fixed}. \quad (17.3)$$

Indeed, for $|z| \geq r_0$ one has $|z^{-p} - \bar{z}^p| = |z|^{-p} - |z|^p \leq C_{p,r_0}(1 - |z|)$; for the remaining points use $|z|^{-p} + |z|^p \leq 2r_N^{-p}$. Under P , this proves

$$S_{-p,N}/N \rightarrow 0 \quad (1 \leq p \leq 2k),$$

on A_N , since $\alpha p < 1$. For $p < k$, induction further gives

$$S_{-p,N} = O_P(\log(N+1) + r_N^{-p}) \quad \text{on } A_N. \quad (17.4)$$

Under Q , write $y_i = T_{N,t}(z_i)$; the configuration y has law P . Uniform convergence $T_{N,t} \rightarrow \text{id}$ implies tightness of every fixed compact-disk count for z and also $S_{j,N}(z)/N \rightarrow 0$ for every fixed positive j . The maps and their inverses are uniformly bilipschitz and map the unit circle onto itself. Consequently

$$1 - |z| \leq C(1 - |T_{N,t}(z)|),$$

with a constant independent of large N . Therefore $\mathcal{D}_N(z) = O_Q(\log(N+1))$. Formula (17.3) now proves $S_{-p,N}(z)/N \rightarrow 0$ on A_N under Q for $p \leq 2k$.

The lower positive modes are tight under Q as well. To see the stronger fact needed later, for every fixed positive j use the uniform Taylor expansion from Lemma 16.6, on A_N :

$$S_{j,N}(y) - S_{j,N}(z) = \frac{j}{k} (a_N S_{j-k,N}(z) - \bar{a}_N S_{j+k,N}(z)) + O_{j,k,t}(N|a_N|^2 r_N^{-2k}). \quad (17.5)$$

For $j < k$, the negative index has absolute value at most $k-1$; the convergence $S_{-(k-j),N}(z)/N \rightarrow 0$ under Q was proved in the preceding paragraph without assuming tightness of any lower positive mode under Q . Both linear terms in (17.5) are therefore $o_Q(1)$. The deterministic error is $O(N^{-1+2\alpha k}) = o(1)$. It follows that

$$S_{j,N}(y) - S_{j,N}(z) \rightarrow 0 \quad \text{in } Q\text{-probability,} \quad j < k. \quad (17.6)$$

Thus the lower positive modes are tight under Q . Applying (17.3) once more proves (17.4) under Q too.

The deterministic likelihood expansion from the transport lemma now applies under both measures. The linear terms other than $-(\beta N/k) \text{Re}(\bar{a}_N S_{k,N})$ contain only single $S_{\pm k,N}$ or products $S_{p,N} S_{k-p,N}$ and $S_{-p,N} S_{-(k-p),N}$ with $1 \leq p < k$. After multiplication by $a_N = O(N^{-1})$, all are $o(1)$ in probability: for the negative products use (17.4) and $\alpha k < 1$. All pure quadratic terms vanish by macroscopic positive and negative moment bounds. The mixed quadratic term has limit $-\beta N^2 |a_N|^2 / (2k) = -2k|t|^2 / \beta$. The total deterministic remainder is bounded by

$$C_{k,t,\beta} \left(N^2 |a_N|^3 r_N^{-3k} + N |a_N|^2 r_N^{-2k} \right) = o(1).$$

Together with (17.2), this proves

$$\log R_{N,t} = -2 \text{Re}(\bar{t} S_{k,N}) - \frac{2k}{\beta} |t|^2 + o(1) \quad (17.7)$$

in probability under *each* of P and $Q_{N,t}$. No assertion of integrability of the error is used.

This identity first gives tightness of $S_{k,N}$ under P . For example, use $t = -1$, let $c = 2k/\beta$, and denote the probabilistic error by E_N . Markov's inequality and $\mathbb{E}_P R_{N,-1} = 1$ imply

$$P\{\text{Re } S_{k,N} > L\} \leq P\{|E_N| > 1\} + e^{-2L+c+1}.$$

The choices $t = 1, i, -i$ control the other three tails. Hence $S_{k,N}$ is tight. It is tight under $Q_{N,t}$ too: on A_N the exact identity $f_a(z)^k = (z^k + a)/(1 + \bar{a}z^k)$ gives

$$S_{k,N}(y) - S_{k,N}(z) = a_N N - \bar{a}_N S_{2k,N}(z) + O_t(N|a_N|^2) = \frac{2kt}{\beta} + o_Q(1).$$

The y mode has the just-proved baseline tightness, proving the claim.

Now (17.7) under Q shows that $\log R_{N,t}$ is tight under Q . This proves uniform integrability of the exact likelihood ratios, since

$$\mathbb{E}_P[R_{N,t} \mathbf{1}_{\{R_{N,t} > K\}}] = Q_{N,t}\{\log R_{N,t} > \log K\} \rightarrow 0 \quad (17.8)$$

as $K \rightarrow \infty$, uniformly in all sufficiently large N . The finitely many excluded N do not affect uniform integrability. This argument is why the small patch near the origin causes no missing tail estimate.

Let $Y_N = (\Xi_N, S_{1,N}, \dots, S_{k-1,N})$, and write Y_N^T for the same observables after transport. Under Q , the vague distance between the two configurations tends to zero: the maps tend uniformly to the identity, and compact-disk counts are tight under Q . Equation (17.6) deals with the remaining coordinates. Thus $Y_N^T - Y_N \rightarrow 0$ in the product-space metric in Q -probability. Tightness under Q implies, for every $F \in C_b$ of this product space,

$$\mathbb{E}_Q[F(Y_N^T) - F(Y_N)] \rightarrow 0.$$

The exact pullback identity consequently gives

$$\mathbb{E}_P F(Y_N) - \mathbb{E}_P[R_{N,t} F(Y_N)] \rightarrow 0. \quad (17.9)$$

Take any jointly convergent subsequence of $(Y_N, S_{k,N})$, with limit (Y, G_k) . Such subsequences exist by tightness. By (17.7) under P and the continuous mapping theorem,

$$(Y_N, S_{k,N}, R_{N,t}) \Rightarrow \left(Y, G_k, \exp \left[-2 \operatorname{Re}(\bar{t} G_k) - \frac{2k}{\beta} |t|^2 \right] \right).$$

Uniform integrability (17.8) permits passage to the expectation in (17.9). We obtain

$$\mathbb{E} \left[F(Y) e^{-2 \operatorname{Re}(\bar{t} G_k)} \right] = e^{2k|t|^2/\beta} \mathbb{E} F(Y), \quad t \in \mathbb{C}. \quad (17.10)$$

In particular, the limiting exponential moments are finite. We have not assumed, and do not need, uniform integrability of the corresponding *unmodified* prelimit exponential statistic on the small-hole event. The later Theorem 18.3 proves the stronger prelimit exponential-moment bounds by a separate fixed-radius argument.

Taking $F = 1$, uniqueness of the moment-generating function in $\mathbb{R}^2$ identifies G_k as the centered circular complex Gaussian of variance $2k/\beta$. For nonnegative F with positive expectation, divide (17.10) by that expectation and apply the same uniqueness theorem to the F -weighted probability law of G_k . It equals the unweighted Gaussian law. Bounded continuous functions therefore show that G_k is independent of Y . The induction hypothesis identifies the lower coordinates of Y as independent Gaussians independent of the limiting interior process. This completes the induction and proves the stated convergences. $\square$

Corollary 17.2 (Harmonic polynomials and the countable mode vector). *If $h(z) = \operatorname{Re} \sum_{j=1}^m b_j z^j$, then*

$$\sum_{i=1}^N h(z_i) \Longrightarrow \mathcal{N} \left(0, \frac{1}{\beta} \sum_{j=1}^m j |b_j|^2 \right).$$

Its limiting variance equals $(\pi\beta)^{-1} \int_{\mathbb{D}} |\nabla h|^2 dA$. The entire countable vector $(S_{j,N})_{j \geq 1}$ converges in the product topology of $\mathbb{C}^{\mathbb{N}}$ to $(G_j)_{j \geq 1}$. Along every subsequence on which Ξ_N converges, this convergence is joint with Ξ_N , and the limiting countable Gaussian vector is independent of the interior limit.

Proof. The first statement is a real linear image of the Gaussian vector in Theorem 17.1. Circular symmetry gives $\operatorname{Var} \operatorname{Re}(b_j G_j) = j |b_j|^2 / \beta$, and independence makes the variances add. Polar integration and orthogonality give $\int_{\mathbb{D}} |\nabla h|^2 dA = \pi \sum_j j |b_j|^2$. For countable product convergence, each coordinate family is tight. Given $\varepsilon > 0$, choose finite bounds L_j such that $\sup_N \mathbb{P}(|S_{j,N}| > L_j) \leq \varepsilon 2^{-j}$. The compact product $\prod_j \{|z| \leq L_j\}$ then has probability at least $1 - \varepsilon$ for every N . This proves tightness in the countable product, also jointly with any tight family of interior configurations. Finite-dimensional distributions, already identified by the theorem, determine every possible subsequential limit in that product. They also prove independence from the interior configuration. $\square$

18 Uniform exponential moments of every fixed harmonic mode

Write $\mathcal{D}_N = \sum_{i=1}^N (1 - |z_i|)$ and $S_{j,N} = \sum_{i=1}^N z_i^j$ for $j \geq 1$.

Lemma 18.1 (A quadratic exponential bound for the radial deficit). *For every $c > 0$ and $\beta > 0$,*

$$\sup_{N \geq 1} \mathbb{E}_N \exp\{c\mathcal{D}_N^2/N\} < \infty.$$

Proof. Use the established simultaneous radial domination with independent exponentials E_j of rates $a_j = 2j + \beta j(j-1)/2$. It gives a coupling under which the ordered depths satisfy $-\log R_k \leq \sum_{j=k}^N E_j$. Fix an integer $J \geq 1$. Since $1 - s \leq -\log s$ for $0 < s \leq 1$, this coupling gives

$$\mathcal{D}_N \leq J + Y_N, \quad Y_N = \sum_{j=J+1}^N (j - J)E_j,$$

where an empty sum is zero. Put $\mu_N = \mathbb{E}Y_N$. The elementary estimate $a_j \geq \beta j(j-1)/2$ shows $\mu_N \leq C_\beta \log(N+1)$.

For each fixed $\lambda > 0$, choose J so large that $\lambda(j-J)/a_j \leq 1/2$ for every $j > J$; this is possible because $a_j \geq \beta j(j-1)/2$ and $(j-J)/(j(j-1)) \leq 1/(j-1) \leq 1/J$. Independence and $-\log(1-u) - u \leq u^2$ for $0 \leq u \leq 1/2$ give

$$\mathbb{E}e^{\lambda(Y_N - \mu_N)} = \prod_{j=J+1}^N \frac{\exp[-\lambda(j-J)/a_j]}{1 - \lambda(j-J)/a_j} \leq \exp\left(\lambda^2 \sum_{j=J+1}^{\infty} \frac{(j-J)^2}{a_j^2}\right) < \infty,$$

uniformly in N . The displayed infinite series converges by comparison with $\sum j^{-2}$.

Take $\lambda = 2c$ and set $b_N = J + \mu_N$ and $X_N = (\mathcal{D}_N - b_N)_+$. On the coupling, $X_N \leq (Y_N - \mu_N)_+$; also $X_N \leq \mathcal{D}_N \leq N$. Consequently

$$\frac{\mathcal{D}_N^2}{N} \leq \frac{2b_N^2}{N} + \frac{2X_N^2}{N} \leq \frac{2b_N^2}{N} + 2X_N.$$

The first term is bounded uniformly in N , while $\mathbb{E}e^{2cX_N} \leq 1 + \mathbb{E}e^{2c(Y_N - \mu_N)}$ is uniformly bounded by the preceding product estimate. This proves the claim. $\square$

Lemma 18.2 (A global bound for a fixed-radius transport). *Fix $k \geq 1$, $t \in \mathbb{C}$, and $0 < r < 1/4$. For sufficiently large N , put $a = 2kt/(\beta N)$ and let $T = T_{a,r}$ be the map of Lemma 16.1. Let R denote its exact change-of-variables likelihood under $\mathbb{P}_{N,\beta}$. Put $M = \#\{i : |z_i| < 2r\}$. There is a constant C , independent of N and the configuration of distinct points, such that*

$$|\log R + 2 \operatorname{Re}(t S_{k,N})| \leq C \left[1 + M + \frac{\mathcal{D}_N^2 + M^2 + \sum_{p=1}^{k-1} |S_{p,N}|^2}{N} \right]. \quad (18.1)$$

Proof. All constants below may depend on k, t, r, β . The derivative estimate for the patched map is $\|DT - I\|_\infty \leq C/N$. Convexity of the disk therefore implies

$$\left| \log \left| \frac{T(z) - T(w)}{z - w} \right| \right| \leq C/N, \quad |\log \det DT(z)| \leq C/N$$

for distinct z, w and sufficiently large N . There are at most NM pairs meeting $\{|z| < 2r\}$. Their total contribution to $\log R$ is bounded in absolute value by CM , and the entire Jacobian contribution is bounded by C .

Let $I = \{i : |z_i| \geq 2r\}$, $n = |I| = N - M$, and $U_j = \sum_{i \in I} z_i^j$ for integers j . For pairs with both indices in I , the map agrees with f_a . The first-order part of Proposition 16.3, with n in place of N , gives

$$\begin{aligned} \sum_{\substack{i < j \\ i, j \in I}} \log \frac{f_a(z_i) - f_a(z_j)}{z_i - z_j} &= \frac{a}{2k} \left[(k-1)U_{-k} - \sum_{p=1}^{k-1} U_{-p}U_{-(k-p)} \right] \\ &\quad - \frac{\bar{a}}{2k} \left[(2n-k-1)U_k + \sum_{p=1}^{k-1} U_p U_{k-p} \right] + O(1). \end{aligned}$$

The remainder is uniformly $O(n^2|a|^2) = O(1)$ because r is fixed; this also follows by summing the second-order error in the individual divided-difference expansion. The identity is valid when $n = 0$ or 1 as well, with the corresponding empty pair sum.

For every fixed $1 \leq p \leq k$, the mean-value theorem on $[2r, 1]$ gives

$$|U_{-p} - \bar{U}_p| \leq C \sum_{i \in I} (1 - |z_i|) \leq C\mathcal{D}_N, \quad |U_p - S_{p,N}| \leq M.$$

Thus, for $p < k$, each positive or negative product in the preceding display is bounded by $C[\mathcal{D}_N^2 + M^2 + \sum_{q=1}^{k-1} |S_{q,N}|^2]$. The isolated U_{-k} term, multiplied by a , is bounded because $|U_{-k}| \leq N(2r)^{-k}$. Finally,

$$\left| \frac{2n-k-1}{N} U_k - 2S_{k,N} \right| \leq C(1+M),$$

using $|S_{k,N}| \leq N$, $|U_k - S_{k,N}| \leq M$, and $M \leq N$. Taking real parts, multiplying the pair expression by β , and using $a = 2kt/(\beta N)$ proves (18.1). $\square$

Theorem 18.3 (Genuine exponential-moment bounds and convergence). *For every $\beta > 0$, integer $m \geq 1$, and $c > 0$,*

$$\sup_{N \geq 1} \mathbb{E}_N \exp \left(c \sum_{j=1}^m |S_{j,N}| \right) < \infty. \quad (18.2)$$

Consequently all mixed moments of every fixed finite vector of harmonic modes converge to those of the independent Gaussian vector identified in Theorem 17.1. Along every interior-convergent subsequence, the Gaussian independence identity extends from bounded tests in the modes to every continuous mode test of at most exponential growth, multiplied by a bounded continuous function of the interior.

Proof. We prove the moment bound by induction on k . The base $k = 1$ follows from the already proved Möbius exponential-moment estimate; alternatively the following argument works for $k = 1$ with no lower-mode terms. Fix $t \in \mathbb{C}$, $0 < r < 1/4$, and use the preceding map and likelihood for all sufficiently large N . Define Q by $dQ = R dP$, where $P = \mathbb{P}_{N,\beta}$. Under Q , the variables $y_i = T(z_i)$ have law P . The deterministic transport bounds give, for each fixed $p \geq 1$,

$$|S_{p,N}(z) - S_{p,N}(y)| \leq C_p, \quad \mathcal{D}_N(z) \leq 2\mathcal{D}_N(y), \quad M(z) \leq \#\{i : |y_i| < 3r\}.$$

The first bound uses $\|T - \text{id}\|_\infty = O(N^{-1})$ and $|z^p - y^p| \leq p|z - y|$ on the closed disk; the others follow from Lemma 16.1.

Every constant multiple of the right-hand side of (18.1) has a uniformly bounded exponential moment under Q . Here are the individual verifications. The compact count has all

exponential moments uniformly in N by the established Gaussian count tail. Since $M \leq N$, $M^2/N \leq M$. The deficit term is controlled by Lemma 18.1 and the transport bound. For $p < k$, $|S_{p,N}(y)| \leq N$ yields $|S_{p,N}(y)|^2/N \leq |S_{p,N}(y)|$, so the induction hypothesis and the bounded transport shift control each lower-mode term. Hölder's inequality combines these finitely many estimates, whose coefficients may be arbitrarily large fixed constants.

The exact change of variables now gives

$$\begin{aligned} \mathbb{E}_P e^{-2 \operatorname{Re}(\bar{t} S_{k,N})} &= \mathbb{E}_Q \exp\{-\log R - 2 \operatorname{Re}(\bar{t} S_{k,N})\} \\ &\leq \mathbb{E}_Q \exp\left\{C \left[1 + M + \frac{\mathcal{D}_N^2 + M^2 + \sum_{p=1}^{k-1} |S_{p,N}|^2}{N}\right]\right\} \leq C_{k,t,r,\beta}. \end{aligned}$$

The finitely many small values of N are harmless because $|S_{k,N}| \leq N$. Using t in the four real and imaginary directions, and $|w| \leq |\operatorname{Re} w| + |\operatorname{Im} w|$, proves uniform exponential moments of $|S_{k,N}|$. A further application of Hölder's inequality proves (18.2).

The established Gaussian convergence and (18.2) imply uniform integrability of every continuous test of at most exponential growth: use the same bound with a larger coefficient. Truncation and weak convergence prove the moment and independence assertions. $\square$

19 Analytic harmonic statistics: tightness and Laplace transforms

This section upgrades the finite-mode central limit theorem to analytic harmonic statistics and to convergence of their exponential moments. The argument uses a smooth rational vector field. Its singularity is removed on a fixed compact disk; the resulting error is controlled by that disk's particle count.

Let f be holomorphic on a neighborhood of $\bar{\mathbb{D}}$, with $f(0) = 0$. Define

$$Y_N(f) = 2 \operatorname{Re} \sum_{i=1}^N f(z_i), \quad Z_N(f) = \mathbb{E}_{N,\beta} e^{Y_N(f)}, \quad dP_N^f = Z_N(f)^{-1} e^{Y_N(f)} dP_{N,\beta}.$$

Here $Z_N(f)$ is a normalized moment-generating function, not the unweighted particle partition function. Also put

$$S_{k,N} = \sum_i z_i^k, \quad X_N(u) = \sum_i \frac{u z_i}{1 - u z_i} = \sum_{k \geq 1} u^k S_{k,N}, \quad |u| < 1,$$

and $D_N = \sum_i (1 - |z_i|)$. Throughout this section, $B_R = \{z : |z| < R\}$.

Lemma 19.1 (Uniform radial estimates under holomorphic tilts). *The radial stochastic comparison for the unweighted disk gas holds, with the same comparison rates*

$$a_j = 2j + \frac{\beta}{2} j(j-1),$$

under every law P_N^f . Consequently, for every $R < 1$ there are constants $C_R, c_R > 0$, depending on β, R but not on N or f , such that

$$P_N^f\{\Xi_N(B_R) \geq m\} \leq C_R e^{-c_R m^2}, \quad m \geq 0. \quad (19.1)$$

Moreover, $\mathbb{E}_N^f D_N = O_\beta(\log(N+1))$. For each fixed f and $s < 1$,

$$\frac{1}{N} \sup_{|u| \leq s} |X_N(u)| \longrightarrow 0 \quad \text{in } P_N^f\text{-probability.} \quad (19.2)$$

Proof. We explain why the previously proved radial comparison applies without change. In the ordered-radius proof, fix all angular variables except a common rotation of an inner block, and scale that block by a complex parameter q . The factors involving that block have the form

$$|P(q)|^\beta \exp \left(2 \operatorname{Re} \sum_{i \text{ in the block}} f(qw_i) \right) = \left| P(q) \exp \left(\frac{2}{\beta} \sum_{i \text{ in the block}} f(qw_i) \right) \right|^\beta,$$

where P is the same holomorphic cross-interaction polynomial as in the unweighted proof. The expression is subharmonic in q . Its circular mean is therefore nondecreasing in $|q|$. The radial Jacobian and the homogeneous Vandermonde exponents are unchanged. Thus the same product association argument gives exactly the same stochastic domination of radial gaps by independent exponentials of rates a_j . This proves (19.1) by the established count-tail estimate. It also gives

$$\mathbb{E}_N^f D_N \leq \mathbb{E}_N^f \sum_i (-\log |z_i|) \leq \sum_{j=1}^N \frac{j}{a_j} = O_\beta(\log(N+1)).$$

Rotation invariance of the unweighted gas and $f(0) = 0$ give $\mathbb{E}_{N,\beta} Y_N(f) = 0$. Jensen's inequality therefore gives $Z_N(f) \geq 1$. Hence, for any event A ,

$$P_N^f(A) \leq e^{2N\|f\|_{\infty, \mathbb{D}}} P_{N,\beta}(A).$$

The already proved macroscopic Fourier concentration bound, whose logarithm is $-c_{k,\varepsilon} N^2 + O_\beta(N \log N)$ for the event $\{|S_{k,N}| > \varepsilon N\}$, implies $S_{k,N}/N \rightarrow 0$ under P_N^f for each fixed k . Finally,

$$\frac{1}{N} \sup_{|u| \leq s} |X_N(u)| \leq \sum_{k=1}^K s^k \frac{|S_{k,N}|}{N} + \frac{s^{K+1}}{1-s}.$$

First choose K and then let $N \rightarrow \infty$. This proves (19.2). $\square$

Lemma 19.2 (A rational transport with a compact-count error). *Fix $0 < s < R_0 < R_1 < 1$ and a fixed holomorphic tilt f as above. For every u with $|u| \leq s$ and every $t \in \{1, -1, i, -i\}$ there is a smooth diffeomorphism $T_{N,u,t} : \mathbb{D} \rightarrow \mathbb{D}$. Its likelihood ratio $R_{N,u,t}^f$ under change of variables in P_N^f satisfies*

$$\mathbb{E}_N^f R_{N,u,t}^f = 1$$

and the deterministic bound

$$\left| \log R_{N,u,t}^f + \beta \operatorname{Re}(\bar{t} X_N(u)) \right| \leq C \left(1 + M_N + \frac{|X_N(u)|^2 + D_N^2 + M_N^2}{N} \right), \quad (19.3)$$

where $M_N = \Xi_N(B_{R_1})$ and C does not depend on N, u, t or the particle configuration. The assertion is needed only off the collision diagonals, which have probability zero.

Proof. For u fixed write

$$x_u(z) = \frac{uz}{1-uz}, \quad y_u(z) = \frac{\bar{u}}{z-\bar{u}}, \quad A_u(z) = zy_u(z), \quad B_u(z) = zx_u(z).$$

Choose a smooth real radial cutoff χ , equal to zero on B_{R_0} and equal to one on $\mathbb{D} \setminus B_{R_1}$. Define

$$v_{u,t}(z) = \chi(z) \{tA_u(z) - \bar{t}B_u(z)\},$$

with the zero definition throughout B_{R_0} . This removes the pole of A_u , since $|u| \leq s < R_0$. The vector fields and their derivatives of any fixed finite order are uniformly bounded in u, t . On $|z| = 1$ one has $y_u(z) = \overline{x_u(z)}$ and therefore

$$\operatorname{Re}(\bar{z}v_{u,t}(z)) = \operatorname{Re}\{\overline{tx_u(z)} - \bar{t}x_u(z)\} = 0.$$

Their flows preserve the disk and its boundary. Let $T_{N,u,t}$ be the flow at time $1/N$.

We record explicitly the uniform expansion used for this flow. If v is one of these vector fields and T_h is its flow, then

$$\log \frac{|T_h(z) - T_h(w)|}{|z - w|} = h \operatorname{Re} \frac{v(z) - v(w)}{z - w} + O(h^2), \quad z \neq w,$$

uniformly on $\mathbb{D}^2$. To see this, the flow equation gives $T_h = \operatorname{id} + hv + O(h^2)$ in the C^1 norm. The $O(h^2)$ remainder is Lipschitz with constant $O(h^2)$, so its divided difference has the same bound; expanding $\log |1 + q|$ proves the formula. Also $\log J_{T_h}(z) = O(h)$ uniformly. Since f has bounded first derivative, $|Y_N(f)(T_h z) - Y_N(f)(z)| = O(Nh)$. The exact density and Jacobian change of variables consequently give, at $h = 1/N$,

$$\log R_{N,u,t}^f = \frac{\beta}{N} \operatorname{Re} \sum_{i < j} \frac{v_{u,t}(z_i) - v_{u,t}(z_j)}{z_i - z_j} + O(1). \quad (19.4)$$

The total pairwise remainder is $O(N^2 h^2) = O(1)$, so this estimate is valid uniformly even for very close particles. Exact change of variables gives $\mathbb{E}_N^f R_{N,u,t}^f = 1$.

Let $J = \{i : |z_i| \geq R_1\}$ and $n = |J| = N - M_N$. Since $v_{u,t}$ is uniformly Lipschitz, deleting all pairs with at least one endpoint outside J changes the sum in (19.4), after division by N , by at most CM_N . For points indexed by J , the exact identities

$$\frac{A_u(z) - A_u(w)}{z - w} = -y_u(z)y_u(w), \quad \frac{B_u(z) - B_u(w)}{z - w} = x_u(z) + x_u(w) + x_u(z)x_u(w)$$

apply. With $X_J = \sum_{i \in J} x_u(z_i)$ and $Y_J = \sum_{i \in J} y_u(z_i)$, the remaining sum is

$$-\frac{t}{2} \left(Y_J^2 - \sum_{i \in J} y_u(z_i)^2 \right) - \bar{t} \left((n-1)X_J + \frac{1}{2} \left(X_J^2 - \sum_{i \in J} x_u(z_i)^2 \right) \right).$$

All single-particle x_u values, and all y_u values on J , are bounded by fixed constants. Thus the diagonal terms contribute $O(1)$ after division by N . Replacing $(n-1)X_J/N$ by $X_N(u)$ costs at most $C(M_N + 1)$. Finally, on $|z| \geq R_1$,

$$y_u(z) - \overline{x_u(z)} = \frac{\bar{u}(1 - |z|^2)}{(z - \bar{u})(1 - \bar{u}z)}, \quad |y_u(z) - \overline{x_u(z)}| \leq C(1 - |z|).$$

It follows that $|Y_J - \overline{X_J}| \leq CD_N$, and that

$$|Y_J|^2 + |X_J|^2 \leq C\{|X_N(u)|^2 + D_N^2 + M_N^2\}.$$

Substitution proves (19.3). □

Proposition 19.3 (Tightness of the analytic resolvent under all fixed tilts). *For every fixed f holomorphic near $\mathbb{D}$, the laws of X_N under P_N^f are tight in the space $\mathcal{H}(\mathbb{D})$ of holomorphic functions, with the topology of uniform convergence on compact subsets.*

Proof. Fix $s < R_0 < R_1 < 1$ as in the preceding lemma. Let C be its error constant. Choose positive δ, ε sufficiently small that

$$C\delta + C\varepsilon + C\delta^2\varepsilon < \frac{\beta}{2\sqrt{2}}.$$

Define the single event, common to every parameter $|u| \leq s$,

$$G_N = \left\{ D_N \leq N^{1/3}, \quad \sup_{|u| \leq s} |X_N(u)| \leq \varepsilon N \right\}.$$

Lemma 19.1 and Markov's inequality imply $P_N^f(G_N) \rightarrow 1$.

For every complex number x one of $t \in \{1, -1, i, -i\}$ satisfies $-\operatorname{Re}(\bar{t}x) \geq |x|/\sqrt{2}$. On $G_N \cap \{M_N \leq \delta|X_N(u)|\}$, the error in (19.3) is at most

$$C_0 + (C\delta + C\varepsilon + C\delta^2\varepsilon)|X_N(u)|.$$

Here $D_N^2/N \leq 1$, $|X_N(u)|^2/N \leq \varepsilon|X_N(u)|$, and $M_N^2/N \leq \delta^2\varepsilon|X_N(u)|$. For the indicated choice of t we consequently have

$$\log R_{N,u,t}^f \geq c|X_N(u)| - C_0, \quad c = \frac{\beta}{2\sqrt{2}} > 0.$$

By Markov's inequality, $\mathbb{E}_N^f R_{N,u,t}^f = 1$, a union bound over the four choices of t , and (19.1),

$$P_N^f(G_N \cap \{|X_N(u)| > L\}) \leq 4e^{C_0 - cL} + C_1 e^{-c_1 L^2}, \quad L \geq 0, \quad (19.5)$$

with constants uniform in N and $|u| \leq s$. In the second term we used that $M_N > \delta|X_N(u)|$ and $|X_N(u)| > L$ imply $M_N > \delta L$. Integrating the tail estimate gives

$$\sup_{N, |u| \leq s} \mathbb{E}_N^f[\mathbf{1}_{G_N} |X_N(u)|] < \infty. \quad (19.6)$$

For $0 < r < s$, Cauchy's integral formula gives, configuration by configuration,

$$\sup_{|u| \leq r} |X_N(u)| \leq \frac{s}{s-r} \frac{1}{2\pi} \int_0^{2\pi} |X_N(se^{i\theta})| d\theta.$$

Multiplying by the indicator of the *same* event G_N , taking expectations, and applying Fubini's theorem and (19.6) bounds the resulting expectation uniformly in N . Since $P_N^f(G_N^c) \rightarrow 0$, the local suprema are tight. This holds for every $r < 1$. On an increasing sequence of such disks, choose bounds with summable exceptional probabilities. Montel's theorem then supplies compact subsets of $\mathcal{H}(\mathbb{D})$ carrying arbitrarily high probability, proving tightness. $\square$

Lemma 19.4 (Tight exponential families give uniform integrability). *Let W_N be real random variables, with $\mathbb{E}e^{tW_N} < \infty$ for all real t . Suppose that for every fixed real t , the variables W_N are tight under the tilted laws*

$$dQ_N^t = \frac{e^{tW_N}}{\mathbb{E}e^{tW_N}} dP_N.$$

Then $\sup_N \mathbb{E}e^{tW_N} < \infty$ for every t , and $(e^{tW_N})_N$ is uniformly integrable for every t . If also $W_N \Rightarrow W$, then $\mathbb{E}e^{tW_N} \rightarrow \mathbb{E}e^{tW}$ for every real t .

Proof. For t fixed, tightness under Q_N^t provides $A < \infty$ with $Q_N^t(|W_N| \leq A) \geq 1/2$ for all N . Consequently

$$\frac{1}{2} \mathbb{E} e^{tW_N} \leq \mathbb{E}[e^{tW_N} \mathbf{1}_{\{|W_N| \leq A\}}] \leq e^{|t|A}.$$

Applying the same bound at pt for any $p > 1$ proves a uniform L^p bound for e^{tW_N} , hence uniform integrability. Convergence in law and uniform integrability give the last assertion, for instance by bounded continuous truncation. These bounds also show that all the asserted limiting exponential moments are finite. $\square$

Corollary 19.5 (Uniform analytic insertion bounds without a central limit theorem). *For every f holomorphic near $\overline{\mathbb{D}}$ with $f(0) = 0$, and every real t , one has*

$$\sup_N \mathbb{E}_{N,\beta} e^{tY_N(f)} < \infty.$$

In particular, for every real q and $R < 1$,

$$\sup_N \sup_{|a| \leq R} \mathbb{E}_{N,\beta} \left| \prod_{i=1}^N (1 - az_i) \right|^q < \infty, \quad \sup_N \sup_{|z| \leq R} \mathbb{E}_{N,\beta} \prod_{i=1}^N |z - z_i|^\beta < \infty. \quad (19.7)$$

These bounds use no central limit theorem or identification of any limiting distribution.

Proof. Write $f(z) = \sum_{k \geq 1} f_k z^k$ and choose $a > 1$ within its disk of holomorphy. Cauchy's estimate gives $|f_k| \leq C a^{-k}$, after decreasing a if necessary. Choose $a^{-1} < r < 1$. For $h(u) = \sum h_k u^k$ in $\mathcal{H}(\mathbb{D})$, another Cauchy estimate gives

$$\left| \sum_{k \geq 1} f_k h_k \right| \leq C \sup_{|u| \leq r} |h(u)| \sum_{k \geq 1} (ar)^{-k}.$$

Thus $h \mapsto 2 \operatorname{Re} \sum f_k h_k$ is a continuous real linear functional on $\mathcal{H}(\mathbb{D})$, and it takes X_N to $Y_N(f)$. Proposition 19.3 applied to every tilt tf proves that $Y_N(f)$ is tight under its own exponential tilt by $e^{tY_N(f)}$. Lemma 19.4 proves the first assertion.

Apply it to $f(z) = (q/2) \log(1 - az)$ for each fixed $|a| < 1$, with the holomorphic logarithm vanishing at zero. This proves the harmonic-product bound for each fixed parameter. As a function of the parameter a , the integrand is the squared modulus of a holomorphic function, for every real q . Its circular means are therefore nondecreasing. Rotation invariance of the gas makes its expectation radial, so the supremum on $|a| \leq R$ is its value at $a = R$, already bounded uniformly in N . Finally, $|z - w| \leq |1 - \bar{z}w|$ on the disk gives the insertion bound. $\square$

Theorem 19.6 (Analytic Gaussian fluctuations and harmonic Laplace limits). *Let $G_1, G_2, \dots$ be independent centered circular complex Gaussians with $\mathbb{E}|G_k|^2 = 2k/\beta$. Then*

$$X_N \Rightarrow X, \quad X(u) = \sum_{k \geq 1} G_k u^k,$$

in $\mathcal{H}(\mathbb{D})$. Along every subsequence for which $\Xi_{N_j} \Rightarrow \Xi$ vaguely in the open disk,

$$(\Xi_{N_j}, X_{N_j}) \Rightarrow (\Xi, X),$$

where X and Ξ are independent.

For $f(z) = \sum_{k \geq 1} f_k z^k$ holomorphic near $\overline{\mathbb{D}}$,

$$\lim_{N \rightarrow \infty} \mathbb{E}_{N,\beta} \exp \left(2 \operatorname{Re} \sum_i f(z_i) \right) = \exp \left(\frac{2}{\beta} \sum_{k \geq 1} k |f_k|^2 \right). \quad (19.8)$$

Moreover, for every bounded continuous functional F on the space of locally finite counting measures and every subsequence as above,

$$\lim_{j \rightarrow \infty} \mathbb{E}_{N_j, \beta} [F(\Xi_{N_j}) e^{Y_{N_j}(f)}] = \mathbb{E} F(\Xi) \exp \left(\frac{2}{\beta} \sum_{k \geq 1} k |f_k|^2 \right). \quad (19.9)$$

Proof. Proposition 19.3, with $f = 0$, proves tightness in $\mathcal{H}(\mathbb{D})$. Taylor coefficient evaluation is continuous in that space. The already proved unweighted finite-mode theorem therefore identifies every subsequential limit with the indicated Gaussian series. That series converges locally uniformly almost surely: for each $r < 1$,

$$\mathbb{E} \sum_{k \geq 1} |G_k| r^k \leq \sum_{k \geq 1} (2k/\beta)^{1/2} r^k < \infty.$$

The finite-mode theorem also identifies joint coefficients with the interior limit and gives independence for every finite coefficient vector. Taylor coefficients generate the Borel sigma field of $\mathcal{H}(\mathbb{D})$ (they determine each function, and their partial sums recover it locally). Hence the full Gaussian analytic function is independent of Ξ , and the joint convergence follows.

Choose $R > 1$ such that f is holomorphic on $\{|z| < R\}$, and choose $R^{-1} < r < 1$. If $h(u) = \sum_{k \geq 0} h_k u^k$ is holomorphic in $\mathbb{D}$, then

$$L_f(h) = \sum_{k \geq 1} f_k h_k$$

is a continuous linear functional on $\mathcal{H}(\mathbb{D})$. Indeed, choose R' with $1/r < R' < R$. Cauchy's estimates give $|f_k| \leq C(R')^{-k}$ and $|h_k| \leq r^{-k} \sup_{|u| \leq r} |h(u)|$, so the series is bounded by a convergent geometric series times that local supremum. In particular, $Y_N(f) = 2 \operatorname{Re} L_f(X_N)$. The analytic convergence proves

$$Y_N(f) \Rightarrow Y(f) = 2 \operatorname{Re} \sum_{k \geq 1} f_k G_k, \quad \operatorname{Var} Y(f) = \frac{4}{\beta} \sum_{k \geq 1} k |f_k|^2,$$

with joint independence from the subsequential interior limit.

For each real t , the exponential tilt by $e^{tY_N(f)}$ is exactly P_N^{tf} . Proposition 19.3 and continuity of L_f show that $Y_N(f)$ is tight under this tilted law. Lemma 19.4 consequently proves uniform integrability of $e^{Y_N(f)}$, as well as of every fixed positive power of it. Passing to the limit gives (19.8), using the elementary real Gaussian moment-generating function. Joint convergence, boundedness of F , and the same uniform integrability give (19.9). $\square$

Corollary 19.7 (Tilted finite modes and invariance of interior limits). *Under P_N^f , every finite mode vector converges to independent circular complex Gaussians with means*

$$m_k = \frac{2k}{\beta} \overline{f_k}$$

and variances $2k/\beta$. Along any unweighted interior-convergent subsequence, the interior process under P_N^f converges to that same interior limit; it is independent of the shifted Gaussian analytic function in the joint limit.

Proof. The preceding proof gives joint convergence of the interior, the analytic resolvent, and $Y_N(f)$, as well as uniform integrability of $e^{Y_N(f)}$. Exponential change of measure is therefore legitimate. Completing the square in each Gaussian coordinate yields the stated means and unchanged variances. The tilt depends only on the Gaussian analytic function, which was independent of the interior; the marginal interior law is therefore unchanged. Infinite-dimensional statements follow first on finite coefficient vectors and then by the already established analytic tightness under P_N^f . $\square$

Corollary 19.8 (Uniform bounds for harmonic products and insertions). *For $|u| < 1$ set*

$$H_N(u) = \sum_{i=1}^N \log |1 - uz_i|.$$

For every real q ,

$$\lim_{N \rightarrow \infty} \mathbb{E}_{N,\beta} e^{qH_N(u)} = (1 - |u|^2)^{-q^2/(2\beta)}. \quad (19.10)$$

For every $r < 1$,

$$\sup_N \sup_{|u| \leq r} \mathbb{E}_{N,\beta} e^{qH_N(u)} < \infty. \quad (19.11)$$

Consequently the insertion functions

$$I_N(z) = \mathbb{E}_{N,\beta} \prod_{i=1}^N |z - z_i|^\beta$$

satisfy $\sup_N \sup_{|z| \leq r} I_N(z) < \infty$ for every $r < 1$, and

$$\limsup_{N \rightarrow \infty} I_N(z) \leq (1 - |z|^2)^{-\beta/2}.$$

Proof. Apply (19.8) to $f(z) = (q/2) \log(1 - uz)$, using the holomorphic logarithm which vanishes at $z = 0$. Its coefficients are $f_k = -qu^k/(2k)$, which give (19.10).

For completeness, pointwise convergence also gives the uniform bound (19.11). For each fixed configuration, $\exp(qH_N(u))$ is the squared modulus of the holomorphic function

$$\exp \left(\frac{q}{2} \sum_i \log(1 - uz_i) \right).$$

Its angular mean in u is nondecreasing with $|u|$, by subharmonicity. After expectation, rotation invariance of the unweighted gas makes $u \mapsto \mathbb{E} e^{qH_N(u)}$ radial, so this function itself is nondecreasing with $|u|$. Its supremum on $|u| \leq r$ is thus its value at the positive real number r , and those values are bounded by (19.10).

Finally,

$$|1 - \bar{z}w|^2 - |z - w|^2 = (1 - |z|^2)(1 - |w|^2) \geq 0, \quad z, w \in \mathbb{D}.$$

Therefore $I_N(z) \leq \mathbb{E} e^{\beta H_N(\bar{z})}$. The claimed bounds follow from (19.10)–(19.11). $\square$

Corollary 19.9 (Exponential bounds for analytic local suprema). *For every $r < 1$ and $\lambda \geq 0$,*

$$\sup_N \mathbb{E}_{N,\beta} \exp \left(\lambda \sup_{|u| \leq r} |X_N(u)| \right) < \infty.$$

In particular, all fixed moments of these local suprema are uniformly integrable, and every finite collection of analytic linear statistics has convergence of all joint polynomial moments.

Proof. Choose $r < s < 1$. The analytic Laplace theorem, applied to constant complex multiples of $z \mapsto sz/(1 - sz)$, bounds the moment-generating function of every fixed real linear projection of $X_N(s)$ uniformly in N . Since, for $a \geq 0$,

$$e^{a|x|} \leq e^{a(|\operatorname{Re} x| + |\operatorname{Im} x|)} \leq \sum_{\varepsilon, \eta \in \{-1, 1\}} e^{a(\varepsilon \operatorname{Re} x + \eta \operatorname{Im} x)},$$

it follows that $\sup_N \mathbb{E} e^{a|X_N(s)|} < \infty$ for every a . Rotation invariance makes the distribution of $X_N(se^{i\theta})$ independent of θ . Cauchy's estimate from the preceding proof and Jensen's inequality now give

$$\begin{aligned} \mathbb{E} \exp \left(\lambda \sup_{|u| \leq r} |X_N(u)| \right) &\leq \frac{1}{2\pi} \int_0^{2\pi} \mathbb{E} \exp \left(\frac{\lambda s}{s-r} |X_N(se^{i\theta})| \right) d\theta \\ &= \mathbb{E} \exp \left(\frac{\lambda s}{s-r} |X_N(s)| \right). \end{aligned}$$

This is uniformly bounded. Higher exponential bounds imply uniform integrability of every power. Finally, the continuous coefficient pairings L_f used above are bounded by a constant times one such local supremum, so the same conclusion holds for their finite products. $\square$

20 Harmonic boundary transport and uniqueness for $\beta > 4$

Write

$$P_N(w) = \prod_{i=1}^N (1 - wz_i), \quad M_{N,\beta}(r) = \mathbb{E}_{N,\beta} |P_N(r)|^\beta, \quad Q_{N,\beta} = Z_{N,\beta}/N!.$$

The following result concerns the original gas with flat area measure in the disk. In particular, no change of model or of the inverse temperature is involved.

Theorem 20.1 (Boundary insertion and the empty limit above 4). *For every fixed $\beta > 0$, there are positive constants c_β, C_β such that, for all sufficiently large N ,*

$$M_{N,\beta}(1) \geq c_\beta \frac{N^{\beta/2}}{(\log N)^\beta}, \quad (20.1)$$

$$\frac{Q_{N+1,\beta}}{Q_{N,\beta}} \geq c_\beta \frac{N^{\beta/2-2}}{(\log N)^\beta}. \quad (20.2)$$

Consequently, for every $\beta > 4$, the full sequence of point processes converges in probability, for the vague topology on the open disk, to the empty configuration. More quantitatively, for every $R < 1$,

$$\sup_{|z| \leq R} \rho_{N,\beta}(z) + \mathbb{P}_{N,\beta} \{ \Xi_N(\{|z| \leq R\}) > 0 \} \leq C_{\beta,R} N^{2-\beta/2} (\log N)^\beta. \quad (20.3)$$

The restriction to each such disk converges in total variation to the point mass at the empty configuration, with the same upper bound up to a constant.

We prove the theorem by a deterministic boundary perturbation and a relative-entropy estimate. The use of a harmonic extension, instead of rotating each particle by a function of its angle alone, is essential: the resulting quadratic cost involves the analytic modes $S_{j,N} = \sum_i z_i^j$ directly.

Lemma 20.2 (Two estimates used in the transport argument). *For every fixed $\beta > 0$, with $B_N = \sum_i -\log |z_i|$, one has*

$$\mathbb{E} B_N \leq C_\beta \log(eN), \quad \sum_{j=1}^N \frac{\mathbb{E} |S_{j,N}|^2}{j} \leq C_\beta N \log(eN). \quad (20.4)$$

Proof. The radial-gap domination bounds B_N in a coupling by $\sum_{k=1}^N kE_k$, where the E_k are independent exponentials with rates $a_k = 2k + \frac{\beta}{2}k(k-1)$. Its mean is $\sum_{k \leq N} k/a_k = O_\beta(\log(eN))$.

For the second assertion let $q = 1 - 1/N$ and $L_N = \log |\Delta_N|$. The identity

$$|1 - q^2 z \bar{w}|^2 - q^2 |z - w|^2 = (1 - q^2 |z|^2)(1 - q^2 |w|^2) \geq 0$$

and the logarithm power series yield

$$\sum_{j \geq 1} \frac{q^{2j}}{j} |S_{j,N}|^2 \leq -2L_N - N(N-1) \log q - N \log(1 - q^2).$$

Relative entropy with respect to normalized product area gives

$$\beta \mathbb{E} L_N \geq \log(Z_{N,\beta}/\pi^N) = \left(\frac{\beta}{2} - 1\right) N \log N + O_\beta(N),$$

where the last equality is the established partition estimate. The expected right-hand side of the preceding inequality is therefore $O_\beta(N \log(eN))$. For $1 \leq j \leq N$, $q^{2j} \geq q^{2N} \geq 1/16$ when $N \geq 2$. This proves the claim; the case $N = 1$ is absorbed into the constant. $\square$

Lemma 20.3 (Harmonic extension of a boundary perturbation). *Let $r \in (1/2, 1)$, $m = (1 - r)^{-1}$, and*

$$\psi_r(\theta) = \sum_{k \geq 1} \frac{r^k}{k} \sin(k\theta), \quad \ell_r = -\log(1 - r^2).$$

Let $\varepsilon > 0$, and let T be the harmonic Poisson extension to $\mathbb{D}$ of

$$e^{i\theta} \mapsto e^{i\theta} e^{i\varepsilon \psi_r(\theta)}.$$

Write $T(z) = z + V(z)$ and write the Fourier coefficients of its boundary displacement as

$$e^{i\theta} (e^{i\varepsilon \psi_r(\theta)} - 1) = \sum_{n \in \mathbb{Z}} c_n e^{in\theta}.$$

There are absolute constants $c, C > 0$ such that, if $\varepsilon m \leq c$, then T is a smooth diffeomorphism of the disk, extending to a homeomorphism of its closure, and

$$\|V\|_\infty \leq C\varepsilon, \quad \|DV\|_\infty \leq C\varepsilon m, \quad (20.5)$$

$$\sum_{n \in \mathbb{Z}} |n| |c_n|^2 \leq \varepsilon^2 (\ell_r/2 + C), \quad |c_1| \leq C\varepsilon^2, \quad (20.6)$$

$$\sum_{n \in \mathbb{Z}} n^2 |c_n|^2 e^{|n|/(2m)} \leq C\varepsilon^2 m. \quad (20.7)$$

Here DV is the real derivative matrix. The coefficient c_1 is real and nonpositive.

Proof. Since $\psi_r = -\text{Arg}(1 - re^{i\theta})$ on the circle,

$$\|\psi_r\|_\infty \leq \pi/2, \quad \|\psi_r'\|_\infty \leq m, \quad \|\psi_r\|_2^2 = \frac{1}{2} \sum_{k \geq 1} r^{2k}/k^2 \leq C.$$

The boundary displacement has modulus at most $\varepsilon\pi/2$; the Poisson representation therefore gives the first bound in (20.5). Moreover,

$$c_1 = \int_0^{2\pi} (e^{i\varepsilon \psi_r(\theta)} - 1) \frac{d\theta}{2\pi}.$$

Oddness of ψ_r makes this real, and the inequalities $0 \leq 1 - \cos x \leq x^2/2$ give the stated bounds on c_1 .

Set $u = e^{i\varepsilon\psi_r} - 1$ and denote its Fourier coefficients by u_k . The Fourier identity

$$\sum_{k \in \mathbb{Z}} |k| |u_k|^2 = \iint \frac{|u(\theta) - u(\phi)|^2}{4 \sin^2((\theta - \phi)/2)} \frac{d\theta}{2\pi} \frac{d\phi}{2\pi}$$

follows by Fourier orthogonality and the Fejér identity $\int (1 - \cos(k\theta))/(1 - \cos \theta) d\theta/(2\pi) = |k|$. It is first proved for trigonometric polynomials and then by smooth approximation; here all functions are analytic on a strip, so convergence in every Sobolev norm is available. Since $|e^{i\varepsilon a} - e^{i\varepsilon b}| \leq \varepsilon|a - b|$ for real a, b , the displayed identity gives

$$\sum_k |k| |u_k|^2 \leq \varepsilon^2 \sum_k |k| |\widehat{\psi_r}(k)|^2 = \varepsilon^2 \ell_r/2.$$

As $c_n = u_{n-1}$ and $|n| \leq |n-1| + 1$, adding $\|u\|_2^2 \leq \varepsilon^2 \|\psi_r\|_2^2$ proves (20.6).

For completeness, we prove the exponentially weighted estimate rather than using a derivative bound for a singular boundary kernel. Set $h = 1/(4m)$. The series for ψ_r extends holomorphically to $|\operatorname{Im} \theta| \leq h$, because $re^h \leq e^{-3/(4m)} < 1$. On either line $\theta = x \pm ih$, Fourier orthogonality gives

$$\|\psi_r(\cdot \pm ih)\|_2 \leq C, \quad \|\psi_r'(\cdot \pm ih)\|_2 \leq C\sqrt{m}, \quad \|\psi_r(\cdot \pm ih)\|_\infty \leq C \log(em).$$

The condition $\varepsilon m \leq c$ bounds $e^{\varepsilon|\operatorname{Im} \psi_r|}$ by an absolute constant. Differentiating $b(\theta) = e^{i\theta}(e^{i\varepsilon\psi_r(\theta)} - 1)$ and using $|e^w - 1| \leq |w|e^{|w|}$ therefore shows

$$\|b'(\cdot \pm ih)\|_2^2 \leq C\varepsilon^2 m.$$

Parseval's identity on the two lines, followed by addition, yields (20.7). The harmonic extension is explicitly

$$V(z) = c_0 + \sum_{n \geq 1} c_n z^n + \sum_{n \leq -1} c_n \bar{z}^{|n|}.$$

Cauchy–Schwarz and (20.7) give

$$\|DV\|_\infty \leq \sum_n |n| |c_n| \leq \left(\sum_n n^2 |c_n|^2 e^{|n|/(2m)} \right)^{1/2} \left(\sum_{n \neq 0} e^{-|n|/(2m)} \right)^{1/2} \leq C\varepsilon m.$$

Shrinking c if necessary makes this smaller than $1/2$.

The boundary map is a circle diffeomorphism, since its angular derivative is $1 + \varepsilon\psi_r' > 0$. The Poisson extension maps the disk strictly inside the disk: it is an average of unit vectors, and equality in the triangle inequality would force the nonconstant boundary map to be constant. Also, on the convex closed disk,

$$|T(z) - T(w)| \geq (1 - \|DV\|_\infty)|z - w|.$$

Thus T is injective and has everywhere invertible real derivative. Its image of the open disk is open by the inverse function theorem. It is also closed relative to the open disk: if $T(z_n) \rightarrow y \in \mathbb{D}$, a convergent subsequence in the closed disk gives $T(z) = y$, and z cannot be on the boundary because T maps that boundary to the unit circle. Connectedness of the disk proves surjectivity. The inverse function theorem and compactness now give all claimed regularity properties. $\square$

Lemma 20.4 (A pairwise bound involving only analytic modes). *For $p \geq 1$ put $D_p(z, w) = (z^p - w^p)/(z - w)$, extended polynomially to the diagonal. For any $z_1, \dots, z_N \in \mathbb{D}$,*

$$\sum_{i < j} |D_p(z_i, z_j)|^2 \leq \frac{pN^2}{2} + \sum_{j=1}^{p-1} (p-j) |S_{j,N}|^2. \quad (20.8)$$

Consequently, for the harmonic displacement in Lemma 20.3, rotation invariance gives

$$\mathbb{E} \sum_{i < j} \left| \frac{V(z_i) - V(z_j)}{z_i - z_j} \right|^2 \leq \frac{N^2}{2} \sum_n |n| |c_n|^2 + \sum_{j \geq 1} K_j \mathbb{E} |S_{j,N}|^2, \quad (20.9)$$

where $K_j = \sum_{|n| > j} (|n| - j) |c_n|^2$.

Proof. For each i , let ζ_i be an independent point on the unit circle with Poisson distribution based at z_i . Its density is $(1 - |z_i|^2)/|e^{i\theta} - z_i|^2$ relative to $d\theta/(2\pi)$, and its moments are $\mathbb{E} \zeta_i^j = z_i^j$ for $j \geq 0$. These statements follow directly by expanding the Poisson kernel. Since D_p is holomorphic in each variable,

$$D_p(z_i, z_l) = \mathbb{E}[D_p(\zeta_i, \zeta_l)], \quad i \neq l,$$

and Jensen's inequality bounds its squared modulus by the corresponding second moment. On the circle, the Fejér identity gives

$$\sum_{i < l} |D_p(\zeta_i, \zeta_l)|^2 = \frac{pN(N-1)}{2} + \sum_{j=1}^{p-1} (p-j) \left(\left| \sum_i \zeta_i^j \right|^2 - N \right).$$

Independence and the Poisson moments imply

$$\mathbb{E} \left| \sum_i \zeta_i^j \right|^2 = |S_{j,N}|^2 + \sum_i (1 - |z_i|^{2j}).$$

Substitution and deletion of nonpositive terms prove (20.8).

For the second assertion, apply a common rotation $z_i \mapsto e^{i\alpha} z_i$. In the divided difference of V , the positive Fourier mode $n \geq 1$ has rotation frequency $n - 1$, while the negative mode $n = -p \leq -1$ has frequency $-p - 1$. These frequencies are all distinct. Orthogonality therefore makes the rotational average of its squared modulus equal to

$$\sum_{n \neq 0} |c_n|^2 |D_{|n|}(z_i, z_l)|^2.$$

The harmless unimodular factor in an antiholomorphic divided difference disappears in its squared modulus. Use rotation invariance, sum (20.8) with these nonnegative weights, and interchange nonnegative sums to obtain (20.9). $\square$

Proposition 20.5 (Mesoscopic moment lower bound by harmonic transport). *Suppose $m_N \rightarrow \infty$ and $m_N \log(eN)/N \rightarrow 0$, and put $r_N = 1 - 1/m_N$. Then, for fixed $\beta > 0$,*

$$\log M_{N,\beta}(r_N) \geq \frac{\beta}{2} \log \frac{1}{1 - r_N^2} - C_\beta - o(1). \quad (20.10)$$

Proof. Suppress subscripts on r, m and put $\varepsilon = 2/N$. Use the map $T = \text{id} + V$ of Lemma 20.3, and let $\mathbf{Q}$ be the pushforward of the gas law $\mathbf{P}$ under T in all coordinates. Set $\eta = \|DV\|_\infty = o(1)$ and

$$W(z) = \beta \sum_i \log |1 - rz_i|.$$

The Gibbs variational inequality, which follows from nonnegativity of relative entropy with respect to $e^W \mathbf{P} / \mathbb{E}_{\mathbf{P}} e^W$, says

$$\log M_{N,\beta}(r) \geq \mathbb{E}_{\mathbf{Q}} W - \text{Ent}(\mathbf{Q} \mid \mathbf{P}). \quad (20.11)$$

All changes of variables are legitimate: T is a diffeomorphism, its Jacobian is bounded above and below, and $|Tz - Tw|/|z - w|$ lies between $1 - \eta$ and $1 + \eta$.

Entropy cost. For any complex q with $|q| \leq \eta < 1$, Taylor's formula gives

$$\log |1 + q| \geq \text{Re } q - \frac{|q|^2}{2(1 - \eta)^2}.$$

Likewise, the real matrix logarithm, or Taylor's formula for the determinant in dimension two, gives

$$\log \det(I + DV) \geq \text{div } V - C|DV|^2$$

when $\eta \leq 1/2$. The exact density change hence implies

$$\begin{aligned} \text{Ent}(\mathbf{Q} \mid \mathbf{P}) &\leq -\mathbb{E} \sum_i \text{div } V(z_i) + C \mathbb{E} \sum_i |DV(z_i)|^2 \\ &\quad - \beta \mathbb{E} \text{Re} \sum_{i < j} \frac{V(z_i) - V(z_j)}{z_i - z_j} + \frac{\beta}{2(1 - \eta)^2} \mathbb{E} \sum_{i < j} \left| \frac{V(z_i) - V(z_j)}{z_i - z_j} \right|^2. \end{aligned} \quad (20.12)$$

Global rotational averaging gives

$$\mathbb{E} \sum_i \text{div } V(z_i) = 2N \text{Re } c_1, \quad \mathbb{E} \sum_{i < j} \frac{V(z_i) - V(z_j)}{z_i - z_j} = \frac{N(N - 1)}{2} c_1.$$

Both linear costs are therefore $O_\beta(N^2 \varepsilon^2) = O_\beta(1)$. The explicit harmonic series also gives, by angular averaging at each radius and then expectation,

$$\mathbb{E} \sum_i |DV(z_i)|^2 \leq CN \sum_n n^2 |c_n|^2 \leq CN \varepsilon^2 m.$$

This estimate holds for the Frobenius matrix norm and hence for any fixed choice of real matrix norm, after changing the absolute constant.

To control the final term in (20.12), notice from (20.7) that

$$\sup_{j \geq 1} j K_j \leq \frac{1}{4} \sum_n n^2 |c_n|^2 \leq C \varepsilon^2 m,$$

because $j(p - j) \leq p^2/4$. Also,

$$\sum_{j > N} K_j \leq \frac{1}{2} \sum_{|n| > N} n^2 |c_n|^2 \leq C \varepsilon^2 m e^{-N/(2m)}.$$

Lemma 20.2 for $j \leq N$, and the deterministic bound $|S_{j,N}| \leq N$ for $j > N$, consequently give

$$\sum_{j \geq 1} K_j \mathbb{E} |S_{j,N}|^2 \leq C_\beta \varepsilon^2 m N \log(eN) + CN^2 \varepsilon^2 m e^{-N/(2m)}. \quad (20.13)$$

Combine (20.9), (20.6), and (20.13). Since $\eta \ell_r = O(m \log(eN)/N) = o(1)$, this proves

$$\text{Ent}(\mathbf{Q} \mid \mathbf{P}) \leq \frac{\beta \varepsilon^2 N^2}{8} \ell_r + C_\beta + o(1) = \frac{\beta}{2} \ell_r + C_\beta + o(1). \quad (20.14)$$

Here the tail error tends to zero: the hypothesis implies $N/m \gg \log N$, and therefore $me^{-N/(2m)} \rightarrow 0$.

Gain in the inserted potential. Let v be the harmonic extension of $ie^{i\theta}\psi_r(\theta)$. It is given explicitly by

$$v(z) = \frac{1}{2} \sum_{k \geq 1} \frac{r^k}{k} (z^{k+1} - \bar{z}^{k-1}).$$

The boundary inequality $|e^{i\epsilon x} - 1 - i\epsilon x| \leq \epsilon^2 x^2/2$ and the Poisson representation show

$$\|V - \epsilon v\|_\infty \leq C\epsilon^2, \quad \|v\|_\infty \leq \pi/2. \quad (20.15)$$

Use the holomorphic logarithm $F(z) = \beta \log(1 - rz)$, with $F(0) = 0$, so that $W = \operatorname{Re} \sum_i F(z_i)$. Rotation invariance implies $\mathbb{E}_{\mathbf{P}} W = 0$. Expanding $F'(z) = -\beta \sum_{k \geq 1} r^k z^{k-1}$ and averaging a common rotation gives the exact identity

$$\epsilon \mathbb{E} \operatorname{Re} \sum_i F'(z_i) v(z_i) = \frac{\epsilon \beta}{2} \sum_{k \geq 1} \frac{r^{2k}}{k} \mathbb{E} \sum_i |z_i|^{2k-2}.$$

For $k \geq 1$, $1 - u^{2k-2} \leq 2k(-\log u)$ on $0 < u \leq 1$. Lemma 20.2 therefore bounds the preceding quantity below by

$$\frac{\epsilon \beta N}{2} \ell_r - C_\beta \epsilon m \log(eN) = \beta \ell_r - o(1).$$

The linear error from $V - \epsilon v$ is at most $CN\|F'\|_\infty \epsilon^2 \leq C_\beta N \epsilon^2 m = o(1)$.

The quadratic Taylor remainder also has this latter bound; the following averaged estimate avoids the incorrect larger bound of order $N\epsilon^2 m^2$. For $0 \leq s \leq 1$, the uniform displacement estimate and $\epsilon m = o(1)$ imply

$$|1 - r(z + sV(z))| \geq (1 - C\epsilon m)|1 - rz| \geq \frac{1}{2}|1 - rz|.$$

Consequently $|F''(z + sV(z))| \leq 4\beta r^2/|1 - rz|^2$. At any fixed radius $u \leq 1$,

$$\int_0^{2\pi} \frac{d\theta/(2\pi)}{|1 - rue^{i\theta}|^2} = \frac{1}{1 - r^2 u^2} \leq \frac{1}{1 - r^2} \leq m.$$

Using $\|V\|_\infty \leq C\epsilon$, Taylor's formula and rotation invariance thus bound the expected total absolute remainder by $C_\beta N \epsilon^2 m = o(1)$. We have proved

$$\mathbb{E}_{\mathbf{Q}} W \geq \beta \ell_r - o(1). \quad (20.16)$$

Equations (20.11), (20.14), and (20.16) prove the proposition. $\square$

Proof of Theorem 20.1. Take $m_N = N/(\log N)^2$ and $r_N = 1 - 1/m_N$ for all sufficiently large N . Proposition 20.5 applies and gives

$$M_{N,\beta}(r_N) \geq c_\beta (1 - r_N^2)^{-\beta/2} \geq c_\beta m_N^{\beta/2} = c_\beta \frac{N^{\beta/2}}{(\log N)^\beta}.$$

For each configuration, $w \mapsto |P_N(w)|^\beta$ is subharmonic, so its circular means are nondecreasing. Rotation invariance identifies the expectation of those means with $M_{N,\beta}(r)$. Thus $M_{N,\beta}(1) \geq M_{N,\beta}(r_N)$, proving (20.1).

We include the exact boundary insertion recursion. Choose the particle of largest radius in an $(N + 1)$ -particle configuration, write it as $te^{i\theta}$, and express the remaining particles

as $te^{i\theta}w_i$, $w_i \in \mathbb{D}$. Ties have zero probability. The polar Jacobian and homogeneity of the Vandermonde give

$$Z_{N+1,\beta} = \frac{2\pi(N+1)}{a_{N+1}} Z_{N,\beta} M_{N,\beta}(1), \quad a_{N+1} = 2(N+1) + \frac{\beta}{2}N(N+1).$$

Indeed, the radial integral is $\int_0^1 t^{a_{N+1}-1} dt = a_{N+1}^{-1}$, and the common angle contributes 2π . Dividing by factorials yields

$$\frac{Q_{N+1,\beta}}{Q_{N,\beta}} = \frac{2\pi}{a_{N+1}} M_{N,\beta}(1),$$

and proves (20.2).

For the conclusion about the process, Corollary 19.5 has already proved, without using a central limit theorem,

$$\sup_n \sup_{|a| \leq R} \mathbb{E}_{n,\beta} \left| \prod_{i=1}^n (1 - az_i) \right|^\beta < \infty \quad (R < 1).$$

The disk identity

$$|1 - \bar{z}w|^2 - |z - w|^2 = (1 - |z|^2)(1 - |w|^2) \geq 0$$

and the one-point insertion formula imply

$$\begin{aligned} \rho_{N,\beta}(z) &= \frac{Q_{N-1,\beta}}{Q_{N,\beta}} \mathbb{E}_{N-1,\beta} \prod_i |z - w_i|^\beta \\ &\leq \frac{Q_{N-1,\beta}}{Q_{N,\beta}} \mathbb{E}_{N-1,\beta} \left| \prod_i (1 - \bar{z}w_i) \right|^\beta. \end{aligned}$$

The uniform harmonic-moment bound and (20.2) give the intensity assertion in (20.3). Integrating over the disk of radius R and applying Markov's inequality gives the probability assertion. This tends to zero when $\beta > 4$.

For a probability law ν on configurations, its total variation distance to the point mass at the empty configuration is $1 - \nu(\{\emptyset\})$, using the convention $\|\nu - \mu\|_{\text{TV}} = \sup_A |\nu(A) - \mu(A)|$. This proves the local total variation assertion. Finally, every compact subset of the open disk lies in a disk of radius $R < 1$. For each compactly supported continuous test function, its integral against Ξ_N is therefore zero with probability tending to one. Equivalently, on any countable family of test functions defining a metric for vague convergence, finite unions and then the summable tail of the metric show convergence in probability to the empty measure. This proves the full-sequence assertion and hence uniqueness of the limiting law for $\beta > 4$. $\square$

21 Quantitative consequences of the empty limit

Fix $\beta > 4$ and put

$$b_N = N^{2-\beta/2} (\log(eN))^\beta.$$

Thus $b_N \rightarrow 0$. In this section we use the adjacent-ratio lower bound (20.2), in the equivalent form

$$\frac{Q_{N-1,\beta}}{Q_{N,\beta}} \leq C_\beta b_N \tag{21.1}$$

for all sufficiently large N , and the uniform harmonic-product bound of Corollary 19.8.

Corollary 21.1 (Decay of every correlation and compact count moment). *For every fixed integer $p \geq 1$ and $r < 1$, there is a constant $C_{\beta,p,r} < \infty$ such that, for all sufficiently large N ,*

$$\rho_{p,N}(z_1, \dots, z_p) \leq C_{\beta,p,r} |\Delta_p(z_1, \dots, z_p)|^\beta b_N^p, \quad |z_1|, \dots, |z_p| \leq r. \quad (21.2)$$

Here $\Delta_1 = 1$. Consequently, for every relatively compact Borel set $B \subset \mathbb{D}$, with $M_N = \Xi_N(B)$, and every fixed integer $p \geq 1$,

$$\mathbb{E}(M_N)_p = O_{\beta,p,B}(b_N^p), \quad \mathbb{P}(M_N \geq p) = O_{\beta,p,B}(b_N^p), \quad \mathbb{E}M_N^p = O_{\beta,p,B}(b_N). \quad (21.3)$$

For each fixed $t \geq 0$, one also has

$$\mathbb{E}(e^{tM_N} - 1) = O_{\beta,t,B}(b_N). \quad (21.4)$$

In particular, the local counting generating functions converge to 1 uniformly on compact subsets of the complex parameter plane, at rate $O(b_N)$.

Proof. The exact p -point insertion formula, for $N \geq p$, is

$$\begin{aligned} \rho_{p,N}(z_1, \dots, z_p) &= \frac{Q_{N-p,\beta}}{Q_{N,\beta}} |\Delta_p(z_1, \dots, z_p)|^\beta \\ &\quad \times \mathbb{E}_{N-p,\beta} \prod_{i=1}^p \prod_{j=1}^{N-p} |z_i - w_j|^\beta. \end{aligned}$$

Indeed, the combinatorial factor $N!/(N-p)!$ in the correlation function cancels against the factorials in the definition $Q_n = Z_n/n!$. The disk inequality $|z - w| \leq |1 - \bar{z}w|$ and Hölder's inequality give

$$\begin{aligned} \mathbb{E}_{N-p,\beta} \prod_{i=1}^p \prod_j |z_i - w_j|^\beta &\leq \mathbb{E}_{N-p,\beta} \prod_{i=1}^p e^{\beta H_{N-p}(\bar{z}_i)} \\ &\leq \prod_{i=1}^p \left(\mathbb{E}_{N-p,\beta} e^{p\beta H_{N-p}(\bar{z}_i)} \right)^{1/p} \leq C_{\beta,p,r}. \end{aligned}$$

The last bound is uniform in the tuple on the specified compact disk. For fixed p , multiplying (21.1) at $N, N-1, \dots, N-p+1$ gives

$$\frac{Q_{N-p,\beta}}{Q_{N,\beta}} \leq C_{\beta,p} \prod_{j=0}^{p-1} b_{N-j} \leq C'_{\beta,p} b_N^p.$$

The last inequality uses $b_{N-j}/b_N \rightarrow 1$ for fixed j . This proves (21.2).

Choose $r < 1$ with $B \subset B_r$. Integrating the correlation bound over B^p proves the factorial-moment assertion; the Vandermonde is bounded on this compact product. Since $(M_N)_p \geq p!$ on $\{M_N \geq p\}$, Markov's inequality proves the probability bound. The identity

$$M_N^p = \sum_{j=1}^p \left\{ \begin{matrix} p \\ j \end{matrix} \right\} (M_N)_j$$

with the Stirling numbers of the second kind gives the ordinary-moment bound, because $b_N \rightarrow 0$.

Finally, the earlier Gaussian compact-count bound gives $\sup_N \mathbb{E}e^{2tM_N} < \infty$. Splitting at counts zero, one, and at least two, and applying Cauchy–Schwarz on the last event, yields

$$\begin{aligned} \mathbb{E}(e^{tM_N} - 1) &\leq (e^t - 1)\mathbb{P}(M_N = 1) + \mathbb{E}[e^{tM_N} \mathbf{1}_{\{M_N \geq 2\}}] \\ &\leq (e^t - 1)\mathbb{E}M_N + (\mathbb{E}e^{2tM_N})^{1/2} \mathbb{P}(M_N \geq 2)^{1/2} = O(b_N). \end{aligned}$$

For complex w with $|w| \leq t$, the elementary inequality $|e^{wM_N} - 1| \leq e^{tM_N} - 1$ proves the locally uniform assertion. $\square$

Corollary 21.2 (A quantitative description of the rare interior points). *Let $B \in \mathbb{D}$ be Borel and put $\mu_N = \int_B \rho_{1,N}(z) dA(z)$. For all sufficiently large N , $0 \leq \mu_N \leq 1$. Let $\mathbf{B}_N$ be the probability law which is empty with probability $1 - \mu_N$ and has one point with intensity measure $\mathbf{1}_B(z) \rho_{1,N}(z) dA(z)$. Then*

$$d_{\text{TV}}(\mathcal{L}(\Xi_N|_B), \mathbf{B}_N) \leq \mathbb{E}[M_N \mathbf{1}_{\{M_N \geq 2\}}] \leq \mathbb{E}(M_N)_2 = O_{\beta,B}(b_N^2).$$

This approximation uses the actual finite- N intensity; it does not assert convergence of a rescaled intensity profile.

Proof. The first corollary gives $\mu_N = O(b_N)$, so $\mathbf{B}_N$ is well-defined for large N . On one-point configurations, the actual probability measure is dominated by the intensity measure defining $\mathbf{B}_N$. Its deficit in total mass is $\mu_N - \mathbb{P}(M_N = 1) = \mathbb{E}[M_N \mathbf{1}_{\{M_N \geq 2\}}]$. On the empty configuration the actual law has excess mass $\mu_N - \mathbb{P}(M_N \geq 1)$, and on configurations with at least two points it has excess mass $\mathbb{P}(M_N \geq 2)$. The latter two masses sum to the one-point deficit. The total variation distance, with the convention $d_{\text{TV}}(P, Q) = \sup_A |P(A) - Q(A)|$, therefore equals that deficit. Since $m \leq m(m-1)$ for integers $m \geq 2$, the claimed bound follows from (21.3) at $p = 2$. $\square$

22 Adjacent partition ratios on a set of full natural density

Set $c_\beta = \beta/2 - 2$ and

$$R_{n,\beta} = \frac{Q_{n+1,\beta}}{Q_{n,\beta}}, \quad n \geq 1.$$

The next theorem is weaker than a full-sequence ratio asymptotic. It makes no assertion about uniqueness of the interior point process.

Theorem 22.1 (Full-density exponent for adjacent ratios). *For every $\beta > 0$ and every $\varepsilon > 0$,*

$$\#\left\{2 \leq n \leq N : \left| \frac{\log R_{n,\beta}}{\log n} - c_\beta \right| > \varepsilon\right\} = O_{\beta,\varepsilon}\left(\frac{N \log \log(e^\varepsilon N)}{\log(eN)}\right). \quad (22.1)$$

There exists a set $\mathcal{D}_\beta \subset \mathbb{N}$ of natural density one, meaning $\#(\mathcal{D}_\beta \cap \{1, \dots, N\})/N \rightarrow 1$, such that

$$R_{n,\beta} = n^{c_\beta + o(1)}, \quad n \rightarrow \infty, \quad n \in \mathcal{D}_\beta. \quad (22.2)$$

Proof. The boundary-transport estimate (20.2) gives, after increasing a constant to cover finitely many indices,

$$s_n := \log R_{n,\beta} - c_\beta \log n \geq -C_\beta - \beta \log \log(en), \quad n \geq 2. \quad (22.3)$$

On the other hand, telescoping and $\log Q_{N,\beta} = c_\beta N \log N + O_\beta(N)$ imply

$$\sum_{n=2}^N s_n = O_\beta(N). \quad (22.4)$$

Indeed, the sum of the log ratios is $\log Q_{N+1,\beta} - \log Q_{2,\beta}$, whereas $\sum_{n=2}^N \log n = \log(N!) = N \log N - N + O(\log N)$.

Let $A_{N,\varepsilon} = \{2 \leq n \leq N : s_n > \varepsilon \log n\}$. Combining (22.3) and (22.4) gives

$$\varepsilon \sum_{n \in A_{N,\varepsilon}} \log n \leq C_\beta N \log \log(e^\varepsilon N).$$

At most $\sqrt{N}$ of these indices are below $\sqrt{N}$, and every remaining logarithm is at least $\frac{1}{2} \log N$. Consequently

$$\#A_{N,\varepsilon} \leq \sqrt{N} + \frac{C_\beta N \log \log(e^e N)}{\varepsilon \log N}.$$

Equation (22.3) rules out $s_n < -\varepsilon \log n$ for all sufficiently large n , so this proves (22.1).

We give an explicit full-density selection. For all sufficiently large n , let

$$\delta_n = \sqrt{\frac{\log \log n}{\log n}}.$$

This is positive, decreases eventually, and tends to zero. Include an index n in $\mathcal{D}_\beta$ precisely when $s_n \leq \delta_n \log n$, allowing an arbitrary choice at finitely many small indices. If $n \leq N$ is a sufficiently large excluded index, then $s_n > \delta_N \log n$. The preceding counting argument, with $\varepsilon = \delta_N$, therefore gives

$$\#(\{1, \dots, N\} \setminus \mathcal{D}_\beta) \leq O_\beta(1) + \sqrt{N} + C_\beta N \sqrt{\frac{\log \log N}{\log N}} = o(N).$$

On $\mathcal{D}_\beta$, the upper bound $s_n / \log n \leq \delta_n \rightarrow 0$ and the lower bound (22.3) squeeze $s_n / \log n$ to zero. This proves (22.2). $\square$

Corollary 22.2 (Diverging total radial depth below 4 on a full-density set). *Fix $0 < \beta < 4$ and put*

$$B_n = \sum_{i=1}^n -\log |z_i|, \quad d_\beta = \frac{2}{\beta} - \frac{1}{2} > 0.$$

On the full-density set $\mathcal{D}_\beta$ supplied by the preceding theorem, for every $0 \leq a < d_\beta$,

$$\mathbb{P}_{n,\beta}\{B_n \leq a \log n\} \leq n^{\beta a + c_\beta + o(1)} \rightarrow 0, \quad n \rightarrow \infty, \quad n \in \mathcal{D}_\beta. \quad (22.5)$$

In particular, $B_n \rightarrow \infty$ in probability along this set of indices.

Proof. The central insertion identity at size $n+1$ gives

$$\mathbb{E}_{n,\beta} e^{-\beta B_n} = \rho_{n+1,\beta}(0) R_{n,\beta}.$$

The already established compact correlation estimates imply $\sup_n \rho_{n+1,\beta}(0) < \infty$. Hence (22.2) gives

$$\mathbb{E}_{n,\beta} e^{-\beta B_n} \leq n^{c_\beta + o(1)} \quad (n \rightarrow \infty, n \in \mathcal{D}_\beta).$$

On the event $B_n \leq a \log n$, the integrand is at least $n^{-\beta a}$. This proves (22.5); its exponent is negative exactly when $a < -c_\beta/\beta = d_\beta$. Choosing any fixed positive $a < d_\beta$ also proves divergence in probability. $\square$

Limitation. The total radial depth is not a compactly supported observable on the open disk. Its divergence can be caused by particles approaching the boundary. Therefore the last corollary does not prove that an interior subsequential limit is nonempty, nor does it identify that limit. Likewise, the full-density ratio theorem does not control the exceptional indices and cannot be upgraded to a full-sequence asymptotic without a further regularity argument for the actual adjacent partition ratios.

23 Boundary-vanishing statistics and changes of potential

Write $d(z) = 1 - |z|$. The distinction between d^p for $p > 1$ and the borderline weight d is useful when analyzing the exterior field. Here we obtain exponential, rather than only polynomial, control of the first class of statistics.

Lemma 23.1 (Uniform exponential moments of scaled disk counts). *For every $\beta > 0$ and $\lambda \geq 0$,*

$$\sup_{N \geq 1} \sup_{0 < t < 1} \mathbb{E}_{N,\beta} \exp\{\lambda t M_N(1-t)\} < \infty.$$

The same assertion holds for every subsequential limit, uniformly over all such limits.

Proof. Put $M = M_N(1-t)$ and $a = \beta/16$. The compact-count bound, together with $\log(1/(1-t)) \geq t$, gives

$$\mathbb{P}(M \geq m) \leq \exp\{m - \frac{\beta}{4}tm(m-1)\}.$$

If $m \geq K_t := \lceil 8/(\beta t) \rceil + 2$, the exponent is at most $-atm^2$. For $q = \lambda t$, summation by parts gives

$$\mathbb{E}e^{qM} = 1 + (e^q - 1) \sum_{m \geq 1} e^{q(m-1)} \mathbb{P}(M \geq m).$$

This is first an identity for truncated integer-valued variables and then follows in general by monotone convergence. The terms below K_t contribute at most e^{qK_t} , which is bounded uniformly in t . For the rest, use $e^q - 1 \leq \lambda e^\lambda t$ and

$$\lambda tm - atm^2 \leq \frac{\lambda^2 t}{2a} - \frac{a}{2}tm^2.$$

The elementary integral bound $\sum_{m \geq 1} e^{-atm^2/2} \leq C_a/\sqrt{t}$ shows that the remaining contribution is bounded uniformly as well. The limiting count-tail bound is the same, so the proof applies verbatim to every limit. $\square$

Theorem 23.2 (Exponential control and convergence of weighted statistics). *Fix $p > 1$ and define*

$$U_p(\xi) = \sum_{z \in \xi} d(z)^p, \quad V_{p,\delta}(\xi) = \sum_{\substack{z \in \xi \\ d(z) < \delta}} d(z)^p, \quad 0 < \delta < 1.$$

For every $c \geq 0$ there is a constant $C_{\beta,p,c}$ such that

$$\sup_N \mathbb{E} e^{cU_p(\Xi_N)} \leq C_{\beta,p,c}, \tag{23.1}$$

$$\sup_N \mathbb{E} \exp\{c\delta^{1-p}V_{p,\delta}(\Xi_N)\} \leq C_{\beta,p,c}, \quad 0 < \delta < 1. \tag{23.2}$$

These bounds also hold for every subsequential limit with the same constants. In particular, for every fixed $q \geq 1$,

$$\sup_N \|V_{p,\delta}(\Xi_N)\|_{L^q} + \sup_{\Xi} \|V_{p,\delta}(\Xi)\|_{L^q} \leq C_{\beta,p,q}\delta^{p-1}.$$

If $g : \mathbb{D} \rightarrow \mathbb{R}$ is continuous and $|g(z)| \leq Cd(z)^p$, then, along any subsequence $\Xi_{N_j} \Rightarrow \Xi$,

$$(\Xi_{N_j}, \Xi_{N_j}(g)) \Rightarrow (\Xi, \Xi(g)), \quad \mathbb{E} e^{t\Xi_{N_j}(g)} \longrightarrow \mathbb{E} e^{t\Xi(g)} \quad (t \in \mathbb{R}).$$

The series defining $\Xi(g)$ is absolutely convergent almost surely. The assertions extend jointly to any finite collection of such functions.

Proof. Layer integration, justified by Tonelli's theorem, gives

$$U_p(\Xi_N) = \int_0^1 p t^{p-1} M_N(1-t) dt.$$

The weight $w(t) = (p-1)t^{p-2}$ integrates to one. Jensen's inequality therefore yields

$$e^{cU_p(\Xi_N)} \leq \int_0^1 w(t) \exp \left\{ \frac{cp}{p-1} t M_N(1-t) \right\} dt.$$

Take expectations and apply Lemma 23.1. For the boundary tail, the pointwise inequality

$$V_{p,\delta}(\Xi_N) \leq \int_0^\delta p t^{p-1} M_N(1-t) dt$$

and the probability weight $w_\delta(t) = (p-1)t^{p-2}/\delta^{p-1}$ on $(0, \delta)$ give the second bound by the same argument. The scaled-count lemma for the limit and Tonelli give identical estimates for Ξ ; one may first truncate the integrals away from zero and then use monotone convergence. Exponential moments imply the stated L^q bounds, for example by $x^q \leq C_q e^x$ for $x \geq 0$.

Choose continuous cutoffs χ_δ with $0 \leq \chi_\delta \leq 1$, equal to one on $\{|z| \leq 1-\delta\}$ and zero sufficiently close to the unit circle. Then $g_\delta = \chi_\delta g$ has compact support, and

$$|\xi(g) - \xi(g_\delta)| \leq C V_{p,\delta}(\xi).$$

Vague convergence gives joint convergence of the process and its g_δ statistic. The uniform L^1 tail bound just proved, first at finite N and then for Ξ , permits $\delta \downarrow 0$ and proves the claimed joint convergence. This standard approximation step can also be verified directly on bounded Lipschitz tests: the replacement error is bounded by the L^1 tail, and the middle term converges for each fixed cutoff. Bound (23.1), applied with an arbitrarily larger coefficient, makes $e^{t\Xi_N(g)}$ uniformly integrable. Truncation proves convergence of its expectation. Finite collections are treated by the same cutoff and the sum of their individual bounds. $\square$

Corollary 23.3 (Explicit limiting changes of potential). *Let g satisfy the preceding theorem, and let f be holomorphic near $\overline{\mathbb{D}}$ with $f(0) = 0$. Define a perturbed law by*

$$d\mathbf{P}_N^{f,g} = \frac{\exp\{Y_N(f) + \Xi_N(g)\}}{\mathbb{E} \exp\{Y_N(f) + \Xi_N(g)\}} d\mathbf{P}_{N,\beta}, \quad Y_N(f) = 2 \operatorname{Re} \sum_i f(z_i).$$

Along every subsequence $\Xi_{N_j} \Rightarrow \Xi$, the normalizer tends to

$$\exp \left\{ \frac{2}{\beta} \sum_{k \geq 1} k |f_k|^2 \right\} \mathbb{E} e^{\Xi(g)},$$

and the interior law under $\mathbf{P}_{N_j}^{f,g}$ tends to the law

$$d\mathbf{P}^g(\xi) = \frac{e^{\xi(g)}}{\mathbb{E} e^{\Xi(g)}} d\mathcal{L}(\Xi)(\xi).$$

Its joint analytic Gaussian limit has means $(2k/\beta)\overline{f_k}$, variances $2k/\beta$, and is independent of that tilted interior process. For $\beta > 4$, the full perturbed sequence still tends to the empty configuration. In this range, $U_p(\Xi_N) \rightarrow 0$ in probability and in every fixed positive moment.

Proof. The analytic Gaussian theorem gives joint convergence of Ξ_N and X_N , with independence in the limit. The cutoff argument in the preceding proof adjoins $\Xi_N(g)$ to that joint convergence. The limit of this last statistic is the measurable function $\Xi(g)$ of the interior process. Hölder's inequality and the separate exponential bounds for $Y_N(f)$ and $U_p(\Xi_N)$ give uniform integrability of every fixed positive power of $e^{Y_N(f)+\Xi_N(g)}$. Passing to expectations against bounded continuous tests proves both the normalizer formula and the limiting change of measure. Gaussian completion of the square gives the stated mode means and variances. Independence follows because the limiting weight factors into a function of the interior and a function of the independent Gaussian field.

For $\beta > 4$, the unweighted full interior limit is empty, so all subsequences have the same perturbed empty limit by the preceding result. Alternatively, uniform integrability transfers the vanishing compact hitting probabilities directly. Apply the weighted-statistic theorem with $g(z) = d(z)^p$ to see that $U_p(\Xi_N) \Rightarrow 0$; the exponential bound gives convergence of every positive moment, and convergence to the constant zero in law is convergence in probability. $\square$

The strict inequality $p > 1$ is used in the integrability of t^{p-2} at zero. The theorem gives no estimate for the first-order boundary field weighted by $1 - |z|$, which remains relevant to uniqueness below and at 4.

24 Inserted charges and harmonic screening

Fix points $a_1, \dots, a_p \in \mathbb{D}$, exponents $\gamma_1, \dots, \gamma_p > 0$, and a function f holomorphic near $\overline{\mathbb{D}}$, with $f(0) = 0$. Repeated points are permitted. Define

$$U(z) = 2 \operatorname{Re} f(z) + \sum_{j=1}^p \gamma_j \log |z - a_j|,$$

and let P_N^U be the probability law proportional to

$$|\Delta_N(z)|^\beta \prod_{i=1}^N e^{U(z_i)} dA(z_i)$$

on $\mathbb{D}^N$. Values at the finitely many zeros of the weight are irrelevant to this density. The normalizing integral is finite and positive. This section identifies its harmonic fluctuations without asserting uniqueness of its interior point-process limit.

Theorem 24.1 (Screening of finitely many inserted charges). *Under P_N^U , the analytic functions*

$$X_N(u) = \sum_i \frac{uz_i}{1 - uz_i} = \sum_{k \geq 1} S_{k,N} u^k$$

converge in law in $\mathcal{H}(\mathbb{D})$ to

$$X^U(u) = \sum_{k \geq 1} (G_k + m_k) u^k, \quad m_k = \frac{2k}{\beta} \overline{f_k} - \frac{1}{\beta} \sum_{j=1}^p \gamma_j a_j^k.$$

Here $f(z) = \sum_{k \geq 1} f_k z^k$, and the G_k are independent centered circular complex Gaussians with $\mathbb{E}|G_k|^2 = 2k/\beta$. The laws of the interior point processes under P_N^U are tight. Along every subsequence on which the interior converges, the displayed analytic convergence is joint with the interior, and the Gaussian analytic function is independent of that interior limit.

For every $h(z) = \sum_{k \geq 1} h_k z^k$ holomorphic near $\overline{\mathbb{D}}$,

$$\lim_{N \rightarrow \infty} \mathbb{E}_N^U e^{2 \operatorname{Re} \sum_i h(z_i)} = \exp \left\{ \frac{2}{\beta} \sum_{k \geq 1} k |h_k|^2 + \frac{4}{\beta} \operatorname{Re} \sum_{k \geq 1} k h_k \overline{f_k} - \frac{2}{\beta} \sum_{j=1}^p \gamma_j \operatorname{Re} h(a_j) \right\}. \quad (24.1)$$

The convergence also holds after multiplication by a bounded continuous interior functional along an interior-convergent subsequence, with its limiting expectation as an additional factor.

Proof. We give the details needed to extend the preceding analytic-tilt argument across the zeros of the present weight.

Uniform local and macroscopic bounds. The ordered-radius comparison remains valid with the same rates a_k . Indeed, in the inner-block scaling argument the extra factors are products of $|q w_i - a_j|^{\gamma_j}$ and a nonvanishing holomorphic exponential. Their product with the Vandermonde cross factors is subharmonic in the complex scaling parameter. One way to verify this last assertion is to take its logarithm off the finitely many zeros: it is a sum of positive multiples of logarithms of holomorphic moduli and a harmonic function. The product is continuous at those zeros and extends subharmonically there. Its circular means are therefore nondecreasing, exactly as required by the original comparison proof. Thus compact counts have the same uniform Gaussian upper tails and

$$\mathbb{E}_N^U \sum_i (-\log |z_i|) = O_\beta(\log(eN)).$$

The one-point density is subharmonic for the same reason. The disk submean inequality and the compact count bound therefore bound that density uniformly on each compact subset of $\mathbb{D}$. This also proves point-process tightness.

Write $\Gamma = \sum_j \gamma_j$. If A_N denotes the normalizing expectation of $e^{\sum_i U(z_i)}$ under the unweighted gas, then Jensen's inequality, rotation invariance, and the angular mean formula give

$$\begin{aligned} \log A_N &\geq \sum_j \gamma_j \mathbb{E}_{N,\beta} \sum_i \log |z_i - a_j| \\ &= \sum_j \gamma_j \mathbb{E}_{N,\beta} \sum_i \log \max(|z_i|, |a_j|) \geq -C_{\beta,\Gamma} \log(eN). \end{aligned}$$

The expectation of the analytic part vanishes because $f(0) = 0$. Since $|z - a_j| \leq 2$ in the disk, the density of P_N^U relative to the unweighted gas is bounded above by $e^{CN} (eN)^C$ for a fixed constant. The unweighted macroscopic Fourier concentration estimate consequently implies $S_{k,N}/N \rightarrow 0$ under P_N^U for each fixed k . As in Lemma 19.1, geometric truncation then proves

$$N^{-1} \sup_{|u| \leq s} |X_N(u)| \rightarrow 0 \quad (s < 1).$$

Together with the radial bound, these statements imply convergence of the normalized empirical measure on the closed disk to uniform measure on its boundary.

Analytic tightness before identifying the limit. The rational transport proof of Proposition 19.3 applies directly. For a parameter disk $|u| \leq s$, choose its fixed cutoff disk to contain both that disk and all the a_j , with a strictly larger radius less than one. The vector field is identically zero on a neighborhood of all the inserted charges. The change of U along its time-1/ N flow is therefore uniformly bounded by C/N per particle: on the region where the vector field is nonzero, all the logarithmic derivatives of U are bounded; on its zero region

the points remain fixed. Thus the additional log likelihood is $O(1)$, deterministically. The rational divided-difference estimate, the four-direction tail argument on the common good event, and Fubini–Cauchy apply without any other change. They prove tightness of X_N in $\mathcal{H}(\mathbb{D})$ under P_N^U . In particular all its fixed positive modes are tight. This argument works with $f + th$ in place of f , for any fixed real t and fixed analytic h .

The small transport and the logarithmic singularities. Fix $k \geq 1$ and $t \in \mathbb{C}$. Put

$$r_N = N^{-1/(100k)}, \quad b_N = \frac{2kt}{\beta N},$$

and use the already constructed patched branched-Möbius map $T_N = T_{b_N, r_N}$. Let $P = P_N^U$, and let Q have density R_N relative to P , where R_N is the exact change-of-variables likelihood. Thus $(T_N)_*Q = P$ and $\mathbb{E}_P R_N = 1$. The uniform displacement is $O(N^{-1}r_N^{1-k}) = o(r_N)$. The compact density bound just proved shows that the event that a point lies within $4r_N$ of the origin or of one of the inserted charges has probability $O(r_N^2)$ under P . It also has probability $O(r_N^2)$ under Q : its image under T_N is contained in the corresponding union of disks of radius $5r_N$, for all sufficiently large N . Denote the complementary event by A_N .

Compact counts are tight under Q , since $T_N \rightarrow \text{id}$ uniformly and its pushforward has law P . The boundary-distance estimate for T_N also gives $\sum_i (1 - |z_i|) = O_Q(\log(eN))$. Every fixed positive mode is tight under Q . To see this directly, write $y_i = T_N(z_i)$ and split particles at a fixed radius $r_0 > 0$. Outside that radius $|T_N(z) - z| \leq C_{r_0}/N$, while the number of points inside is tight and the global displacement tends to zero. It follows that $|S_{j,N}(y) - S_{j,N}(z)| = O_Q(1)$ for each fixed j ; the y mode is tight under P . The empirical measures under Q have the same uniform boundary limit, again by uniform convergence of T_N to the identity.

On A_N , the negative-mode bound used in the original proof applies: for every fixed $q \geq 1$ and any fixed $r_0 \in (0, 1)$,

$$|S_{-q,N} - \overline{S_{q,N}}| \leq C_{q,r_0} \sum_i (1 - |z_i|) + C_q r_N^{-q} \Xi_N(B_{r_0}).$$

Consequently all the inputs to the deterministic branched-transport likelihood expansion hold under both P and Q . Specifically, $S_{-q,N} = O(\log(eN) + r_N^{-q})$ in probability for $q < k$, $S_{\pm q,N}/N \rightarrow 0$ for $1 \leq q \leq 2k$, and every lower positive mode is tight. Products of lower modes, multiplied by N^{-1} , are therefore negligible. The deterministic remainders are $O(N^{-1}r_N^{-3k} + N^{-1}r_N^{-2k}) = o(1)$. The unweighted part of the log likelihood is thus

$$-2 \operatorname{Re}(\bar{t} S_{k,N}) - \frac{2k}{\beta} |t|^2 + o(1)$$

in probability under both laws. This is exactly the deterministic expansion already established in the finite-mode proof; it has not been integrated over the exceptional small disks.

We now compute the extra change $\sum_i [U(T_N z_i) - U(z_i)]$. Choose a fixed $r_0 < 1$ whose disk contains all the inserted charges and the origin. On A_N , for particles in that disk the logarithmic singularities are at distance at least $4r_N$. The mean value theorem and the displacement bound make the absolute change of U per such particle at most $C/(Nr_N^k) = o(1)$. There are only a tight number of such particles, under either law. Their total contribution is negligible. Outside that fixed disk, U is smooth with bounded derivatives and

$$N(T_N(z) - z) \longrightarrow v_{k,t}(z) = \frac{2}{\beta} (tz^{1-k} - \bar{t}z^{1+k})$$

uniformly. The boundary empirical limit therefore shows that the total extra change converges in probability, under both laws, to the boundary average of the directional derivative $dU(z)[v_{k,t}(z)]$.

On $|z| = 1$ the vector $v_{k,t}$ is tangent to the boundary, and

$$U(z) = 2 \operatorname{Re} g(z), \quad g(z) = f(z) + \frac{1}{2} \sum_{j=1}^p \gamma_j \log(1 - \bar{a}_j z),$$

with all logarithms zero at $z = 0$. If $g(z) = \sum_{j \geq 1} g_j z^j$, then Fourier orthogonality gives that directional average as $(4k/\beta) \operatorname{Re}(tg_k)$. Thus

$$\log R_N = -2 \operatorname{Re}(\bar{t} S_{k,N}) - \frac{2k}{\beta} |t|^2 + \frac{4k}{\beta} \operatorname{Re}(tg_k) + o(1) \quad (24.2)$$

in probability under both P and Q .

Identification and independence. For every fixed positive j , the existing mode-shift expansion gives, on A_N ,

$$S_{j,N}(T_N z) - S_{j,N}(z) = \frac{j}{k} \{b_N S_{j-k,N} - \bar{b}_N S_{j+k,N}\} + o(1).$$

The negative-mode estimates and the empirical limit imply that this converges in Q -probability to zero if $j \neq k$, and to $2kt/\beta$ if $j = k$. The interior configuration also changes by a quantity tending to zero in the vague metric, by uniform convergence of T_N and tight compact counts.

The right side of (24.2) is tight under Q , so the exact likelihoods R_N are uniformly integrable under P :

$$\mathbb{E}_P[R_N \mathbf{1}_{\{R_N > K\}}] = Q(\log R_N > \log K) \longrightarrow 0$$

uniformly in sufficiently large N , as $K \rightarrow \infty$. Let Y_N comprise the interior configuration and any finite collection of modes other than k . Its transported version differs from it by $o_Q(1)$. Exact change of variables gives, for $F \in C_b$,

$$\mathbb{E}_P F(Y_N) - \mathbb{E}_P[R_N F(Y_N)] \longrightarrow 0.$$

On any jointly convergent subsequence, denote the limiting k th mode by Z_k and the remaining variables by Y . Uniform integrability and (24.2) yield

$$\mathbb{E}[F(Y) e^{-2 \operatorname{Re}(\bar{t} Z_k)}] = \exp \left\{ \frac{2k}{\beta} |t|^2 - \frac{4k}{\beta} \operatorname{Re}(tg_k) \right\} \mathbb{E} F(Y).$$

The uniqueness theorem for real Gaussian moment-generating functions identifies Z_k as a circular complex Gaussian of mean $(2k/\beta) \bar{g}_k$ and variance $2k/\beta$, independent of Y . This holds for each k and every finite collection of the other modes. It therefore identifies the full finite-dimensional joint law and its independence from the interior. Analytic tightness, continuity of Taylor coefficients, and their determination of a holomorphic function prove the asserted convergence in $\mathcal{H}(\mathbb{D})$. Finally,

$$\frac{2k}{\beta} \bar{g}_k = \frac{2k}{\beta} \bar{f}_k - \frac{1}{\beta} \sum_j \gamma_j a_j^k,$$

which is the stated mean.

Laplace transforms. The continuous analytic pairing from Theorem 19.6 gives convergence in law of $2 \operatorname{Re} \sum_i h(z_i)$ to the corresponding real Gaussian pairing of X^U . Under its exponential tilt by any fixed real multiple, the model has the same inserted charges and analytic part $f + th$. The analytic tightness established above still applies. Thus Lemma 19.4 proves

uniform integrability of every fixed exponential and allows passage to its expectation. The Gaussian variance is $(4/\beta) \sum_k k|h_k|^2$, and its mean is

$$\frac{4}{\beta} \operatorname{Re} \sum_k k h_k \overline{f_k} - \frac{2}{\beta} \sum_j \gamma_j \operatorname{Re} h(a_j).$$

This proves (24.1). Joint convergence and independence, with the same uniform integrability, prove the mixed assertion. $\square$

Corollary 24.2 (Palm screening and relative correlation invariance). *Fix distinct $a_1, \dots, a_p \in \mathbb{D}$. The reduced Palm law of the $(N+p)$ -particle gas at these locations is the N -particle law with $f = 0$ and $\gamma_j = \beta$ above. Its k th harmonic mode therefore has limiting mean $-\sum_j a_j^k$ and variance $2k/\beta$. For every analytic h as above,*

$$\lim_{N \rightarrow \infty} \mathbb{E}_N^{\text{Palm}(a)} e^{2 \operatorname{Re} \sum_i h(z_i)} = \exp \left\{ \frac{2}{\beta} \sum_{k \geq 1} k |h_k|^2 - 2 \operatorname{Re} \sum_{j=1}^p h(a_j) \right\}.$$

If $\rho_{p,N}^h$ denotes the p -point density of the gas tilted by $e^{2 \operatorname{Re} \sum_i h(z_i)}$, then

$$\frac{\rho_{p,N}^h(a_1, \dots, a_p)}{\rho_{p,N}(a_1, \dots, a_p)} \rightarrow 1.$$

This is a relative statement and does not require a nonzero limiting intensity.

Proof. The Palm density follows immediately by fixing the p inserted coordinates in the Vandermonde density and canceling their mutual Vandermonde factor in normalization. The first two conclusions are Theorem 24.1. All finite- N denominators in the last display are positive for a distinct tuple. Exact insertion and exponential tilting give

$$\frac{\rho_{p,N}^h(a)}{\rho_{p,N}(a)} = e^{2 \operatorname{Re} \sum_j h(a_j)} \frac{\mathbb{E}_{N-p}^{\text{Palm}(a)} e^{2 \operatorname{Re} \sum_i h(z_i)}}{\mathbb{E}_{N,\beta} e^{2 \operatorname{Re} \sum_i h(z_i)}}.$$

The Palm Laplace limit and (19.8) cancel the two charge terms and the common Gaussian variance term, proving the limit one. $\square$

25 The exact Möbius identity and interior symmetries

For $t \in \mathbb{C}$ let $v_t(z) = t - \bar{t}z^2$, viewed as a real vector field. A smooth local cylinder functional means a function

$$F(\eta) = \Phi \left(\int h_1 d\eta, \dots, \int h_q d\eta \right),$$

where $h_j \in C_c^1(\mathbb{D}; \mathbb{R})$ and $\Phi \in C_b^1(\mathbb{R}^q; \mathbb{R})$ has bounded first derivatives. Define its directional derivative by

$$\mathcal{L}_t F(\eta) = \sum_{j=1}^q \partial_j \Phi \left(\int h_1 d\eta, \dots, \int h_q d\eta \right) \int dh_j(z) [v_t(z)] d\eta(z).$$

Proposition 25.1 (An exact local Ward identity and its first-order limit). *For every such F and every $t \in \mathbb{C}$,*

$$\mathbb{E}_{N,\beta} \mathcal{L}_t F(\Xi_N) = 2c_N \operatorname{Cov}_{N,\beta} (F(\Xi_N), \operatorname{Re}(\bar{t}S_{1,N})), \quad c_N = 2 + \frac{\beta}{2}(N-1). \quad (25.1)$$

In particular, the covariance is $O_{\beta,F,t}(N^{-1})$. If $\Xi_{N_j} \Rightarrow \Xi$ along a subsequence, then

$$N_j \operatorname{Cov}_{N_j,\beta} (F(\Xi_{N_j}), \operatorname{Re}(\bar{t}S_{1,N_j})) \rightarrow \frac{1}{\beta} \mathbb{E} \mathcal{L}_t F(\Xi). \quad (25.2)$$

Proof. Use the exact disk automorphism $T_a(z) = (z + a)/(1 + \bar{a}z)$ and its likelihood

$$R_{N,a} = (1 - |a|^2)^{c_N N} \prod_i |1 + \bar{a}z_i|^{-2c_N}.$$

Its change-of-variables identity is $\mathbb{E}[F((T_a)_* \Xi_N) R_{N,a}] = \mathbb{E}F(\Xi_N)$. Differentiating at $a = st$, $s = 0$, gives

$$\left. \frac{d}{ds} T_{st}(z) \right|_{s=0} = v_t(z), \quad \left. \frac{d}{ds} R_{N,st} \right|_{s=0} = -2c_N \operatorname{Re}(\bar{t} S_{1,N}).$$

For each fixed N all derivatives are bounded on a sufficiently small parameter interval, so differentiation under the integral is legitimate. Rotation invariance gives $\mathbb{E}S_{1,N} = 0$, proving (25.1).

For some compact $K \Subset \mathbb{D}$ and a fixed constant C , $|\mathcal{L}_t F(\eta)| \leq C\eta(K)$. The compact-count bounds therefore bound its expectation uniformly in N , proving the covariance estimate. The functional $\mathcal{L}_t F$ is continuous for the vague topology: it is a finite sum of products of continuous cylinder functions and integrals of continuous compactly supported functions. Uniform moments of the compact count give uniform integrability. Thus $\mathbb{E}\mathcal{L}_t F(\Xi_{N_j}) \rightarrow \mathbb{E}\mathcal{L}_t F(\Xi)$. Finally, $N/(2c_N) \rightarrow 1/\beta$, which proves (25.2). $\square$

Corollary 25.2 (A sufficient symmetry criterion and the resulting intensity). *If, along an interior-convergent subsequence, the covariances in (25.2) are $o(N_j^{-1})$ for every smooth local cylinder F and every $t \in \mathbb{C}$, then the limiting law is invariant under all conformal automorphisms of $\mathbb{D}$. Its intensity is necessarily*

$$\rho_1(z) = \frac{c}{(1 - |z|^2)^2}, \quad c \geq 0.$$

The case $c = 0$ is exactly the almost surely empty process.

Proof. The proposition gives $\mathbb{E}\mathcal{L}_t F(\Xi) = 0$ for all the specified tests. Let φ_s be the flow of v_t on the disk. It consists of disk automorphisms. For bounded s -intervals the functionals $F \circ (\varphi_s)_*$ remain smooth local cylinders whose supports lie in a common compact set. Their derivatives are dominated by a fixed constant times its particle count. Hence

$$\frac{d}{ds} \mathbb{E}F((\varphi_s)_* \Xi) = \mathbb{E}\mathcal{L}_t(F \circ (\varphi_s)_*)(\Xi) = 0.$$

Thus the law is invariant under each of these flows. Together with the already established rotation invariance, these flows generate every disk automorphism: the flow with $t = e^{i\theta}$ sends zero to $e^{i\theta} \tanh s$, and an automorphism fixing zero is a rotation. Smooth local cylinders determine the point-process law, for example by approximating compactly supported continuous Laplace tests, so the claimed invariance follows.

The limiting intensity is continuous by the earlier correlation theorem. For $T_a(0) = a$ the derivative has modulus $|T'_a(0)| = 1 - |a|^2$. Invariance of the intensity measure and change of variables therefore imply $\rho_1(a)(1 - |a|^2)^2 = \rho_1(0)$, proving the formula with $c = \rho_1(0)$. If $c = 0$, every compact count has expectation zero, and a countable exhaustion shows that the configuration is empty almost surely. The converse is immediate. $\square$

Remark 25.3 (The scale missing from an argument based only on Gaussian independence). The proved bound in Proposition 25.1 is $O(N^{-1})$, whereas the symmetry criterion requires $o(N^{-1})$. The first-order limit in (25.2) explains this difference exactly: its value is the interior response to the Möbius flow. Leading-order Gaussian independence does not determine that response.

A fixed nontrivial disk automorphism also cannot be inserted directly into the fixed-analytic-tilt invariance theorem. Its exact likelihood contains the exponent $2c_N = 4 + \beta(N-1)$, which grows with N . The previously established Laplace theorem applies to each fixed analytic potential. Substituting this growing exponent would require a new uniform estimate. Neither the symmetry criterion nor that new estimate is asserted here, and the displayed hyperbolic intensity formula is a conditional conclusion, not a formula established for all inverse temperatures.

26 An exact Gaussian Hankel representation at $\beta = 4$

The endpoint $\beta = 4$, outside the scope of the quantitative boundary-transport theorem, admits the following exact representation. The endpoint is settled later by the force argument; the representation here does not, by itself, prove convergence of adjacent partition ratios or of the point process.

Proposition 26.1 (Gaussian Hankel determinant identity). *Let $a_0, a_1, \dots$ be independent standard circular complex Gaussian random variables, normalized by $\mathbb{E}|a_j|^2 = 1$. Define the nested complex symmetric Hankel matrices*

$$H_N = \left(\frac{a_{i+j}}{\sqrt{i+j+1}} \right)_{0 \leq i, j < N}.$$

Then, with ordinary area in the gas partition function,

$$\frac{Q_{N,4}}{\pi^N} = \mathbb{E} |\det H_N|^2. \quad (26.1)$$

In particular,

$$\frac{Q_{N+1,4}}{Q_{N,4}} = \pi \frac{\mathbb{E} |\det H_{N+1}|^2}{\mathbb{E} |\det H_N|^2}. \quad (26.2)$$

Proof. Put $d\mu = dA/\pi$ on $\mathbb{D}$. The matrix-entry covariances are

$$\mathbb{E}[H_{ij} \overline{H_{kl}}] = \frac{\mathbf{1}_{\{i+j=k+l\}}}{i+j+1} = \int_{\mathbb{D}} z^{i+j} \bar{z}^{k+l} d\mu(z), \quad \mathbb{E}[H_{ij} H_{kl}] = 0.$$

The complex Gaussian Wick formula says that a product of N unbarred entries and N barred entries has expectation equal to the sum, over all bijections between them, of the products of their covariances. For clarity, this formula follows by differentiating the identity

$$\mathbb{E} \exp \left(\sum_j u_j a_j + \sum_j v_j \bar{a}_j \right) = \exp \left(\sum_j u_j v_j \right),$$

first for finitely many independent Gaussians and then for their linear combinations. All expressions used here involve finitely many variables.

Index permutations of $\{0, \dots, N-1\}$ by σ, τ, π . Expanding the two determinants and applying Wick gives

$$\mathbb{E} |\det H_N|^2 = \sum_{\sigma, \tau, \pi \in S_N} \text{sgn}(\sigma) \text{sgn}(\tau) \prod_{i=0}^{N-1} \int_{\mathbb{D}} z_i^{i+\sigma(i)} \bar{z}_i^{\pi(i)+\tau(\pi(i))} d\mu(z_i).$$

Set $\kappa = \tau \circ \pi$, so that $\text{sgn}(\tau) = \text{sgn}(\kappa) \text{sgn}(\pi)$. Write

$$V_N(z) = \det[z_i^j]_{0 \leq i, j < N}, \quad |V_N(z)| = |\Delta_N(z)|.$$

Collecting the three permutation sums inside the product integral yields

$$\mathbb{E}|\det H_N|^2 = \int_{\mathbb{D}^N} \left(\prod_{i=0}^{N-1} z_i^i \right) V_N(z) \overline{V_N(z)}^2 d\mu^{\otimes N}(z). \quad (26.3)$$

Finally symmetrize this integral over all permutations of the variables. The factor $\overline{V_N}^2$ is symmetric, while V_N changes by the sign of the permutation. Moreover,

$$\sum_{\lambda \in S_N} \text{sgn}(\lambda) \prod_{i=0}^{N-1} z_{\lambda(i)}^i = V_N(z).$$

Thus (26.3) equals

$$\frac{1}{N!} \int_{\mathbb{D}^N} |\Delta_N(z)|^4 d\mu^{\otimes N}(z) = \frac{Z_{N,4}}{N! \pi^N} = \frac{Q_{N,4}}{\pi^N}.$$

This proves (26.1), and division proves (26.2). $\square$

Normalization check. For $N = 2$,

$$\det H_2 = \frac{a_0 a_2}{\sqrt{3}} - \frac{a_1^2}{2}, \quad \mathbb{E}|\det H_2|^2 = \frac{1}{3} + \frac{\mathbb{E}|a_1|^4}{4} = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}.$$

This agrees with the direct disk integral $Q_{2,4} = 5\pi^2/6$.

What remains to be proved by this route. The matrices H_N are nested, but their determinant moments are not ordinary leading principal minors of a deterministic positive matrix. Consequently the Hadamard–Fischer determinant inequality does not imply log-concavity of the sequence in (26.1). A convergence or monotonicity theorem for the adjacent determinant-moment ratios in (26.2) is still needed. Even such a ratio theorem would have to be combined with convergence of the local insertion functions to establish uniqueness of the interior point process at $\beta = 4$ by this determinant route. The independent score proof below does not need either adjacent-ratio convergence or such an insertion limit.

Proposition 26.2 (Explicit exponential bounds at $\beta = 4$). *For every $N \geq 1$,*

$$\pi^N \frac{(2N-1)!!}{N^N} \leq Q_{N,4} \leq \pi^N. \quad (26.4)$$

Consequently the radius of convergence λ_c of $\sum_{N \geq 0} Q_{N,4} \lambda^N$ satisfies

$$\frac{1}{\pi} \leq \lambda_c \leq \frac{e}{2\pi}. \quad (26.5)$$

These bounds do not assert convergence of the adjacent ratios.

Proof. We first compute the circle integral needed in the argument, without using a Selberg or Dyson integral formula. On the unit circle,

$$|\Delta_N(z)|^4 = \frac{\Delta_N(z)^4}{\prod_i z_i^{2N-2}}.$$

Its integral under normalized product angular measure is therefore the coefficient of $\prod_i z_i^{2N-2}$ in Δ_N^4 . The confluent Vandermonde identity expresses Δ_N^4 as the determinant with two rows for each z_i , namely

$$(1, z_i, \dots, z_i^{2N-1}), \quad (0, 1, 2z_i, \dots, (2N-1)z_i^{2N-2}).$$

This identity follows from the ordinary Vandermonde determinant on $(z_1, z_1 + h_1, \dots, z_N, z_N + h_N)$ by subtracting each first row from its following row, dividing by $\prod_i h_i$, and letting all $h_i \rightarrow 0$.

Expand the confluent determinant in its two-row blocks. Choosing columns $p < q$, numbered from 0 to $2N - 1$, in one block contributes $(q - p)z_i^{p+q-1}$. The required exponent forces $p + q = 2N - 1$. Thus the chosen column pairs must be exactly $\{p, 2N - 1 - p\}$, $0 \leq p < N$, assigned in $N!$ ways to the row blocks. All their signs are positive: in increasing order of p , the column shuffle has $N(N - 1)$ inversions, an even number, and exchanging any two two-column blocks preserves the sign. Their common product is $\prod_{p=0}^{N-1} (2N - 1 - 2p) = (2N - 1)!!$. We have proved

$$\int_{\mathbb{T}^N} |\Delta_N(z)|^4 dm^{\otimes N}(z) = N!(2N - 1)!! \quad (26.6)$$

Let $\tilde{H}_N = (a_{i+j})_{0 \leq i, j < N}$. The Wick-and-symmetrization proof of Proposition 26.1 applies verbatim with normalized circle measure in place of normalized area, because $\mathbb{E}[a_{i+j}\bar{a}_{k+l}] = \mathbf{1}_{\{i+j=k+l\}}$ is now the corresponding monomial covariance. Hence (26.6) gives

$$\mathbb{E}|\det \tilde{H}_N|^2 = (2N - 1)!!.$$

Group the determinant polynomial into Gaussian monomials,

$$\det \tilde{H}_N = \sum_{\mathbf{m}} b_{\mathbf{m}} \prod_{s=0}^{2N-2} a_s^{m_s}.$$

Distinct monomials are orthogonal under complex Gaussian measure, with squared norm $\prod_s m_s!$. Replacing a_s by $a_s/\sqrt{s+1}$ therefore gives

$$\mathbb{E}|\det H_N|^2 = \sum_{\mathbf{m}} |b_{\mathbf{m}}|^2 \frac{\prod_s m_s!}{\prod_s (s+1)^{m_s}}, \quad \sum_{\mathbf{m}} |b_{\mathbf{m}}|^2 \prod_s m_s! = (2N - 1)!!.$$

Every nonzero monomial arises from at least one determinant permutation σ . Its denominator is $\prod_s (s+1)^{m_s} = \prod_{i=0}^{N-1} (i + \sigma(i) + 1)$. The sum of these N factors is N^2 , so the arithmetic-geometric mean inequality bounds their product above by N^N . For its lower bound, whenever $i < j$ and $\sigma(i) = b > \sigma(j) = a$, replacing this inversion by the ordered pair changes the two-factor product from $(i+b+1)(j+a+1)$ to $(i+a+1)(j+b+1)$, a decrease of $(j-i)(b-a) > 0$. Removing all inversions shows that the minimum is attained at the identity permutation and equals $\prod_i (2i+1) = (2N - 1)!!$. Thus every denominator in the nonnegative sum lies between $(2N - 1)!!$ and N^N . The last two displays and (26.1) prove (26.4).

Finally, $(2N - 1)!! = (2N)!/(2^N N!)$ and Stirling's formula give $((2N - 1)!!/N^N)^{1/N} \rightarrow 2/e$. The root test for the power series proves (26.5). $\square$

27 An exact outermost-particle transfer and the remaining covariance estimate

Throughout this section $\beta > 0$ is fixed, $Z_N = Z_{N,\beta}$, and $\mathbb{E}_N$ denotes expectation under the unweighted disk gas with N particles. We use the previously proved tightness and the compact-count estimate

$$B_{\beta,r} := \sup_{N \geq 1} \mathbb{E}_N[M_N(r)^2] < \infty, \quad 0 < r < 1. \quad (27.1)$$

For completeness, an explicit permissible value of $B_{\beta,r}$ is

$$\sum_{k=1}^{\infty} (2k-1)b_k, \quad b_1 = 1, \quad b_k = \min \left\{ 1, \exp \left(k - \frac{\beta}{4} k(k-1) \log \frac{1}{r} \right) \right\} \quad (k \geq 2).$$

This follows from $\mathbb{E}X^2 = \sum_{k \geq 1} (2k-1)\mathbb{P}(X \geq k)$ for nonnegative integer-valued X and the established count-tail bound.

The following transfer formula is exact. It is important that the remaining particles acquire a nonconstant weight after deletion: their law is not asserted to be the original $(N-1)$ -particle law.

Proposition 27.1 (Deletion and rescaling of the outermost particle). *For $N \geq 2$, let*

$$R_N = \max_{1 \leq j \leq N} |z_j|, \quad a_N = 2N + \frac{\beta}{2}N(N-1).$$

Almost surely there is a unique maximizing index J_N , and $R_N > 0$. Write $z_{J_N} = R_N e^{i\Theta_N}$, with $\Theta_N \in [0, 2\pi)$, and let $W_N = (w_1, \dots, w_{N-1})$ be the remaining labeled particles divided by R_N , retaining their relative order of labels. Define the point configuration

$$\hat{\Xi}_N = \sum_{j=1}^{N-1} \delta_{w_j}$$

and, for $n \geq 1$,

$$G_n(w_1, \dots, w_n) = \frac{1}{2\pi} \int_0^{2\pi} \prod_{j=1}^n |e^{i\theta} - w_j|^\beta d\theta. \quad (27.2)$$

Then R_N is independent of the pair (W_N, Θ_N) and has density $a_N r^{a_N-1} \mathbf{1}_{(0,1)}(r)$. Moreover,

$$\mathcal{L}(W_N)(dw) = \frac{G_{N-1}(w)}{\mathbb{E}_{N-1} G_{N-1}} \mathbb{P}_{N-1, \beta}(dw), \quad (27.3)$$

and

$$Z_N = \frac{2\pi N}{a_N} Z_{N-1} \mathbb{E}_{N-1} G_{N-1}. \quad (27.4)$$

The same tilted-law formula holds for the corresponding unlabeled point configurations.

Proof. The original law is absolutely continuous with respect to area measure on $\mathbb{D}^N$. Each set $\{|z_i| = |z_j|\}$, $i \neq j$, and the set on which all particles are zero have area measure zero. Thus J_N is unique and $R_N > 0$ almost surely.

On the event that label N is the maximizing label, make the change of variables

$$z_N = r e^{i\theta}, \quad z_j = r w_j \quad (1 \leq j \leq N-1), \quad 0 < r < 1, \quad w \in \mathbb{D}^{N-1}.$$

Its area Jacobian is $r^{2(N-1)+1}$. One way to see this without ignoring the dependence on r is to order the coordinates as $(z_N, z_1, \dots, z_{N-1})$: the derivative matrix is block triangular, with polar Jacobian r for z_N and two-dimensional dilation Jacobian r^2 for each remaining coordinate. Also,

$$|\Delta_N(z)|^\beta = r^{\beta N(N-1)/2} |\Delta_{N-1}(w)|^\beta \prod_{j=1}^{N-1} |e^{i\theta} - w_j|^\beta.$$

Consequently, for every bounded measurable function H of (r, w, θ) , exchangeability and summation over the possible maximizing labels give

$$\begin{aligned} \mathbb{E}_N H(R_N, W_N, \Theta_N) &= \frac{N}{Z_N} \int_0^1 r^{a_N-1} dr \int_{\mathbb{D}^{N-1}} |\Delta_{N-1}(w)|^\beta \prod_{j=1}^{N-1} dA(w_j) \\ &\quad \times \int_0^{2\pi} \prod_{j=1}^{N-1} |e^{i\theta} - w_j|^\beta H(r, w, \theta) d\theta. \end{aligned} \quad (27.5)$$

There is no additional ordering restriction on w : the maximizing particle was selected by its radius, while the remaining original labels are still exchangeable.

Taking $H = 1$ and using $\int_0^1 r^{a_N-1} dr = a_N^{-1}$ proves (27.4). The factorization in (27.5) proves the asserted independence and radial density. Integrating over θ proves (27.3). The denominator is finite and strictly positive: $G_n \leq 2^{\beta n}$, and its integrand is positive whenever all $w_j \in \mathbb{D}$. Pushing the labeled law forward to its counting measure proves the final assertion. $\square$

We next quantify the difference between the rescaled remaining configuration and the original configuration on a fixed compact set. For a nonnegative $f \in C_c^1(\mathbb{D})$, write

$$F_f(\xi) = \exp[-\xi(f)], \quad L_f = \|\nabla f\|_\infty,$$

where f is extended by zero to $\mathbb{C}$. This extension is C^1 because f has compact support in $\mathbb{D}$.

Proposition 27.2 (A summable error for compact Laplace functionals). *Choose $0 < r < r_0 < 1$ such that $\text{supp } f \subset B_r$. For every $N \geq 2$,*

$$\left| \mathbb{E}_N F_f(\Xi_N) - \frac{\mathbb{E}_{N-1}[G_{N-1} F_f(\Xi_{N-1})]}{\mathbb{E}_{N-1} G_{N-1}} \right| \leq r_0^{a_N} + \frac{L_f r}{r_0} \sqrt{B_{\beta, r}} \sqrt{\frac{2}{(a_N + 1)(a_N + 2)}}. \quad (27.6)$$

In particular, the left-hand side is $O_{\beta, f, r, r_0}(N^{-2})$.

Proof. Use the coupling in Proposition 27.1. On $\{R_N \geq r_0\}$, the deleted particle lies outside $\text{supp } f$. For every remaining particle, if $|z_j| \geq r$, both $f(z_j)$ and $f(z_j/R_N)$ vanish. If $|z_j| < r$, the global Lipschitz bound on f gives

$$|f(z_j/R_N) - f(z_j)| \leq L_f |z_j| \left(\frac{1}{R_N} - 1 \right) \leq \frac{L_f r}{r_0} (1 - R_N).$$

Therefore

$$|\widehat{\Xi}_N(f) - \Xi_N(f)| \leq \frac{L_f r}{r_0} (1 - R_N) M_N(r) \quad \text{on } \{R_N \geq r_0\}.$$

The function $s \mapsto e^{-s}$ is 1-Lipschitz on $[0, \infty)$. On the complementary event the two Laplace functionals differ by at most 1. It follows that

$$\begin{aligned} |\mathbb{E}_N F_f(\widehat{\Xi}_N) - \mathbb{E}_N F_f(\Xi_N)| &\leq \mathbb{P}_N(R_N < r_0) + \frac{L_f r}{r_0} \mathbb{E}_N[(1 - R_N) M_N(r)] \\ &\leq r_0^{a_N} + \frac{L_f r}{r_0} \sqrt{B_{\beta, r}} \sqrt{\mathbb{E}_N(1 - R_N)^2}. \end{aligned}$$

Here the second step uses Cauchy–Schwarz, not independence between R_N and $M_N(r)$; such independence is not being asserted. Direct integration of the radial density gives

$$\mathbb{E}_N(1 - R_N)^2 = a_N \int_0^1 (1 - s)^2 s^{a_N-1} ds = \frac{2}{(a_N + 1)(a_N + 2)}.$$

Equation (27.3) now proves (27.6). Finally, for $N \geq 2$, $a_N \geq \beta N^2/4$. Thus the second term in (27.6) is $O(N^{-2})$, while its first term decays at least as fast as $\exp[-(\beta/4) \log(1/r_0) N^2]$. $\square$

The preceding two propositions identify an explicit estimate that would suffice to prove uniqueness at any fixed temperature where it holds.

Proposition 27.3 (A one-sided covariance criterion for uniqueness). *For $n \geq 1$ and nonnegative $f \in C_c^1(\mathbb{D})$, put*

$$c_n(f) = \frac{\text{Cov}_n(F_f(\Xi_n), G_n)}{\mathbb{E}_n G_n}, \quad x_- := \max\{-x, 0\}.$$

Suppose that, for every nonnegative $f \in C_c^1(\mathbb{D})$,

$$\sum_{n=1}^{\infty} [c_n(f)]_- < \infty. \quad (27.7)$$

Then the entire sequence Ξ_N converges in distribution in $\mathcal{M}_p(\mathbb{D})$ to a unique law. The same conclusion follows if (27.7) is verified only on a countable family of nonnegative smooth compactly supported functions whose Laplace functionals determine laws on $\mathcal{M}_p(\mathbb{D})$.

For each test function satisfying (27.7), the sequence $\mathbb{E}_N F_f(\Xi_N)$ in fact has finite total variation, and $\sum_n |c_n(f)| < \infty$.

Proof. Fix f and put $x_n = \mathbb{E}_n F_f(\Xi_n) \in [0, 1]$. By Proposition 27.2,

$$x_{n+1} - x_n = c_n(f) + \varepsilon_n(f), \quad \sum_{n=1}^{\infty} |\varepsilon_n(f)| < \infty. \quad (27.8)$$

Consequently the sum of the negative parts of the increments is finite:

$$\sum_{n \geq 1} (x_n - x_{n+1})_+ \leq \sum_{n \geq 1} [c_n(f)]_- + \sum_{n \geq 1} |\varepsilon_n(f)| < \infty.$$

Writing $p_n = (x_{n+1} - x_n)_+$ and $d_n = (x_n - x_{n+1})_+$, telescoping gives

$$\sum_{n=1}^m p_n = x_{m+1} - x_1 + \sum_{n=1}^m d_n \leq 1 + \sum_{n \geq 1} d_n.$$

Thus $\sum_n |x_{n+1} - x_n| < \infty$, so x_n converges. Equation (27.8) also gives $\sum_n |c_n(f)| < \infty$. The one-sided form of the hypothesis may therefore be convenient to establish, but is not claimed here to be strictly weaker than absolute covariance summability in the presence of (27.8) and $0 \leq x_n \leq 1$.

By the already established tightness, every subsequence of particle numbers tending to infinity has a further subsequence with a weak limit. The function F_f is bounded and vaguely continuous. Hence all such limits have the same Laplace functional at every test for which x_n converges.

We recall why these functionals determine the limiting law. Choose a countable collection $(g_j)_{j \geq 1} \subset C_c^\infty(\mathbb{D})$ of nonnegative functions separating nonnegative Radon measures; it can be obtained from countable uniform approximation families on a compact exhaustion of $\mathbb{D}$. Include all finite sums $\sum_j m_j g_j$ with nonnegative integer coefficients. Equality of Laplace functionals for these sums gives equality of all mixed moments of every finite vector

$$(e^{-\xi(g_1)}, \dots, e^{-\xi(g_k)}) \in [0, 1]^k.$$

Polynomials are uniformly dense in $C([0, 1]^k)$, so those moments determine each finite-dimensional distribution. The coordinates $\xi \mapsto \xi(g_j)$ form a countable measure-determining family for the Borel sigma field of the vague topology on locally finite Radon measures. Thus equality of their joint distributions determines the law of ξ . This also explains the stated countable-family version of the criterion.

It follows that all weak subsequential limits have the same law. Tightness then implies convergence of the full sequence: otherwise a subsequence outside some neighborhood of that law would have a further weakly convergent subsequence, giving a contradiction. $\square$

Remark 27.4 (What remains unproved). The transfer formula and the $O(N^{-2})$ comparison error above are unconditional consequences of the finite-particle density and the proved compact-count bounds. In contrast, (27.7) has not been established here for general β . In particular, neither independence of R_N and $\widehat{\Xi}_N$ nor positivity of the boundary weight G_n implies the required covariance estimate. No uniqueness theorem for general β is being deduced from this conditional criterion. Theorems 20.1 and 36.1 prove uniqueness for $\beta \geq 4$ without verifying that covariance estimate. If the criterion is eventually verified, the previously proved subsequential local total variation result will also upgrade to the full sequence: a subsequence violating that conclusion would, by tightness, have a further weakly convergent subsequence, whose limit must be the unique limit and along which local total variation convergence is already known.

28 All positive harmonic modes can miss an exterior field

The following deterministic example isolates a difficulty in identifying the Gibbs conditionals of a limit. It is not a construction of different subsequential limits of the gas.

Proposition 28.1 (Boundary information beyond the global harmonic modes). *Fix an integer $k \geq 1$ and a real number b . There exist configurations $\eta_n = \sum_{j=0}^{n-1} \delta_{w_{j,n}}$ of distinct points in $\mathbb{D}$ such that*

$$\begin{aligned} \eta_n &\longrightarrow 0 \quad \text{vaguely on } \mathbb{D}, \\ \frac{1}{n} \eta_n &\Longrightarrow \frac{d\theta}{2\pi} \quad \text{weakly on } \overline{\mathbb{D}}, \\ \sum_{j=0}^{n-1} w_{j,n}^\ell &\longrightarrow 0 \quad \text{for every fixed integer } \ell \geq 1, \\ \sum_{j=0}^{n-1} (-\log |w_{j,n}|)^p &\longrightarrow 0 \quad \text{for every real } p > 1, \end{aligned}$$

whereas, locally uniformly for $z \in \mathbb{D}$,

$$\sum_{j=0}^{n-1} \log \left| 1 - \frac{z}{w_{j,n}} \right| \longrightarrow b \operatorname{Re}(z^k). \quad (28.1)$$

Consequently, for every $0 < R < 1$, the probability densities on B_R proportional to $\prod_j |z - w_{j,n}|^\beta$ converge uniformly to

$$\frac{\exp\{\beta b \operatorname{Re}(z^k)\}}{\int_{B_R} \exp\{\beta b \operatorname{Re}(u^k)\} dA(u)}. \quad (28.2)$$

These limiting densities vary with b , although all the limits in the first display are unchanged.

Proof. Choose a constant $c > |b|$. For all sufficiently large n , put

$$\theta_j = \frac{2\pi j}{n}, \quad w_{j,n} = e^{-c/n} e^{i\theta_j} \exp\left(\frac{b}{n} e^{ik\theta_j}\right).$$

Their moduli and arguments satisfy

$$-\log |w_{j,n}| = \frac{c - b \cos(k\theta_j)}{n} > 0, \quad \arg w_{j,n} = \theta_j + \frac{b}{n} \sin(k\theta_j) \pmod{2\pi}.$$

The latter circle map is strictly increasing when $n > |b|k$, proving distinctness. The moduli tend uniformly to one and the angular perturbations tend uniformly to zero. This proves both the vague limit and the weak empirical limit. It also gives

$$\sum_j (-\log |w_{j,n}|)^p \leq (c + |b|)^p n^{1-p} \rightarrow 0 \quad (p > 1).$$

Fix a positive integer ℓ . Taylor expansion with a uniform error gives

$$w_{j,n}^\ell = e^{i\ell\theta_j} \left(1 - \frac{\ell c}{n} + \frac{\ell b}{n} e^{ik\theta_j} + O_{\ell,b,c}(n^{-2}) \right).$$

For $n > \ell + k$, the sums of both $e^{i\ell\theta_j}$ and $e^{i(\ell+k)\theta_j}$ vanish. Summing the display therefore gives $\sum_j w_{j,n}^\ell = O_{\ell,b,c}(n^{-1})$. This proves the claim simultaneously for each fixed positive harmonic mode, without requiring any relation between ℓ and a growing particle number.

Fix $R < 1$. For $|z| \leq R$, write

$$v_j = ze^{-i\theta_j}, \quad \delta_j = \frac{c - be^{ik\theta_j}}{n}, \quad \frac{z}{w_{j,n}} = v_j e^{\delta_j}.$$

Choose R' strictly between R and 1. For all sufficiently large n , $|v_j e^{s\delta_j}| \leq R'$ for $0 \leq s \leq 1$. The branch of $\log(1 - v)$ that is zero at $v = 0$ is consequently defined throughout the expansion. Its second derivative in δ is bounded uniformly there. Taylor's theorem in the complex parameter gives

$$\log(1 - v_j e^{\delta_j}) = \log(1 - v_j) - \delta_j \frac{v_j}{1 - v_j} + O_{R,b,c}(n^{-2}).$$

The product over roots of unity, with equality of branches fixed at $z = 0$, gives $\sum_j \log(1 - v_j) = \log(1 - z^n)$. Expanding $v_j/(1 - v_j) = \sum_{h \geq 1} z^h e^{-ih\theta_j}$ and summing characters gives, for $n > k$,

$$\frac{1}{n} \sum_j \frac{v_j}{1 - v_j} = \frac{z^n}{1 - z^n}, \quad \frac{1}{n} \sum_j e^{ik\theta_j} \frac{v_j}{1 - v_j} = \frac{z^k}{1 - z^n}.$$

Thus, uniformly for $|z| \leq R$,

$$\sum_j \log(1 - z/w_{j,n}) = \log(1 - z^n) - c \frac{z^n}{1 - z^n} + b \frac{z^k}{1 - z^n} + O_{R,b,c}(n^{-1}).$$

Taking real parts and letting $n \rightarrow \infty$ proves (28.1).

Finally, write

$$\prod_j |z - w_{j,n}|^\beta = \left(\prod_j |w_{j,n}|^\beta \right) \exp \left\{ \beta \sum_j \log |1 - z/w_{j,n}| \right\}.$$

The first factor cancels upon normalization. Uniform convergence on $\overline{B_R}$ proves (28.2). The limiting densities for distinct b differ, since $\operatorname{Re}(z^k)$ is not constant on B_R . $\square$

For a tagged particle conditioned to lie in B_R , with all remaining particles fixed at $w_{j,n}$, the finite-gas conditional density is exactly the density used in the proposition. These boundary configurations are not asserted to be typical for the remaining-particle marginal of the $(n + 1)$ -particle gas. The proposition therefore does not disprove uniqueness. It proves that the displayed limiting data, even including *every fixed positive harmonic sum*, do not determine the exterior logarithmic field. A uniqueness argument using conditional Gibbs laws still has to control that field under the actual gas probabilities.

29 An exact hyperbolic reformulation and a high-temperature obstruction

This section records an exact reformulation. It does not assert uniqueness of an infinite-volume Gibbs measure. In particular, the elementary integrable-interaction criterion for a high-temperature cluster expansion does not apply to the interaction obtained below.

Put

$$g(z, w) = \log \frac{|1 - z\bar{w}|}{|z - w|}, \quad d\lambda(z) = \frac{dA(z)}{(1 - |z|^2)^2}.$$

Here g is the nonnegative Green function of the disk normalized by $-\Delta_z g(z, w) = 2\pi\delta_w$, with zero boundary values and the convention $g(z, z) = +\infty$. Indeed $\log |1 - z\bar{w}|$ is harmonic on the disk, while $\Delta_z \log |z - w| = 2\pi\delta_w$; the latter distributional identity follows by integrating the normal derivative $1/\varepsilon$ on a circle of radius ε around w and then letting $\varepsilon \downarrow 0$. The identity

$$|1 - z\bar{w}|^2 - |z - w|^2 = (1 - |z|^2)(1 - |w|^2)$$

proves nonnegativity and the zero boundary values. Both g and λ are invariant under disk automorphisms. Our measure λ is one quarter of the area measure for the curvature -1 metric $4|dz|^2/(1 - |z|^2)^2$.

Proposition 29.1 (Exact decomposition). *For any distinct $z_1, \dots, z_N \in \mathbb{D}$, put $S_k = \sum_{j=1}^N z_j^k$. Then*

$$|\Delta_N(z)|^\beta \prod_{j=1}^N dA(z_j) = \exp \left\{ -\beta \sum_{i < j} g(z_i, z_j) - \frac{\beta}{2} \sum_{k=1}^{\infty} \frac{|S_k|^2}{k} \right\} \prod_{j=1}^N (1 - |z_j|^2)^{2-\beta/2} d\lambda(z_j). \quad (29.1)$$

All series in this identity converge absolutely.

Proof. Writing $r_* = \max_j |z_j| < 1$ gives $|S_k| \leq N r_*^k$, which proves the convergence assertion. The absolutely convergent Taylor series for $\log(1 - \zeta)$ gives

$$\begin{aligned} \sum_{i < j} \log |1 - z_i \bar{z}_j| &= -\frac{1}{2} \sum_{k \geq 1} \frac{1}{k} \left(|S_k|^2 - \sum_i |z_i|^{2k} \right) \\ &= -\frac{1}{2} \sum_{k \geq 1} \frac{|S_k|^2}{k} - \frac{1}{2} \sum_i \log(1 - |z_i|^2). \end{aligned}$$

Use $\log |z_i - z_j| = -g(z_i, z_j) + \log |1 - z_i \bar{z}_j|$ and $dA(z) = (1 - |z|^2)^2 d\lambda(z)$. $\square$

Proposition 29.2 (Failure of absolute Mayer integrability). *For every $\beta > 0$ and every $z \in \mathbb{D}$,*

$$\int_{\mathbb{D}} \left(1 - e^{-\beta g(z, w)} \right) d\lambda(w) = +\infty, \quad \int_{\mathbb{D}} \left(1 - e^{-\beta g(z, w)} \right)^2 d\lambda(w) < \infty.$$

The second integral is independent of z and equals

$$\pi \int_0^1 \frac{(1 - t^{\beta/2})^2}{(1 - t)^2} dt.$$

Proof. For $a \in \mathbb{D}$, the map $\varphi_a(w) = (w - a)/(1 - \bar{a}w)$ satisfies

$$1 - |\varphi_a(w)|^2 = \frac{(1 - |a|^2)(1 - |w|^2)}{|1 - \bar{a}w|^2}, \quad |\varphi'_a(w)| = \frac{1 - |a|^2}{|1 - \bar{a}w|^2}.$$

These identities prove invariance of λ ; the corresponding pseudohyperbolic-distance identity proves invariance of g . Thus it suffices to set $z = 0$, in which case $e^{-\beta g(0,w)} = |w|^\beta$. Polar coordinates and $t = |w|^2$ reduce the first integral to

$$\pi \int_0^1 \frac{1 - t^{\beta/2}}{(1 - t)^2} dt.$$

Since $(1 - t^{\beta/2})/(1 - t) \rightarrow \beta/2$ as $t \uparrow 1$, this integral diverges logarithmically. Squaring the numerator instead gives a bounded integrand near 1, and an integrand tending to 1 near 0. This proves all assertions. $\square$

There is a separate issue in turning (29.1) into a limiting specification: the positive harmonic moments S_k do not by themselves determine the centered Green field created by particles approaching the boundary. The following identity locates this issue without making any limiting interchange.

Lemma 29.3 (The centered exterior Green field). *Let $0 < r < R < 1$. For $|z| \leq r$ and $R < |w| < 1$,*

$$g(z, w) - g(0, w) = \operatorname{Re} \sum_{k=1}^{\infty} \frac{z^k}{k} (w^{-k} - \bar{w}^k).$$

The series converges uniformly in these z, w . In particular, for any finite configuration ξ supported on $\{|w| > R\}$,

$$\sum_{w \in \xi} (g(z, w) - g(0, w)) = \operatorname{Re} \sum_{k=1}^{\infty} \frac{z^k}{k} \sum_{w \in \xi} (w^{-k} - \bar{w}^k).$$

Furthermore,

$$|g(z, w) - g(0, w)| \leq C_{r,R}(1 - |w|).$$

Proof. Use

$$g(z, w) - g(0, w) = \log |1 - z\bar{w}| - \log |1 - z/w|$$

and the Taylor series for both logarithms. Uniform convergence follows from $|z/w| \leq r/R < 1$ and $|z\bar{w}| \leq r < 1$. For the final bound write $w = se^{i\theta}$. Since

$$s^{-k} - s^k = \int_s^1 k(t^{-k-1} + t^{k-1}) dt \leq k(1 - s)(R^{-k-1} + R^{-1}),$$

the sum of the absolute values is at most

$$(1 - s) \sum_{k \geq 1} r^k (R^{-k-1} + R^{-1}) \leq C_{r,R}(1 - s).$$

$\square$

The last bound involves the first boundary deficit $\sum_w (1 - |w|)$, rather than a power strictly larger than one. Uniform control of higher powers of the boundary deficit therefore does not justify discarding this field. Similarly, the first proposition of this section cannot be combined with an independent Gaussian harmonic limit by simply factoring expectations: the Green factor is a statistic of particles throughout the disk. Establishing the required boundary-tail control, or a different uniqueness principle that avoids it, remains necessary for this approach.

30 A precise variance criterion for the exterior inverse field

The result in this section is conditional. It isolates a quantitative estimate that would control the exterior field which is invisible to vague convergence. It does not assert that the estimate holds for the gas, and it does not by itself assert uniqueness.

For $k \geq 1$ and $s < 1$, define the angular count

$$A_{k,N}(s) = \sum_{|z_i| \leq s} e^{-ik \arg z_i}.$$

The convention for the argument at zero is immaterial, since a particle is at zero with probability zero. Consider the following hypothesis: there exist $s_* \in (0, 1)$, $C < \infty$, $b \geq 0$, and $0 \leq \gamma < 2$ such that

$$\sup_N \mathbb{E} |A_{k,N}(s)|^2 \leq C k^b (1-s)^{-\gamma}, \quad k \geq 1, \quad s_* \leq s < 1. \quad (30.1)$$

The previously proved count bound only gives an estimate with exponent $\gamma = 2$. Thus (30.1) is a genuine additional assumption.

Lemma 30.1 (Verification at the determinantal point). *For $\beta = 2$, hypothesis (30.1) holds with $b = 0$ and $\gamma = 1$.*

Proof. Let $f(z) = \mathbf{1}_{\{|z| \leq s\}} e^{-ik \arg z}$. Rotation invariance gives $\mathbb{E} \Xi_N(f) = 0$. The determinantal one- and two-point formulas proved in Section 12 give

$$\mathbb{E} |\Xi_N(f)|^2 = \int |f(z)|^2 \rho_{1,N}(z) dA(z) - \iint f(z) \overline{f(w)} |K_N(z, w)|^2 dA(z) dA(w).$$

The second integral is nonnegative: expanding the finite orthonormal basis of K_N writes it as a sum of quantities $|\int f(z) \phi_i(z) \overline{\phi_j(z)} dA(z)|^2$. Consequently

$$\mathbb{E} |A_{k,N}(s)|^2 \leq \mathbb{E} M_N(s) = \sum_{j=1}^N s^{2j} \leq \frac{s^2}{1-s^2} \leq \frac{1}{1-s},$$

uniformly in k, N, s . This is the claimed hypothesis at 2. $\square$

Proposition 30.2 (Vanishing boundary tails of the inverse correction). *Assume (30.1), and put $\delta = 1 - \gamma/2 > 0$. For $R < 1$ define*

$$C_{k,N}(R) = \sum_{|z_i| > R} (z_i^{-k} - \overline{z_i}^k).$$

For every $R_0 \in (\max\{s_, 0\}, 1)$, every $R \geq R_0$, and every $k \geq 1$,*

$$\sup_N \|C_{k,N}(R)\|_{L^2} \leq C_{R_0} k^{1+b/2} R_0^{-k-1} (1-R)^\delta. \quad (30.2)$$

In particular, if $q < R_0$ and

$$H_{N,R}(z) = \sum_{k \geq 1} \frac{z^k}{k} C_{k,N}(R), \quad |z| < R,$$

then

$$\sup_N \left\| \sup_{|z| \leq q} |H_{N,R}(z)| \right\|_{L^2} \leq C_{q,R_0} (1-R)^\delta. \quad (30.3)$$

Proof. Writing $z_i = u_i e^{i\theta_i}$ and using

$$u^{-k} - u^k = \int_u^1 k(s^{-k-1} + s^{k-1}) ds,$$

finite summation and Fubini give the exact identity

$$C_{k,N}(R) = \int_R^1 k(s^{-k-1} + s^{k-1})[A_{k,N}(s) - A_{k,N}(R)] ds. \quad (30.4)$$

There is no boundary atom at a deterministic radius, and changing the endpoint convention does not affect this integral. The coefficient in the integrand is at most $2kR_0^{-k-1}$. Minkowski's integral inequality and (30.1) bound the L^2 norm by

$$Ck^{1+b/2}R_0^{-k-1} \int_R^1 [(1-s)^{-\gamma/2} + (1-R)^{-\gamma/2}] ds,$$

which is (30.2). Another application of Minkowski, now to the power series, gives (30.3), since $\sum_{k \geq 1} k^{b/2}(q/R_0)^k < \infty$. All series used before this bound converge absolutely for each finite configuration when $|z| < R$. $\square$

Proposition 30.3 (Identification from the limiting configuration). *Assume (30.1) and $\Xi_{N_j} \Rightarrow \Xi$. Fix $r \in (0, 1)$. There is a measurable random holomorphic function on B_r ,*

$$H_r^\Xi(z) = \lim_{R \uparrow 1} \sum_{r < |w| \leq R, w \in \Xi} \sum_{k \geq 1} \frac{z^k}{k} (w^{-k} - \bar{w}^k), \quad (30.5)$$

where the limit exists in L^2 for the supremum norm on every smaller closed disk. It is measurable with respect to the exterior configuration $\Xi|_{\{|w| > r\}}$. Jointly with the point process and its positive Gaussian harmonic modes,

$$\sum_{|z_i| > r} \sum_{k \geq 1} \frac{z_i^k}{k} (z_i^{-k} - \bar{z}_i^k) \implies H_r^\Xi(z) \quad \text{in } \mathcal{H}(B_r).$$

The convergence is in the topology of uniform convergence on compact subsets; the statement about the positive modes uses their already proved joint analytic convergence and independence from Ξ .

Proof. First truncate the sums to $r < |w| \leq R < 1$. They are continuous functions of the configuration with values in $\mathcal{H}(B_r)$ at every configuration with no point on the two delimiting circles. The limiting process has this property almost surely, by its continuous intensity. Compact count moments are uniformly bounded. Consequently convergence on each finite annulus holds jointly with the process and its positive analytic modes, and also gives convergence of second moments of each bounded-annulus sum in any smaller-disk supremum norm. For the latter assertion the supremum is bounded by a fixed constant times the number of particles in that annulus, so uniform fourth count moments provide uniform integrability.

For $R < S < 1$, the finite-particle annulus sum between R and S equals $H_{N,R} - H_{N,S}$. Proposition 30.2 bounds its L^2 supremum norm by $C(1-R)^\delta$, uniformly in S, N , once R exceeds a fixed radius larger than the parameter disk. Passing to the limit on the finite annulus gives the same bound for Ξ . The truncated sums for Ξ are therefore Cauchy in L^2 on every smaller disk. Their limits are compatible and holomorphic; for example, take an almost surely uniformly convergent subsequence on a countable compact exhaustion. They are exterior-measurable because every truncated sum is. The uniform finite- N tail bound and the corresponding limiting tail bound now justify removal of the truncation in the joint convergence, by the usual bounded-Lipschitz test argument: fix R , pass to the limit, and then let $R \uparrow 1$. This proves the assertions. $\square$

Proposition 30.4 (Same-count conditional density ratios). *Under the same hypothesis, fix $r \in (0, 1)$ and an integer $m \geq 1$ with $\mathbb{P}\{\Xi(B_r) = m\} > 0$. Condition the finite processes on $\Xi_{N_j}(B_r) = m$. For their exterior configurations η_N , let $p_N(x \mid \eta_N, m)$ be the conditional density of a uniformly randomly labeled inside tuple $x \in B_r^m$. For distinct-coordinate tuples x, y , one has the exact finite- N identity*

$$\frac{p_N(x \mid \eta_N, m)}{p_N(y \mid \eta_N, m)} = \frac{|\Delta_m(x)|^\beta}{|\Delta_m(y)|^\beta} \exp \left\{ -\beta \operatorname{Re} \left[\sum_{i=1}^m E_N(x_i) - \sum_{i=1}^m E_N(y_i) \right] \right\}, \quad (30.6)$$

where

$$E_N(z) = \sum_{k \geq 1} \frac{z^k}{k} \sum_{w \in \eta_N} w^{-k}.$$

As random holomorphic functions on B_r , these exterior fields converge jointly with the conditioned point process to

$$E_r(z) = \sum_{k \geq 1} \frac{z^k}{k} \left(\bar{G}_k - \sum_{w \in \Xi \cap B_r} \bar{w}^k \right) + H_r^\Xi(z). \quad (30.7)$$

Here Ξ has its law conditioned on $\Xi(B_r) = m$, and the Gaussian vector (G_k) is independent of it. In particular, the random ratios in (30.6) converge in law, locally uniformly in (x, y) away from collision diagonals, to the same expression with E_r in place of E_N .

Proof. Conditional on the exterior and the number of inside particles, the finite density is proportional to

$$|\Delta_m(x)|^\beta \prod_{i=1}^m \prod_{w \in \eta_N} |x_i - w|^\beta.$$

Its normalizer is finite and positive: there are finitely many exterior points, all strictly outside the closed radius- r disk almost surely, and the displayed product is bounded and positive off its collision diagonals. In the ratio of two such densities the normalizer cancels. For $|z| < r < |w|$,

$$\log |z - w| = \log |w| - \operatorname{Re} \sum_{k \geq 1} \frac{z^k}{k w^k}.$$

The constant exterior terms cancel because both tuples have m entries. This proves (30.6).

The exact decomposition

$$\sum_{w \in \eta_N} w^{-k} = \overline{S_{k,N}} - \sum_{w \in \Xi_N \cap B_r} \bar{w}^k + C_{k,N}(r)$$

and Proposition 30.3 identify the limit field. The Gaussian series converges locally uniformly almost surely by its second moments and geometric weights; the inside sum is finite. Conditioning causes no loss of convergence: the count event has a positive limiting probability, its indicator is almost surely continuous at the limiting configuration, and joint convergence hence passes to the conditional laws. Independence of the Gaussian vector from Ξ is preserved by this conditioning. Finally, exponentiation and evaluation on compact tuple sets are continuous for local uniform convergence of analytic functions, proving the ratio convergence. $\square$

Remark 30.5 (The remaining two issues). Neither (30.1) nor a uniqueness conclusion is proved in this section. The limiting ratio formula includes the additional Gaussian field and the actual inside power sums. Convergence of these random conditional-density ratios must not be identified, without a separate argument, with convergence of conditional laws given the limiting

exterior configuration alone. Moreover, fixing the number of inside particles removes the activity factor from a Gibbs density: it gives no rule for selecting the conditional distribution of that number. These are distinct issues from the vanishing-tail estimate. Sections 41 and 42 later give an unconditional canonical mixture theorem and prove that the additional harmonic-field randomness cannot simply be discarded. The complete count-resolved equation is obtained at $\beta = 2$ in Section 46.

31 A positive radial mixture at every even inverse temperature

This section uses the polynomial structure available when $\beta = 2m$, where m is a positive integer. It is a different finite-dimensional representation from the ordered-gap expansion. In particular, it improves some of the general bounds on the total radial depth.

Proposition 31.1 (Unordered radial mixture and majorization). *Expand the polynomial*

$$\Delta_N(z)^m = \sum_{\mathbf{k}} c_{\mathbf{k}} z_1^{k_1} \cdots z_N^{k_N}.$$

All indices in this sum satisfy $k_i \geq 0$ and $\sum_i k_i = mN(N-1)/2$. There is a probability distribution

$$p_N(\mathbf{k}) = \frac{|c_{\mathbf{k}}|^2 \prod_i (k_i + 1)^{-1}}{\sum_{\ell} |c_{\ell}|^2 \prod_i (\ell_i + 1)^{-1}} \quad (31.1)$$

such that, conditionally on $\mathbf{k}$, the variables $|z_i|^2$ are independent and have densities $(k_i + 1)x^{k_i} \mathbf{1}_{(0,1)}(x) dx$. Every $\mathbf{k}$ with nonzero coefficient is majorized by $\mathbf{b} = (0, m, 2m, \dots, m(N-1))$.

Proof. Write $z_i = \sqrt{x_i} e^{i\theta_i}$. Orthogonality of angular monomials gives

$$\int_{[0,2\pi]^N} |\Delta_N(z)|^{2m} \prod_i \frac{d\theta_i}{2\pi} = \sum_{\mathbf{k}} |c_{\mathbf{k}}|^2 \prod_i x_i^{k_i}.$$

Since $dA(z_i) = \pi dx_i d\theta_i / (2\pi)$, integrating the x_i proves the asserted mixture, including its normalization.

The ordinary determinant expansion writes Δ_N as a signed sum of monomials whose exponent vectors are permutations of $(0, 1, \dots, N-1)$. A monomial appearing in its m -th power therefore has exponent vector $\mathbf{k}$ equal to a sum of m such permutations. Cancellation can delete exponent vectors, but cannot create new ones. For a set I of q coordinate indices,

$$\sum_{i \in I} k_i \leq m((N-q) + (N-q+1) + \cdots + (N-1)).$$

Taking the largest q coordinates gives the majorization inequalities; the equality of total sums was already observed. $\square$

Theorem 31.2 (Sharp extremal product bounds for the radial mixture). *Put $B_N = \sum_{i=1}^N -\log |z_i|$ and $\beta = 2m$ as above. For every $\gamma > 0$,*

$$\prod_{j=0}^{N-1} \frac{mj+1}{mj+1+\gamma/2} \leq \mathbb{E}_{N,2m} e^{-\gamma B_N} \leq \left(\frac{1+m(N-1)/2}{1+m(N-1)/2+\gamma/2} \right)^N. \quad (31.2)$$

For $0 < \theta < 2$ one also has

$$\mathbb{E}_{N,2m} e^{\theta B_N} \leq \prod_{j=0}^{N-1} \frac{mj+1}{mj+1-\theta/2}. \quad (31.3)$$

Consequently

$$\mathbb{E}_{N,2m} B_N \leq \frac{1}{2} \sum_{j=0}^{N-1} \frac{1}{mj+1} = \frac{1}{\beta} \log N + O_\beta(1), \quad (31.4)$$

and, for each $0 < \theta < 2$, a finite constant $C_{\beta,\theta}$ satisfies

$$\mathbb{P}_{N,\beta} \left\{ B_N \geq \frac{1}{\beta} \log N + t \right\} \leq C_{\beta,\theta} e^{-\theta t}, \quad t \geq 0. \quad (31.5)$$

In the charge-matched case there is the particularly simple bound

$$\mathbb{E}_{N,\beta} \prod_i |z_i|^\beta \geq \frac{1}{mN+1} = \frac{1}{\beta N/2 + 1}. \quad (31.6)$$

Proof. Conditioning on the mixture index gives

$$\mathbb{E}[e^{-\gamma B_N} \mid \mathbf{k}] = \prod_i \frac{k_i + 1}{k_i + 1 + \gamma/2}, \quad \mathbb{E}[e^{\theta B_N} \mid \mathbf{k}] = \prod_i \frac{k_i + 1}{k_i + 1 - \theta/2}.$$

For $a > 0$, the function $g_a(x) = \log(x+1) - \log(x+1+a)$ is concave on $[0, \infty)$. The majorization inequalities imply $\sum_i g_a(k_i) \geq \sum_i g_a(b_i)$, while Jensen's inequality gives $\sum_i g_a(k_i) \leq N g_a(m(N-1)/2)$. In the present setting this use of majorization needs no general theorem: there are permutations $\sigma_1, \dots, \sigma_m$ such that $\mathbf{k} = m^{-1} \sum_{\ell=1}^m \sigma_\ell \mathbf{b}$. Coordinatewise Jensen, followed by summation in the coordinate index, gives $\sum_i g_a(k_i) \geq m^{-1} \sum_\ell \sum_i g_a((\sigma_\ell \mathbf{b})_i) = \sum_i g_a(b_i)$. These bounds hold for each index, and hence after mixing.

For $a = \theta/2 \in (0, 1)$, the function $h_a(x) = \log(x+1) - \log(x+1-a)$ is convex. Coordinatewise Jensen in the same average-of-permutations representation reverses the preceding inequality and proves (31.3). The function $x \mapsto 1/(x+1)$ is also convex. The conditional expectation of B_N is $\frac{1}{2} \sum_i (k_i + 1)^{-1}$, so the same argument gives (31.4).

Finally,

$$\sum_{j=0}^{N-1} \log \frac{mj+1}{mj+1-\theta/2} = \frac{\theta}{2m} \log N + O_{m,\theta}(1).$$

Indeed the $j = 0$ term is finite, and the Taylor remainder after the linear term for $j \geq 1$ is bounded by a constant times $(mj+1)^{-2}$; these remainders are summable. Exponential Markov gives (31.5). When $\gamma = 2m$, the lower product in (31.2) telescopes to $1/(mN+1)$. $\square$

Remark 31.3 (Scope of the improvement). At $\beta = 2$, the only exponent vectors are permutations of $(0, 1, \dots, N-1)$, so both product formulas before their Jensen upper bound are exact. At $\beta = 4$ the charge-matched estimate reads $\mathbb{E} \prod_i |z_i|^4 \geq (2N+1)^{-1}$. These are finite- N bounds; they provide no lower bound on B_N tending to infinity, and in particular do not prove that the critical process is empty.

32 An exact Bergman intensity bound at even inverse temperature

Let $\mathcal{B} = A^2(\mathbb{D}, dA)$ be the Bergman Hilbert space with reproducing kernel

$$K(z, w) = \frac{1}{\pi(1 - z\bar{w})^2}.$$

For a closed subspace $\mathcal{H} \subset \mathcal{B}$, write $K_{\mathcal{H}}$ for its reproducing kernel. Point evaluation remains continuous on every such subspace.

Lemma 32.1 (Exclusion for a reproducing-kernel direction). *Let $F \in \mathcal{H}^{\otimes N}$ be a holomorphic function of nonzero norm, and suppose F vanishes whenever any two coordinates coincide. Give the labeled points density $|F|^2/\|F\|^2$ with respect to dA^N , and let ρ_F be the intensity of their counting measure. Then*

$$\rho_F(a) \leq K_{\mathcal{H}}(a, a), \quad a \in \mathbb{D}. \quad (32.1)$$

No permutation symmetry assumption is necessary.

Proof. If $K_{\mathcal{H}}(a, a) = 0$, evaluation at a vanishes identically on $\mathcal{H}$, so every marginal density at a is zero. Otherwise let $u_a = K_{\mathcal{H}}(\cdot, a)/\sqrt{K_{\mathcal{H}}(a, a)}$, and let P_i be the orthogonal rank-one projection onto u_a in tensor coordinate i , acting as the identity in the other coordinates. Put $\Psi = F/\|F\|$. The reproducing identity gives

$$\|P_i \Psi\|^2 = \frac{1}{K_{\mathcal{H}}(a, a)} \int_{\mathbb{D}^{N-1}} |\Psi(z_1, \dots, z_{i-1}, a, z_{i+1}, \dots, z_N)|^2 dA^{N-1}.$$

The projections commute, and $P_i P_j \Psi = 0$ for $i \neq j$, because the corresponding double evaluation is zero. For $Q = \sum_i P_i$,

$$\langle \Psi, Q^2 \Psi \rangle = \langle \Psi, Q \Psi \rangle =: q.$$

Cauchy–Schwarz implies $q^2 \leq \|\Psi\|^2 \|Q \Psi\|^2 = q$; as $q \geq 0$, this gives $q \leq 1$. Summing the preceding marginal identity gives $q = \rho_F(a)/K_{\mathcal{H}}(a, a)$, proving the result. $\square$

Theorem 32.2 (Finite- N bounds for the gas and its reduced Palm laws). *Let $\beta = 2m$ with m a positive integer. For every $N \geq 1$,*

$$\rho_{1,N}(z) \leq \frac{1}{\pi(1-|z|^2)^2}, \quad \mathbb{E} \Xi_N(B_r) \leq \frac{r^2}{1-r^2}. \quad (32.2)$$

For distinct fixed points $a_1, \dots, a_p$ with $p < N$, define

$$\mathcal{H}_{\mathbf{a},m} = \{f \in \mathcal{B} : f^{(j)}(a_i) = 0, \ 1 \leq i \leq p, \ 0 \leq j < m\}.$$

The reduced Palm intensity, defined by the finite- N correlation ratio, satisfies

$$\frac{\rho_{p+1,N}(a_1, \dots, a_p, z)}{\rho_{p,N}(a_1, \dots, a_p)} \leq K_{\mathcal{H}_{\mathbf{a},m}}(z, z). \quad (32.3)$$

In particular, with $t = |(z-a)/(1-\bar{a}z)|^2$,

$$\rho_{2,N}(a, z) \leq \rho_{1,N}(a) \frac{t^m(m+1-mt)}{\pi(1-|z|^2)^2}. \quad (32.4)$$

These bounds pass to every subsequential point-process limit through the already proved locally uniform correlation convergence.

Proof. Use $F = \Delta_N^m$ and $\mathcal{H} = \mathcal{B}$ in the lemma; the density is exactly the gas density at $\beta = 2m$. Integrating the kernel diagonal over B_r gives $r^2/(1-r^2)$.

Conditioning on p distinct marked points leaves the normalized squared modulus of the polynomial

$$F_{\mathbf{a}}(w) = \Delta_{N-p}(w)^m \prod_{j=1}^{N-p} \prod_{i=1}^p (w_j - a_i)^m.$$

Each coordinate belongs to $\mathcal{H}_{\mathbf{a},m}$, and the polynomial still vanishes on every collision of the remaining coordinates. Applying the lemma proves the Palm bound. The denominator correlation is strictly positive for distinct marked points, since its insertion integral is positive.

For a single marked point at zero, the subspace consists of Bergman functions whose first m Taylor coefficients vanish. Its kernel diagonal is

$$K_{\mathcal{H}_{0,m}}(z, z) = \frac{1}{\pi} \sum_{k=m}^{\infty} (k+1) |z|^{2k} = \frac{|z|^{2m} (m+1 - m|z|^2)}{\pi(1 - |z|^2)^2}.$$

The disk automorphism $\varphi_a(z) = (z - a)/(1 - \bar{a}z)$ induces a unitary Bergman operator $f \mapsto \varphi'_a(f \circ \varphi_a)$, as is seen directly by changing variables in the area integral. It maps the subspace vanishing to order m at zero onto the subspace vanishing to order m at a . Consequently its kernel diagonal is $|\varphi'_a(z)|^2 K_{\mathcal{H}_{0,m}}(\varphi_a(z), \varphi_a(z))$. The identity $|\varphi'_a(z)|/(1 - |\varphi_a(z)|^2) = 1/(1 - |z|^2)$ proves the displayed pair bound. The limit assertion follows by passage to the inequalities on compact sets; for the Palm bound it is enough to retain the undivided form $\rho_{p+1} \leq \rho_p K_{\mathcal{H}}$ if the limiting denominator vanishes. $\square$

Corollary 32.3 (A determinantal upper bound and uniform count moments). *For every even $\beta = 2m$, every $p \geq 1$, and all distinct points,*

$$\rho_{p,N}(z_1, \dots, z_p) \leq \det[K(z_i, z_j)]_{i,j=1}^p \leq \prod_{i=1}^p K(z_i, z_i). \quad (32.5)$$

If $A \in \mathbb{D}$ is Borel and $\mu_A = \int_A K(z, z) dA(z)$, then for every $t \geq 0$,

$$\mathbb{E} e^{t\Xi_N(A)} \leq \exp\{\mu_A(e^t - 1)\}. \quad (32.6)$$

All these inequalities hold also for subsequential limits and for their images under arbitrary disk automorphisms. They are inequalities for correlations and moments; they do not assert stochastic domination of point-process laws.

Proof. Iterate (32.3). The subspace vanishing to order m at each of the preceding marks is contained in the subspace vanishing to order one there, so its kernel diagonal is smaller. The order-one subspace is the orthogonal complement of the span of the corresponding reproducing vectors. Its kernel diagonal at the next point is the squared norm of the next vector after orthogonal projection off that span. The product of these successive squared norms is the Gram determinant $\det[K(z_i, z_j)]$. The vectors at distinct points are linearly independent, since polynomial interpolation separates their evaluations. This proves the first inequality for $p \leq N$; for $p > N$ the left side is zero. Hadamard's inequality, or the same projection description, gives the second inequality.

Integrating it over A^p bounds the p -th factorial moment by μ_A^p . For an integer-valued $M \geq 0$, $e^{tM} = \sum_{p \geq 0} (M)_{\underline{p}} (e^t - 1)^p / p!$. All terms are nonnegative for $t \geq 0$, so monotone convergence and the factorial moment bounds prove the exponential estimate. The locally uniform correlation convergence proves the limiting inequalities; the same moment argument applies to the limiting process. Finally the Bergman kernel and its vanishing subspaces transform unitarily under disk automorphisms, as in the preceding proof. Both sides of the correlation inequalities acquire the same product Jacobian factor. This proves their covariance, and hence the assertion for transformed laws. $\square$

Remark 32.4 (What the bound does not settle). At $\beta = 4$, the bound gives the exact finite- N estimate $\pi(1 - |z|^2)^2 \rho_{1,N}(z) \leq 1$. It does not show that its left side tends to zero by itself; the subsequent critical emptiness proof combines it with the canonical-score estimates and a second limiting procedure. The rank-one argument bounds occupation of each coherent kernel direction; it does not assert domination of the entire one-body operator by the identity for arbitrary symmetric holomorphic states.

33 Electric forces at particles and an augmented limiting Ward identity

The estimates in this section concern the full electric force evaluated at an actual particle. They do not assume control of the exterior inverse field. In particular, the marks constructed below are not asserted to be measurable functions of the limiting configuration.

For a distinct finite tuple define the complex force

$$C_{i,N} = \sum_{j \neq i} \frac{1}{z_i - z_j}.$$

The real directional derivative of the log density in a direction $v \in \mathbb{C}$ at the i th coordinate is $\beta \operatorname{Re}(v C_{i,N})$. Thus the real force vector, in the usual identification of $\mathbb{R}^2$ with $\mathbb{C}$, is $\beta \bar{C}_{i,N}$. Write $\partial = (\partial_x - i\partial_y)/2$ and $\Delta = \partial_x^2 + \partial_y^2$.

Proposition 33.1 (Exact Campbell force identities). *For every fixed N and $\beta > 0$, $\rho_{1,N}$ is C^2 in $\mathbb{D}$. For every $\phi \in C_c^\infty(\mathbb{D})$,*

$$\beta^2 \mathbb{E} \sum_i \phi(z_i) |C_{i,N}|^2 = \int_{\mathbb{D}} \rho_{1,N}(z) \Delta \phi(z) dA(z), \quad (33.1)$$

$$\rho_{1,N}(a) \mathbb{E}_N^! C_{a,N} = \frac{2}{\beta} \partial \rho_{1,N}(a). \quad (33.2)$$

Here $\mathbb{E}_N^!$ is the reduced Palm expectation at a , and $C_{a,N} = \sum_{w \in \Xi_N} 1/(a - w)$ under that reduced Palm law. The first identity is valid also for signed ϕ . Moreover,

$$\Delta \log \rho_{1,N}(a) = \beta^2 \left(\mathbb{E}_N^! |C_{a,N}|^2 - |\mathbb{E}_N^! C_{a,N}|^2 \right). \quad (33.3)$$

Proof. Use the finite positive measure ν_N from the insertion formula, so that

$$\rho_{1,N}(a) = \int |P_y(a)|^\beta d\nu_N(y), \quad P_y(a) = \prod_{j=1}^{N-1} (a - y_j).$$

For every polynomial P , in the sense of distributions,

$$\Delta |P|^\beta = \beta^2 |P|^\beta |P'|/P|^2. \quad (33.4)$$

At a zero of multiplicity q , the right side is locally bounded by a constant times $|a - y|^{q\beta-2}$ and hence is integrable. To verify that there is no additional point mass, integrate by parts outside radius- ε circles around the zeros. The two boundary terms are $O(\varepsilon^{q\beta})$ and $O(\varepsilon^{q\beta+1})$, respectively, and tend to zero. Away from the zeros the identity follows by two ordinary derivatives of $\exp(\beta \log |P|)$.

For nonnegative ϕ , integrate (33.4) against ϕ , and then against ν_N . Tonelli applies on the nonnegative right side; the other side is absolutely bounded by $\|\Delta \phi\|_\infty \int_{\operatorname{supp} \phi} \rho_{1,N} dA$. This proves the first formula and its local integrability. Signed tests follow by domination with a compactly supported nonnegative cutoff. One distributional derivative similarly gives

$$\partial \rho_{1,N}(a) = \frac{\beta}{2} \int |P_y(a)|^\beta \sum_j (a - y_j)^{-1} d\nu_N(y).$$

Here the density of ν_N with respect to area on $\mathbb{D}^{N-1}$ is bounded, with a bound allowed to depend on N . Each integrand in a first or second real derivative is bounded in absolute value by a finite sum of constant multiples of

$$|a - y_j|^{\beta-2} \prod_{\ell \neq j} |a - y_\ell|^\beta \quad \text{and} \quad |a - y_j|^{\beta-1} |a - y_\ell|^{\beta-1} \prod_{h \notin \{j, \ell\}} |a - y_h|^\beta.$$

These expressions are locally integrable uniformly in a on compact sets. Their integrals over the sets where any singular denominator is within ε of zero tend to zero uniformly in a : for the first expression use $\int_{|u|<\varepsilon} |u|^{\beta-2} dA(u) = 2\pi\varepsilon^\beta/\beta$, and for the second use the analogous integral with exponent $\beta-1$. If an exponent is nonnegative, ordinary boundedness suffices. Off those sets, dominated convergence applies. Thus the distributional first and second derivatives are continuous, which proves C^2 regularity and the pointwise Palm formulas. Finally $\rho_{1,N} > 0$, and differentiating its logarithm and using $|\nabla \rho_{1,N}|^2 = 4|\partial \rho_{1,N}|^2$ gives (33.3). $\square$

Corollary 33.2 (Uniform local force energy and Fisher information). *For every compact $K \Subset \mathbb{D}$ there is $C_{\beta,K} < \infty$ such that*

$$\sup_N \mathbb{E} \sum_{z_i \in K} |C_{i,N}|^2 \leq C_{\beta,K}. \quad (33.5)$$

More explicitly, for $0 \leq r < 1$,

$$\sup_N \mathbb{E} \sum_{|z_i| \leq r} |C_{i,N}|^2 \leq C_\beta (1-r)^{-3}. \quad (33.6)$$

For every nonnegative $\phi \in C_c^\infty(\mathbb{D})$,

$$\int \phi |\nabla \sqrt{\rho_{1,N}}|^2 dA \leq \frac{1}{4} \int \rho_{1,N} \Delta \phi dA. \quad (33.7)$$

Consequently every limiting intensity satisfies $\sqrt{\rho_1} \in H_{\text{loc}}^1(\mathbb{D})$ and the same inequality, with the right side interpreted distributionally.

Proof. Choose a nonnegative smooth cutoff equal to one on K and apply (33.1), using the uniform compact count bound. For the explicit version use a cutoff equal to one on B_r , supported in B_R , where $R = (1+r)/2$, whose second derivatives are bounded by $C(1-r)^{-2}$. Such cutoffs also exist for r near zero, by choosing the cutoff constant on a fixed smaller centered disk. The radial count estimate $\mathbb{E} M_N(R) \leq (1-R^{\beta/2})^{-1}$, and $1-R^{\beta/2} \geq c_\beta(1-R)$, give the stated power. By Jensen in the Palm law,

$$|\nabla \sqrt{\rho_{1,N}}|^2 = \frac{\beta^2}{4} \rho_{1,N} |\mathbb{E}_N^! C_{a,N}|^2 \leq \frac{\beta^2}{4} \rho_{1,N} \mathbb{E}_N^! |C_{a,N}|^2.$$

Integration proves (33.7). Along a process-convergent subsequence the intensities converge locally uniformly, hence their square roots converge in local L^2 . Cutoffs equal to one on compact sets give a locally bounded sequence in H^1 . Weak compactness and lower semicontinuity, including after multiplication of gradients by $\sqrt{\phi}$, prove the assertion for the limiting intensity. $\square$

Theorem 33.3 (A force-marked subsequential limit). *Suppose $\Xi_{N_j} \Rightarrow \Xi$. There is a further subsequence along which the particles and their force marks have a joint limit*

$$\left(\Xi_{N_j}, \sum_i \delta_{(z_i, C_{i,N_j})} \right) \Longrightarrow \left(\Xi, \sum_{z \in \Xi} \delta_{(z, C_z)} \right).$$

The marked measures have the vague topology on $\mathbb{D} \times \mathbb{C}$. Every particle of Ξ has exactly one finite complex mark C_z . For nonnegative $\phi \in C_c^\infty(\mathbb{D})$,

$$\beta^2 \mathbb{E} \sum_{z \in \Xi} \phi(z) |C_z|^2 \leq \int \rho_1 \Delta \phi dA. \quad (33.8)$$

Let $v \in C_c^1(\mathbb{D}; \mathbb{C})$ be a real vector field and let

$$F(\eta) = \Phi \left(\int h_1 d\eta, \dots, \int h_q d\eta \right), \quad h_j \in C_c^1(\mathbb{D}; \mathbb{R}), \quad \Phi \in C_b^1(\mathbb{R}^q; \mathbb{R}),$$

with bounded first derivatives. Define

$$\mathcal{L}_v F(\eta) = \sum_j \partial_j \Phi \left(\int h_1 d\eta, \dots, \int h_q d\eta \right) \int dh_j(z) [v(z)] d\eta(z).$$

The augmented limiting law satisfies

$$\mathbb{E} \left[\mathcal{L}_v F(\Xi) + F(\Xi) \sum_{z \in \Xi} \operatorname{div} v(z) + \beta F(\Xi) \operatorname{Re} \sum_{z \in \Xi} v(z) C_z \right] = 0. \quad (33.9)$$

All three expectations are absolutely convergent.

Proof. On a compact base set K , the number of marks is bounded by $\Xi_N(K)$, whose moments are uniformly bounded. This gives tightness of the marked measures on compact subsets of $\mathbb{D} \times \mathbb{C}$, jointly with Ξ_N . Moreover, (33.5) gives

$$\mathbb{E} \# \{i : z_i \in K, |C_{i,N}| > T\} \leq C_{\beta,K} T^{-2}.$$

Thus no spatial particle is lost through escape of its mark to infinity. To formalize the identification of the projection, test against a continuous function with compact support in the base, multiply it by a continuous mark cutoff equal to one on $|c| \leq T$, pass to a joint subsequential limit, and then let $T \rightarrow \infty$. The displayed bound makes the discrepancy tend to zero in L^1 . The projection is therefore Ξ almost surely. Since Ξ is simple, there is exactly one mark over each of its particles. Continuous compact mark cutoffs followed by monotone approximation of $|c|^2$ yield (33.8) from (33.1) and local uniform intensity convergence.

At finite N , integration by parts in the i th coordinate, then summation in i , gives

$$\mathbb{E} \left[\mathcal{L}_v F(\Xi_N) + F(\Xi_N) \sum_i \operatorname{div} v(z_i) + \beta F(\Xi_N) \operatorname{Re} \sum_i v(z_i) C_{i,N} \right] = 0. \quad (33.10)$$

There is no boundary term on $\partial\mathbb{D}$ because v is compactly supported. For the collision diagonals this can either be justified by deleting small circles and observing the boundary factors $O(\varepsilon^{\beta+1})$, followed by integration in the other coordinates, or by using the locally integrable first distributional derivative of the density. Its absolute value is bounded by a finite sum of products containing one factor $|z_i - z_j|^{\beta-1}$; these are integrable for $\beta > 0$.

The first two terms in (33.10) pass to the limit by vague continuity and uniform compact count moments. For the last term use a continuous mark cutoff at level T . The cut off statistic is continuous and bounded by a constant times $T\Xi_N(K)$, so its expectation also converges. The omitted tail is bounded in absolute expectation, uniformly in N , by

$$\|F\|_\infty \|v\|_\infty \mathbb{E} \sum_{z_i \in K} |C_{i,N}| \mathbf{1}_{\{|C_{i,N}| > T\}} \leq C_{\beta,F,v,K}/T.$$

The limiting energy bound gives the same tail bound for the limit. Letting $T \rightarrow \infty$ proves (33.9) and absolute integrability. The statements about tightness and subsequences here can also be obtained directly by exhausting both spaces by compact sets, bounding their point counts, and applying diagonal compactness of bounded finite point measures on each compact set. $\square$

Remark 33.4 (What the force marks do and do not resolve). The theorem constructs finite, locally square-integrable forces at the particles of every subsequential limit and supplies an unconditional integration-by-parts identity on this augmented probability space. It does not show that C_z is determined by Ξ , identify it with a particular summation of the infinite Coulomb force, or establish uniqueness of either the unmarked or the augmented law. A boundary harmonic contribution may still survive in the marks. In particular, the angular-count variance hypothesis in Section 30 has not been deduced from the local force-energy bound.

Corollary 33.5 (A canonical configuration-measurable logarithmic derivative). *For every subsequential limiting law there is a measurable complex function $b(z, \eta)$, defined for $z \in \eta$ up to Campbell-null sets, which satisfies*

$$\beta^2 \mathbb{E} \sum_{z \in \Xi} \phi(z) |b(z, \Xi)|^2 \leq \int \rho_1 \Delta \phi dA, \quad 0 \leq \phi \in C_c^\infty(\mathbb{D}),$$

and, for the cylinder functions and vector fields of Theorem 33.3,

$$\mathbb{E} \left[\mathcal{L}_v F(\Xi) + F(\Xi) \sum_{z \in \Xi} \operatorname{div} v(z) + \beta F(\Xi) \operatorname{Re} \sum_{z \in \Xi} v(z) b(z, \Xi) \right] = 0.$$

Among locally Campbell-integrable functions, this identity determines b uniquely up to Campbell-null sets. Thus the real vector field $\beta \overline{b(z, \eta)}$ is a canonical logarithmic derivative of the limiting law. This assertion does not identify it with a prescribed summation of the infinite Coulomb force.

Proof. Consider the Campbell measure of the augmented law on triples (η, z, c) :

$$\widehat{\Gamma}(H) = \mathbb{E} \sum_{z \in \Xi} H(\Xi, z, C_z).$$

Its projection to (η, z) is the ordinary Campbell measure Γ , and both measures are sigma-finite, with finite mass over compact base sets. The configuration space and the other factors are standard Borel. Disintegrate $\widehat{\Gamma}$ with respect to Γ and denote the conditional probability kernel of the mark by $\kappa_{\eta, z}(dc)$. This can, equivalently, be done after multiplication by a strictly positive integrable weight depending only on z , which turns the Campbell measures into finite measures without changing the conditional kernel. Set

$$b(z, \eta) = \int c \kappa_{\eta, z}(dc).$$

The local second-moment bound of Theorem 33.3 makes this integral finite Γ -almost everywhere. Conditional Jensen gives the asserted energy bound. Conditional expectation of the augmented Ward identity gives the unmarked identity, because its force coefficient $F(\eta)v(z)$ is measurable with respect to (η, z) .

For uniqueness, let b_1, b_2 be two locally Campbell-integrable functions satisfying the identity. Subtract their identities. For every admissible real cylinder function F and every complex vector field $v \in C_c^1(\mathbb{D}; \mathbb{C})$ this gives

$$\operatorname{Re} \int F(\eta) v(z) (b_1 - b_2)(z, \eta) d\Gamma(\eta, z) = 0.$$

Using iv as well gives the same identity without the real part. The bounded smooth cylinder functions form a unital algebra generating the Borel sigma-algebra of the configuration space:

integrals against a countable separating family in $C_c^1(\mathbb{D})$ generate the vague sigma-algebra, and bounded smooth scalar transforms recover their measurability. For each fixed v , the monotone-class theorem therefore extends the last equality to every bounded Borel function F . On each compact base region the measure $(b_1 - b_2) d\Gamma$ has finite total variation. Continuous compactly supported spatial tests, approximated by C_c^1 tests, determine finite measures. Hence the equality for all product tests $F(\eta)v(z)$ implies $(b_1 - b_2) d\Gamma = 0$ there. Exhausting the disk proves uniqueness. $\square$

34 Gaussian force regression and a lower bound on intensity curvature

The independent Gaussian harmonic field forces a specific component of the electric-force fluctuations. This gives a constraint on every limiting intensity, and a second proof of emptiness above 4. The proof in this section uses compactness, compact correlation bounds, the finite-mode Gaussian theorem and its independence from the limiting configuration, and Section 33. It does not use the boundary transport estimate for adjacent partition functions.

Theorem 34.1 (Regression of force marks on the Gaussian modes). *Refine the subsequence of Theorem 33.3 so that the force-marked configuration converges jointly with every fixed positive harmonic mode. Let $(G_k)_{k \geq 1}$ be their limiting independent circular Gaussian variables, independent of Ξ , with $\mathbb{E}|G_k|^2 = 2k/\beta$. Let $b(z, \Xi)$ be the canonical score of Corollary 33.5. Then, in the sense of conditional expectations under the Campbell measure,*

$$\mathbb{E}[C_z \mid \Xi, z, G_1, \dots, G_m] = b(z, \Xi) - \sum_{k=1}^m \overline{G_k} z^{k-1}. \quad (34.1)$$

In particular, for every nonnegative $\phi \in C_c^\infty(\mathbb{D})$,

$$\int \rho_1 \Delta \phi dA \geq \beta^2 \mathbb{E} \sum_{z \in \Xi} \phi(z) |b(z, \Xi)|^2 + 2\beta \int \frac{\phi(z) \rho_1(z)}{(1 - |z|^2)^2} dA(z). \quad (34.2)$$

The infinite series in (34.1), with $m = \infty$, also converges in local Campbell L^2 and gives the conditional expectation given all the G_k .

Proof. Tightness of the marked configurations and of every finite collection of harmonic modes gives the asserted joint subsequence by diagonal extraction. The marginal law of $(\Xi, (G_k))$ retains the proved Gaussian independence. In particular its Campbell measure factors as the Campbell measure of Ξ times the Gaussian probability law.

Take a cylinder function F and vector field v as in Theorem 33.3, and a real bounded C^1 function h on $\mathbb{C}^m \simeq \mathbb{R}^{2m}$ with bounded derivatives. Apply the finite integration-by-parts calculation to $F(\Xi_N)h(S_{1,N}, \dots, S_{m,N})$. Although the power sums are not local, at fixed N they and their derivatives are bounded, and the vector field remains compactly supported. In addition to the three terms of (33.10), multiplied by h , the derivative of h contributes

$$F(\Xi_N) Dh(S_{1,N}, \dots, S_{m,N}) \left[\left(k \sum_i v(z_i) z_i^{k-1} \right)_{k=1}^m \right].$$

All terms pass to the joint limit: the first and derivative terms are bounded by constants times compact particle counts, and the force term is handled by the mark truncation and L^2 tail bound in the proof of Theorem 33.3.

For a circular complex Gaussian of variance $2k/\beta$, ordinary real Gaussian integration by parts, or its Wirtinger form, gives

$$\mathbb{E} \partial_{G_k} h(G) = \frac{\beta}{2k} \mathbb{E} [h(G) \overline{G_k}].$$

Indeed its density is proportional to $\exp(-\beta|g|^2/(2k))$. Since $Dh(G)[d] = 2 \operatorname{Re} \sum_k \partial_{G_k} h(G) d_k$, conditioning first on Ξ turns the preceding derivative term into

$$\beta \mathbb{E} \left[F(\Xi) h(G) \operatorname{Re} \sum_{z \in \Xi} v(z) \sum_{k=1}^m \overline{G_k} z^{k-1} \right].$$

Write $T_m(z) = \sum_{k=1}^m \overline{G_k} z^{k-1}$. The other two nonforce terms factor by independence of G and Ξ . Subtract the unmarked Ward identity for b , multiplied by $\mathbb{E} h(G)$. The result is

$$\mathbb{E} \left[F(\Xi) h(G) \operatorname{Re} \sum_{z \in \Xi} v(z) (C_z + T_m(z) - b(z, \Xi)) \right] = 0.$$

Replacing v by iv removes the real-part restriction. Product tests $F(\eta)h(g)v(z)$ determine the locally finite Campbell measure: bounded smooth cylinder functions and bounded smooth functions of g are generating algebras, and compactly supported smooth spatial tests determine finite measures. The monotone-class argument used in Corollary 33.5 therefore proves the conditional-expectation identity.

Conditional Jensen and Gaussian independence now give

$$\mathbb{E} \sum_{z \in \Xi} \phi(z) |C_z|^2 \geq \mathbb{E} \sum_{z \in \Xi} \phi(z) |b(z, \Xi)|^2 + \frac{2}{\beta} \int \phi(z) \rho_1(z) \sum_{k=1}^m k |z|^{2k-2} dA(z).$$

There is no cross term because the Gaussian variables have zero mean and are independent of the unmarked Campbell measure. Combine this with (33.8), let $m \rightarrow \infty$, and use $\sum_{k \geq 1} kx^{k-1} = (1-x)^{-2}$ to obtain (34.2). On a compact spatial set, the same geometric series proves T_m is Cauchy in Campbell L^2 . Conditional expectations converge in L^2 to the expectation given the sigma-algebra generated by all the modes. One can apply this usual probability-space assertion after restriction to any finite-mass compact Campbell window; these restrictions exhaust the disk. $\square$

Theorem 34.2 (Curvature and boundary growth of a nonempty intensity). *Every subsequential limiting intensity satisfies the measure inequality*

$$\Delta \rho_1 \geq \frac{|\nabla \rho_1|^2}{\rho_1} dA + \frac{2\beta \rho_1}{(1-|z|^2)^2} dA, \quad (34.3)$$

where the first quotient is assigned zero on $\{\rho_1 = 0\}$. On every open set where $\rho_1 > 0$,

$$\Delta \log \rho_1 \geq \frac{2\beta}{(1-|z|^2)^2} dA. \quad (34.4)$$

If the limiting process is not almost surely empty, then

$$r \mapsto (1-r^2)^{\beta/2} \rho_1(r) \quad \text{is nondecreasing on } (0, 1). \quad (34.5)$$

Consequently, for $0 < r_0 \leq r < 1$,

$$\rho_1(r) \geq \rho_1(r_0) \left(\frac{1-r_0^2}{1-r^2} \right)^{\beta/2}. \quad (34.6)$$

Proof. Disintegrate the unmarked Campbell measure with respect to its spatial marginal $\rho_1(z)dA(z)$, and write $m_b(z)$ for the mean of $b(z, \Xi)$ under this conditional law, and put $J(z) = \rho_1(z)m_b(z)$. The Ward identity for $F = 1$ gives, distributionally,

$$\partial_x \rho_1 = \beta \operatorname{Re} J, \quad \partial_y \rho_1 = -\beta \operatorname{Im} J.$$

Local boundedness of ρ_1 and the Campbell L^2 bound imply $J \in L^2_{\text{loc}}$ by conditional Jensen. Thus $\rho_1 \in H^1_{\text{loc}}$, and another conditional Jensen inequality gives

$$\frac{|\nabla \rho_1(z)|^2}{\rho_1(z)} \leq \beta^2 \rho_1(z) \mathbb{E}[|b(z, \Xi)|^2 \mid z]$$

almost everywhere, with the stated zero convention. Combining this with (34.2) proves (34.3). The measure $\Delta \rho_1$ is positive and locally finite, also directly from subharmonicity.

Here is a weak chain-rule justification of the logarithmic inequality. On a relatively compact subset of $\{\rho_1 > 0\}$ the continuous function ρ_1 is bounded above and below by positive constants. For a nonnegative smooth test ϕ supported there, the function $\psi = \phi/\rho_1$ is continuous, compactly supported and in H^1 . Nonnegative mollifications approximate it uniformly and in H^1 , with support in a common positive-intensity region. Testing the Radon measure $\Delta \rho_1$ against these mollifications and passing to the limit gives

$$\int \frac{\phi}{\rho_1} d(\Delta \rho_1) = - \int \nabla \rho_1 \cdot \nabla (\phi/\rho_1) dA = \langle \Delta \log \rho_1, \phi \rangle + \int \phi \frac{|\nabla \rho_1|^2}{\rho_1^2} dA.$$

Inserting (34.3) cancels the gradient term and proves (34.4).

For a nonempty limiting law, the intensity dichotomy gives $\rho_1(r) > 0$ for every $r > 0$. Since $\Delta \log(1 - |z|^2) = -4/(1 - |z|^2)^2$, the radial function

$$h(z) = \log \rho_1(z) + \frac{\beta}{2} \log(1 - |z|^2)$$

is subharmonic on the punctured disk. It is locally bounded above near zero, since ρ_1 is continuous and locally bounded. The removable singularity property gives a subharmonic extension across zero, possibly with value $-\infty$ there. Therefore $t \mapsto h(e^t)$ is convex and nondecreasing, by the radial subharmonicity argument used earlier for the intensity. Exponentiating proves the last two claims. $\square$

Corollary 34.3 (An independent empty-limit proof and the critical constraint). *For every $\beta > 4$ the full process sequence converges to the empty configuration. At $\beta = 4$, any subsequential limit that is not almost surely empty must satisfy*

$$\lim_{r \uparrow 1} (1 - r^2)^2 \rho_1(r) = c, \quad 0 < c \leq 1/\pi. \quad (34.7)$$

The critical assertion is a necessary condition for a nonempty limit, not an assertion that such a limit exists.

Proof. Radial monotonicity and the count comparison, applied with $R = (1 + r)/2$, give

$$\rho_1(r) \pi(R^2 - r^2) \leq \mathbb{E} \Xi(B_R) \leq \frac{1}{1 - R^{\beta/2}},$$

and hence $\rho_1(r) \leq C_\beta (1 - r)^{-2}$. For $\beta > 4$, this contradicts (34.6) if the limit is nonempty. Every subsequential limit is therefore empty, and tightness implies convergence of the full sequence. This argument gives no rate in N .

At $\beta = 4$, the weighted intensity is positive, nondecreasing, and bounded above, and thus has a finite positive limit c . Moreover,

$$\mathbb{E}\Xi(B_r) = 2\pi \int_0^r s\rho_1(s) ds, \quad \mathbb{E}\Xi(B_r) \leq (1 - r^2)^{-1}.$$

The assumed positive limit of $(1 - r^2)^2\rho_1(r)$ implies $(1 - r^2)\mathbb{E}\Xi(B_r) \rightarrow \pi c$, by integrating upper and lower bounds $(c \pm \varepsilon)/(1 - s^2)^2$ on an outer annulus. Therefore $\pi c \leq 1$. $\square$

Proposition 34.4 (The variance that remains at the critical value). *For a nonempty limiting law, let*

$$V(z) = \text{Var}(b(z, \Xi) \mid z)$$

be the conditional variance under disintegration of the Campbell measure with respect to its spatial coordinate. On $\mathbb{D} \setminus \{0\}$, the curvature bound strengthens to

$$\Delta \log \rho_1 \geq \left(\frac{2\beta}{(1 - |z|^2)^2} + \beta^2 V(z) \right) dA(z). \quad (34.8)$$

At $\beta = 4$, for every $r_0 \in (0, 1)$,

$$\int_{r_0 < |z| < 1} (1 - |z|) V(z) dA(z) < \infty. \quad (34.9)$$

Thus a lower bound $V(z) \geq c(1 - |z|)^{-2}$ almost everywhere on an entire outer annulus, for any $c > 0$, would rule out a nonempty critical limit. The proof below uses boundary translation and repulsion to obtain the contradiction without first establishing this particular lower bound.

Proof. In the proof of Theorem 34.2, retain the conditional variance instead of discarding it in Jensen:

$$\beta^2 \rho_1 \mathbb{E}[|b|^2 \mid z] = \frac{|\nabla \rho_1|^2}{\rho_1} + \beta^2 \rho_1 V.$$

The same weak chain rule gives (34.8).

At $\beta = 4$ put $h(r) = \log[(1 - r^2)^2 \rho_1(r)]$. Corollary 34.3 shows that h has a finite limit at 1, and h is radial subharmonic on the punctured disk. Let $g(t) = h(e^t)$, which is convex for $t < 0$. Denote its distributional second-derivative measure by μ_g , which is nonnegative. For any fixed $T < 0$ and $T < t < 0$, the integral representation of a convex function gives

$$g(t) = g(T) + (t - T)g'_+(T) + \int_{(T, t]} (t - s) d\mu_g(s).$$

Letting $t \uparrow 0$ and using the finite boundary value proves $\int_{(T, 0)} (-s) d\mu_g(s) < \infty$. The radial Laplacian relation gives

$$\int_{e^T < |z| < 1} \log(1/|z|) d(\Delta h)(z) = 2\pi \int_{(T, 0)} (-s) d\mu_g(s) < \infty.$$

Since $\Delta h \geq 16VdA$ and $\log(1/r) \geq 1 - r$, this implies (34.9). The stated hypothetical lower bound would make its integral diverge logarithmically. $\square$

35 Uniform local disk bounds and boundary blowup compactness

The radial comparison is local in a useful sense: conditioning on the exterior of any disk produces a holomorphic tilt of the same disk gas. It therefore supplies estimates independent of both the location and the radius of that disk.

Proposition 35.1 (Domination in every interior disk). *Let $B(a, R) \subset \mathbb{D}$ be an open Euclidean disk, and let $0 < q < 1$. For every $\beta > 0$, every N , and every integer $n \geq 2$,*

$$\mathbb{P}\{\Xi_N(B(a, qR)) \geq n\} \leq \min \left\{ 1, \exp \left(n - \frac{\beta}{4} n(n-1) \log \frac{1}{q} \right) \right\}. \quad (35.1)$$

The same estimate holds conditionally on the exterior configuration and the number of particles in $B(a, R)$, for almost every conditioning configuration. It also holds with Ξ_N replaced by any subsequential limit Ξ . In particular, for every $t \geq 0$ and $s > 0$ there are finite constants depending only on β, q, t, s such that

$$\sup_{N, a, R} \mathbb{E} e^{t \Xi_N(B(a, qR))} < \infty, \quad \sup_{N, a, R} \mathbb{E} \Xi_N(B(a, qR))^s < \infty,$$

and the same uniform bounds hold for all subsequential limits.

Proof. Fix the disk and condition on its exterior configuration $y_1, \dots, y_{N-m}$ and the inside count m . The joint density is absolutely continuous, so almost surely no particle lies on the conditioning circle. Consequently $|y_j - a| > R$ for every exterior particle. After rescaling the inside coordinates as $z_i = a + Ru_i$, their conditional density on $\mathbb{D}^m$ is proportional to

$$|\Delta_m(u)|^\beta \prod_{i=1}^m \prod_{j=1}^{N-m} |a + Ru_i - y_j|^\beta.$$

Each function $u \mapsto a + Ru - y_j$ is nonzero on a neighborhood of $\overline{\mathbb{D}}$ and has a holomorphic logarithm there, possibly after shrinking the neighborhood. There are finitely many exterior particles, so a common neighborhood suffices. Hence the second factor is $\exp(2 \operatorname{Re} \sum_i f(u_i))$, where

$$f(u) = \frac{\beta}{2} \sum_{j=1}^{N-m} \log(a + Ru - y_j)$$

is holomorphic on that neighborhood. Subtracting $f(0)$ does not change the normalized conditional law. The radial comparison of Lemma 19.1 applies with constants independent of this holomorphic function and of m . The exponential-gap calculation leading to (1.2) therefore proves (35.1) conditionally. The case $m < n$ is automatic. Integrating the conditional inequality proves the unconditional one.

For a subsequential limit, the map $\eta \mapsto \eta(B(a, qR))$ is lower semicontinuous for the vague topology, because the disk is open and relatively compact in $\mathbb{D}$. Its integer-valued upper-level event $\{\eta(B(a, qR)) \geq n\}$ is open. The portmanteau inequality transfers the same upper bound to Ξ . Summation of the displayed Gaussian tail proves all the claimed moment bounds. This argument applies to every deterministic disk contained in $\mathbb{D}$, so its constants remain uniform in its center and radius. $\square$

Theorem 35.2 (Boundary blowup compactness and scaled repulsion). *Let Ξ be any subsequential limit at inverse temperature $\beta > 0$, and let*

$$\mathbb{H} = \{w \in \mathbb{C} : \operatorname{Re} w < 0\}.$$

For $0 < \delta < 1/2$ define the trimmed boundary blowup

$$\hat{\Xi}_\delta = \sum_{\substack{z \in \Xi \\ |z| < 1 - \delta^2}} \delta_{(z-1)/\delta}. \quad (35.2)$$

These are finite point processes on $\mathbb{H}$, and their laws are tight as $\delta \downarrow 0$. For every compact $K \Subset \mathbb{H}$ there are $C_K, c_K > 0$ such that

$$\sup_{0 < \delta < 1/2} \mathbb{P}\{\widehat{\Xi}_\delta(K) \geq n\} \leq C_K e^{-c_K n^2}, \quad n \geq 0. \quad (35.3)$$

In particular all local count moments and exponential moments are uniformly bounded.

Write $\alpha = \min\{1, \beta\}$ and let h_p be the normalized correlations of Ξ from Theorem 4.1. For each compact $K \Subset \mathbb{H}$ and each $p \geq 1$, there are $C_{K,p} < \infty$ and $\delta_{K,p} > 0$ such that, for $0 < \delta < \delta_{K,p}$, the functions

$$H_{p,\delta}(w) = \delta^{2p+\beta p(p-1)/2} h_p(1 + \delta w_1, \dots, 1 + \delta w_p)$$

are bounded by $C_{K,p}$ and have α -Hölder seminorm at most $C_{K,p}$ on K^p . In particular,

$$\delta^{2p} \rho_p(1 + \delta w_1, \dots, 1 + \delta w_p) \leq C_{K,p} |\Delta_p(w)|^\beta, \quad w \in K^p. \quad (35.4)$$

These are exactly the correlations of $\widehat{\Xi}_\delta$ on K^p for all sufficiently small δ .

Every boundary blowup subsequential limit is simple and has continuous correlation densities $\varrho_p = |\Delta_p|^\beta H_p$, where H_p is nonnegative, plurisubharmonic and locally α -Hölder. Along a process-convergent blowup subsequence, both normalized and ordinary correlations converge locally uniformly. Convergence also holds in total variation on every relatively compact window.

Proof. First record a minor issue concerning the ambient space. The literal rescaling of an infinite configuration in $\mathbb{D}$ need not be locally finite on all of $\mathbb{H}$: its particles may accumulate at other points of the rescaled circular boundary, which lie inside this half-plane. The trimming in (35.2) avoids this issue. It removes particles in a region that is eventually disjoint from every fixed compact subset of $\mathbb{H}$ after scaling. Indeed, for compact $K \Subset \mathbb{H}$ choose $d > 0$ and $L < \infty$ with $\operatorname{Re} w \leq -d$ and $|w| \leq L$ on K . Then

$$|1 + \delta w|^2 \leq 1 - 2d\delta + L^2\delta^2 < (1 - \delta^2)^2$$

for sufficiently small δ . Thus the trimmed and literal rescalings agree on K eventually. Equivalently, the theorem gives local convergence on compact subsets of the expanding domains $\{w : |1 + \delta w| < 1\}$ without choosing a trim.

Every compact subset of $\mathbb{H}$ is contained in a Euclidean disk whose closure lies in $\mathbb{H}$. To verify this, place the center at $-L_0$ on the real axis, choose $0 < \varepsilon < d$, and take radius $L_0 - \varepsilon$. The inequality

$$|w + L_0|^2 - (L_0 - \varepsilon)^2 = 2L_0(\operatorname{Re} w + \varepsilon) + |w|^2 - \varepsilon^2 < 0$$

holds on the compact set when L_0 is large. We may therefore choose concentric disks

$$K \subset B(c, r), \quad 0 < r < s < t < R, \quad \overline{B(c, R)} \subset \mathbb{H}.$$

For sufficiently small δ , the physical disk $B(1 + \delta c, \delta R)$ lies in $\mathbb{D}$, since its rescaled closure is a compact subset of $\mathbb{H}$. Proposition 35.1, with $q = r/R$, bounds the count in the physical image of K by a Gaussian tail independent of δ . Trimming only decreases counts. For the remaining $\delta \geq \delta_0 > 0$, all retained particles lie in $\{|z| < 1 - \delta_0^2\}$, whose count has a Gaussian tail by the original radial estimate. Combining the two bounds proves (35.3). Compact exhaustion and the usual point-measure tightness criterion now prove tightness on $\mathbb{H}$.

We prove the correlation bounds first at finite N . On the rescaled domain define

$$H_{p,N,\delta}(w) = \delta^{2p+\beta p(p-1)/2} h_{p,N}(1 + \delta w_1, \dots, 1 + \delta w_p).$$

These functions are nonnegative and plurisubharmonic. The insertion formula (4.6) also writes each of them as an integral $\int |F_y(w)|^\beta d\mu_{N,\delta}(y)$ with positive finite measure and holomorphic polynomial F_y . Their normalization is chosen so that

$$|\Delta_p(w)|^\beta H_{p,N,\delta}(w) = \delta^{2p} \rho_{p,N}(1 + \delta w_1, \dots, 1 + \delta w_p),$$

$$\int_{B(c,t)^p} |\Delta_p(w)|^\beta H_{p,N,\delta}(w) dA^p(w) = \mathbb{E}(\Xi_N(B(1 + \delta c, \delta t)))_p.$$

The right side is bounded uniformly in N and small δ by Proposition 35.1, applied inside the larger physical disk of radius δR .

Translate and rescale the separated-annular-average inequality of Lemma 4.2 from the unit disk to the fixed nested disks $B(c, s) \subset B(c, t)$. All change-of-variable constants depend only on these fixed disks and on p, β . It follows that $H_{p,N,\delta}$ is uniformly bounded on $B(c, s)^p$. Applying this argument on slightly larger intermediate disks as needed gives a bound on a neighborhood of that compact polydisk. For completeness, the subsequent Hölder estimate uses only this bound and the positive holomorphic-integral representation. Submean and Cauchy estimates give

$$\int \left(\sup_{B(c,r)^p} |F_y|^\beta + \sum_{j=1}^p \sup_{B(c,r)^p} |\partial_j F_y|^\beta \right) d\mu_{N,\delta}(y) \leq C_{K,p}.$$

One can use an intermediate radius between r and s for the suprema to control differences on the closed smaller polydisk. For $\beta \leq 1$, use $||u|^\beta - |v|^\beta| \leq |u - v|^\beta$; for $\beta > 1$, use $||u|^\beta - |v|^\beta| \leq \beta|u - v|(|u| + |v|)^{\beta-1}$ and Hölder's inequality. Integration then gives the asserted uniform exponent α , exactly as in Lemma 4.3. This proves all the bounds uniformly at finite N and at small scales.

Now hold δ fixed and take the original process-convergent subsequence $N_j \rightarrow \infty$. The locally uniform correlation convergence of Theorem 4.1, on the fixed physical compact disks, passes these same bounds to $H_{p,\delta}$ and proves (35.4). The initial observation about the trim identifies these densities with those of $\hat{\Xi}_\delta$ on every fixed compact window once δ is sufficiently small.

Finally let a sequence $\delta_j \downarrow 0$ have a process limit on $\mathbb{H}$. The uniform Hölder bounds give locally uniform subsequential limits of each H_{p,δ_j} . The factorial-measure partition identity (4.9) and the uniform compact count moments identify the resulting densities with the factorial moment measures of the limiting point process. This identifies each normalized density away from the diagonals, and continuity identifies it everywhere; hence convergence holds along the full process-convergent subsequence. Nonnegativity, plurisubharmonicity and the Hölder estimates persist under local uniform convergence. The second factorial density assigns zero mass to the diagonal, so the limiting process is simple. Finally the Janossy expansion and proof of Theorem 5.3 use only local uniform correlation convergence and exponential compact count bounds, both of which have just been established. They therefore give the stated local total variation convergence. $\square$

Lemma 35.3 (Deterministic logarithmic derivatives are incompatible with repulsion). *Let $U \subset \mathbb{R}^2$ be connected and open. Suppose a simple point process on U has all local count moments, a strictly positive smooth intensity λ , and a continuous second correlation density ρ_2 that vanishes on the diagonal. Assume its Ward identity is*

$$\mathbb{E} [\mathcal{L}_v F(\Xi) + F(\Xi) \Xi(\operatorname{div} v + v \cdot \nabla \log \lambda)] = 0 \quad (35.5)$$

for every compactly supported C^1 real vector field and every bounded smooth local cylinder function F . Then $\rho_2 \equiv 0$, so the process has at most one particle almost surely. In particular, no such process exists if $\int_U \lambda dA > 1$.

Proof. The local moment assumptions extend the identity to $F(\eta) = \eta(h)$ for every real $h \in C_c^1(U)$. Explicitly, approximate the identity function on $\mathbb{R}$ by smooth bounded functions χ_M that agree with it on $[-M, M]$, satisfy $|\chi_M(s)| \leq C|s|$ and $|\chi'_M(s)| \leq C$, and converge with their derivatives pointwise. Apply the assumed identity to $\chi_M(\Xi(h))$ and use domination by a fixed compact count for the derivative term and by the square of a compact count for the other term.

Put $q = \operatorname{div} v + v \cdot \nabla \log \lambda$. The one- and two-point formulas yield

$$0 = \int_U \nabla h(a) \cdot v(a) \lambda(a) dA(a) + \int_U h(a) q(a) \lambda(a) dA(a) + \iint_{U^2} h(w) q(a) \rho_2(a, w) dA(a) dA(w).$$

The first two integrals sum to $\int \operatorname{div}(hv\lambda) dA = 0$. As h and v are arbitrary, the remaining integral states distributionally that

$$\nabla_a \rho_2(a, w) = \rho_2(a, w) \nabla \log \lambda(a).$$

Interchanging the two coordinates gives the analogous identity in w . Dividing by the smooth positive function $\lambda(a)\lambda(w)$ shows that the continuous function $Q(a, w) = \rho_2(a, w)/(\lambda(a)\lambda(w))$ has all distributional first derivatives zero on U^2 . Such a function is constant on a connected open set: convolution makes it locally a smooth function with zero gradient, and overlapping balls identify the constants. Since U^2 is connected, Q is constant there. Its value on the diagonal is zero, hence $\rho_2 \equiv 0$.

Every relatively compact set K consequently satisfies $\mathbb{E}[\Xi(K)(\Xi(K) - 1)] = 0$, and a countable exhaustion gives $\Xi(U) \leq 1$ almost surely. Its mean is therefore at most one, in contradiction with $\int_U \lambda dA > 1$ when that condition holds. $\square$

36 The critical gas also converges to the empty process

Theorem 36.1 (Empty limit at the critical inverse temperature). *At $\beta = 4$, every subsequential limit of Ξ_N is almost surely empty. Consequently the full sequence converges to the empty configuration, vaguely in law and locally in total variation.*

Proof. Suppose instead that a subsequential limit Ξ is not almost surely empty, and write ρ for its intensity. Let b be its canonical complex score, and let $m(z) = \mathbb{E}[b(z, \Xi) \mid z]$ and $V(z) = \mathbb{E}[|b(z, \Xi) - m(z)|^2 \mid z]$ denote the conditional mean and variance under its Campbell measure. We use the conclusions already proved at $\beta = 4$:

$$(1 - r^2)^2 \rho(r) \longrightarrow c \in (0, 1/\pi], \quad (36.1)$$

$$h(r) := \log[(1 - r^2)^2 \rho(r)] \text{ is nondecreasing and } h(e^t) \text{ is convex,} \quad (36.2)$$

$$\int_{r_0 < |z| < 1} (1 - |z|) V(z) dA(z) < \infty \quad (r_0 > 0), \quad (36.3)$$

$$m(z) = \frac{2}{\beta} \partial \log \rho(z) \quad \text{almost everywhere where } \rho > 0. \quad (36.4)$$

The last identity is the one-point consequence of the canonical Ward identity. Rotation invariance and uniqueness of the canonical score give $b(e^{i\theta}z, (e^{i\theta})_*\eta) = e^{-i\theta}b(z, \eta)$ up to Campbell-null sets: the rotated expression satisfies the same Ward identity, which determines the score uniquely. Consequently $V(z) = V(|z|)$ almost everywhere.

Boundary rescaling. Set $H = \{w = x + iy : x > 0\}$ and, for $\varepsilon > 0$, put

$$T_\varepsilon(z) = \frac{1 - z}{\varepsilon}, \quad \eta_\varepsilon = \sum_{z \in \Xi: |z| \leq 1 - \varepsilon^2} \delta_{T_\varepsilon(z)}.$$

The trimmed configuration is finite almost surely and hence defines a point process on all of H . This trim is necessary: an untrimmed image could accumulate on its circular boundary inside H . Every fixed compact subset of H lies inside $T_\varepsilon(B_{1-\varepsilon^2})$ for sufficiently small ε , so trimming changes none of the local statistics used below. Theorem 35.2, with the reflected coordinate $w \mapsto -w$, gives tightness of η_ε on H , uniform moments of every compact count, and locally uniform correlation convergence along subsequences. Its subsequential limits are simple and have continuous second correlation density vanishing on the diagonal. Indeed these are exactly the estimates applied after this translation and dilation, with constants depending only on the chosen compact subset of H .

By (36.1), the rescaled intensities obey

$$\rho_\varepsilon(w) = \varepsilon^2 \rho(1 - \varepsilon w) \longrightarrow \lambda(w) := \frac{c}{4x^2} \quad \text{locally uniformly on } H. \quad (36.5)$$

Here $1 - |1 - \varepsilon w|^2 = 2\varepsilon x - \varepsilon^2 |w|^2$. In particular ρ_ε is uniformly bounded on each compact subset of H , and every rescaled limiting process has intensity λ .

The rescaled score becomes deterministic. Attach to each retained point the transformed original score and its spatial conditional mean:

$$b_\varepsilon(w) = -\varepsilon b(1 - \varepsilon w, \Xi), \quad m_\varepsilon(w) = -\varepsilon m(1 - \varepsilon w).$$

The marks b_ε are defined on the original probability space; they are not asserted to be functions of the trimmed configuration. The minus sign is the derivative of $z = 1 - \varepsilon w$. For a compact set $K \subseteq H$, take constants $a, B > 0$ such that $a \leq x \leq B$ and $|y| \leq B$ on K . For small ε , $r = |1 - \varepsilon w|$ lies between $1 - C_K \varepsilon$ and $1 - c_K \varepsilon$, and

$$\left| \frac{\partial r}{\partial x} \right| = \varepsilon \frac{1 - \varepsilon x}{r} \in [\varepsilon/2, 2\varepsilon].$$

Using rotation invariance of V , Campbell's formula, and the uniform intensity bound on K , gives

$$\begin{aligned} \mathbb{E} \sum_{w \in \eta_\varepsilon \cap K} |b_\varepsilon(w) - m_\varepsilon(w)|^2 &= \int_K \rho_\varepsilon(w) \varepsilon^2 V(|1 - \varepsilon w|) dA(w) \\ &\leq C_K \varepsilon \int_{1-C_K \varepsilon}^{1-c_K \varepsilon} V(r) dr \leq C'_K \int_{1-C_K \varepsilon}^{1-c_K \varepsilon} (1-r) V(r) dr \longrightarrow 0. \end{aligned} \quad (36.6)$$

The last step follows from (36.3); r is bounded away from zero on these annuli.

We next show, in essential supremum on every compact subset of H ,

$$m_\varepsilon(w) \longrightarrow m_\infty(w) := -\frac{2}{\beta x}. \quad (36.7)$$

Let $\ell = \lim_{r \uparrow 1} h(r) = \log c$. Convexity of $h(e^t)$ implies, almost everywhere and with $r_+ = (1+r)/2$,

$$0 \leq h'(r) \leq \frac{h(r_+) - h(r)}{r \log(r_+/r)} \leq \frac{C[\ell - h(r)]}{1-r}$$

on an outer annulus. Therefore $\varepsilon h'(|1 - \varepsilon w|) \rightarrow 0$ in essential supremum on K . The weak derivative chain rule and (36.4) give

$$m_\varepsilon(w) = -\frac{4\varepsilon \overline{(1 - \varepsilon w)}}{\beta(1 - |1 - \varepsilon w|^2)} - \frac{\varepsilon \overline{(1 - \varepsilon w)}}{\beta|1 - \varepsilon w|} h'(|1 - \varepsilon w|).$$

The first term tends uniformly to $-2/(\beta x)$ and the second tends to zero essentially uniformly, proving (36.7). Notice that $\beta \overline{m_\infty(w)}$, as a real vector, equals $\nabla \log \lambda(w) = (-2/x, 0)$.

Passage of the Ward identity and contradiction. Choose a convergent subsequence of the rescaled processes and call its limit η . For each compactly supported vector field v on H and bounded smooth local cylinder F , the canonical Ward identity transforms exactly into the following identity on the original probability space. For fixed local tests their spatial supports lie strictly inside the trimming disk for all sufficiently small ε , so the trim has no effect on their derivatives:

$$\mathbb{E} \left[\mathcal{L}_v F(\eta_\varepsilon) + F(\eta_\varepsilon) \eta_\varepsilon (\operatorname{div} v) + \beta F(\eta_\varepsilon) \operatorname{Re} \sum_{w \in \eta_\varepsilon} v(w) b_\varepsilon(w) \right] = 0$$

for small enough ε . Its force term may be replaced by the deterministic m_∞ . To see this for the centered remainder, apply Cauchy–Schwarz to the Campbell measure over a compact set containing $\operatorname{supp} v$:

$$\mathbb{E} \sum_{w \in \eta_\varepsilon \cap K} |b_\varepsilon - m_\varepsilon| \leq (\mathbb{E} \eta_\varepsilon(K))^{1/2} \left(\mathbb{E} \sum_{w \in \eta_\varepsilon \cap K} |b_\varepsilon - m_\varepsilon|^2 \right)^{1/2} \longrightarrow 0.$$

The difference between m_ε and m_∞ is handled by (36.7) and the bounded mean count. The remaining local statistics pass to the limit by uniform compact count moments. Hence η satisfies

$$\mathbb{E} [\mathcal{L}_v F(\eta) + F(\eta) \eta (\operatorname{div} v + v \cdot \nabla \log \lambda)] = 0.$$

The deterministic-score rigidity lemma applies: η has all local count moments, positive smooth intensity λ , and a continuous repulsive second correlation density. It follows that $\eta(H) \leq 1$ almost surely. This is impossible because $\int_H c/(4x^2) dA = \infty$; equivalently, a sufficiently long compact rectangle with $1 \leq x \leq 2$ already has expected count greater than one. The contradiction proves that no nonempty critical subsequential limit exists.

Finally tightness gives full vague convergence to the empty process. For every compact K , compact-count uniform integrability and convergence to zero imply $\mathbb{E} \Xi_N(K) \rightarrow 0$. Thus $\mathbb{P}\{\Xi_N(K) > 0\} \rightarrow 0$, which is exactly convergence in total variation of the restriction to the empty law. No quantitative rate at $\beta = 4$ is asserted. $\square$

Corollary 36.2 (Qualitative consequences at and above the endpoint). *For every fixed $\beta \geq 4$ and every integer $p \geq 1$,*

$$\rho_{p,N} \longrightarrow 0, \quad h_{p,N} \longrightarrow 0 \quad \text{locally uniformly on } \mathbb{D}^p.$$

For every relatively compact Borel set $A \subset \mathbb{D}$, every $s > 0$, and every real t ,

$$\mathbb{E} \Xi_N(A)^s \longrightarrow 0, \quad \mathbb{E} e^{t \Xi_N(A)} \longrightarrow 1.$$

For every $q > 1$, the boundary-weight statistic $U_{q,N} = \sum_i (1 - |z_i|)^q$ converges to zero in every positive moment, and $\mathbb{E} e^{t U_{q,N}} \rightarrow 1$ for every real t . No rate in these statements is asserted at $\beta = 4$.

Proof. The full process limit is empty by Theorem 36.1 and the theorem above 4. The locally uniform hierarchy convergence of Theorem 4.1 applies to this full convergent sequence; the empty process has zero factorial densities. The continuous normalized densities are zero as well, since they vanish away from the collision diagonals, a dense subset of $\mathbb{D}^p$.

Choose a compactly supported continuous function at least one on $\overline{A}$. Its statistic tends to zero in probability, so $\Xi_N(A) \rightarrow 0$ in probability. Uniform compact count moments of all

orders and uniform exponential moments at every real parameter give uniform integrability of each displayed random variable, proving the count assertions. For example, the exponential moment at $2t$ controls uniform integrability when $t > 0$; for $t \leq 0$ the integrand is bounded.

For $U_{q,N}$, the boundary truncation estimates and limiting-statistic theorem in Section 23 give convergence to the corresponding statistic of the empty process, namely zero. Their uniform exponential bounds give uniform integrability of every power and every fixed exponential, proving the remaining claims. $\square$

37 Critical emptiness through disk automorphisms

This proof uses the exact Bergman bounds from Theorem 32.2 and Corollary 32.3. The other inputs are the canonical logarithmic derivative, the critical residual-variance estimate, and the deterministic-score rigidity lemma proved earlier. All limiting processes in the argument live on the same open disk.

Theorem 37.1 (Uniqueness at the critical value). *At $\beta = 4$, the unscaled disk gas converges along the full sequence to the empty point process. On every compact window the convergence is in total variation.*

Proof. Let Ξ be any subsequential limit. We show that it is empty. Suppose otherwise, and write ρ for its one-point intensity. The already established critical curvature consequence says that

$$c(r) = (1 - r^2)^2 \rho(r) > 0, \quad g(t) = \log c(e^t), \quad -\infty < t < 0,$$

is nondecreasing and convex as a function of t . The Bergman bound gives $c(r) \leq 1/\pi$. Thus

$$c(r) \longrightarrow c_* \in (0, 1/\pi] \quad \text{as } r \uparrow 1. \quad (37.1)$$

Only radii sufficiently close to one are needed in the displays below; the same argument applies if positivity and the convexity assertion were initially formulated only on such an outer annulus.

Let $b(z, \eta)$ be the canonical complex logarithmic derivative from Corollary 33.5, and put $B(z, \eta) = 4b(z, \eta)$, regarded as a real two-dimensional vector. Disintegrate the Campbell measure over z and let $V(z) = \text{Var}(b(z, \Xi) \mid z)$. The canonical identity with constant cylinder test implies

$$\mathbb{E}[B(z, \Xi) \mid z] = \nabla \log \rho(z).$$

The residual-variance result gives, for some $r_0 < 1$,

$$\int_{r_0 < |z| < 1} (1 - |z|) V(z) dA(z) < \infty. \quad (37.2)$$

We may choose V radial up to area-null sets. Indeed rotations preserve the law, and transforming the canonical score under a rotation gives another score for the same law. Its uniqueness identifies the two versions up to Campbell-null sets, so their conditional variances agree. Since ρ is positive on the annulus, this is also an area-almost-everywhere assertion there.

For real $a \uparrow 1$, define

$$\varphi_a(w) = \frac{a + w}{1 + aw}, \quad \Xi_a = (\varphi_a^{-1})_* \Xi, \quad \varepsilon = 1 - a.$$

The transformed intensity is exactly

$$\rho_a(w) = |\varphi'_a(w)|^2 \rho(\varphi_a(w)) = \frac{c(|\varphi_a(w)|)}{(1 - |w|^2)^2} \longrightarrow \lambda(w) := \frac{c_*}{(1 - |w|^2)^2}, \quad (37.3)$$

locally uniformly. Here we used $1 - |\varphi_a(w)|^2 = (1 - a^2)(1 - |w|^2)/|1 + aw|^2$. The exact all-correlation Bergman bounds remain valid for Ξ_a . Consequently these processes are tight, all their compact count moments are uniformly bounded, and along a sequence $a \uparrow 1$ they converge to a process Υ with intensity λ . Their two-point bounds also give uniform repulsion of order four on compact sets.

We record why the limiting second correlation may be taken continuous, as required by the score-rigidity lemma. At finite N , each even-temperature correlation is the diagonal of a positive holomorphic kernel: replace the second copy of the marked coordinates in the squared-polynomial insertion integral by independent complex coordinates. Cauchy–Schwarz bounds that kernel at $(\mathbf{z}, \overline{\mathbf{w}})$ by $\sqrt{\rho_{p,N}(\mathbf{z})\rho_{p,N}(\mathbf{w})}$. The Bergman diagonal bounds make these kernels locally uniformly bounded. Montel’s theorem and Cauchy estimates therefore give locally convergent holomorphic kernels along subsequences, both in the original $N \rightarrow \infty$ limit and after the subsequent automorphisms. In the latter case the transformed kernel acquires the holomorphic factor $\prod_i \varphi'_a(z_i)\overline{\varphi'_a(w_i)}$. Thus the limiting correlations are real analytic on their diagonals, in particular continuous. Their integrals agree with the factorial moments of the limiting process by uniform compact count moments. The repulsion bound passes to this continuous version, giving $\rho_{2,\Upsilon}(w, w) = 0$. It also ensures simplicity of Υ .

The canonical real score for Ξ_a is

$$B_a(w, \eta) = D\varphi_a(w)^T B(\varphi_a(w), (\varphi_a)_*\eta) + \nabla \log |\varphi'_a(w)|^2. \quad (37.4)$$

This follows directly by changing variables in the Campbell Ward identity; the Jacobian contributes the second term. Its conditional mean is $d_a = \nabla \log \rho_a$, and conformality gives the centered variance

$$\mathbb{E}[|B_a - d_a|^2 \mid w] = 16|\varphi'_a(w)|^2 V(\varphi_a(w)).$$

For every compact $K \Subset \mathbb{D}$, elementary bounds on the displayed automorphism give positive constants c_K, C_K such that on K ,

$$c_K \varepsilon \leq 1 - |\varphi_a(w)| \leq C_K \varepsilon, \quad |\varphi'_a(w)| \leq C_K \varepsilon.$$

Moreover $\varphi_a(K)$ is contained in a sector of angular width $C_K \varepsilon$ about the point 1: indeed $\varphi_a(w) - 1 = (a - 1)(1 - w)/(1 + aw)$. Using the Bergman intensity bound and changing variables,

$$\begin{aligned} \mathbb{E} \sum_{w \in \Xi_a \cap K} |B_a(w, \Xi_a) - d_a(w)|^2 &= 16 \int_{\varphi_a(K)} |\varphi'_a(\varphi_a^{-1}(z))|^2 \rho(z) V(z) dA(z) \\ &\leq C_K \int_{\varphi_a(K)} V(z) dA(z) \\ &\leq C_K \varepsilon \int_{1-C_K \varepsilon}^{1-c_K \varepsilon} V(r) dr \longrightarrow 0. \end{aligned}$$

For the last convergence, r is bounded away from zero and $1 - r$ is comparable to ε on the integration interval; thus the last expression is bounded by a fixed constant times the integral of $(1 - r)V(r)r dr$ over that interval. It tends to zero by (37.2) and absolute continuity of the integral.

We also have locally uniform convergence of the mean scores,

$$d_a(w) \longrightarrow d(w) := \nabla \log \lambda(w) = \frac{4w}{1 - |w|^2}. \quad (37.5)$$

Here is an estimate that does not require interchanging a limit and a derivative. For a nondecreasing convex function g with a finite limit at zero, the convex secant inequality gives

$$0 \leq (-t)g'_+(t) \leq 2(g(t/2) - g(t)) \longrightarrow 0, \quad t \uparrow 0.$$

For w in a fixed compact set, $-\log |\varphi_a(w)|$ is comparable to ε , and $|\varphi'_a| = O_K(\varepsilon)$. The chain rule in $\log \rho_a = \log c(|\varphi_a|) - 2\log(1 - |w|^2)$ therefore shows that the gradient of the first term tends uniformly to zero. The possible one-sided formulation is harmless; the intensities are real analytic where positive by the kernel argument above, so the ordinary derivative is available on the annuli in use.

Let F be a bounded smooth local cylinder function and v a compactly supported C^1 vector field. The Ward identity for Ξ_a reads

$$\mathbb{E} \left[\mathcal{L}_v F(\Xi_a) + F(\Xi_a) \sum_{w \in \Xi_a} \{ \operatorname{div} v(w) + v(w) \cdot B_a(w, \Xi_a) \} \right] = 0.$$

Replacing B_a by d_a changes this by a quantity tending to zero: Campbell Cauchy–Schwarz bounds its absolute value by

$$\|F\|_\infty \left(\mathbb{E} \sum_{w \in \Xi_a} |v(w)|^2 \right)^{1/2} \left(\mathbb{E} \sum_{w \in \Xi_a \cap \operatorname{supp} v} |B_a - d_a|^2 \right)^{1/2}.$$

The first factor involving the process is uniformly bounded and the second tends to zero. Uniform compact count moments, vague convergence, and (37.5) then allow passage to the limit in every remaining term. We obtain

$$\mathbb{E} [\mathcal{L}_v F(\Upsilon) + F(\Upsilon) \Upsilon (\operatorname{div} v + v \cdot \nabla \log \lambda)] = 0.$$

All hypotheses of Lemma 35.3 now hold on the connected disk. Hence Υ has at most one point almost surely. This contradicts $\int_{\mathbb{D}} \lambda dA = \infty$, since $c_* > 0$. The assumed nonempty subsequential limit cannot exist.

Finally, tightness of the original sequence and uniqueness of its only possible subsequential limit imply full convergence to the empty process. Uniform compact count moments imply convergence of compact count means to zero, so the probability of a nonempty compact window tends to zero. This is precisely total variation convergence on that window. $\square$

38 Empty or infinite limits for $\beta \geq 2$

The local force identity can be restricted to an event prescribing the *total* number of limiting particles. This restriction is useful because total particle count is invariant under compactly supported motions, although it is not a continuous function in the vague topology.

Theorem 38.1 (Empty or infinite when $\beta \geq 2$). *Let $\beta \geq 2$, and let Ξ be any subsequential limit of the disk gas. Then*

$$\mathbb{P}\{\Xi(\mathbb{D}) \in \{0, \infty\}\} = 1.$$

Equivalently, every event $\{\Xi(\mathbb{D}) = m\}$ with an integer $m \geq 1$ has probability zero. This does not assert that the process is nonempty: a mixture of the empty configuration and infinite configurations is not excluded by the theorem.

Proof. Fix an integer $m \geq 1$ and write $E_m = \{\Xi(\mathbb{D}) = m\}$. We work on a joint limiting probability space carrying Ξ , its force marks, and the Gaussian modes from Theorem 34.1. It is enough to show that $p_m = \mathbb{P}(E_m) > 0$ is impossible when $\beta \geq 2$.

Local approximation of the total-count event. Choose smooth radial functions $\chi_\ell \in C_c^\infty(\mathbb{D}; [0, 1])$ increasing pointwise to one, with $\chi_\ell = 1$ on disks whose radii increase to one. Such a choice follows, for example, by applying a fixed nonincreasing smooth cutoff to $\ell(1 - |z|^2)$ and taking its complement. Choose $f \in C_c^\infty(\mathbb{R}; [0, 1])$ with $f(m) = 1$ and support in $(m - 1/2, m + 1/2)$, and set

$$H_\ell(\eta) = f\left(\int \chi_\ell d\eta\right).$$

Then H_ℓ is a bounded continuous cylinder function and

$$H_\ell(\eta) \longrightarrow \mathbf{1}_{\{\eta(\mathbb{D})=m\}}$$

for every locally finite configuration η , by monotone convergence of $\int \chi_\ell d\eta$ to its integer-valued or infinite total mass. For every fixed compactly supported vector field v , one has $\mathcal{L}_v H_\ell = 0$ for all sufficiently large ℓ , because χ_ℓ is identically one on a neighborhood of $\text{supp } v$.

A restricted force-energy identity. Let $0 \leq \phi \in C_c^\infty(\mathbb{D})$, and choose ℓ large enough that $\chi_\ell = 1$ on a neighborhood of $\text{supp } \phi$. At finite N ,

$$\beta^2 \mathbb{E} \left[H_\ell(\Xi_N) \sum_i \phi(z_i) |C_{i,N}|^2 \right] = \mathbb{E} \left[H_\ell(\Xi_N) \sum_i \Delta \phi(z_i) \right]. \quad (38.1)$$

Indeed, in the insertion variable a in this neighborhood,

$$H_\ell \left(\delta_a + \sum_j \delta_{y_j} \right) = f \left(1 + \sum_j \chi_\ell(y_j) \right),$$

which is independent of a and lies between zero and one. Multiplying the insertion measure by this nonnegative bounded factor preserves the polynomial distributional identity (33.4). Integrating that identity against ϕ proves (38.1), with exactly the same collision-integrability justification as in Proposition 33.1.

Pass along the joint convergent subsequence with ℓ fixed. Continuous compact mark cutoffs and the uniform compact count moments justify convergence of each bounded-mark term on the left. Monotone approximation of $|c|^2$ then gives a lower bound for its full limiting expectation. The right side converges by bounded continuity of H_ℓ and uniform integrability of compact particle counts. Thus

$$\beta^2 \mathbb{E} \left[H_\ell(\Xi) \sum_{z \in \Xi} \phi(z) |C_z|^2 \right] \leq \mathbb{E} \left[H_\ell(\Xi) \sum_{z \in \Xi} \Delta \phi(z) \right].$$

Let $\ell \rightarrow \infty$. Dominated convergence applies on the left by the unrestricted local force-energy bound and on the right by the local count bound. We obtain

$$\beta^2 \mathbb{E} \left[\mathbf{1}_{E_m} \sum_{z \in \Xi} \phi(z) |C_z|^2 \right] \leq \mathbb{E} \left[\mathbf{1}_{E_m} \sum_{z \in \Xi} \Delta \phi(z) \right]. \quad (38.2)$$

Define the restricted intensity u_m by

$$\mathbb{E} [\mathbf{1}_{E_m} \Xi(da)] = u_m(a) dA(a).$$

This density exists and satisfies $0 \leq u_m \leq \rho_1$ almost everywhere, because its intensity measure is dominated by the unrestricted one. In particular it is locally bounded, and

$$\int_{\mathbb{D}} u_m dA = mp_m < \infty. \quad (38.3)$$

The measure and its density are rotation invariant.

Restricted Ward identity and Gaussian force variance. Apply the unmarked Ward identity for the canonical score b to the cylinder function H_ℓ and a fixed compactly supported v . For all large ℓ its derivative term vanishes. The local Campbell L^2 bound for b and the count bound imply absolute integrability, so dominated convergence gives

$$\mathbb{E} \left[\mathbf{1}_{E_m} \sum_{z \in \Xi} (\operatorname{div} v(z) + \beta \operatorname{Re}(v(z)b(z, \Xi))) \right] = 0. \quad (38.4)$$

Disintegrate the Campbell measure restricted to E_m over its spatial marginal $u_m(a)dA(a)$. Denote the conditional mean of b by $\mathbf{b}_m(a)$, and set $J_m = u_m \mathbf{b}_m$. Formula (38.4) says

$$\partial_x u_m = \beta \operatorname{Re} J_m, \quad \partial_y u_m = -\beta \operatorname{Im} J_m$$

in the sense of distributions. Conditional Jensen and local boundedness of u_m show $J_m \in L^2_{\text{loc}}$; hence $u_m \in H^1_{\text{loc}}$. Moreover,

$$\int \phi \frac{|\nabla u_m|^2}{u_m} dA \leq \beta^2 \mathbb{E} \left[\mathbf{1}_{E_m} \sum_{z \in \Xi} \phi(z) |b(z, \Xi)|^2 \right], \quad (38.5)$$

where the quotient is assigned zero on $\{u_m = 0\}$. The gradient is zero almost everywhere on that zero set, as also follows directly from the displayed distributional derivative formulas.

The conditional regression (34.1) may be multiplied by $\mathbf{1}_{E_m}$, since this event is determined by Ξ . The Gaussian modes remain independent of the restricted unmarked Campbell measure. Conditional Jensen, followed by Gaussian integration of the square and then monotone convergence in the number of modes, therefore gives

$$\begin{aligned} & \mathbb{E} \left[\mathbf{1}_{E_m} \sum_{z \in \Xi} \phi(z) |C_z|^2 \right] \\ & \geq \mathbb{E} \left[\mathbf{1}_{E_m} \sum_{z \in \Xi} \phi(z) |b(z, \Xi)|^2 \right] + \frac{2}{\beta} \int \frac{\phi(a) u_m(a)}{(1 - |a|^2)^2} dA(a). \end{aligned}$$

Combining this bound with (38.2) and (38.5) proves the measure inequality

$$\Delta u_m \geq \frac{|\nabla u_m|^2}{u_m} dA + \frac{2\beta u_m}{(1 - |a|^2)^2} dA. \quad (38.6)$$

In particular Δu_m is a positive Radon measure.

Boundary growth contradicts finite mass. There is a continuous radial representative of u_m on every annulus compactly contained in the punctured disk. To see this without assuming regularity of the event E_m , write $U_m(t) = u_m(e^t)$, $t < 0$. Radial H^1 regularity implies $U_m \in H^1_{\text{loc}}((-\infty, 0))$, so U_m has a continuous representative. The positivity of Δu_m , tested against radial functions $\zeta(\log |a|)$, implies $U_m'' \geq 0$ distributionally. Thus U_m is convex. Since $u_m \leq \rho_1$ almost everywhere and ρ_1 is bounded near zero, U_m is bounded above as $t \rightarrow -\infty$.

Convexity then forces U_m to be nondecreasing: a negative supporting slope would contradict that upper bound. These representatives agree on intersecting annuli.

If $p_m > 0$, equation (38.3) implies u_m is not identically zero. By radial monotonicity it is strictly positive on some outer annulus $r_* < |a| < 1$. On compact subsets of this annulus it is continuous and bounded below by a positive constant. The weak logarithmic chain rule from Theorem 34.2 therefore applies to (38.6) and gives

$$\Delta \log u_m \geq \frac{2\beta}{(1 - |a|^2)^2} \quad \text{on } \{r_* < |a| < 1\}.$$

More explicitly, testing Δu_m against ϕ/u_m is justified by uniform and H^1 approximation on each such compact subset; the chain rule subtracts $\phi|\nabla u_m|^2/u_m^2$, which cancels the first term on the right of (38.6). Consequently

$$a \mapsto \log u_m(a) + \frac{\beta}{2} \log(1 - |a|^2)$$

is radial subharmonic on that annulus. Its expression as a function of $t = \log |a|$ is a finite convex function on $(\log r_*, 0)$. A supporting affine function at any interior point supplies a lower bound as $t \uparrow 0$. Exponentiating gives constants $c > 0$ and $r_0 < 1$ such that

$$u_m(r) \geq c(1 - r^2)^{-\beta/2}, \quad r_0 < r < 1.$$

But

$$2\pi \int_{r_0}^1 r(1 - r^2)^{-\beta/2} dr = \infty \quad \text{when } \beta \geq 2,$$

contradicting (38.3). Thus $p_m = 0$. Taking the union over all integers $m \geq 1$ proves the theorem. $\square$

39 Schur coordinates and an explicit reference process

This section gives a change of variables for the exact disk gas, an independently solvable reference measure, and the precise likelihood relating the two. The reference measure is the truncated circular beta ensemble of Killip and Kozhan [7]. It is not the disk gas when $\beta \neq 2$. The identities and convergence argument needed here are proved directly; no claim of bibliographic novelty is made for the reference limit.

39.1 The exact coefficient density

For a monic polynomial P of degree n , write $P^*(z) = z^n \overline{P(1/\bar{z})}$. Given $\alpha_0, \dots, \alpha_{N-1} \in \mathbb{D}$, define

$$P_0 = 1, \quad P_{j+1}(z) = zP_j(z) - \overline{\alpha_j}P_j^*(z). \quad (39.1)$$

Proposition 39.1 (Exact Schur density). *Recursion (39.1) is a real analytic bijection between $\mathbb{D}^N$ and the coefficient vectors of monic polynomials of degree N whose roots lie in $\mathbb{D}$. Its real Jacobian is*

$$J_N(\alpha) = \prod_{j=0}^{N-1} (1 - |\alpha_j|^2)^j. \quad (39.2)$$

Consequently, if $b = \beta/2$, the Schur parameters of the exact gas have joint density

$$\frac{1}{Q_{N,\beta}} |\text{Disc } P_N|^{b-1} \prod_{j=0}^{N-1} (1 - |\alpha_j|^2)^j \prod_{j=0}^{N-1} dA(\alpha_j). \quad (39.3)$$

Here $|\text{Disc } P_N| = |\Delta_N|^2$, and the density is understood almost everywhere. In particular, at $\beta = 2$ the parameters are independent, with respective densities $(j+1)(1 - |\alpha_j|^2)^j/\pi$.

Proof. If the roots of P_j lie in the disk, then on its boundary $|zP_j(z)| = |P_j^*(z)|$. Rouché's theorem proves that P_{j+1} has all $j+1$ roots in the disk. Conversely, the constant coefficient of a stable monic polynomial P_n determines $\alpha_{n-1} = -\overline{P_n(0)} \in \mathbb{D}$, and

$$P_{n-1}(z) = \frac{P_n(z) + \overline{\alpha_{n-1}}P_n^*(z)}{z(1 - |\alpha_{n-1}|^2)}.$$

The numerator has a zero at zero and, again by Rouché's theorem, exactly n zeros in the disk. The quotient is monic, of degree $n-1$, with all roots in the disk. This proves bijectivity and the stated regularity of the inverse.

Write $P_{n-1}(z) = z^{n-1} + a_1z^{n-2} + \cdots + a_{n-1}$. With $\alpha = \alpha_{n-1}$ fixed, its first $n-1$ new coefficients are $a_k - \overline{\alpha}a_{n-k}$, whereas the constant coefficient is $-\overline{\alpha}$. The real determinant of the first map is $(1 - |\alpha|^2)^{n-1}$: after a scalar unitary change of coordinates its antilinear involution has $n-1$ real eigenvalues 1 and $n-1$ real eigenvalues -1 . The constant-coefficient map has absolute Jacobian one, and block triangularity proves (39.2) by induction.

The map from labeled roots to coefficients has complex Jacobian equal, up to sign, to the Vandermonde and has multiplicity $N!$ off the collision set. Thus its real Jacobian is $|\Delta_N|^2$ and its pushforward density is $N!|\Delta_N|^{\beta-2}/Z_{N,\beta}$. Combining the two changes of variables gives (39.3). The integrability for $0 < \beta < 2$, including near the discriminant set, follows directly from this change of variables; it is not an additional assumption. The independent normalization at $\beta = 2$ follows from $\int_{\mathbb{D}} (1 - |a|^2)^j dA(a) = \pi/(j+1)$. $\square$

39.2 A Gram determinant is the entire change of measure

Put

$$R_N(z) = \prod_{i,j=1}^N (1 - z_i \bar{z}_j) > 0, \quad K(z, w) = \frac{1}{1 - z\bar{w}}.$$

Proposition 39.2 (Resultant and Gram identities). *For the polynomial in (39.1),*

$$R_N = \prod_{j=0}^{N-1} (1 - |\alpha_j|^2)^{j+1}. \quad (39.4)$$

For distinct points in the disk,

$$D_N(z) := \det[K(z_i, z_j)]_{i,j=1}^N = \frac{|\Delta_N(z)|^2}{R_N(z)} = \prod_i \frac{1}{1 - |z_i|^2} \prod_{i < j} \left| \frac{z_i - z_j}{1 - z_i \bar{z}_j} \right|^2. \quad (39.5)$$

Let $R_{N,b}$ be the law obtained from independent Schur parameters of densities

$$\frac{b(j+1)}{\pi} (1 - |a|^2)^{b(j+1)-1} dA(a), \quad 0 \leq j < N. \quad (39.6)$$

Its density on labeled roots is

$$\frac{b^N}{\pi^N} |\Delta_N(z)|^2 R_N(z)^{b-1} \prod_i dA(z_i). \quad (39.7)$$

The exact disk gas satisfies

$$\frac{dP_{N,2b}}{dR_{N,b}}(z) = \frac{D_N(z)^{b-1}}{\mathbb{E}_{R_{N,b}} D_N^{b-1}}, \quad \mathbb{E}_{R_{N,b}} D_N^{b-1} = \frac{b^N Z_{N,2b}}{\pi^N}. \quad (39.8)$$

Moreover, adjoining one point a changes the determinant by exactly

$$\frac{D_{N+1}(z_1, \dots, z_N, a)}{D_N(z_1, \dots, z_N)} = \frac{1}{1 - |a|^2} \prod_{j=1}^N \left| \frac{a - z_j}{1 - a \bar{z}_j} \right|^2. \quad (39.9)$$

Proof. Write $\text{Res}(A, B)$ for the polynomial resultant. Since P_N is monic, $\text{Res}(P_N, P_N^*) = \prod_i P_N^*(z_i) = R_N$. The reversed recursion gives $P_n^* = P_{n-1}^* - \alpha_{n-1} z P_{n-1}$, and hence

$$P_n^* = (1 - |\alpha_{n-1}|^2) P_{n-1}^* - \alpha_{n-1} P_n.$$

The elementary multiplicative and remainder properties of the resultant imply

$$\begin{aligned} \text{Res}(P_n, P_n^*) &= (1 - |\alpha_{n-1}|^2)^n \text{Res}(P_n, P_{n-1}^*) \\ &= (1 - |\alpha_{n-1}|^2)^n \text{Res}(z P_{n-1}, P_{n-1}^*) \\ &= (1 - |\alpha_{n-1}|^2)^n \text{Res}(P_{n-1}, P_{n-1}^*), \end{aligned}$$

since $P_{n-1}^*(0) = 1$. These identities follow either by evaluating products over roots or by polynomial continuity when degrees drop. Induction proves (39.4).

The Cauchy determinant identity gives $\det[(1 - x_i y_j)^{-1}] = \prod_{i < j} (x_i - x_j)(y_i - y_j) / \prod_{i,j} (1 - x_i y_j)$. For completeness, after multiplication by the denominator the numerator is alternating in each family, has degree at most $N - 1$ in each individual variable, and is therefore a constant multiple of the two Vandermondes. Comparing a successive residue at $x_N y_N = 1$ shows inductively that the constant is one. Taking $y_j = \bar{z}_j$ proves the first equality in (39.5); separating diagonal factors proves the second. Positivity also follows because K is a Gram kernel: $K(z, w) = \sum_{k \geq 0} z^k \bar{w}^k$.

Dividing the product density (39.6) by the Jacobian (39.2), and then multiplying by $|\Delta_N|^2 / N!$, gives (39.7). The difference of the exponents in this division is $(b - 1)(j + 1)$, so (39.4) applies. Division of the actual root density by (39.7) gives (39.8). Finally, cancel factors in (39.5) to obtain (39.9). $\square$

39.3 The reference process has the Bergman limit for every parameter

For P_n from the reference construction, the function $B_n = P_n / P_n^*$ is a finite Blaschke product, and its zeros are exactly the roots of P_n . It satisfies

$$B_0 = 1, \quad B_{n+1}(z) = \frac{z B_n(z) - \bar{\alpha}_n}{1 - \alpha_n z B_n(z)}. \quad (39.10)$$

Theorem 39.3 (Solvable reference limit). *Fix $b > 0$, and use the independent parameters (39.6). In the topology of uniform convergence on compact subsets of $\mathbb{D}$,*

$$\sqrt{N} B_N(z) \implies -\frac{1}{\sqrt{b}} \sum_{\ell=0}^{\infty} \xi_{\ell} z^{\ell}, \quad (39.11)$$

where the ξ_{ℓ} are independent standard circular complex Gaussians. Consequently the unscaled zero process of $R_{N,b}$ converges to the Bergman process for every $b > 0$. The identification of the limiting zero process follows by comparison with $b = 1$, where the reference is exactly the $\beta = 2$ disk gas.

Proof. Direct integration of (39.6) gives

$$\mathbb{E} |\alpha_n|^2 = \frac{1}{b(n+1) + 1}, \quad \mathbb{E} |\alpha_n| \leq C_b (n+1)^{-1/2}. \quad (39.12)$$

If $w \in \mathbb{D}$ and a has uniform argument conditional on $|a| = s$, a geometric-series integration shows that

$$\mathbb{E} \left[\left| \frac{w - \bar{a}}{1 - a w} \right|^2 \middle| |a| = s \right] = |w|^2 + \frac{s^2(1 - |w|^2)^2}{1 - s^2|w|^2} \leq |w|^2 + s^2.$$

Independence of α_n from the preceding parameters yields, for $|z| \leq r < 1$,

$$\mathbb{E}|B_{n+1}(z)|^2 \leq r^2 \mathbb{E}|B_n(z)|^2 + \frac{C_b}{n+1}.$$

Iteration, splitting the resulting geometric sum at $n/2$, gives

$$\sup_{|z| \leq r} \mathbb{E}|B_n(z)|^2 \leq \frac{C_{b,r}}{n+1}. \quad (39.13)$$

If $r < R < 1$, Cauchy's formula and Cauchy-Schwarz imply

$$\sup_{|z| \leq r} |B_n(z)|^2 \leq C_{r,R} \int_0^{2\pi} |B_n(Re^{i\theta})|^2 \frac{d\theta}{2\pi}.$$

Thus (39.13) holds with the supremum inside the expectation as well, after changing its constant.

Write the nonlinear remainder as

$$E_n(z) := B_{n+1}(z) - zB_n(z) + \overline{\alpha_n} = \frac{\alpha_n z B_n(z)(zB_n(z) - \overline{\alpha_n})}{1 - \alpha_n z B_n(z)}.$$

The denominator has modulus at least $1 - r$ on $|z| \leq r$. Since $|B_n| \leq 1$, independence, (39.12), and the supremum version of (39.13) give

$$\mathbb{E} \sup_{|z| \leq r} |E_n(z)| \leq C_{b,r} (n+1)^{-3/2}. \quad (39.14)$$

Iterating the linearized recursion gives the exact identity

$$B_N(z) = z^N - \sum_{\ell=0}^{N-1} z^\ell \overline{\alpha_{N-1-\ell}} + \sum_{n=0}^{N-1} z^{N-1-n} E_n(z).$$

By (39.14), the expectation of the supremum of the last sum, multiplied by $\sqrt{N}$, tends to zero on every compact disk: for $n \geq N/2$ its bound is $C_{b,r}/N$, and for $n < N/2$ exponential decay of r^{N-1-n} dominates its bounded sum. The term $\sqrt{N}z^N$ tends to zero as well.

For every fixed L , the independent variables $\sqrt{N}\alpha_{N-1}, \dots, \sqrt{N}\alpha_{N-L}$ converge jointly to independent complex Gaussians of variance $1/b$. This follows immediately from their rescaled densities

$$\frac{b(N-\ell)}{\pi N} \left(1 - \frac{|u|^2}{N}\right)^{b(N-\ell)-1} \mathbf{1}_{\{|u| < \sqrt{N}\}}, \quad 0 \leq \ell < L.$$

The remaining series tails vanish uniformly in the required order. Indeed, independence and zero means give for $|z| = R < 1$

$$\mathbb{E} \left| \sqrt{N} \sum_{\ell=L}^{N-1} z^\ell \overline{\alpha_{N-1-\ell}} \right|^2 = N \sum_{\ell=L}^{N-1} \frac{R^{2\ell}}{b(N-\ell)+1}.$$

The part $\ell \leq N/2$ is at most $C_{b,R} R^{2L}$, while the other part is at most $C_b N R^N \log(N+1)$ and tends to zero. A second application of Cauchy's formula gives the same tail control for the supremum on $|z| \leq r < R$. Finite-dimensional convergence followed by this tail estimate proves (39.11).

The limiting analytic function is almost surely not identically zero. For analytic functions converging locally uniformly to a nonzero analytic function, the zero counting measures

converge vaguely: on any small disk whose boundary avoids the limit zeros, Rouché's theorem preserves the number of zeros, with multiplicity; finitely many such disks and uniform approximation of compactly supported continuous tests prove the assertion. The continuous mapping theorem therefore proves convergence of the zero processes. The scalar $-1/\sqrt{b}$ does not alter the limiting zero set. At $b = 1$, (39.7) is the exact $\beta = 2$ gas, whose Bergman limit was proved earlier. Therefore the zero set of the limiting analytic function, and hence the reference limit at every $b > 0$, is the Bergman process. $\square$

39.4 Why direct consistency fails for the exact gas

Proposition 39.4 (Failure of forward Schur consistency). *For every $\beta > 0$ with $\beta \neq 2$, the marginal law of α_0 under the two-particle exact gas is not uniform on $\mathbb{D}$. Consequently the exact finite-dimensional Schur laws (39.3) are not the initial-coordinate marginals of one probability measure on $\mathbb{D}^{\mathbb{N}}$.*

Proof. At $N = 1$, the parameter α_0 is uniform. At $N = 2$, put $b = \beta/2$. The polynomial and its discriminant are

$$P_2(z) = z^2 + (\overline{\alpha_1}\alpha_0 - \overline{\alpha_0})z - \overline{\alpha_1}, \quad \text{Disc } P_2 = (\overline{\alpha_1}\alpha_0 - \overline{\alpha_0})^2 + 4\overline{\alpha_1}.$$

Up to a common normalizing constant the marginal density at $\alpha_0 = 0$ is

$$I_0 = 4^{b-1} \int_{\mathbb{D}} (1 - |a|^2) |a|^{b-1} dA(a) = \frac{\pi 4^b}{(b+1)(b+3)}. \quad (39.15)$$

Its radial boundary limit is

$$I_1 = \int_{\mathbb{D}} (1 - |a|^2) |1 + a|^{2b-2} dA(a) = \frac{\pi \Gamma(2b+1)}{\Gamma(b+2)^2}. \quad (39.16)$$

Here is a direct evaluation of the last integral. The binomial series and angular orthogonality give

$$\frac{I_1}{\pi} = \sum_{k \geq 0} \frac{((1-b)_k)^2}{(k!)^2 (k+1)(k+2)} = \frac{1}{2} {}_2F_1(1-b, 1-b; 3; 1).$$

For $0 < b < 1$, Euler's beta integral, first at an argument below one and then by monotone convergence of this nonnegative series, evaluates the last expression as $\Gamma(2b+1)/\Gamma(b+2)^2$. Both the integral and the gamma ratio are holomorphic in b on $\Re b > 0$ (use domination on compact subsets near the single boundary point $a = -1$), so the identity theorem proves the formula for all real $b > 0$.

We justify the boundary limit used in (39.16). It suffices by rotation to take $\alpha_0 = t \uparrow 1$. Write $s = (1 - \sqrt{1-t^2})/(1 + \sqrt{1-t^2})$. Its discriminant factors as

$$t^2(a-1)^2 + 4a = \frac{4}{(1+s)^2} (a+s)(1+sa).$$

For $b \geq 1$ bounded convergence applies. If $0 < b < 1$, set $p = 1 - b < 1$. Since $|a+s| \leq |1+sa|$ for $|a| < 1$,

$$|(a+s)(1+sa)|^{-p} \leq |a+s|^{-2p}.$$

For some $q > 1$ one has $2pq < 2$; hence these integrands have uniformly bounded $L^q(\mathbb{D})$ norms as $s \uparrow 1$. Uniform integrability and pointwise convergence prove the boundary limit. The same argument with an exponent strictly below two gives continuity of the marginal near $\alpha_0 = 0$. Thus differing values I_0, I_1 really imply a nonconstant density, rather than only a difference of versions on null sets.

The gamma duplication identity reduces their ratio to

$$\frac{I_0}{I_1} = \frac{\sqrt{\pi}(b+1)\Gamma(b+1)}{(b+3)\Gamma(b+1/2)}.$$

This ratio is strictly increasing for $b > 0$: its logarithmic derivative is

$$\frac{1}{b+1} - \frac{1}{b+3} + \psi(b+1) - \psi(b+1/2) > 0,$$

where ψ is the logarithmic derivative of Γ and is strictly increasing because $\psi'(x) = \sum_{k=0}^{\infty} (x+k)^{-2} > 0$ for $x > 0$. At $b = 1$ the ratio is one, so it differs from one for every $b \neq 1$. This proves the proposition. $\square$

Remark 39.5 (What this representation does and does not settle). The Gram likelihood (39.8) is exact. It gives an explicit reference construction with a proved full-sequence limit, and (39.9) isolates the interaction created by an additional interior point. Its normalized likelihood is not shown here to have a limit, or to be uniformly integrable jointly with the zero process. Those estimates cannot be omitted: without them, Theorem 39.3 says nothing about convergence of the exact disk gas at $\beta \neq 2$. Proposition 39.4 rules out direct forward consistency; it does not rule out convergence of terminal Schur coordinates, nor a successful reverse-recursion argument.

39.5 Only logarithmically many terminal parameters are needed locally

Proposition 39.6 (Uniform reverse contraction). *Consider two recursions of the form (39.10) which use the same last $L \geq 1$ parameters but may have arbitrary initial values in the closed unit disk. Their terminal functions B and $\tilde{B}$ satisfy*

$$\sup_{|z| \leq r} |B(z) - \tilde{B}(z)| \leq 2 \operatorname{arctanh}(r) r^{L-1}, \quad 0 < r < 1. \quad (39.17)$$

Proof. Use the Poincaré distance whose infinitesimal metric is $2|dw|/(1-|w|^2)$. The map $w \mapsto (w - \bar{\alpha})/(1 - \alpha w)$ is an isometry. Multiplication by z , with $|z| \leq r$, has Lipschitz constant at most r in this metric, because its infinitesimal norm is

$$\frac{|z|(1-|w|^2)}{1-|zw|^2} \leq |z| \leq r.$$

After the first shared step the distance is at most $4 \operatorname{arctanh}(r)$, since before its isometry both values lie in the closed disk of radius r . Each remaining step contracts that distance by at most r . Poincaré distance is at least twice Euclidean distance, giving (39.17). Boundary initial values cause no difficulty after multiplication by z ; alternatively one can take a limit from inside the disk. $\square$

Theorem 39.7 (Logarithmic terminal representation of the exact gas). *Fix $\beta > 0$ and $0 < r < 1$. There is a constant $C_{\beta,r} < \infty$ with the following property. Obtain the Schur parameters of the exact N -particle gas by Proposition 39.1, and put*

$$L_N = \min\{N, \lceil C_{\beta,r} \log(N+1) \rceil\}.$$

Restart recursion (39.10) with value zero, using only $\alpha_{N-L_N}, \dots, \alpha_{N-1}$, and let $\hat{\Xi}_N$ be the zero counting measure of its terminal rational function in $\mathbb{D}$. Then, for every continuous function f with compact support in B_r ,

$$\int f d\Xi_N - \int f d\hat{\Xi}_N \longrightarrow 0 \quad \text{in probability.} \quad (39.18)$$

Thus the interior process is approximated, in its natural topology, by a function of $O(\log N)$ terminal Schur parameters.

Proof. We first record the consequence of the radial comparison already proved for the exact gas. If $H_N = \sum_{i=1}^N -\log |z_i|$, it is stochastically bounded above by $Y_N = \sum_{j=1}^N jE_j$, where the E_j are independent exponentials of rates $a_j = 2j + \beta j(j-1)/2$. Hence

$$\mathbb{E}Y_N = \frac{2}{\beta} \log N + O_\beta(1), \quad \text{Var } Y_N = \sum_{j=1}^N \frac{j^2}{a_j^2} \leq C_\beta.$$

For every $A > 2/\beta$ it follows that

$$\mathbb{P}\{H_N > A \log N\} \longrightarrow 0. \quad (39.19)$$

No coupling between Schur parameters and the dominating exponentials is required for this probabilistic assertion.

Choose $r < R < R_1 < 1$. For $|z| \leq R$ and $|w| > R_1$, the disk Green function obeys

$$-\log \left| \frac{z-w}{1-z\bar{w}} \right| \leq C_{R,R_1} (-\log |w|). \quad (39.20)$$

Indeed, the pseudohyperbolic distance in this expression is bounded below by $d = (R_1 - R)/(1 + R) > 0$, while

$$1 - \left| \frac{z-w}{1-z\bar{w}} \right|^2 = \frac{(1-|z|^2)(1-|w|^2)}{|1-z\bar{w}|^2} \leq \frac{2(1-|w|)}{(1-R)^2}.$$

The inequality $-\log x \leq (1-x^2)/(2d^2)$ for $d \leq x \leq 1$ proves (39.20). For an interior root with $|w| \leq R_1$ and $|z-w| \geq \delta$, the same Green function is at most $\log(2/\delta)$. Since B_N is the finite Blaschke product of the roots, this gives on any such contour

$$|B_N(z)| \geq \exp\{-C_{R,R_1} H_N - M_N(R_1) \log(2/\delta)\}. \quad (39.21)$$

Here are explicit contours sufficient to compare point measures. Fix a mesh $\eta > 0$ small enough that every square of side η intersecting B_r is contained in B_R . Put $M = M_N(R_1)$ and $\delta = \eta/[8(M+1)]$. Choose the horizontal and vertical shifts of the square grid so that every grid line has distance at least δ from the corresponding coordinate of each of these M roots. Such shifts exist: in either shift interval of length η , the forbidden intervals have total length at most $2M\delta < \eta/4$. Consequently every boundary point of the grid squares under consideration has distance at least δ from every root in B_{R_1} .

Compact-count tightness implies $M_N(R_1) \leq \sqrt{\log N}$ with probability tending to one. For this fixed mesh, therefore,

$$M_N(R_1) \log \frac{16(M_N(R_1) + 1)}{\eta} = o(\log N)$$

on events whose probabilities tend to one. Combining with (39.19) and (39.21), the modulus on all the chosen square boundaries is at least $N^{-C_{R,R_1} A - o(1)}$ with probability tending to one. On the other hand, Proposition 39.6 bounds the difference between B_N and the restarted function by $2 \operatorname{arctanh}(R) R^{L_N-1}$ on B_R . Choose $C_{\beta,r}$ so large that

$$C_{\beta,r} |\log R| > C_{R,R_1} A + 2.$$

This choice is independent of the mesh. Rouché's theorem then shows, with probability tending to one, that the original and restarted functions have the same number of zeros

in each selected square. Both functions are analytic in the disk; the restarted one is not identically zero almost surely, since its value at zero is $-\overline{\alpha_{N-1}}$ and the original product of root moduli is almost surely positive.

On this event the zeros can be paired within individual squares, including multiplicities. If ω_f is the modulus of continuity of f extended by zero, the difference of its two integrals is at most $\omega_f(\sqrt{2}\eta)M_N(R)$. The latter counts are tight uniformly in N . First take $N \rightarrow \infty$ at fixed mesh, then let $\eta \downarrow 0$. This proves (39.18). $\square$

39.6 Failure of uniform integrability above the empty-limit threshold

Corollary 39.8 (Asymptotic singularity of the Gram change of measure). *For $b \geq 2$, write*

$$W_N = \frac{D_N^{b-1}}{\mathbb{E}_{R_{N,b}} D_N^{b-1}}.$$

Then

$$\|P_{N,2b} - R_{N,b}\|_{\text{TV}} \rightarrow 1, \quad W_N \rightarrow 0 \quad \text{in } R_{N,b}\text{-probability.} \quad (39.22)$$

In particular, the mean-one likelihoods W_N are not uniformly integrable.

Proof. Let A_R be the event of no points in B_R . Since $W_N dR_{N,b} = dP_{N,2b}$,

$$\mathbb{E}_{R_{N,b}} \min(W_N, 1) \leq R_{N,b}(A_R) + P_{N,2b}(A_R^c).$$

For fixed $R < 1$, the second term tends to zero by the empty-limit theorem for $2b \geq 4$, including Theorem 36.1. The first tends to the Bergman hole probability, by Theorem 39.3; the circle $|z| = R$ has zero probability of containing a Bergman point, so the count convergence indeed implies this hole convergence. These Bergman hole probabilities tend to zero as $R \uparrow 1$, since that process has infinitely many points almost surely. Thus $\mathbb{E} \min(W_N, 1) \rightarrow 0$. For a likelihood of mean one,

$$1 - \|P_{N,2b} - R_{N,b}\|_{\text{TV}} = \mathbb{E}_{R_{N,b}} \min(W_N, 1),$$

where total variation is the supremum over measurable events. Also, for every $\varepsilon > 0$,

$$R_{N,b}\{W_N > \varepsilon\} \leq \frac{\mathbb{E} \min(W_N, 1)}{\min(\varepsilon, 1)} \rightarrow 0.$$

Uniform integrability would then force $\mathbb{E} W_N \rightarrow 0$, contrary to its constant value one. $\square$

40 A lowest-weight decomposition at the critical parameter

We give an exact representation of the one-particle radial distribution at $\beta = 4$. It separates a simple, explicit representation-theoretic kernel from a probability distribution on primary sectors. The latter distribution contains the finer asymptotic information that the explicit kernel alone does not determine.

Use normalized area $d\mu = dA/\pi$ and the polynomial Bergman inner product on $\mathbb{D}^n$. On polynomials define

$$D_n = \sum_{i=1}^n \partial_{w_i}, \quad L_n = \sum_{i=1}^n (w_i^2 \partial_{w_i} + 2w_i), \quad H_n = \sum_{i=1}^n w_i \partial_{w_i} + n.$$

All adjoint statements below concern polynomials, so no assertion about domains of closed unbounded operators is needed. Monomial orthogonality and $\|w^j\|^2 = (j+1)^{-1}$ give

$$D_n^* = L_n, \quad [D_n, L_n] = 2H_n, \quad [H_n, L_n] = L_n. \quad (40.1)$$

For example, in one variable, $\langle \partial w^{j+1}, w^j \rangle = 1 = \langle w^{j+1}, (w^2 \partial + 2w)w^j \rangle$; all other monomial matrix elements follow from orthogonality. The commutators follow by differentiation.

Let $\mathcal{I}_n$ be the space of symmetric polynomials divisible by Δ_n^2 , with $\Delta_1 = 1$. Write $\mathcal{I}_{n,d}$ for its homogeneous part of degree d and

$$\mathcal{P}_{n,d} = \mathcal{I}_{n,d} \cap \ker D_n.$$

Translation invariance of Δ_n^2 gives $D_n \Delta_n^2 = 0$. The pairwise derivative identity

$$\sum_i w_i^2 \partial_i \Delta_n^2 = 2(n-1) \left(\sum_i w_i \right) \Delta_n^2$$

shows that D_n, L_n, H_n all preserve $\mathcal{I}_n$. In particular, $\mathcal{I}_{n,d} = \{0\}$ for $d < n(n-1)$.

Lemma 40.1 (Orthogonal decomposition into descendants). *For every degree d ,*

$$\mathcal{I}_{n,d} = \bigoplus_{\substack{\ell \leq d \\ \ell \geq n(n-1)}}^\perp L_n^{d-\ell} \mathcal{P}_{n,\ell}. \quad (40.2)$$

If $v \in \mathcal{P}_{n,\ell}$, $\kappa = n + \ell$, and $K \geq 0$, then

$$\frac{\|D_n^k L_n^K v\|^2}{\|L_n^K v\|^2} = (K)_{\underline{k}} (2\kappa + K - 1)_{\underline{k}}, \quad 0 \leq k \leq K, \quad (40.3)$$

where $(a)_{\underline{k}} = a(a-1) \cdots (a-k+1)$; the numerator is zero for $k > K$.

Proof. The commutators give, by induction,

$$D_n L_n^j v = j(2\kappa + j - 1) L_n^{j-1} v, \quad \|L_n^j v\|^2 = j!(2\kappa)_j \|v\|^2,$$

where the last Pochhammer symbol is rising. Iterating the first identity and dividing the second proves the norm formula.

In each finite-dimensional homogeneous space, the orthogonal complement of $L_n \mathcal{I}_{n,d-1}$ is exactly $\mathcal{P}_{n,d}$, by the adjoint identity and preservation of the ideal. This proves that the descendants span, by induction on degree. They are mutually orthogonal: for primitive vectors u, v of different degrees, transfer powers of L_n across the inner product using its adjoint D_n . The lowering formula reduces the inner product to a scalar multiple of an inner product between a primitive vector and a positive descendant. Transferring one remaining raising operator kills that primitive vector. Within a fixed primitive degree, the displayed norm identity preserves orthogonality of primitive vectors. This proves the orthogonal direct sum. $\square$

Fix $N \geq 2$, set $n = N - 1$ and $d = N(N - 1)$, and expand

$$F_N(z, w) := \Delta_N(z, w)^2 = \sum_{k=0}^{2N-2} z^k f_{k,N}(w).$$

Then

$$f_{0,N}(w) = \left(\prod_{i=1}^n w_i^2 \right) \Delta_n(w)^2, \quad f_{k,N} = \frac{(-D_n)^k}{k!} f_{0,N}. \quad (40.4)$$

The second assertion follows by equating coefficients in $(\partial_z + D_n)F_N = 0$. By the lemma there are unique orthogonal vectors

$$f_{0,N} = \sum_{K=0}^{2N-2} h_{K,N}, \quad h_{K,N} \in L_n^K \mathcal{P}_{n,d-K}.$$

Set $b_{K,N} = \|h_{K,N}\|^2$, allowing zero sectors, and put

$$T_N = 2N^2 - 2, \quad A_N(K) = \frac{1}{K+1} \binom{T_N}{K}, \quad p_N(K) = \frac{b_{K,N} A_N(K)}{\|F_N\|^2}. \quad (40.5)$$

Theorem 40.2 (An explicit hypergeometric kernel for the radial law). *The numbers $p_N(K)$ are a probability distribution on $\{0, \dots, 2N-2\}$. Define*

$$a_{k,N} = \frac{\|f_{k,N}\|^2}{(k+1)\|F_N\|^2}. \quad (40.6)$$

These numbers also sum to one and satisfy

$$a_{k,N} = \sum_{K=k}^{2N-2} p_N(K) \frac{\binom{K+1}{k+1} \binom{T_N-K-1}{k}}{\binom{T_N}{K}}. \quad (40.7)$$

Equivalently, first sample K from p_N ; then sample J from the hypergeometric distribution obtained by drawing K objects without replacement from a population of T_N , of which $K+1$ are marked; finally put $k = K - J$. The one-point intensity with respect to ordinary area is

$$\rho_{1,N}(z) = \frac{N}{\pi} \sum_{k=0}^{2N-2} (k+1) a_{k,N} |z|^{2k}. \quad (40.8)$$

In particular,

$$0 \leq \pi \rho_{1,N}(0) - N p_N(0) \leq \frac{N}{N^2 - 1}. \quad (40.9)$$

Thus $\rho_{1,N}(0) \rightarrow 0$ if and only if $N p_N(0) \rightarrow 0$. More generally, for every fixed nonnegative integer k ,

$$N a_{k,N} = N p_N(k) + O_k(N^{-1}) \quad (N \rightarrow \infty). \quad (40.10)$$

Proof. In the sector indexed by K , the primitive weight is $\kappa = n + (d - K) = N^2 - 1 - K$. The lowering norm formula, together with orthogonality after lowering and (40.4), gives

$$\|F_N\|^2 = \sum_K b_{K,N} \sum_{k=0}^K \frac{(K)_{\underline{k}} (T_N - K - 1)_{\underline{k}}}{k!^2 (k+1)}.$$

The inner summand equals

$$\frac{1}{K+1} \binom{K+1}{k+1} \binom{T_N - K - 1}{k}.$$

Substitute $j = K - k$ and use the elementary binomial convolution: selecting K objects from disjoint sets of sizes $K+1$ and $T_N - K - 1$ gives

$$\sum_{k=0}^K \binom{K+1}{k+1} \binom{T_N - K - 1}{k} = \binom{T_N}{K}.$$

This proves both the normalization $A_N(K)$ and the probability formula. The same counting interpretation gives the stated hypergeometric law.

The coefficient functions $f_{k,N}$ are homogeneous of different degrees. Their inner products vanish under simultaneous rotation of the w_i . Integrating $|F_N(z, w)|^2$ in w , and remembering that normalized area is dA/π , therefore proves (40.8).

At $k = 0$, the conditional probability is $(K + 1)/\binom{T_N}{K}$. It equals one for $K = 0$. For $1 \leq K \leq 2N - 2$, it is at most $2/T_N$: the ratio of its values at $K + 1$ and K is $(K + 2)/(T_N - K)$, which is at most one in the relevant range (including $N = 2$). Summing against p_N proves (40.9).

To check the last statement, write the conditional kernel as $w_N(k, K)$. For fixed k ,

$$w_N(k, k) = \frac{\binom{T_N - k - 1}{k}}{\binom{T_N}{k}} = 1 + O_k(T_N^{-1}).$$

For $K \geq k$, direct cancellation of factorials gives

$$\frac{w_N(k, K + 1)}{w_N(k, K)} = \frac{K + 2}{K + 1 - k} \frac{T_N - K - 1 - k}{T_N - K - 1} \frac{K + 1}{T_N - K}.$$

Uniformly over $k \leq K \leq 2N - 3$, this is $O_k(N^{-1})$. Thus, for all sufficiently large N , the kernel is decreasing for $K \geq k + 1$, and its largest value in that range is

$$w_N(k, k + 1) = (k + 2) \frac{\binom{T_N - k - 2}{k}}{\binom{T_N}{k + 1}} = O_k(T_N^{-1}).$$

The mixture formula now gives $|a_{k,N} - p_N(k)| = O_k(T_N^{-1})$, which proves (40.10). $\square$

Corollary 40.3 (An exact primary-sector criterion for critical emptiness). *For the unweighted disk gas at $\beta = 4$, the following are equivalent:*

1. Ξ_N converges to the empty point process on $\mathbb{D}$;
2. for every fixed nonnegative integer k , $Np_N(k) \rightarrow 0$.

When these conditions hold, convergence on each compact window is in total variation. The criterion uses the previously proved uniform compact count bounds, but does not presume uniqueness of a nonempty limit.

Proof. Write $n_{k,N} = Na_{k,N}$. Integrating (40.8) gives the nonnegative series

$$\mathbb{E}\Xi_N(B_r) = \sum_{k=0}^{2N-2} n_{k,N} r^{2k+2}, \quad 0 < r < 1.$$

If Ξ_N converges to the empty process, the compact counts converge to zero in probability. Their uniform second-moment bounds imply convergence of their means to zero. Thus $n_{k,N} r^{2k+2} \leq \mathbb{E}\Xi_N(B_r) \rightarrow 0$ for each fixed k . Equation (40.10) implies the second condition.

Conversely, that equation shows that $n_{k,N} \rightarrow 0$ for each fixed k . For $0 < r < R < 1$ and an integer $M \geq 0$, positivity gives

$$\sum_{k > M} n_{k,N} r^{2k+2} \leq (r/R)^{2M+4} \mathbb{E}\Xi_N(B_R).$$

The right side tends to zero uniformly in N as $M \rightarrow \infty$ by the compact-count bound. The finite sum over $k \leq M$ tends to zero with N . Hence $\mathbb{E}\Xi_N(B_r) \rightarrow 0$, and Markov's inequality gives $\mathbb{P}\{\Xi_N(B_r) > 0\} \rightarrow 0$. This is exactly total variation convergence of the process in B_r to the empty configuration. A countable exhaustion of the disk proves vague convergence in law. $\square$

Combining this criterion with the independently proved critical emptiness theorem, Theorem 37.1, gives $Np_N(k) \rightarrow 0$ for every fixed k . This is a qualitative consequence; the representation does not presently give a decay rate for these primary-sector probabilities.

Small-size checks. For $N = 2$ the one-variable primitive space consists only of constants, so $p_2(2) = 1$. Here $T_2 = 6$, and the formula gives $(a_{0,2}, a_{1,2}, a_{2,2}) = (1/5, 3/5, 1/5)$. This agrees with direct integration of $|z - w|^4$ and gives $\pi\rho_{1,2}(0) = 2/5$. For an additional check, direct expansion of Δ_3^2 gives

$$\|F_3\|^2 = \frac{49}{12}, \quad (a_{0,3}, \dots, a_{4,3}) = \left(\frac{23}{245}, \frac{44}{245}, \frac{108}{245}, \frac{10}{49}, \frac{4}{49} \right).$$

The only sectors are $K = 0, 2, 4$, since a symmetric translation-invariant polynomial in two variables is a polynomial in $(w_1 - w_2)^2$. Substitution in the triangular formula yields

$$(p_3(0), p_3(2), p_3(4)) = \left(\frac{81}{1001}, \frac{128}{273}, \frac{104}{231} \right),$$

whose entries sum to one and reproduce all five orbital probabilities.

Remark 40.4 (Primary-sector decay and the remaining quantitative question). The binomial kernel in (40.7) is completely explicit. The independent critical emptiness argument proves qualitative decay of the normalized mass of the primary projection $h_{0,N} \in \ker D_{N-1}$ of $\prod_i w_i^2 \Delta_{N-1}^2$. It proves $Np_N(0) \rightarrow 0$, but no direct norm estimate or quantitative rate for this projection has been obtained from the representation alone. The condition that a holomorphic state vanish on pairwise collisions does not alone select an infinite-volume state: every finite-particle Laughlin polynomial already has that property. A uniqueness argument based on a positive contact Hamiltonian would therefore need an additional, justified condition selecting the boundary state.

41 Unconditional canonical quasi-Gibbs laws on interior disks

This section gives a conditional-law statement for every subsequential limit and every $\beta > 0$. It requires no estimate on the exterior inverse-angular field. The extra harmonic field first appears on an augmented probability space. Averaging this field gives a genuine conditional density for the unmarked process, but that density need not factor into one-body harmonic weights.

Fix an open Euclidean disk $B = B(a_0, R)$ with $\overline{B} \subset \mathbb{D}$. Write $M = \Xi(B)$ and $\eta = \Xi|_{\mathbb{D} \setminus \overline{B}}$. The limiting intensity is locally bounded, so $\Xi(\partial B) = 0$ almost surely. Thus η records the entire exterior up to a null event. Whenever $M = m \geq 1$, order the m interior points uniformly at random, independently of all other randomness, and denote the resulting tuple by $X = (X_1, \dots, X_m)$. Conditional densities below refer to this uniform ordering and ordinary area measure.

For a holomorphic function q on B define

$$Z_{B,m}(q) = \int_{B^m} |\Delta_m(\mathbf{z})|^\beta \prod_{i=1}^m |q(z_i)|^2 dA^m(\mathbf{z}), \quad m \geq 1.$$

In particular this definition has no factor $1/m!$.

Lemma 41.1 (A normal-family bound for normalized exterior factors). *Put*

$$d_{m,\beta} = 2m + \frac{\beta}{2}m(m-1), \quad J_{m,\beta} = \int_{\mathbb{D}^m} |\Delta_m(\mathbf{u})|^\beta dA^m(\mathbf{u}).$$

Then $0 < J_{m,\beta} < \infty$. If q is holomorphic on B and $Z_{B,m}(q) = 1$, then, whenever $B(a, r) \subset B$,

$$|q(a)|^{2m} \leq J_{m,\beta}^{-1} r^{-d_{m,\beta}}. \quad (41.1)$$

Consequently the functions with $Z_{B,m}(q) = 1$ form a locally uniformly bounded family in the space $\mathcal{H}(B)$ of holomorphic functions with its locally uniform topology. The closure of the subfamily that is zero-free and satisfies $q(a_0) > 0$ is compact. Every member of this closure is either identically zero or zero-free; a nonzero member satisfies $q(a_0) > 0$. Every member also satisfies $Z_{B,m}(q) \leq 1$.

Proof. Finiteness and positivity of $J_{m,\beta}$ follow because its integrand is continuous and bounded on the closed polydisk and strictly positive off the collision diagonals. For fixed $\mathbf{u} \in \mathbb{D}_r^m$, the function

$$\lambda \mapsto \prod_{i=1}^m q(a + \lambda u_i)$$

is holomorphic on a neighborhood of the closed unit disk. The mean-value inequality for its squared modulus gives

$$\frac{1}{2\pi} \int_0^{2\pi} \prod_i |q(a + e^{i\theta} u_i)|^2 d\theta \geq |q(a)|^{2m}.$$

The weight $|\Delta_m(\mathbf{u})|^\beta$ and the domain $\mathbb{D}_r^m$ are invariant under a common rotation of all coordinates. Integrating the last inequality with this weight therefore gives

$$1 \geq \int_{B(a,r)^m} |\Delta_m(\mathbf{z})|^\beta \prod_i |q(z_i)|^2 dA^m(\mathbf{z}) \geq J_{m,\beta} r^{d_{m,\beta}} |q(a)|^{2m}.$$

This proves the bound. Applying it with a radius uniformly positive on each compact subset of B proves local boundedness. The Cauchy integral formula then bounds all derivatives on smaller compact sets; Arzelà–Ascoli and a diagonal exhaustion give compactness of the closure in $\mathcal{H}(B)$. The elementary zero-free alternative for locally uniform limits follows from Hurwitz’s theorem. For completeness, if a nonzero limit had a zero, a small circle enclosing only that zero and no zero on its boundary would, by uniform convergence and Rouché’s theorem, force a zero of the approximating functions. Finally Fatou’s lemma gives $Z_{B,m}(q) \leq 1$, and the phase assertion follows by evaluation at a_0 and the zero-free alternative. $\square$

Theorem 41.2 (An augmented canonical DLR equation). *Let Ξ be any subsequential limit of the disk gas at a fixed $\beta > 0$. Fix B as above and $m \geq 1$ such that $\mathbb{P}(M = m) > 0$. On an extension of the probability space under the conditional law given $M = m$, there is a random zero-free holomorphic function q on B satisfying*

$$q(a_0) > 0, \quad Z_{B,m}(q) = 1,$$

such that

$$\mathbb{P}\{X \in d\mathbf{z} \mid \eta, q, M = m\} = |\Delta_m(\mathbf{z})|^\beta \prod_{i=1}^m |q(z_i)|^2 dA^m(\mathbf{z}). \quad (41.2)$$

The joint law can be obtained by a further subsequential limit of the actual finite- N exterior factors. In particular $h = (2/\beta) \log |q|$ is a finite harmonic function on B , and the right side is the usual canonical log-gas density with the additional one-body term $\beta \sum_i h(z_i)$. On the same augmented limit, the original finite- N forces at all interior points converge jointly, and are given by

$$C_{X_i} = \sum_{j \neq i} (X_i - X_j)^{-1} + \frac{2}{\beta} \frac{q'(X_i)}{q(X_i)}. \quad (41.3)$$

The theorem does not assert that q or h is measurable with respect to η , or even with respect to the complete unmarked configuration.

Proof. Write $M_N = \Xi_N(B)$ and $\eta_N = \Xi_N|_{\mathbb{D} \setminus \bar{B}}$. Almost surely there is no finite- N particle on ∂B . For a realized finite exterior tuple $\mathbf{w}$, choose the unique holomorphic logarithm L_w on B such that

$$e^{L_w(z)} = \frac{z - w}{a_0 - w}, \quad L_w(a_0) = 0.$$

It exists because B is simply connected and the quotient has no zero there. Set

$$q_N^{\text{raw}}(z) = \exp \left(\frac{\beta}{2} \sum_{w \in \eta_N} L_w(z) \right), \quad q_N(z) = \frac{q_N^{\text{raw}}(z)}{Z_{B,m}(q_N^{\text{raw}})^{1/(2m)}} \quad \text{on } \{M_N = m\}.$$

The sum is finite. Its normalization is positive and finite: the raw factor is bounded on B , and is positive in modulus there. Factoring the original joint density into interior–interior, exterior–exterior and cross terms proves exactly

$$\mathbb{P}\{X_N \in d\mathbf{z} \mid \eta_N, M_N = m\} = |\Delta_m(\mathbf{z})|^\beta \prod_i |q_N(z_i)|^2 dA^m(\mathbf{z}). \quad (41.4)$$

The factorial arising from uniform ordering cancels in the conditional normalization.

Let N_j be the original process-convergent subsequence. Since the limit has no particle on ∂B , the restrictions and counts converge jointly, and $\mathbb{P}(M_{N_j} = m) \rightarrow \mathbb{P}(M = m) > 0$. Thus the laws conditioned on this event converge on unmarked configurations. Under these conditional laws the interior tuples are tight in B^m . One direct verification is to bound the probability that a coordinate lies in a thin interior collar of ∂B by the expected particle count in that collar divided by $\mathbb{P}(M_{N_j} = m)$. The compact intensity bound makes this uniformly small as the collar shrinks. On each remaining compact set, ordinary finite-dimensional compactness applies. Uniform random ordering gives the stated limiting ordering.

Lemma 41.1 places q_N in a fixed compact subset of $\mathcal{H}(B)$ after taking its closure. Hence, on a further subsequence, (η_N, X_N, q_N) converges jointly under the law conditioned on $M_N = m$ to (η, X, q) . At this stage the possible limit $q \equiv 0$ has not yet been excluded.

For $\psi \in C_c(B^m)$ define

$$T_\psi(q) = \int_{B^m} \psi(\mathbf{z}) |\Delta_m(\mathbf{z})|^\beta \prod_i |q(z_i)|^2 dA^m(\mathbf{z}).$$

On the compact normal family this is bounded and continuous. At finite N , equation (41.4) implies, for every bounded continuous function H of exterior configurations and holomorphic functions,

$$\mathbb{E}[H(\eta_N, q_N) \psi(X_N) \mid M_N = m] = \mathbb{E}[H(\eta_N, q_N) T_\psi(q_N) \mid M_N = m].$$

Joint convergence passes this identity to the limit. Bounded continuous functions determine finite measures on these metrizable spaces, so the identity extends to bounded Borel H . Equivalently,

$$\mathbb{E}[\psi(X) \mid \eta, q, M = m] = T_\psi(q).$$

Choose continuous compactly supported functions $0 \leq \psi_j \uparrow 1$ on B^m . Since X has all m coordinates in B , conditional monotone convergence on the left and ordinary monotone convergence on the right yield $1 = Z_{B,m}(q)$ almost surely. Thus q is not identically zero, and the zero-free alternative proves all the assertions about q . The conditional identity now extends to all bounded Borel ψ by the monotone-class theorem, proving the claimed probability kernel. At finite N , logarithmic differentiation of q_N gives exactly

$$C_{X_{i,N}, N} = \sum_{j \neq i} (X_{i,N} - X_{j,N})^{-1} + \frac{2}{\beta} \frac{q'_N(X_{i,N})}{q_N(X_{i,N})}.$$

Locally uniform convergence of holomorphic functions implies locally uniform convergence of their derivatives. Because the limiting q is zero-free, all limiting coordinates lie in B , and the limit is simple, the right side is continuous at almost every limiting tuple. The continuous-mapping theorem proves the force assertion. $\square$

Theorem 41.3 (An unconditional canonical quasi-Gibbs law). *For every $\beta > 0$, every subsequential limit and every disk $\overline{B} \subset \mathbb{D}$, the following holds. For almost every exterior configuration η , and every integer $m \geq 1$ for which $p_m(\eta) := \mathbb{P}(M = m \mid \eta) > 0$, the conditional law of the uniformly ordered interior tuple has a density*

$$\mathbb{P}\{X \in d\mathbf{z} \mid \eta, M = m\} = |\Delta_m(\mathbf{z})|^\beta g_{\eta,m}(\mathbf{z}) dA^m(\mathbf{z}). \quad (41.5)$$

Here $g_{\eta,m}$ is symmetric, strictly positive and real analytic on B^m , including the collision diagonals. Its logarithm is plurisubharmonic. For each compact $K \subset B$ there are a deterministic finite constant $C_{B,K,m,\beta}$ and a measurable random constant $c_{\eta,K,m,\beta} > 0$ such that

$$c_{\eta,K,m,\beta} \leq g_{\eta,m}(\mathbf{z}) \leq C_{B,K,m,\beta}, \quad \mathbf{z} \in K^m. \quad (41.6)$$

Thus the conditional law is equivalent to $|\Delta_m|^\beta dA^m$ on B^m and has full support on the space of m distinct interior points. For $m = 0$, the conditional law is simply the point mass at the empty configuration.

Proof. For each m with $\mathbb{P}(M = m) > 0$, disintegrate the augmented law of Theorem 41.2 with respect to $(\eta, M = m)$, and let $\nu_{\eta,m}$ be the resulting conditional law of q . For almost every such exterior it is supported on zero-free functions with $Z_{B,m}(q) = 1$ and $q(a_0) > 0$. Put

$$g_{\eta,m}(\mathbf{z}) = \int \prod_i |q(z_i)|^2 \nu_{\eta,m}(dq). \quad (41.7)$$

Taking conditional expectation in the augmented DLR equation proves (41.5). The same construction can be made for all relevant m simultaneously, since there are countably many values. Exterior sets null under the law given $M = m$ are null under $p_m(\eta) \mathbb{P}_\eta(d\eta)$, so this proves the stated almost-every-exterior quantifier on $\{p_m > 0\}$.

Let $r > 0$ be small enough that $B(a, r) \subset B$ for every $a \in K$. The normal-family bound gives, for every admissible q ,

$$\prod_i |q(z_i)|^2 \leq J_{m,\beta}^{-1} r^{-d_{m,\beta}}, \quad \mathbf{z} \in K^m.$$

This proves the deterministic upper bound. Zero-freeness and compactness of K give $\min_K |q| > 0$ for every admissible q , and

$$c_{\eta,K,m,\beta} := \int (\min_K |q|)^{2m} \nu_{\eta,m}(dq) > 0$$

gives the lower bound. The minimum is a Borel function of q in the locally uniform topology; the integral is finite by the same upper bound. An exhaustion of B by compact disks gives one full-measure set of exteriors on which these assertions and strict positivity hold throughout B^m .

For regularity, write $F_q(\mathbf{z}) = \prod_i q(z_i)$ and define

$$K_{\eta,m}(\mathbf{z}, \mathbf{w}) = \int F_q(\mathbf{z}) \overline{F_q(\mathbf{w})} \nu_{\eta,m}(dq).$$

The deterministic bounds on slightly larger compact sets and the Cauchy integral formula permit differentiation under the integral to all orders. More explicitly, the Taylor series of F_q

on any polydisk strictly contained in B^m has coefficients uniformly bounded by the same geometric majorant; integrating the products of these series proves that $K_{\eta,m}$ is holomorphic in $\mathbf{z}$ and antiholomorphic in $\mathbf{w}$. Its diagonal $g_{\eta,m}(\mathbf{z}) = K_{\eta,m}(\mathbf{z}, \mathbf{z})$ is therefore real analytic.

On any complex line and with a prime denoting its holomorphic directional derivative, differentiation and Cauchy–Schwarz give

$$\partial\bar{\partial}\log g = \frac{(\int |F'_q|^2 d\nu)(\int |F_q|^2 d\nu) - |\int F'_q \overline{F_q} d\nu|^2}{g^2} \geq 0.$$

This proves log-plurisubharmonicity. Finally strict positivity of g and positivity of the Vandermonde off the area-null collision set prove equivalence and full support. $\square$

Corollary 41.4 (Conditional Poisson absolute continuity and local moves). *Conditioned on the exterior, the law in B is absolutely continuous with respect to a unit-intensity Poisson process on B . It is equivalent to that Poisson law if and only if $p_m(\eta) > 0$ for every integer $m \geq 0$. No assertion that these count probabilities are all positive is made here.*

For any C^1 diffeomorphism $T : B \rightarrow B$ that equals the identity in a neighborhood of ∂B , moving all interior points by T and fixing the exterior produces a law equivalent to the original limiting law. On the event $M = m \geq 1$, a version of its Radon–Nikodym derivative at the interior tuple $\mathbf{x}$ is

$$\left| \frac{\Delta_m(T^{-1}\mathbf{x})}{\Delta_m(\mathbf{x})} \right|^\beta \frac{g_{\eta,m}(T^{-1}\mathbf{x})}{g_{\eta,m}(\mathbf{x})} \prod_{i=1}^m |\det DT^{-1}(x_i)|. \quad (41.8)$$

It equals one when $m = 0$.

Proof. The unit-intensity Poisson law has the expansion

$$\mathbb{E}_{\text{Pois}} F = e^{-|B|} \sum_{m=0}^{\infty} \frac{1}{m!} \int_{B^m} F \left(\sum_i \delta_{z_i} \right) dA^m.$$

Thus the conditional density relative to this law, on configurations with $m \geq 1$ points, is $e^{|B|} m! p_m(\eta) |\Delta_m|^\beta g_{\eta,m}$; for $m = 0$ it is $e^{|B|} p_0(\eta)$. The equivalence criterion follows from strict positivity in each allowed sector. A diffeomorphism preserves the count, the exterior and distinctness. The ordinary change-of-variables formula in each conditional density gives (41.8), which is finite and strictly positive almost surely. Integrating over counts and exteriors proves mutual absolute continuity of the full laws. $\square$

Proposition 41.5 (The score of the canonical conditional density). *Let b be the canonical logarithmic derivative from Corollary 33.5. For almost every exterior and each allowed count $m \geq 1$, at almost every distinct interior tuple under its conditional law,*

$$b(z_i, \eta + \sum_j \delta_{z_j}) = \sum_{j \neq i} \frac{1}{z_i - z_j} + \frac{2}{\beta} \partial_{z_i} \log g_{\eta,m}(\mathbf{z}). \quad (41.9)$$

The last term is the posterior average of the holomorphic exterior force in the augmented DLR equation:

$$\frac{2}{\beta} \partial_{z_i} \log g_{\eta,m}(\mathbf{z}) = \frac{2}{\beta} \int \frac{q'(z_i)}{q(z_i)} \frac{\prod_j |q(z_j)|^2}{g_{\eta,m}(\mathbf{z})} \nu_{\eta,m}(dq).$$

This is an identification with a conditional-density score, not with an exterior-measurable infinite Coulomb sum.

Proof. Away from collisions, differentiate the positive smooth conditional density in Theorem 41.3:

$$\frac{2}{\beta} \partial_{z_i} \log(|\Delta_m(\mathbf{z})|^\beta g_{\eta,m}(\mathbf{z})) = \sum_{j \neq i} (z_i - z_j)^{-1} + \frac{2}{\beta} \partial_{z_i} \log g_{\eta,m}(\mathbf{z}).$$

Compactly supported integration by parts in the interior variables identifies this with the conditional logarithmic derivative. The collision singularities are locally integrable against the density, since their worst radial factor is $|z_i - z_j|^{\beta-1}$ and $\beta > 0$. The factor g and all its derivatives are bounded on compact sets. Here are details justifying the disintegration of the global Ward identity. Fix a vector field v compactly supported in B . Choose $\chi_\ell \in C_c^\infty(\mathbb{D})$ converging pointwise off ∂B to $\mathbf{1}_B$, equal to one near $\text{supp } v$, with their gradients supported in shrinking collars of ∂B . Then $\Xi(\chi_\ell) \rightarrow M$ almost surely, and $\mathcal{L}_v \Xi(\chi_\ell) = 0$ exactly. If a bounded smooth scalar function ϑ_m equals one at the integer m and zero at all other nonnegative integers, then $\vartheta_m(\Xi(\chi_\ell)) \rightarrow \mathbf{1}_{\{M=m\}}$ boundedly, also with zero derivative in the Ward identity. Exterior cylinders use test functions supported in $\mathbb{D} \setminus \bar{B}$ and likewise have zero derivative. The already proved local Campbell L^2 bound on b permits passage to bounded pointwise limits of these multipliers. A monotone-class argument therefore allows arbitrary bounded Borel functions of the exterior and the indicator of the inside count, without introducing any derivative of these factors.

To identify individual coordinates, cover the set of distinct interior tuples by countably many coordinate charts made of m pairwise disjoint disks with rational data, compactly contained in B . On a chart the configuration has exactly one point in each disk and no others in B . Local count selectors for these disks are obtained by the same collar approximation; for a vector field supported in one chosen disk, all their derivatives vanish. Products of coordinate tests are represented on the chart by products of linear statistics with their test functions supported in the respective disjoint disks. Bounded smooth truncations put these tests in the original cylinder class. Consequently the global identity, after disintegration, determines the distributional derivative in each individual coordinate on every such chart. The conditional density is strictly positive and smooth there, so its logarithmic derivative must equal the restriction of b . Countable exhaustion proves the assertion almost everywhere. This argument uses the integrability of the existing b and does not presuppose a global integrability bound on the candidate $\partial \log g$. The posterior formula follows by differentiating (41.7). Its integrability follows from the deterministic compact bounds on q and q' after cancellation of the displayed factor $q(z_i)$ with its product weight. $\square$

Corollary 41.6 (Strict positivity of every nonzero repulsion factor). *For each $p \geq 1$, let h_p be the continuous function from Theorem 4.1, so that $\rho_p = |\Delta_p|^\beta h_p$. Exactly one of the following alternatives holds:*

1. $\mathbb{P}\{\Xi(\mathbb{D}) \geq p\} = 0$, in which case $h_p \equiv 0$;
2. $\mathbb{P}\{\Xi(\mathbb{D}) \geq p\} > 0$, in which case $h_p(\mathbf{z}) > 0$ for every $\mathbf{z} \in \mathbb{D}^p$, including collision diagonals.

In the second case, every nonempty open $U \subset \mathbb{D}$ has $\mathbb{P}\{\Xi(U) \geq p\} > 0$, and the leading constants in Corollary 4.6 are strictly positive at every center. In particular every limit that is not almost surely empty satisfies $\rho_1(z) > 0$ throughout $\mathbb{D}$, including at $z = 0$.

Proof. The first alternative follows immediately from the factorial moment identity, followed by continuity of h_p . In the second alternative, fix $\mathbf{a} \in \mathbb{D}^p$. By exhaustion there is a disk $\bar{B} \subset \mathbb{D}$ containing all its coordinates and satisfying $\mathbb{P}\{\Xi(B) \geq p\} > 0$. Since $\Xi(B)$ is finite almost surely, there is $m \geq p$ with $P_m := \mathbb{P}\{\Xi(B) = m\} > 0$. Average the canonical mixture

factor over the exterior under the law given this count, and write

$$\bar{g}_{B,m}(\mathbf{z}) = \mathbb{E}[g_{\eta,m}(\mathbf{z}) \mid \Xi(B) = m].$$

The same deterministic compact bounds as in the theorem show that this function is continuous and strictly positive throughout B^m . The contribution to the p -point factorial density from the event $\{\Xi(B) = m\}$ gives, for almost every $\mathbf{z} \in B^p$,

$$\rho_p(\mathbf{z}) \geq P_m(m)_p \int_{B^{m-p}} |\Delta_m(\mathbf{z}, \mathbf{w})|^\beta \bar{g}_{B,m}(\mathbf{z}, \mathbf{w}) dA^{m-p}(\mathbf{w}).$$

When $m > p$, choose $m - p$ pairwise disjoint closed disks of positive radius in B avoiding the finite set of coordinates of $\mathbf{a}$, and restrict the integral to their product. Divide by $|\Delta_p(\mathbf{z})|^\beta$ off the collision diagonals. The resulting restricted integral, including its cross-distance factors, is a continuous strictly positive function in a neighborhood of $\mathbf{a}$, even when some coordinates of $\mathbf{a}$ coincide. When $m = p$ the same statement holds with no integral. Continuity of h_p extends the resulting lower bound from almost every distinct tuple to that entire neighborhood, proving $h_p(\mathbf{a}) > 0$. Integrating the now positive correlation density over distinct tuples in U^p proves the local count assertion. The previously established small-cluster formulas then have strictly positive coefficients. $\square$

Remark 41.7 (The remaining DLR problem). The unmarked equation (41.5) is an unconditional canonical conditional-law theorem. It identifies the repulsion exponent, proves equivalence of all interior locations at each allowed count, and gives a mixture representation of its smooth factor. It does not compute that factor from a prescribed regularized interaction with the exterior, determine the conditional count probabilities, or prove uniqueness of the limiting law. In particular, one may not replace $g_{\eta,m}$ by $\exp(\beta \sum_i h_\eta(z_i))$ merely because every finite- N conditional density has that form: a nontrivial mixture over limiting harmonic fields generally fails to have such a product form.

42 A Gaussian lower bound on the curvature of conditional laws

The harmonic mixture in the augmented DLR equation is necessarily nontrivial. Its posterior fluctuations contain the independent Gaussian harmonic field. This observation strengthens mere log-plurisubharmonicity to an explicit matrix inequality for every canonical conditional density.

For $\mathbf{x} = (x_1, \dots, x_m) \in B^m$ define

$$\mathcal{K}(\mathbf{x})_{ij} = \frac{1}{(1 - x_i \bar{x}_j)^2}, \quad \mathcal{R}_m(\mathbf{x}) = \prod_{i,j=1}^m (1 - x_i \bar{x}_j).$$

The second expression is real and strictly positive, since it equals $\prod_i (1 - |x_i|^2) \prod_{i < j} |1 - x_i \bar{x}_j|^2$. For Hermitian matrices, $A \succeq B$ means that $A - B$ is positive semidefinite. We use the Wirtinger convention $\partial_z = (\partial_x - i\partial_y)/2$, so $\Delta_z = 4\partial_z \partial_{\bar{z}}$.

Theorem 42.1 (Strict conditional Levi curvature). *In Theorem 41.3, for almost every exterior η and every allowed count $m \geq 1$, the positive real-analytic factor $g_{\eta,m}$ satisfies*

$$\left[\partial_{x_i} \partial_{\bar{x}_j} \log g_{\eta,m}(\mathbf{x}) \right]_{i,j=1}^m \succeq \frac{\beta}{2} \mathcal{K}(\mathbf{x}), \quad \mathbf{x} \in B^m. \quad (42.1)$$

Consequently

$$\log(g_{\eta,m} \mathcal{R}_m^{\beta/2}) \text{ is plurisubharmonic on } B^m. \quad (42.2)$$

For the ordinary-area conditional density $p_{\eta,m} = |\Delta_m|^\beta g_{\eta,m}$, every coordinate satisfies

$$\Delta_{x_i} \log p_{\eta,m}(\mathbf{x}) \geq \frac{2\beta}{(1 - |x_i|^2)^2} \quad (42.3)$$

off the collision diagonals. The matrix on the right of (42.1) is positive definite whenever the coordinates are distinct.

Proof. We first describe the required joint realization. In the construction of Theorem 41.2, retain the force-marked configuration and all positive harmonic modes when extracting the subsequence. Their joint tightness was established in Theorems 33.3 and 34.1. For the fixed event $M = m$, the normalized exterior factors belong to a fixed compact normal family. They can be set equal to zero off $\{M_N = m\}$ while taking the joint subsequence; then condition on that event, whose probability converges to a positive number. The proof of the augmented DLR theorem applies without change. Its force identity (41.3) therefore holds for marks that also obey the Gaussian regression (34.1). Uniform ordering can be supplied by auxiliary randomness independent of this entire construction.

Fix an exterior and an allowed count for which the conclusions of the augmented and unmarked DLR theorems hold, and abbreviate the mixing measure by $\nu = \nu_{\eta,m}$. Put

$$A_q(\mathbf{x}) = \prod_{i=1}^m q(x_i), \quad g(\mathbf{x}) = \int |A_q(\mathbf{x})|^2 \nu(dq).$$

The posterior distribution of q given the complete unmarked configuration, written as $(\eta, m, \mathbf{x})$ with its uniform ordering, is

$$\nu_{\mathbf{x}}(dq) = \frac{|A_q(\mathbf{x})|^2}{g(\mathbf{x})} \nu(dq). \quad (42.4)$$

This is Bayes' formula applied to (41.2); the Vandermonde cancels because it does not depend on q . All statements made initially almost everywhere below can be taken with respect to this posterior.

Let $L_i(q, \mathbf{x}) = q'(x_i)/q(x_i)$. The zero-free property makes this well defined. Differentiation under the mixing integral yields

$$\partial_{x_i} \partial_{\overline{x_j}} \log g = \text{Cov}_{\nu_{\mathbf{x}}}(L_i, \overline{L_j}). \quad (42.5)$$

For clarity, the covariance on the right means $\mathbb{E}[L_i \overline{L_j}] - \mathbb{E}[L_i] \mathbb{E}[\overline{L_j}]$. The differentiation is justified even without a uniform bound on q'/q . Indeed $A_q L_i = \partial_{x_i} A_q$, and the deterministic normal-family bounds, followed by the Cauchy integral formula on slightly larger compact polydisks, bound both A_q and all its first derivatives uniformly in q . Thus all integrals of $|A_q|^2 L_i \overline{L_j}$ are absolutely convergent, locally uniformly in $\mathbf{x}$, and the asserted derivative formula follows.

Off collisions, the force identity gives

$$C_i = \sum_{j \neq i} (x_i - x_j)^{-1} + \frac{2}{\beta} L_i.$$

The first term is fixed by the unmarked configuration. Hence (42.5) becomes

$$\left[\partial_{x_i} \partial_{\overline{x_j}} \log g \right]_{ij} = \frac{\beta^2}{4} \text{Cov}((C_i)_{i=1}^m \mid \Xi, \mathbf{x}). \quad (42.6)$$

Here the notation $\mathbf{x}$ in the conditioning only retains the independent ordering of points already contained in Ξ .

For any integer $L \geq 1$, Gaussian regression gives simultaneously at all the finitely many interior particles

$$\mathbb{E}[C_i \mid \Xi, \mathbf{x}, G_1, \dots, G_L] = b(x_i, \Xi) - \sum_{k=1}^L \overline{G_k} x_i^{k-1}.$$

The Campbell almost-everywhere assertion in that theorem holds simultaneously at these particles: the expected number of exceptions in any compact subset is zero, and a countable exhaustion covers B . The modes are independent of Ξ and of its independent ordering. Conditional orthogonal projection, applied to every complex linear combination of the C_i , therefore gives

$$\text{Cov}((C_i)_{i=1}^m \mid \Xi, \mathbf{x}) \succeq \frac{2}{\beta} \left[\sum_{k=1}^L k x_i^{k-1} \overline{x_j}^{k-1} \right]_{ij}.$$

The conditional second moments are finite almost surely, either by local Campbell integrability or directly from the preceding normal-family derivative bound. Letting $L \rightarrow \infty$, the geometric series converges locally uniformly and gives $\mathcal{K}$. Combining this with (42.6) proves (42.1) initially almost everywhere under the conditional density. That density is strictly positive off collisions, so the same assertion holds Lebesgue-almost everywhere there. Both matrix sides are continuous on all of B^m ; a countable dense set of complex test vectors then extends the inequality everywhere, including the diagonals.

Direct differentiation gives

$$\partial_{x_i} \partial_{\overline{x_j}} \log \mathcal{R}_m = -\mathcal{K}(\mathbf{x})_{ij}.$$

This proves (42.2). The Vandermonde logarithm is pluriharmonic off its zero set, so its Levi matrix there vanishes. Taking the diagonal in (42.1) and multiplying by four proves (42.3). Finally $\mathcal{K}$ is the Gram matrix of the vectors $(\sqrt{k} x_i^{k-1})_{k \geq 1}$. For distinct coordinates the first m entries already form an invertible Vandermonde matrix. Thus this Gram matrix is positive definite. $\square$

Corollary 42.2 (A harmonic exterior alone cannot describe the conditional law). *For any allowed count $m \geq 1$, the unmarked conditional density cannot have the form*

$$p_{\eta, m}(\mathbf{x}) = Z^{-1} |\Delta_m(\mathbf{x})|^\beta \exp \left\{ \beta \sum_{i=1}^m h_\eta(x_i) \right\}$$

with h_η harmonic on B . In the augmented DLR construction, the exterior force q'/q cannot be measurable with respect to the complete unmarked configuration on any event of positive probability that is measurable with respect to that configuration and on which $M \geq 1$.

Proof. In the displayed product representation the logarithm of the smooth factor has zero Levi matrix, contradicting the strictly positive diagonal in (42.1). For the second assertion, conditional on the unmarked configuration the covariance of q'/q at each interior point is at least $\beta/(2(1 - |x_i|^2)^2) > 0$, by (42.5) and the theorem. A measurable force would have zero conditional variance. The contradiction remains valid after restriction to any positive-probability event measurable with respect to the unmarked configuration and contained in $M \geq 1$. $\square$

Proposition 42.3 (A necessary background bound for a compensated Green DLR law). *Let a nonempty subsequential limit satisfy a canonical Green DLR specification relative to $d\lambda(x) = (1 - |x|^2)^{-2} dA(x)$, with conditional densities*

$$p_{\eta, m}(\mathbf{x}) = Z_{\eta, m}^{-1} \exp \left\{ -\beta \sum_{i < j} g_{\mathbb{D}}(x_i, x_j) - \beta \sum_i H_\eta(x_i) \right\} \prod_i (1 - |x_i|^2)^{-2}, \quad (42.7)$$

where $g_{\mathbb{D}}(x, y) = -\log |(x - y)/(1 - x\bar{y})|$. Assume that the finite exterior field is twice continuously differentiable in B and satisfies

$$\Delta H_{\eta}(x) = \frac{2\pi c}{(1 - |x|^2)^2}, \quad x \in B,$$

for a constant $c \geq 0$. Then necessarily

$$c \leq \frac{4 - \beta}{\pi\beta}. \quad (42.8)$$

In particular, if c is the intensity with respect to hyperbolic area in an isometry-invariant compensated DLR description, that intensity must obey this bound. This is a necessary condition for such a description; no existence of the assumed field or DLR specification is inferred from it.

Proof. Since the law is not almost surely empty, some relatively compact disk has positive probability of an allowed count $m \geq 1$. On such a sector, $g_{\mathbb{D}}(x_i, x_j)$ is harmonic in x_i away from x_j . Moreover $\Delta[-2\log(1 - |x|^2)] = 8/(1 - |x|^2)^2$. Taking the coordinate Laplacian of the logarithm of (42.7) therefore gives

$$\Delta_{x_i} \log p_{\eta, m}(\mathbf{x}) = \frac{8 - 2\pi\beta c}{(1 - |x_i|^2)^2}$$

off collisions. Comparison with (42.3) proves $8 - 2\pi\beta c \geq 2\beta$, which is exactly (42.8). $\square$

Remark 42.4 (Why the compensated field has this Laplacian). For an exterior configuration supported in B^c , finite sums $\sum_w g_{\mathbb{D}}(x, w)$ are harmonic in B . A compensated field formed by subtracting $c \int_E g_{\mathbb{D}}(x, w) d\lambda(w)$, with E containing B , has distributional Laplacian $2\pi c(1 - |x|^2)^{-2}$ there, because $-\Delta_x g_{\mathbb{D}}(x, w) = 2\pi\delta_w$. If such compensated fields converge in local $L^1(B)$, their limit retains this identity. Subtracting the smooth particular solution $-(\pi c/2)\log(1 - |x|^2)$ leaves a distributionally harmonic function, which has a smooth harmonic representative. Thus the regularity and Laplacian assumptions of the proposition follow from this specific local convergence, if it can be proved. The required convergence has not been established here for general β .

Corollary 42.5 (Saturation determines the residual force). *Suppose the canonical Green DLR assumptions of Proposition 42.3 hold on all relatively compact interior disks, with $c = (4/\beta - 1)/\pi$. In any joint force-and-Gaussian limit as in Theorem 34.1, almost surely every particle satisfies*

$$C_z = b(z, \Xi) - \sum_{k=1}^{\infty} \overline{G_k} z^{k-1}. \quad (42.9)$$

The series converges locally in Campbell L^2 . Thus these force marks are determined by the unmarked configuration and its independent Gaussian harmonic field, with no additional force randomness. At $\beta = 2$ this conclusion is unconditional: the Bergman DLR law of Theorem 46.3 and the potential regularity in Corollary 46.5 verify the hypotheses with $c = 1/\pi$.

Proof. For $i \neq j$, the smooth factor in (42.7) has mixed Levi entry

$$\partial_{x_i} \partial_{\bar{x}_j} \log g_{\eta, m} = \frac{\beta}{2(1 - x_i \bar{x}_j)^2}.$$

Indeed the pair contribution is $-\beta \log |1 - x_i \bar{x}_j|$; its holomorphic–antiholomorphic part has coefficient $-\beta/2$. No one-body term contributes to these mixed entries. The diagonal entry is

$$\frac{2 - \pi\beta c/2}{(1 - |x_i|^2)^2} = \frac{\beta}{2(1 - |x_i|^2)^2}$$

at the specified value of c . Thus equality holds throughout (42.1). Equation (42.6) shows that the conditional force covariance is exactly $(2/\beta)\mathcal{K}$.

Write $T(z) = \sum_{k \geq 1} \overline{G}_k z^{k-1}$. The infinite-mode regression gives $\mathbb{E}[C_z \mid \Xi, (G_k)] = b(z, \Xi) - T(z)$. Conditional orthogonal projection therefore decomposes each force variance into the variance $2/\{\beta(1 - |z|^2)^2\}$ of $T(z)$ and the nonnegative mean square of $C_z - b(z, \Xi) + T(z)$. The preceding covariance equality makes that mean square zero at every interior particle, in the almost-sure conditional sense. Restricting first to compact Campbell windows justifies all L^2 computations. A countable exhaustion by interior disks proves (42.9) at every particle almost surely. The claimed series convergence is exactly that established in Theorem 34.1. $\square$

43 A consistent augmentation by a meromorphic electric field

The one-disk augmented equations can be made simultaneous and consistent. The field retained here is additional random data. Its existence does not assert that the exterior of the unmarked point process determines the conditional interaction.

Choose once and for all $0 < r_1 < r_2 < \dots \uparrow 1$ and put $B_j = \mathbb{D}_{r_j}$. Almost surely the limiting configuration has no point on any of the circles ∂B_j .

Proposition 43.1 (Simultaneous fields and meromorphic gluing). *For every subsequential limit Ξ there is an augmentation by random functions $q_j \in \mathcal{H}(B_j)$ with the following properties. Writing $m_j = \Xi(B_j)$, set $q_j = 0$ if $m_j = 0$; if $m_j \geq 1$, then q_j is zero-free, $q_j(0) > 0$, and $Z_{B_j, m_j}(q_j) = 1$. The augmented canonical DLR equation of Theorem 41.2 holds for every j and every allowed positive value of m_j on this single probability space.*

For $j < k$ with $m_j \geq 1$, one has on B_j

$$\frac{2 q'_j(z)}{\beta q_j(z)} = \frac{2 q'_k(z)}{\beta q_k(z)} + \sum_{w \in \Xi \cap (B_k \setminus B_j)} \frac{1}{z - w}. \quad (43.1)$$

Consequently, on the event $\Xi \neq \emptyset$, the meromorphic functions

$$\mathcal{C}_j(z) = \sum_{w \in \Xi \cap B_j} \frac{1}{z - w} + \frac{2 q'_j(z)}{\beta q_j(z)}, \quad m_j \geq 1, \quad (43.2)$$

glue to one meromorphic function $\mathcal{C}$ on $\mathbb{D}$. It has exactly the particles of Ξ as its poles, each simple with residue one. Its regular part at a particle equals the corresponding augmented force mark. On $\{\Xi = \emptyset\}$ we may set $\mathcal{C} = 0$.

If $m_j = 0$ but $\Xi \neq \emptyset$, its missing exterior field is recovered by restricting this $\mathcal{C}$ to B_j ; it is holomorphic there. More generally

$$\mathcal{E}_j(z) = \mathcal{C}(z) - \sum_{w \in \Xi \cap B_j} \frac{1}{z - w} \quad (43.3)$$

extends holomorphically to a neighborhood of $\overline{B}_j$. For $m_j \geq 1$, it is $(2/\beta)q'_j/q_j$ on B_j .

Proof. At finite N , define $q_{j,N}$ by the normalization in the proof of Theorem 41.2 on $\{\Xi_N(B_j) = m \geq 1\}$, and set it to zero when the count is zero. For each fixed j the count laws are tight. For any finite collection of allowed counts, the corresponding $q_{j,N}$ belong to a finite union of compact normal-family closures from Lemma 41.1. This proves joint tightness of the count and function. A countable diagonal extraction therefore retains Ξ and all the functions q_j jointly. Conditional on any positive-probability event $\{m_j = m \geq 1\}$, the proof of the one-disk theorem applies without alteration: testing only the exterior and

q_j still gives its exact conditional density. The argument with compactly supported test functions increasing to one proves $Z_{B_j, m}(q_j) = 1$ on that event. Countably many j, m give one probability-one set on which all assertions hold simultaneously.

Before taking limits, logarithmic differentiation of the exterior products gives (43.1) exactly. The annulus between two fixed circles is relatively compact in $\mathbb{D}$. Its particles converge as a finite configuration and none lies on its boundary in the limit. On $\{m_j \geq 1\}$ both normalized holomorphic functions have nonzero zero-free limits, so their logarithmic derivatives converge locally uniformly. The finite annular Cauchy sum converges locally uniformly on B_j . This proves the displayed compatibility. The countable collection of identities holds on one probability-one set.

If $\Xi \neq \emptyset$, there is some first disk with positive count, and all larger disks also have positive count. The compatibility identity says exactly that their functions $\mathcal{C}_j$ agree on overlaps. They therefore define $\mathcal{C}$ on their union, which is all of $\mathbb{D}$. Each local expression has simple poles of residue one at the interior particles and is holomorphic elsewhere. No cancellation can remove a pole, because the other terms in that local expression are holomorphic near that particle. Its regular part is the expression in (41.3).

This also explains the construction for any smaller zero-count disk: use a larger positive-count disk to define $\mathcal{C}$ and restrict it. The result is independent of the chosen larger disk. Finally, local finiteness and the absence of particles on ∂B_j give an open neighborhood of $\bar{B}_j$ containing no particles other than the finitely many inside B_j . Subtracting their simple pole terms removes every pole there, proving the asserted extension of $\mathcal{E}_j$. The construction is measurable: the first positive count disk is a measurable integer, and values on a countable dense subset, away from the almost surely absent fixed-point poles, determine the glued meromorphic function. Equivalently one may use the usual Borel space of meromorphic functions with the locally uniform spherical topology. $\square$

Theorem 43.2 (A consistent canonical specification on the augmented space). *On the augmented law of Proposition 43.1, the conditional distribution in B_j given*

$$\Xi|_{\mathbb{D} \setminus \bar{B}_j}, \quad m_j, \quad \mathcal{E}_j|_{B_j}$$

is the following canonical kernel. If $m_j = 0$, it fixes the state. If $m_j = m \geq 1$, let q be the unique zero-free holomorphic function on B_j with

$$\frac{2}{\beta} \frac{q'}{q} = \mathcal{E}_j, \quad q(0) > 0, \quad Z_{B_j, m}(q) = 1.$$

Sample a uniformly ordered tuple $\mathbf{x} \in B_j^m$ with density

$$|\Delta_m(\mathbf{x})|^\beta \prod_i |q(x_i)|^2 dA^m(\mathbf{x}). \quad (43.4)$$

Keep the exterior fixed and replace the meromorphic field by

$$\mathcal{C}^{\text{new}}(z) = \mathcal{C}(z) + \sum_{i=1}^m \frac{1}{z - x_i} - \sum_{w \in \Xi \cap B_j} \frac{1}{z - w}. \quad (43.5)$$

Denote this kernel on augmented states by γ_j . Its result satisfies the same admissibility conditions as the original state, and the augmented law $\hat{P}$ satisfies

$$\hat{P}\gamma_j = \hat{P} \quad \text{for every } j.$$

For $j < k$, the canonical kernels are consistent:

$$\gamma_j \gamma_k = \gamma_k \gamma_j = \gamma_k. \quad (43.6)$$

These are count-preserving equations. They do not determine the probabilities of the different inside counts, and their conditioning includes an exterior-field mark in addition to the unmarked exterior.

Proof. A holomorphic function on a disk has a holomorphic primitive. Exponentiating $(\beta/2)$ times the primitive of $\mathcal{E}_j$, chosen zero at the origin, determines q up to a positive constant; the displayed phase and normalization determine that constant uniquely. It exists and is finite because $\mathcal{E}_j$ extends to a neighborhood of $\overline{B}_j$. Thus, when $m_j \geq 1$, knowing $(m_j, \mathcal{E}_j)$ is equivalent to knowing (m_j, q_j) . The one-disk augmented DLR equation proves the asserted conditional distribution of the interior tuple.

One must also check that this describes the conditional law of the whole augmented state, rather than just of the point positions. Given the exterior, count, field $\mathcal{E}_j$ and an interior tuple, the formula

$$\mathcal{C}(z) = \mathcal{E}_j(z) + \sum_{i=1}^{m_j} (z - x_i)^{-1} \quad \text{on } B_j$$

determines the global meromorphic function uniquely by the identity theorem. Existence of that continuation is part of the original augmentation. For a resampled tuple it is supplied explicitly by (43.5). The meromorphic continuation, on the set where it exists, is measurable: restriction of meromorphic functions from $\mathbb{D}$ to B_j is an injective Borel map between standard Borel spaces, and hence has a Borel inverse on its image. Thus no further random data in the augmented state need be specified once these quantities and the tuple are known. This proves $\hat{P}\gamma_j = \hat{P}$.

The update cancels precisely the old interior pole terms and inserts precisely the new ones, preserving the residue-one condition and leaving $\mathcal{E}_j$ unchanged. The sampled tuple is distinct and avoids all the countably many circles almost surely by absolute continuity. For any smaller disk, subtracting its new interior pole terms leaves a function holomorphic near its closed disk; for any larger disk its exterior field is unchanged. All required positive finite normalizations therefore continue to exist. This verifies stability of the admissible augmented state space.

For consistency, first observe that a move in $B_j \subset B_k$ preserves the exterior of B_k , its total count m_k , and its exterior field $\mathcal{E}_k$: the two finite Cauchy-sum changes cancel in (43.3). Consequently γ_k is the same kernel before and after applying γ_j , proving $\gamma_j\gamma_k = \gamma_k$.

For the reverse order, suppose $m_k \geq 1$ and condition within the γ_k density on the particles in $B_k \setminus \overline{B}_j$ and on the number m_j of remaining particles. The finite Vandermonde factorization gives a conditional density proportional to

$$|\Delta_{m_j}(\mathbf{x})|^\beta \prod_{i=1}^{m_j} |q_k(x_i)|^2 \prod_{\substack{i \leq m_j \\ w \text{ in the annulus}}} |x_i - w|^\beta.$$

Its holomorphic logarithmic derivative is exactly the nested field in (43.1); its normalization is therefore (43.4) for B_j . Conditionally resampling from this density leaves the γ_k law unchanged, by ordinary conditional expectation. The field update is the same finite Cauchy-sum update for both constructions. This proves $\gamma_k\gamma_j = \gamma_k$. If either relevant inside count is zero, the corresponding inner resampling is the identity and the same argument applies. In particular when $m_k = 0$ both kernels are the identity on the interior. This proves all the stated consistency relations. $\square$

Remark 43.3 (What is still extra information). The field $\mathcal{C}$ contains the particle positions as its poles, but may contain an additional holomorphic part coming from the boundary. Conditioning on $\mathcal{E}_j$ retains that part. The consistent augmented specification is therefore

not an assertion that the unmarked exterior alone determines a single harmonic interaction. Integrating out the field gives precisely the canonical quasi-Gibbs mixture from Theorem 41.3.

44 Hyperbolic covariance of the score and of the Gaussian force

Write $\lambda(dz) = (1 - |z|^2)^{-2} dA(z)$. This differs by a constant factor from the area measure of the curvature -1 metric; the group of isometries is unchanged. In this section an automorphism means an orientation-preserving conformal automorphism of $\mathbb{D}$. Reflection invariance, which is already known for every limiting law, then supplies all orientation-reversing isometries as well.

Proposition 44.1 (The exact geometric transformation of a score). *Let P be a point-process law having the locally Campbell-integrable canonical score b in the normalization of Corollary 33.5. For an automorphism φ , let P^φ be the law of $\varphi_*^{-1}\Xi$, where Ξ has law P . Then a canonical score for P^φ is*

$$b^\varphi(w, \eta) = \varphi'(w)b(\varphi(w), \varphi_*\eta) + \frac{2}{\beta} \frac{\varphi''(w)}{\varphi'(w)}. \quad (44.1)$$

Define the score relative to hyperbolic area by

$$b_H(z, \eta) = b(z, \eta) - \frac{4}{\beta} \frac{\bar{z}}{1 - |z|^2}.$$

It transforms without the inhomogeneous term:

$$b_H^\varphi(w, \eta) = \varphi'(w)b_H(\varphi(w), \varphi_*\eta). \quad (44.2)$$

Equivalently, for a compactly supported vector field v , the Ward identity is

$$\mathbb{E} \left[\mathcal{L}_v F(\Xi) + F(\Xi) \sum_{z \in \Xi} \{ \operatorname{div}_\lambda v(z) + \beta \operatorname{Re}(v(z)b_H(z, \Xi)) \} \right] = 0, \quad (44.3)$$

where $\operatorname{div}_\lambda v = \operatorname{div} v + 4 \operatorname{Re}(v\bar{z})/(1 - |z|^2)$.

Proof. The real score is $B = \beta \bar{b}$. Change variables by $z = \varphi(w)$ in its Campbell integration-by-parts identity. The chain rule and the change of area give the transformed real score

$$B^\varphi(w, \eta) = D\varphi(w)^T B(\varphi(w), \varphi_*\eta) + \nabla_w \log |\varphi'(w)|^2.$$

This calculation can first be made with the spatial support contained in a fixed compact set. The map and its inverse have bounded derivatives there, so all integrations and their changes of variables are justified by the assumed local integrability. In complex notation, $D\varphi^T B = \overline{\varphi'} B$ and $\nabla \log |\varphi'|^2 = 2\overline{\varphi''}/\varphi'$. Conjugation and division by β prove (44.1). Uniqueness of the score identifies this version with the canonical version whenever the latter has already been constructed.

The identity

$$\frac{|\varphi'(w)|^2}{(1 - |\varphi(w)|^2)^2} = \frac{1}{(1 - |w|^2)^2}$$

and its logarithmic derivative give

$$\varphi'(w) \frac{4}{\beta} \frac{\overline{\varphi(w)}}{1 - |\varphi(w)|^2} + \frac{2}{\beta} \frac{\varphi''(w)}{\varphi'(w)} = \frac{4}{\beta} \frac{\bar{w}}{1 - |w|^2}.$$

Subtracting this identity proves (44.2). Substitution of the definition of b_H in the original Ward identity proves (44.3). $\square$

Proposition 44.2 (The Gaussian force already has full isometry covariance). *Let G_k be the independent circular Gaussians of Theorem 34.1, and define*

$$T(z) = \sum_{k \geq 1} \overline{G_k} z^{k-1}.$$

This series defines a random holomorphic function on $\mathbb{D}$. For every automorphism φ ,

$$(\varphi'(z)T(\varphi(z)))_{z \in \mathbb{D}} \stackrel{\text{law}}{=} (T(z))_{z \in \mathbb{D}}. \quad (44.4)$$

Thus the independent Gaussian summand in the force regression is already a covariant holomorphic one-form. This assertion does not assume, or prove, that the point-process law is invariant.

Proof. For every $r < 1$, $\sum_k \mathbb{E}|G_k| r^{k-1} < \infty$, since $\mathbb{E}|G_k| \leq \sqrt{2k/\beta}$. Tonelli's theorem gives almost sure absolute and uniform convergence on $\{|z| \leq r\}$, and a countable exhaustion gives holomorphy. The field is centered circular Gaussian with covariance

$$\mathbb{E}[T(z)\overline{T(w)}] = \frac{2}{\beta} \sum_{k \geq 1} k(z\overline{w})^{k-1} = \frac{2/\beta}{(1-z\overline{w})^2}, \quad \mathbb{E}[T(z)T(w)] = 0.$$

The elementary automorphism identity

$$\frac{\varphi'(z)\overline{\varphi'(w)}}{(1-\varphi(z)\overline{\varphi(w)})^2} = \frac{1}{(1-z\overline{w})^2}$$

therefore proves equality of all finite-dimensional Gaussian laws. Evaluations on a countable dense set generate the Borel sigma-algebra of $\mathcal{H}(\mathbb{D})$, proving the asserted equality of random holomorphic functions. $\square$

Theorem 44.3 (Necessary consequences of hyperbolic invariance). *Suppose a subsequential limiting law P is invariant under every automorphism and is not concentrated on the empty configuration. Then there are constants $c > 0$ and $v > 0$ such that*

$$\rho_1(z) = \frac{c}{(1-|z|^2)^2}, \quad (44.5)$$

$$\mathbb{E}[b(z, \Xi) \mid z] = \frac{4}{\beta} \frac{\overline{z}}{1-|z|^2}, \quad (44.6)$$

$$\text{Var}(b(z, \Xi) \mid z) = \frac{v}{(1-|z|^2)^2} \quad \text{for area-almost every } z, \quad (44.7)$$

where

$$0 < v \leq \frac{8-2\beta}{\beta^2}. \quad (44.8)$$

The conditional expectations refer to the spatial disintegration of the Campbell measure. Moreover $P(0 < \Xi(\mathbb{D}) < \infty) = 0$.

Proof. Invariance and the continuous intensity imply $\rho_1(w) = |\varphi'(w)|^2 \rho_1(\varphi(w))$. Choosing an automorphism that sends w to zero proves (44.5), with $c = \rho_1(0)$; if $c = 0$, the process is empty almost surely by compact exhaustion. The Ward identity with $F = 1$ says $\beta \mathbb{E}[b(z, \Xi) \mid z] = \nabla \log \rho_1(z)$, which proves (44.6).

By score uniqueness, invariance identifies the score in (44.1) with b . The deterministic second term does not change conditional variances; therefore, with $V(z) = \text{Var}(b(z, \Xi) \mid z)$,

$$V(w) = |\varphi'(w)|^2 V(\varphi(w)) \quad \text{for almost every } w.$$

In particular $V(z)dA(z)$ is an invariant locally finite measure. Here local finiteness follows from the Campbell L^2 bound and the positive lower bound of ρ_1 on compact sets. For completeness, an invariant locally integrable density V must be $v(1 - |z|^2)^{-2}$: differentiate its invariance against a compactly supported smooth test along the flows of $1 - z^2$ and $i(1 + z^2)$. In distributions on $z = x + iy$ this gives

$$\begin{pmatrix} 1 - x^2 + y^2 & -2xy \\ -2xy & 1 + x^2 - y^2 \end{pmatrix} \begin{pmatrix} \partial_x V \\ \partial_y V \end{pmatrix} = 4V \begin{pmatrix} x \\ y \end{pmatrix}.$$

The matrix determinant is $1 - |z|^4 > 0$. Its inverse gives $\nabla V = 4(x, y)V/(1 - |z|^2)$, hence $\nabla[(1 - |z|^2)^2 V] = 0$. A locally integrable function with zero distributional gradient on the connected disk is constant: convolve on relatively compact disks, where the convolution is constant, and let the smoothing parameter tend to zero. This proves (44.7) with $v \geq 0$.

The residual-variance inequality (34.8), together with $\Delta \log \rho_1 = 8/(1 - |z|^2)^2$, gives $8 \geq 2\beta + \beta^2 v$. If $v = 0$, then $\beta \overline{b(z, \Xi)} = \nabla \log \rho_1(z)$ Campbell-almost everywhere. The canonical Ward identity would consequently have a deterministic score. The continuous two-point correlation and its vanishing on the diagonal, proved earlier for every subsequential limit, allow application of Lemma 35.3. That lemma implies that the process has at most one point almost surely. This contradicts $\mathbb{E}\Xi(\mathbb{D}) = \int_{\mathbb{D}} c(1 - |z|^2)^{-2} dA = \infty$. Thus $v > 0$ and (44.8) follows.

Finally, the event $\{\Xi(\mathbb{D}) = m\}$ is invariant for every finite m . If it had positive probability for $m \geq 1$, the conditional law on that event would have an invariant intensity measure of total mass m . This measure has a locally integrable density, since it is bounded by the unconditional intensity divided by that positive probability. The preceding invariant-density argument makes it a constant multiple of λ . This is incompatible with finite positive total mass. Taking the union over $m \geq 1$ proves the final assertion. $\square$

Remark 44.4 (Why a score identity does not determine the DLR count weights). Even an explicit canonical score does not in general determine the activity. For every $a > 0$, the Poisson process with intensity $a\lambda$ has the same score $b(z, \eta) = 4\bar{z}/[\beta(1 - |z|^2)]$. Indeed, on a compact window its conditional density given its count m is proportional to $\prod_{j=1}^m (1 - |z_j|^2)^{-2}$; ordinary integration by parts in each coordinate gives this score. The factors $a^m/m!$ affect the count law and disappear under the spatial derivative. These Poisson laws are not candidates for the repulsive limits considered here, but they give an exact demonstration of the information that a spatial Ward identity omits. A full DLR theorem must also identify the conditional probabilities of the different counts.

Remark 44.5 (What remains unproved). The invariance of T and the transformation law for b_H are unconditional statements. Invariance of P is an additional hypothesis in Theorem 44.3. For $\beta = 2$ the Bergman kernel proves it, and for $\beta \geq 4$ it follows from the empty-limit theorem. For $0 < \beta < 4$, $\beta \neq 2$, neither the score covariance calculation nor the Gaussian-force regression removes the missing $o(N^{-1})$ estimate in Corollary 25.2. A fixed automorphism still produces a holomorphic tilt with a coefficient of order N .

45 Why an invariant DLR equation must be renormalized

The distinction between a Green interaction and a compensated Green interaction is essential. The following obstruction applies to every isometry-invariant point process, without a moment assumption. Write

$$\delta(a, w) = \left| \frac{a - w}{1 - a\bar{w}} \right|, \quad g(a, w) = -\log \delta(a, w), \quad d\lambda(a) = \frac{dA(a)}{(1 - |a|^2)^2}.$$

Theorem 45.1 (No nonempty invariant Blaschke configuration). *Let Ξ be a locally finite point process on $\mathbb{D}$ whose law is invariant under all disk automorphisms. Almost surely, precisely one of the following alternatives holds:*

$$\Xi = \emptyset; \quad \sum_{w \in \Xi} g(a, w) = +\infty \quad \text{for every } a \in \mathbb{D}.$$

Consequently a nonempty invariant process cannot be supported on Blaschke configurations, that is, configurations for which $\sum_w (1 - |w|) < \infty$.

Proof. For a deterministic locally finite configuration ξ define

$$p_\xi(a) = \prod_{w \in \xi} \delta(a, w) = \lim_{R \uparrow 1} \prod_{w \in \xi: |w| \leq R} \delta(a, w) = \exp \left\{ - \sum_{w \in \xi} g(a, w) \right\}.$$

All factors belong to $[0, 1]$, so this limit exists, is independent of the order of multiplication, and lies in $[0, 1]$. Each finite product is the modulus of a holomorphic finite Blaschke product, hence is subharmonic. These subharmonic functions decrease as R increases. Their bounded nonnegative limit is upper semicontinuous, and passing to the limit in the submean inequality shows that p_ξ is itself subharmonic. In particular its distributional Laplacian is nonnegative.

The invariance of pseudohyperbolic distance gives $p_{\varphi\xi}(\varphi(a)) = p_\xi(a)$ for every disk automorphism φ . By transitivity and invariance in law, $\mathbb{E}p_\Xi(a) = c$ is independent of a . For every nonnegative $\chi \in C_c^\infty(\mathbb{D})$, Fubini's theorem, justified by $0 \leq p_\Xi \leq 1$, gives

$$0 \leq \mathbb{E}\langle \Delta p_\Xi, \chi \rangle = \int_{\mathbb{D}} c \Delta \chi dA = 0.$$

Thus $\langle \Delta p_\Xi, \chi \rangle = 0$ almost surely. Choose a countable family of nonnegative smooth compactly supported functions such that every compact subset of the disk is contained in the strictly positive set of one of them. Since Δp_Ξ is a positive measure, it follows on one probability-one event that $\Delta p_\Xi = 0$ throughout the disk. A locally integrable function with zero distributional Laplacian has a harmonic representative. The subharmonic representative p_Ξ agrees with this harmonic function everywhere: its averages over shrinking disks converge to its value, by the submean inequality and upper semicontinuity.

If $w \in \Xi$, then $p_\Xi(w) = 0$ because one of the product factors is zero. The minimum principle for the nonnegative harmonic function p_Ξ therefore implies $p_\Xi \equiv 0$. If Ξ is empty, the product equals one identically. This proves the dichotomy. Finally, $g(0, w) = -\log |w|$ is comparable to $1 - |w|$ near the boundary; finitely many interior points do not change convergence of the corresponding series away from its logarithmic poles. This gives the last assertion. $\square$

Corollary 45.2 (The uncompensated invariant Gibbs equation has no solution). *Fix $\beta > 0$ and an activity $z > 0$. For a relatively compact open $B \subset \mathbb{D}$ and an exterior configuration $Y \subset B^c$, consider the unrenormalized grand-canonical kernel*

$$\gamma_B(d\eta \mid Y) \propto \sum_{m=0}^{\infty} \frac{z^m}{m!} \exp \left\{ -\beta \sum_{i < j} g(a_i, a_j) - \beta \sum_{i=1}^m \sum_{w \in Y} g(a_i, w) \right\} \prod_{i=1}^m d\lambda(a_i), \quad (45.1)$$

where $(a_1, \dots, a_m) \in B^m$ and the $m = 0$ term is one. There is no disk-automorphism-invariant locally finite probability law satisfying the DLR equations for these kernels.

Proof. The kernels are well defined: nonnegativity of g bounds their normalizing constants between 1 and $\exp\{z\lambda(B)\}$. Suppose an invariant law P satisfied the DLR equations. By

Theorem 45.1, its nonempty configurations have infinite Green potential everywhere. Such a configuration is necessarily infinite. Removing its finitely many points in B leaves an exterior Y for which $\sum_{w \in Y} g(a, w) = \infty$ for Lebesgue-almost every $a \in B$: the removed finite sum is finite outside its finite set of poles. For such an exterior, every term with $m \geq 1$ in (45.1) vanishes. Hence $\gamma_B(\cdot \mid Y)$ is concentrated on the empty configuration.

If $\Xi \cap B$ is nonempty, then Ξ is nonempty and infinite, so its exterior is of the preceding type. The DLR equation consequently gives $P(\Xi \cap B \neq \emptyset) = 0$. Applying this to a countable exhaustion by relatively compact disks yields $P(\Xi = \emptyset) = 1$. But for the empty exterior the $m = 1$ term in (45.1) has positive mass $z\lambda(B)$ whenever B has positive area. Thus the empty law does not satisfy the same DLR equation, a contradiction. $\square$

Remark 45.3. The corollary rules out only the displayed *uncompensated* specification. It does not rule out invariant processes, compensated Gibbs equations, or canonical equations restricted to a fixed number of particles. In particular, the Bergman process is nonempty and invariant, and its correct conditional equation uses a regularized Green field.

45.1 The deterministic term in an anchored Green field

Here is an elementary calculation that locates one normalization issue in such regularizations. Let

$$d\mu(w) = \frac{dA(w)}{\pi(1 - |w|^2)^2}, \quad I_R(a) = \int_{|w| < R} g(a, w) d\mu(w).$$

For $|a| < R < 1$,

$$I_R(a) = -\frac{R^2 \log R}{1 - R^2} - \frac{1}{2} \log(1 - R^2) + \frac{1}{2} \log(1 - |a|^2). \quad (45.2)$$

Indeed, the circular average of $\log |1 - a\bar{w}|$ is zero, whereas the circular average of $\log |a - w|$ is $\log \max\{|a|, |w|\}$. Thus

$$I_R(a) = \int_0^R \frac{-2s \log \max\{|a|, s\}}{(1 - s^2)^2} ds.$$

Integration at $a = 0$ gives the first two terms in (45.2). Differentiating the difference with respect to $u = |a|$ gives $-u/(1 - u^2)$, and the difference vanishes at $u = 0$; integration yields the last term.

Suppose a point process has intensity μ and its centered Green statistics

$$H_R(a) = \sum_{w \in \Xi: |w| < R} g(a, w) - I_R(a)$$

converge in L^2 to $H(a)$ for each fixed a . Then

$$\lim_{R \uparrow 1} \sum_{w \in \Xi: |w| < R} (g(a, w) - g(0, w)) = H(a) - H(0) + \frac{1}{2} \log(1 - |a|^2) \quad (45.3)$$

in L^2 . The position-dependent last term must be retained when using origin-centered cutoffs and anchoring by subtraction at zero. Full centering by $I_R(a)$ removes this term and, when the centered limit is independent of the exhaustion, gives an isometry-covariant field.

For completeness, the required L^2 convergence holds for the Bergman process by an elementary variance estimate. Its determinantal correlations imply, for real compactly supported $f \geq 0$,

$$\text{Var} \left(\sum_{w \in \Xi} f(w) \right) = \int f(w)^2 d\mu(w) - \iint f(w)f(v) |K(w, v)|^2 dA(w)dA(v) \leq \int f(w)^2 d\mu(w).$$

For $R < S$, apply this inequality to $f(w) = g(a, w)\mathbf{1}_{\{R < |w| < S\}}$. The logarithmic pole is square integrable on compact sets; near the unit circle, $g(a, w) = O_a(1 - |w|)$. Consequently $\int g(a, w)^2 d\mu(w) < \infty$, and the annular square integral tends to zero as $R \uparrow 1$. This proves the Cauchy property of $H_R(a)$. The same argument compares arbitrary increasing exhaustions, since each of the two set differences can be treated separately. Hence the fully centered Bergman Green field is independent of the exhaustion.

45.2 A precise compensated specification and its missing input

The following conditional construction separates algebraic consistency from the probabilistic assertion that a limit law satisfies the resulting equations.

Proposition 45.4 (A finite-addition criterion for a DLR specification). *Let Ω be a measurable class of locally finite configurations, stable under finite additions and deletions. Suppose there is a measurable real-valued function $H(a, Y)$, defined for $Y \in \Omega$ and $a \notin Y$, with the finite-addition identity*

$$H(a, Y \cup \{b\}) = H(a, Y) + g(a, b) \quad (a \neq b, a, b \notin Y). \quad (45.4)$$

Assume that for each relatively compact open B and each $Y \in \Omega$ supported on B^c , the function $a \mapsto H(a, Y)$ is bounded below on B . Fix $z > 0$ and $\beta > 0$. Then

$$\gamma_B^{H, z}(d\eta \mid Y) = \frac{1}{Z_B(Y)} \sum_{m=0}^{\infty} \frac{z^m}{m!} \exp \left\{ -\beta \sum_{i < j} g(a_i, a_j) - \beta \sum_{i=1}^m H(a_i, Y) \right\} \prod_{i=1}^m d\lambda(a_i) \quad (45.5)$$

defines finite, normalized, consistent kernels on B ; diagonals are irrelevant to the Lebesgue integrals. If Ω is invariant under disk automorphisms and

$$H(\varphi(a), \varphi(Y)) = H(a, Y),$$

the kernels are covariant under these automorphisms as well.

Proof. If $h_B(Y) = \inf_{a \in B} H(a, Y) > -\infty$, nonnegativity of g gives

$$1 \leq Z_B(Y) \leq \exp \left\{ ze^{-\beta h_B(Y)} \lambda(B) \right\} < \infty.$$

For $B \subset C$, divide a configuration in C into $\eta \subset B$ and $\zeta \subset C \setminus B$. The energy in (45.5) splits into the energy of ζ in the exterior Y , plus the energy of η in the exterior $Y \cup \zeta$, because

$$\sum_{a \in \eta} H(a, Y \cup \zeta) = \sum_{a \in \eta} H(a, Y) + \sum_{a \in \eta} \sum_{b \in \zeta} g(a, b).$$

The factorial reference measure splits over the two disjoint sets. Consequently the conditional law of η , given ζ , under the C -kernel is exactly the B -kernel, proving consistency. Covariance follows by change of variables from invariance of g and λ and the assumed covariance of H . $\square$

A candidate field associated with a constant hyperbolic background $c\lambda$ is

$$H(a, Y) = \lim_{n \rightarrow \infty} \left\{ \sum_{w \in Y \cap E_n} g(a, w) - c \int_{E_n} g(a, w) d\lambda(w) \right\}, \quad (45.6)$$

where E_n exhausts the disk. Whenever this defines a measurable field with the stated lower bounds and with a limit independent of the exhaustion, finite addition proves (45.4) and

change of variables proves isometry covariance. Thus the preceding proposition applies. These are hypotheses in the present discussion for general β , not established properties of all gas limits.

There is also a useful necessary identity that retains the particle count information. If a law P supported on Ω satisfies these DLR equations, then for every nonnegative measurable F supported in a relatively compact set in its first argument,

$$\mathbb{E}_P \sum_{a \in \Xi} F(a, \Xi \setminus \{a\}) = z \int_{\mathbb{D}} \mathbb{E}_P \left[e^{-\beta H(a, \Xi)} F(a, \Xi) \right] d\lambda(a). \quad (45.7)$$

To prove it, condition on the exterior of a bounded set containing the support of the first argument, substitute (45.5), and distinguish one particle in the term with m particles. The factor $m/m!$ becomes $1/(m-1)!$, while (45.4) gives precisely the exponential on the right. All integrands are nonnegative, so Tonelli's theorem justifies every rearrangement.

The activity z cancels from the conditional density when the number of particles in B is fixed. Consequently, convergence of density ratios for configurations with the same number of particles cannot, by itself, identify (45.7) or the full DLR equation. A proof for the original gas must in addition identify the relative weights of the different particle counts.

45.3 A conditional compatibility test for hyperbolic invariance

This subsection states an implication, not a new existence or invariance theorem for the gas. It records a normalization test that any proposed canonical DLR derivation should pass.

Proposition 45.5 (Covariance forbids an uncompensated one-body factor). *Suppose P is a nonempty, disk-automorphism-invariant point-process law. Suppose its conditional law in each relatively compact open set B , given the exterior Y and the inside count m , has a strictly positive density away from collisions of the form*

$$\frac{1}{Z_{B,Y,m}} \prod_{i < j} \delta(a_i, a_j)^\beta \prod_{i=1}^m e^{-\beta H(a_i, Y)} (1 - |a_i|^2)^q d\lambda(a_i), \quad (45.8)$$

where q is deterministic and H is isometry covariant. Then $q = 0$.

Proof. Some relatively compact open disk B has $P(\Xi(B) \geq 1) > 0$. Fix a nonrotation disk automorphism φ . Invariance of P identifies the push-forward of its conditional law on B , given (Y, m) , with its conditional law on $\varphi(B)$, given $(\varphi(Y), m)$, almost surely. The Green interaction, H , and λ are covariant, so the ratio between the two displayed densities, pulled back to B^m , is proportional to

$$\prod_{i=1}^m \left(\frac{1 - |\varphi(a_i)|^2}{1 - |a_i|^2} \right)^q = \prod_{i=1}^m |\varphi'(a_i)|^q.$$

The equality of the normalized conditional laws requires this product to be constant almost everywhere on B^m , for a conditional event with some $m \geq 1$. Their densities are strictly positive there away from collisions, so continuity makes the product constant everywhere off the collision diagonals. Fixing all coordinates but one shows that $|\varphi'|^q$ is constant on an open set. For a nonrotation automorphism $|\varphi'(a)| = (1 - |u|^2)/|1 - \bar{u}a|^2$ with $u \neq 0$, which is nonconstant on every open set. Hence $q = 0$. $\square$

To see why this test is relevant to the original gas, suppose, in addition to assumptions not yet proved for general β , that its canonical limiting density were established in the anchored form

$$\frac{1}{\tilde{Z}_{B,Y,m}} \prod_{i < j} \delta(a_i, a_j)^\beta \prod_{i=1}^m (1 - |a_i|^2)^{2-\beta/2} e^{-\beta C_Y(a_i)} d\lambda(a_i). \quad (45.9)$$

Here B contains the origin, and C_Y denotes the limit of the origin-anchored exterior sums $\sum_{w \in Y, |w| < R} (g(a, w) - g(0, w))$. Suppose the invariant law has intensity $c\lambda$, with $0 < c < \infty$, and suppose its fully centered field H exists independently of the exhaustion. Formula (45.2), with the intensity multiplied by $c\pi$, then gives

$$C_Y(a) = H(a, Y) - H(0, Y) + \frac{c\pi}{2} \log(1 - |a|^2).$$

The factor involving $H(0, Y)$ is constant when m is fixed. Thus (45.9) becomes (45.8) with

$$q = 2 - \frac{\beta}{2} - \frac{\beta c\pi}{2}.$$

If this same canonical law is established on all relatively compact sets, covariance consequently forces

$$c = \frac{4/\beta - 1}{\pi}. \quad (45.10)$$

At $\beta = 2$ this agrees with the Bergman intensity. At $\beta = 4$ it vanishes. Equation (45.10) is only a compatibility consequence of the explicitly stated canonical-DLR and field hypotheses. Neither those hypotheses nor hyperbolic invariance at other $0 < \beta < 4$ have been proved here.

46 A complete compensated DLR equation at $\beta = 2$

At the exactly solvable temperature the conditional law can be identified including its conditional particle number. We give a derivation, because it also identifies the interaction that a proposed limiting DLR equation must reproduce. Throughout this section $\mathbf{B}$ denotes the Bergman process, and

$$d\nu(z) = \frac{dA(z)}{\pi(1 - |z|^2)^2}, \quad \delta(z, w) = \left| \frac{z - w}{1 - z\bar{w}} \right|, \quad g(z, w) = -\log \delta(z, w).$$

Thus ν is precisely the first correlation measure of $\mathbf{B}$. Our normalization of hyperbolic area includes the factor $1/\pi$.

46.1 An intrinsic compensated potential

Lemma 46.1 (Centered potentials and exponential integrability). *For every $q \in \mathbb{D}$ there is a real random variable H_q such that, for any increasing exhaustion $E_j \uparrow \mathbb{D}$ by relatively compact Borel sets,*

$$H_q(X) = \lim_{j \rightarrow \infty} \left\{ \sum_{x \in X \cap E_j} g(q, x) - \int_{E_j} g(q, z) d\nu(z) \right\} \quad \text{in } L^2(\mathbf{B}). \quad (46.1)$$

The result is independent of the exhaustion. There is a jointly measurable version for Lebesgue-almost every q and $\mathbf{B}$ -almost every X . Put $Y_q = -2H_q$. For every $p > 0$, the exponentials of the corresponding truncated pY_q converge in L^1 . In particular $\exp(Y_q)$ is strictly positive and integrable.

Proof. For a real compactly supported test function f the determinantal variance identity gives

$$\text{Var} \sum_{x \in X} f(x) = \int f^2 d\nu - \iint f(z)f(w)|K(z, w)|^2 dA(z)dA(w) \leq \int f^2 d\nu.$$

The second term is nonnegative: for bounded tests it is the squared Hilbert–Schmidt norm of the self-adjoint operator KfK , since K is an orthogonal projection. Truncation extends the bound to the required $L^2(\nu)$ tests. Hyperbolic invariance and polar integration give

$$\int_{\mathbb{D}} |\log \delta(q, z)|^2 d\nu(z) = \int_0^1 \frac{(\log s)^2}{(1-s)^2} ds = 2 \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

The variance bound therefore proves the Cauchy property and independence of the exhaustion. Applying the same bound after integrating q over a compact set yields jointly measurable L^2 limits, consistently on a countable exhaustion of the q variable.

For $f \leq 0$ of compact support the determinantal generating functional and $\log \det(I+A) \leq \text{Tr } A$ give

$$\mathbb{E} \exp \left(p \left\{ \sum_{x \in X} f(x) - \int f d\nu \right\} \right) \leq \exp \left(\int (e^{pf} - 1 - pf) d\nu \right). \quad (46.2)$$

One may first replace f by $\max(f, -M)$ and then pass to the limit; the sums are finite on its compact support and the deterministic integrals converge. Since $0 \leq e^{pu} - 1 - pu \leq p^2 u^2 / 2$ for $u \leq 0$, the right-hand side is bounded uniformly over truncations of $f(z) = \log \delta(q, z)^2$. Applying the bound with any larger exponent proves uniform integrability. Convergence in probability, already supplied by L^2 convergence of the centered statistics, then proves the asserted L^1 convergence of their exponentials. $\square$

In particular the compensation in (46.1) is intrinsic. A different exhaustion does not change the centered random variable. Removing finitely many points is a different operation: the compensating integral is retained, so the potential changes by the ordinary finite sum of their pair potentials.

Proposition 46.2 (Normalized Palm likelihood). *Let $\gamma = \lim_{n \rightarrow \infty} (\sum_{k=1}^n k^{-1} - \log n)$. The reduced Palm law $\mathbf{B}^{!q}$ at q is equivalent to $\mathbf{B}$, and its Radon–Nikodym derivative is*

$$\Psi_q(X) := \frac{d\mathbf{B}^{!q}}{d\mathbf{B}}(X) = e^{-\gamma} \exp(-2H_q(X)). \quad (46.3)$$

For finite additions or removals, wherever these expressions are defined,

$$\Psi_q(X \cup \{p_1, \dots, p_m\}) = \Psi_q(X) \prod_{j=1}^m \delta(q, p_j)^2. \quad (46.4)$$

Moreover, under $\mathbf{B}$, the single random variable Ψ_q has the exponential distribution of mean one.

Proof. It suffices to start with $q = 0$. Write $t = r^2$ and let $f_t(z) = \mathbf{1}_{\{|z|^2 < t\}} \log |z|^2$. Direct integration gives

$$I(t) := \int f_t d\nu = \frac{t \log t}{1-t} + \log(1-t). \quad (46.5)$$

Set $h_t = e^{f_t}$. In the orthonormal Bergman basis $e_k(z) = \sqrt{(k+1)/\pi} z^k$, the operator $K(1-h_t)K$ is diagonal, with eigenvalues

$$a_k(t) = (k+1) \int_0^t (1-s) s^k ds, \quad 0 \leq a_k(t) \leq \frac{1}{k+2}.$$

Its trace is $-\log(1-t)$. Consequently

$$\log \mathbb{E} \exp \left(\sum_{x \in X} f_t(x) - I(t) \right) = -\frac{t \log t}{1-t} + \sum_{k \geq 0} \{\log(1-a_k(t)) + a_k(t)\}.$$

The summands are dominated in absolute value by a constant times $(k+2)^{-2}$. Their limiting sum is

$$\lim_{M \rightarrow \infty} \left\{ -\log(M+2) + \sum_{k=0}^M \frac{1}{k+2} \right\} = \gamma - 1.$$

Since $-t \log t / (1-t) \rightarrow 1$, Lemma 46.1 proves $\mathbb{E}e^{Y_0} = e^\gamma$.

We also identify the tilted law directly. Weighting the Bergman process by $\prod_{x \in X} h_t(x)$ and normalizing gives a determinantal process with kernel

$$K_t(z, w) = \sqrt{h_t(z)h_t(w)} \sum_{k \geq 0} \frac{k+1}{\pi(1-a_k(t))} z^k \bar{w}^k. \quad (46.6)$$

For completeness, the finite-rank version follows by multiplying each row and column of the joint determinant by $\sqrt{h_t}$ and normalizing the resulting orthogonal functions; radially makes their Gram matrix diagonal. Passing through the increasing Bergman projections gives (46.6). The passage is justified by convergence of their Fredholm generating functionals on compact sets: $1 - h_t$ has compact support, while $1 - a_k(t) \geq 1/2$ bounds all inverse Gram coefficients. The same bound gives uniform convergence of the kernel series on compact sets and convergence of its diagonal integrals.

As $t \uparrow 1$, the kernel in (46.6) converges locally, including its diagonal integrals, to

$$\frac{|z||w|}{\pi} \sum_{k \geq 0} (k+2) z^k \bar{w}^k.$$

After the harmless gauge multiplication by $(z/|z|)(\bar{w}/|w|)$, this is

$$\frac{z\bar{w}}{\pi} \sum_{k \geq 0} (k+2) z^k \bar{w}^k = K(z, w) - \frac{1}{\pi}.$$

The latter is the reduced Palm kernel at zero, as is seen by applying the Schur-complement identity to each correlation determinant with one argument fixed at zero. On the other hand, Lemma 46.1 shows that the normalized densities of these tilted laws converge in $L^1(\mathbf{B})$ to $e^{-\gamma}e^{Y_0}$. This proves (46.3) at zero. The Bergman kernel identity

$$K(\varphi z, \varphi w) \varphi'(z) \overline{\varphi'(w)} = K(z, w)$$

for disk automorphisms, together with the exhaustion independence of the centered statistic, proves the assertion at every q . Positivity of the exponential proves equivalence. Finite multiplicativity follows already at the truncated-product level.

Here is also a check of the complete one-point distribution of the likelihood. For $p > 0$, radial diagonalization gives

$$\mathbb{E}e^{pY_0} = \prod_{n=1}^{\infty} \frac{e^{p/n}}{1+p/n} = e^{\gamma p} \Gamma(1+p). \quad (46.7)$$

To justify the first limit without a formal infinite-product interchange, set $b_{n,t} = \int_0^1 (e^{p f_t(\sqrt{s})} - 1) n s^{n-1} ds$. Then $|b_{n,t}| \leq p/(n+p)$ and

$$0 \leq b_{n,t} - p \int_0^1 f_t(\sqrt{s}) n s^{n-1} ds \leq \frac{p^2}{2} \int_0^1 (\log s)^2 n s^{n-1} ds = \frac{p^2}{n^2}.$$

The identity $\log(1+b) = b + O(b^2)$ bounds the remaining tail uniformly by a summable multiple of n^{-2} ; the finitely many small n cause no difficulty. Dominated convergence therefore gives the first product in (46.7). The second equality is Euler's product for the gamma function, equivalently obtained by applying the beta integral and letting its integer parameter increase. It follows that $\mathbb{E}\Psi_q^m = m!$ for every integer $m \geq 0$. These moments determine a probability law: for $0 < s < 1$, monotone convergence gives $\mathbb{E}e^{s\Psi_q} = \sum_{m \geq 0} s^m = (1-s)^{-1}$. They therefore identify the exponential law of mean one. $\square$

46.2 The full conditional law, including the number of points

Let $B \Subset \mathbb{D}$ be a nonempty open set and $C = \mathbb{D} \setminus B$. For an exterior configuration $Y = X \cap C$, define $H_q(Y)$ by the same centered expression as in (46.1), with the sum over Y and the integral still over the whole truncating set. Equivalently, if $X = Y \cup \xi$ is a typical full configuration,

$$H_q(Y) = H_q(X) - \sum_{p \in \xi} g(q, p), \quad q \in B.$$

This definition is measurable and independent of the chosen completion for exterior-almost every Y and ν -almost every q . Indeed, the truncated exterior expressions depend only on (q, Y) . Apply the joint L^2 construction in Lemma 46.1 along a subsequence and the exact finite-deletion identity. Their almost sure limit therefore also depends only on (q, Y) ; disintegration gives the stated almost-everywhere assertion. Set

$$W_Y(q) = e^{-\gamma} \exp(-2H_q(Y)).$$

Theorem 46.3 (Compensated grand-canonical Bergman specification). *For $\mathbb{B}|_C$ -almost every Y , the conditional law of $X \cap B$ is*

$$\Gamma_B(d\xi \mid Y) = \frac{1}{Z_B(Y)} \sum_{m=0}^{\infty} \frac{1}{m!} \left[\prod_{i=1}^m W_Y(q_i) \prod_{1 \leq i < j \leq m} \delta(q_i, q_j)^2 \right] \prod_{i=1}^m d\nu(q_i), \quad (46.8)$$

where the m -th term is pushed forward by $(q_1, \dots, q_m) \mapsto \{q_1, \dots, q_m\}$, the $m = 0$ term is a unit mass at the empty configuration, and $1 \leq Z_B(Y) < \infty$. Every integer $m \geq 0$ has strictly positive conditional probability. The conditional law is therefore equivalent to a Poisson process with intensity $\nu|_B$. In particular the Bergman process is not number rigid.

Conditionally also on $\#(X \cap B) = m$, its symmetric ordered density with respect to $\nu^{\otimes m}$ is proportional to

$$\exp \left[-2 \sum_{i < j} g(q_i, q_j) - 2 \sum_{i=1}^m H_{q_i}(Y) \right]. \quad (46.9)$$

Thus the pair potential in this conditional law is the hyperbolic Green potential. Both its divergent exterior interaction and its conditional particle-number normalization are explicitly accounted for.

Proof. The reduced Campbell identity and Proposition 46.2 give the following Georgii–Nguyen–Zessin identity, first for nonnegative measurable tests F and then for integrable ones:

$$\mathbb{E} \sum_{q \in X} F(q, X \setminus \{q\}) = \int_{\mathbb{D}} \mathbb{E}[F(q, X) \Psi_q(X)] d\nu(q). \quad (46.10)$$

Indeed both sides equal the integral of F against the reduced Campbell measure, after disintegration at its distinguished point. Restrict q to B , multiply the test by any bounded measurable function of $Y = X \cap C$, and disintegrate over Y . For almost every Y the conditional finite process Q_Y on B then has insertion rate, relative to ν , given by

$$c_Y(q, \xi) = W_Y(q) \prod_{p \in \xi} \delta(q, p)^2.$$

This follows from (46.4). The passage can be performed first for a countable class of cylinder tests that generates the relevant sigma fields, and then for all nonnegative tests by a monotone-class argument.

We spell out why this identity determines also the probabilities of the different counts. Let $J_{Y,m}$ be the symmetric Janossy measure: its total mass is $m!Q_Y(\#B = m)$. The conditional Campbell identity implies the recursion, in the sense of measures,

$$J_{Y,m+1}(dq, dq_1 \cdots dq_m) = c_Y(q, \{q_1, \dots, q_m\}) d\nu(q) J_{Y,m}(dq_1 \cdots dq_m). \quad (46.11)$$

For example, this follows by using a test that vanishes unless the remaining configuration has exactly m points and symmetrizing its arguments; the factors $m!$ cancel. It also shows inductively that all the Janossy measures have densities once $J_{Y,0}$ is specified. If $J_{Y,0} = 0$, the recursion makes every $J_{Y,m}$ zero, contradicting the fact that Q_Y is supported on finite configurations. Hence $J_{Y,0} > 0$. Iteration gives

$$J_{Y,m}(dq_1 \cdots dq_m) = J_{Y,0} \prod_i W_Y(q_i) \prod_{i < j} \delta(q_i, q_j)^2 \prod_i d\nu(q_i).$$

Summing its mass divided by $m!$ gives $Z_B(Y) = J_{Y,0}^{-1} < \infty$ and proves (46.8). Every integrand is positive off the diagonals, for $\nu^{\otimes m}$ -almost every tuple. As $0 < \nu(B) < \infty$, every term has strictly positive integral. The same formula proves equivalence to Poisson measure and the canonical conditional law. Nonrigidity follows because an exterior-measurable count would have a conditionally deterministic distribution, whereas here every possible finite count has positive conditional probability. $\square$

Corollary 46.4 (Correctly normalized L -ensemble). *The positive operator on $L^2(B, \nu)$ with kernel*

$$L_Y(z, w) = \frac{\sqrt{(1 - |z|^2)W_Y(z)} \sqrt{(1 - |w|^2)W_Y(w)}}{1 - z\bar{w}} \quad (46.12)$$

is trace class and satisfies

$$Z_B(Y) = \det(I + L_Y), \quad Q_Y(\#B = 0) = \det(I + L_Y)^{-1}.$$

The conditional process is determinantal, with kernel $L_Y(I + L_Y)^{-1}$. In particular $L_Y(z, z) = W_Y(z)$; this diagonal identity fixes the normalization unambiguously.

Proof. The Cauchy determinant identity gives

$$\det(L_Y(q_i, q_j))_{i,j=1}^m = \prod_i W_Y(q_i) \prod_{i < j} \delta(q_i, q_j)^2.$$

Positivity follows also from the power-series expansion of $(1 - z\bar{w})^{-1}$. The $m = 1$ coefficient of the already finite partition function shows $\int_B W_Y d\nu < \infty$. Writing L_Y as the sum of the rank-one kernels generated by $\sqrt{(1 - |z|^2)W_Y(z)}z^k$, the sum of their traces is exactly $\int_B W_Y d\nu$. Thus the sum converges in trace norm and defines a positive trace-class operator. Its Fredholm expansion is (46.8); the generating functional then identifies the correlation kernel $L_Y(I + L_Y)^{-1}$. $\square$

Corollary 46.5 (A curvature obstruction to a naive conditional limit). *For exterior-almost every Y , the compensated potential has a smooth representative on B , and*

$$h_Y(q) := H_q(Y) + \frac{1}{2} \log(1 - |q|^2)$$

is harmonic there. The one-particle conditional density p_Y with respect to ordinary area, after additionally conditioning on $\#B = 1$, therefore satisfies

$$p_Y(q) = \frac{1}{Z_Y} \frac{e^{-2h_Y(q)}}{1 - |q|^2}, \quad \Delta \log p_Y(q) = \frac{4}{(1 - |q|^2)^2}. \quad (46.13)$$

If B is connected, this conditional probability cannot be obtained as a weak limit on B of the finite- N , $\beta = 2$ conditional one-particle laws with any sequence of deterministic exterior configurations in $\mathbb{D} \setminus B$.

Proof. Use centered disks for the truncation. The distributional identity $\Delta_q g(q, z) = -2\pi\delta_z(q)$ gives, for compactly supported tests in B and all truncating disks containing $\bar{B}$,

$$\Delta_q \left\{ \sum_{y \in Y \cap E_j} g(q, y) - \int_{E_j} g(q, z) d\nu(z) \right\} = \frac{2}{(1 - |q|^2)^2}.$$

Use the joint L^2 construction and finite deletion to pass to the limit against the test functions. Explicitly, on each compact q -window the deleted sum has finite joint second moment: expand its square, use the compact bounds on the first two correlation functions, and the local integrability of $\log^2 |q - p|$. Thus finite deletion preserves the required joint L^2 control. Since $\Delta \log(1 - |q|^2) = -4/(1 - |q|^2)^2$, the displayed h_Y is distributionally harmonic. Weyl's lemma gives its smooth harmonic representative. Substitution into the $m = 1$ term of (46.8), including the factor $d\nu/dA$, proves (46.13).

For a finite exterior configuration $\{y_1, \dots, y_n\}$, the conditional one-particle density is $|F_n(q)|^2$, where

$$F_n(q) = \frac{\prod_{j=1}^n (q - y_j)}{\left(\int_B \prod_{j=1}^n |u - y_j|^2 dA(u) \right)^{1/2}}.$$

Each F_n is holomorphic and has no zero in B , and $\int_B |F_n|^2 dA = 1$. The submean inequality makes this family locally bounded. Montel's theorem supplies a locally uniformly convergent subsequence with holomorphic limit F . If the probability measures $|F_n|^2 dA$ converge weakly to a probability measure on B , testing against compactly supported continuous functions identifies that measure as $|F|^2 dA$, so F is not identically zero. Hurwitz's theorem implies that F has no zero in connected B . Its limiting log density is consequently harmonic, in contradiction with (46.13). $\square$

This obstruction does not contradict convergence of the point processes. It demonstrates that conditioning on the full exterior and taking the infinite-particle limit do not commute in the required way; the correct limiting conditional law retains an effect of boundary randomness.

46.3 Isometry covariance and the scope of this specification

Proposition 46.6 (Covariance under every hyperbolic isometry). *The Bergman process is invariant under every isometry of the hyperbolic disk, including orientation-reversing ones. For each such isometry φ , compatible almost-everywhere versions satisfy*

$$H_{\varphi q}(\varphi X) = H_q(X), \quad \Psi_{\varphi q}(\varphi X) = \Psi_q(X),$$

and the conditional kernels satisfy

$$\Gamma_{\varphi B}(\varphi A \mid \varphi Y) = \Gamma_B(A \mid Y)$$

for exterior-almost every Y and every measurable event A of configurations in B .

Proof. For orientation-preserving isometries, invariance follows from the Bergman kernel transformation identity used above, after changing the area variables in every correlation determinant. Reflection follows from $K(\bar{z}, \bar{w}) = \overline{K(z, w)}$: the determinants are real. Together

these generate all hyperbolic isometries. Both g and ν are invariant. Transform the exhaustion in (46.1); exhaustion independence gives the equality of centered potentials in L^2 and hence of their compatible almost-everywhere versions. Finite deletion gives the same relation for exterior configurations. Changing variables in each term of (46.8) proves the claimed covariance, including equality of its partition functions. $\square$

The formulas above identify the conditional laws of one specific, already constructed process. They do not prove uniqueness among all infinite-volume solutions of a compensated Green specification, and they do not identify a specification for the nonsolvable disk gases. In particular, finite-volume Euclidean pair factors and their conditional exterior fields cannot simply be passed term by term to the limit: already at $\beta = 2$, the factor

$$\prod_{i < j} \delta(q_i, q_j)^2 = |\Delta_m(q)|^2 \prod_{i < j} |1 - q_i \bar{q}_j|^{-2}$$

contains a two-body image term absent from the finite canonical conditional density with a fixed exterior configuration.

Remark 46.7 (Relation to the literature). The conditional Bergman law was explicitly studied by A. I. Bufetov [8]. The preceding self-contained derivation fixes the regularization, reference measure, factorial convention, and L -kernel normalization used here. The Palm likelihood has expectation one, and the L -kernel has diagonal W_Y .

47 Interpolation lower bounds and nonempty limits for $0 < \beta \leq 1$

This section gives a lower bound, rather than a compactness or conditional-law argument. Its only temperature restriction comes from the elementary inequality $|u + v|^\beta \leq |u|^\beta + |v|^\beta$ for $0 < \beta \leq 1$. In particular, it proves that an empty subsequential limit is impossible throughout this interval.

Theorem 47.1 (An interpolation lower bound). *Fix $0 < \beta \leq 1$. If f is a nonzero polynomial of degree strictly less than N , then, for every $a \in \mathbb{D}$,*

$$\rho_{1,N}(a) \geq \frac{|f(a)|^\beta}{\int_{\mathbb{D}} |f(z)|^\beta dA(z)}. \quad (47.1)$$

More generally, let $1 \leq p \leq N$, and let F be a nonzero alternating polynomial in p variables, of degree strictly less than N in each individual variable. Then

$$\rho_{p,N}(\mathbf{a}) \geq \frac{p! |F(\mathbf{a})|^\beta}{\int_{\mathbb{D}^p} |F(\mathbf{z})|^\beta dA^p(\mathbf{z})} \quad (\mathbf{a} \in \mathbb{D}^p). \quad (47.2)$$

In particular,

$$\rho_{p,N}(\mathbf{a}) \geq \frac{p!}{Z_{p,\beta}} |\Delta_p(\mathbf{a})|^\beta \quad (N \geq p). \quad (47.3)$$

Proof. For a configuration $\mathbf{z} = (z_1, \dots, z_N)$ of distinct points, write

$$\ell_i(a; \mathbf{z}) = \prod_{j \neq i} \frac{a - z_j}{z_i - z_j}.$$

Polynomial interpolation gives $f(a) = \sum_{i=1}^N \ell_i(a; \mathbf{z}) f(z_i)$ whenever $\deg f < N$. Subadditivity therefore gives the pointwise inequality

$$|f(a)|^\beta \leq \sum_{i=1}^N |\ell_i(a; \mathbf{z})|^\beta |f(z_i)|^\beta.$$

All summands are nonnegative. Multiplication by the gas density and Tonelli's theorem are therefore legitimate without a preliminary integrability assumption. For each i , cancellation of the factors involving z_i in the Vandermonde shows that

$$\begin{aligned} \mathbb{E}_{N,\beta} \sum_{i=1}^N |\ell_i(a; \mathbf{z})|^\beta |f(z_i)|^\beta \\ &= \frac{N}{Z_{N,\beta}} \left(\int_{\mathbb{D}} |f(z)|^\beta dA(z) \right) \int_{\mathbb{D}^{N-1}} |\Delta_{N-1}(\mathbf{y})|^\beta \prod_{j=1}^{N-1} |a - y_j|^\beta dA^{N-1}(\mathbf{y}) \\ &= \rho_{1,N}(a) \int_{\mathbb{D}} |f(z)|^\beta dA(z). \end{aligned}$$

This proves (47.1). The collision set discarded at the start has probability zero.

For completeness, the exact multilinear identity behind the p -point version is

$$F(\mathbf{a}) = \sum_{\substack{I \subset \{1, \dots, N\} \\ |I|=p}} \det[\ell_{i_k}(a_j; \mathbf{z})]_{j,k=1}^p F(z_{i_1}, \dots, z_{i_p}), \quad (47.4)$$

where $i_1 < \dots < i_p$ orders I . To prove it, apply the one-variable interpolation formula successively in all p variables. Terms with repeated indices vanish because F is alternating. Grouping the remaining ordered indices by their underlying set gives the determinant in (47.4).

The Cauchy determinant identity, or direct cancellation in the Lagrange formulas, gives in absolute value

$$\left| \det[\ell_{i_k}(a_j; \mathbf{z})]_{j,k=1}^p \right| = \frac{|\Delta_p(\mathbf{a})|}{|\Delta_p(\mathbf{z}_I)|} \frac{\prod_{j=1}^p \prod_{k \notin I} |a_j - z_k|}{\prod_{i \in I} \prod_{k \notin I} |z_i - z_k|}. \quad (47.5)$$

Initially the formula can be read away from all zero denominators; the determinant formula extends it everywhere else by continuity. Subadditivity applied to (47.4), followed by integration against $|\Delta_N|^\beta / Z_{N,\beta}$, cancels both the cross factors and the Vandermonde on $\mathbf{z}_I$. The variables indexed by I consequently contribute exactly $\int_{\mathbb{D}^p} |F|^\beta dA^p$. The remaining variables contribute the integral in the definition of $h_{p,N}$. As there are $\binom{N}{p}$ sets I , the resulting identity is

$$\begin{aligned} \mathbb{E}_{N,\beta} \sum_{|I|=p} |\det[\ell_{i_k}(a_j; \mathbf{z})]|^\beta |F(\mathbf{z}_I)|^\beta \\ &= \frac{\rho_{p,N}(\mathbf{a})}{p!} \int_{\mathbb{D}^p} |F(\mathbf{z})|^\beta dA^p(\mathbf{z}). \end{aligned}$$

This proves (47.2) for distinct a_j , and continuity proves it on the diagonals as well. Choosing $F = \Delta_p$, whose degree in each variable is $p-1$, proves (47.3). $\square$

Corollary 47.2 (Every limiting intensity dominates the Bergman intensity). *Let $0 < \beta \leq 1$, and let Ξ be any subsequential limit of the disk gas. Its one-point density satisfies*

$$\rho_1(a) \geq \frac{1}{\pi(1 - |a|^2)^2}, \quad a \in \mathbb{D}. \quad (47.6)$$

In particular, Ξ is not the almost surely empty process, and $\mathbb{E}\Xi(\mathbb{D}) = +\infty$. Its factorial densities satisfy

$$\rho_p(\mathbf{a}) \geq \frac{p!}{Z_{p,\beta}} |\Delta_p(\mathbf{a})|^\beta \quad (p \geq 1). \quad (47.7)$$

Every nonempty open $U \subset \mathbb{D}$ consequently has $\mathbb{P}\{\Xi(U) \geq p\} > 0$ for every positive integer p .

Proof. Fix $a \in \mathbb{D}$. The zero-free function

$$f_a(z) = (1 - \bar{a}z)^{-4/\beta}$$

uses the holomorphic logarithm equal to zero at $z = 0$ and is holomorphic on a neighborhood of $\bar{\mathbb{D}}$. Its Taylor polynomials converge uniformly on $\bar{\mathbb{D}}$. For each fixed polynomial, first pass to the subsequential limit in (47.1), using the local uniform convergence of correlation densities. Then pass to f_a by uniform convergence. The change of variables by a disk automorphism, or angular integration of the geometric series, gives

$$\int_{\mathbb{D}} |1 - \bar{a}z|^{-4} dA(z) = \frac{\pi}{(1 - |a|^2)^2}.$$

Also $|f_a(a)|^\beta = (1 - |a|^2)^{-4}$. Their quotient proves (47.6). Its integral over $\mathbb{D}$ is infinite.

Pass to the limit directly in (47.3) to obtain (47.7). Any nonempty open U contains p disjoint disks with compact closures in U . The Vandermonde is nonzero on their product, which has positive area. Thus the integral of ρ_p over that product is strictly positive. This is an expectation of a nonnegative factorial count and vanishes if $\Xi(U) < p$ almost surely. The last assertion follows. $\square$

Remark 47.3 (What is and is not concluded). The conclusion that the limiting law is not concentrated on the empty configuration does not by itself say that its empty-configuration probability is zero. Similarly, an infinite expected total count does not by itself prove an almost surely infinite total count. No monotonicity in N , no uniqueness, and no hyperbolic invariance is used or asserted here. The proof applies to $0 < \beta \leq 1$; extending its subadditivity step to $1 < \beta < 2$ is not legitimate.

48 Nonempty limits throughout $0 < \beta < 2$

The interpolation argument extends beyond its pointwise subadditivity range by using a boundary weight. The important input is a uniform radial bound under holomorphic product weights, including weights with zeros. The argument also allows the boundary weight to be centered at any point in the hyperbolic metric.

Lemma 48.1 (Radial domination with an arbitrary holomorphic weight). *Let $n \geq 1$, $\beta > 0$, and let h be holomorphic on a neighborhood of $\bar{\mathbb{D}}$, not identically zero. Give $\mathbb{D}^n$ density proportional to*

$$|\Delta_n(\mathbf{z})|^\beta \prod_{j=1}^n |h(z_j)|^\beta. \quad (48.1)$$

The ordered radial depths are jointly stochastically dominated by

$$T_j^{(n)} = \sum_{k=j}^n E_k, \quad E_k \text{ independent exponential of rate } a_k = 2k + \frac{\beta}{2}k(k-1).$$

The bound is independent of h . Consequently, for $p > 1$ and $q > 0$, there is a finite constant $A_{\beta,p,q}$, independent of n, h , such that

$$\mathbb{E} \left[\sum_{j=1}^n (1 - |z_j|^2)^p \right]^q \leq A_{\beta,p,q}. \quad (48.2)$$

The same bound holds with $|z_j|$ replaced by $|\phi(z_j)|$, where ϕ is any automorphism of $\mathbb{D}$.

Proof. Use the ordered ratios $t_k = r_k/r_{k+1}$, $r_{n+1} = 1$, as in the radial comparison proof. The density is an angular factor times $\prod_k t_k^{a_k-1}$. To check that this factor is nondecreasing in t_k , rotate the first k angles together. The factors crossing from these k particles to the others, together with $\prod_{i \leq k} h(z_i)$, are the modulus to the power β of a holomorphic function of the common variable $t_k e^{i\theta}$. Every remaining factor is nonnegative and independent of this variable. The circular submean inequality therefore gives the same monotonicity as before. For $k = n$ there are no crossing factors, but the product of the h factors is still holomorphic under the common dilation and rotation. Zeros of h cause no difficulty: $|h|^\beta$ is subharmonic for every $\beta > 0$. Product association now proves the asserted domination, with exactly the same beta reference law.

For precision, the moment bound can be written with the finite constant

$$A_{\beta,p,q} = \mathbb{E} \left[\sum_{j=1}^{\infty} (1 - e^{-2T_j})^p \right]^q, \quad T_j = \sum_{k=j}^{\infty} E_k.$$

Its finiteness follows from the already established weighted-depth moment estimate for this reference process. Alternatively, for $j \geq 2$ the exponential-moment estimate $\log \mathbb{E} e^{\lambda T_j} \leq 4\lambda/[\beta(j-1)]$ when $0 \leq \lambda \leq \beta j(j-1)/4$ gives $\mathbb{E} T_j^s \leq C_{\beta,s} j^{-s}$ for every $s > 0$. For $q \geq 1$, Minkowski's inequality and $1 - e^{-2t} \leq 2t$ then give

$$\left\| \sum_j (1 - e^{-2T_j})^p \right\|_{L^q} \leq C_{\beta,p,q} \sum_j j^{-p} < \infty;$$

for $0 < q < 1$, use Jensen's inequality after the first-moment bound. The term $j = 1$ is bounded by one and does not require a moment bound on T_1 of a different type.

Finally, after a disk automorphism $z = \phi^{-1}(u)$, the new density has the form (48.1), with a new holomorphic product weight. Indeed,

$$|\phi^{-1}(u) - \phi^{-1}(v)| = |u - v| |(\phi^{-1})'(u)|^{1/2} |(\phi^{-1})'(v)|^{1/2}.$$

The derivative factor in each variable is consequently $|(\phi^{-1})'|^{\beta(n-1)/2+2}$. Since the derivative is zero-free on a neighborhood of $\overline{\mathbb{D}}$, its required real power is the modulus of a holomorphic function, using a holomorphic logarithm on a slightly larger disk. Multiplying this function by $h \circ \phi^{-1}$ gives the new weight. The bound just proved does not depend on that weight or on ϕ . $\square$

Theorem 48.2 (A positive hyperbolic intensity bound below 2). *For every $0 < \beta < 2$ there exists $c_\beta > 0$ such that every subsequential limiting process of the unweighted disk gas satisfies*

$$\rho_1(a) \geq \frac{c_\beta}{\pi(1 - |a|^2)^2}, \quad a \in \mathbb{D}. \quad (48.3)$$

For $0 < \beta \leq 1$ one can take $c_\beta = 1$. For $1 < \beta < 2$, put $q = \beta - 1$ and choose any $p > 1$ such that $pq < 1$. One may take

$$c_\beta = \left(1 + \frac{A_{\beta,p,q}}{1 - pq} \right)^{-1} > 0, \quad (48.4)$$

with the constant from Lemma 48.1. In particular, no subsequence can converge to the almost surely empty process, and every limiting law has infinite expected total count.

Proof. The case $\beta \leq 1$ is Corollary 47.2. Suppose $1 < \beta < 2$, and fix $a \in \mathbb{D}$. Let ϕ_a be a disk automorphism taking a to zero and set

$$w_a(z) = (1 - |\phi_a(z)|^2)^p, \quad W_a(\mathbf{z}) = \sum_{i=1}^N w_a(z_i).$$

For every polynomial f of degree less than N , Lagrange interpolation and weighted Hölder give the pointwise inequality

$$|f(a)|^\beta \leq W_a(\mathbf{z})^q \sum_{i=1}^N \frac{|\ell_i(a; \mathbf{z}) f(z_i)|^\beta}{w_a(z_i)^q}. \quad (48.5)$$

All weights are strictly positive in the open disk. To check the Hölder step, write each summand u_i as $(u_i/w_i^{q/\beta})w_i^{q/\beta}$ and use exponents β and β/q .

The exact cancellation in the one-point interpolation proof now gives

$$\begin{aligned} |f(a)|^\beta &\leq \rho_{1,N}(a) \mathbb{E}_N^{!a} \int_{\mathbb{D}} |f(z)|^\beta \frac{(W_a(\mathbf{y}) + w_a(z))^q}{w_a(z)^q} dA(z) \\ &\leq \rho_{1,N}(a) \left(\int_{\mathbb{D}} |f(z)|^\beta dA(z) + A_{\beta,p,q} \int_{\mathbb{D}} |f(z)|^\beta w_a(z)^{-q} dA(z) \right). \end{aligned} \quad (48.6)$$

Here $\mathbb{P}_N^{!a}$ is the reduced Palm distribution at a , which is the $(N-1)$ -particle gas with holomorphic weight $h(z) = z - a$. Its normalizer is positive. The first line follows directly by Tonelli and cancellation; no Palm regularity theorem is needed. The second line uses $0 < q < 1$, hence $(x+y)^q \leq x^q + y^q$, and the last assertion of Lemma 48.1 under this Palm law. For $N = 1$ the exterior sum is zero and the same calculation applies. The second integral is finite because $pq < 1$ and ϕ_a is a smooth diffeomorphism of the closed disk.

Choose the holomorphic branch

$$f_a(z) = \phi'_a(z)^{2/\beta}.$$

It is holomorphic on a neighborhood of $\overline{\mathbb{D}}$ and satisfies $|f_a(z)|^\beta = |\phi'_a(z)|^2$. Its Taylor polynomials converge uniformly on $\overline{\mathbb{D}}$. For each fixed Taylor polynomial, first pass to the chosen particle subsequence in (48.6). Then pass to f_a . Dominated convergence applies to the second integral because w_a^{-q} is integrable. Changing variables $u = \phi_a(z)$ gives

$$\int_{\mathbb{D}} |f_a(z)|^\beta dA(z) = \pi, \quad \int_{\mathbb{D}} |f_a(z)|^\beta w_a(z)^{-q} dA(z) = \int_{\mathbb{D}} (1 - |u|^2)^{-pq} dA(u) = \frac{\pi}{1 - pq}.$$

Since $|f_a(a)|^\beta = (1 - |a|^2)^{-2}$, this proves (48.3) with (48.4). $\square$

Remark 48.3 (Scope of this lower bound). The intensity estimate does not by itself exclude an empty atom. The exterior-only interpolation argument below will prove almost sure infiniteness throughout the same interval. Neither argument proves uniqueness or hyperbolic invariance. The present weighted Hölder estimate requires both $p > 1$ and $p(\beta - 1) < 1$, which are incompatible for $\beta \geq 2$.

48.1 Strict positivity of every factorial correlation below 2

Theorem 48.4 (Uniform lower bounds for every fixed order). *Fix $0 < \beta < 2$ and a positive integer k . There is a constant $b_{\beta,k} > 0$ such that*

$$\rho_{k,N}(\mathbf{a}) \geq b_{\beta,k} |\Delta_k(\mathbf{a})|^\beta \quad (N \geq k, \mathbf{a} \in \mathbb{D}^k). \quad (48.7)$$

The same bound holds for every subsequential limiting factorial density. Consequently, for every nonempty open $U \subset \mathbb{D}$ and every $k \geq 1$, every limiting law satisfies $\mathbb{P}\{\Xi(U) \geq k\} > 0$.

Proof. For $\beta \leq 1$, use (47.3). Suppose $1 < \beta < 2$, put $q = \beta - 1$, and fix $s > 1$ with $sq < 1$. Set

$$v(z) = (1 - |z|^2)^s, \quad W(\mathbf{z}) = \sum_i v(z_i), \quad V_I = \prod_{i \in I} v(z_i).$$

The elementary symmetric polynomial satisfies

$$S_k(\mathbf{z}) := \sum_{|I|=k} V_I \leq W(\mathbf{z})^k / k!.$$

Weighted Hölder applied to the alternating interpolation identity (47.4) gives

$$|F(\mathbf{a})|^\beta \leq S_k(\mathbf{z})^q \sum_{|I|=k} \frac{|\det[\ell_{i_t}(a_j; \mathbf{z})] F(\mathbf{z}_I)|^\beta}{V_I^q}$$

for every alternating polynomial F of degree less than N in each variable. As in the preceding cancellation calculation, expectation of the right side equals

$$\frac{\rho_{k,N}(\mathbf{a})}{k!} \int_{\mathbb{D}^k} |F(\mathbf{z})|^\beta \prod_{i=1}^k v(z_i)^{-q} \mathbb{E}_N^{\mathbf{a}} S_k(\mathbf{y}, \mathbf{z})^q dA^k(\mathbf{z}). \quad (48.8)$$

This identity is first read at distinct anchors a_j . The reduced Palm law in this formula is the $(N - k)$ -particle gas with holomorphic product weight $h(z) = \prod_{j=1}^k (z - a_j)$. Lemma 48.1 therefore bounds every moment of $W(\mathbf{y})$, uniformly in N and in the anchors. Since $v \leq 1$,

$$S_k(\mathbf{y}, \mathbf{z})^q \leq (k!)^{-q} (W(\mathbf{y}) + k)^{kq}.$$

Consequently the Palm expectation in (48.8) is at most a finite constant $B_{\beta,k,s}$, independent of all anchors, the removed variables, and N . The same assertion is immediate for $N = k$, when $\mathbf{y}$ is empty.

Take $F = \Delta_k$. The integral

$$J_{\beta,k,s} := \int_{\mathbb{D}^k} |\Delta_k(\mathbf{z})|^\beta \prod_{i=1}^k (1 - |z_i|^2)^{-sq} dA^k(\mathbf{z})$$

is finite and positive: the Vandermonde is bounded on the closed polydisk and each one-variable boundary weight is integrable because $sq < 1$. Thus $b_{\beta,k} = k! / (B_{\beta,k,s} J_{\beta,k,s})$ is a valid strictly positive constant. Continuity extends the inequality to colliding anchors. Local uniform convergence of factorial densities proves the limiting statement. Integrating over a product of k disjoint disks inside U , as before, proves the final assertion. $\square$

48.2 A sharp pathwise upper bound on boundary counts

Proposition 48.5 (Sharp leading upper bound for the comparison count). *For every subsequential limit at any $\beta > 0$, its centered-disk counts $M(r) = \Xi(\{z : |z| < r\})$ satisfy*

$$\limsup_{r \uparrow 1} (1-r)M(r) \leq \frac{2}{\beta} \quad \text{almost surely.} \quad (48.9)$$

Moreover, the family $\{(1-r)M(r) : 1/2 < r < 1\}$ is uniformly integrable.

Proof. For the reference depths $T_m = \sum_{k=m}^{\infty} E_k$ one has

$$\mathbb{E}T_m = \frac{2}{\beta m} + O_{\beta}(m^{-2}), \quad \text{Var}(T_m) = O_{\beta}(m^{-3}).$$

These follow by summing the expansions of a_k^{-1} and a_k^{-2} , with $a_k = 2k + \beta k(k-1)/2$. Choose any $\delta > 0$ and indices $m_j = \lceil (1+\delta)^j \rceil$. Chebyshev's inequality shows that the probabilities of $|m_j T_{m_j} - m_j \mathbb{E}T_{m_j}| > \varepsilon$ are summable. Borel–Cantelli therefore gives $m_j T_{m_j} \rightarrow 2/\beta$ almost surely. Monotonicity of T_m , followed by a countable choice of $\delta \downarrow 0$, proves

$$mT_m \rightarrow 2/\beta \quad \text{almost surely.} \quad (48.10)$$

The joint ordered-depth domination passes to limiting processes. Here is one precise way to make this passage. For every finite N , product association gives stochastic domination of the entire ordered depth vector by $(T_1, \dots, T_N)$, after padding the actual vector with zeros. The usual monotone coupling theorem on countable products gives a coupling with this coordinatewise inequality. Take a subsequential weak limit of these couplings, using tightness of the point processes and the fixed reference law. At every rational radius, the count inequality passes almost surely through a Skorokhod representation: both the actual limiting process and the reference process almost surely avoid that circle. The latter assertion follows since each T_m has a continuous distribution. The resulting count inequality, first at countably many rational radii and then at all radii by monotonicity, is

$$M(r) \leq L(r) := \#\{m : T_m > \log(1/r)\}.$$

Alternatively, it follows directly by passing the finite-dimensional stochastic order of the rational-radius count vectors to the limit. Equation (48.10) implies $(1-r)L(r) \rightarrow 2/\beta$, proving (48.9).

For uniform integrability one does not need the coupling. The reference exponential bound already established in the radial comparison gives

$$\mathbb{P}\{M(r) \geq m\} \leq \exp\left(m - \frac{\beta}{4}m(m-1)\log(1/r)\right) \quad (m \geq 2).$$

Put $s = 1-r \in (0, 1/2)$ and use $\log(1/r) \geq s$. For all t larger than a constant depending only on β , taking $m = \lfloor t/s \rfloor$ shows that

$$\sup_{1/2 < r < 1} \mathbb{P}\{(1-r)M(r) > t\} \leq C_{\beta} e^{-c_{\beta} t^2}.$$

For example, once $t \geq \max(4, 32/\beta)$, one has $m \geq t/(2s)$, $m-1 \geq m/2$, and the negative quadratic term absorbs the positive m term. This uniform Gaussian tail proves uniform integrability, and indeed uniform moments of every order. $\square$

48.3 Almost sure infiniteness below 2

The preceding interpolation inequalities can also be applied to a polynomial which depends on the particles in a fixed interior disk. The dependence is legitimate because those particles make zero contributions to the interpolation sum. Every surviving term removes and reinserts an exterior particle, leaving the polynomial unchanged.

Theorem 48.6 (Every limiting process is almost surely infinite below 2). *For every $0 < \beta < 2$ and every subsequential limiting process,*

$$\mathbb{P}\{\Xi(\mathbb{D}) = \infty\} = 1. \quad (48.11)$$

More quantitatively, set $\tau = 1$ when $0 < \beta \leq 1$. When $1 < \beta < 2$, let τ be any number satisfying $0 < \tau < 2 - \beta$. There is a finite constant $C_{\beta,\tau}$ such that, for every integer $m \geq 0$, every $R \in (0, 1)$, and every $N > m$,

$$\mathbb{E}_{N,\beta} \left[\mathbf{1}_{\{M_N(R) \leq m\}} \prod_{|z_j| < R} |z_j|^\beta \right] \leq C_{\beta,\tau} 2^{m\beta} (1 - R^2)^\tau. \quad (48.12)$$

The same inequality holds for every limiting process, with its finite product over $\Xi \cap B_R$. Empty products equal one.

Proof. Fix m , R , and $N > m$. On the event $M_N(R) \leq m$, form the monic polynomial

$$P_{R,\mathbf{z}}(t) = \prod_{j: |z_j| < R} (t - z_j).$$

Its degree is at most $m < N$, so Lagrange interpolation at zero gives

$$P_{R,\mathbf{z}}(0) = \sum_{i: |z_i| \geq R} \ell_i(0; \mathbf{z}) P_{R,\mathbf{z}}(z_i). \quad (48.13)$$

Indeed every omitted term contains the zero $P_{R,\mathbf{z}}(z_i) = 0$. Fixed circles and collision sets have zero probability, so the distinction between strict and non-strict inequalities in the exterior index set is immaterial.

Suppose first $0 < \beta \leq 1$. Apply subadditivity to (48.13), multiply by $\mathbf{1}_{\{M_N(R) \leq m\}}$, and take expectations. In the summand indexed by i , the variable z_i ranges only over $\mathbb{D} \setminus B_R$. The polynomial and the count event therefore depend only on the other variables $\mathbf{y}$, and are unchanged when z_i is integrated over this exterior region. The cancellation already used for deterministic polynomials is thus still exact. It gives

$$\begin{aligned} & \mathbb{E}_{N,\beta} \left[\mathbf{1}_{\{M_N(R) \leq m\}} |P_{R,\mathbf{z}}(0)|^\beta \right] \\ & \leq \rho_{1,N}(0) \mathbb{E}_N^{10} \left[\mathbf{1}_{\{M_{\mathbf{y}}(R) \leq m\}} \int_{\mathbb{D} \setminus B_R} |P_{R,\mathbf{y}}(z)|^\beta dA(z) \right]. \end{aligned} \quad (48.14)$$

On the event inside this expectation, $|P_{R,\mathbf{y}}(z)| \leq 2^m$ for all $z \in \mathbb{D}$. The uniform compact one-point bound gives $\sup_N \rho_{1,N}(0) < \infty$. The right side of (48.14) is consequently at most $C_\beta 2^{m\beta} (1 - R^2)$, proving (48.12) in this range.

Now suppose $1 < \beta < 2$. Set $q = \beta - 1$ and choose $s > 1$ so that $sq = 1 - \tau < 1$. Put

$$v(z) = (1 - |z|^2)^s, \quad W(\mathbf{z}) = \sum_j v(z_j).$$

Weighted Hölder on the surviving exterior summands gives

$$|P_{R,\mathbf{z}}(0)|^\beta \leq W(\mathbf{z})^q \sum_{i:|z_i| \geq R} \frac{|\ell_i(0; \mathbf{z}) P_{R,\mathbf{z}}(z_i)|^\beta}{v(z_i)^q}.$$

Using the larger sum W in place of the exterior weight sum only increases the right side. The same exterior-only cancellation now gives the upper bound

$$\rho_{1,N}(0) \mathbb{E}_N^{!0} \left[\mathbf{1}_{\{M_{\mathbf{y}}(R) \leq m\}} \int_{\mathbb{D} \setminus B_R} |P_{R,\mathbf{y}}(z)|^\beta \frac{(W(\mathbf{y}) + v(z))^q}{v(z)^q} dA(z) \right].$$

The reduced Palm law has holomorphic product weight $h(z) = z$. Lemma 48.1 bounds $\mathbb{E}_N^{!0} W^q$ uniformly. Using $(W + v)^q \leq W^q + v^q$ and $|P_{R,\mathbf{y}}(z)| \leq 2^m$ on the count event therefore gives

$$\begin{aligned} & \mathbb{E}_{N,\beta} \left[\mathbf{1}_{\{M_N(R) \leq m\}} |P_{R,\mathbf{z}}(0)|^\beta \right] \\ & \leq C_{\beta,s} 2^{m\beta} \left(\pi(1 - R^2) + \int_{\mathbb{D} \setminus B_R} (1 - |z|^2)^{-sq} dA(z) \right) \\ & = C_{\beta,s} 2^{m\beta} \left(\pi(1 - R^2) + \frac{\pi}{\tau} (1 - R^2)^\tau \right). \end{aligned}$$

Absorb the constants to prove (48.12).

For each fixed R , the functional on the left of (48.12) is bounded by one and is continuous at every locally finite configuration avoiding ∂B_R . Every limiting process almost surely avoids this circle. Thus convergence in law passes the inequality to the limiting process.

Finally let $R \uparrow 1$. On $\{\Xi(\mathbb{D}) > m\}$ the count indicator is eventually zero. On $\{\Xi(\mathbb{D}) \leq m\}$ all particles are eventually in B_R , and the functional converges to $\prod_{z \in \Xi} |z|^\beta$. Dominated convergence gives

$$\mathbb{E} \left[\mathbf{1}_{\{\Xi(\mathbb{D}) \leq m\}} \prod_{z \in \Xi} |z|^\beta \right] = 0.$$

On the indicated finite-count event this finite product is strictly positive almost surely: the process has no point at the fixed location zero. Hence $\mathbb{P}\{\Xi(\mathbb{D}) \leq m\} = 0$. A countable union over m proves (48.11). $\square$

Corollary 48.7 (An almost sure logarithmic lower growth bound). *Let $0 < \beta < 2$, and put*

$$\kappa_\beta = \begin{cases} 1, & 0 < \beta \leq 1, \\ 2 - \beta, & 1 < \beta < 2. \end{cases}$$

For every subsequential limiting process, the innermost radius $r_ := \inf\{|z| : z \in \Xi\}$ belongs to $(0, 1)$ almost surely, and*

$$\liminf_{r \uparrow 1} \frac{M(r)}{\log(1/(1 - r^2))} \geq \frac{\kappa_\beta}{\beta \log(2/r_*)} > 0 \quad \text{almost surely.} \quad (48.15)$$

Proof. Almost sure infiniteness gives $r_* < 1$. If $r_* = 0$, there are either infinitely many points in a fixed compact disk or a point at zero; both alternatives have probability zero. Thus $r_* > 0$. Fix $0 < \varepsilon < 1$, and let τ be allowed in Theorem 48.6. On the event $\{r_* \geq \varepsilon, M(R) \leq m\}$ the finite product in (48.12) is at least $\varepsilon^{m\beta}$. Consequently

$$\mathbb{P}\{r_* \geq \varepsilon, M(R) \leq m\} \leq C_{\beta,\tau} (2/\varepsilon)^{m\beta} (1 - R^2)^\tau.$$

Take $1 - R_n^2 = e^{-n}$ and $m_n = \lfloor an \rfloor$, with $0 < a < \tau/[\beta \log(2/\varepsilon)]$. The upper bounds are summable in n , since their exponential rate is at most $-\tau + a\beta \log(2/\varepsilon) < 0$. Borel–Cantelli shows that, on $\{r_* \geq \varepsilon\}$, eventually $M(R_n) > an$. Monotonicity of M between R_n and R_{n+1} then gives a lower bound a for the liminf in (48.15). Make these conclusions simultaneously for countably many rational ε, a and for a countable sequence of allowed τ increasing to κ_β . On the resulting probability-one event, let a tend to its upper endpoint and then let rational ε increase to r_* . This proves the claimed bound. $\square$

Corollary 48.8 (A lower tail for the number of interior particles). *Under the hypotheses and notation of Theorem 48.6, for every fixed $m \geq 0$ there is a finite constant $C_{\beta, \tau, m}$ such that*

$$\mathbb{P}\{M(R) \leq m\} \leq C_{\beta, \tau, m}(1 - R^2)^{2\tau/(m\beta+2)}. \quad (48.16)$$

The same bound holds at finite $N > m$. In particular the hole probability is $O_{\beta, \tau}((1 - R^2)^\tau)$.

Proof. For $0 < \varepsilon < 1/2$, the uniform compact intensity bound gives

$$\mathbb{P}\{\Xi(B_\varepsilon) > 0\} \leq C_\beta \varepsilon^2,$$

and likewise for every finite gas. On its complement, if $M(R) \leq m$, the product in (48.12) is at least $\varepsilon^{m\beta}$. Consequently

$$\mathbb{P}\{M(R) \leq m\} \leq C_\beta \varepsilon^2 + C_{\beta, \tau} 2^{m\beta} \varepsilon^{-m\beta} (1 - R^2)^\tau.$$

For R sufficiently close to one, choose $\varepsilon = (1 - R^2)^{\tau/(m\beta+2)}$. This proves the bound near one; increasing its constant covers the remaining R . For $m = 0$ the product is identically one on the hole event, so (48.12) already gives the stated hole bound directly. $\square$

49 A non-Blaschke theorem for every limit below two

The random interpolation polynomial can be improved by approximating a finite Blaschke product. This removes the factor exponential in the interior count and proves a boundary-density statement substantially stronger than almost sure infiniteness.

Lemma 49.1 (Uniform polynomial approximation with prescribed interior zeros). *Fix $0 < R < 1$, an integer $m \geq 0$, and $\varepsilon > 0$. There is an integer $D \geq m$ such that the following holds for every $0 \leq n \leq m$ and every tuple $\mathbf{a} = (a_1, \dots, a_n)$ in B_R . Set*

$$P_{\mathbf{a}}(t) = \prod_{j=1}^n (t - a_j), \quad Q_{\mathbf{a}}(t) = \prod_{j=1}^n (1 - \overline{a_j}t),$$

and let $T_{D-n}(Q_{\mathbf{a}}^{-1})$ be the Taylor polynomial at zero through degree $D - n$. The polynomial

$$f_{\mathbf{a}, D}(t) = P_{\mathbf{a}}(t) T_{D-n}(Q_{\mathbf{a}}^{-1})(t)$$

has degree at most D , vanishes at each a_j , satisfies $f_{\mathbf{a}, D}(0) = P_{\mathbf{a}}(0)$, and obeys

$$\sup_{|t| \leq 1} |f_{\mathbf{a}, D}(t)| \leq 1 + \varepsilon.$$

The choice of D is uniform over the indicated tuples.

Proof. For $n \geq 1$, the coefficient of t^k in $Q_{\mathbf{a}}^{-1}$ has modulus at most $\binom{k+n-1}{n-1} R^k \leq \binom{k+m-1}{m-1} R^k$. This follows by expanding each factor as a geometric series and counting the weak compositions of k into n parts. Since $R < 1$,

$$\sup_{|t| \leq 1} \left| Q_{\mathbf{a}}^{-1}(t) - T_{D-n}(Q_{\mathbf{a}}^{-1})(t) \right| \leq \sum_{k > D-m} \binom{k+m-1}{m-1} R^k \rightarrow 0$$

uniformly in the tuples as $D \rightarrow \infty$. On the closed unit disk, $|P_{\mathbf{a}}| \leq 2^m$. Each factor $(t - a_j)/(1 - \bar{a}_j t)$ has modulus at most one there, so $|P_{\mathbf{a}}/Q_{\mathbf{a}}| \leq 1$. Choose D to make the Taylor error at most $2^{-m}\varepsilon$ and multiply by $P_{\mathbf{a}}$. The remaining assertions follow immediately from the definition and $Q_{\mathbf{a}}^{-1}(0) = 1$. When $n = 0$ take $f_{\emptyset, D} = 1$; the case $m = 0$ is included. $\square$

Theorem 49.2 (Uniform Laplace decay for accumulated radial depth). *Let $0 < \beta < 2$ and let Ξ be any subsequential limiting process of the original disk gas. Write*

$$H(R) = \sum_{z \in \Xi \cap B_R} -\log |z|.$$

There is almost surely no point at zero, so this is finite for every $R < 1$. Put $\kappa_\beta = 1$ if $0 < \beta \leq 1$, and $\kappa_\beta = 2 - \beta$ if $1 < \beta < 2$. For $\beta \leq 1$ take $\tau = 1$; for $1 < \beta < 2$ take any $0 < \tau < 2 - \beta$. There is a finite constant $C_{\beta, \tau}$, uniform over all subsequential limits, such that

$$\mathbb{E} e^{-\beta H(R)} \leq C_{\beta, \tau} (1 - R^2)^\tau, \quad 0 < R < 1. \quad (49.1)$$

Consequently,

$$\sum_{z \in \Xi} (1 - |z|) = \infty \quad \text{almost surely,} \quad (49.2)$$

and the more quantitative pathwise bound is

$$\liminf_{R \uparrow 1} \frac{H(R)}{\log(1/(1 - R^2))} \geq \frac{\kappa_\beta}{\beta} \quad \text{almost surely.} \quad (49.3)$$

Proof. Fix R, m, ε , choose D from Lemma 49.1, and take $N > D$. On $\{M_N(R) \leq m\}$ use $f_{\mathbf{a}, D}$ with $\mathbf{a}$ the tuple of particles in B_R . Lagrange interpolation at zero is valid because the polynomial has degree less than N . Its interior-particle terms vanish, leaving

$$P_{\mathbf{a}}(0) = \sum_{i: |z_i| \geq R} \ell_i(0; \mathbf{z}) f_{\mathbf{a}, D}(z_i).$$

For each surviving term, the particle indexed by i lies outside B_R . Removing it or integrating it over $\mathbb{D} \setminus B_R$ therefore leaves the tuple $\mathbf{a}$, its count, its polynomial, and its Taylor truncation degree unchanged. Thus the exact Palm cancellation used in the proof of Theorem 48.6 applies without alteration. The polynomial depends measurably and symmetrically on the interior tuple, and all integrands after the absolute-value inequality are nonnegative, so Tonelli justifies the integration.

For $0 < \beta \leq 1$, subadditivity and that cancellation give

$$\begin{aligned} & \mathbb{E}_{N, \beta} \left[\mathbf{1}_{\{M_N(R) \leq m\}} \prod_{|z_j| < R} |z_j|^\beta \right] \\ & \leq \rho_{1, N}(0) \mathbb{E}_N^{\text{lo}} \left[\mathbf{1}_{\{M_{\mathbf{y}}(R) \leq m\}} \int_{\mathbb{D} \setminus B_R} |f_{\mathbf{a}(\mathbf{y}), D}(z)|^\beta dA(z) \right] \\ & \leq C_\beta (1 + \varepsilon)^\beta (1 - R^2). \end{aligned}$$

Here $\sup_N \rho_{1,N}(0) < \infty$ is the proved compact intensity bound.

For $1 < \beta < 2$, put $q = \beta - 1$ and $s = (1 - \tau)/q > 1$, and use the weights $v(z) = (1 - |z|^2)^s$, $W(\mathbf{y}) = \sum_j v(y_j)$. Weighted Hölder followed by the same cancellation bounds the preceding left side by

$$\rho_{1,N}(0)(1 + \varepsilon)^\beta \mathbb{E}_N^{!0} \int_{\mathbb{D} \setminus B_R} \frac{(W(\mathbf{y}) + v(z))^q}{v(z)^q} dA(z).$$

The reduced Palm law has holomorphic product weight $h(z) = z$. Lemma 48.1 bounds its q -th moment of W uniformly in N . Since $0 < q < 1$, $(W + v)^q \leq W^q + v^q$. The last display is therefore at most

$$C_{\beta,\tau}(1 + \varepsilon)^\beta \left((1 - R^2) + \int_{\mathbb{D} \setminus B_R} (1 - |z|^2)^{-sq} dA(z) \right) \leq C'_{\beta,\tau}(1 + \varepsilon)^\beta (1 - R^2)^\tau,$$

because $sq = 1 - \tau$ and the integral equals $\pi(1 - R^2)^\tau/\tau$. These constants do not depend on m , D , or ε .

For fixed R, m, D , pass along the chosen particle-number subsequence. The bounded functional on the left is continuous at locally finite configurations with no particle on ∂B_R , an event of full limiting probability. Next let $\varepsilon \downarrow 0$, choosing D separately before each particle-number limit. Finally let $m \rightarrow \infty$. Local finiteness and monotone convergence give (49.1). This order of limits is essential: the polynomial degree need not be bounded uniformly in m or R .

As $R \uparrow 1$, $H(R)$ increases to the total radial depth. Dominated convergence in (49.1) shows that the exponential of minus β times that total depth is zero almost surely. Thus $\sum_z -\log |z| = \infty$. There are only finitely many particles with $|z| \leq 1/2$, none at zero, and on $1/2 \leq r < 1$ one has $1 - r \leq -\log r \leq 2(1 - r)$. This proves (49.2).

For the quantitative assertion, take $1 - R_n^2 = e^{-n}$ and $0 < a < \tau/\beta$. Markov's inequality gives

$$\mathbb{P}\{H(R_n) \leq an\} \leq e^{\beta an} \mathbb{E} e^{-\beta H(R_n)} \leq C_{\beta,\tau} e^{-(\tau - \beta a)n}.$$

These bounds are summable. Borel–Cantelli and monotonicity of H between consecutive radii imply a lower bound a for the liminf in (49.3). Take countably many $a \uparrow \tau/\beta$, and, when $\beta > 1$, countably many $\tau \uparrow 2 - \beta$. The asserted bound follows. $\square$

Corollary 49.3 (Necessary boundary density and divergence of the Green field). *Every limit in Theorem 49.2 satisfies*

$$\frac{\kappa_\beta}{\beta} \leq \limsup_{R \uparrow 1} (1 - R) \Xi(B_R) \leq \frac{2}{\beta} \quad \text{almost surely.} \quad (49.4)$$

Moreover, on one probability-one event,

$$\sum_{z \in \Xi} g(a, z) = \infty \quad \text{for every } a \in \mathbb{D}, \quad g(a, z) = -\log \left| \frac{a - z}{1 - \bar{a}z} \right|. \quad (49.5)$$

No invariance assumption is needed in either assertion.

Proof. The upper bound in (49.4) is Proposition 48.5. For the lower bound, write $M(R) = \Xi(B_R)$. Since the innermost point has a strictly positive radius, Stieltjes integration by parts gives

$$H(R) = (-\log R)M(R) + \int_0^R \frac{M(t)}{t} dt.$$

If $\limsup_{R \uparrow 1} (1 - R)M(R) < c$, then eventually $M(R) \leq c/(1 - R)$. The displayed formula yields $\limsup_{R \uparrow 1} H(R)/\log(1/(1 - R^2)) \leq c$. Any $c < \kappa_\beta/\beta$ contradicts (49.3), proving the lower bound.

Finally, for $a, z \in \mathbb{D}$, the automorphism identity gives

$$1 - \left| \frac{a - z}{1 - \bar{a}z} \right|^2 = \frac{(1 - |a|^2)(1 - |z|^2)}{|1 - \bar{a}z|^2} \geq \frac{1 - |a|}{1 + |a|}(1 - |z|^2).$$

The inequality $-\log r \geq (1 - r^2)/2$ for $0 < r \leq 1$ and (49.2) show that every such Green sum diverges. The reasoning is deterministic for each configuration satisfying that divergence, and therefore holds for all a simultaneously. A pole only makes the corresponding sum infinite immediately. $\square$

Corollary 49.4 (Uncompensated grand-canonical equations cannot describe these limits). *For $0 < \beta < 2$, no actual subsequential disk-gas limit satisfies the uncompensated grand-canonical Green specification (45.1) at a positive activity. This conclusion does not assume hyperbolic invariance.*

Proof. For every compact disk B , the process has only finitely many particles in B . Removing them leaves an exterior whose Green potential is infinite at every point of B other than possibly those finitely many removed particles, by (49.5). Thus every positive-count term in the uncompensated kernel vanishes almost everywhere on its tuple space, and that kernel assigns count zero with probability one. The DLR identity would give $\Xi(B) = 0$ almost surely for every disk in an exhaustion. This contradicts Theorem 49.2. $\square$

49.1 Local equivalence of all nontrivial candidate limits

Proposition 49.5 (Every local particle count occurs). *Let P be a simple, locally finite point-process law on $\mathbb{D}$ satisfying the strictly positive canonical densities of Theorem 41.3. Suppose that its total particle count has unbounded support, in the precise sense that $P\{\Xi(\mathbb{D}) \geq m\} > 0$ for every integer $m \geq 1$. Then, for every nonempty disk B with $\bar{B} \subset \mathbb{D}$,*

$$P\{\Xi(B) = m\} > 0 \quad \text{for every integer } m \geq 0. \quad (49.6)$$

The restriction of P to B is mutually absolutely continuous with the unit-area Poisson process on B .

Consequently every subsequential limit of the disk gas that is not concentrated on the empty configuration has this local equivalence property. In particular it holds for every limit when $0 < \beta < 2$, and for the Bergman limit at $\beta = 2$. Any two nontrivial candidate limits, even at different inverse temperatures, have mutually absolutely continuous restrictions to every relatively compact disk.

Proof. Fix $m \geq 0$. If $m \geq 1$, choose a disk D_0 compactly contained in $\mathbb{D}$, containing $\bar{B}$ in its interior, such that $P\{\Xi(D_0) \geq m\} > 0$. Such a disk exists by monotone convergence along a centered exhaustion containing $\bar{B}$ and the unbounded-support assumption. If $m = 0$, choose any such D_0 . Local finiteness yields an integer $n \geq m$ with $P\{\Xi(D_0) = n\} > 0$; for $m = 0$ this follows simply because the probabilities over all finite n sum to one.

Conditional on the exterior of D_0 and this count n , the interior law has a strictly positive density away from collisions on D_0^n . The set of tuples with exactly m coordinates in B and $n - m$ coordinates in $D_0 \setminus \bar{B}$ has positive area when $n > 0$, because both regions have positive area; if $n = 0$ the desired event already holds. Hence the conditional probability of $\Xi(B) = m$ is strictly positive for almost every exterior occurring with count n . Integrating proves (49.6).

For each $m \geq 1$, condition now on the exterior of B and $\Xi(B) = m$. The canonical density is equivalent to area measure on B^m , since the collision set has zero area and the remaining factor is strictly positive. Averaging this density over the exterior shows that the unconditional m th Janossy measure is equivalent to area measure on B^m : a set of positive

area has positive conditional probability for every admissible exterior, and the unconditional mass of this count sector is positive. The empty sector also has positive mass by the first part. The Poisson law has exactly these count sectors, each with a strictly positive constant Janossy density. Its countable decomposition therefore proves mutual absolute continuity of the complete local laws.

For an actual subsequential limit, the canonical hypothesis is Theorem 41.3. For $0 < \beta < 2$, its total count is infinite almost surely by Theorem 48.6. For $\beta \geq 2$, Theorem 38.1 says that its total count belongs to $\{0, \infty\}$ almost surely. Unless the law is the empty point mass, it thus has positive probability of infinite total count and satisfies the same unbounded-support hypothesis. The final assertion follows by transitivity of local equivalence. $\square$

Remark 49.6 (Local overlap does not exclude different phases). The proposition concerns unconditional local laws. It does not prove that every count has positive probability after conditioning on a fixed exterior. Nor does local mutual absolute continuity imply global mutual absolute continuity or equality of point-process laws. If distinct candidate states exist, the proposition says they cannot be separated by an event of probability zero in a fixed compact window. Their probabilities of positive-probability local events may still differ, and they may be globally singular because of tail information.

50 A Blaschke interpolation criterion for the interval below 4

The next result is a conditional reduction. It isolates a single uniform boundary estimate which would prove nonemptiness in the remaining interval $2 < \beta < 4$. The estimate itself is not proved here. Ordinary Blaschke interpolation suffices; a Hermite formula, which would introduce derivatives of the exterior field, is unnecessary.

Lemma 50.1 (The total radial depth cannot vanish). *For every fixed $\beta > 0$ there is a finite C_β such that*

$$\mathbb{P}_{N,\beta} \left\{ \sum_{i=1}^N -\log |z_i| \leq u \right\} \leq (C_\beta u)^N, \quad N \geq 1, \quad u > 0. \quad (50.1)$$

Consequently, for every $b > 0$ there is $d_{\beta,b} > 0$ such that

$$\inf_{N \geq 1} \mathbb{E}_{N,\beta} \left(1 - \prod_{i=1}^N |z_i|^2 \right)^b \geq d_{\beta,b}. \quad (50.2)$$

Proof. The established partition asymptotic implies, after decreasing its constant to cover the finitely many small values of N , that

$$Z_{N,\beta} \geq c_\beta^N N^{(\beta/2-1)N}$$

for some $c_\beta > 0$ and every $N \geq 1$. Hadamard's inequality gives $|\Delta_N|^\beta \leq N^{\beta N/2}$ on the disk. Put $t_i = -\log |z_i|$. After angular integration the area element is $(2\pi)^N e^{-2 \sum_i t_i} dt_1 \cdots dt_N$. The simplex $\{t_i \geq 0 : \sum_i t_i \leq u\}$ has volume $u^N/N!$. Therefore

$$\mathbb{P} \left\{ \sum_i t_i \leq u \right\} \leq \frac{(2\pi)^N N^{\beta N/2} u^N}{N! c_\beta^N N^{(\beta/2-1)N}} \leq \left(\frac{2\pi e}{c_\beta} u \right)^N,$$

where $N! \geq (N/e)^N$. Choose $u_\beta > 0$ with $C_\beta u_\beta \leq 1/2$. On the complementary event, $1 - \prod_i |z_i|^2 \geq 1 - e^{-2u_\beta}$. Thus $d_{\beta,b} = \frac{1}{2}(1 - e^{-2u_\beta})^b$ is valid. $\square$

For $n \geq 0$, let $\mathbb{P}_{n,\beta}^{[0]}$ denote the probability law on $\mathbb{D}^n$ with density proportional to

$$|\Delta_n(\mathbf{y})|^\beta \prod_{j=1}^n |y_j|^\beta.$$

It is exactly the reduced Palm distribution at zero of the $(n+1)$ -particle unweighted gas. For $n=0$ it is the empty configuration. Define its reflected characteristic polynomial by

$$Q_{\mathbf{y}}(z) = \prod_{j=1}^n (1 - z\overline{y_j}), \quad z \in \mathbb{D}.$$

Theorem 50.2 (A sufficient uniform boundary moment estimate). *Fix $1 < \beta < 4$. Suppose that there is $K_\beta < \infty$ such that*

$$\sup_{n \geq 0} \mathbb{E}_{n,\beta}^{[0]} |Q_{\mathbf{y}}(z)|^\beta \leq K_\beta (1 - |z|^2)^{-\beta/2}, \quad z \in \mathbb{D}. \quad (50.3)$$

Then there is $c > 0$ such that $\rho_{1,N}(0) \geq c$ for every $N \geq 1$. In particular every subsequential limiting law is different from the almost surely empty law.

The hypothesis is a bound uniform both in particle number and as $|z| \uparrow 1$. Pointwise convergence of the left side for fixed z , or uniform bounds on fixed compact subdisks, do not imply it.

Proof. For a configuration of distinct points put

$$P(\zeta) = \prod_{i=1}^N (\zeta - z_i), \quad Q(\zeta) = \prod_{i=1}^N (1 - \zeta\overline{z_i}), \quad B(\zeta) = P(\zeta)/Q(\zeta).$$

The polynomial part of the partial-fraction expansion of Q/P is the constant $\overline{B(0)}$. Its residue at z_i is $1/B'(z_i)$. Hence, initially away from the zeros of P ,

$$1 - \overline{B(0)}B(a) = \sum_{i=1}^N \frac{B(a)}{(a - z_i)B'(z_i)}.$$

Equivalently, writing the Lagrange polynomials as before,

$$1 - \overline{B(0)}B(a) = \sum_{i=1}^N \ell_i(a; \mathbf{z}) \frac{Q(z_i)}{Q(a)}. \quad (50.4)$$

Both sides extend continuously to all $a \in \mathbb{D}$. At $a=0$, $Q(0)=1$ and the left side is $1 - \prod_i |z_i|^2$.

Set $q = \beta - 1 > 0$. Choose

$$1 < p < \frac{\beta + 2}{2(\beta - 1)}. \quad (50.5)$$

Such a choice is possible precisely because $\beta < 4$. Put $v(z) = (1 - |z|^2)^p$ and $W(\mathbf{z}) = \sum_i v(z_i)$. Weighted Hölder applied to (50.4) gives

$$\left(1 - \prod_i |z_i|^2\right)^\beta \leq W(\mathbf{z})^q \sum_i \frac{|\ell_i(0; \mathbf{z})Q(z_i)|^\beta}{v(z_i)^q}.$$

No subadditivity assumption on q is used here.

In the summand indexed by i , let $z = z_i$ and let $\mathbf{y}$ be the other particles. Then $Q(z_i) = (1 - |z|^2)Q_{\mathbf{y}}(z)$. Cancelling the Lagrange denominators against the gas density and summing over i gives, by Tonelli,

$$\begin{aligned} \mathbb{E}_{N,\beta} \left(1 - \prod_i |z_i|^2 \right)^\beta \\ \leq \rho_{1,N}(0) \int_{\mathbb{D}} (1 - |z|^2)^{\beta-pq} \mathbb{E}_{N-1,\beta}^{[0]} \left[(W(\mathbf{y}) + v(z))^q |Q_{\mathbf{y}}(z)|^\beta \right] dA(z). \end{aligned} \quad (50.6)$$

The auxiliary moment of W is harmless. Indeed, tilting $\mathbb{P}_{n,\beta}^{[0]}$ by $|Q_{\mathbf{y}}(z)|^\beta$ produces a gas with holomorphic product weight

$$h_z(w) = w(1 - \bar{z}w).$$

Lemma 48.1 therefore implies

$$\mathbb{E}_{n,\beta}^{[0]} \left[(W + v(z))^q |Q_{\mathbf{y}}(z)|^\beta \right] \leq A'_{\beta,p,q} \mathbb{E}_{n,\beta}^{[0]} |Q_{\mathbf{y}}(z)|^\beta, \quad (50.7)$$

with a finite constant independent of n, z . For example, take $A'_{\beta,p,q}$ to be a uniform bound for $\mathbb{E}(1 + W)^q$ under arbitrary holomorphic product weights. The tilted normalizers are positive, and the case $n = 0$ is immediate.

By the hypothesis, the right side of (50.6) is at most

$$\rho_{1,N}(0) A'_{\beta,p,q} K_\beta \int_{\mathbb{D}} (1 - |z|^2)^{\beta/2-pq} dA(z) = \rho_{1,N}(0) A'_{\beta,p,q} K_\beta \frac{\pi}{1 + \beta/2 - pq}.$$

The denominator is strictly positive by (50.5). Lemma 50.1, with $b = \beta$, bounds the left side away from zero. This proves the asserted uniform lower bound on $\rho_{1,N}(0)$. Local uniform convergence of the correlation densities then proves the limiting conclusion. $\square$

Corollary 50.3 (Any uniform boundary exponent strictly below 2 suffices). *In Theorem 50.2, the hypothesis can be replaced by*

$$\sup_{n \geq 0} \mathbb{E}_{n,\beta}^{[0]} |Q_{\mathbf{y}}(z)|^\beta \leq K(1 - |z|^2)^{-\kappa}, \quad z \in \mathbb{D}, \quad (50.8)$$

for any finite K and any $\kappa < 2$. The same uniform positive lower bound on $\rho_{1,N}(0)$ follows.

Proof. Choose

$$1 < p < \frac{\beta + 1 - \kappa}{\beta - 1},$$

which is possible because $\kappa < 2$ and $\beta > 1$. The proof of the theorem, through (50.7), is unchanged. The final area integral is now

$$\int_{\mathbb{D}} (1 - |z|^2)^{\beta-p(\beta-1)-\kappa} dA(z) = \frac{\pi}{1 + \beta - p(\beta - 1) - \kappa} < \infty.$$

The same numerator lower bound completes the proof. $\square$

Remark 50.4 (A Gaussian coefficient contraction already fails for one particle). Write $b = \beta/2$. For the one-particle charged law $\mathbb{P}_{1,\beta}^{[0]}$, direct radial integration gives

$$\mathbb{E}_{1,\beta}^{[0]} |y|^2 = \frac{\beta + 2}{\beta + 4} = \frac{b + 1}{b + 2}.$$

The coefficient of z in $(1 - z\bar{y})^b$ is $-b\bar{y}$. Rotational symmetry therefore gives

$$\mathbb{E}_{1,\beta}^{[0]}|1 - z\bar{y}|^\beta = 1 + \frac{b^2(b+1)}{b+2}|z|^2 + O(|z|^4).$$

By comparison, $(1 - |z|^2)^{-b} = 1 + b|z|^2 + O(|z|^4)$. The first nonconstant coefficient of the former is larger when $b(b+1)/(b+2) > 1$, equivalently $b > \sqrt{2}$. Thus the proposed Gaussian domination with multiplicative constant exactly one is false already for this charged one-particle law when $2\sqrt{2} < \beta < 4$. This does not disprove either boundary-moment hypothesis with an unspecified finite multiplicative constant.

Remark 50.5 (The unresolved step). The criterion is a proved implication, not a proof of (50.3) for $2 < \beta < 4$. The reduction avoids any exterior electric-field estimate and identifies an explicit charged characteristic-polynomial moment as a possible next target. It does not prove uniqueness, hyperbolic invariance, or almost sure infiniteness in that remaining interval.

51 The corrected meromorphic field is independent of the Gaussian modes

The independence of the harmonic modes from the point configuration can be strengthened to include a corrected electric field. The correction is intrinsic at finite particle number: it is the derivative of the uncompensated hyperbolic Green potential. The required small-transport estimate uses its vanishing at the boundary, rather than an estimate on inverse angular statistics.

For a finite configuration ξ put

$$J_w(z) = \frac{1}{z - w} + \frac{\bar{w}}{1 - z\bar{w}} = \frac{1 - |w|^2}{(z - w)(1 - z\bar{w})}, \quad \mathcal{A}_\xi(z) = \sum_{w \in \xi} J_w(z).$$

Write $\mathcal{A}_N = \mathcal{A}_{\Xi_N}$ and

$$\mathcal{C}_N(z) = \sum_{w \in \Xi_N} \frac{1}{z - w}, \quad T_N(z) = \sum_{w \in \Xi_N} \frac{\bar{w}}{1 - z\bar{w}} = \sum_{k \geq 1} \overline{S_{k,N}} z^{k-1}.$$

Thus $\mathcal{A}_N = \mathcal{C}_N + T_N$ exactly. These are meromorphic functions on the disk, with their poles interpreted in the usual sense.

Lemma 51.1 (Stability of the corrected field under small boundary transports). *Let $T_{N,t}$ be any one of the mode transports used in the proof of Theorem 17.1, including the small Möbius transport for the first mode. For every finite list of deterministic points $u_1, \dots, u_l \in \mathbb{D}$,*

$$(\mathcal{A}_{(T_{N,t})_*\Xi_N}(u_i) - \mathcal{A}_{\Xi_N}(u_i))_{i=1}^l \longrightarrow 0$$

in probability under both the original law P_N and its exact pullback law $Q_{N,t}$. Each evaluation is finite almost surely.

Proof. Fix $\max_i |u_i| < R < 1$. On a slightly larger outer annulus the maps and their inverses have C^1 distance $O(N^{-1})$ from the identity. This follows directly from their displayed formula outside the shrinking patch near zero; for the first mode it follows from the Möbius formula. They map the unit circle onto itself. Globally, $\epsilon_N := \sup |T_{N,t} - \text{id}| \rightarrow 0$.

For $|w| \geq R$, all derivatives in w up to order two of $J_w(u_i)$ are bounded uniformly. For N large the same is true along the image of the annulus. The real C^1 norm of

$$F_{N,i}(w) = J_{T_{N,t}(w)}(u_i) - J_w(u_i)$$

is consequently $O(N^{-1})$. Moreover $F_{N,i} = 0$ on $|w| = 1$, because $J_w(u_i) = 0$ there and the transport preserves that circle. Integration of its radial derivative gives

$$|F_{N,i}(w)| \leq \frac{C}{N}(1 - |w|), \quad |w| \geq R. \quad (51.1)$$

The outer-particle contribution is at most CD_N/N , where $D_N = \sum_w(1 - |w|)$. It tends to zero in probability under both laws: $D_N = O_{P_N}(\log(eN))$, and the same assertion under $Q_{N,t}$ was proved in the mode-transport argument using its exact pullback and uniform preservation of boundary distance.

For the inner particles, fix $\epsilon > 0$ and exclude the event that some particle lies within 2ϵ of an evaluation point. Under P_N its probability is $O(\epsilon^2)$ by the compact intensity bound. Under $Q_{N,t}$ the same bound, with an error tending to zero, follows by transporting to P_N and enlarging the finitely many balls by ϵ_N . Away from these balls, the derivatives of $J_w(u_i)$ are bounded by a constant depending on R, ϵ , so the remaining inner contribution is at most $C_{R,\epsilon} \epsilon_N \Xi_N(\mathbb{D}_R)$. The compact count is tight under both laws, hence this tends to zero in probability. First let $N \rightarrow \infty$, then $\epsilon \downarrow 0$. The finite list of evaluations causes no change to the argument. $\square$

Theorem 51.2 (Gaussian separation of the corrected meromorphic field). *Every subsequential point-process limit admits a simultaneous augmentation by its meromorphic field $\mathcal{C}$ on $\{\Xi \neq \emptyset\}$ and the Gaussian sequence $G = (G_k)_{k \geq 1}$ such that*

$$\mathcal{A} = \mathcal{C} + T_G \quad \text{on } \{\Xi \neq \emptyset\}, \quad T_G(z) = \sum_{k \geq 1} \overline{G_k} z^{k-1},$$

has the following properties. Set $\mathcal{A} = 0$ on the empty event. Then

$$(\Xi, \mathcal{A}) \text{ is independent of } G. \quad (51.2)$$

On the nonempty event, $\mathcal{A}$ is a meromorphic function with simple residue-one poles exactly at Ξ . It is obtained as the limit of $\mathcal{A}_N$ on every positive-count localization. The assertion does not say that $\mathcal{A}$ is measurable from Ξ .

Proof. Retain all normalized exterior factors on the fixed countable disk exhaustion in Proposition 43.1, along with all the harmonic modes. Theorem 19.6 also gives $T_N \Rightarrow T_G$ in $\mathcal{H}(\mathbb{D})$: conjugate its coefficients and divide the analytic resolvent by its argument, with the removable value at zero. For each positive-count disk the zero-free limits of the normalized factors identify the locally meromorphic limit of $\mathcal{C}_N$. Gluing therefore gives the asserted limit of $\mathcal{A}_N$ on the nonempty event.

We give the localization needed to avoid assuming tightness of $\mathcal{C}_N$ on the empty branch. Fix a radius $r < 1$ from the exhaustion, a finite deterministic evaluation list $\mathbf{u}$, and write $E_{r,N} = \{\Xi_N(\mathbb{D}_r) \geq 1\}$. Introduce a cemetery symbol $*$, give $\mathbb{C}^l \sqcup \{*\}$ its Polish disjoint-union topology, and define the censored vector

$$A_{r,N} = \begin{cases} (\mathcal{A}_N(u_1), \dots, \mathcal{A}_N(u_l)), & E_{r,N}, \\ *, & E_{r,N}^c. \end{cases}$$

Its law jointly with Ξ_N is tight along the chosen subsequence. Indeed, on every fixed positive count the preceding simultaneous normal-family construction gives a nonzero normalized limit on a larger disk containing the evaluation list; the remaining high counts have arbitrarily small probability by the uniform count bounds. No deterministic evaluation point is a limiting particle, by absolute continuity of the limiting intensity. Thus the limiting censored vector is

$(\mathcal{A}(u_i))_i$ on $E_r = \{\Xi(\mathbb{D}_r) \geq 1\}$ and $*$ otherwise. These conclusions follow for all countably many radii and all finite lists from a fixed countable dense set on a single diagonal extraction.

Under the mode transport, $E_{r,N}$ changes with probability tending to zero under both P_N and $Q_{N,t}$. A change requires a particle in a shrinking collar of $\partial\mathbb{D}_r$; the compact intensity bound and exact pullback control this probability. Lemma 51.1 shows that the censored field vector is also unchanged in probability. The point configuration is unchanged in the vague metric in probability. For every fixed mode j other than the transported mode k , its increment tends to zero in probability as well. For $j < k$ this is (17.6); for $j > k$ the same expansion (17.5) has only positive indices and the previously proved macroscopic mode bounds apply. For the first mode the identical calculation follows by expanding the Möbius map. All these assertions hold under $Q_{N,t}$.

Consequently, one may add $A_{r,N}$ and any finite list of modes other than k to the unaltered observables Y_N in the proof of (17.9). The tightness just established and the exact pullback imply the same tightness under $Q_{N,t}$, because the transported observables have their original P_N laws and the differences tend to zero in probability. Bounded continuous tests therefore give, in the limit,

$$\mathbb{E} \left[F(\Xi, A_r, (G_j)_{j \neq k}) e^{-2 \operatorname{Re}(\bar{i} G_k)} \right] = e^{2k|t|^2/\beta} \mathbb{E} F(\Xi, A_r, (G_j)_{j \neq k}), \quad (51.3)$$

where initially only finitely many of the other modes are tested. Precisely the same uniform integrability of the exact likelihood ratios as in (17.8) justifies this passage; no bound on exponential moments of the meromorphic mark is required.

Uniqueness of the Gaussian Laplace transform identifies the conditional distribution of G_k given these observables as its unconditional Gaussian distribution. Bounded continuous tests extend to bounded Borel tests by the monotone-class theorem. Applying the identity successively to each coordinate of a finite Gaussian vector shows that this entire vector is independent of (Ξ, A_r) : at each step the remaining mode tests and the bounded test of (Ξ, A_r) are admissible. Countable-product uniqueness then gives independence from the full sequence G .

Finally, on E_r the censored mark is the actual field evaluation. Let $r \uparrow 1$ through the exhaustion and use bounded convergence to obtain independence of all finite field evaluations together with Ξ , restricted to $\{\Xi \neq \emptyset\}$. On the empty event the field is assigned zero, and its contribution factors because Ξ is already known to be independent of G . Evaluations on a countable dense set determine a meromorphic function and generate its Borel sigma-algebra. This proves (51.2). $\square$

Remark 51.3 (The convention on the empty branch). The earlier gluing construction set $\mathcal{C} = 0$ on the empty configuration. The present theorem instead specifies $\mathcal{A} = 0$ there, so it is convenient to set $\mathcal{C} = -T_G$ on that branch when the relation $\mathcal{A} = \mathcal{C} + T_G$ is desired everywhere. There are no particle force marks on that event, and every canonical resampling has count zero. Thus this convention changes none of the previous force identities or positive-count DLR equations. No convergence of the uncensored $\mathcal{C}_N$ on that branch is claimed.

52 Eliminating the independent Gaussian field from the canonical equations

The independence statement of Theorem 51.2 is stronger than regression of the force at the particles. It permits exact Gaussian elimination in the augmented canonical equations. The resulting kernels retain a meromorphic field, but no longer retain the independent Gaussian modes. Their pair interaction is the hyperbolic Green interaction.

Write

$$s_k(\mathbf{x}) = \sum_{i=1}^m x_i^k, \quad T_g(z) = \sum_{k \geq 1} \overline{g_k} z^{k-1}, \quad J_p(z) = \frac{1}{z-p} + \frac{\bar{p}}{1-z\bar{p}}.$$

Thus $J_p = \partial_z \log \delta(z, p)^2$, where $\delta(z, p) = |(z-p)/(1-z\bar{p})|$. Let $\mathcal{A} = \mathcal{C} + T_G$ on the nonempty event, with the empty branch defined as in the independence theorem for $\mathcal{A}$. For $B = B_j$ set

$$\mathcal{R}_B(z) = \mathcal{A}(z) - \sum_{p \in \Xi \cap B} J_p(z). \quad (52.1)$$

This function is holomorphic on a neighborhood of $\bar{B}$. On the nonempty event, if $H_{B,k} = G_k - s_k(\Xi \cap B)$, then

$$\mathcal{R}_B = \mathcal{E}_B + T_{H_B}. \quad (52.2)$$

Here T_{H_B} converges locally uniformly: the Gaussian coefficients have at most polynomial growth almost surely, and the subtracted finite power sums decay geometrically.

Lemma 52.1 (Retaining the exterior positive modes). *In the simultaneous construction of the meromorphic augmentation, all the exterior sequences H_B can also be retained. For every positive allowed count m , the conditional interior density given $(\eta, m, \mathcal{E}_B, H_B)$ is the original canonical density with harmonic field $\mathcal{E}_B$. Consequently, given $(\eta, m, \mathcal{R}_B, H_B = h)$ it is proportional to*

$$|\Delta_m(\mathbf{x})|^\beta \exp \left\{ \beta \operatorname{Re} \sum_i V_{\mathcal{R}_B}(x_i) - \beta \operatorname{Re} \sum_{k \geq 1} \frac{\bar{h}_k s_k(\mathbf{x})}{k} \right\} dA^m(\mathbf{x}), \quad (52.3)$$

where $V'_{\mathcal{R}_B} = \mathcal{R}_B$ and $V_{\mathcal{R}_B}(0) = 0$.

Proof. At finite N the quantities $H_{B,k,N} = S_{k,N} - s_k(\Xi_N \cap B)$ depend only on the exterior. The finite conditional density therefore remains exactly the same when any finite number of them is included in its conditioning. In the normal-family proof of Theorem 41.2, multiply the exterior test function by a bounded continuous function of any finite list of these coordinates. Their joint tightness follows from the Gaussian-mode tightness and the local count bounds. Along the simultaneous extraction, the finite interior power sums converge because ∂B contains no limiting particle. Hence their limits are precisely the displayed $H_{B,k}$.

The compact-test identity passes to the limit just as in that theorem: the integral against an interior compactly supported test is continuous in the holomorphic factor, and the additional bounded continuous exterior-mode test causes no change. Exhausting the interior tuple space proves that the limiting normalizing integral is one. The monotone-class theorem, first in each finite list of mode coordinates and then in their countable union, gives the asserted conditional identity with the whole sequence. Countably many disks and counts can be treated on the same extraction.

On the nonempty event, knowing $(\mathcal{R}_B, H_B)$ is equivalent to knowing $(\mathcal{E}_B, H_B)$ by (52.2). A primitive of T_h vanishing at zero is $\sum_{k \geq 1} \bar{h}_k z^k / k$. Substitution into the harmonic canonical density proves (52.3). $\square$

Lemma 52.2 (Gaussian Bayes formula on a compact tuple space). *Let $X \in B^m$, where $B = \mathbb{D}_r$, $r < 1$, and let Z be any random variable taking values in a standard Borel space. Suppose that $G = (G_k)_{k \geq 1}$ is independent of (X, Z) , with independent circular complex coordinates of variance $2k/\beta$. Put $H = G - s(X)$. Fix a deterministic tuple $\mathbf{a} \in B^m$ and write*

$$\|s(\mathbf{x})\|_*^2 = \sum_{k \geq 1} \frac{|s_k(\mathbf{x})|^2}{k}.$$

The Gaussian law of H given $X = \mathbf{x}$ has density, relative to the law $\gamma_{-s(\mathbf{a})}$ of $G - s(\mathbf{a})$,

$$r(\mathbf{x}, h) = \exp \left\{ -\beta \operatorname{Re} \sum_{k \geq 1} \frac{\overline{h_k} [s_k(\mathbf{x}) - s_k(\mathbf{a})]}{k} - \frac{\beta}{2} (\|s(\mathbf{x})\|_*^2 - \|s(\mathbf{a})\|_*^2) \right\}. \quad (52.4)$$

For almost every h this is continuous in $\mathbf{x}$ on $\overline{B}^m$, strictly positive, and bounded above and below by positive finite constants depending on h . If Π_Z denotes the conditional law of X given Z , then

$$\mathbb{P}(X \in d\mathbf{x} \mid Z, H = h) = \frac{r(\mathbf{x}, h) \Pi_Z(d\mathbf{x})}{\int r(\mathbf{y}, h) \Pi_Z(d\mathbf{y})}. \quad (52.5)$$

Proof. For finitely many coordinates, divide their explicit Gaussian densities. This gives the finite truncation of (52.4). All shifts belong to the Gaussian Cameron–Martin space because $|s_k(\mathbf{x})| \leq m r^k$ and $\sum_k r^{2k}/k < \infty$. For completeness, under the reference law write $h = G - s(\mathbf{a})$. The logarithm of the truncated density is a centered real Gaussian linear statistic minus half its variance. The variances are uniformly bounded, since they equal $\beta \sum_{k \leq L} |s_k(\mathbf{x}) - s_k(\mathbf{a})|^2/k$. Its exponentials therefore converge in L^1 , for example by their uniform L^2 bound, and the limit has expectation one. Agreement on all finite-coordinate cylinder events proves that the limit is the Radon–Nikodym density of the full translated product measure.

Almost surely $|G_k| \leq k^2$ eventually, by the Gaussian tail estimate and the summability of those tails. Consequently the series involving h in (52.4) converges absolutely and uniformly for $\mathbf{x} \in \overline{B}^m$. The norm series does too. The stated continuity and positive finite bounds follow. Independence of G and (X, Z) now expresses their joint conditional law, given Z , as $\Pi_Z(d\mathbf{x}) r(\mathbf{x}, h) \gamma_{-s(\mathbf{a})}(dh)$. Disintegration gives (52.5); its denominator is finite and strictly positive by the bounds just proved. $\square$

Theorem 52.3 (A Gaussian-free augmented Green specification). *For every subsequential limit, use the augmentation from Theorem 51.2, for which $(\Xi, \mathcal{A})$ is independent of all the Gaussian modes G_k , together with the retained-mode canonical identity of Lemma 52.1. For every $B = B_j$ and every allowed count $m \geq 1$, the interior conditional density given $(\eta, m, \mathcal{R}_B)$ is*

$$\frac{1}{Z_{B,m}(\mathcal{R}_B)} \prod_{i < j} \delta(x_i, x_j)^\beta \prod_{i=1}^m \frac{\exp\{\beta \operatorname{Re} V_{\mathcal{R}_B}(x_i)\}}{(1 - |x_i|^2)^{\beta/2}} dA^m(\mathbf{x}). \quad (52.6)$$

The normalizing constant is strictly positive and finite. Resample this tuple and update the meromorphic field by

$$\mathcal{A}^{\text{new}}(z) = \mathcal{A}(z) + \sum_{i=1}^m J_{x_i}(z) - \sum_{p \in \Xi \cap B} J_p(z). \quad (52.7)$$

For $m = 0$ use the identity kernel. These kernels preserve the augmented law of $(\Xi, \mathcal{A})$ and form a consistent canonical specification on the chosen nested disks.

Equivalently, their ordinary-area densities are proportional to

$$|\Delta_m(\mathbf{x})|^\beta \exp \left\{ \beta \operatorname{Re} \sum_i V_{\mathcal{R}_B}(x_i) + \frac{\beta}{2} \|s(\mathbf{x})\|_*^2 \right\}. \quad (52.8)$$

No hypothesis about convergence of an inverse exterior field is used in this elimination. The hypothesis of independence of the corrected field is essential; independence of Ξ and G alone would not justify conditioning additionally on $\mathcal{R}_B$.

Proof. On the event $\Xi(B) = m$, use the auxiliary ordering independent of all other variables and put $Z = (\eta, m, \mathcal{R}_B)$. This is a measurable function of $(\Xi, \mathcal{A})$ and the fixed count. The asserted Gaussian independence therefore remains valid under this count conditioning, and Lemma 52.2 applies.

For almost every Z and almost every h in its conditional law, (52.5) agrees with (52.3). The latter is a strictly positive density off the diagonals. Since $r(\mathbf{x}, h)$ is positive and bounded above and below, it follows first that Π_Z is absolutely continuous and then that its density is proportional to that density divided by $r(\mathbf{x}, h)$. Substitution cancels every term containing h . The terms containing only the reference tuple $\mathbf{a}$ are part of the normalizing constant. What remains is exactly (52.8). This also shows directly that the resulting density does not depend on the selected good value of h .

The elementary series $\sum_{k \geq 1} u^k/k = -\log(1 - u)$ gives

$$\|s(\mathbf{x})\|_*^2 = -\sum_i \log(1 - |x_i|^2) - 2 \sum_{i < j} \log |1 - x_i \bar{x}_j|.$$

This proves (52.6). The primitive $V_{\mathcal{R}_B}$ is bounded on $\bar{B}$, as is $(1 - |x|^2)^{-\beta/2}$, while $0 \leq \delta(x, y) \leq 1$ on the disk. Thus the normalizing integral is finite. It is positive because the integrand is positive on the open set of distinct tuples.

Given the exterior, its count, $\mathcal{R}_B$ and a resampled tuple, the global meromorphic field is uniquely determined by its restriction $\mathcal{R}_B + \sum_i J_{x_i}$ to B . Its continuation exists by (52.7), and is measurable by the same Borel restriction argument used for the original meromorphic augmentation. This proves that the displayed kernel is the conditional law of the entire augmented state and hence preserves it.

For nested disks $B_j \subset B_k$ one has

$$\mathcal{R}_{B_j}(z) = \mathcal{R}_{B_k}(z) + \sum_{p \in \Xi \cap (B_k \setminus B_j)} J_p(z).$$

A move in B_j leaves the exterior, total count, and field $\mathcal{R}_{B_k}$ unchanged, by (52.7). Conversely, conditioning the B_k kernel on its annular points factors its Green pair products into the inner pair product and the cross factors $\prod_{i,p} \delta(x_i, p)^\beta$. Each cross factor is the exponential of the real part of a primitive of βJ_p on B_j . The fixed one-body weight $(1 - |x|^2)^{-\beta/2}$ is unchanged. Therefore the inner conditional density is precisely the B_j kernel. These two facts prove both orders of canonical consistency, including zero-count cases as in Theorem 43.2. $\square$

Proposition 52.4 (Isometry covariance of the Gaussian-free kernels). *For this covariance statement, use the ambient admissible state space in which an empty configuration may carry any holomorphic field $\mathcal{A}$. The law constructed above is supported on the particular choice $\mathcal{A} = 0$ on its empty event. That convention need not be preserved by the following action. For a holomorphic automorphism φ of $\mathbb{D}$, transform an augmented state by*

$$\Xi^\varphi = \varphi^{-1}\Xi, \quad \mathcal{A}^\varphi(z) = \varphi'(z)\mathcal{A}(\varphi(z)) + \left(\frac{2}{\beta} - \frac{1}{2}\right) \frac{\varphi''(z)}{\varphi'(z)}. \quad (52.9)$$

The kernels (52.6) transform covariantly between a relatively compact disk and its image or preimage under φ . The statement applies to any such disk for which the kernel is defined by its exterior field, regardless of the particular nested exhaustion used above. It is covariance of the specification; it does not assert invariance of its Gibbs laws. This affine action is chosen to compensate the one-body reference measure in the kernel. It is distinct from the pure one-form identity obeyed by the finite physical sum $\mathcal{A}_N = \sum_p J_p$ under direct transport of all its particles. No identification of those two actions is asserted.

Proof. Hyperbolic distance invariance gives

$$\varphi'(z)J_{\varphi(p)}(\varphi(z)) = J_p(z).$$

Consequently the exterior field transforms by the same affine formula as (52.9). A primitive of the additional term is $(2/\beta - 1/2)\log \varphi'$, up to an irrelevant constant. The identity $1 - |\varphi(z)|^2 = |\varphi'(z)|(1 - |z|^2)$ shows that the one-body area measure in the kernel transforms as

$$\frac{dA(\varphi(z))}{(1 - |\varphi(z)|^2)^{\beta/2}} = |\varphi'(z)|^{2-\beta/2} \frac{dA(z)}{(1 - |z|^2)^{\beta/2}}.$$

The logarithm of this extra factor is exactly β times the real part of the additional primitive. The pair factors are invariant, and normalization then proves the claim. Reflection is handled by complex conjugation, so the same conclusion holds for the full isometry group with the corresponding conjugated field. $\square$

Remark 52.5 (The remaining information in the field). Relative to hyperbolic area $d\lambda = (1 - |z|^2)^{-2}dA$, the one-body factor in (52.6) is $(1 - |z|^2)^{2-\beta/2} \exp\{\beta \operatorname{Re} V_{\mathcal{R}_B}(z)\}$. Its deterministic curvature is the one associated with a local neutralizing density $(4/\beta - 1)/\pi$ relative to λ . This identifies the background coefficient in the augmented equations. It neither proves that this coefficient is the particle intensity nor identifies $\mathcal{R}_B$ as a compensated sum determined by the unmarked exterior. Such an identification, or a uniqueness theorem for these augmented equations with specified boundary behavior and count weights, would be an additional result.

Corollary 52.6 (Positive-kernel factorization of the unmarked conditional law). *For almost every exterior η and every allowed count $m \geq 1$, let $g_{\eta,m}$ be the analytic factor in Theorem 41.3. There is a positive Hermitian kernel $\mathcal{H}_{\eta,m}(\mathbf{x}, \mathbf{y})$, holomorphic in $\mathbf{x} \in B^m$ and antiholomorphic in $\mathbf{y} \in B^m$, such that*

$$g_{\eta,m}(\mathbf{x}) = \exp \left\{ \frac{\beta}{2} \|s(\mathbf{x})\|_*^2 \right\} \mathcal{H}_{\eta,m}(\mathbf{x}, \mathbf{x}). \quad (52.10)$$

Equivalently, the polarized positive kernel with diagonal $g_{\eta,m}$ factors as

$$\mathcal{G}_{\eta,m}(\mathbf{x}, \mathbf{y}) = \exp \left\{ \frac{\beta}{2} \sum_{k \geq 1} \frac{s_k(\mathbf{x}) \overline{s_k(\mathbf{y})}}{k} \right\} \mathcal{H}_{\eta,m}(\mathbf{x}, \mathbf{y}). \quad (52.11)$$

In particular the quotient by the displayed Gaussian kernel remains positive as a kernel, a stronger assertion than positivity of its logarithmic Levi form.

Proof. Disintegrate the conditional Green kernels over the law $\mu_{\eta,m}$ of $\mathcal{R}_B$ given (η, m) . Put

$$F_R(\mathbf{x}) = \mathcal{Z}_{B,m}(R)^{-1/2} \exp \left\{ \frac{\beta}{2} \sum_i V_R(x_i) \right\}, \quad \mathcal{H}_{\eta,m}(\mathbf{x}, \mathbf{y}) = \int F_R(\mathbf{x}) \overline{F_R(\mathbf{y})} d\mu_{\eta,m}(R).$$

The diagonal identity follows from (52.8), initially almost everywhere. The deterministic compact upper bounds for $g_{\eta,m}$ from the canonical quasi-Gibbs theorem and the positivity of the Gaussian factor bound these diagonal integrals locally almost everywhere. The mean-value inequality for each $|F_R|^2$, integrated in R and then over a slightly larger polydisk, upgrades this to a bound at every point of a smaller polydisk. Cauchy–Schwarz bounds the corresponding off-diagonal integrals. Local integration of holomorphic functions, or the mean-value formula followed by Fubini on smaller polydisks, proves joint holomorphic–antiholomorphic regularity. It also extends the diagonal identity to every tuple, including the diagonals. Positivity of $\mathcal{H}$ follows directly by integrating the square of any finite linear combination of the F_R . The exponential kernel in (52.11) has diagonal equal to the Gaussian factor in (52.10); its

positivity follows from its power-series expansion as an exponential of an inner-product kernel. The product therefore has diagonal $g_{\eta,m}$ and is positive. Uniqueness of polarization of a real analytic diagonal gives (52.11). $\square$

Corollary 52.7 (A conditionally independent Gaussian component of every force). *Given the unmarked configuration, the joint force law at any finite list of distinct particles is the convolution of a probability measure and a nondegenerate circular complex Gaussian law. More precisely, if $\mathbf{z} = (z_1, \dots, z_m)$ are distinct particles and A_i is the regular part of $\mathcal{A}$ at z_i , then*

$$(C_{z_i})_{i=1}^m = (A_i)_{i=1}^m - (T_G(z_i))_{i=1}^m,$$

and the two vectors on the right are conditionally independent given Ξ . The Gaussian covariance matrix is

$$\Sigma_{ij} = \frac{2}{\beta} \frac{1}{(1 - z_i \bar{z}_j)^2}.$$

Thus this conditional force law has a strictly positive smooth density on $\mathbb{C}^m$, bounded above by $(\pi^m \det \Sigma)^{-1}$. Its characteristic function satisfies

$$\left| \mathbb{E} \left[e^{i \operatorname{Re} \sum_i \bar{t}_i C_{z_i}} \mid \Xi \right] \right| \leq \exp \left\{ -\frac{1}{2\beta} \sum_{i,j} \bar{t}_i \frac{t_j}{(1 - z_i \bar{z}_j)^2} \right\}.$$

Statements at selected particles may be interpreted under finite Campbell windows, or using any measurable ordering of the particles.

Proof. Adding the holomorphic function T_G changes the regular part at a pole by its value at that pole, so the displayed decomposition follows from the definition of $\mathcal{A}$. The independence of $(\Xi, \mathcal{A})$ and G proves the conditional independence. The covariance follows by summing $\sum_{k \geq 1} k(z_i \bar{z}_j)^{k-1}$. For distinct points this matrix is positive definite: a vanishing quadratic form would give $\sum_i \bar{t}_i z_i^{k-1} = 0$ for every k , and the first m equations form an invertible Vandermonde system.

The Gaussian density is everywhere positive and bounded by its value at zero. Integrating its translates gives the asserted positive bounded density. Each derivative of a nondegenerate Gaussian density is bounded, so differentiation under the probability integral proves smoothness of all orders. Finally a circular complex Gaussian with covariance Σ has real-linear characteristic function $\exp(-t^* \Sigma t / 4)$. The other independent factor has modulus at most one, yielding the stated bound. $\square$

53 Why the corrected canonical equations still need a state-selection condition

The Gaussian elimination gives substantially more information than a spatial Ward equation. Nevertheless, the resulting canonical equations alone cannot imply either uniqueness or hyperbolic invariance. The following explicit family demonstrates this point for $0 < \beta < 2$. Each fixed- m law below is excluded as an original-gas limit by Theorem 48.2: that theorem gives infinite expected total count, whereas the example has total count m .

Proposition 53.1 (Distinct finite-particle solutions of the corrected equations). *Suppose $0 < \beta < 2$, and let $m \geq 1$ be an integer. There is a rotationally invariant probability law $P_\beta^{[m]}$ with exactly m particles in $\mathbb{D}$, whose uniformly ordered tuple has density*

$$\frac{1}{Z_{\beta,m}} \prod_{1 \leq i < j \leq m} \delta(z_i, z_j)^\beta \prod_{i=1}^m (1 - |z_i|^2)^{-\beta/2} dA^m(\mathbf{z}). \quad (53.1)$$

Augment it by

$$\mathcal{A}(z) = \sum_{p \in \Xi} J_p(z).$$

This augmented law satisfies all the Gaussian-free canonical kernels of Theorem 52.3, on every disk $B \Subset \mathbb{D}$. These laws are distinct for distinct m . No one of them is invariant under all disk automorphisms.

More generally, one may replace the density in (53.1) by its product with

$$\exp\left\{\beta \operatorname{Re} \sum_i h(z_i)\right\},$$

where h is any holomorphic function on a neighborhood of $\overline{\mathbb{D}}$, and set

$$\mathcal{A} = h' + \sum_{p \in \Xi} J_p.$$

The same canonical assertion holds. Arbitrary probability mixtures over the integer $m \geq 0$ of the laws with $h = 0$, with the empty state at $m = 0$, satisfy these equations as well.

Proof. Put $b = \beta/2 < 1$. Since $0 \leq \delta(z, w) \leq 1$,

$$Z_{\beta, m} \leq \left(\int_{\mathbb{D}} (1 - |z|^2)^{-b} dA(z) \right)^m = \left(\frac{\pi}{1 - b} \right)^m < \infty.$$

The integral is strictly positive, because its integrand is positive on the nonempty open set of distinct tuples. The density is symmetric, absolutely continuous, and invariant under simultaneous rotations. It therefore defines the stated simple point process.

Fix a disk $B \Subset \mathbb{D}$. Its boundary has no particles almost surely. Given its exterior η and its inside count n , factor the finite pair product in (53.1). The conditional density of its inside tuple $\mathbf{x}$ is proportional to

$$\prod_{i < j} \delta(x_i, x_j)^\beta \prod_i (1 - |x_i|^2)^{-b} \prod_{\substack{i \leq n \\ p \in \eta}} \delta(x_i, p)^\beta.$$

The retained exterior field is $\mathcal{R}_B = \sum_{p \in \eta} J_p$. Each function $\log \delta(z, p)$ is harmonic on B , and $\partial_z \log \delta(z, p)^2 = J_p(z)$. Hence the last product is, up to a constant independent of $\mathbf{x}$, the exponential of $\beta \operatorname{Re} \sum_i V_{\mathcal{R}_B}(x_i)$. This is exactly (52.6). Replacing the inside tuple changes $\mathcal{A}$ by precisely (52.7). Thus the assertion holds for the entire augmented state, not only for its position marginal. The argument applies without alteration when the additional field h' is retained. That density remains normalizable because $\operatorname{Re} h$ is bounded on $\overline{\mathbb{D}}$.

The laws for different total counts have disjoint supports. A finite nonzero invariant point process cannot exist here: its intensity would be a nonzero finite invariant measure on $\mathbb{D}$. In the present case this measure has a continuous density, and invariance would force that density to be $c(1 - |z|^2)^{-2}$ for some $c > 0$ by the change of variables under an automorphism sending zero to a specified point. Such a measure has infinite mass. This contradicts its total mass m . Finally, the specification preserves the total count and is the same kernel on each of these invariant sectors, so integration of the stationarity identities against arbitrary mixture weights proves the mixture assertion. $\square$

Example 53.2 (The one-particle solution saturates the proved curvature bound). For $m = 1$, the density in the proposition is explicitly

$$\rho_\beta^{[1]}(z) = \frac{1 - \beta/2}{\pi} (1 - |z|^2)^{-\beta/2}.$$

It satisfies

$$\Delta \log \rho_\beta^{[1]}(z) = \frac{2\beta}{(1 - |z|^2)^2},$$

so the unconditional Gaussian curvature inequality alone does not exclude it. Its intensity is not proportional to hyperbolic area. The example shows concretely that the remaining state-selection issue is not merely a missing formulation of the spatial equations.

Remark 53.3 (Adding back the independent Gaussian does not remove the examples). Attach to any of the preceding laws an independent Gaussian sequence with $\mathbb{E}|G_k|^2 = 2k/\beta$, and set $\mathcal{C} = \mathcal{A} - T_G$. This gives exactly the asserted independence of $(\Xi, \mathcal{A})$ from G . For completeness, these augmented laws also satisfy the earlier Euclidean harmonic canonical equations. On a disk B , its exterior fixes $\mathcal{R}_B$, and write $H = G - s(\Xi \cap B)$. Conditional on the inside tuple $\mathbf{x}$, the translated Gaussian density in H contributes the factor

$$\exp \left\{ -\beta \operatorname{Re} \sum_{k \geq 1} \frac{\overline{H_k} s_k(\mathbf{x})}{k} - \frac{\beta}{2} \|s(\mathbf{x})\|_*^2 \right\},$$

up to a factor independent of $\mathbf{x}$; this is the Gaussian Bayes formula of Lemma 52.2. The norm term cancels the positive norm term in (52.8). The resulting density is $|\Delta_n(\mathbf{x})|^\beta$ times the exponential of a harmonic one-body potential with derivative $\mathcal{E}_B = \mathcal{R}_B - T_H$. The field H is determined by $\mathcal{E}_B$ and the unmarked exterior in these examples: $\mathcal{R}_B$ is already determined by that exterior, and the restriction of T_H to B uniquely determines its holomorphic continuation to $\mathbb{D}$, whose Taylor coefficients determine H . This continuation is measurable by the Borel restriction argument used for the meromorphic augmentation. Thus conditioning on the original exterior-field mark gives exactly the original canonical kernel. The count probabilities and the admissible boundary behavior of the corrected field remain additional information to be selected by the finite- N model.

Corollary 53.4 (All these finite-count mixtures are excluded as gas limits). *For $0 < \beta < 2$, no probability mixture $\sum_{m \geq 0} p_m P_\beta^{[m]}$ of the preceding finite-particle laws can be a subsequential limit of the original disk gas, even if the mixture has infinite expected total count.*

Proof. Every such mixture has finite total count almost surely, whereas Theorem 48.6 proves that every actual subsequential limit has infinite total count almost surely. $\square$

54 Infinitely many infinite solutions of the canonical Green equations

The finite-count examples do not address the almost sure infinitude of actual limits for $0 < \beta < 2$. The following construction does. For every $0 < \beta < 4$ it produces distinct, almost surely infinite, rotationally invariant position laws satisfying exactly the same augmented canonical Green equations as the disk-gas limits. These are counterexamples to uniqueness of that specification, not counterexamples to uniqueness of the original finite- N limit. The almost sure non-Blaschke property of actual limits will in fact make every new example mutually singular with every original-gas limit when $0 < \beta \leq 2$.

Throughout this section put

$$d(z) = 1 - |z|^2, \quad g(z, w) = -\log \delta(z, w), \quad b = \beta/2.$$

A grand-canonical kernel permits both insertion and deletion of particles; its additional parameter $\lambda > 0$ is the activity. Conditioning such a kernel on its inside count removes this parameter.

54.1 A construction with summable Green tails

Lemma 54.1 (Existence with a summable boundary weight). *Let $\beta > 0$, and let w be a strictly positive continuous function on $\mathbb{D}$ such that*

$$\int_{\mathbb{D}} d(z)w(z) dA(z) < \infty. \quad (54.1)$$

For every $\lambda > 0$ there exists a simple locally finite point process $P_{\lambda, w}$ satisfying the following assertions.

1. *Its factorial moment densities satisfy, almost everywhere,*

$$\rho_p(z_1, \dots, z_p) \leq \lambda^p \prod_{i=1}^p w(z_i), \quad p \geq 1. \quad (54.2)$$

In particular,

$$\mathbb{E} \sum_{p \in \Xi} d(p) \leq \lambda \int_{\mathbb{D}} d(z)w(z) dA(z) < \infty. \quad (54.3)$$

2. *For every disk $B \Subset \mathbb{D}$, given $\eta = \Xi \cap B^c$, its interior grand-canonical density, with the usual unordered reference $\sum_{m \geq 0} dA^m/m!$, is*

$$\frac{1}{Z_B(\eta)} \lambda^m \prod_{i=1}^m w(x_i) \exp \left\{ -\beta \sum_{i < j} g(x_i, x_j) - \beta \sum_{i=1}^m \sum_{p \in \eta} g(x_i, p) \right\}. \quad (54.4)$$

The exterior sums are finite, uniformly on $\bar{B}$, almost surely, and $1 \leq Z_B(\eta) \leq \exp\{\lambda \int_B w dA\}$.

3. *Its insertion identity is*

$$\mathbb{E} \sum_{x \in \Xi} F(x, \Xi \setminus \{x\}) = \lambda \mathbb{E} \int_{\mathbb{D}} F(x, \Xi) e^{-\beta \sum_{p \in \Xi} g(x, p)} w(x) dA(x) \quad (54.5)$$

for every nonnegative measurable F , with equality in $[0, \infty]$.

4. *If $\int_{\mathbb{D}} w dA = \infty$, then $\#\Xi = \infty$ almost surely. For a fixed w , states obtained with different activities have distinct position laws, regardless of the choices made in taking limits.*

The process may be obtained as a subsequential limit of the free grand-canonical laws on concentric disks. Consequently it may be chosen rotationally invariant whenever w is rotationally invariant.

Proof. Write $\sigma(dz) = w(z)dA(z)$ and $D_R = \{|z| < R\}$. On D_R use the probability density proportional to

$$\lambda^n e^{-\beta \sum_{i < j} g(z_i, z_j)} \prod_i w(z_i)$$

relative to $\sum_{n \geq 0} dA^n/n!$. The partition function lies between one and $e^{\lambda \sigma(D_R)}$, since $g \geq 0$. Its ordered factorial density is

$$\lambda^p \prod_i w(z_i) e^{-\beta \sum_{i < j} g(z_i, z_j)} \frac{Z_R[z_1, \dots, z_p]}{Z_R},$$

where the numerator inserts into every remaining-particle integrand the factor

$$\exp\{-\beta \sum_{i,j} g(z_i, y_j)\} \leq 1.$$

This proves (54.2) uniformly in R . These bounds imply local tightness: for every compact K and every $t \geq 0$,

$$\mathbb{E}(1+t)^{\Xi_R(K)} = \sum_{p \geq 0} \frac{t^p}{p!} \int_{K^p} \rho_{p,R} dA^p \leq e^{\lambda t \sigma(K)}.$$

The second factorial bound excludes collisions in every limiting process: on any compact set its integral over ordered pairs at distance less than ε tends to zero as $\varepsilon \downarrow 0$. The first bound implies that deterministic area-null sets contain no limiting particle. A diagonal extraction therefore gives a simple locally finite limit. Testing factorial measures on compact sets and using the exponential count bound passes (54.2) to it. This also proves (54.3) and the bounds

$$\sup_R \mathbb{E} \sum_{p \in \Xi_R \setminus D_s} d(p) \leq \lambda \int_{\mathbb{D} \setminus D_s} d d\sigma \longrightarrow 0 \quad (s \uparrow 1), \quad (54.6)$$

with the same inequality for the limit.

We give the details needed to pass the conditional kernels, because the infinite mass of σ prevents replacing this step by an assertion of global Poisson finiteness. For each compact $K \Subset \mathbb{D}$ there are $s_K < 1$ and $C_K < \infty$ for which

$$\sup_{z \in K} g(z, p) \leq C_K d(p), \quad |p| > s_K. \quad (54.7)$$

Indeed,

$$1 - \delta(z, p)^2 = \frac{d(z)d(p)}{|1 - z\bar{p}|^2} \leq C'_K d(p),$$

and $-\log(1-u) \leq 2u$ for $0 \leq u \leq 1/2$. The identity $g = -\frac{1}{2} \log \delta^2$ proves the claim. Thus the total contribution of points outside D_s to each exterior potential on K tends to zero in expectation, uniformly in the finite-volume index, by (54.6).

One can consequently carry out the extraction in a topology that records both vague convergence of configurations and their finite weighted measures $\sum_p d(p)\delta_p$ on the closed disk. To justify this without any additional compactness assumption, these finite measures have uniformly bounded total mass; their restrictions outside D_s have uniformly vanishing expected mass by (54.6). Tightness of finite measures on the compact closed disk and local configuration tightness give a joint extraction. The first moment tail bound forces the limiting weighted measure to put no mass on the boundary and to equal $\sum_p d(p)\delta_p$ in the interior. Along a Skorokhod representation of this joint extraction, convergence of weighted total masses and of weighted measures on each $\overline{D}_s$ whose boundary is empty implies

$$\lim_{s \uparrow 1} \limsup_{R \uparrow 1} \sum_{p \in \Xi_R \setminus D_s} d(p) = 0 \quad \text{almost surely.}$$

Here and below $R \uparrow 1$ refers to the extracted sequence.

Fix $B \Subset \mathbb{D}$. Its boundary is empty almost surely. There are only finitely many exterior points in any larger compact disk, and their positive distance from $\overline{B}$ is positive. Matching these points under vague convergence and using (54.7) shows that their exterior potentials converge uniformly on $\overline{B}$. Each fixed- m density in (54.4) therefore converges almost everywhere. It is bounded by $\lambda^m \prod_i w(x_i)$, whose summed integrals divided by $m!$ equal $e^{\lambda \sigma(B)}$. Dominated convergence gives convergence of the normalizing constants and of all bounded

interior test integrals. The original finite-volume conditional identity can now be multiplied by a bounded continuous exterior cylinder function and passed to the limit. A monotone-class argument proves the conditional identity for the full exterior sigma field. This works for every specified disk B ; a countable determining family can be handled simultaneously. It proves the second assertion, including uniform finiteness of its exterior sums.

For completeness, conditional on the exterior of a disk, shift the summation index m in the left side of (54.5) and single out one of its m coordinates. The factor $m/m!$ becomes $1/(m-1)!$, and the singled-out particle contributes precisely $\lambda w(x) \exp\{-\beta \sum_p g(x, p)\}$. This proves the insertion identity for nonnegative F supported in that disk in its first coordinate. Monotone convergence over disks proves the full identity.

Suppose now $\sigma(\mathbb{D}) = \infty$. For each deterministic finite configuration η , the function

$$a_\eta(x) = \prod_{p \in \eta} \delta(x, p)^\beta$$

tends uniformly to one as $|x| \uparrow 1$. Since σ is finite on compact subsets, $\int a_\eta d\sigma = \infty$. Apply (54.5) with $F(x, \eta) = \mathbf{1}_{\{\#\eta=m\}}$. Its left side is $(m+1)\mathbb{P}(\#\Xi = m+1)$ and is finite; its right side is infinite if $\mathbb{P}(\#\Xi = m) > 0$. Therefore $\mathbb{P}(\#\Xi = m) = 0$ for every integer $m \geq 0$.

Finally take a nonempty relatively compact disk B . Its insertion integral

$$I_w(\Xi) = \int_B e^{-\beta \sum_p g(x, p)} w(x) dA(x)$$

is bounded by $\sigma(B)$ and is strictly positive almost surely. The strict positivity follows from finite weighted depth, the Green tail bound, and local finiteness: the potential is finite at every x except possibly the countable set of particles. The identity gives $\mathbb{E}\Xi(B) = \lambda \mathbb{E}I_w(\Xi)$. Both expectations are functionals of the position law, and the second is strictly positive. The same position law therefore cannot satisfy the identity with two different activities. Rotation invariance, when available in finite volume, passes to the extracted law. $\square$

Theorem 54.2 (Distinct infinite canonical states at every $0 < \beta < 4$). *For each $0 < \beta < 4$ there are infinitely many distinct, almost surely infinite, rotationally invariant position laws which admit an augmentation satisfying the Gaussian-free canonical specification (52.6)–(52.7). Their corrected meromorphic fields can be taken in the particularly restrictive form*

$$\mathcal{A}(z) = h'(z) + \sum_{p \in \Xi} J_p(z), \quad \sum_{p \in \Xi} d(p) < \infty \quad \text{almost surely}, \quad (54.8)$$

where the sum converges absolutely and locally uniformly off the particles. For $2 \leq \beta < 4$ one may take $h = 0$. For $0 < \beta < 2$ one may take a single random boundary point ξ , uniformly distributed on the unit circle, and

$$h_\xi(z) = -\frac{2 - \beta/2}{\beta} \log(1 - \bar{\xi}z), \quad h'_\xi(z) = \frac{2 - \beta/2}{\beta} \frac{\bar{\xi}}{1 - \bar{\xi}z}. \quad (54.9)$$

The branch of the logarithm vanishes at zero. In the latter construction, ξ is almost surely the unique accumulation point of Ξ on the boundary; in particular it is determined by the positions.

None of these rotationally invariant states is invariant under the full group of disk automorphisms. For $0 < \beta \leq 2$, each of them is mutually singular with every subsequential limit of the original unweighted disk gas.

Proof. For a holomorphic function h on $\mathbb{D}$ set

$$w_h(z) = d(z)^{-b} e^{\beta \operatorname{Re} h(z)}.$$

For $1 \leq b < 2$, $h = 0$ gives

$$\int_{\mathbb{D}} w_h dA = \infty, \quad \int_{\mathbb{D}} d w_h dA = \frac{\pi}{2-b} < \infty.$$

The preceding lemma therefore supplies infinite rotationally invariant states, distinct for different λ .

Suppose $0 < b < 1$, put $s = 2 - b \in (1, 2)$, and first fix $\xi = 1$. The weight is $w_h(z) = d(z)^{-b}|1 - z|^{-s}$. It is integrable outside every neighborhood of 1. Near 1 use $z = (1 - t)e^{i\theta}$, with $t > 0$ and $|\theta|$ small. Uniform comparability of the corresponding positive quantities gives

$$d(z) \asymp t, \quad |1 - z|^2 \asymp t^2 + \theta^2, \quad dA(z) \asymp dt d\theta.$$

Since $s > 1$, for fixed small $\varepsilon > 0$ and $0 < t < \varepsilon/2$,

$$\int_{-\varepsilon}^{\varepsilon} (t^2 + \theta^2)^{-s/2} d\theta \asymp t^{1-s};$$

substitution $\theta = tu$ proves both bounds by comparison with the positive finite integral over the real line. Consequently the local integral of $d^q w_h$ has the same finiteness as

$$\int_0^{\varepsilon/2} t^{q-b+1-s} dt = \int_0^{\varepsilon/2} t^{q-1} dt.$$

It diverges for $q = 0$ and converges for $q = 1$. The weight thus meets all hypotheses of the lemma. Its point process has infinitely many points, but only finitely many outside every neighborhood of 1, by the first moment bound. It follows that 1 is its unique boundary accumulation point. This point is a measurable function of the process. Indeed, for a fixed sequence $r_n \uparrow 1$, on this event it equals

$$\lim_{n \rightarrow \infty} \frac{\sum_{p \in \Xi \cap D_{r_n}} p}{\Xi(D_{r_n})};$$

the denominators tend to infinity, and all but finitely many summands lie in any prescribed neighborhood of the unique accumulation point. Terms with zero denominator can be assigned the value zero. Rotate a chosen such law by ξ and average over uniform ξ . This produces a rotationally invariant law, with the same finite expected weighted depth and the asserted measurable boundary parameter. The law at $\xi = 1$ may also be chosen invariant under complex conjugation, since all its finite-volume laws have that symmetry; the rotational mixture is then invariant under reflection as well.

To verify the canonical equations in all cases, note the algebraic bound

$$|J_p(z)| = \frac{d(p)}{|z - p| |1 - z\bar{p}|} \leq C_K d(p), \quad z \in K, \quad |p| > s_K,$$

for every compact $K \Subset \mathbb{D}$ and suitable $s_K < 1$. The finite expected depth proves absolute local uniform convergence of (54.8). The series is meromorphic with a simple pole of residue one at each particle. Its retained exterior field is

$$\mathcal{R}_B = h' + \sum_{p \in \eta} J_p.$$

The grand-canonical density of the preceding lemma, conditioned on its count m , is

$$\prod_{i < j} \delta(x_i, x_j)^\beta \prod_i d(x_i)^{-b} \exp \left\{ \beta \operatorname{Re} \sum_i h(x_i) - \beta \sum_{i, p \in \eta} g(x_i, p) \right\}.$$

As J_p is the holomorphic derivative corresponding to $\log \delta(\cdot, p)$, this is precisely the required kernel with field $\mathcal{R}_B$, up to a constant independent of the inside tuple. Replacing that tuple updates $\mathcal{A}$ by the required finite sum. For fixed h the exterior field is already a function of the exterior. In the rotational mixture, h_ξ is also a function of the exterior, because deleting finitely many particles leaves the unique boundary accumulation point unchanged. Thus conditioning additionally on the field does not change the displayed conditional density.

For the rotational mixture with $b < 1$, distinctness still holds at the level of position laws. Its insertion identity is

$$\mathbb{E} \sum_{x \in \Xi} F(x, \Xi \setminus \{x\}) = \lambda \mathbb{E} \int F(x, \Xi) e^{-\beta \sum_p g(x, p)} w_{h_\xi(\Xi)}(x) dA(x).$$

This follows by averaging the identities at fixed ξ ; finite insertion or deletion does not change $\xi(\Xi)$. On a fixed compact disk the integral on the right, without λ , is positive and bounded, uniformly in ξ . The same argument as in the lemma identifies λ from the position law. This proves distinctness for the rotational mixtures too.

All constructed laws have finite expected weighted depth. If one were invariant under every disk automorphism, its locally finite intensity would be $c dA/d^2$, by transitivity and change of variables. The insertion identity makes this measure nonzero, so $c > 0$. But its expected weighted depth would then be $c \int_{\mathbb{D}} dA/d = \infty$, a contradiction. Finally, Theorem 49.2 proves that every actual limit for $0 < \beta < 2$ has $\sum_p d(p) = \infty$ almost surely. At $\beta = 2$ the unique original limit is Bergman; its full hyperbolic invariance, almost sure nonemptiness, and Theorem 45.1 imply $\sum_p g(0, p) = \infty$ almost surely. Since $g(0, p) = -\log |p|$ is comparable to $d(p)$ outside any fixed compact disk, and local finiteness accounts for the remaining finitely many points, Bergman also has $\sum_p d(p) = \infty$ almost surely. (The origin is almost surely not a particle.) All the constructed laws instead satisfy $\sum_p d(p) < \infty$ almost surely. These disjoint measurable events prove the asserted mutual singularity. $\square$

54.2 An explicit determinantal family at $\beta = 2$

Proposition 54.3 (Activity interpolates from dilute infinite states to Bergman). *At $\beta = 2$ and $h = 0$ the preceding grand-canonical disk exhaustion has a unique limit for each fixed activity $\lambda > 0$. It is the determinantal process, relative to ordinary area, with kernel*

$$K_\lambda(z, w) = \sum_{n=0}^{\infty} \frac{\lambda(n+1)}{n+1+\pi\lambda} (z\bar{w})^n. \quad (54.10)$$

It satisfies the same augmented canonical Green equations as the Bergman process, has infinitely many points almost surely, and satisfies

$$\rho_\lambda(0) = \frac{\lambda}{1+\pi\lambda}, \quad \rho_\lambda(z) \leq \frac{\lambda}{1-|z|^2}, \quad \mathbb{E} \sum_{p \in \Xi} d(p) \leq \pi\lambda.$$

The parameter λ therefore gives explicit distinct infinite solutions of those equations even at the exactly solvable temperature. As $\lambda \uparrow \infty$ these position laws converge locally to the Bergman process.

Proof. The Cauchy determinant identity gives

$$\det \left[\frac{1}{1 - z_i \bar{z}_j} \right]_{i,j=1}^m = \prod_i d(z_i)^{-1} \prod_{i < j} \delta(z_i, z_j)^2. \quad (54.11)$$

For example, multiplying the determinant by $\prod_{i,j} (1 - z_i \bar{z}_j)$ yields a polynomial alternating in the z_i and separately in their conjugates. Its degrees force it to be the product of the two

Vandermondes; comparing a coefficient fixes the sign and constant, and rearranging gives the displayed positive identity.

On D_R the grand-canonical density is consequently an L -ensemble with $L(z, w) = \lambda/(1 - z\bar{w})$. The functions

$$e_{n,R}(z) = \sqrt{\frac{n+1}{\pi R^{2n+2}}} z^n$$

are orthonormal, and the nonzero eigenvalues of L are $\ell_{n,R} = \lambda\pi R^{2n+2}/(n+1)$. They are summable when $R < 1$. Expanding the determinant in this orthonormal basis and integrating each term proves

$$Z_R = \prod_{n \geq 0} (1 + \ell_{n,R}).$$

The same expansion, with a multiplicative test function inserted, or with finitely many coordinates left unintegrated, shows that its correlation kernel is $L(I + L)^{-1}$. More explicitly,

$$K_{\lambda,R}(z, w) = \sum_{n \geq 0} \frac{\lambda(n+1)}{n+1 + \lambda\pi R^{2n+2}} (z\bar{w})^n.$$

These calculations can first be performed with a finite orthonormal sum; passage to the displayed formulas is justified by positivity of the Cauchy–Binet expansions and the summability of the eigenvalues.

The coefficients are at most λ , so the kernels converge uniformly on compact subsets of $\mathbb{D} \times \mathbb{D}$ to (54.10). All fixed correlation functions converge locally uniformly. Hadamard’s inequality bounds their absolute values on a compact set K by $[\lambda/(1 - \sup_K |z|^2)]^p$. The factorial expansion of each compactly supported Laplace functional therefore converges absolutely, and proves uniqueness and convergence of the position law. It agrees with the state constructed above and inherits its infinite total count and canonical equations. Evaluation at zero and summation of the coefficient bound give the asserted intensity estimates and, by integration, the depth estimate.

Lastly, as $\lambda \uparrow \infty$ each coefficient tends to $(n+1)/\pi$ and is bounded by that quantity. Hence K_λ converges locally uniformly to

$$K_{\text{Berg}}(z, w) = \frac{1}{\pi(1 - z\bar{w})^2}.$$

The same locally dominated Laplace-functional expansion proves local convergence to the Bergman point process. $\square$

Remark 54.4 (What this establishes about nonuniqueness). The canonical equations, almost sure infinitude, rotational invariance, and even an absolutely convergent corrected exterior field do not select a unique state. This failure is proved here for every $0 < \beta < 4$. At $\beta = 2$ the explicit states also show why this failure is compatible with convergence of the original model to a unique limit: the original model selects Bergman, whereas every finite-activity state in (54.10) is mutually singular with Bergman. The same almost sure boundary-mass separation holds between every constructed state and every original-gas limit for $0 < \beta < 2$. No two different subsequential limits of the original unweighted disk gas are constructed by this argument. Moreover none of these counterstates is fully hyperbolically invariant. The question of uniqueness within a correctly selected invariant Gibbs class remains separate.

55 Canonical tails, extremal states, and the precise nonuniqueness question

There are three logically different questions: uniqueness of the augmented canonical Gibbs state, uniqueness of its position marginal, and uniqueness of the subsequential limit selected

by the original finite-particle sequence. The following results keep these questions separate. They also give an exact test for whether a decomposition into Gibbs states changes the position law.

Fix $\beta > 0$ and nested disks $B_j \uparrow \mathbb{D}$. On the admissible augmented state space of pairs $\omega = (\Xi, \mathcal{A})$, write K_j for the Gaussian-free canonical kernel in Theorem 52.3, and set

$$I_j(\omega) = (\Xi|_{B_j^c}, \Xi(B_j), \mathcal{R}_{B_j}), \quad \mathcal{F}_j = \sigma(I_j).$$

We work on a common measurable state space on which these kernels and invariants are defined. A law P is called a Gibbs law for these kernels if $K_j F$ is a version of $\mathbb{E}_P[F \mid \mathcal{F}_j]$ for every bounded measurable F and every j . The conditional-kernel and consistency assertions already proved give this property to the constructed augmented subsequential limits. Boundary-null assumptions are understood for the chosen countable exhaustion.

Lemma 55.1 (The canonical information decreases along an exhaustion). *For $j < k$, one has $\mathcal{F}_k \subseteq \mathcal{F}_j$.*

Proof. The exterior of B_j determines both the exterior of B_k and all the particles in $B_k \setminus B_j$. Moreover,

$$\Xi(B_k) = \Xi(B_j) + \Xi(B_k \setminus B_j), \quad \mathcal{R}_{B_k} = \mathcal{R}_{B_j} - \sum_{p \in \Xi \cap (B_k \setminus B_j)} J_p$$

on B_j . The latter restriction determines the exterior field on B_k by its unique holomorphic continuation; the restriction and continuation maps on the admissible function spaces are measurable, as established for the augmentation. Thus I_k is a measurable function of I_j . $\square$

For a Gibbs law P , let

$$\mathcal{T}_P = \bigcap_{j \geq 1} \mathcal{F}_j^P,$$

where the superscript denotes completion under P . This is a *canonical* tail: unlike the ordinary point-configuration tail, it retains the count information and corrected field information in each I_j .

Theorem 55.2 (An exact canonical-tail criterion). *Let P be a Gibbs law for the kernels K_j .*

1. *For every bounded measurable real F ,*

$$K_j F \longrightarrow H_F := \mathbb{E}_P[F \mid \mathcal{T}_P] \quad \text{almost surely and in } L^1(P) \text{ and } L^2(P).$$

2. *The law P is an extreme point of the convex set of Gibbs laws for these same kernels if and only if $\mathcal{T}_P$ is P -trivial. Distinct extreme Gibbs laws are mutually singular on the augmented state space.*
3. *If $F = F(\Xi)$ is a bounded real position observable and $\text{Var}_P(H_F) > 0$, there are two Gibbs laws obtained by conditioning P on complementary canonical-tail events whose position marginals differ. Conversely, if $H_F = \mathbb{E}_P F$ almost surely for every bounded position observable F , then every probability law of the form fP , with f nonnegative, $\mathcal{T}_P$ -measurable and $\mathbb{E}_P f = 1$, has the same position marginal as P .*

Proof. The first assertion is the reverse martingale convergence theorem for the decreasing completed sigma-fields in Lemma 55.1. Boundedness gives both claimed norm convergences.

If $A \in \mathcal{T}_P$ has probability strictly between zero and one, then $P(\cdot \mid A)$ and $P(\cdot \mid A^c)$ are Gibbs laws: for bounded F , proper conditioning gives

$$\mathbb{E}_P[\mathbf{1}_A K_j F] = \mathbb{E}_P[\mathbf{1}_A F]$$

for every j . The same identity with an additional bounded $\mathcal{F}_j$ -measurable multiplier proves the conditional-kernel assertion, including for completed representatives of A . They decompose P nontrivially, so P is not extreme.

Conversely, suppose $P = tQ + (1 - t)R$, where $0 < t < 1$ and Q, R are Gibbs laws for the same kernels. The density $f = dQ/dP$ is bounded by $1/t$. For every bounded measurable F , stationarity of Q and the conditional-kernel identity under P give

$$\begin{aligned}\mathbb{E}_P[fF] &= \mathbb{E}_Q F = \mathbb{E}_Q K_j F = \mathbb{E}_P[fK_j F] \\ &= \mathbb{E}_P[\mathbb{E}_P(f \mid \mathcal{F}_j) K_j F] = \mathbb{E}_P[\mathbb{E}_P(f \mid \mathcal{F}_j) F].\end{aligned}$$

Consequently $f = \mathbb{E}_P(f \mid \mathcal{F}_j)$ almost surely for every j , so f is $\mathcal{T}_P$ -measurable. If that tail is trivial, $f = 1$ almost surely, and $Q = P = R$.

For mutual singularity, take two extreme laws Q, R and put $P = (Q + R)/2$. The preceding argument shows that $f = dQ/dP$ is measurable with respect to every $\mathcal{F}_j^P$, hence also to the corresponding completed canonical tails under Q and R . Extremality makes f constant Q -almost surely and constant R -almost surely. If these two constants agree, then $f = 1$ P -almost surely and $Q = R$. If they differ, their respective level sets separate Q and R , proving singularity.

For the third assertion, choose a real c such that $A = \{H_F > c\}$ has probability strictly between zero and one; this is possible because H_F is not constant. Then

$$\mathbb{E}_P[F \mid A] = \mathbb{E}_P[H_F \mid A] > c \geq \mathbb{E}_P[H_F \mid A^c] = \mathbb{E}_P[F \mid A^c].$$

The conditional laws are Gibbs by the preceding argument, and their position expectations differ. In the converse situation, conditioning on $\mathcal{T}_P$ gives

$$\mathbb{E}_P[fF(\Xi)] = \mathbb{E}_P[f\mathbb{E}_P(F(\Xi) \mid \mathcal{T}_P)] = \mathbb{E}_P F(\Xi).$$

This holds for every bounded position observable and proves equality of the position marginals. The same conditioning argument shows that fP remains a Gibbs law; truncation justifies it for an unbounded integrable density f . $\square$

Proposition 55.3 (An explicit family of distinct extreme states). *Let $0 < \beta < 2$. Every law of Proposition 53.1 with fixed finite total count m and fixed function h holomorphic on a neighborhood of $\overline{\mathbb{D}}$ is extreme for the augmented canonical kernels. In particular, for $m = 1$ the choices $h(z) = tz$, $t \in \mathbb{R}$, give distinct extreme laws at the same temperature and the same total count. Their position laws are mutually absolutely continuous, whereas their augmented laws are mutually singular.*

Proof. Write $P = P_{\beta, m, h}$ and let $E_j = \{\Xi(B_j) = m\}$. The events E_j increase to the entire space almost surely. On E_j , the invariant I_j has the single value $(\emptyset, m, h')$. Therefore, for bounded F , the conditional expectation $K_j F$ equals the deterministic number $\mathbb{E}_P[F \mid E_j]$ on E_j , whenever $P(E_j) > 0$. That number tends to $\mathbb{E}_P F$, since $P(E_j) \rightarrow 1$. Every configuration lies in E_j eventually, so $K_j F \rightarrow \mathbb{E}_P F$ almost surely. The first part of Theorem 55.2 now implies that the canonical tail is trivial: apply the conclusion to the indicator of any completed tail event, using a measurable version. The extremality conclusion follows.

For $m = 1$ and $h(z) = tz$, the position density is proportional to

$$(1 - |z|^2)^{-\beta/2} e^{\beta t \operatorname{Re} z}.$$

Distinct real t give distinct densities; their ratio is strictly positive and bounded above and below. The augmented state, however, satisfies the exact identity $\mathcal{A} - J_z = t$ everywhere away from its pole. This identity provides disjoint measurable supports for distinct t . $\square$

Remark 55.4 (Symmetry of a mixture does not imply extremality). For example, average the one-particle states with $h_\theta(z) = te^{-i\theta}z$, $t > 0$, over a uniform angle. The result is a rotation-invariant Gibbs law. It is not extreme: the nonconstant global field $\mathcal{A} - J_z = te^{-i\theta}$ is retained by every canonical resampling. Every component is extreme by the same proof as the proposition. All these finite-count states are excluded as actual gas limits by Theorem 48.6.

Proposition 55.5 (A sufficient criterion for uniqueness of position laws). *Let $\mathcal{G}$ be a class of Gibbs laws for the same kernels and suppose a measurable set Ω_0 has full measure under every member. Let (F_ℓ) be a countable family of bounded position observables that determines point-process laws. If*

$$\lim_{j \rightarrow \infty} \sup_{\omega, \omega' \in \Omega_0} |K_j F_\ell(\omega) - K_j F_\ell(\omega')| = 0 \quad \text{for every } \ell,$$

then all laws in $\mathcal{G}$ have the same position marginal.

Proof. For $P, Q \in \mathcal{G}$, the Gibbs identity gives $\mathbb{E}_P F_\ell = \mathbb{E}_P K_j F_\ell$ and $\mathbb{E}_Q F_\ell = \mathbb{E}_Q K_j F_\ell$. Their difference is bounded in absolute value by the displayed oscillation. It therefore vanishes. The determining property proves equality of the position laws. $\square$

Remark 55.6 (What would establish nonuniqueness for the original sequence). The finite-particle gas is invariant under rotations and reflections, and these actions are continuous for vague convergence. Every actual subsequential position law therefore has these symmetries. Asymmetric extreme states can be constituents of an invariant law without making the original sequence nonconvergent. Likewise, different isometry transforms of a Gibbs state, when distinct, solve covariantly related boundary problems; their existence alone does not show that the specified finite-particle sequence selects both of them.

A proof of multiple subsequential limits must exhibit a bounded continuous local position observable and two particle-number subsequences on which its expectations have different limits. A canonical-tail decomposition supplies candidate states, but does not supply those subsequences. Conversely, proving tail triviality for each candidate separately does not prove they coincide, as the explicit extreme states above demonstrate. The oscillation condition in Proposition 55.5 is a condition that would compare different candidates; it has not been proved for the actual infinite-particle states.

56 An independent perturbative route: exact first response at two

This section studies perturbation in the inverse temperature itself. The order of operations is essential: we first differentiate the exact finite-particle gas at $\beta = 2$, and then let N increase. We obtain a unique limit of every such correlation response, with full hyperbolic covariance. No interchange with a fixed nonzero temperature perturbation is asserted.

Write

$$U_N(z) = \sum_{i < j} \log |z_i - z_j|, \quad K_N(u, v) = \frac{1}{\pi} \sum_{k=0}^{N-1} (k+1)(u\bar{v})^k, \quad K(u, v) = \frac{1}{\pi(1-u\bar{v})^2}.$$

The p -point correlation density with respect to $dA^{\otimes p}$ is denoted by $\rho_{N,\beta}^{(p)}$. All tuples below have distinct coordinates.

Theorem 56.1 (The intensity response is explicitly hyperbolic). *For every $0 < r < 1$ and $0 < \varepsilon < 1$,*

$$\sup_{|a| \leq r} \left| \partial_\beta \rho_{N,\beta}^{(1)}(a) \right|_{\beta=2} + \frac{1}{\pi(1-|a|^2)^2} \leq C_{r,\varepsilon} N^{-1+\varepsilon}. \quad (56.1)$$

At the center the finite- N answer is exactly

$$\partial_\beta \rho_{N,\beta}^{(1)}(0) \Big|_{\beta=2} = \frac{1}{\pi} \left(-1 + \frac{1}{2N} + \frac{1}{N+1} \right). \quad (56.2)$$

Consequently, for every compactly supported bounded measurable f ,

$$\lim_{N \rightarrow \infty} \partial_\beta \mathbb{E}_{N,\beta} \sum_i f(z_i) \Big|_{\beta=2} = - \int_{\mathbb{D}} \frac{f(a)}{\pi(1-|a|^2)^2} dA(a).$$

We first prove the integral estimate needed to justify passage through the noncompact pair-energy integral.

Lemma 56.2 (A weighted kernel-tail estimate). *For $0 < \varepsilon < 1$,*

$$\iint_{\mathbb{D}^2} (1 + |\log |u - v||) |K(u, v)| dA(u) dA(v) < \infty, \quad (56.3)$$

$$\iint_{\mathbb{D}^2} (1 + |\log |u - v||) |K_N(u, v) - K(u, v)| dA(u) dA(v) \leq C_\varepsilon N^{-1+\varepsilon}. \quad (56.4)$$

Proof. Put $t = u\bar{v}$. Summing the geometric derivative gives

$$\pi(K - K_N) = \frac{t^N}{(1-t)^2} + \frac{Nt^N}{1-t}.$$

The region where $\min(|u|, |v|) \leq 1/2$ contributes at most $C(N+1)2^{-N}$ to the second assertion: the denominators are bounded below, and $\sup_u \int_{\mathbb{D}} (1 + |\log |u - v||) dA(v) < \infty$. It contributes finitely to the first assertion as well.

On the remaining region write $u = re^{i\theta_1}$, $v = se^{i\theta_2}$, let $\theta \in [-\pi, \pi]$ be their angle difference, and put $d = 1 - rs$. There are absolute positive constants such that

$$|1 - rse^{i\theta}| \geq c(d + |\theta|), \quad |re^{i\theta} - s| \geq c|\theta|.$$

The second inequality holds because $|re^{i\theta} - s|^2 = (r - s)^2 + 2rs(1 - \cos \theta)$ and $r, s \geq 1/2$. The elementary bound $1 + |\log x| \leq C_\varepsilon(1 + x^{-\varepsilon})$ for $0 < x \leq 2$ therefore bounds the logarithmic weight by $C_\varepsilon(1 + |\theta|^{-\varepsilon})$. Splitting the angular integrals at $|\theta| = d$ gives

$$\int_{-\pi}^{\pi} \frac{1 + |\theta|^{-\varepsilon}}{(d + |\theta|)^2} d\theta \leq C_\varepsilon d^{-1-\varepsilon}, \quad \int_{-\pi}^{\pi} \frac{1 + |\theta|^{-\varepsilon}}{d + |\theta|} d\theta \leq C_\varepsilon d^{-\varepsilon}.$$

If $x = 1 - r$ and $y = 1 - s$, then d is comparable with $x + y$ and $(rs)^N \leq e^{-cN(x+y)}$. Integrating any nonnegative function of $x + y$ over $[0, 1/2]^2$ is bounded above by its integral against $(x + y) d(x + y)$ on $[0, 1]$. Thus the two tail terms are bounded by

$$C_\varepsilon \int_0^1 e^{-cNh} h^{-\varepsilon} dh + C_\varepsilon N \int_0^1 e^{-cNh} h^{1-\varepsilon} dh \leq C_\varepsilon N^{-1+\varepsilon}.$$

Without the factors $(rs)^N$, the first angular estimate reduces the integrability assertion to $\int_0^1 h^{-\varepsilon} dh < \infty$. $\square$

Lemma 56.3 (Finite Palm response). *For a distinct tuple $\mathbf{a} = (a_1, \dots, a_p)$ and $N \geq p$, let $P_{N,\mathbf{a}}$ be the orthogonal projection in $L^2(\mathbb{D}, dA)$ onto the span of $K_N(\cdot, a_1), \dots, K_N(\cdot, a_p)$. Its rank is p . Set $P_N = P_{N,\mathbf{a}}$ and*

$$V_N(u) = \int_{\mathbb{D}} \log |u - v| K_N(v, v) dA(v) = \frac{1}{2} \sum_{n=1}^N \frac{|u|^{2n} - 1}{n}.$$

Then

$$\begin{aligned}
R_{N,p}(\mathbf{a}) &:= \frac{\partial_\beta \rho_{N,\beta}^{(p)}(\mathbf{a}) \big|_{\beta=2}}{\rho_{N,2}^{(p)}(\mathbf{a})} \\
&= \sum_{i < j} \log |a_i - a_j| + \sum_i V_N(a_i) - \int_{\mathbb{D}} V_N(u) P_N(u, u) dA(u) \\
&\quad - \sum_i \int_{\mathbb{D}} \log |a_i - u| P_N(u, u) dA(u) \\
&\quad + \frac{1}{2} \iint_{\mathbb{D}^2} \log |u - v| \{P_N(u, u) P_N(v, v) - |P_N(u, v)|^2\} dA(u) dA(v) \\
&\quad + \operatorname{Re} \iint_{\mathbb{D}^2} \log |u - v| K_N(u, v) \overline{P_N(u, v)} dA(u) dA(v). \tag{56.5}
\end{aligned}$$

Proof. For fixed N differentiation under the finite-dimensional integrals is valid on every compact positive- β interval. Indeed $|\Delta|^\beta (1 + |\log |\Delta||)$ is bounded on $\mathbb{D}^N$ uniformly on such an interval, after using $|\Delta| \leq 2^{N(N-1)/2}$ and $\sup_{0 < x \leq 1} x^c |\log x| < \infty$. The same observation applies when a distinct tuple is fixed and only the remaining particles are integrated. Differentiation of that quotient gives

$$R_{N,p}(\mathbf{a}) = \sum_{i < j} \log |a_i - a_j| + \mathbb{E}_{N,2}^{\mathbf{a}} \left[\sum_i \sum_{u \in X} \log |a_i - u| + U_{N-p}(X) \right] - \mathbb{E}_{N,2} U_N.$$

The reduced Palm kernel is $K_N - P_N$: expanding correlation determinants with the distinguished tuple by the Schur-complement identity proves this directly. Its one-point function is $K_N(u, u) - P_N(u, u)$. Expanding the difference of its two-point function and that of K_N gives

$$\begin{aligned}
&- K_N(u, u) P_N(v, v) - P_N(u, u) K_N(v, v) \\
&\quad + P_N(u, u) P_N(v, v) + 2 \operatorname{Re} \{K_N(u, v) \overline{P_N(u, v)}\} - |P_N(u, v)|^2.
\end{aligned}$$

Integration against $\frac{1}{2} \log |u - v|$ proves the stated formula. Finally angular averaging gives $\int \log |u - v| (n/\pi) |v|^{2n-2} dA(v) = (|u|^{2n} - 1)/(2n)$, proving the expression for V_N . $\square$

Theorem 56.4 (All first correlation responses have an invariant limit). *Let $P_{\mathbf{a}}$ be the rank- p projection onto the span of $K(\cdot, a_1), \dots, K(\cdot, a_p)$, and put*

$$v(u) = -\frac{1}{2} \log(1 - |u|^2).$$

Define $R_p(\mathbf{a})$ by the right side of (56.5), replacing P_N, K_N, V_N by $P_{\mathbf{a}}, K, v$, respectively. Its integrals are absolutely convergent. Uniformly on compact subsets of the distinct-tuple space,

$$R_{N,p}(\mathbf{a}) \longrightarrow R_p(\mathbf{a}), \quad \partial_\beta \rho_{N,\beta}^{(p)}(\mathbf{a}) \big|_{\beta=2} \longrightarrow \det[K(a_i, a_j)]_{i,j=1}^p R_p(\mathbf{a}). \tag{56.6}$$

For every hyperbolic isometry φ ,

$$R_p(\varphi a_1, \dots, \varphi a_p) = R_p(a_1, \dots, a_p). \tag{56.7}$$

In particular the signed limiting correlation response in (56.6) transforms as an invariant p -point correlation measure.

Proof. On a compact set of distinct tuples, the Gram matrices $[K_N(a_i, a_j)]$ converge uniformly to strictly positive definite matrices. Strict positivity follows, for example, from polynomial interpolation at the distinct points. Their inverses are uniformly bounded. Moreover $K_N(u, a_i)$

converges to $K(u, a_i)$ uniformly for all $u \in \mathbb{D}$, with a geometrically small error, because the a_i remain in a compact subdisk. The finite-rank projection formula therefore gives bounded kernels P_N , uniformly on $\mathbb{D}^2$, and their uniform convergence to $P_{\mathbf{a}}$.

Since $\int P_N(u, u) dA(u) = p$, the additive constant $-H_N/2$ in V_N cancels exactly in $\sum_i V_N(a_i) - \int V_N P_N(u, u) dA$. Its remaining part $v_N(u) = \frac{1}{2} \sum_{n \leq N} |u|^{2n}/n$ increases to $v(u)$ and

$$\int_{\mathbb{D}} (v - v_N) dA = \frac{\pi}{2} \sum_{n > N} \frac{1}{n(n+1)} = \frac{\pi}{2(N+1)}.$$

This proves convergence of the V_N terms. Logarithmic kernels are uniformly integrable when one argument ranges over a compact set, and integrable on $\mathbb{D}^2$, so all terms containing only P_N converge. Lemma 56.2 treats the last term and also proves its absolute convergence. These estimates prove both limits and, for $p = 1$, the rate $O(N^{-1+\varepsilon})$.

The limiting signed correlation response is locally integrable also across partial collisions. Indeed, for $|a_i| \leq r < 1$, every function in the range of $P_{\mathbf{a}}$ has the form $f = P/Q_{\mathbf{a}}$, where

$$Q_{\mathbf{a}}(u) = \prod_{i=1}^p (1 - u\bar{a}_i)^2, \quad \deg P \leq 2p - 2.$$

On the entire disk this denominator is bounded above and below by positive constants depending only on p, r . Equivalence of norms on the finite-dimensional polynomial space therefore gives $\|f\|_{L^\infty(\mathbb{D})} \leq C_{p,r} \|f\|_{L^2(\mathbb{D})}$. An orthonormal basis of the rank- p range then proves $|P_{\mathbf{a}}(u, v)| \leq C_{p,r}$, uniformly even as some nodes coalesce. The absolutely convergent integral formula consequently shows

$$\left| R_p(\mathbf{a}) - \sum_{i < j} \log |a_i - a_j| \right| \leq C_{p,r}.$$

The logarithmic terms are locally integrable, and the Bergman correlation determinant is locally bounded, as required.

We verify invariance rather than infer it from formal cancellation of divergent energies. First take an orientation-preserving isometry and put $\ell(u) = \frac{1}{2} \log |\varphi'(u)|$. The function ℓ is bounded and harmonic on $\mathbb{D}$, and

$$\log |\varphi u - \varphi v| = \log |u - v| + \ell(u) + \ell(v), \quad v(\varphi u) = v(u) - \ell(u).$$

The Bergman unitary $f \mapsto \varphi'(f \circ \varphi)$ carries the projection associated with $\varphi \mathbf{a}$ to that associated with $\mathbf{a}$. It follows that all projection and kernel factors in the defining integrals have the required change-of-variables covariance. Set $T = \int_{\mathbb{D}} \ell(u) P_{\mathbf{a}}(u, u) dA(u)$ and $A = \sum_i \ell(a_i)$. The five lines of the response formula change, respectively, by

$$(p-1)A, \quad -A+T, \quad -pA-pT, \quad (p-1)T, \quad 2T.$$

For the fourth change use $P^2 = P$ and $\text{Tr } P = p$; for the last use $KP = P$. The sum is $2(T - A)$.

For completeness $T = A$. If h is bounded holomorphic, multiplication by h has adjoint mapping $K(\cdot, a_i)$ to $\overline{h(a_i)} K(\cdot, a_i)$. On their finite-dimensional span its adjoint compression therefore has trace $\sum_i \overline{h(a_i)}$, and taking conjugates gives

$$\int_{\mathbb{D}} h(u) P_{\mathbf{a}}(u, u) dA(u) = \sum_i h(a_i).$$

Here ℓ is the real part of a bounded holomorphic function (use the explicit logarithm of the derivative of a disk automorphism). Taking real parts proves $T = A$. Thus the response is unchanged. Complex conjugation leaves every term unchanged and accounts for the orientation-reversing isometries. $\square$

Proof of Theorem 56.1. For $p = 1$ the fourth line of (56.5) vanishes, because P_N has rank one. The preceding theorem proves that its limiting response R_1 is invariant, hence constant. It remains to compute that constant, and the same computation gives the exact finite- N central response.

At $a = 0$ one has $P_N(u, v) = 1/\pi$. Direct radial integration gives

$$V_N(0) - \frac{1}{\pi} \int_{\mathbb{D}} V_N(u) dA(u) = -\frac{1}{2} + \frac{1}{2(N+1)}, \quad \frac{1}{\pi} \int_{\mathbb{D}} \log |u| dA(u) = -\frac{1}{2}.$$

For the last term, angular averaging gives

$$\frac{1}{\pi^2} \iint_{\mathbb{D}^2} \log |u - v| dA(u) dA(v) = -\frac{1}{4},$$

and for every integer $k \geq 1$,

$$\begin{aligned} \frac{1}{\pi^2} \iint_{\mathbb{D}^2} \log |u - v| u^k \bar{v}^k dA(u) dA(v) &= -\frac{4}{k} \int_0^1 r dr \int_0^r s^{2k+1} ds \\ &= -\frac{1}{k(k+1)(k+2)}. \end{aligned}$$

Indeed the k -th angular Fourier coefficient of $\log |re^{i\theta} - s|$ is $-(2k)^{-1}(\min(r, s)/\max(r, s))^k$; this follows by expanding $\log(1 - z)$ for $|z| < 1$, and the equality of radii is a null set. Thus the last term is

$$-\frac{1}{4} - \sum_{k=1}^{N-1} \frac{1}{k(k+2)} = -1 + \frac{1}{2N} + \frac{1}{2(N+1)}.$$

Adding the three terms gives $R_{N,1}(0) = -1 + (2N)^{-1} + (N+1)^{-1}$ and hence $R_1 \equiv -1$. Since $\rho_{N,2}^{(1)}(0) = 1/\pi$, this proves the exact central formula. The rate already established in the preceding proof, together with locally geometric convergence of $K_N(a, a)$ to $K(a, a)$, proves (56.1). Integrating against f proves the final assertion. $\square$

Corollary 56.5 (Explicit response at particle fusion). *Let $H_n = \sum_{j=1}^n j^{-1}$. If the distinct coordinates of $\mathbf{a} \in \mathbb{D}^p$ all tend to zero, then*

$$R_p(\mathbf{a}) - \sum_{i < j} \log |a_i - a_j| \longrightarrow C_p, \quad C_p = \frac{2p-1}{4} H_p - \frac{2p+1}{2} H_{2p} + p. \quad (56.8)$$

In particular

$$R_2(0, a) = \log |a| - \frac{25}{12} + o(1), \quad a \rightarrow 0.$$

At an arbitrary common limiting point $q \in \mathbb{D}$, the corresponding constant, after subtracting the Euclidean logarithmic Vandermonde, is $C_p - \frac{1}{2}p(p-1)\log(1 - |q|^2)$.

Proof. As all nodes tend to zero, the projection $P_{\mathbf{a}}$ converges uniformly on $\mathbb{D}^2$ to K_p . One way to verify this without an assumption on the relative speeds of the nodes is to replace the spanning functions $K(\cdot, a_i)$ by their Newton divided differences in the variables $\bar{a}_i$. For a holomorphic function F on a circle containing all those variables, the divided difference has the contour formula

$$F[t_0, \dots, t_j] = \frac{1}{2\pi i} \int_{|\zeta|=r_0} \frac{F(\zeta)}{\prod_{i=0}^j (\zeta - t_i)} d\zeta.$$

For $F(\zeta) = \pi^{-1}(1 - u\zeta)^{-2}$, $|u| < 1$ and fixed $r_0 < 1$, this formula proves convergence uniformly in u to $(j+1)u^j/\pi$. The limiting Gram matrix is nonsingular, so the projection convergence

follows. The integral estimates used in Theorem 56.4 now permit passage to the limit after removing the explicit logarithmic Vandermonde.

We evaluate the remaining terms. Write $n = k + 1$ for the normalized monomial of degree k . Its expectations of $v(u)$ and $\log |u|$ are respectively $H_n/2$ and $-1/(2n)$. Thus the second and third terms of the response formula contribute

$$-\frac{1}{2} \sum_{n=1}^p H_n + \frac{p}{2} H_p.$$

The fourth term is $\mathbb{E}_{p,2} U_p$, and the last is $-\sum_{n=1}^p H_n$. Here are direct computations of both, including the infinite sum in the last term. If

$$J_{kl} = \frac{1}{\pi^2} \iint_{\mathbb{D}^2} \log |u - v| u^k \bar{u}^l v^l \bar{v}^k dA(u) dA(v),$$

angular averaging followed by radial integration gives

$$(k+1)(l+1)J_{kl} = \begin{cases} -1/[4(k+1)], & k = l, \\ -\min(k+1, l+1)/[|k-l|(k+l+2)], & k \neq l. \end{cases}$$

For fixed $n = l + 1$ the sum over $k \geq 0$ is $-H_n$: separate $k = l$ and use

$$\sum_{j>n} \frac{n}{j^2 - n^2} = \frac{1}{2} H_{2n}, \quad \sum_{j<n} \frac{j}{n^2 - j^2} = \frac{1}{2} (H_{n-1} + H_n - H_{2n-1}).$$

The sum is absolutely convergent. Equivalently it is the limit of the kernel integral, justified by Lemma 56.2. This proves the value of the last term.

The radial product contribution to $\mathbb{E}_{p,2} U_p$ is $-\frac{1}{4} \sum_{n,m=1}^p (n+m)^{-1}$, because the larger of two independent squared radii with respective beta parameters n, m has beta parameter $n+m$. Subtracting the finite kernel term gives

$$\begin{aligned} \mathbb{E}_{p,2} U_p &= -\frac{1}{4} \sum_{n,m=1}^p \frac{1}{n+m} + \frac{1}{8} H_p + \sum_{1 \leq n < j \leq p} \frac{n}{j^2 - n^2} \\ &= \frac{6p+5}{4} H_p - \frac{2p+1}{2} H_{2p} - \frac{p}{2}. \end{aligned}$$

The last equality follows by substituting

$$\sum_{n,m=1}^p \frac{1}{n+m} = (2p+1)H_{2p} - 2(p+1)H_p$$

and the preceding partial-fraction sum; for the latter simplification one can use $\sum_{j=1}^p H_{2j-1} = (p + \frac{1}{2})H_{2p} - \frac{1}{4}H_p - p$. Combining all four contributions and using $\sum_{n=1}^p H_n = (p+1)H_p - p$ proves the value of C_p . For fusion at q , apply a disk automorphism taking q to zero. Its derivative there has modulus $(1 - |q|^2)^{-1}$; the assertion then follows from (56.7). $\square$

Remark 56.6 (What the response settles and what it leaves open). The one-point response has exactly the invariant shape, with density slope $-1/\pi$. Thus failure of hyperbolic invariance cannot be detected at the first perturbative order about $\beta = 2$; the same statement holds for every fixed correlation order. If full convergence on a temperature interval and an exchange of the temperature derivative with that limit were separately established, the derivative of the limiting intensity at two would necessarily be the value proved above. Neither assumption follows from the first-response theorem. In particular no fixed $\beta \neq 2$ is settled merely by a Taylor formula whose second derivative has not been bounded uniformly in N .

Remark 56.7 (A separate monotonicity route). For this unscaled problem, proving that all correlation functions increase with N for a fixed $0 < \beta < 2$ would establish uniqueness using the already proved local bounds and correlation-moment determination. Monotonicity at $\beta = 2$ follows from the increasing projection kernels. The broader $\beta < 2$ statement was explicitly conjectured by Hedenmalm in his 2015 lectures, and is not assumed here; see <https://irma.math.unistra.fr/~klevtsov/past/Hedenmalm.pdf>, slide 6. The exact central first response above is compatible with this direction, but does not prove monotonicity at another temperature.

57 A uniform second temperature response at the determinantal point

The next estimate controls a second connected energy diagram. It still concerns derivatives at the single temperature $\beta = 2$; it does not give a uniform interval of analyticity.

Theorem 57.1 (Uniform second response of local correlation logarithms). *Let $p \geq 1$ and let A be a compact subset of the space of distinct tuples in $\mathbb{D}^p$. There is a finite constant C_A such that*

$$\sup_{N \geq p} \sup_{\mathbf{a} \in A} \left| \partial_\beta^2 \log \rho_{N,\beta}^{(p)}(\mathbf{a}) \right|_{\beta=2} \leq C_A. \quad (57.1)$$

Consequently the second derivatives of the correlation densities themselves are uniformly bounded on A at $\beta = 2$.

We use ordinary area and the notation of Section 56. Write $\mathcal{H} = L^2(\mathbb{D}, dA)$, $\phi_n(z) = \sqrt{(n+1)/\pi} z^n$, K_N also for its orthogonal projection, and $Q_N = I - K_N$. Put $f(u, v) = \log |u - v|$. For a finite-rank projection J define

$$\begin{aligned} V_J(u) &= \int_{\mathbb{D}} f(u, v) J(v, v) dA(v), \\ (A_J h)(u) &= \int_{\mathbb{D}} f(u, v) J(u, v) h(v) dA(v), \quad F_J = M_{V_J} - A_J. \end{aligned}$$

Here M_s denotes multiplication by s . These expressions below are applied to the finite occupied subspaces, where all their matrix elements and norms are finite. All logarithmic singularities are square integrable with respect to the bounded finite-volume correlation densities.

Lemma 57.2 (The Slater variance identity). *Let J be a finite-rank orthogonal projection onto bounded functions, and let $Q = I - J$. For its determinantal process, the variance of*

$$\sum_{i < j} f(z_i, z_j) + \sum_i s(z_i)$$

is

$$\|Q(F_J + M_s)J\|_{\text{HS}}^2 + \|Q^{\wedge 2} M_f J^{\wedge 2}\|_{\text{HS}}^2. \quad (57.2)$$

The second term is an operator between the antisymmetric tensor squares, with their usual Hilbert norms.

Proof. Choose orthonormal bases of $J\mathcal{H}$ and $Q\mathcal{H}$. The normalized determinant of the occupied basis is a unit vector in the antisymmetric tensor power of $\mathcal{H}$, and its squared modulus is the determinantal joint density. Multiplication by the displayed symmetric statistic changes at most two occupied basis vectors. Expand the resulting vector in the orthonormal

determinant basis. Its component on the original determinant is its mean. The components with exactly one occupied vector replaced are the matrix elements of $Q(F_J + M_s)J$: direct integration over each unchanged orbital gives the direct term V_J and the exchange term $-A_J$. Components with exactly two replacements are the matrix elements of $Q^{\wedge 2}M_f J^{\wedge 2}$. There are no other components. Parseval's identity after removing the mean gives (57.2). To justify the expansion one can first use bounded f, s and a finite ambient orthonormal basis, then pass through its increasing spans and bounded truncations. The square-integrability just stated gives convergence of the original random statistic in L^2 , hence of its orthogonal components. $\square$

Lemma 57.3 (A uniformly bounded particle-hole operator). *The operators*

$$T_N = Q_N F_{K_N} K_N$$

satisfy

$$\sup_N \|T_N\|_{\text{op}} \leq \frac{\pi}{\sqrt{6}} + \frac{1}{4}. \quad (57.3)$$

After adding the irrelevant scalar $H_N/2$ to F_{K_N} , one also has

$$\|(F_{K_N} + H_N/2)\phi_n\|_2 \leq C \log(n+2), \quad 0 \leq n < N. \quad (57.4)$$

Proof. Rotation covariance makes $F_{K_N}\phi_n$ a radial multiplier of ϕ_n . Put $t = |u|^2$. Angular integration as in Section 56 gives, for $0 \leq n < N$,

$$\frac{A_{K_N}\phi_n(u)}{\phi_n(u)} = -\frac{1}{2}H_{n+1} - \frac{1}{2} \sum_{j=1}^{N-n-1} \frac{t^j}{j} + \frac{1}{2(n+1)} \sum_{j=N-n}^N t^j.$$

The quotient is interpreted by its radial continuous formula at zero. For clarity, the summand of A_{K_N} indexed by k contributes

$$\begin{cases} -1/[2(n-k)] + t^{k+1}/[2(n+1)], & k < n, \\ (t^{n+1} - 1)/[2(n+1)], & k = n, \\ -(k+1)t^{k-n}/[2(k-n)(n+1)] + t^{k+1}/[2(n+1)], & k > n. \end{cases}$$

Their sum is the preceding identity. Since $V_{K_N} = -H_N/2 + \frac{1}{2} \sum_{j=1}^N t^j/j$, the multiplier after discarding $-H_N/2$ is

$$b_{n,N}(t) = \frac{1}{2}H_{n+1} + \frac{1}{2} \sum_{j=1}^N \frac{t^j}{j} + \frac{1}{2} \sum_{j=1}^{N-n-1} \frac{t^j}{j} - \frac{1}{2(n+1)} \sum_{j=N-n}^N t^j. \quad (57.5)$$

The last term has range in $[-1/2, 0]$ and hence standard deviation at most $1/4$. Under $|\phi_n(u)|^2 dA(u)$, the squared radius has density $(n+1)t^n dt$. For this law

$$\mathbb{E}[-\log(1-t)] = H_{n+1}, \quad \text{Var}[-\log(1-t)] = \sum_{j=1}^{n+1} \frac{1}{j^2} \leq \frac{\pi^2}{6}.$$

These identities follow by differentiating twice the elementary beta integral $\int_0^1 t^n(1-t)^s dt$ at $s = 0$. For each partial sum $a_M(t) = \sum_{j \leq M} t^j/j$,

$$|a_M(t) - a_M(s)| \leq |\log(1-t) - \log(1-s)|,$$

because its derivative is at most $(1-t)^{-1}$. The independent-copy identity for variance therefore bounds its standard deviation by $\pi/\sqrt{6}$. Taking standard deviations in (57.5) proves

the claimed bound on each centered radial multiplier. The resulting vectors for different n have different angular frequencies and are orthogonal; Q_N centers precisely that radial multiplier. This proves the operator norm bound. The same formula, its means, and the variance bound give (57.4). $\square$

Lemma 57.4 (Two-body vertices with one controlled analytic orbital). *Let $h = \sum_{l=0}^{N-1} c_l \phi_l$ and put $\|h\|_* = \sum_l |c_l| \sqrt{l+1}$. There is an absolute constant C such that*

$$\|(Q_N \otimes Q_N)M_f(h \otimes K_N)\|_{\text{HS}} \leq C\|h\|_*, \quad (57.6)$$

$$\|(\langle h| \otimes Q_N)M_f(K_N \otimes K_N)\|_{\text{HS}} \leq C\|h\|_*. \quad (57.7)$$

Here $h \otimes K_N$ maps v to $h \otimes K_N v$, and the bra in the second formula integrates the first output coordinate against $\bar{h}$.

Proof. Decompose $f(u, v) = \log |1 - u\bar{v}| - g(u, v)$, where g is the positive hyperbolic Green function. Since $K_N(v, v) \leq K(v, v)$,

$$\int_{\mathbb{D}} g(u, v)^2 K_N(v, v) dA(v) \leq \int_{\mathbb{D}} g(u, v)^2 d\nu(v) = \frac{\pi^2}{12}.$$

The last equality follows by a disk automorphism and the radial integral $\frac{1}{4} \int_0^1 (\log t)^2 (1-t)^{-2} dt$. For the loss estimate one can omit the output projections, and the Hilbert–Schmidt norm squared is bounded by the last constant times $\|h\|_2^2$. For the gain estimate, fix the second input point v . The norm squared of the first-coordinate functional on the occupied subspace is $\|K_N(g(\cdot, v)h)\|_2^2 \leq \int g(u, v)^2 |h(u)|^2 dA(u)$. Integration against $K_N(v, v) dA(v)$ proves the same bound.

For the image kernel use its angular series

$$\log |1 - u\bar{v}| = -\frac{1}{2} \sum_{k=1}^{\infty} \frac{u^k \bar{v}^k + \bar{u}^k v^k}{k}.$$

First take $h = \phi_l$, $l < N$. Terms with different angular frequencies are orthogonal in the squared norms below. In the loss estimate the first series survives Q_N in the first coordinate only for $k \geq N - l$. Its squared norm is at most

$$\frac{1}{4} \sum_{k \geq N-l} \frac{l+1}{k^2(l+k+1)} \sum_{n=0}^{N-1} \frac{n+1}{n+k+1} \leq C(l+1).$$

Indeed the inner sum is at most N , $l+k+1 \geq N+1$, and $\sum_{k \geq N-l} k^{-2} \leq C/(N-l) \leq C$. For the second series the second-coordinate condition is $n+k \geq N$. Its squared norm is at most

$$\frac{1}{4} \sum_{k \geq 1} \frac{l+1}{k^2(l+k+1)} \sum_{n=\max(0, N-k)}^{N-1} \frac{n+1}{n+k+1} \leq \frac{l+1}{4} \sum_{k \geq 1} \frac{1}{k(l+k+1)} \leq C(l+1).$$

Projecting the other coordinate can only decrease these bounds.

For the gain estimate the first series has nonzero first-coordinate matrix element only when its occupied index is $l-k$, with $1 \leq k \leq l$. The resulting squared norm is

$$\frac{1}{4} \sum_{k=1}^l \frac{l-k+1}{k^2(l+1)} \|Q_N M_{\bar{z}^k} K_N\|_{\text{HS}}^2 \leq \frac{1}{4} \sum_{k=1}^l \frac{1}{k} \leq C(l+1).$$

Here $\|Q_N M_{\bar{z}^k} K_N\|_{\text{HS}}^2 \leq k$. To see this, the squared column norm for ϕ_n equals $(n+1)/(n+k+1)$ for $n < k$, and $k^2/[(n+1)(n+k+1)]$ for $n \geq k$. Summing all $n \geq 0$ telescopes to k ;

restricting to $n < N$ decreases the sum. The second image series has first occupied index $l + k < N$ and second index n with $n + k \geq N$. Its squared norm is bounded by

$$\frac{1}{4} \sum_{k=1}^{N-l-1} \frac{l+1}{k^2(l+k+1)} \sum_{n=\max(0, N-k)}^{N-1} \frac{n+1}{n+k+1} \leq C(l+1).$$

The triangle inequality between the two series gives both desired bounds for ϕ_l . Their sums converge in the indicated Hilbert–Schmidt spaces; alternatively start with an angular Abel factor and pass to the limit. Finally apply the triangle inequality to the expansion of h , proving the asserted norm $\|h\|_*$. $\square$

Lemma 57.5 (The screened Palm perturbation is summable). *For $\mathbf{a} \in A$, let $P_N = P_{N,\mathbf{a}}$ and $P = P_{\mathbf{a}}$ be the projections in Lemma 56.3. Set*

$$s_{\mathbf{a}}(u) = \sum_i \log |u - a_i|, \quad B_N = M_{s_{\mathbf{a}} - V_{P_N}} + A_{P_N}.$$

Then, uniformly over $\mathbf{a} \in A$ and $N \geq p$,

$$\sum_{n=0}^{N-1} \|B_N \phi_n\|_2 \leq C_A. \quad (57.8)$$

There are orthonormal bases $h_{j,N}$ of the ranges of P_N such that $\|h_{j,N}\|_* \leq C_A$ and

$$\|(F_{K_N} + H_N/2)h_{j,N}\|_2 \leq C_A. \quad (57.9)$$

Proof. The Gram-matrix argument in Theorem 56.4 gives orthonormal bases whose coefficients in the ϕ_l basis satisfy $|c_{j,l,N}| \leq C_A \sqrt{l+1} r^l$ for some $r < 1$ independent of N and $\mathbf{a} \in A$. After enlarging the constants, the same bounds hold for all $N \geq p$. They prove the last two assertions using (57.4). They also give $\|P_N - P\|_{L^\infty(\mathbb{D}^2)} \leq C_A r_1^N$ for some $r_1 < 1$.

Put $S = s_{\mathbf{a}} - V_P$. The harmonic trace identity proved in Theorem 56.4 implies

$$V_P(u) = \sum_i \log |u - a_i|, \quad |u| > 1.$$

For example expand the logarithm in inverse powers of u , use $\int P(v, v) dA(v) = p$ and $\int v^k P(v, v) dA(v) = \sum_i a_i^k$, and then use continuity at the boundary. Thus S vanishes identically outside the disk. The potential of the smooth bounded density $P(v, v)$ on the disk is continuously differentiable across its boundary, because the gradient of the logarithmic kernel is locally integrable. The first normal derivative of S is consequently zero there as well. Inside a fixed boundary annulus, S solves $\Delta S = -2\pi P(u, u)$ with zero boundary data. The smooth Dirichlet regularity for the disk (equivalently its Poisson and Green integral formulas differentiated in boundary charts) gives a bounded second derivative up to the boundary. All density derivatives and source distances are uniform over A , so

$$|S(u)| \leq C_A(1 - |u|^2)^2 \quad \text{on a fixed boundary annulus.} \quad (57.10)$$

On its complementary compact disk, S has only logarithmic singularities, with uniformly bounded L^2 norm. Consequently

$$\sum_{n \geq 0} \|S \phi_n\|_2 < \infty$$

uniformly over A : on the annulus use $(n+1) \int_0^1 t^n (1-t)^4 dt = O((n+1)^{-4})$ and then take its square root; on the compact disk use the geometric factor $\sqrt{n+1} r^n$.

It remains to treat A_P . Write its kernel as $f(u, v) \sum_j h_j(u) \overline{h_j(v)}$, where the finitely many h_j extend holomorphically to a fixed disk of radius greater than one and have uniformly bounded analytic norms there. If L is the integral operator with kernel f , then $A_P = \sum_j M_{h_j} L M_{\bar{h}_j}$. We claim

$$\sum_{n \geq 0} \|L(\bar{z}^k \phi_n)\|_2 \leq C(k+1). \quad (57.11)$$

For $n > k$, direct angular integration gives

$$L(\bar{z}^k \phi_n)(u) = -\frac{\sqrt{\pi(n+1)} u^{n-k}}{2(n-k)} \left\{ \frac{1 - |u|^{2k+2}}{k+1} + \frac{|u|^{2k+2}}{n+1} \right\}.$$

If $n \geq 2k+1$, its norm is at most $C(n+1)^{-2}$, since the bracket is bounded by $(1-|u|^2) + (n+1)^{-1}$ and the relevant radial beta integrals have that order. For $n \leq 2k$, the Hilbert–Schmidt norm of L bounds each column by an absolute constant. This proves (57.11). Expanding the analytic functions h_j in their geometrically decaying Taylor series and summing proves $\sum_n \|A_P \phi_n\|_2 \leq C_A$.

The replacement of P by P_N has error at most $C_A r_1^N$ in the sup norm of its kernel. The associated potential error is bounded in sup norm by the same order, using the uniform integrability of $|\log|u-v||$; the exchange operator error is bounded in operator norm by its Hilbert–Schmidt norm of that order. Summing its first N column bounds adds at most $C_A N r_1^N$. Combining these bounds proves (57.8). $\square$

Proof of Theorem 57.1. Differentiation of the fixed-tuple partition-function quotient twice gives

$$\partial_\beta^2 \log \rho_{N,\beta}^{(p)}(\mathbf{a}) \Big|_{\beta=2} = \text{Var}_{K_N - P_N} \left(U_{N-p} + \sum_{u \in X} s_{\mathbf{a}}(u) \right) - \text{Var}_{K_N}(U_N).$$

The deterministic pair energy of the distinguished tuple drops out. The differentiations are legitimate by the same logarithmic domination as before, now with its square.

Abbreviate $K = K_N$, $P = P_N$, $J = K - P$, $Q = I - K$ and $B = B_N$. Then $I - J = Q + P$ and $F_J + M_{s_{\mathbf{a}}} = F_K + B$. In all one-body expressions below replace F_K by $F = F_K + H_N/2$, which does not change a matrix element between an occupied and an unoccupied subspace. The difference of the single-replacement variances is exactly

$$- \|QFP\|_{\text{HS}}^2 + 2 \text{Re} \langle QFJ, QBJ \rangle_{\text{HS}} + \|QBJ\|_{\text{HS}}^2 + \|P(F+B)J\|_{\text{HS}}^2.$$

The first term is bounded by $p \sup_N \|T_N\|_{\text{op}}^2$. The column sum in (57.8) bounds the trace norm of BK and hence of BJ , so the cross term is bounded by $2 \|T_N\|_{\text{op}} \|BJ\|_1$. It bounds the third term as well. Finally F and B are self-adjoint on the vectors in question, so

$$\|P(F+B)J\|_{\text{HS}} = \|J(F+B)P\|_{\text{HS}} \leq \|FP\|_{\text{HS}} + \|BP\|_{\text{HS}} \leq C_A$$

by Lemma 57.5. Thus the entire single-replacement difference is uniformly bounded.

For the double-replacement term use the orthogonal decompositions

$$\wedge^2 K\mathcal{H} = \wedge^2 J\mathcal{H} \oplus (J\mathcal{H} \wedge P\mathcal{H}) \oplus \wedge^2 P\mathcal{H},$$

and the corresponding decomposition of $\wedge^2(Q+P)\mathcal{H}$. Its variance difference is the sum of four signed squared norms: the negative blocks have input $J \wedge P$ or $\wedge^2 P$ and output $\wedge^2 Q$, while the positive blocks have input $\wedge^2 J$ and output $Q \wedge P$ or $\wedge^2 P$. Each negative block is bounded by the loss estimate in Lemma 57.4, summed over an orthonormal basis of $P\mathcal{H}$. The positive block with output $Q \wedge P$ is bounded by its gain estimate, enlarging the input

projection from J to K . The remaining block has output $\wedge^2 P$; by taking the adjoint and omitting its input projection, it is bounded by $\|M_f P^{\wedge 2}\|_{\text{HS}}$. This is uniformly finite since the kernel of P is uniformly bounded on $\mathbb{D}^2$ and f^2 is integrable there. Passing between tensor and antisymmetric tensor norms costs only an absolute factor. The uniform analytic norm bounds on that basis give C_A for the sum of all four squared norms. This proves (57.1).

The identity $\partial_\beta^2 \rho = \rho \{ \partial_\beta^2 \log \rho + (\partial_\beta \log \rho)^2 \}$, the first-response estimates, and boundedness of the Bergman determinants on A prove the last assertion. $\square$

Remark 57.6 (The remaining analytic step). The preceding theorem bounds the first two temperature derivatives at two after the bulk contributions have canceled. It does not bound derivatives on a neighborhood of two, and it does not give factorial bounds at all orders. Either a uniform remainder estimate on such a neighborhood or a convergent connected expansion would be needed to deduce convergence at a fixed nearby temperature. In particular, this estimate alone supplies no additional fixed-temperature uniqueness range.

58 Global temperature singularity and the failure of direct interpolation

The following conclusion concerns the full finite-particle laws. It does not assert singularity of their restrictions to a compact disk, or of their limiting interior processes.

Theorem 58.1 (Different temperatures are globally distinguishable). *Let $\beta_0, \beta_1 > 0$ be distinct. Then*

$$\|P_{N,\beta_1} - P_{N,\beta_0}\|_{\text{TV}} \longrightarrow 1, \quad (58.1)$$

where total variation is the supremum of differences of event probabilities. The same assertion holds for the origin-charged laws

$$d\hat{P}_{n,\beta}(y) = \hat{Z}_{n,\beta}^{-1} |\Delta_n(y)|^\beta \prod_{j=1}^n |y_j|^\beta dA(y_j).$$

For either family let L_N be the likelihood of temperature β_1 with respect to temperature β_0 . Then

$$L_N \longrightarrow 0 \quad \text{in probability under temperature } \beta_0, \quad \mathbb{E}_{N,\beta_0} L_N^p \longrightarrow \infty \quad \text{for every } p > 1. \quad (58.2)$$

The likelihood nevertheless has expectation one for every N .

Proof. For each fixed m , the already proved harmonic-mode theorem gives

$$W_{N,k} := \sqrt{\frac{\beta_0}{2k}} S_{k,N}, \quad 1 \leq k \leq m,$$

converging under temperature β_0 to independent standard circular complex Gaussians, and under temperature β_1 to independent circular complex Gaussians of variance $r = \beta_0/\beta_1 \neq 1$. For the charged family the same statement follows from Theorem 24.1: the charge is at zero, so every limiting harmonic mean is zero.

Put $T_{N,m} = m^{-1} \sum_{k=1}^m |W_{N,k}|^2$ and $d = |1 - r|/3$. Under the two limiting Gaussian laws this statistic has respective means $1, r$ and variances $1/m, r^2/m$. Chebyshev's inequality shows that the event $|T_{N,m} - r| < d$, after $N \rightarrow \infty$ with m fixed, has probability at least $1 - r^2/(md^2)$ under the second temperature and at most $1/(4md^2)$ under the first. The interval endpoints have zero probability under both limiting laws, so convergence in distribution indeed gives those event probabilities. Taking first $N \rightarrow \infty$ and then $m \rightarrow \infty$ proves (58.1).

For equivalent probability measures with likelihood L , $\mathbb{E} \min(1, L) = 1 - \|LP - P\|_{\text{TV}}$. Consequently

$$\mathbb{P}_{N, \beta_0}(L_N > \varepsilon) \leq \frac{\mathbb{E}_{N, \beta_0} \min(1, L_N)}{\min(1, \varepsilon)} \longrightarrow 0.$$

If the p -th moments were bounded along an infinite subsequence, they would make the likelihoods uniformly integrable on that subsequence. Convergence to zero in probability would then force their expectations to tend to zero, contradicting expectation one. Thus every p -th moment diverges along the full sequence. $\square$

Proposition 58.2 (An explicit finite-mode lower bound). *Suppose $b = \beta_1/\beta_0 > 1$ and $p > 1$. For either family in Theorem 58.1, and every integer $m \geq 1$,*

$$\liminf_{N \rightarrow \infty} \mathbb{E}_{N, \beta_0} L_N^p \geq \left(\frac{b^p}{1 + p(b-1)} \right)^m. \quad (58.3)$$

The quantity in parentheses is strictly greater than one.

Proof. Let $q = p/(p-1)$. For any bounded continuous nonnegative function h of the first m normalized harmonic modes, Hölder's inequality gives

$$\mathbb{E}_{N, \beta_0} L_N^p \geq \frac{(\mathbb{E}_{N, \beta_1} h)^p}{(\mathbb{E}_{N, \beta_0} h^q)^{p-1}}.$$

The limiting likelihood of the second Gaussian vector with respect to the first is

$$\ell_m(w) = b^m \exp \left(-(b-1) \sum_{k=1}^m |w_k|^2 \right).$$

Choose $h = \ell_m^{p-1}$, which is bounded and continuous because $b > 1$, and pass to the Gaussian limits in the displayed inequality. The resulting ratio equals

$$\mathbb{E}_{\text{CN}(0, I_m)} \ell_m^p = \left(\frac{b^p}{1 + p(b-1)} \right)^m.$$

Strict convexity of $x \mapsto x^p$, applied to the nonconstant mean-one likelihood ℓ_1 , proves strict inequality over one. $\square$

Corollary 58.3 (The exact obstruction in a global Hölder bound). *Fix $\beta > 2$, $p > 1$, $q = p/(p-1)$, and set $\hat{A}_n(t) = \log \hat{Z}_{n,t}$. The likelihood factor in the direct interpolation from the charged temperature-two law is*

$$\mathcal{C}_{n, \beta, p} = \exp \left\{ \frac{\hat{A}_n(2 + p(\beta-2)) + (p-1)\hat{A}_n(2) - p\hat{A}_n(\beta)}{p} \right\} \longrightarrow \infty. \quad (58.4)$$

In particular the otherwise valid estimate

$$\hat{\mathbb{E}}_{n, \beta} |Q_y(z)|^\beta \leq \mathcal{C}_{n, \beta, p} \left(\hat{\mathbb{E}}_{n, 2} |Q_y(z)|^{\beta q} \right)^{1/q}, \quad Q_y(z) = \prod_j (1 - z\bar{y}_j),$$

does not supply an n -uniform moment bound with finite amplitude. More strongly, for each fixed $0 < |z| < 1$,

$$\inf_{p \geq 1} \left\{ \|\hat{L}_{n, \beta}\|_{L^p(\hat{\mathbb{P}}_{n, 2})} \| |Q_y(z)|^\beta \|_{L^{p/(p-1)}(\hat{\mathbb{P}}_{n, 2})} \right\} \longrightarrow \infty. \quad (58.5)$$

Thus even optimizing the Hölder exponent separately for every n fails to give a uniform bound at a single nonzero interior point.

Proof. The exact likelihood under the charged law is

$$\hat{L}_{n,\beta} = \frac{\hat{Z}_{n,2}}{\hat{Z}_{n,\beta}} \exp \left((\beta - 2) \left\{ \log |\Delta_n(y)| + \sum_j \log |y_j| \right\} \right).$$

Thus $\mathcal{C}_{n,\beta,p} = (\hat{\mathbb{E}}_{n,2} \hat{L}_{n,\beta}^p)^{1/p}$ exactly. The theorem proves divergence. Hölder gives the stated moment inequality. To prove the stronger adaptive assertion, note that for fixed $s > 0$ and $|z| < 1$, the charged harmonic exponential-moment theorem gives

$$\lim_{n \rightarrow \infty} \hat{\mathbb{E}}_{n,2} |Q_y(z)|^s = (1 - |z|^2)^{-s^2/4}.$$

Indeed apply (24.1) with the analytic function $h(u) = (s/2) \log(1 - \bar{z}u)$; its value at zero is zero. It follows, for each fixed $q_0 > 1$, that

$$\lim_{n \rightarrow \infty} \| |Q_y(z)|^\beta \|_{L^{q_0}} = (1 - |z|^2)^{-\beta^2 q_0/4}.$$

For $0 < |z| < 1$ these limits grow without bound as q_0 increases. Given a target bound, choose q_0 so that this limit exceeds it, and put $p_0 = q_0/(q_0 - 1)$. If $1 < p \leq p_0$, monotonicity of norms makes the second Hölder factor at least its q_0 -norm, while the likelihood norm is at least one. If $p \geq p_0$, its likelihood factor is at least the divergent fixed p_0 -norm, while the second factor is at least its L^1 norm, whose limit is the positive number $(1 - |z|^2)^{-\beta^2/4}$. This proves (58.5), without applying a fixed-exponent limit theorem to moving exponents. $\square$

Remark 58.4 (Scope of the obstruction). Different limiting variances in infinitely many boundary harmonic modes cause the global singularity. Those modes are independent of every interior limit already constructed. The theorem is therefore compatible with local absolute continuity, with local moment bounds, and with full convergence on an interval of temperatures. It rules out the particular use of a globally uniformly integrable change of temperature. A successful perturbation would have to integrate out or renormalize this boundary likelihood before estimating its local effect.

59 The hyperbolic spectral gap and the boundary modes it does not remove

Failure of absolute integrability is not the whole story for the Green interaction. Its integral operator is bounded on a natural Hilbert space. The following results establish that fact, identify an exact obstruction to using it as a uniqueness proof, and give a separate uniqueness statement for the choice of renormalized potential. Throughout this section,

$$g(z, w) = -\log \left| \frac{z - w}{1 - z\bar{w}} \right|, \quad d\lambda(z) = \frac{dA(z)}{(1 - |z|^2)^2}, \quad (Gf)(z) = \int_{\mathbb{D}} g(z, w) f(w) d\lambda(w).$$

59.1 A bounded Green operator despite the divergent row integral

Proposition 59.1 (The sharp L^2 Green bound). *The operator G , initially defined on bounded compactly supported functions, extends to a bounded positive self-adjoint operator on $L^2(\lambda)$, with*

$$\|G\|_{L^2(\lambda) \rightarrow L^2(\lambda)} = 2\pi. \quad (59.1)$$

For every $\beta > 0$, the operator with kernel $k_\beta(z, w) = 1 - e^{-\beta g(z, w)}$ is also bounded on this space and satisfies

$$\|K_\beta\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \leq 2\pi\beta. \quad (59.2)$$

The second assertion is an operator norm bound; it does not assert positive definiteness of k_β .

Proof. We first prove the sharp inequality

$$\int_{\mathbb{D}} |\nabla u|^2 dA \geq \int_{\mathbb{D}} |u|^2 d\lambda, \quad u \in C_c^\infty(\mathbb{D}). \quad (59.3)$$

Under a conformal map to the upper half-plane $\mathbb{H} = \{x + iy : y > 0\}$, Dirichlet energy is unchanged and $d\lambda = dx dy / (4y^2)$. The identity

$$0 \leq \int_0^\infty \left| \partial_y v - \frac{v}{2y} \right|^2 dy = \int_0^\infty |\partial_y v|^2 dy - \frac{1}{4} \int_0^\infty \frac{|v|^2}{y^2} dy$$

holds for compactly supported smooth v ; integration by parts proves the equality, for real or complex v . Integrating in x and adding the nonnegative horizontal derivative term proves (59.3).

Here the constant one is optimal. Choose nonzero real functions $\chi, \psi \in C_c^\infty((-1, 1))$, put $A_L = e^{2L}$, and use the upper-half-plane test functions

$$v_L(x, y) = \chi(x/A_L) y^{1/2} \psi((\log y)/L).$$

Writing $C_\chi = \int \chi^2$, $C_\psi = \int \psi^2$, and using the same notation with primes for the corresponding derivative integrals gives

$$\|v_L\|_{L^2(\lambda)}^2 = \frac{A_L L}{4} C_\chi C_\psi, \quad \int |\partial_y v_L|^2 dx dy = A_L L C_\chi \left(\frac{C_\psi}{4} + \frac{C_{\psi'}}{L^2} \right).$$

The mixed term in the latter equality is zero because $\int \psi \psi' = 0$. Moreover,

$$\int |\partial_x v_L|^2 dx dy = \frac{C_{\chi'}}{A_L} \int_0^\infty y \psi((\log y)/L)^2 dy \leq \frac{C_{\chi'}}{A_L} L e^{2L} C_\psi.$$

Thus their Rayleigh quotients tend to one.

Let A be the positive self-adjoint operator associated with the closure of the Dirichlet form $\mathcal{E}(u, u) = \int |\nabla u|^2 dA$ on $L^2(\lambda)$. The inequality just proved says $A \geq I$, and the test functions say that the lower endpoint of its spectrum is one. Consequently its bounded inverse has norm one. We now identify $2\pi A^{-1}$ with the displayed Green kernel, so no separate boundary condition is being assumed in this argument.

For a nonnegative smooth compactly supported f , let u_R be the zero-boundary solution on $\{|z| < R\}$ of

$$-\Delta u_R = 2\pi f(1 - |z|^2)^{-2}, \quad u_R(z) = \int_{|w| < R} g_R(z, w) f(w) d\lambda(w),$$

where R contains the support of f and g_R is the disk Green function. Its extension by zero belongs to the form domain. Testing the equation with u_R and applying (59.3) gives

$$\|u_R\|_2^2 \leq \mathcal{E}(u_R, u_R) = 2\pi \langle f, u_R \rangle \leq 2\pi \|f\|_2 \|u_R\|_2.$$

The Green kernels increase to g as $R \uparrow 1$. Monotone convergence and the preceding bounds show that u_R tends in $L^2(\lambda)$ to Gf . Weak compactness in the form domain and the weak equation identify $Gf = 2\pi A^{-1}f$. Taking differences, then using density, proves this identity for the extended operator. The spectral conclusion yields (59.1).

Finally, $0 \leq k_\beta \leq \beta g$ pointwise. For a compactly supported f ,

$$|K_\beta f| \leq \beta G|f|, \quad \|K_\beta f\|_2 \leq 2\pi \beta \|f\|_2.$$

Density proves (59.2). □

59.2 Exact positive solutions outside the contraction space

Proposition 59.2 (Boundary modes survive strict Hilbert-space contraction). *For $\xi \in \partial\mathbb{D}$ and $0 < s < 1$, define*

$$u_{s,\xi}(z) = \left(\frac{1 - |z|^2}{|\xi - z|^2} \right)^s.$$

For every $z \in \mathbb{D}$ the following integral is finite, and

$$Gu_{s,\xi}(z) = \frac{\pi}{2s(1-s)} u_{s,\xi}(z). \quad (59.4)$$

None of these functions belongs to $L^2(\lambda)$. In particular, for every $0 < t < 1/(2\pi)$ there are nonzero positive smooth solutions of $u = tGu$, although $\|tG\|_{2 \rightarrow 2} < 1$.

Proof. The conformal map from the disk to $\mathbb{H}$ which sends ξ to infinity identifies the Poisson kernel $(1 - |z|^2)/|\xi - z|^2$ with the height coordinate y , up to a positive constant. The Laplace–Beltrami operator for λ is $(1 - |z|^2)^2 \Delta$ in the disk and $4y^2 \Delta$ on $\mathbb{H}$. Since $4y^2 \Delta y^s = -4s(1-s)y^s$, it follows that

$$-\Delta u_{s,\xi} = 4s(1-s)u_{s,\xi}(1 - |z|^2)^{-2}.$$

Also $y^s \notin L^2(dx dy/(4y^2))$, already because its integral over $x \in \mathbb{R}$ and $1 < y < 2$ is infinite. This proves the nonintegrability assertion.

We justify the Green representation, including the absence of an additional harmonic boundary term. Rotating, take $\xi = 1$, and write $\varepsilon = 1 - R$. For $R \geq 1/2$ and $-\pi \leq \theta \leq \pi$,

$$\frac{1 - R^2}{|1 - Re^{i\theta}|^2} \leq C \frac{\varepsilon}{\varepsilon^2 + \theta^2}.$$

Therefore the circular average of $u_{s,1}(Re^{i\theta})$ is at most a constant times

$$\varepsilon^s \int_0^\pi (\varepsilon^2 + \theta^2)^{-s} d\theta.$$

This tends to zero: it is $O(\varepsilon^s)$ if $s < 1/2$, $O(\varepsilon^{1/2} \log(1/\varepsilon))$ if $s = 1/2$, and $O(\varepsilon^{1-s})$ if $s > 1/2$.

The Green representation on the disk of radius $R > |z|$ gives

$$u_{s,1}(z) = h_R(z) + \frac{4s(1-s)}{2\pi} \int_{|w| < R} g_R(z, w) u_{s,1}(w) d\lambda(w),$$

where h_R is the harmonic extension of the boundary values of $u_{s,1}$. For fixed z , the normalized Poisson kernel of that disk is bounded uniformly as $R \uparrow 1$. The circular-average estimate implies $h_R(z) \rightarrow 0$. Monotone convergence of g_R now proves both finiteness of the integral and (59.4). For the final assertion choose either root in $(0, 1)$ of $4s(1-s) = 2\pi t$. $\square$

This is a concrete obstruction to a proposed proof, not a nonuniqueness theorem for the gas. An L^2 operator contraction does rule out L^2 differences. It cannot rule out the displayed positive boundary modes. In particular, a proof based on such a contraction must first place the actual differences of Gibbs states in the contraction space, or establish a boundary condition which excludes these modes. A homogeneous nonzero intensity difference is itself outside $L^2(\lambda)$ because $\lambda(\mathbb{D}) = \infty$.

There is a stronger issue: finite variance at each point, even together with full isometry invariance, does not supply the spatial L^2 hypothesis needed for the contraction.

Proposition 59.3 (Invariant random modes below the spatial spectral gap). *For every $0 < \kappa < 1$, there is a real smooth centered Gaussian field U on the disk such that*

1. its law is invariant under every hyperbolic isometry and $\mathbb{E}U(z)^2 = 1$;
2. $-(1 - |z|^2)^2 \Delta U = \kappa U$ almost surely;
3. for every fixed z ,

$$\lim_{R \uparrow 1} \int_{|w| < R} g(z, w) U(w) d\lambda(w) = \frac{2\pi}{\kappa} U(z) \quad \text{in } L^2 \text{ of probability.} \quad (59.5)$$

Thus these fields are nonzero solutions of $U = tGU$ in the displayed truncated, mean-square sense, where $t = \kappa/(2\pi)$ and $\|tG\|_{L^2(\lambda) \rightarrow L^2(\lambda)} = \kappa < 1$.

Proof. Choose $1/2 < s < 1$ with $\kappa = 4s(1 - s)$, put $\alpha = 2(1 - s) \in (0, 1)$, and let σ be normalized arc length on the unit circle. The kernel

$$k(\xi, \eta) = |\xi - \eta|^{-\alpha}$$

is integrable on the product circle and is positive definite. For the latter assertion use, off the diagonal,

$$|\xi - \eta|^{-\alpha} = \frac{1}{\Gamma(\alpha/2)} \int_0^\infty t^{\alpha/2-1} e^{-t|\xi-\eta|^2} dt.$$

Each Gaussian kernel in this integral is positive definite: writing circle points as vectors in $\mathbb{R}^2$, expand $e^{-t|\xi-\eta|^2} = e^{-2t} \sum_{n \geq 0} (2t)^n (\xi \cdot \eta)^n / n!$, and each summand is the Gram kernel of a tensor power. For bounded functions, absolute integrability of k justifies integration of the quadratic forms. This proves the claim, with the diagonal irrelevant to the arc-length integrals.

Let

$$J = \iint k(\xi, \eta) d\sigma(\xi) d\sigma(\eta) \in (0, \infty), \quad [f, h]_s = J^{-1} \iint f(\xi) h(\eta) k(\xi, \eta) d\sigma(\xi) d\sigma(\eta).$$

Complete the continuous real functions modulo their zero-norm subspace in this positive semidefinite inner product. There is a centered real Gaussian linear map W on the resulting separable Hilbert space with $\mathbb{E}[W(f)W(h)] = [f, h]_s$. Explicitly, choose an orthonormal basis (e_n) and independent standard real Gaussians (Z_n) and set $W(f) = \sum_n [f, e_n]_s Z_n$, with convergence in mean square. Define

$$P_z(\xi) = \frac{1 - |z|^2}{|\xi - z|^2}, \quad U(z) = W(P_z^s).$$

In particular $\mathbb{E}U(0)^2 = [1, 1]_s = 1$.

We verify invariance directly. If φ is a disk automorphism and $j(\xi) = |\varphi'(\xi)|$ is its boundary Jacobian, then

$$P_{\varphi(z)}(\varphi(\xi)) = \frac{P_z(\xi)}{j(\xi)}, \quad |\varphi(\xi) - \varphi(\eta)| = \sqrt{j(\xi)j(\eta)} |\xi - \eta|.$$

Changing variables in $[P_{\varphi(z)}^s, P_{\varphi(w)}^s]_s$ therefore gives a factor $j(\xi)^{1-s-\alpha/2} j(\eta)^{1-s-\alpha/2} = 1$. The covariance is invariant, so the centered Gaussian law is invariant. Complex conjugation preserves the same covariance and provides the orientation-reversing isometries. In particular all pointwise variances equal one.

For each compact disk, the functions $P_z(\xi)^s$ and their z -derivatives of every order are bounded uniformly in ξ . Because k is integrable, their Hilbert-space derivatives are bounded as well. Gaussian moment estimates and the continuity criterion, applied to the corresponding derivative fields on a countable exhaustion, give a smooth version of U . Equivalently, one

can first obtain a continuous version from $\mathbb{E}|U(z) - U(w)|^{2m} \leq C_{K,m}|z - w|^{2m}$ on compact sets and then use the weak equation below and interior elliptic regularity. The calculation in Proposition 59.2 shows that $-(1 - |z|^2)^2 \Delta P_z(\xi)^s = \kappa P_z(\xi)^s$. It may be passed through W against smooth compactly supported test functions, since W is continuous into L^2 of probability. On a countable dense set of test functions it holds on one probability-one event, and then on all test functions by continuity. This proves the asserted almost sure differential equation.

Finally, for fixed z define the boundary functions

$$h_R(\xi) = \int_{|w| < R} g(z, w) P_w(\xi)^s d\lambda(w).$$

By Proposition 59.2,

$$0 \leq h_R(\xi) \uparrow \frac{2\pi}{\kappa} P_z(\xi)^s.$$

The limit is bounded uniformly on the boundary because z is fixed. Dominated convergence in the double integral defining $[\cdot, \cdot]_s$ shows convergence to this limit in the boundary Hilbert space. The logarithmic pole of $g(z, w)$ is integrable on compact sets, so stochastic Fubini, or approximation by finite sums in the Hilbert space, gives

$$W(h_R) = \int_{|w| < R} g(z, w) U(w) d\lambda(w).$$

The continuity of W now proves (59.5). $\square$

The proposition does not construct two Gibbs states: its charge source is a smooth Gaussian field rather than a point measure. It proves a narrower but useful negative assertion. Spatial $L^2(\lambda)$ and the space of invariant random fields with finite one-point variance are different spaces, and the spectral-gap contraction cannot be transferred from the former to the latter.

59.3 Uniqueness of an invariant renormalized field up to a constant

Lemma 59.4 (An integrable invariant harmonic field is spatially constant). *Let U be a random real harmonic function on $\mathbb{D}$. Suppose its law is invariant under precomposition by every disk automorphism, and $\mathbb{E}|U(0)| < \infty$. Then almost surely U is a constant function. The constant may be random.*

Proof. Let $\Phi(t) = \sqrt{1 + t^2}$. Transitivity and invariance imply that $\mathbb{E}\Phi(U(z))$ is a finite constant independent of z . On each sample, harmonicity gives

$$\Delta\Phi(U) = \Phi''(U)|\nabla U|^2 \geq 0, \quad \Phi''(t) = (1 + t^2)^{-3/2} > 0.$$

For nonnegative $\chi \in C_c^\infty(\mathbb{D})$, Fubini is justified by the constant finite expectation and gives

$$0 \leq \mathbb{E}\langle \Delta\Phi(U), \chi \rangle = \int \mathbb{E}\Phi(U(z)) \Delta\chi(z) dA(z) = 0.$$

Using a countable collection of such functions positive on a compact exhaustion proves $\Delta\Phi(U) = 0$ everywhere almost surely. The displayed identity and strict positivity of Φ'' imply $\nabla U = 0$. Connectedness completes the proof. $\square$

Proposition 59.5 (Ambiguity of an integrable covariant Green field). *Let P be an isometry-invariant point-process law. On a measurable configuration class Ω of full P -measure, stable under finite additions and deletions, suppose $H_1(a, X), H_2(a, X)$ are measurable real functions off the points of X , define locally integrable distributions in a , and are taken in their smooth off-pole representatives. Suppose they satisfy all the following conditions:*

1. They obey the same distributional equation

$$\Delta H_j = 2\pi(c\lambda - \Xi), \quad j = 1, 2,$$

for every $X \in \Omega$, where measures are identified with distributions relative to dA . Their difference has a measurable harmonic representative U_X on the whole disk for every $X \in \Omega$.

2. Their joint law is covariant under isometries, and their difference is represented by a random harmonic function U with $\mathbb{E}|U(0)| < \infty$.

3. On Ω , their finite-addition identities hold pointwise off the poles:

$$H_j(a, X \cup \{b\}) = H_j(a, X) + g(a, b), \quad a \neq b.$$

Then $H_1(a, X) - H_2(a, X) = C(X)$ almost surely, independent of a . The random variable C is measurable with respect to the tail σ -field of X . In particular, if that tail is trivial, C is deterministic. The two resulting canonical specifications are the same, wherever their finite-volume normalizing constants are finite. Their grand-canonical specifications agree after replacing the activity z for H_1 by $ze^{-\beta C}$ for H_2 .

Proof. The first two hypotheses and Lemma 59.4 make U spatially constant. The third makes its harmonic representative unchanged by any finite modification: subtract the identities off their finitely many poles, then extend the difference harmonically across those points. For every relatively compact B , deleting the finitely many points in B therefore gives $C(X) = C(X|_{B^c})$. The right side is measurable in the exterior configuration. A countable exhaustion proves tail measurability. For a fixed inside count m , the change of potential multiplies every unnormalized weight by the same factor $e^{-\beta m C}$, which cancels on normalization. Across all counts it changes z^m to $(ze^{-\beta C})^m$, as claimed. $\square$

Remark 59.6 (What this uniqueness statement does and does not identify). The last proposition concerns two fields defined over the same invariant point-process law. It does not compare two different point-process laws, prove tail triviality, or establish existence of the compensated field for the original gas. Its value is to isolate the remaining ambiguity: under the stated integrability and covariance assumptions it is a tail-dependent activity, rather than an arbitrary harmonic function of the insertion point.

60 A direct test for different subsequential limits

The existence of several solutions of a limiting DLR specification is different from the existence of several subsequential limits of the particular finite-volume sequence in (1.1). This section gives exact tests for the latter question. It also identifies the additional estimate needed to exclude a simple alternation between two different limits. No such estimate is asserted without proof.

Write P_n for the law of Ξ_n . We use the boundary insertion weight G_n of (27.2) and put $\hat{G}_n = G_n/\mathbb{E}_n G_n$. The exact outermost-particle identity is already proved in Proposition 27.1. In particular, for $f \in C_c^1(\mathbb{D})$ with $f \geq 0$, Proposition 27.2 gives

$$L_{n+1}(f) - L_n(f) = c_n(f) + e_n(f), \quad \sum_{n \geq 1} |e_n(f)| < \infty, \quad (60.1)$$

where

$$L_n(f) = \mathbb{E}_n e^{-\Xi_n(f)}, \quad c_n(f) = \mathbb{E}_n [e^{-\Xi_n(f)} (\hat{G}_n - 1)].$$

These are explicit finite-dimensional integral ratios. Namely, if

$$\begin{aligned}\mathcal{Z}_n(f) &= \int_{\mathbb{D}^n} |\Delta_n(z)|^\beta e^{-\sum_j f(z_j)} dA^{\otimes n}(z), \\ \mathcal{B}_n(f) &= \frac{1}{2\pi} \int_0^{2\pi} \int_{\mathbb{D}^n} |\Delta_n(z)|^\beta e^{-\sum_j f(z_j)} \prod_j |e^{it} - z_j|^\beta dA^{\otimes n}(z) dt,\end{aligned}$$

then

$$c_n(f) = \frac{\mathcal{B}_n(f)}{\mathcal{B}_n(0)} - \frac{\mathcal{Z}_n(f)}{\mathcal{Z}_n(0)}. \quad (60.2)$$

Proposition 60.1 (An exact oscillation criterion for nonuniqueness). *Choose a countable collection $\mathcal{F} \subset C_c^1(\mathbb{D})$ of nonnegative functions whose Laplace functionals determine point-process laws. The following statements are equivalent:*

1. *the sequence P_n has at least two distinct subsequential limits;*
2. *for some $f \in \mathcal{F}$, the bounded partial sums $\sum_{n=1}^m c_n(f)$ do not converge as $m \rightarrow \infty$;*
3. *for some $f \in \mathcal{F}$,*

$$\lim_{K \rightarrow \infty} \sup_{q \geq p \geq K} \left| \sum_{n=p}^q c_n(f) \right| > 0. \quad (60.3)$$

For any test witnessing these equivalent statements, both $\sum_n [c_n(f)]_+$ and $\sum_n [c_n(f)]_-$ diverge. In particular, $\limsup_n |c_n(f)| > 0$ for even one such f is a sufficient condition for two distinct subsequential limits.

Proof. Summing (60.1) shows that $\sum_{n=1}^m c_n(f)$ differs from $L_{m+1}(f)$ by a convergent sequence. It is consequently bounded, and it converges if and only if $L_m(f)$ converges. Cauchy's criterion gives equivalence with (60.3). If every $L_m(f)$, $f \in \mathcal{F}$, converges, tightness implies that all subsequential limits have the same determining Laplace functionals, hence are equal. Conversely, distinct subsequential limits differ on one test in $\mathcal{F}$, so the corresponding $L_m(f)$ does not converge.

If a bounded sequence of partial sums has a finite sum of positive parts or a finite sum of negative parts, the other sum is finite too: otherwise the partial sums tend to minus or plus infinity. Both sums being finite would imply convergence. This proves the divergence assertion. Finally, if $c_n(f)$ does not tend to zero, its partial sums cannot converge. Equivalently, select indices along which it is bounded away from zero in absolute value; joint extraction of P_n and P_{n+1} and the vanishing transfer error gives two limits with different f -Laplace functionals. $\square$

Proposition 60.2 (All fixed regularizations are screened). *For $0 \leq s < 1$, set*

$$G_{n,s}(z) = \frac{1}{2\pi} \int_0^{2\pi} \prod_{j=1}^n |1 - se^{-it} z_j|^\beta dt, \quad \hat{G}_{n,s} = \frac{G_{n,s}}{\mathbb{E}_n G_{n,s}}.$$

For each fixed $s < 1$,

$$\mathbb{E}_n G_{n,s} \longrightarrow (1 - s^2)^{-\beta/2}, \quad (60.4)$$

$$\mathbb{E}_n [F(\Xi_n)(\hat{G}_{n,s} - 1)] \longrightarrow 0 \quad (60.5)$$

for every bounded continuous F . If the additional estimate

$$\lim_{s \uparrow 1} \limsup_{n \rightarrow \infty} \mathbb{E}_n |\hat{G}_n - \hat{G}_{n,s}| = 0 \quad (60.6)$$

holds, then P_n and P_{n+1} have asymptotically zero distance in every metric for weak convergence on their relatively compact closure. Estimate (60.6) is a sufficient hypothesis, not a proved statement about the gas.

Proof. For fixed $s < 1$ and t , the function $h_{s,t}(z) = (\beta/2) \log(1 - se^{-it}z)$ is holomorphic on a neighborhood of $\overline{\mathbb{D}}$ and vanishes at zero. Its coefficients are $-(\beta/2)s^k e^{-ikt}/k$. Theorem 19.6 gives

$$\mathbb{E}_n e^{Y_n(h_{s,t})} \longrightarrow \exp \left(\frac{\beta}{2} \sum_{k \geq 1} \frac{s^{2k}}{k} \right) = (1 - s^2)^{-\beta/2}.$$

Rotation invariance makes the expectation on the left independent of t , proving (60.4). Along an arbitrary subsequence on which $\Xi_n \Rightarrow \Xi$, the mixed Laplace identity in the same theorem gives the additional factor $\mathbb{E}F(\Xi)$. Integration in t is legitimate: for fixed s , the absolute value of each mixed expectation is bounded by $\|F\|_\infty \mathbb{E}_n e^{Y_n(h_{s,0})}$, uniformly in n, t . Dominated convergence proves (60.5) along that subsequence. Tightness and a subsequence contradiction prove it along the whole sequence.

Finally, the difference between the unregularized and regularized mixed expectations is bounded by $\|F\|_\infty \mathbb{E}_n |\widehat{G}_n - \widehat{G}_{n,s}|$. The additional hypothesis, followed by the fixed- s result, gives vanishing normalized covariance for every F . Applying this to compact Laplace functionals and using (60.1) gives the asserted adjacent-size conclusion by tightness. $\square$

Proposition 60.3 (A connectedness consequence of adjacent-size screening). *Suppose that P_n and P_{n+1} have asymptotically zero weak distance. Then the set of subsequential limits of P_n is compact and connected. If this set is not a singleton, it contains continuum many different probability laws. Thus under this hypothesis a two-phase parity alternation is impossible, although full convergence does not follow.*

Proof. The closure of the sequence is compact by tightness, and its limit set C is a nonempty compact set. If $C = C_0 \cup C_1$ were a disconnection into nonempty disjoint compact sets, their distance would be positive. Choose disjoint neighborhoods of C_0, C_1 at positive distance. All sufficiently late terms of the sequence lie in their union, since otherwise a subsequence outside the union would have a limit point outside C . Both neighborhoods are visited infinitely often. Thus infinitely many consecutive terms cross from one neighborhood to the other, contradicting the vanishing of their distances. This proves connectedness.

If $x, y \in C$ are distinct, the continuous function $z \mapsto d(x, z)$ maps C onto a connected subset of the real line containing 0 and $d(x, y) > 0$, and hence onto a set containing an interval. Therefore C has at least continuum cardinality. A separable metric space has at most continuum cardinality, which proves the final assertion. $\square$

What would constitute an actual disproof of uniqueness. A positive oscillation in (60.3) would be a disproof for the original disk gas, rather than merely a multiplicity statement for an auxiliary DLR equation. Convergence to zero of each individual covariance rules out adjacent-size phase switches but still allows slow variation through a connected family of limits. The fixed- s calculation does not permit interchange of $n \rightarrow \infty$ and $s \uparrow 1$. The missing estimate concerns fluctuations at boundary scales tending to zero with n ; it is not supplied by convergence of every fixed harmonic mode or by the leading partition-function asymptotic.

60.1 An independent sufficient monotonicity criterion

For the fixed-potential planar gas, monotonicity of all correlation measures in particle number is known at $\beta = 2$; Hedenmalm [9, slide 6] proposed the analogous statement for $\beta < 2$. We do not assume that proposal holds for the disk gas. The following implication explains precisely why a disk-specific proof would settle uniqueness below 2.

Proposition 60.4. *Fix $\beta > 0$. If, for every $p \geq 1$, the correlation measures $\rho_{p,N} dA^p$ are nondecreasing in $N \geq p$, then the disk gas has a unique subsequential limit and hence converges along the full sequence.*

Proof. Uniform local bounds make the increasing correlation measures converge to locally finite measures, uniquely for every p . Every subsequential point-process limit has these measures by the already proved local correlation convergence. They determine its law: for a compactly supported nonnegative bounded function f , expand $\prod_{z \in \Xi} (1 + (e^{-f(z)} - 1))$ into factorial sums. The absolute sum is bounded by $2^{\Xi(\text{supp } f)}$, which is integrable by the established compact exponential moments. The resulting Laplace functional is therefore uniquely specified by the factorial measures. Laplace functionals determine point-process laws. Tightness then upgrades uniqueness of every subsequential limit to full convergence. $\square$

This is a sufficient criterion, not a monotonicity theorem for $\beta \neq 2$. The interpolation inequalities above bound correlations from below; they do not compare successive particle numbers.

61 An exact second temperature response at the center

The intensity conjecture has a second, nontrivial perturbative check. The order of limits in the following theorem is part of its statement.

Theorem 61.1 (Exact second response at the center). *For the unweighted disk gas, with intensity measured against ordinary area,*

$$\lim_{N \rightarrow \infty} \partial_\beta^2 \log \rho_{N,\beta}^{(1)}(0) \Big|_{\beta=2} = 0, \quad (61.1)$$

$$\lim_{N \rightarrow \infty} \partial_\beta^2 \rho_{N,\beta}^{(1)}(0) \Big|_{\beta=2} = \frac{1}{\pi}. \quad (61.2)$$

Thus the first two finite-volume temperature responses at the center agree, after passage to the limit, with the conjectured function $(4/\beta - 1)/\pi$. No interchange of a temperature derivative with an infinite-volume limit is asserted.

We give all cancellations, including a boundary contribution that would be lost by simply replacing finite Bergman projections by the infinite one. For this proof use normalized area $d\mu = dA/\pi$ and orthonormal functions $h_n(z) = \sqrt{n+1} z^n$. Let K_N project onto their first N terms, let K be the full Bergman projection, and put $Q_N = I - K_N$, $Q = I - K$, and $P = |h_0\rangle\langle h_0|$. All operator norms and integrals in this section use μ . Changing the reference measure multiplies the intensity by π and does not change its logarithmic temperature derivatives. Write $\zeta_2 = \sum_{j \geq 1} j^{-2}$ and

$$f(u, v) = \log |u - v| = I(u, v) - g(u, v), \quad I(u, v) = \log |1 - u\bar{v}|, \quad g(u, v) = \log \left| \frac{1 - u\bar{v}}{u - v} \right|.$$

For $m < n$ put $\psi_{mn} = h_m \wedge h_n$, with the convention $h_m \wedge h_n = (h_m \otimes h_n - h_n \otimes h_m)/\sqrt{2}$.

Lemma 61.2 (The single-replacement contribution). *The difference between the single-replacement variances in the Slater identity, for the Palm process at zero with statistic $U_{N-1} + \sum \log |z_i|$ and for the original N -particle process, converges to*

$$S = \frac{47}{16} - \frac{7}{4}\zeta_2.$$

Proof. The Palm occupied space is $J_N = K_N - P$. The screened perturbation $B = M_{\log|z|-V_P} + A_P$ is independent of N . For $n \geq 1$, with $t = |z|^2$, its action is

$$Bh_n = \left\{ \frac{1}{2} \log t + \frac{1}{2}(1 - 1/n) - \frac{n}{2(n+1)}t \right\} h_n.$$

Indeed $V_P = (t - 1)/2$ and $A_P h_n / h_n = -1/(2n) + t/[2(n+1)]$, by direct angular integration. The Hartree–Fock multiplier of Lemma 57.3, after removing its scalar, converges for each fixed n in $L^2((n+1)t^n dt)$ to

$$b_n(t) = \frac{1}{2}H_{n+1} - \log(1 - t).$$

The uniform bound for the centered multipliers in that lemma gives strong convergence of the single-replacement operators on the Bergman space. The trace-norm column bound for BK in Lemma 57.5 then justifies passage to the limit in their cross term. More explicitly, approximate BK by its first M columns, pass to the limit for those columns, and bound the remainder by the uniform operator norm times $\sum_{n>M} \|Bh_n\|_2$. The latter tends to zero. The quadratic B term passes by the same argument in Hilbert–Schmidt norm. The lost h_0 column passes by its L^2 convergence. The $P(F + B)J_N$ term is zero, since its input and output have different angular frequencies. Consequently the limit is

$$-1 + \sum_{a=2}^{\infty} \left\{ \text{Var}_{at^{a-1}dt} \left[-\log(1 - t) + \frac{1}{2} \log t - \frac{a-1}{2a}t \right] - \text{Var}_{at^{a-1}dt} [-\log(1 - t)] \right\}.$$

The series is absolutely convergent by the same trace-norm argument. The elementary beta integral and its first two derivatives give

$$\begin{aligned} \text{Var}(\log t) &= a^{-2}, \quad \text{Var}(t) = \frac{a}{(a+1)^2(a+2)}, \quad \text{Cov}(\log t, t) = \frac{1}{(a+1)^2}, \\ \text{Cov}(-\log(1 - t), \log t) &= \sum_{j>a} \frac{1}{j^2}, \quad \text{Cov}(-\log(1 - t), t) = \frac{a}{(a+1)^2}. \end{aligned}$$

The summand therefore equals

$$\sum_{j>a} \frac{1}{j^2} + \frac{5}{8a} + \frac{1}{4a^2} - \frac{1}{2(a+1)} - \frac{9}{8(a+2)}.$$

For completeness,

$$\sum_{a=2}^M \sum_{j>a} j^{-2} = H_M - H_M^{(2)} + M \sum_{j>M} j^{-2} - \zeta_2 + 1,$$

where $H_M^{(2)} = \sum_{j \leq M} j^{-2}$ and $M \sum_{j>M} j^{-2} \rightarrow 1$ by integral comparison. Summing the rational terms and then letting M increase gives $S = -1 + (2 - 2\zeta_2) + (31/16 + \zeta_2/4)$, as claimed. $\square$

Lemma 61.3 (The infinite Green replacement terms). *Define*

$$L_g = \sum_{n \geq 1} \|Q^{\wedge 2} M_g \psi_{0n}\|_2^2, \quad G_g = \|\text{Proj}_{h_0 \wedge Q\mathcal{H}} M_g (K - P)^{\wedge 2}\|_{\text{HS}}^2.$$

Both are finite and

$$L_g = \frac{1}{2} - \frac{1}{4}\zeta_2, \quad G_g = \frac{3}{2}\zeta_2 - \frac{39}{16}.$$

Proof. Set

$$A = \sum_{n \geq 1} \|M_g \psi_{0n}\|_2^2, \quad B = \sum_{n \geq 1} \|(K \otimes I)M_g \psi_{0n}\|_2^2, \quad C = \sum_{n \geq 1} \|(K \otimes K)M_g \psi_{0n}\|_2^2.$$

Since the vectors are antisymmetric, $L_g = A - 2B + C$. The Bergman kernel against μ is $(1 - u\bar{v})^{-2}$. Summing the nonnegative squared differences defining A gives

$$A = \iint g(u, v)^2 \{K(v, v) - \operatorname{Re} K(u, v)\} d\mu(u) d\mu(v) = \frac{\zeta_2}{2} - \frac{1}{2}.$$

Here $\int g(u, v)^2 K(v, v) d\mu(v) = \zeta_2/2$ by disk automorphisms and $\frac{1}{4} \int_0^1 (\log t)^2 (1-t)^{-2} dt = \zeta_2/2$. For the other integral use $v = (u - w)/(1 - \bar{u}w)$. The product $K(u, v) d\mu(v)$ transforms to $(1 - \bar{u}w)^{-2} d\mu(w)$, whose angular average is one; thus its integral against $\log^2 |w|$ is $1/2$. All integrations and limits are justified, for example, by $|K(u, v)| \leq \sqrt{K(u, u)K(v, v)}$ and Cauchy–Schwarz with weight g^2 , since each corresponding diagonal integral is finite.

To evaluate B , let $T_v = KM_{g(\cdot, v)}K$ on the Bergman space. Conformal covariance makes T_v unitarily equivalent to T_0 , and direct radial integration gives

$$T_0 h_n = \frac{h_n}{2(n+1)}, \quad \|T_v\|_{\text{HS}}^2 = \frac{1}{4} \zeta_2.$$

The reproducing vector at v is an eigenvector of T_v with eigenvalue $1/2$: a disk automorphism taking zero to v transports the constant eigenvector to a nonzero multiple of that vector. Hence $(T_v^2 h_0)(v) = 1/4$. Finally,

$$(T_v h_0)(u) = \frac{1 - |v|^2}{2(1 - u\bar{v})}, \quad K(v, v) \|T_v h_0\|_2^2 = \frac{-\log(1 - |v|^2)}{4|v|^2},$$

with the continuous interpretation at zero. Expanding the squared contractions in the first coordinate now gives

$$\begin{aligned} B &= \frac{1}{2} \int \{K(v, v) \|T_v h_0\|_2^2 + \|T_v\|_{\text{HS}}^2 - 2 \operatorname{Re}(T_v^2 h_0)(v)\} d\mu(v) \\ &= \frac{1}{2} \left(\frac{\zeta_2}{4} + \frac{\zeta_2}{4} - \frac{1}{2} \right) = \frac{\zeta_2 - 1}{4}. \end{aligned}$$

These are finite quantities, so the expansion is legitimate by Parseval and Cauchy–Schwarz.

For C and G_g it is useful to record the elementary Green integral

$$\int g(u, v) u^l \bar{u}^k d\mu(u) = \frac{H_{l-k}(v) - v^{l+1} \bar{v}^{k+1}}{2(l+1)(k+1)}, \quad (61.3)$$

where $H_j(v) = v^j$ for $j \geq 0$ and $H_j(v) = \bar{v}^{-j}$ for $j < 0$. One proof applies $-\Delta_v$ to both sides and uses zero boundary values; uniqueness follows from the maximum principle. Alternatively the angular expansion of g gives the same expression directly. For $k \geq 0$, contraction of $g\psi_{0n}$ against h_k in its first coordinate is

$$D_{kn}(v) = \frac{(n+1 - n|v|^2)v^n \bar{v}^k - H_{n-k}(v)}{2\sqrt{2(k+1)(n+1)}}.$$

Only $0 \leq k \leq n$ survive the second Bergman projection. Their squared norms after that projection sum to

$$C_n = \sum_{k=0}^n \frac{(n-2k)^2}{8(k+1)(n+1)(n-k+1)(n+2)^2} = \frac{H_{n+1}}{4(n+1)(n+2)} - \frac{1}{2(n+2)^2}.$$

Using $\sum_{a \geq 1} H_a/[a(a+1)] = \zeta_2$, which follows by interchanging positive sums, gives $C = 1/2 - \zeta_2/4$. Combining A, B, C proves the formula for L_g .

For $1 \leq m < n$, contraction of $g\psi_{mn}$ against h_0 is

$$\frac{n-m}{2\sqrt{2(m+1)(n+1)}} v^{m+n}(1-|v|^2).$$

Projecting off the analytic monomial of degree $m+n$, and accounting for both antisymmetric coordinates, gives the squared norm

$$\frac{(n-m)^2}{4(m+1)(n+1)(m+n+2)^2(m+n+3)}.$$

Summation with $s = m+n+2$ therefore yields

$$\begin{aligned} G_g &= \sum_{s=4}^{\infty} \left\{ \frac{H_{s-2}-1}{4s(s+1)} - \frac{s-3}{2s^2(s+1)} \right\} \\ &= \frac{3}{2}\zeta_2 - \frac{39}{16}. \end{aligned}$$

To check the last equality directly, the two required sums are $\sum_{s \geq 4} H_{s-2}/[s(s+1)] = 2/3$ and $\sum_{s \geq 4} (s-3)/[s^2(s+1)] = 61/12 - 3\zeta_2$; both follow by telescoping and interchanging positive harmonic sums. $\square$

Lemma 61.4 (The finite-projection boundary contributions cancel). *The difference of the finite- N double-replacement variances tends to $G_g - L_g$.*

Proof. The difference in question is the gain with input $J_N^{\wedge 2}$ and output $h_0 \wedge Q_N \mathcal{H}$, minus the loss with input $h_0 \wedge J_N \mathcal{H}$ and output $Q_N^{\wedge 2}$. The Green parts of these two operators converge in Hilbert–Schmidt norm to the corresponding infinite-projection operators. For the loss, the unprojected input operator is Hilbert–Schmidt by the finite value of A above. For the gain, even the unprojected contracted operator $(\langle h_0 | \otimes I)M_g(K \otimes K)$ is Hilbert–Schmidt. Indeed, Bessel’s inequality gives the bound

$$\sum_{m,n \geq 0} \left\| h_n(v) \int g(u,v) h_m(u) d\mu(u) \right\|_{L^2(d\mu(v))}^2 \leq \iint g(u,v)^2 K(v,v) d\mu(u) d\mu(v) < \infty.$$

Strong convergence of the input and output projections applied to these fixed Hilbert–Schmidt operators proves the assertion.

The image kernel contributes finite norms that must be retained. Its angular series shows that the squared loss norm is exactly

$$L_{I,N} = \frac{1}{4} \sum_{n=1}^{N-1} \sum_{k \geq N-n} \frac{n+1}{n+k+1} \int_0^1 t^k \left(\frac{1}{k} - \frac{t^n}{n+k} \right)^2 dt. \quad (61.4)$$

For example the output with angular frequencies $(-k, n+k)$ and its antisymmetric partner have radial coefficient proportional to $\bar{u}^k v^{n+k} (1/k - |u|^{2n}/(n+k))$. These are exactly the surviving terms: the analytic output index must be at least N . Different such angular pairs are orthogonal. For each fixed k , the k indices $n = N-k, \dots, N-1$ give a total tending to $1/[4k(k+1)]$. Their total is always at most $1/[4k(k+1)]$, since the integrand is bounded by t^k/k^2 and there are at most k indices. Dominated convergence gives $L_{I,N} \rightarrow 1/4$.

For the gain, contraction of $I\psi_{mn}$ against h_0 , with $1 \leq m < n < N$ and $s = m+n$, is the analytic function

$$-\frac{(n-m)(s+1)}{2\sqrt{2}mn\sqrt{(m+1)(n+1)}} v^s.$$

It survives Q_N exactly when $s \geq N$. Thus

$$G_{I,N} = \sum_{\substack{1 \leq m < n < N \\ m+n \geq N}} \frac{(n-m)^2(m+n+1)}{4m^2n^2(m+1)(n+1)}. \quad (61.5)$$

For each fixed m , the m indices $n = N-m, \dots, N-1$ give a total tending to $1/[4m(m+1)]$. For all N , the corresponding total is at most $1/[2m(m+1)]$: there are at most m indices, $(n-m)^2/n^2 \leq 1$, and $(m+n+1)/(n+1) \leq 2$. Hence $G_{I,N} \rightarrow 1/4$ as well.

Finally the Green-image cross terms tend to zero. Regard the gain and loss operators as operators between fixed tensor spaces by including their projections. Their image parts have bounded Hilbert–Schmidt norm by the preceding two formulas and converge weakly to zero in that space. To see weak convergence, first test against operators with finitely many analytic input columns and finitely many output vectors. In the loss, each fixed input column has image norm tending to zero directly from (61.4); in the gain each fixed input pair is annihilated once $N > m+n$. Finite-rank tests are dense, and the uniform Hilbert–Schmidt bound completes the argument. Their inner products with the strongly convergent Green parts therefore vanish. Since $f = I - g$, the limiting double-replacement difference is $(G_g + 1/4) - (L_g + 1/4) = G_g - L_g$. $\square$

Proof of Theorem 61.1. Differentiating the finite-volume Palm partition quotient twice, as in Theorem 57.1, gives a difference of the two energy variances. The Slater identity splits it into the single- and double-replacement terms evaluated above. Their sum tends to

$$\left(\frac{47}{16} - \frac{7}{4}\zeta_2\right) + \left(\frac{3}{2}\zeta_2 - \frac{39}{16} - \frac{1}{2} + \frac{1}{4}\zeta_2\right) = 0.$$

The original ordinary-area intensity satisfies $\rho_{N,2}(0) = 1/\pi$. The exact first-response formula gives $\partial_\beta \log \rho_{N,\beta}(0)|_2 \rightarrow -1$. Using $\rho'' = \rho\{(\log \rho)'' + ((\log \rho)')^2\}$ proves (61.2). $\square$

62 The exact second response at every interior anchor

Theorem 62.1 (Locally uniform second intensity response). *For every $0 < r < 1$,*

$$\lim_{N \rightarrow \infty} \sup_{|a| \leq r} \left| \partial_\beta^2 \log \rho_{N,\beta}^{(1)}(a) \Big|_{\beta=2} \right| = 0, \quad (62.1)$$

$$\lim_{N \rightarrow \infty} \sup_{|a| \leq r} \left| \partial_\beta^2 \rho_{N,\beta}^{(1)}(a) \Big|_{\beta=2} - \frac{1}{\pi(1-|a|^2)^2} \right| = 0. \quad (62.2)$$

These statements take the finite-volume derivatives before the particle-number limit.

Continue with normalized area, projections, and kernels from Section 61. Put $L(z) = -\log(1-|z|^2)$, $U_k = M_{z^k}$, $H_k = Q M_{\bar{z}^k} K$, and $T_L = K M_L K$ on its natural domain. The normalized coherent vector and its finite version are

$$h_a(z) = \frac{1-|a|^2}{(1-\bar{a}z)^2}, \quad h_{a,N}(z) = \frac{K_N(z,a)}{\sqrt{K_N(a,a)}}.$$

Their projections are denoted by P_a and $P_{a,N}$.

Lemma 62.2 (An image-kernel boundary formula). *Let $h_N \in K_N \mathcal{H}$ be unit vectors converging to a unit analytic vector h in the norm $\|\sum_l c_l h_l\|_* = \sum_l |c_l| \sqrt{l+1}$. Suppose h extends*

holomorphically to a neighborhood of the closed disk. Write $P_N = |h_N\rangle\langle h_N|$, $J_N = K_N - P_N$, $P = |h\rangle\langle h|$, and $J = K - P$. Then the image-kernel loss and gain satisfy

$$\|Q_N^{\wedge 2} M_I(h_N \wedge K_N)\|_{\text{HS}}^2 \longrightarrow \frac{1}{4}, \quad (62.3)$$

$$\|\text{Proj}_{h_N \wedge Q_N \mathcal{H}} M_I J_N^{\wedge 2}\|_{\text{HS}}^2 \longrightarrow \|\text{Proj}_{h \wedge Q \mathcal{H}} M_I J^{\wedge 2}\|_{\text{HS}}^2 + \frac{1}{4} \left\{ \int L|h|^2 d\mu - \sum_{k \geq 1} \frac{|\mu_k(h)|^2}{k} \right\}, \quad (62.4)$$

where $\mu_k(h) = \int z^k |h(z)|^2 d\mu(z)$. The first operator is viewed as a map from $K_N \mathcal{H}$ by $v \mapsto h_N \wedge v$. Its kernel contains h_N , so this has precisely the loss norm on $h_N \wedge J_N \mathcal{H}$.

The image loss converges weakly to zero in Hilbert–Schmidt space. The image gain is the sum of a strongly convergent operator, whose limit is its displayed infinite-projection counterpart, and a bounded operator converging weakly to zero. For the coherent vectors $h = h_a$, the infinite-projection gain is zero and the additional term in braces equals one.

Proof. We first prove the claims for a fixed unit polynomial h of degree at most d and $N > d$. Define the analytic tail operator $R_{k,N} = Q_N U_k K_N$. Since multiplication by $\bar{z}^k$ lowers the angular index,

$$Q_N M_{\bar{z}^k} K_N = H_k K_N, \quad \|R_{k,N}\|_{\text{HS}}^2 = \sum_{n=\max(0, N-k)}^{N-1} \frac{n+1}{n+k+1} \leq \min(k, N).$$

For fixed k this last quantity tends to k . Moreover H_k is Hilbert–Schmidt and $\|H_k\|_{\text{HS}}^2 = k$, as follows by summing the squared column norms

$$\|H_k h_l\|_2^2 = \begin{cases} k^2 / [(l+1)(l+k+1)], & k \leq l, \\ (l+1)/(l+k+1), & k > l. \end{cases}$$

The same formulas, summed first over k for each fixed l , give the bounded-operator identity

$$\sum_{k \geq 1} \frac{H_k^* H_k}{k} = I \quad \text{on the Bergman space.} \quad (62.5)$$

Indeed the summands are diagonal in h_l and their diagonal sum is

$$\sum_{k=1}^l \frac{k}{(l+1)(l+k+1)} + \sum_{k>l} \frac{l+1}{k(l+k+1)} = 1.$$

The image angular expansion is $M_I = -\frac{1}{2} \sum_{k \geq 1} (U_k \otimes U_k^* + U_k^* \otimes U_k)/k$. The loss, apart from a term tending to zero, is therefore

$$-\frac{1}{2} \sum_{k \geq 1} \frac{(H_k h) \wedge (R_{k,N} K_N)}{k}.$$

For fixed k its squared Hilbert–Schmidt norm tends to $\|H_k h\|_2^2 / (4k)$. Terms with different k are orthogonal on each input monomial: their high analytic output indices differ, and their other output indices are at most $d < N$. The sum of their squared norms is bounded by $\frac{1}{4} \sum_k \|H_k h\|_2^2 / k$. The omitted term contains $R_{k,N} h$ and can occur only when $k \geq N - d$. Its squared norm is $O_d(N^{-1})$. For clarity, $\|U_k h\|_2^2 \leq C_h / (k+1)$ and $\|H_k K_N\|_{\text{HS}}^2 \leq \min(k, N)$; summing these bounds against k^{-2} over $k \geq N - d$ gives $O_h(N^{-1})$. Cross terms in this

omitted sum cost at most $2d+1$, because equal high output angular indices require $|k-k'| \leq d$. Equation (62.5) proves the loss limit. Each fixed input column tends to zero, so boundedness gives its weak Hilbert–Schmidt convergence to zero.

For the gain first allow input $K_N^{\wedge 2}$. Identify the output $h \wedge Q_N \mathcal{H}$ with $Q_N \mathcal{H}$ by contraction. For an input $h_m \wedge h_n$, $m < n$, its output is

$$-\frac{1}{2} \sum_{k \geq 1} \frac{1}{k} \{a_k(m)H_k h_n - a_k(n)H_k h_m + b_k(m)R_{k,N} h_n - b_k(n)R_{k,N} h_m\},$$

where $a_k(m) = \langle h, U_k h_m \rangle$ and $b_k(m) = \langle h, U_k^* h_m \rangle$; use the inner product linear in its second entry. The a part has $k \leq d$ and converges strongly in Hilbert–Schmidt norm, since each H_k is Hilbert–Schmidt. It is the infinite-projection gain operator.

The b part converges weakly to zero. For finitely many k , its nonzero low input indices satisfy $m \leq d+k$, whereas the tail output comes from $n \geq N-k$; these two ranges are disjoint for sufficiently large N . Orthogonality of the high output angular indices then gives its limiting squared norm

$$\frac{1}{4} \sum_{k \geq 1} \frac{\|U_k h\|_2^2}{k} = \frac{1}{4} \int L|h|^2 d\mu.$$

To justify the infinite sum, write h in its $d+1$ monomials and use Cauchy–Schwarz for that finite sum. For a fixed monomial, different output angular indices are orthogonal; the two terms that interchange m, n cost at most a factor two. Hence the squared norm of the part $k > K_0$ is at most

$$C_h \sum_{k > K_0} \frac{1}{k^2} \frac{\min(k, N)}{k+1} \leq C_h \sum_{k > K_0} \frac{1}{k(k+1)}.$$

This bound is uniform in N and tends to zero with K_0 . It proves both passage to the infinite sum and the asserted weak convergence. Its cross term with the strongly convergent a part therefore vanishes.

It remains to remove the input space $h \wedge K_N \mathcal{H}$. Its restriction is obtained by putting h in one input slot. The H_k terms again converge strongly; the sole surviving tail term is

$$-\frac{1}{2} \sum_{k \geq 1} \frac{\overline{\mu_k(h)}}{k} R_{k,N} K_N.$$

For polynomial h , the moments vanish when $k > d$. Orthogonality on each input column shows that the squared norm of this term tends to $\frac{1}{4} \sum_k |\mu_k(h)|^2/k$; it converges weakly to zero. The term containing $R_{k,N} h$ has squared norm $O_h(N^{-1})$ by the same large- k estimate used for the loss. Subtract the squared norms of this orthogonal input block from those with full input $K_N^{\wedge 2}$. This proves the gain formula and the strong-plus-weak description for polynomial h .

We extend the result to the stated analytic vectors by polynomial approximation. The image versions of the two vertex bounds in Lemma 57.4 bound the full loss and full gain operators by $C\|h\|_*$, uniformly in N ; the same estimates bound their differences when h is replaced by another vector. Restriction to $h \wedge K_N$ changes continuously as well, by the triangle inequality and these bounds. Approximate h by normalized Taylor polynomials, first let N increase, then let the polynomial degree increase. The assumption on h_N permits its replacement by the corresponding normalized truncations of h . The scalar term $\int L|h|^2 d\mu$ is continuous in this approximation, since $\|h\|_\infty \leq \|h\|_*$ and L is integrable. Also

$$\sum_{k \geq 1} \frac{|\mu_k(h)|^2}{k} = \iint -\log|1 - u\bar{v}| |h(u)|^2 |h(v)|^2 d\mu(u) d\mu(v)$$

by the angular series, or its Abel regularizations followed by dominated convergence. The kernel is absolutely integrable, so this expression is continuous under uniform convergence of the analytic vectors. This completes the approximation argument.

For $h = h_a$, reproducing gives $\mu_k(h_a) = a^k$. Furthermore the squared coefficients are $(1 - |a|^2)^2(l+1)|a|^{2l}$, whence

$$\int L|h_a|^2 d\mu = (1 - |a|^2)^2 \sum_{l \geq 0} (l+1) H_{l+1} |a|^{2l} = 1 - \log(1 - |a|^2).$$

Thus the expression in braces is one. Finally, if u, v belong to $J_a \mathcal{H}$, then $u(a) = v(a) = 0$. In the angular series for I , contraction against h_a of an analytic factor $z^k u(z)$ vanishes; contraction of an antianalytic factor leaves an analytic function in the other coordinate, which is killed by Q . This proves that the infinite-projection image gain is zero, first for finite angular sums and then by the vertex bounds. $\square$

Lemma 62.3 (Conformal covariance of the limiting single contribution). *The limiting difference of the single-replacement variances at an anchor a exists and equals the constant S in Lemma 61.2.*

Proof. The limiting renormalized Hartree–Fock operator on analytic vectors is

$$FK = (M_L + \tfrac{1}{2}T_L)K.$$

This follows from the radial multiplier formula, since $T_L h_n = H_{n+1} h_n$. Let $P = P_a$, $J = K - P$, and

$$B_a = M_{V_g(P) - g(\cdot, a)} + A_I(P) - A_g(P),$$

where $V_g(P)(u) = \int g(u, v)P(v, v)d\mu(v)$ and $A_j(P)$ has kernel $j(u, v)P(u, v)$. This is precisely the previous screened Palm perturbation: the harmonic reproducing identity gives $V_I(P)(u) = I(u, a)$, so $\log|u - a| - V_f(P) = V_g(P) - g(u, a)$. The finite-Palm trace-norm bounds, local uniform convergence of the coherent vectors, and the uniform bound on $Q_N F_{K_N} K_N$ show that the single contribution converges to

$$S_a = -\|QFP\|_{\text{HS}}^2 + 2\operatorname{Re}\langle QFJ, QB_a J \rangle_{\text{HS}} + \|QB_a J\|_{\text{HS}}^2 + \|P(F + B_a)J\|_{\text{HS}}^2.$$

The cross pairing is well defined because QFK is bounded and $B_a K$ is trace class. Convergence follows by truncating the trace-class factor and the finitely many coherent columns, as in the center proof. The bounds used there hold uniformly on compact sets of anchors.

Take a disk automorphism ϕ with $\phi(0) = a$, and put $d = \log|\phi'|$. Pull back by the unitary $u \mapsto \phi'(u)f(\phi(u))$ on $L^2(\mu)$. It preserves the Bergman space and takes P_a to P_0 . Since $L \circ \phi = L - d$, the pulled-back F on the Bergman space is $F - M_d - T_d/2$, where $T_d = KM_d K$. The Green part of B_a is invariant, whereas $I(\phi(u), \phi(v)) = I(u, v) + (d(u) + d(v))/2$. Thus its pulled-back screened perturbation is

$$B_0 + \tfrac{1}{2}(M_d P_0 + P_0 M_d).$$

Put $P = P_0$, $J = K - P$ for the rest of the proof. Combining these transformations shows that the full single-replacement operator changes from QFK to $QFK - QM_d K$, and the Palm operator changes from $(Q + P)(F + B_0)J$ to that operator minus $(Q + P)M_d J$. For the P output this uses $PT_d J = PM_d J$; its coefficient cancels the extra $PM_d J/2$ from B_0 .

The cross terms in the difference of these two squared norms vanish. This cancellation must be made before pairing: QFK itself need not be Hilbert–Schmidt. Let $T = QFK$ and $H = QM_d K$. At the center $P(F + B_0)J = 0$. The cross difference reduces to

$$\langle QB_0 J, HJ \rangle_{\text{HS}} - \langle TP, HP \rangle_{\text{HS}}.$$

Both pairings are absolutely defined by the trace-class and finite-rank bounds. Both are zero because T and B_0 preserve angular frequency, whereas multiplication by the nonconstant harmonic part of d changes it. Its constant part is annihilated by the off-diagonal projections.

The remaining squared-norm difference is

$$-\|QM_dP\|_{\text{HS}}^2 + \|PM_dJ\|_{\text{HS}}^2 = 0.$$

Indeed write $d = d_0 + \sum_{k \geq 1} (d_k z^k + \bar{d}_k \bar{z}^k)$. The function d extends harmonically past the closed disk, so all relevant series converge absolutely, and the two displayed norms both equal $\sum_{k \geq 1} |d_k|^2 / (k+1)$. The off-diagonal operator QM_dK is Hilbert–Schmidt, with squared norm $\sum_{k \geq 1} k |d_k|^2$. Consequently $S_a = S_0 = S$. $\square$

Proof of Theorem 62.1. Use the single- and double-replacement variance decomposition at the anchor a . The preceding lemma handles the single term. The Green gain and loss converge strongly in Hilbert–Schmidt norm, by the unprojected vertex bounds from the center proof and convergence of the coherent vectors. Their limiting norms equal G_g, L_g , since Green multiplication, K , and P_a transform unitarily under disk automorphisms. Lemma 62.2 shows that the image parts each have limiting squared norm $1/4$ and converge weakly to zero for coherent vectors. Their cross terms with the strongly convergent Green parts vanish. Thus the double difference tends to $G_g - L_g = -S$, proving the first limit.

All passages are uniform for $|a| \leq r < 1$. To see this directly, the coherent vectors and their finite versions have geometrically decaying coefficients in $\|\cdot\|_*$, uniformly on that compact set. They therefore admit simultaneous polynomial approximations. For each fixed polynomial degree and angular cutoff the preceding limits concern finitely many coefficients, uniformly bounded radial integrals, and convergent finite-dimensional expressions; hence they are uniform on this compact parameter set. The discarded tails are controlled uniformly by the vertex bounds and trace-norm bounds used above. The same compactness argument applies to the Green operators, whose unprojected Hilbert–Schmidt kernels depend continuously on a . This proves local uniform convergence. Finally use the locally uniform first-response theorem and $\rho_{N,2}(a) \rightarrow 1/[\pi(1 - |a|^2)^2]$ in $\rho'' = \rho\{(\log \rho)'' + ((\log \rho)')^2\}$. $\square$

63 The conjectured density coefficient as a charge exponent

Let $0 < \beta < 2$ and use ordinary planar area. Set

$$H_N = \sum_{i=1}^N -\log |z_i|, \quad M_N(t) = \mathbb{E}_{N,\beta} e^{-tH_N}, \quad t > -2.$$

These moments are finite: $|\Delta_N|^\beta$ is bounded on $\overline{\mathbb{D}}^N$ and $|z|^t$ is integrable for $t > -2$. The t -charged law is

$$d\mathbb{P}_{N,\beta}^{[t]} = M_N(t)^{-1} e^{-tH_N} d\mathbb{P}_{N,\beta}. \quad (63.1)$$

The charge may be negative, positive, or zero.

Lemma 63.1 (Uniform relative intensities for central charges). *Let $\hat{\rho}_N^{[t]}$ denote the one-point density of $\mathbb{P}_{N,\beta}^{[t]}$ relative to $|a|^t dA(a)$, continued continuously to $a = 0$ by its defining integral. For every compact interval $I \subset (-2, \infty)$ there are constants $0 < c_{\beta,I} \leq C_{\beta,I} < \infty$ such that*

$$c_{\beta,I} \leq \hat{\rho}_N^{[t]}(0) \leq C_{\beta,I}, \quad N \geq 1, \quad t \in I. \quad (63.2)$$

Moreover, with $R_{n,\beta} = Q_{n+1,\beta}/Q_{n,\beta}$,

$$M_N(t) = \frac{M_{N-1}(t + \beta)}{R_{N-1,\beta} \hat{\rho}_N^{[t]}(0)}, \quad N \geq 1, \quad (63.3)$$

where $M_0(u) = 1$.

Proof. The ordered radial-ratio representation of the t -charged gas has exactly the same nondecreasing angular factor as the unweighted gas. Its reference rates are

$$a_k^{[t]} = 2k + \frac{\beta}{2}k(k-1) + tk = k \left(2 + t + \frac{\beta}{2}(k-1) \right) > 0.$$

Thus the radial domination proof applies without alteration. Uniformly for t in a compact subinterval of $(-2, \infty)$, $a_k^{[t]} \geq ca_k$ for every k , with a fixed $c > 0$. The reference independent exponential gaps are consequently bounded by c^{-1} times the original reference gaps in a coordinatewise coupling. The compact-count and superlinear weighted-count moment arguments therefore give uniform bounds on

$$\mathbb{E}_{N,\beta}^{[t]} \Xi_N(B_r) \quad \text{and} \quad \mathbb{E}_{N,\beta}^{[t]} \left[\sum_{i=1}^N (1 - |z_i|^2)^s \right]^q$$

for every fixed $r < 1$, $s > 1$, and $q > 0$. These bounds also hold for the charges in $I + \beta$.

For clarity, the relative intensity is the continuous subharmonic function

$$\hat{\rho}_N^{[t]}(a) = \frac{N}{Z_{N,\beta} M_N(t)} \int_{\mathbb{D}^{N-1}} |\Delta_{N-1}(y)|^\beta \prod_{j=1}^{N-1} |y_j|^t |a - y_j|^\beta dA^{N-1}(y).$$

Continuity and subharmonicity follow by the subharmonicity of powers of polynomial moduli and domination on each compact set for fixed N, t . Its circular mean is nondecreasing in radius. Hence, for $A = \{1/4 < |a| < 1/2\}$,

$$\hat{\rho}_N^{[t]}(0) \leq |A|^{-1} \int_A \hat{\rho}_N^{[t]}(a) dA(a) \leq C_I \mathbb{E}_{N,\beta}^{[t]} \Xi_N(A) \leq C_{\beta,I}.$$

The middle inequality uses a uniform positive lower bound on $|a|^t$ for $a \in A$ and $t \in I$. Using an annulus avoids any division by a vanishing weight at the origin.

For the lower bound, interpolate the constant polynomial at zero. If $\beta \leq 1$, subadditivity and Lagrange cancellation give

$$1 \leq \hat{\rho}_N^{[t]}(0) \int_{\mathbb{D}} |z|^t dA(z).$$

If $1 < \beta < 2$, put $q = \beta - 1$ and choose $s > 1$ with $sq < 1$. Apply weighted Hölder with $w(z) = (1 - |z|^2)^s$ to the same Lagrange identity. The reduced Palm integral at zero has precisely the $(N-1)$ -particle law with central charge $t + \beta$, because insertion at zero multiplies its weight by $|y|^\beta$. The uniform weighted-count bound just proved and $(x + y)^q \leq x^q + y^q$ yield

$$1 \leq \hat{\rho}_N^{[t]}(0) \left\{ \int_{\mathbb{D}} |z|^t dA(z) + C_{\beta,I,s} \int_{\mathbb{D}} |z|^t (1 - |z|^2)^{-sq} dA(z) \right\}.$$

Both integrals are uniformly finite for $t \in I$: the condition $t > -2$ handles zero and $sq < 1$ handles the boundary. This gives the required positive lower bound, including $N = 1$ when the exterior sum is empty.

Finally, evaluating the defining integral at $a = 0$ gives

$$\hat{\rho}_N^{[t]}(0) = \frac{N Z_{N-1,\beta} M_{N-1}(t + \beta)}{Z_{N,\beta} M_N(t)} = \frac{M_{N-1}(t + \beta)}{R_{N-1,\beta} M_N(t)}.$$

This is the claimed recursion, with no asymptotic normalization. $\square$

Theorem 63.2 (Signed charge exponents and the radial-depth law). *Fix $0 < \beta < 2$ and set $d_\beta = 2/\beta - 1/2$. There is a set $\mathcal{E}_\beta \subset \mathbb{N}$ of natural density one such that*

$$\frac{\log M_N(t)}{\log N} \longrightarrow -d_\beta t \quad (63.4)$$

locally uniformly for $t \in (-2, \infty)$, as $N \rightarrow \infty$ through $\mathcal{E}_\beta$. For every fixed $t > -2$ and every finite $q > 0$,

$$\mathbb{E}_{N,\beta}^{[t]} \left| \frac{H_N}{\log N} - d_\beta \right|^q \longrightarrow 0. \quad (63.5)$$

For every $\varepsilon > 0$ and $0 < s < t + 2$,

$$\mathbb{P}_{N,\beta}^{[t]} \left\{ \left| \frac{H_N}{\log N} - d_\beta \right| > \varepsilon \right\} \leq 2N^{-s\varepsilon + o(1)}. \quad (63.6)$$

In particular these statements apply to the original gas, $t = 0$, with any tail exponent $s\varepsilon$ for $0 < s < 2$.

Proof. For every positive integer p , exact confluence of the p anchors in the factorial correlation identity gives

$$h_{p,N+p}(0, \dots, 0) = \frac{Q_{N,\beta}}{Q_{N+p,\beta}} M_N(p\beta). \quad (63.7)$$

Indeed, after division by the Vandermonde of the anchors at distinct locations, its integral contains $\prod_{i,j} |a_i - y_j|^\beta$. Letting all a_i tend to zero gives the charge $p\beta$; the factorial prefactor is $(N+p)!/(N!Z_{N+p,\beta})$. Continuity of $h_{p,N+p}$ justifies this evaluation on the collision diagonal. The hierarchy upper bound and Theorem 48.4 give

$$0 < b_{\beta,p} \leq h_{p,N+p}(0, \dots, 0) \leq B_{\beta,p} < \infty \quad (63.8)$$

uniformly in N . For the lower bound, divide the stated correlation inequality at distinct anchors and then use continuity.

Let $\mathcal{D}_\beta$ be the density-one set of Theorem 22.1. There is a density-one set $\mathcal{E}_\beta$ with the following property: for every fixed integer j , all sufficiently large $N \in \mathcal{E}_\beta$ have $N + j \in \mathcal{D}_\beta$. To verify this diagonal selection, put

$$A_m = \{N \geq m + 1 : N + j \in \mathcal{D}_\beta \text{ for } |j| \leq m\}.$$

These are nested density-one sets. Choose strictly increasing $L_m \rightarrow \infty$ so that $\#([1, x] \setminus A_m) \leq x/m$ for every integer $x \geq L_m$. For $L_m \leq N < L_{m+1}$ retain N exactly when $N \in A_m$. If $L_m \leq x < L_{m+1}$, all excluded indices between L_1 and x lie outside A_m by nesting. Their number is at most x/m ; adding the fixed initial segment proves density one. The eventual shift property is immediate.

It follows, for each fixed integer $p \geq 1$, that

$$\frac{\log(Q_{N+p,\beta}/Q_{N,\beta})}{\log N} \longrightarrow p \left(\frac{\beta}{2} - 2 \right) = -p\beta d_\beta.$$

Equations (63.7) and (63.8) therefore prove $\log M_N(p\beta)/\log N \rightarrow -p\beta d_\beta$. At $p = 0$ this is exact. The functions $f_N(u) = \log M_N(u)/\log N$ are convex by Hölder's inequality. For $p\beta \leq u \leq (p+1)\beta$, convexity gives an upper bound by the chord at those endpoints and a lower bound by the line through $((p+1)\beta, f_N((p+1)\beta))$ with slope $[f_N((p+2)\beta) - f_N((p+1)\beta)]/\beta$. Both lines converge uniformly on the interval to $-d_\beta u$. Thus convergence is uniform on every bounded subinterval of $[0, \infty)$. The same proof applies with N replaced by $N - \ell$ for any fixed integer ℓ , since the chosen set has the eventual property for every integer shift of $\mathcal{D}_\beta$.

Now fix a compact interval $I \subset (-2, \infty)$. Choose a fixed integer $\ell \geq 0$ such that $t + \ell\beta \geq 0$ for all $t \in I$. Iterating (63.3) gives

$$\begin{aligned} \log M_N(t) &= \log M_{N-\ell}(t + \ell\beta) - \sum_{j=0}^{\ell-1} \log R_{N-j-1,\beta} \\ &\quad - \sum_{j=0}^{\ell-1} \log \hat{\rho}_{N-j}^{[t+j\beta]}(0). \end{aligned}$$

The final sum is uniformly $O_{\beta,I}(1)$ by Lemma 63.1. After division by $\log N$, the other terms converge uniformly in $t \in I$ to $-d_\beta(t + \ell\beta) + \ell\beta d_\beta$. This proves (63.4).

For fixed $t > -2$ and $0 < s < t + 2$, exponential Markov gives

$$\begin{aligned} \mathbb{P}_{N,\beta}^{[t]} \{H_N > (d_\beta + \varepsilon) \log N\} &\leq N^{-s(d_\beta + \varepsilon)} \frac{M_N(t-s)}{M_N(t)} = N^{-s\varepsilon + o(1)}, \\ \mathbb{P}_{N,\beta}^{[t]} \{H_N < (d_\beta - \varepsilon) \log N\} &\leq N^{s(d_\beta - \varepsilon)} \frac{M_N(t+s)}{M_N(t)} = N^{-s\varepsilon + o(1)}. \end{aligned}$$

The lower event is empty when its threshold is nonpositive. This proves the probability estimate. To justify all moments, for sufficiently large retained N the first inequality with an arbitrary threshold u gives

$$\mathbb{P}_{N,\beta}^{[t]} \{H_N / \log N > u\} \leq N^{-s(u-d_\beta)+1} \leq 2^{-s(u-d_\beta)+1}, \quad u \geq d_\beta + 2/s.$$

This common exponential upper tail proves uniform integrability of every finite power. The finitely many earlier $N \geq 2$ also have all moments, because $M_N(t-s) < \infty$ and $x^q \leq C_{q,s} e^{sx}$. Convergence in probability now proves the asserted absolute-moment convergence. $\square$

Remark 63.3 (A coefficient is identified, but the central prefactor is not). The conjectured intensity coefficient is $(4/\beta - 1)/\pi = 2d_\beta/\pi$. The theorem identifies d_β in the actual uncharged finite-particle gas and in all fixed charged laws with $t > -2$, without assuming hyperbolic invariance or uniqueness. It still does not identify the local limiting intensity. The exact central insertion formula is

$$\rho_{N+1,\beta}(0) = \frac{M_N(\beta)}{Q_{N+1,\beta}/Q_{N,\beta}}.$$

The logarithmic asymptotics of numerator and denominator cancel. Their finite prefactor ratio is not determined by (63.4). The theorem also retains an exceptional set of particle numbers of natural density zero.

63.1 The determinantal charge benchmark

At $\beta = 2$ the signed-charge depth statement has a full-sequence version and an exact finite prefactor.

Proposition 63.4 (All central charges at the determinantal point). *For $t > -2$ and $\beta = 2$,*

$$M_N(t) = \prod_{k=1}^N \frac{2k}{2k+t} = \frac{\Gamma(N+1)\Gamma(1+t/2)}{\Gamma(N+1+t/2)}.$$

Consequently $\log M_N(t)/\log N \rightarrow -t/2$ locally uniformly on $(-2, \infty)$. Under each fixed t -charged law, $H_N/\log N \rightarrow 1/2$ in every positive absolute moment along the full sequence. The deviation estimate of Theorem 63.2 holds with $d_2 = 1/2$ and any $0 < s < t + 2$.

Proof. Expand the two Vandermonde determinants and integrate all angles. Orthogonality eliminates the terms with different monomial indices, and the remaining radial norms are

$$\int_{\mathbb{D}} |z|^{2(k-1)+t} dA(z) = \frac{2\pi}{2k+t}.$$

Thus the charged partition function is $N!$ times the product of these norms. Division by its value at $t = 0$ proves the product formula; repeated use of $\Gamma(x+1) = x\Gamma(x)$ proves the gamma-function form. Uniformly for t in a fixed compact subset of $(-2, \infty)$, Taylor expansion for all sufficiently large k gives $\log(1 + t/(2k)) = t/(2k) + O(k^{-2})$. The finitely many earlier terms are uniformly bounded, so $\log M_N(t) = -(t/2) \log N + O(1)$ uniformly on that compact set. The exponential Markov inequalities and the uniform exponential tail argument in the preceding theorem now apply without an exceptional set of particle numbers and prove the remaining claims. $\square$

In particular, $M_N(2) = 1/(N+1)$ and $R_{N,2} = \pi/(N+1)$, so the exact insertion quotient is $\rho_{N+1,2}(0) = 1/\pi$. Away from $\beta = 2$ the logarithmic charge asymptotic alone does not identify this quotient's finite prefactor.

63.2 The same radial-depth coefficient at every hyperbolic center

For a deterministic $a \in \mathbb{D}$ put

$$H_N(a) = \sum_{w \in \Xi_N} g(a, w), \quad g(a, w) = -\log \left| \frac{a-w}{1-a\bar{w}} \right|.$$

Thus $H_N(0) = \sum_w -\log |w|$. Each $H_N(a)$ is finite almost surely, although as a function of a it has logarithmic singularities at the particle positions.

Lemma 63.5 (Tightness of anchored potential differences below 2). *For $0 < \beta < 2$ and each fixed $a \in \mathbb{D}$, the random variables $H_N(a) - H_N(0)$ form a tight family. The same is true jointly for any finite list of deterministic anchors.*

Proof. Take an arbitrary sequence of particle numbers. By tightness of the point processes, the almost-sure infinitude of every limiting process below 2, and Theorem 51.2, it has a further subsequence with a simultaneous augmentation by the nonempty limiting configuration Ξ and the corrected meromorphic field $\mathcal{A}$. The normal-family construction underlying that theorem and the meromorphic gluing proposition give locally uniform convergence of $\mathcal{A}_N$ away from the limiting poles, on each positive-count localization. These localizations exhaust an event of probability one because Ξ is nonempty. Equivalently one may retain the normalized holomorphic exterior factors in the countable disk exhaustion used in that construction.

Use a Skorohod representation of these jointly convergent Polish-space objects. Almost surely none of the finitely many deterministic anchors or the origin belongs to Ξ , because its intensity is absolutely continuous. On such a realization, choose for each anchor a piecewise smooth path from 0 to that anchor whose compact image avoids Ξ . Such a path exists since the configuration is locally finite. The path can be covered by finitely many compact subsets of the complement of the poles. The convergences of the configurations and fields imply that for all sufficiently large retained N it also avoids the finite- N poles, and $\mathcal{A}_N$ converges uniformly on that path. The identity $2\partial_z g(z, w) = -J_w(z)$ gives

$$H_N(a) - H_N(0) = -\operatorname{Re} \int_0^a \mathcal{A}_N(z) dz.$$

The real part is independent of the avoiding path: the residue of each pole is one, so the real part of every closed contour integral is zero. Uniform convergence on the chosen paths

therefore gives finite limits for all the anchored differences. This proves subsequential tightness along every original sequence and hence tightness of the original families. Random selection of the paths in this argument creates no measurability issue: the left-hand sides are measurable random variables and the almost-sure limits are their pointwise limits; the paths only verify finiteness and convergence on each realization. $\square$

Corollary 63.6 (A center-independent depth coefficient). *Fix $0 < \beta < 2$ and let $\mathcal{E}_\beta$ be a natural-density-one set on which the uncharged depth conclusion of Theorem 63.2 holds. Put $d_\beta = 2/\beta - 1/2$. For every fixed $a \in \mathbb{D}$ and every $p > 0$,*

$$\mathbb{E} \left| \frac{H_N(a)}{\log N} - d_\beta \right|^p \longrightarrow 0, \quad N \rightarrow \infty, \quad N \in \mathcal{E}_\beta. \quad (63.9)$$

The convergence holds jointly in probability for finitely many anchors. Moreover, for every compact $K \Subset \mathbb{D}$ and every $p > 0$,

$$\mathbb{E} \int_K \left| \frac{H_N(a)}{\log N} - d_\beta \right|^p dA(a) \longrightarrow 0 \quad (N \in \mathcal{E}_\beta). \quad (63.10)$$

This statement concerns deterministic anchors and integrated spatial norms; it asserts no supremum bound over a compact set containing random particle positions.

Proof. The tightness lemma shows $(H_N(a) - H_N(0))/\log N \rightarrow 0$ in probability. Combined with the uncharged depth law at zero, it proves the asserted pointwise and finite joint convergence in probability.

We verify uniform integrability and its uniformity over compact sets of anchors. Fix $K \Subset \mathbb{D}$ and choose $\sup_{a \in K} |a| < R < 1$. There is a finite constant C_K such that

$$g(a, w) \leq C_K(-\log |w|), \quad a \in K, \quad |w| \geq R. \quad (63.11)$$

Indeed, the ratio of the two positive functions is continuous on the closed outer annulus after extension to $|w| = 1$. Its boundary value is $(1 - |a|^2)/|a - w|^2$, uniformly bounded for $a \in K$. The extension follows either by differentiating radially at the unit circle or from

$$1 - \left| \frac{a - w}{1 - a\bar{w}} \right|^2 = \frac{(1 - |a|^2)(1 - |w|^2)}{|1 - a\bar{w}|^2}$$

and $-\log x \sim 1 - x$ as $x \uparrow 1$. On the inner disk,

$$g(a, w) \leq \log 2 + \log^+ \frac{1}{|a - w|}.$$

Consequently

$$H_N(a) \leq C_K H_N(0) + L_N(a), \quad L_N(a) = \sum_{w \in \Xi_N \cap B_R} \left(\log 2 + \log^+ \frac{1}{|a - w|} \right). \quad (63.12)$$

For every $q > 0$,

$$\sup_N \sup_{a \in K} \mathbb{E} L_N(a)^q < \infty. \quad (63.13)$$

To prove this, first take an integer $k \geq \max(1, q)$. Expand the k th power of the nonnegative sum according to which of its k indices coincide. Each resulting term is an integral against a factorial density $\rho_{j,N}$ on B_R^j , where $1 \leq j \leq k$, of a product of powers of $\log 2 + \log^+(1/|a - w|)$ of total degree k . The densities have uniform bounds on this compact polydisk. Each logarithmic power has an area integral bounded uniformly in $a \in K$, because all singularities

are translates of the integrable functions $(\log^+(1/|w|))^l$ on a bounded planar set. There are finitely many index partitions, which proves the claim for k . Hölder's inequality then proves it for q .

The uncharged depth theorem gives bounded moments of every order for $H_N(0)/\log N$ on $\mathcal{E}_\beta$, with $N \geq 2$. Equations (63.12) and (63.13) therefore imply, for every $q > 0$,

$$\sup_{\substack{N \in \mathcal{E}_\beta \\ N \geq 2}} \sup_{a \in K} \mathbb{E} \left(\frac{H_N(a)}{\log N} \right)^q < \infty.$$

Choosing $q > p$ proves uniform integrability to the p th power. Pointwise convergence in probability now proves (63.9). Finally its integrand's expectation is bounded uniformly in $a \in K$ and retained N . Dominated convergence in a , followed by Tonelli's theorem, proves (63.10). $\square$

Remark 63.7 (The order-one term remains undetermined). The preceding result fixes the leading coefficient $d_\beta \log N$ of the positive Green potential at every center. The intensity is encoded by the order-one spatial variation of its expectation, namely the exact mean defect of Proposition 64.8. Thus the center-independent leading coefficient, by itself, does not prove that the mean defect vanishes or determine the local intensity.

64 Two exact reductions of the intensity conjecture

The conjectural formula separates into a spatial-shape problem and a coefficient problem. The first reduction below uses only a one-point Palm statistic. The second determines the coefficient from symmetry of a meromorphic field, without assuming that this field is measurable from the particle configuration or constructing a centered Green potential. Neither symmetry hypothesis is proved for the original gas.

64.1 The exact Palm dipole controlling the radial profile

Write $\rho_N(a) = \rho_{1,N}(a)$ and $c_N = 2 + \beta(N-1)/2$. Let $\mathbb{E}_N^a$ denote expectation under the reduced Palm law of the N -particle gas at $a \in \mathbb{D}$, and put

$$d_N(a) = a + \mathbb{E}_N^a \sum_{w \in \Xi_N} w.$$

Thus the sum in this display contains the other $N-1$ particles. Write $\rho_N(a) = f_N(|a|^2)$.

Proposition 64.1 (A finite-particle Palm dipole identity). *For every $N \geq 1$ and $\beta > 0$, f_N is positive and continuously differentiable on $[0, 1)$. For $a \neq 0$,*

$$d_N(a) = \frac{a}{c_N} \left(2 - (1 - |a|^2) \frac{f'_N(|a|^2)}{f_N(|a|^2)} \right). \quad (64.1)$$

In particular, the quotient $d_N(a)/a$ is real and radial and has a continuous extension at zero. For $0 \leq r < 1$,

$$\log \frac{(1 - r^2)^2 \rho_N(r)}{\rho_N(0)} = -c_N \int_0^{r^2} \frac{d_N(\sqrt{s})}{\sqrt{s}(1-s)} ds, \quad (64.2)$$

with that extension understood at $s = 0$.

Proof. The finite- N insertion formula expresses $\rho_N(a)$ as a positive constant times the integral of $\prod_{j=1}^{N-1} |a - w_j|^\beta |\Delta_{N-1}(\mathbf{w})|^\beta$. First and second derivatives in a are locally integrable, because the worst single-factor singularity has modulus a constant times $|a - w_j|^{\beta-2}$, integrable in two dimensions for $\beta > 0$. Terms with two differentiated factors have exponents $\beta - 1$ in two separate variables and are likewise integrable. The other factors are bounded on the finite product of disks. Splitting small balls around a from their complements justifies differentiation and the continuity of the resulting derivatives. Thus ρ_N is C^2 . Rotation invariance gives the asserted regularity of f_N , including at zero by the second-order radial Taylor formula. Strict positivity follows from the insertion formula.

Take a real $h \in C_c^1(\mathbb{D})$ and apply the exact Möbius Ward identity to $F(\xi) = \sum_{a \in \xi} h(a)$. This functional is bounded for fixed N ; alternatively it can be truncated outside the possible range of its argument before applying that identity. Since $\mathbb{E}S_{1,N} = 0$, the Campbell formula gives, for every $t \in \mathbb{C}$,

$$\int_{\mathbb{D}} dh(a)[t - \bar{t}a^2]\rho_N(a) dA(a) = 2c_N \int_{\mathbb{D}} h(a)\rho_N(a) \operatorname{Re}(\bar{t}d_N(a)) dA(a).$$

The real divergence of $a \mapsto t - \bar{t}a^2$ equals $-4 \operatorname{Re}(\bar{t}a)$. Since $\rho_N(a) = f_N(|a|^2)$, integration by parts therefore makes the left side equal to

$$\int_{\mathbb{D}} h(a) \operatorname{Re}(\bar{t}a) \left[4f_N(|a|^2) - 2(1 - |a|^2)f'_N(|a|^2) \right] dA(a).$$

The Palm expectation in d_N is continuous in a by its bounded insertion numerator and positive denominator. The arbitrariness of h and t proves (64.1) everywhere. Rearranging that equality gives

$$\frac{d}{ds} \log[(1-s)^2 f_N(s)] = -\frac{c_N}{1-s} \frac{d_N(\sqrt{s})}{\sqrt{s}}.$$

Integration from zero to r^2 proves the second formula. $\square$

Corollary 64.2 (An integrated dipole criterion for the spatial shape). *Suppose $N_j \rightarrow \infty$, the intensities converge locally uniformly to ρ , and $\rho(0) > 0$. Then the following two assertions are equivalent:*

1. $\rho(r) = \rho(0)(1 - r^2)^{-2}$ for every $0 \leq r < 1$;
2. for every $0 \leq r < 1$,

$$c_{N_j} \int_0^{r^2} \frac{d_{N_j}(\sqrt{s})}{\sqrt{s}(1-s)} ds \longrightarrow 0.$$

A sufficient condition is, for every $r < 1$,

$$N_j \int_0^{r^2} \frac{|d_{N_j}(\sqrt{s})|}{\sqrt{s}(1-s)} ds \longrightarrow 0. \quad (64.3)$$

These criteria concern only the first correlation function and do not require Möbius invariance of all local observables.

Proof. The left side of (64.2) converges to $\log[(1 - r^2)^2 \rho(r)/\rho(0)]$ whenever $\rho(r) > 0$. Conversely, if the integral tends to zero, that identity gives $(1 - r^2)^2 \rho_{N_j}(r)/\rho_{N_j}(0) \rightarrow 1$, hence positivity of $\rho(r)$ and the asserted shape. Finally $c_{N_j}/N_j \rightarrow \beta/2$, so (64.3) suffices. $\square$

Remark 64.3 (The missing order of Palm screening). The harmonic Palm theorem proves $d_N(a) \rightarrow 0$ for each fixed a . The exact identity above explains why this leading screening is insufficient: the profile involves a dipole multiplied by c_N , which is of order N . In particular,

even an $O(N^{-1})$ dipole can produce a nonconstant limiting factor multiplying $(1 - |a|^2)^{-2}$. At $\beta = 2$ the criterion can be checked directly: putting $s = |a|^2$,

$$\pi(1 - s)^2 f_N(s) = 1 - (N + 1)s^N + Ns^{N+1},$$

and consequently

$$\frac{d_N(a)}{a} = \frac{N(1 - s)^2 s^{N-1}}{1 - (N + 1)s^N + Ns^{N+1}}, \quad c_N = N + 1.$$

The right side is exponentially small on every fixed compact subdisk as $N \rightarrow \infty$. No corresponding estimate at other temperatures is claimed.

Proposition 64.4 (An exact susceptibility deficit at the center). *Let*

$$V_N = \mathbb{E}_N |S_{1,N}|^2, \quad V_N^{!0} = \mathbb{E}_N^{!0} \left| \sum_w w \right|^2.$$

For every $N \geq 1$ and $\beta > 0$,

$$V_N = \frac{N}{c_N}, \quad c_N^2 (V_N^{!0} - V_N) = \frac{f'_N(0)}{f_N(0)} - 2. \quad (64.4)$$

Consequently $f'_N(0)/f_N(0) \rightarrow 2$ is equivalent to $V_N^{!0} - V_N = o(N^{-2})$. The equality $V_N^{!0} = V_N$ holds at $\beta = 2$ for every $N \geq 2$.

Proof. Applying the unweighted Möbius Ward identity to $S_{1,N}$ and using rotation invariance gives $N = c_N V_N$. For the other identity put $n = N - 1$ and, under the Palm law at zero, write $S = \sum_{j=1}^n w_j$ and $T = \sum_{j=1}^n w_j^{-1}$. For $n = 0$ both sums are zero. For $n > 0$ these variables and ST are integrable, because the Palm density contains $\prod_j |w_j|^\beta$, and a single inverse power is locally integrable against each of these weighted area factors. Differentiating the Palm insertion formula at $a = 0$ gives

$$\lim_{a \rightarrow 0} \frac{d_N(a)}{a} = 1 - \frac{\beta}{2} \mathbb{E}_N^{!0}(ST).$$

Indeed the derivative of the insertion factor with respect to a is $-(\beta/2)T$ there; rotation invariance makes the derivative of its normalizer vanish and removes the conjugate-linear term.

Under the disk flow $w \mapsto (w + t)/(1 + tw)$ with real t , the logarithmic derivative at zero of the Palm density, including its Jacobian, equals $-2c_N \operatorname{Re} S + \beta \operatorname{Re} T$. The differentiated statistic S has mean derivative n . The change-of-variables identity therefore yields

$$0 = n - c_N V_N^{!0} + \frac{\beta}{2} \mathbb{E}_N^{!0}(ST).$$

The derivative is integrable also at the inserted zero, by the integrability noted above; it may equivalently be justified first off a shrinking disk and then by dominated integral estimates. It follows that

$$\lim_{a \rightarrow 0} \frac{d_N(a)}{a} = N - c_N V_N^{!0}.$$

Combine this with (64.1) at zero and $V_N = N/c_N$ to obtain (64.4). The asymptotic equivalence uses $c_N/N \rightarrow \beta/2$. At $\beta = 2$, the explicit density $f_N(s) = \pi^{-1} \sum_{k=0}^{N-1} (k+1)s^k$ gives $f'_N(0)/f_N(0) = 2$ when $N \geq 2$, proving the last assertion. $\square$

Remark 64.5 (A finer cancellation than the harmonic central limit theorem). The common limiting variance $2/\beta$ does not determine the susceptibility difference in (64.4). Even identifying the N^{-1} correction to both variances would leave the N^{-2} scale unresolved. The proposition is an exact statement about finite- N spatial derivatives. Turning it into a necessary criterion for a proposed limiting profile additionally requires convergence of those derivatives; local uniform intensity convergence alone does not justify that step.

64.2 Affine meromorphic-field symmetry fixes the coefficient

Theorem 64.6 (Intensity fixed by an affine field symmetry). *Let $\kappa \in \mathbb{R}$. Let Ξ be a simple locally finite random configuration in $\mathbb{D}$, with no particle at any fixed deterministic point. Let $\mathcal{A}$ be a random meromorphic function whose poles are exactly the points of Ξ , each pole being simple with residue one. Assume that, for every disk automorphism φ , the pair*

$$\left(\varphi^{-1}\Xi, \varphi' \mathcal{A} \circ \varphi + \kappa \frac{\varphi''}{\varphi'} \right) \quad \text{has the same law as } (\Xi, \mathcal{A}). \quad (64.5)$$

Assume also $\mathbb{E}|\mathcal{A}(0)| < \infty$. Then

$$\mathbb{E}\mathcal{A}(z) = \frac{2\kappa\bar{z}}{1-|z|^2}, \quad \mathbb{E}\Xi(dz) = \frac{2\kappa}{\pi(1-|z|^2)^2} dA(z). \quad (64.6)$$

In particular $\kappa \geq 0$. If $\kappa = 0$, the configuration is empty almost surely; if $\kappa > 0$, its total count is infinite almost surely. For the affine action of the Gaussian-eliminated disk-gas specification, $\kappa = 2/\beta - 1/2$, so the asserted intensity is exactly

$$\rho(z) = \frac{4/\beta - 1}{\pi(1-|z|^2)^2}.$$

Proof. Take $\varphi(z) = e^{i\theta}z$ in the symmetry assumption. The random variables $\mathcal{A}(0)$ and $e^{i\theta}\mathcal{A}(0)$ have the same law. Integrability gives $\mathbb{E}\mathcal{A}(0) = 0$. For a fixed $a \in \mathbb{D}$ use $\varphi_a(z) = (z+a)/(1+\bar{a}z)$. Since

$$\varphi'_a(0) = 1 - |a|^2, \quad \frac{\varphi''_a(0)}{\varphi'_a(0)} = -2\bar{a},$$

the symmetry assumption implies the distributional identity

$$\mathcal{A}(0) \stackrel{\text{law}}{=} (1 - |a|^2)\mathcal{A}(a) - 2\kappa\bar{a}.$$

It proves integrability at every deterministic a , the first identity in (64.6), and the locally uniform bound

$$\mathbb{E}|\mathcal{A}(a)| \leq \frac{\mathbb{E}|\mathcal{A}(0)| + 2|\kappa||a|}{1 - |a|^2}.$$

Tonelli therefore gives $\mathbb{E} \int_K |\mathcal{A}(a)| dA(a) < \infty$ for every compact $K \Subset \mathbb{D}$.

Every sample meromorphic function is locally integrable; subtracting its finitely many residue-one principal parts on a neighborhood of a test function's support leaves a holomorphic function there. The distributional identity $\partial_{\bar{z}}(1/(z-a)) = \pi\delta_a$ thus gives, sample by sample,

$$\partial_{\bar{z}}\mathcal{A} = \pi\Xi.$$

The local expected-integrability bound permits expectation to pass through its distributional pairing. It also ensures local finiteness of $\mathbb{E}\Xi$: pair the displayed identity with a nonnegative smooth function equal to one on a given compact set. Consequently

$$\pi\mathbb{E}\Xi = \partial_{\bar{z}}\mathbb{E}\mathcal{A} = \frac{2\kappa}{(1-|z|^2)^2} dA(z),$$

as asserted. Positivity of the left side forces $\kappa \geq 0$. Zero intensity implies the almost surely empty configuration by a countable compact exhaustion. For $\kappa > 0$, integrating over B_r yields $\mathbb{E}\Xi(B_r) = 2\kappa r^2/(1 - r^2)$, which diverges as $r \uparrow 1$. If some finite total count m had positive probability, condition on $\{\Xi(\mathbb{D}) = m\}$. This event is invariant under every action in (64.5), and integrability of $\mathcal{A}(0)$ persists under this conditioning. Applying the already proved intensity identity to that conditional law would give infinite expected total count, contradicting the deterministic count m . All finite counts therefore have probability zero. $\square$

Remark 64.7 (Scope of the field criterion). The proof needs only the corresponding symmetry identities for first moments and the local expected-integrability that they supply. Full joint stationarity is a convenient sufficient hypothesis. Neither a centered Green potential, a count-resolved DLR equation, ergodicity, nor measurability of $\mathcal{A}$ from Ξ is required. Conversely, covariance of the canonical resampling kernels does not prove (64.5) for a chosen Gibbs law. Nor does the proved leading independence of the Gaussian harmonic modes imply this affine symmetry. These remain substantive hypotheses for an actual disk-gas limit.

64.3 An exact normalized mean-potential defect

The coefficient in the proposed intensity can be tested without assuming that the corrected field is measurable from the point configuration, or that its limit has pointwise integrable evaluations. The following identities are finite-particle identities followed by limits of their expectations. They do not interchange an expectation with convergence of an unbounded random field.

Put $q_\beta = 4/\beta - 1$ and $\kappa_\beta = q_\beta/2$. For a finite configuration $\xi \subset \mathbb{D}$ having no particle at 0 or z , define

$$V_\xi(z) = \sum_{w \in \xi} \{g(z, w) - g(0, w)\},$$

$$U_\xi(z) = V_\xi(z) - \kappa_\beta \log(1 - |z|^2), \quad g(z, w) = -\log \left| \frac{z - w}{1 - z\bar{w}} \right|.$$

Write $m_N(r) = \mathbb{E}\Xi_N(B_r)$ and $u_N(z) = \mathbb{E}U_{\Xi_N}(z)$, where $B_r = \{|z| < r\}$.

Proposition 64.8 (The mean defect is determined exactly by the intensity). *For every N , $\beta > 0$, and $z \in \mathbb{D}$, the expectation defining $u_N(z)$ is finite and*

$$u_N(z) = -\int_0^{|z|} \frac{m_N(s)}{s} ds - \kappa_\beta \log(1 - |z|^2). \quad (64.7)$$

Let Ξ_{N_j} converge to a subsequential point-process limit with intensity $\rho(z)dA(z)$, and set $m(r) = 2\pi \int_0^r s\rho(s)ds$. Then u_{N_j} converges locally in $C^2(\mathbb{D})$ to the deterministic radial function

$$u(z) = -\int_0^{|z|} \frac{m(s)}{s} ds - \kappa_\beta \log(1 - |z|^2). \quad (64.8)$$

It satisfies $u(0) = 0$ and

$$\Delta u(z) = 2\pi \left\{ \frac{q_\beta}{\pi(1 - |z|^2)^2} - \rho(z) \right\}. \quad (64.9)$$

Consequently, for this subsequential limit the proposed formula $\rho(z) = q_\beta/[\pi(1 - |z|^2)^2]$ holds everywhere if and only if u vanishes identically.

Proof. At fixed N the density is bounded on $\mathbb{D}^N$: the Vandermonde is bounded there and its partition function is strictly positive. The logarithmic singularities at $w = 0, z$ are integrable with respect to area, while $\log |1 - z\bar{w}|$ is bounded for fixed $z \in \mathbb{D}$. Thus all expectations and the angular integrations below are absolutely convergent at fixed N .

Rotation invariance makes the intensity radial. Jensen's elementary angular formula gives, for $r, s > 0$ with $r, s < 1$,

$$\frac{1}{2\pi} \int_0^{2\pi} \log |r - se^{i\theta}| d\theta = \log \max(r, s), \quad \frac{1}{2\pi} \int_0^{2\pi} \log |1 - rse^{i\theta}| d\theta = 0.$$

For completeness, away from $r = s$ both formulas follow by factoring out the larger modulus and integrating the convergent power series for $\operatorname{Re} \log(1 - ae^{i\theta})$, $|a| < 1$. At $r = s$, the identity follows by direct integration of $\log |1 - e^{i\theta}| = \log(2|\sin(\theta/2)|)$, whose integral is zero; alternatively, dominated convergence after separating a small interval about $\theta = 0$ gives the same conclusion. It follows that the angular mean of $g(r, se^{i\theta}) - g(0, se^{i\theta})$ is $-\log(r/s)$ for $s < r$ and zero for $s \geq r$. Therefore

$$\mathbb{E}V_{\Xi_N}(z) = -2\pi \int_0^r s\rho_N(s) \log(r/s) ds = - \int_0^r \frac{m_N(s)}{s} ds, \quad r = |z|,$$

where Tonelli's theorem applied to the nonnegative integrand establishes the last equality. This proves (64.7).

The proved local uniform convergence of the intensities implies local uniform convergence of $m_N(r)/r^2$ including its continuous value at zero. In particular the integral in (64.7) converges uniformly on compact radial intervals. To check two derivatives without assuming derivatives of ρ_N converge, set $h_N(r) = - \int_0^r m_N(s) ds/s$. For $r > 0$,

$$h'_N(r) = -\frac{m_N(r)}{r}, \quad h''_N(r) = -2\pi\rho_N(r) + \frac{m_N(r)}{r^2}.$$

The corresponding Hessian of $h_N(|z|)$ is a combination of $h''_N(r)$ and $h'_N(r)/r$, both controlled by the locally uniform convergence of ρ_N . At zero both tend to $-\pi\rho_N(0)$; continuity follows from continuity of ρ_N . These formulas prove local C^2 convergence. They also give $\Delta h = -2\pi\rho$. Since $\Delta \log(1 - |z|^2) = -4/(1 - |z|^2)^2$, this proves (64.9).

If the conjectured intensity holds, the right-hand side of (64.9) vanishes. The function u is radial, harmonic on the entire disk, and zero at the origin, so it is zero everywhere. This also follows directly by integrating $m(r) = q_\beta r^2/(1 - r^2)$. The converse follows immediately from (64.9). $\square$

Remark 64.9 (What this does and does not prove). The quantity u is the limit of the deterministic expectations $\mathbb{E}U_{\Xi_N}$. Identifying it with the expectation of a limiting random potential requires a separate uniform-integrability argument. Moreover, tightness or existence of that random potential would not itself show that its mean is zero. Thus the exact background coefficient in the augmented canonical kernels is compatible with the proposed intensity precisely when the mean defect above vanishes; kernel covariance alone does not establish that vanishing.

One can express the same missing information in terms of the corrected meromorphic field $\mathcal{A}_N(z) = \sum_{w \in \Xi_N} J_w(z)$. At finite N its first moment exists at every fixed interior point and angular integration gives

$$\mathbb{E}\mathcal{A}_N(z) = \frac{m_N(|z|)}{z} \quad (z \neq 0), \quad \mathbb{E}\mathcal{A}_N(0) = 0. \quad (64.10)$$

Indeed the angular mean of $(z - w)^{-1}$ is z^{-1} when $|w| < |z|$ and zero when $|w| > |z|$, while the angular mean of $\bar{w}/(1 - z\bar{w})$ is zero. Its singularity is locally integrable, so this calculation

uses ordinary Fubini at each finite N . For $a \in \mathbb{D}$ let $\varphi_a(z) = (a + z)/(1 + \bar{a}z)$. The affine transformation prescribed by the Gaussian-free kernels has mean at zero

$$\begin{aligned} & \mathbb{E} \left[\varphi'_a(0) \mathcal{A}_N(\varphi_a(0)) + \kappa_\beta \frac{\varphi''_a(0)}{\varphi'_a(0)} \right] \\ &= \bar{a} \left\{ \frac{(1 - |a|^2) m_N(|a|)}{|a|^2} - q_\beta \right\}, \quad a \neq 0. \end{aligned} \quad (64.11)$$

Its limit exists along the point-process subsequence by local count convergence. Vanishing of that limit for every a is another exact equivalent of the proposed intensity. This assertion concerns limits of finite-particle means and requires no unproved passage of an expectation through a limiting meromorphic mark.

64.4 Exact changes of geometry and a finite-size prediction at the critical value

The two-dimensional counterion literature contains the same finite disk ensemble, but several limits and changes of geometry must be distinguished. In particular, a finite-size extrapolation at the critical value should be confronted with the qualitative emptiness theorems, rather than regarded as an established limiting formula.

Proposition 64.10 (Inversion and excess charge). *Under the map $w_i = 1/z_i$, the unweighted disk gas has density proportional to*

$$|\Delta_N(w)|^\beta \prod_{i=1}^N |w_i|^{-\beta(N-1)-4} \mathbf{1}_{\{|w_i|>1\}} dA(w_i). \quad (64.12)$$

Writing the external-disk counterion density as $|\Delta_N(w)|^{2\Xi} \prod_i |w_i|^{-2\xi}$, its parameters are

$$\Xi = \frac{\beta}{2}, \quad \xi = \frac{\beta}{2}(N-1) + 2.$$

Consequently the central charge in units of a counterion charge is

$$Q = \frac{\xi}{\Xi} = N - 1 + \frac{4}{\beta}, \quad Q - N = \frac{4}{\beta} - 1.$$

If $\rho_{N,\beta}^{\text{ext}}$ denotes the inverted intensity, then

$$\rho_{N,\beta}^{\text{ext}}(w) = |w|^{-4} \rho_{1,N}(1/w), \quad \lim_{|w| \rightarrow \infty} |w|^4 \rho_{N,\beta}^{\text{ext}}(w) = \rho_{1,N}(0).$$

Proof. The origin has probability zero. The identities $|z_i - z_j| = |w_i - w_j|/(|w_i||w_j|)$ and $dA(z_i) = |w_i|^{-4} dA(w_i)$ give (64.12). Matching the exponents proves the parameter statements. Changing variables in the one-point intensity gives the last identity, and its limit follows from continuity of $\rho_{1,N}$ at zero. $\square$

The external-disk notation above agrees with the Boltzmann factor in equation (4.1) of [Umaña Mallarino–Téllez \(2015\)](#). Thus the conjectured coefficient $\pi\rho_\beta(0) = 4/\beta - 1$ has an exact interpretation as the excess charge in this finite-model reformulation. This matching does not prove equality between the excess charge and the limiting density amplitude. The condensation thresholds and limiting regimes treated there do not supply that equality along $\xi = \Xi(N-1) + 2$ with $N \rightarrow \infty$ and fixed Ξ .

Proposition 64.11 (The single-wall cylinder at coupling four). *Consider the cylinder $\{u = x + iy : x > 0, y \in \mathbb{R}/W\mathbb{Z}\}$, with N counterions, a boundary line charge density $\sigma = N/W$, and coupling $\Gamma > 0$. The density*

$$\exp\left(-\Gamma\pi\sigma\sum_i x_i\right)\prod_{i<j}\left|2\sinh\frac{\pi(u_i-u_j)}{W}\right|^\Gamma\prod_i dx_i dy_i$$

is transformed by $\zeta_i = e^{-2\pi u_i/W}$ into a density proportional to

$$|\Delta_N(\zeta)|^\Gamma\prod_i|\zeta_i|^{\Gamma/2-2}\mathbf{1}_{\{0<|\zeta_i|<1\}}dA(\zeta_i).$$

In particular, at $\Gamma = 4$ this is exactly the original unweighted disk gas at $\beta = 4$.

Proof. Put $r_i = |\zeta_i|$. The wall weight is $r_i^{\Gamma N/2}$. Each pair contributes

$$\left|2\sinh\frac{\pi(u_i-u_j)}{W}\right|^\Gamma = |\zeta_i - \zeta_j|^\Gamma (r_i r_j)^{-\Gamma/2}.$$

Finally $dx_i dy_i = (W/(2\pi))^2 r_i^{-2} dA(\zeta_i)$. The exponent for each r_i is therefore $\Gamma N/2 - \Gamma(N-1)/2 - 2 = \Gamma/2 - 2$. $\square$

In Šamaj–Trizac (2011), Section IV.B, equations (4.21)–(4.23), the dimensionless occupation coefficient at $\Gamma = 4$ satisfies

$$f_{0,N}^{(2)} = 2\pi\rho_{1,N}(0).$$

Indeed their moments are $w_j = 1/(j+1)$ and their cylinder density, after the change of area in the preceding proof, gives $\rho_{1,N}(r) = \pi^{-1}\sum_j\langle\Xi_j\Psi_j\rangle r^{2j}$; their definition is $f_{0,N}^{(2)} = 2w_0\langle\Xi_0\Psi_0\rangle$. Here the symbols Ξ_j, Ψ_j are the Grassmann composites of that paper, not point processes.

The paper extrapolates its exact finite- N data with $N \leq 10$ to $f_{0,N}^{(2)} \rightarrow 0.195$; see Figure 5 and its accompanying discussion. This would predict a positive limiting disk intensity $0.195/(2\pi)$ at the origin. It is a numerical extrapolation, not a proof of that limit. Theorems 36.1 and 37.1, together with locally uniform intensity convergence, instead imply

$$f_{0,N}^{(2)} \rightarrow 0.$$

Neither critical proof supplies a convergence rate, so this qualitative conclusion does not by itself quantify the finite-size error in the published fit.

Remark 64.12 (What the Poisson–Boltzmann comparison establishes). The disk profile in [1, Section 4.2] is $4a^2/[\pi\beta(a^2 - r^2)^2]$, with $a^2 = R^2 + 4R^2/(\beta N)$, within the Poisson–Boltzmann approximation. At $R = 1$ its formal $N \rightarrow \infty$ limit has coefficient $4/(\pi\beta)$. This supports the leading weak-coupling term but does not determine the proposed constant correction $-1/\pi$. Moreover, the planar limit with fixed physical distance from a wall and the limit at fixed relative disk coordinate are distinct limiting operations. Neither interchange is assumed here.

65 Finite-particle numerical checks of the proposed intensity

This section reports finite-particle computations, not additional limit theorems. The proposed intensity would imply

$$\lim_{N \rightarrow \infty} \mathbb{E}M_N(r) = \left(\frac{4}{\beta} - 1\right) \frac{r^2}{1 - r^2}. \quad (65.1)$$

The goal of the computation was to test this prediction across several particle numbers and temperatures, while checking the sampler against the exact finite-particle answer at $\beta = 2$.

65.1 The target law and the transition kernel

The target was exactly the unweighted disk law $Z_{N,\beta}^{-1} |\Delta_N(z)|^\beta \prod_i dA(z_i)$. Write $z_i = e^{-t_i} e^{i\theta_i}$, with $t_i > 0$ and $\theta_i \in \mathbb{R}/(2\pi\mathbb{Z})$. In these coordinates its density, up to a constant, is

$$\exp\left(\beta \sum_{i < j} \log |z_i - z_j| - 2 \sum_i t_i\right).$$

A randomly selected particle was updated by one of three proposals: with probability 0.4, change its angle by an independent normal variable of standard deviation $2\pi/N$; with probability 0.4, multiply its depth t_i by $e^{1.4G}$, where G is standard normal; and with probability 0.2, redraw its angle uniformly and its depth from

$$q_N(t) = 0.85Ne^{-Nt} + 0.3e^{-2t}, \quad t > 0.$$

The logarithm of the Metropolis acceptance ratio contains the change in $\beta \sum_{j \neq i} \log |z_i - z_j|$. Its additional term is respectively

$$0, \quad \log(t'_i/t_i) - 2(t'_i - t_i), \quad -2(t'_i - t_i) + \log q_N(t_i) - \log q_N(t'_i).$$

These terms include the polar-area Jacobian and the proposal-density ratio. For the ideal kernels in exact arithmetic, detailed balance follows directly: if p denotes the target density and q a proposal density, the off-diagonal accepted probability flux is $\min\{p(x)q(x,y), p(y)q(y,x)\}$, which is symmetric in x, y . Thus this computation samples the original finite-particle model, subject to ordinary floating-point and finite-run errors; it introduces no extra radial potential and no spatial scaling.

65.2 Run length, diagnostics, and uncertainty

Four chains were run independently for each reported pair (N, β) , using seeds 93017, 46023, 75263, and 18349. A sweep consisted of N random particle-update attempts. For $N \leq 128$, each chain discarded 10000 sweeps and then recorded every tenth sweep during 100000 further sweeps. For $N = 512$, the corresponding numbers were 15000 discarded and 60000 retained sweeps, again recording every tenth sweep. Initial depths were independent exponential variables of rates $2(i+1)$, $i = 0, \dots, N-1$, and initial angles were independent uniform points in the N consecutive equal angular cells. This initialization was not assumed to have the target distribution.

The errors displayed below are one estimated Monte Carlo standard error. They were computed from the sample variance of nonoverlapping batch means, using batches of 500 recorded observations, or 5000 sweeps. There were 80 such batches in total when $N \leq 128$ and 48 when $N = 512$. For the counts at radii 0.25, 0.5, 0.7, 0.8, 0.9, classical split-chain $\hat{R}$ was below 1.001 in every reported run. Estimated integrated autocorrelation times, measured in recorded observations and using the initial monotone positive sequence of adjacent autocorrelation sums, were between 1 and 2.6. These are empirical diagnostics. No finite-time mixing bound or rigorous coverage statement for the displayed error bars is asserted. In particular, the error bars do not include finite- N bias relative to an infinite-particle limit.

β	N	Estimated $\mathbb{E}M_N(0.5)$	Estimated $\mathbb{E}M_N(0.8)$
1	32	0.91173 ± 0.00452	4.44265 ± 0.00750
1	128	0.97515 ± 0.00419	5.07308 ± 0.00683
1	512	0.99667 ± 0.00480	5.26092 ± 0.01133
1	∞ (conj.)	1.00000	5.33333
2	128	0.33278 ± 0.00268	1.76978 ± 0.00611
2	∞ (exact)	0.33333	1.77778
3	32	0.12258 ± 0.00200	0.70400 ± 0.00475
3	128	0.11253 ± 0.00197	0.62623 ± 0.00410
3	512	0.11096 ± 0.00280	0.61133 ± 0.00545
3	∞ (conj.)	0.11111	0.59259
3.5	32	0.06630 ± 0.00137	0.41600 ± 0.00346
3.5	128	0.05683 ± 0.00144	0.32870 ± 0.00340
3.5	512	0.05671 ± 0.00156	0.30562 ± 0.00515
3.5	∞ (conj.)	0.04762	0.25397

The rows marked “conj.” are evaluations of (65.1); they are not extrapolations from the data.

65.3 Exact calibration and what the experiment does not establish

At $\beta = 2$, the finite-particle kernel is

$$K_N(z, w) = \frac{1}{\pi} \sum_{k=1}^N k (z\bar{w})^{k-1}.$$

Integrating its diagonal over B_r gives the exact finite- N formula

$$\mathbb{E}M_N(r) = \sum_{k=1}^N r^{2k} = \frac{r^2(1 - r^{2N})}{1 - r^2}.$$

Similarly, radial integration of $-\log|z|$ gives

$$\mathbb{E} \sum_{i=1}^N -\log|z_i| = \frac{1}{2} \sum_{k=1}^N \frac{1}{k}.$$

At $N = 128$ the latter exact value is 2.7165735463; the simulation gave 2.717462 ± 0.003440 . The exact count means at radii 0.5 and 0.8 differ negligibly, at the precision displayed, from $1/3$ and $16/9$; both are compatible with the Monte Carlo estimates in the table.

The other temperatures exhibit visible dependence on N . For example, at $\beta = 3.5$ and $N = 512$, the estimated count at radius 0.5 exceeds the conjectured limiting value by about 5.8 estimated Monte Carlo standard errors. This is a discrepancy between a finite- N mean and a proposed limit, so it is not a counterexample to the conjecture. Conversely, the proximity of some of the other entries to their conjectured limits is not evidence sufficient to identify an exact coefficient. The experiment neither supplies a justified bias expansion nor rules out nearby alternative limiting intensities. In particular, fitting a universal $1/N$ correction would be an additional unproved assumption.

The reproducibility bundle includes the C++ sampler, the Python run and analysis scripts, the summary statistics, and the recorded observable traces.

66 Euclidean coordinates for the intensity and its screening defect

The following change of coordinates turns the intensity conjecture into an assertion of constant density relative to ordinary Euclidean area. The interaction and the elliptic operator must

be transformed at the same time. We make both transformations explicit before testing the proposed energy argument.

Proposition 66.1 (Area coordinates and the transformed interaction). *The maps*

$$T(z) = \frac{z}{\sqrt{1-|z|^2}}, \quad F(x) = \frac{x}{\sqrt{1+|x|^2}}$$

are mutually inverse smooth diffeomorphisms between $\mathbb{D}$ and $\mathbb{R}^2$. If $d\lambda(z) = dA(z)/(1-|z|^2)^2$, then $F^*d\lambda = dx$. The image of the finite disk gas has density $\widehat{Z}_{N,\beta}^{-1}e^{-\beta\widehat{H}_{N,\beta}}$ relative to $dx_1 \cdots dx_N$, where

$$\widehat{H}_{N,\beta}(x_1, \dots, x_N) = -\sum_{i < j} \log |F(x_i) - F(x_j)| + \frac{2}{\beta} \sum_i \log(1 + |x_i|^2). \quad (66.1)$$

For any point process with intensity $\rho(z)dA(z)$, its image has intensity

$$\tilde{\rho}(x) = \frac{\rho(F(x))}{(1 + |x|^2)^2}. \quad (66.2)$$

Consequently the proposed disk intensity is equivalent to $\tilde{\rho} \equiv c_\beta = (4/\beta - 1)/\pi$.

Proof. The radial map $R \mapsto R/\sqrt{1+R^2}$ is strictly increasing from $[0, \infty)$ onto $[0, 1)$ and has the stated inverse. At $x \neq 0$, the derivative DF has radial eigenvalue $(1 + |x|^2)^{-3/2}$ and tangential eigenvalue $(1 + |x|^2)^{-1/2}$. Both extend smoothly and positively at zero. Thus $\det DF = (1 + |x|^2)^{-2}$, whereas $1 - |F(x)|^2 = (1 + |x|^2)^{-1}$. These identities prove the area assertion, the transformed Hamiltonian by changing variables in each particle coordinate, and the intensity formula by the Campbell identity. The last conclusion follows by substitution. $\square$

Proposition 66.2 (The transformed Green operator and mean defect). *Define*

$$A(x) = \frac{I + (2 + |x|^2)xx^\top}{1 + |x|^2}, \quad \mathcal{L} = \operatorname{div}(A\nabla).$$

The matrix A has radial and tangential eigenvalues $1 + |x|^2$ and $(1 + |x|^2)^{-1}$, respectively, and determinant one. For smooth functions v, w of compact support in $\mathbb{D}$, writing $V = v \circ F$ and $W = w \circ F$, one has

$$\int_{\mathbb{D}} \nabla v \cdot \nabla w dA = \int_{\mathbb{R}^2} \nabla V \cdot A \nabla W dx. \quad (66.3)$$

The transformed disk Green function

$$\Gamma(x, y) = -\log \left| \frac{F(x) - F(y)}{1 - F(x)\overline{F(y)}} \right|$$

satisfies $-\mathcal{L}_x \Gamma(x, y) = 2\pi\delta_y$ in distributions and tends to zero as $|x| \rightarrow \infty$ for fixed y . For the normalized mean defect u of Proposition 64.8, $U = u \circ F$ satisfies exactly

$$\mathcal{L}U = 2\pi(c_\beta - \tilde{\rho}), \quad U(0) = 0. \quad (66.4)$$

The same identities hold for every finite-particle mean defect.

Proof. The chain rule and change of area show that the matrix in the Dirichlet form is $(\det DF)DF^{-1}DF^{-\top}$. The derivative eigenvalues computed above give the stated matrix and its eigenvalues. This proves (66.3). Apply this identity locally to the weak disk Green equation $-\Delta_z g(z, w) = 2\pi\delta_w$. The logarithmic singularity and its first derivative are locally

integrable, so the change of variables is valid after excising a small circle and then letting its radius decrease to zero. Equivalently it follows directly from the weak equation tested against a smooth compactly supported function composed with T . This proves the asserted point source, including its coefficient. As $|x| \rightarrow \infty$, $|F(x)| \rightarrow 1$ and $g(F(x), F(y)) \rightarrow 0$ uniformly in the angular coordinate, proving the boundary assertion.

In general, for smooth u the same weak change of variables yields $\mathcal{L}(u \circ F) = (\det DF)(\Delta u) \circ F$. Substitute the proved equation $\Delta u = 2\pi(c_\beta(1 - |z|^2)^{-2} - \rho(z))$ and the Jacobian identity. This gives (66.4); the normalization is unchanged because $F(0) = 0$. $\square$

The coefficients are uniformly elliptic on every bounded set, but their eigenvalue ratio grows like $(1 + |x|^2)^2$. Thus ordinary whole-plane Coulomb screening estimates do not transfer without checking the operator and the boundary terms. The next results identify these boundary terms exactly.

66.1 Radial screening energy and the boundary at infinity

The area-flattening coordinates separate an exact local identity from the boundary assumptions needed for a global energy argument. Write

$$F(x) = \frac{x}{\sqrt{1 + |x|^2}}, \quad q = \frac{4}{\beta} - 1, \quad c = \frac{q}{\pi}, \quad \tilde{\rho}(x) = \frac{\rho(F(x))}{(1 + |x|^2)^2}.$$

Let $U(x) = u(F(x))$, with u as in Proposition 64.8, and set

$$A(x) = \frac{I + (2 + |x|^2)xx^\top}{1 + |x|^2}, \quad \mathcal{L} = \operatorname{div}(A\nabla), \quad M(R) = 2\pi \int_0^R s\tilde{\rho}(s) ds.$$

Here and below $U(R)$ denotes the value of a radial function on the circle of Euclidean radius R . The radial eigenvalue of A is $1 + R^2$, so $\mathcal{L}U = R^{-1}(R(1 + R^2)U')'$.

Proposition 66.3 (Exact local energy identity). *For an actual subsequential limit,*

$$D(R) := qR^2 - M(R) = R(1 + R^2)U'(R), \quad (66.5)$$

$$\mathcal{E}(R) := \int_{B_R} \nabla U \cdot A \nabla U dx = 2\pi \int_0^R \frac{D(s)^2}{s(1 + s^2)} ds. \quad (66.6)$$

For every real constant a and every finite $R > 0$,

$$\mathcal{E}(R) = 2\pi\{U(R) - a\}D(R) + 2\pi \int_{B_R} (U - a)(\tilde{\rho} - c) dx. \quad (66.7)$$

In particular, the proposed intensity is equivalent to $\mathcal{E}(R) = 0$ for every finite R .

Proof. The mean-defect equation and the change of variables give $\mathcal{L}U = 2\pi(c - \tilde{\rho})$. Integrating this equation over B_R and using the divergence theorem yields the first identity; the origin contributes no flux since U is C^2 and radial. Polar integration of the energy gives the second identity. Multiply the equation by $U - a$ and integrate by parts on B_R . The boundary integral is $2\pi R(1 + R^2)(U(R) - a)U'(R)$, which is the first term in (66.7). All integrals are finite on a bounded disk, so no assumption at infinity has entered. Finally, vanishing of all the nonnegative energies forces $U' = 0$; the normalization $U(0) = 0$ then forces $U = 0$. The mean-defect proposition identifies this with the proposed intensity. $\square$

A radial finite-energy function has a trace at infinity. Subtracting that trace removes the need to postulate a separate boundary-flux condition. It also changes the sign estimate one would need to prove.

Proposition 66.4 (A finite-energy criterion with the correct constant). *Suppose that U is radial and C^2 on $\mathbb{R}^2$, that $\mathcal{L}U = 2\pi(c - \tilde{\rho})$ with continuous $\tilde{\rho}$, and that*

$$\mathcal{E}(\infty) := \int_{\mathbb{R}^2} \nabla U \cdot A \nabla U \, dx < \infty.$$

Then a finite limit $\ell = \lim_{R \rightarrow \infty} U(R)$ exists. If $\mathcal{E}_{\text{tail}}(R) = \mathcal{E}(\infty) - \mathcal{E}(R)$, then

$$|U(R) - \ell|^2 \leq \frac{\mathcal{E}_{\text{tail}}(R)}{4\pi} \log(1 + R^{-2}), \quad R > 0. \quad (66.8)$$

Fix a smooth function $\chi : [0, \infty) \rightarrow [0, 1]$ that equals one on $[0, 1]$ and zero on $[2, \infty)$, and put $\chi_R(x) = \chi(|x|/R)$. Then

$$\mathcal{E}(\infty) = 2\pi \lim_{R \rightarrow \infty} \int_{\mathbb{R}^2} \chi_R(U - \ell)(\tilde{\rho} - c) \, dx. \quad (66.9)$$

The limit in this formula exists without an assumption of absolute integrability of its integrand over the entire plane. Consequently, the additional estimate

$$\limsup_{R \rightarrow \infty} \int \chi_R(U - \ell)(\tilde{\rho} - c) \, dx \leq 0 \quad (66.10)$$

forces $\tilde{\rho} = c$ everywhere. If $U(0) = 0$, it also forces $U = 0$ and $\ell = 0$.

Proof. For $S > R > 0$, Cauchy–Schwarz gives

$$|U(S) - U(R)|^2 \leq \left(\int_R^S s(1 + s^2) U'(s)^2 \, ds \right) \left(\int_R^S \frac{ds}{s(1 + s^2)} \right).$$

The first factor is bounded by $\mathcal{E}_{\text{tail}}(R)/(2\pi)$, and the second is bounded by $\frac{1}{2} \log(1 + R^{-2})$. Thus $U(R)$ is Cauchy as R tends to infinity. Taking $S \rightarrow \infty$ proves (66.8).

Set $v = U - \ell$. Testing the equation against the compactly supported function $\chi_R v$ gives

$$\int \chi_R \nabla U \cdot A \nabla U = 2\pi \int \chi_R v(\tilde{\rho} - c) - \int v \nabla \chi_R \cdot A \nabla U.$$

The final integral is supported on $R \leq |x| \leq 2R$. For $R \geq 1$, the trace bound, $\log(1 + s^{-2}) \leq s^{-2}$, and $|\chi'_R(s)| \leq \|\chi'\|_\infty/R$ show that

$$\begin{aligned} \int v^2 \nabla \chi_R \cdot A \nabla \chi_R \, dx &= 2\pi \int_R^{2R} v(s)^2 s(1 + s^2) \chi'_R(s)^2 \, ds \\ &\leq C_\chi \mathcal{E}_{\text{tail}}(R). \end{aligned}$$

Another Cauchy–Schwarz inequality therefore bounds the absolute value of the final integral by $C_\chi^{1/2} \mathcal{E}_{\text{tail}}(R)$, which tends to zero. Dominated convergence sends the energy term on the left to $\mathcal{E}(\infty)$ and proves (66.9). Under (66.10), this nonnegative energy is zero. Positive definiteness of A gives $\nabla U = 0$; the differential equation then gives $\tilde{\rho} = c$. $\square$

Remark 66.5 (The constant in the test function matters). Replacing $U - \ell$ by U adds the term $\ell \int \chi_R(\tilde{\rho} - c)$ to the work integral. Finite energy does not, by itself, control this net-charge term. Thus a sign estimate using U and one using $U - \ell$ are not interchangeable without further information. No finite-energy estimate or either sign estimate is asserted here for an actual limit away from $\beta = 2$.

66.2 Energy escapes to infinity even at the determinantal point

Uniform finite-particle control of the energy on the whole plane cannot be the missing estimate: that assertion already fails at the exactly solved temperature.

Proposition 66.6 (Finite-particle obstruction and an exact benchmark). *Use the finite-particle mean defect $U_N = u_N \circ F$ and the corresponding expected count M_N and energy $\mathcal{E}_N$. For every fixed N and every β with $q \neq 0$,*

$$\lim_{R \rightarrow \infty} \frac{\mathcal{E}_N(R)}{R^2} = \pi q^2, \quad \mathcal{E}_N(\infty) = \infty. \quad (66.11)$$

Writing $H_N = \sum_{z \in \Xi_N} -\log |z|$, one also has

$$0 \leq U_N(R) - \frac{q}{2} \log(1 + R^2) + \mathbb{E}H_N \leq \frac{N}{2} \log(1 + R^{-2}). \quad (66.12)$$

If $0 < \beta < 4$ and $a > q^{-1/2}$, then, without using the determinantal structure,

$$\liminf_{N \rightarrow \infty} \frac{\mathcal{E}_N(a\sqrt{N})}{N} \geq \pi \int_{1/q}^{a^2} \left(q - \frac{1}{v} \right)^2 dv > 0. \quad (66.13)$$

At $\beta = 2$, put $t_R = R^2/(1 + R^2)$. Then exactly

$$M_N(R) = R^2(1 - t_R^N), \quad D_N(R) = R^2 t_R^N, \quad (66.14)$$

$$U_N(R) = \frac{1}{2} \sum_{k=N+1}^{\infty} \frac{t_R^k}{k}, \quad (66.15)$$

$$\mathcal{E}_N(R) = \pi \int_0^{R^2} \left(\frac{v}{1+v} \right)^{2N+1} dv. \quad (66.16)$$

Hence $\mathcal{E}_N(R) \rightarrow 0$ for each fixed R , whereas, for each $a > 0$,

$$\lim_{N \rightarrow \infty} \frac{\mathcal{E}_N(a\sqrt{N})}{N} = \pi \int_0^{a^2} e^{-2/v} dv > 0. \quad (66.17)$$

For arbitrary $R > 0$ one has the explicit bound

$$\mathcal{E}_N(R) \leq \pi R^2 \exp \left(-\frac{2N+1}{1+R^2} \right). \quad (66.18)$$

In particular, along any growing radii R_N for which the right-hand side tends to zero, the energy on B_{R_N} vanishes.

Proof. The local energy identity applies unchanged at finite N . Since $0 \leq M_N(R) \leq N$ and $M_N(R) \rightarrow N$, its integrand satisfies

$$\frac{D_N(R)^2}{R(1+R^2)} = q^2 R + O_{N,q}(R^{-1}) \quad (R \rightarrow \infty).$$

Integration proves (66.11). For the potential bound, put $r = R/\sqrt{1+R^2}$. Tonelli's theorem gives $\mathbb{E}H_N = \int_0^1 m_N(s) ds/s < \infty$. The finite mean-defect formula consequently yields

$$U_N(R) = \frac{q}{2} \log(1 + R^2) - \mathbb{E}H_N + \int_r^1 \frac{m_N(s)}{s} ds.$$

The last integral lies between zero and $N(-\log r)$, which is the upper bound in (66.12).

For $q > 0$ and $s \geq \sqrt{N/q}$ the elementary bound $M_N(s) \leq N$ implies $D_N(s) \geq qs^2 - N \geq 0$. Restrict the energy integral to this range and then set $v = s^2/N$. For $a > q^{-1/2}$ this gives

$$\frac{\mathcal{E}_N(a\sqrt{N})}{N} \geq \pi \int_{1/q}^{a^2} \frac{(qv - 1)^2}{v(v + 1/N)} dv.$$

Dominated convergence on this fixed interval proves (66.13).

At $\beta = 2$ the finite Bergman kernel gives $m_N(r) = \sum_{k=1}^N r^{2k} = r^2(1 - r^{2N})/(1 - r^2)$. This proves (66.14). Integrating $m_N(s)/s$ in the mean-defect formula and using $-\log(1 - t) = \sum_{k \geq 1} t^k/k$ proves (66.15). Substitute the discrepancy into (66.6), then set $v = s^2$. This gives (66.16).

For $R = a\sqrt{N}$, set $v = Nw$ in that identity. For each $w > 0$,

$$\left(\frac{Nw}{1 + Nw} \right)^{2N+1} \longrightarrow e^{-2/w}.$$

The integrands lie between zero and one, so dominated convergence on $[0, a^2]$ proves (66.17). Finally, the integrand in (66.16) is increasing in v , and $\log(v/(1 + v)) \leq -1/(1 + v)$. These two facts prove (66.18), which also proves fixed-radius energy convergence. $\square$

66.3 Why the established one-point bounds do not supply the sign

The following examples concern candidate intensity functions only. They are not asserted to be intensities of Gibbs laws or of actual disk-gas limits.

Proposition 66.7 (Localized defects pass the one-point tests). *Fix $0 < \beta < 4$. There exist positive smooth radial intensity functions ρ_ε on $\mathbb{D}$, distinct from $q/[\pi(1 - |z|^2)^2]$, such that:*

1. *they agree with that proposed intensity outside a fixed compact annulus in $\mathbb{D}$;*
2. *their total signed difference from the proposed intensity is zero;*
3. *their normalized mean defects have finite energy, vanish outside a compact annulus, and have zero boundary term in (66.7) for all sufficiently large R ;*
4. *they satisfy $\Delta \log \rho_\varepsilon(z) \geq 2\beta/(1 - |z|^2)^2$ everywhere.*

For each such nonzero defect, the work integral appearing in either global sign criterion is strictly positive.

Proof. Choose a nonzero smooth radial function ψ on $\mathbb{R}^2$ supported in the annulus $1 < |x| < 2$. Set

$$U_\varepsilon(x) = \varepsilon\psi(x), \quad \tilde{\rho}_\varepsilon(x) = c - \frac{\varepsilon}{2\pi}\mathcal{L}\psi(x), \quad \rho_\varepsilon(z) = \frac{\tilde{\rho}_\varepsilon(F^{-1}(z))}{(1 - |z|^2)^2}.$$

Since $c > 0$ and $\mathcal{L}\psi$ is bounded and compactly supported, $\tilde{\rho}_\varepsilon$ is strictly positive for all sufficiently small $|\varepsilon| > 0$. The equations give $\mathcal{L}U_\varepsilon = 2\pi(c - \tilde{\rho}_\varepsilon)$ and $U_\varepsilon(0) = 0$. Uniqueness of a smooth radial solution with this normalization, obtained by integrating its radial flux, shows that U_ε is exactly the mean defect defined from this candidate intensity. Integration of $\mathcal{L}\psi$ over the plane gives zero, because ψ is compactly supported. This proves the first three assertions, including the zero signed mass after the change of variables.

For the last assertion, the unperturbed profile ρ_0 satisfies

$$\Delta \log \rho_0(z) = \frac{8}{(1 - |z|^2)^2}.$$

The difference from the required lower bound is $(8 - 2\beta)/(1 - |z|^2)^2$, strictly positive. On the fixed compact support of the perturbation, $\rho_\varepsilon \rightarrow \rho_0$ in C^2 and all these functions have a

common positive lower bound for sufficiently small $|\varepsilon|$. Thus the strict inequality persists there; outside that support the profiles agree exactly. Finally $\ell = 0$ and integration by parts gives

$$\int U_\varepsilon(\tilde{\rho}_\varepsilon - c) dx = \frac{1}{2\pi} \int \nabla U_\varepsilon \cdot A \nabla U_\varepsilon dx > 0.$$

The strict positivity follows because ψ is nonzero and compactly supported, hence is not constant. $\square$

These examples isolate the logical issue: neither finite energy, vanishing boundary terms, the correct exterior profile, zero total charge discrepancy, nor the proved one-point curvature bound forces the required sign. Any successful sign argument must use additional information about the interacting gas. This proposition does not rule out that additional information or the intensity conjecture.

67 An exact neutral-plasma representation and its energy obstruction

This section gives an exact finite-volume interpretation of the proposed intensity coefficient. The interpretation does not assume screening. It also identifies two obstacles to importing a screening argument: the change from the hyperbolic Green interaction is a critical quadratic tilt, and the finite-volume mean electrostatic energy need not vanish.

Put $\delta(z) = 1 - |z|^2$ and

$$d\lambda(z) = \delta(z)^{-2} dA(z), \quad c_\beta = \frac{4/\beta - 1}{\pi}.$$

The metric with area measure λ is $ds_\lambda^2 = |dz|^2/\delta(z)^2$, of curvature -4 . Its distance is denoted by d_λ . The curvature -1 convention is $ds_\mathbb{H}^2 = 4ds_\lambda^2$; thus $d_\mathbb{H} = 2d_\lambda$ and $dV_\mathbb{H} = 4d\lambda$. In the Euclidean coordinates $x = z/\sqrt{\delta(z)}$, $d\lambda = d^2x$ and $dV_\mathbb{H} = 4d^2x$. Define the two invariant interactions

$$v_s(z, w) = -\log \sinh d_\lambda(z, w) = -\log |z - w| + \frac{1}{2} \log \delta(z) + \frac{1}{2} \log \delta(w), \quad (67.1)$$

$$g(z, w) = -\log \tanh d_\lambda(z, w) = -\log |z - w| + \log |1 - z\bar{w}|. \quad (67.2)$$

Equivalently, the first interaction is $-\log \sinh(d/(2a))$ when the curvature is $-1/a^2$, and $a = 1/2$ in our λ convention.

Proposition 67.1 (Exact neutral one-component plasma). *Let $0 < \beta < 4$, $c = c_\beta$, and*

$$R_N^2 = \frac{N}{N + \pi c}, \quad D_N = \{z : |z| < R_N\}.$$

Consider N unit positive charges in D_N , a fixed background charge $-c \mathbf{1}_{D_N} \lambda$, pair interaction v_s , and reference measure $\lambda^{\otimes N}$. Including the background self-energy, define

$$\begin{aligned} H_N^s(z_1, \dots, z_N) &= \sum_{i < j} v_s(z_i, z_j) - c \sum_i \int_{D_N} v_s(z_i, w) d\lambda(w) \\ &\quad + \frac{c^2}{2} \iint_{D_N^2} v_s(w, w') d\lambda(w) d\lambda(w'). \end{aligned} \quad (67.3)$$

This background has total charge $-N$. There is a finite constant $C_{N,c}^s$, independent of all particle positions, such that

$$H_N^s = -\sum_{i < j} \log |z_i - z_j| - \frac{1 + \pi c}{2} \sum_i \log \delta(z_i) + C_{N,c}^s.$$

Consequently the canonical probability density $e^{-\beta H_N^s}/Z_N^s$ relative to $\lambda^{\otimes N}$ is exactly the unweighted disk gas

$$\frac{1}{Z_{N,\beta}(R_N)} |\Delta_N(z)|^\beta \prod_i 1_{D_N}(z_i) dA(z_i).$$

If $X_{N,\beta}$ denotes the original unit-disk gas, this law is the law of $R_N X_{N,\beta}$. The two sequences have the same subsequential limits in the local topology.

Proof. Direct radial integration gives

$$\lambda(B_R) = \frac{\pi R^2}{1 - R^2},$$

so $c\lambda(D_N) = N$. For $R < 1$ define the finite constants

$$I_R^0 = \int_{B_R} \log |w| d\lambda(w), \quad J_R = \int_{B_R} \log \delta(w) d\lambda(w).$$

The logarithmic singularity is integrable and δ is bounded below on B_R , so both constants are finite. The circular mean identity $\frac{1}{2\pi} \int_0^{2\pi} \log |r - se^{it}| dt = \log \max(r, s)$ implies, for $0 < r < R$,

$$\frac{d}{dr} \int_{B_R} \log |r - w| d\lambda(w) = \frac{\lambda(B_r)}{r} = \frac{\pi r}{1 - r^2}.$$

Integration from zero proves

$$\int_{B_R} \log |z - w| d\lambda(w) = I_R^0 - \frac{\pi}{2} \log \delta(z), \quad |z| < R. \quad (67.4)$$

The pair part of (67.3) contributes

$$-\sum_{i < j} \log |z_i - z_j| + \frac{N-1}{2} \sum_i \log \delta(z_i).$$

The particle-background part, by (67.1) and (67.4), contributes

$$-\frac{\pi c + N}{2} \sum_i \log \delta(z_i) + cN I_{R_N}^0 - \frac{cN}{2} J_{R_N}.$$

Thus one may take

$$C_{N,c}^s = cN I_{R_N}^0 - \frac{cN}{2} J_{R_N} + \frac{c^2}{2} \iint_{D_N^2} v_s(w, w') d\lambda(w) d\lambda(w').$$

This is finite by the same local logarithmic integrability. Multiplying the Boltzmann factor by $\prod_i \delta(z_i)^{-2} dA(z_i)$ leaves the exponent $\beta(1 + \pi c)/2 - 2 = 0$ on every $\delta(z_i)$. In particular, $Z_N^s = e^{-\beta C_{N,c}^s} Z_{N,\beta}(R_N)$.

The substitution $z_i = R_N u_i$ gives

$$Z_{N,\beta}(R_N) = R_N^{2N + \beta N(N-1)/2} Z_{N,\beta}(1),$$

and proves the asserted equality in law. Finally $R_N \rightarrow 1$. For $f \in C_c^1(\mathbb{D})$ choose a compact set $K \Subset \mathbb{D}$ containing both its support and all its inverse images under the scalings $z \mapsto R_N z$, for sufficiently large N . Then

$$|(R_N X_{N,\beta})(f) - X_{N,\beta}(f)| \leq (1 - R_N) \|\nabla f\|_\infty X_{N,\beta}(K).$$

The compact count tightness already proved makes this tend to zero in probability. A countable convergence-determining collection of such f proves the local-limit assertion. $\square$

Remark 67.2 (What neutrality does and does not establish). The proposition identifies the conjectured coefficient with a genuine finite-volume background, rather than with a formal mean-field parameter. It does not establish that the local particle density converges to that background density. Global neutrality says only that the integral of the finite-volume charge density vanishes. The boundary of these hyperbolic disks has size comparable to their volume, so a boundary layer cannot be discarded merely by dividing by volume.

Proposition 67.3 (The Green comparison is a critical quadratic tilt). *At the same c, N, D_N , define H_N^g by replacing v_s by g throughout (67.3). There are finite position-independent constants $C_{N,c}^g$ and $C_{N,c}^s$ such that*

$$H_N^g - C_{N,c}^g = \sum_{i < j} g(z_i, z_j) - \frac{\pi c}{2} \sum_i \log \delta(z_i),$$

$$(H_N^s - C_{N,c}^s) - (H_N^g - C_{N,c}^g) = \frac{1}{2} \sum_{k=1}^{\infty} \frac{|S_k|^2}{k}, \quad S_k = \sum_i z_i^k.$$

If P_N^s and P_N^g are the corresponding canonical probability laws, then

$$\frac{dP_N^g}{dP_N^s} = \frac{\exp\{\frac{\beta}{2} \sum_{k \geq 1} |S_k|^2/k\}}{\mathbb{E}_{P_N^s} \exp\{\frac{\beta}{2} \sum_{k \geq 1} |S_k|^2/k\}}. \quad (67.5)$$

The denominator is finite for every N but tends to infinity as $N \rightarrow \infty$. In fact, for every fixed integer $K \geq 1$,

$$\mathbb{E}_{P_N^s} \exp \left\{ \frac{\beta}{2} \sum_{k=1}^K \frac{|S_k|^2}{k} \right\} \rightarrow \infty. \quad (67.6)$$

Proof. For every $|z|, |w| < 1$, $\log |1 - z\bar{w}| = -\operatorname{Re} \sum_{k \geq 1} z^k \bar{w}^k / k$. The convergence is uniform on $\overline{D}_N^2$. The integral of each positive angular mode of the radial background vanishes. Thus the background potential of g is the negative of (67.4), and one may take

$$C_{N,c}^g = cN I_{R_N}^0 + \frac{c^2}{2} \iint_{D_N^2} g(w, w') d\lambda(w) d\lambda(w').$$

Subtracting the two configuration-dependent energies gives

$$-\sum_{i < j} \log |1 - z_i \bar{z}_j| - \frac{1}{2} \sum_i \log \delta(z_i) = -\frac{1}{2} \sum_{i,j} \log |1 - z_i \bar{z}_j| = \frac{1}{2} \sum_{k \geq 1} \frac{|S_k|^2}{k}.$$

This proves (67.5). For fixed N the sum is bounded above by $N^2 \sum_{k \geq 1} R_N^{2k}/k < \infty$, so its exponential is integrable.

Under P_N^s , the already proved harmonic-mode central limit theorem and the scaling $R_N \rightarrow 1$ give $(S_1, \dots, S_K) \Rightarrow (G_1, \dots, G_K)$, where the G_k are independent circular complex Gaussians with $\mathbb{E}|G_k|^2 = 2k/\beta$. For a single such Gaussian,

$$\mathbb{E} \exp\{\beta |G_k|^2 / (2k)\} = \infty :$$

the positive quadratic exponent cancels its Gaussian density exactly. Apply weak convergence to the bounded continuous truncations of the exponential in (67.6), and then let the truncation level increase. Their limiting expectations are unbounded, proving that the liminf is infinite. The full denominator dominates each truncated-mode moment, and hence also diverges. $\square$

Remark 67.4. Consequently, convergence and asymptotic independence of finitely many Gaussian modes cannot justify replacing the sinh plasma by the Green plasma through (67.5). The required positive quadratic exponential lies exactly at the Gaussian integrability threshold. No assertion about a local difference of the two limiting point processes follows from this failure of uniform integrability.

Proposition 67.5 (Mean boundary energy persists even at the solved temperature). *Take $\beta = 2$, $c = 1/\pi$, and $R_N^2 = N/(N + 1)$. Let σ_N be the signed mean charge measure of the neutral sinh plasma:*

$$\sigma_N = \mathbb{E}_{P_N^s} \sum_i \delta_{z_i} - \pi^{-1} 1_{D_N} \lambda.$$

Its Green energy

$$\mathcal{E}_N = \frac{1}{2} \iint g(z, w) d\sigma_N(z) d\sigma_N(w)$$

is finite and nonnegative, and

$$\liminf_{N \rightarrow \infty} \frac{\mathcal{E}_N}{N} \geq \frac{1}{4} \int_2^3 \left(\frac{1 - e^{-(t-1)}}{t-1} - \frac{1}{t} \right)^2 dt > 0. \quad (67.7)$$

Nevertheless its particle intensity converges locally to $\pi^{-1} \delta(z)^{-2}$.

Proof. The last assertion is the Bergman limit and the deterministic scaling in Proposition 67.1. The mean measure is radial, supported in $\overline{D}_N$, has a bounded density for each fixed N , and has total charge zero. Let $Q_N(r) = \sigma_N(B_r)$ and $V_N(z) = \int g(z, w) d\sigma_N(w)$. The angular mean of the image term in g is zero, so direct angular integration gives $rV_N'(r) = -Q_N(r)$. Because $Q_N(r) = 0$ for $r \geq R_N$, the potential vanishes outside D_N : it is constant there and has zero boundary value at $|z| = 1$. Integration by parts, or first integration on compact subdisks using the bounded density, therefore gives

$$\mathcal{E}_N = \frac{1}{4\pi} \int_{\mathbb{D}} |\nabla V_N|^2 dA = \frac{1}{2} \int_0^{R_N} \frac{Q_N(r)^2}{r} dr. \quad (67.8)$$

Near zero $Q_N(r) = O(r^2)$, so the final integral is finite.

The truncated Bergman kernel on B_{R_N} gives

$$\mathbb{E}_{P_N^s} \# B_r = \sum_{k=1}^N \left(\frac{r^2}{R_N^2} \right)^k, \quad Q_N(r) = \sum_{k=1}^N \left(\frac{r^2}{R_N^2} \right)^k - \frac{r^2}{1 - r^2}, \quad r < R_N.$$

For $t \in [2, 3]$ set $r_N(t)^2 = 1 - t/N$. For all sufficiently large N , these radii lie in $(0, R_N)$, and

$$\frac{r_N(t)^2}{R_N^2} = 1 - \frac{t-1}{N} - \frac{t}{N^2}.$$

The logarithm of this ratio is $-(t-1)/N + O(N^{-2})$, uniformly in $t \in [2, 3]$. A uniform Riemann-sum limit therefore gives

$$\frac{Q_N(r_N(t))}{N} \rightarrow h(t) = \int_0^1 e^{-(t-1)s} ds - \frac{1}{t} = \frac{1 - e^{-(t-1)}}{t-1} - \frac{1}{t}.$$

For $t > 1$, the inequality $\log t < t - 1$ implies $h(t) > 0$. Restrict (67.8) to the annulus parametrized by $2 \leq t \leq 3$, and use $|dr|/r = dt/[2N(1 - t/N)]$. This yields

$$\frac{\mathcal{E}_N}{N} \geq \frac{1}{4} \int_2^3 \frac{(Q_N(r_N(t))/N)^2}{1 - t/N} dt,$$

whose limit is the strictly positive right-hand side of (67.7). $\square$

Remark 67.6 (Consequence for the proposed screening strategy). An argument that tries to prove the vanishing of the entire finite-volume mean-charge energy, or even its vanishing after division by N , cannot work: Proposition 67.5 rules out both conclusions in the explicitly solved case. This does not rule out an energy argument for the limiting defect. Such an argument must first isolate the local limit from the moving boundary layer and justify the limiting energy and boundary terms.

Scope of the primary literature. The algebraic plasma correspondence above is consistent with the Hamiltonian in Section 7.1 of Jancovici and Téllez, *Two-Dimensional Coulomb Systems on a Surface of Constant Negative Curvature*, [arXiv:cond-mat/9801322](#). Their Section 5 assumes screening in deriving the sum rules; the general-temperature argument in Section 4.3 for equality of the sinh and Green bulk correlations likewise uses screening. Their exact calculations establish the special-temperature case. Fantoni, Jancovici and Téllez, *Pressures for a One-Component Plasma on a Pseudosphere*, [arXiv:cond-mat/0207525](#), derive an explicit grand-canonical density at coupling two; equation (4.52) yields the background density deep in the bulk while allowing boundary excess charge. These results do not supply the missing screening estimate at general β .

68 The connected-force correction to the screening equation

The mean Poisson equation does not contain the first Ward equation. Combining the two equations isolates a connected two-point force whose work would control the proposed screening. The formulation below keeps the summation at finite particle number until the limiting force has been defined; no absolutely convergent infinite pair integral is assumed.

Identify a complex number a with the real vector $\mathbf{v}(a) = (\operatorname{Re} a, -\operatorname{Im} a)$. Put $\delta(z) = 1 - |z|^2$, $q = 4/\beta - 1$, and

$$g(z, w) = -\log \left| \frac{z - w}{1 - z\bar{w}} \right|, \quad j(z, w) = -\nabla_z g(z, w) = \mathbf{v} \left(\frac{1}{z - w} + \frac{\bar{w}}{1 - z\bar{w}} \right).$$

Let ρ_N and $\rho_{2,N}$ denote the first and ordered second correlation densities of the unweighted N -particle disk gas. Define

$$\begin{aligned} h_N(z, w) &= \rho_{2,N}(z, w) - \rho_N(z)\rho_N(w), \\ I_N(z) &= \int_{\mathbb{D}} j(z, w) h_N(z, w) dA(w), & F_N(z) &= I_N(z)/\rho_N(z), \\ T_N(z) &= \sum_{w \in \Xi_N} \frac{\bar{w}}{1 - z\bar{w}}, & D_N(z) &= \mathbb{E}_N^z \mathbf{v}(T_N(z)). \end{aligned} \tag{68.1}$$

The superscript z in the last expression denotes the full Palm law, including the particle at z . Thus D_N includes the self contribution $z/\delta(z)$. Each integral in (68.1) is an ordinary absolutely convergent integral at fixed N : the first-order singularity is locally integrable, the correlation densities are bounded at fixed N , and $j(z, w)$ is bounded near the boundary for fixed interior z .

Proposition 68.1 (A finite-particle Ward equation with a connected force). *Let u_N be the normalized mean defect of Proposition 64.8, and put $f_N(z) = \delta(z)^2 \rho_N(z)$. Then, for every N and $\beta > 0$,*

$$\nabla \log f_N = -\beta \nabla u_N + \beta F_N - \beta D_N \quad \text{on } \mathbb{D}. \tag{68.2}$$

Proof. Differentiating the finite insertion integral, justified by the locally integrable powers $|z - w|^{\beta-1}$, gives the first Ward equation

$$\nabla \log \rho_N(z) = \beta \mathbb{E}_N^! z \mathfrak{p} \left(\sum_{w \in \Xi_N} \frac{1}{z - w} \right).$$

The reduced Palm expectation of the complete Green force equals

$$\mathbb{E}_N^! z \sum_w j(z, w) = \int_{\mathbb{D}} j(z, w) \rho_N(w) dA(w) + F_N(z).$$

Rotation invariance and the angular mean computation in (64.10) give

$$a_N(z) := \int_{\mathbb{D}} j(z, w) \rho_N(w) dA(w) = \frac{m_N(|z|)}{|z|^2} z,$$

with value zero at $z = 0$. Removing the image force from the complete Green force therefore gives

$$\nabla \log \rho_N = \beta \left(a_N + F_N + \frac{z}{\delta} - D_N \right).$$

The exact mean-defect formula gives $\nabla u_N = -a_N + qz/\delta$. Finally, $\nabla \log f_N = \nabla \log \rho_N - 4z/\delta$ and $\beta(q + 1) = 4$, which prove the assertion. $\square$

Proposition 68.2 (A limiting connected force without an infinite sum). *Suppose Ξ_{N_j} converges to a subsequential limit with intensity ρ , and let $\Omega = \{z \in \mathbb{D} : \rho(z) > 0\}$. On every compact subset of Ω , F_{N_j} is bounded in L^2 , converges weakly in L^2 to a uniquely determined vector field F , and D_{N_j} converges weakly in L^2 to zero. With $f = \delta^2 \rho$ and u the limiting mean defect,*

$$\nabla \log f = -\beta \nabla u + \beta F \quad \text{weakly on } \Omega. \quad (68.3)$$

This definition does not assert that F equals the result of first passing to limiting correlation functions in (68.1) and then integrating up to $\partial \mathbb{D}$.

Proof. Fix $K \Subset \Omega$ and a slightly larger compact set contained in Ω . Local uniform convergence of the intensities bounds ρ_{N_j} above and below there. Corollary 33.2 and conditional Jensen's inequality give a uniform local bound on $\int |\nabla \log \rho_{N_j}|^2 dA$. The logarithms converge uniformly on compact sets, and consequently their gradients converge weakly in local L^2 to $\nabla \log \rho$. This last statement follows by weak compactness and distributional differentiation; every weak cluster point is the same distributional derivative.

Here is an explicit justification of the analytic supremum moments. Choose $r < s < 1$ with $K \subset \{|z| \leq r\}$. Cauchy's integral formula and Jensen's inequality give, for $p \geq 1$,

$$\sup_{|z| \leq r} |T_N(z)|^p \leq \left(\frac{s}{s-r} \right)^p \frac{1}{2\pi} \int_0^{2\pi} |T_N(se^{i\theta})|^p d\theta.$$

Rotation invariance makes the expectation of the integrand independent of θ . At the real point s , $T_N(s) = \overline{\sum_{w \in \Xi_N} w / (1 - sw)}$. Theorem 19.6, applied to arbitrary real and imaginary scalar multiples of the function $w \mapsto w / (1 - sw)$, bounds the exponential moments of both real components uniformly in N . Hence $\sup_N \mathbb{E} |T_N(s)|^p < \infty$ for every $p \geq 1$. The preceding Cauchy estimate gives the required uniform supremum moments. The compact-count bounds then imply

$$\sup_N \mathbb{E} \sum_{z \in \Xi_N \cap K} |T_N(z)|^2 < \infty$$

by Cauchy–Schwarz. Palm conditional Jensen gives a uniform bound on $\int_K \rho_N |D_N|^2 dA$, hence on $\|D_N\|_{L^2(K)}$. For a continuous compactly supported vector test v ,

$$\int \rho_N(z) v(z) \cdot D_N(z) dA(z) = \mathbb{E} \sum_{z \in \Xi_N} v(z) \cdot \mathbf{v}(T_N(z)).$$

Joint convergence of the analytic harmonic field and the point process, and their proved independence in the limit, make the last expression converge to zero. Indeed, the limiting analytic Gaussian field is centered. The product is uniformly integrable by the same higher-moment and count bounds, so this conclusion concerns expectations, not just convergence in law. To pass from $\rho_N D_N$ to D_N , use the fixed continuous test v/ρ , where v has compact support in K . The resulting expression differs from $\int v \cdot D_N$ by at most

$$\left\| \frac{\rho_N}{\rho} - 1 \right\|_{L^\infty(K)} \|v\|_{L^2(K)} \|D_N\|_{L^2(K)},$$

which tends to zero. Density of continuous tests in L^2 , together with the uniform local L^2 bound, proves that D_N converges weakly in local L^2 to zero.

Now use (68.2). The local C^2 convergence of u_N , the weak convergence of the logarithmic gradients, and the preceding bound on D_N give both the claimed local L^2 bound on F_N and the unique weak limit $F = \nabla u + \beta^{-1} \nabla \log f$. $\square$

68.1 A compact entropy–energy identity in the plane

Let $\Psi(x) = x/\sqrt{1+|x|^2}$ map $\mathbb{R}^2$ onto $\mathbb{D}$, and set

$$p(x) = \frac{\rho(\Psi(x))}{(1+|x|^2)^2}, \quad U(x) = u(\Psi(x)), \quad \mathcal{F}(x) = D\Psi(x)^\top F(\Psi(x)).$$

The letter p here denotes an intensity, not a probability density of total mass one. Define the positive definite conductivity

$$A(x) = \frac{I + (2+|x|^2)xx^\top}{1+|x|^2}.$$

The Jacobian of Ψ is $(1+|x|^2)^{-2}$ and direct change of variables in the Dirichlet form gives $A = (\det D\Psi) D\Psi^{-1} D\Psi^{-\top}$. Thus the two equations are exactly

$$\operatorname{div}(A \nabla U) = 2\pi(c - p), \quad \nabla \log p = -\beta \nabla U + \beta \mathcal{F}, \quad c = \frac{4/\beta - 1}{\pi}. \quad (68.4)$$

Theorem 68.3 (Entropy, mean-field energy, and connected-force work). *Assume $0 < \beta < 4$. Let $0 \leq \chi \in C_c^1(\mathbb{R}^2)$ have support in the positive-intensity set $\{p > 0\}$. Then all the following integrals are finite, and*

$$\begin{aligned} & 2\pi \int \chi(p - c) \log(p/c) dx + \beta \int \chi \nabla U \cdot A \nabla U dx \\ &= \beta \int \chi \mathcal{F} \cdot A \nabla U dx + \int \log(p/c) \nabla \chi \cdot A \nabla U dx. \end{aligned} \quad (68.5)$$

Suppose in addition that $p > 0$ on the plane and there are $0 \leq \chi_k \leq 1$ of compact support with $\chi_k \rightarrow 1$ pointwise, such that

$$\int \log(p/c) \nabla \chi_k \cdot A \nabla U dx \longrightarrow 0, \quad (68.6)$$

$$\limsup_{k \rightarrow \infty} \int \chi_k \mathcal{F} \cdot A \nabla U dx \leq 0. \quad (68.7)$$

Then $p \equiv c$, and hence the proposed disk intensity holds. No prior assumption of finite total energy or finite total entropy is needed for this sufficient criterion.

Proof. On the support of χ , p is continuous and bounded away from zero, $\log(p/c)$ is in H^1 , U is C^2 , and $\mathcal{F}$ is in L^2 . The matrices A and A^{-1} are bounded there. Thus all products are integrable. Taking the scalar product of the second equation in (68.4) with $\chi A \nabla U$ and integrating gives

$$\int \chi \nabla \log(p/c) \cdot A \nabla U + \beta \int \chi \nabla U \cdot A \nabla U = \beta \int \chi \mathcal{F} \cdot A \nabla U.$$

Weak integration by parts in the first term, using the first equation of (68.4), gives

$$\int \chi \nabla \log(p/c) \cdot A \nabla U = 2\pi \int \chi(p-c) \log(p/c) - \int \log(p/c) \nabla \chi \cdot A \nabla U.$$

This proves (68.5). Both integrands on its left are nonnegative, since $c > 0$, A is positive definite, and $(t-c) \log(t/c) \geq 0$ for $t > 0$. Under the two cutoff hypotheses their sum has upper limit at most zero. Fatou's lemma then gives $\int (p-c) \log(p/c) = 0$. Its integrand vanishes only where $p = c$. Continuity proves $p \equiv c$. Equivalently, the energy vanishes, U is constant, and $U(0) = 0$ fixes that constant. $\square$

Remark 68.4 (The correlation estimate which remains unproved). The right side of (68.5) is an exact correction to an entropy–energy screening argument. Neither (68.6) nor (68.7) has been proved for a general disk-gas limit. In particular, $\mathcal{F}$ is not set to zero by the independence of the harmonic Gaussian modes: that independence removed D_N and left the connected Green force F_N .

68.2 Two checks on proposed sign shortcuts

Proposition 66.7 already shows that the normalized mean Poisson equation, the known curvature inequality, and even compact support of the mean defect do not imply the screening sign. The following finite-particle example separately checks that a sign condition on connected correlations alone is also insufficient.

Example 68.5 (Negative connected correlations alone do not determine force work). For the actual one-particle disk gas, $\rho_1 = 1/\pi$ and $h_1(z, w) = -1/\pi^2 < 0$ for all distinct z, w . Nevertheless, $F_1(z) = -z$ and

$$u_1(z) = -\frac{|z|^2}{2} - \frac{q}{2} \log \delta(z), \quad F_1(z) \cdot \nabla u_1(z) = |z|^2 \left(1 - \frac{q}{\delta(z)}\right).$$

For $2 < \beta < 4$, this work is strictly positive whenever $0 < |z|^2 < 1 - q$. Thus even pointwise negative connected correlations do not, by themselves, give a nonpositive connected-force work for this potential. This finite-particle example is not a counterexample to a limiting screening assertion: its full-Palm term D_1 is present. The example only rules out inferring the desired sign directly from $h_N \leq 0$ without an additional argument.

69 Gibbs variation and the correlation contribution to screening

69.1 What Gibbs convexity controls, and the missing fluctuation term

An exponential change of an external potential has a definite response sign. Its comparison density, however, is the density of another Gibbs law. It cannot be replaced by a proposed constant background without an additional theorem.

Proposition 69.1 (The exact response and symmetric entropy identities). *Let P be a point-process law, let v be a bounded compactly supported real function, and put $X = \sum_{x \in \Xi} v(x)$. Assume that $\mathbb{E}_P \exp\{a \sum_{x \in \Xi} |v(x)|\} < \infty$ for every $a > 0$. For $0 \leq t \leq 1$ set*

$$\frac{dP_t}{dP} = \frac{e^{\beta t X}}{\mathbb{E}_P e^{\beta t X}}.$$

Write $\rho_t(x)dx$ for the intensity, when it exists. Then

$$\int v(x) \{\rho_1(x) - \rho_0(x)\} dx = \beta \int_0^1 \text{Var}_{P_t}(X) dt, \quad (69.1)$$

$$\text{Ent}(P_1 | P) + \text{Ent}(P | P_1) = \beta \int v(x) \{\rho_1(x) - \rho_0(x)\} dx. \quad (69.2)$$

The same formulas hold for finite canonical Gibbs laws without any extra moment hypothesis.

Proof. The exponential-moment assumption permits two differentiations of $\psi(t) = \log \mathbb{E}_P e^{\beta t X}$ under the expectation, uniformly on a neighborhood of $[0, 1]$. Thus $\psi'(t) = \beta \mathbb{E}_{P_t} X$ and $\psi''(t) = \beta^2 \text{Var}_{P_t}(X)$. Integrating the second identity proves (69.1), using the Campbell formula. Its absolute integrability follows from the assumption, which also implies $\mathbb{E}_{P_t} \sum_x |v(x)| < \infty$ uniformly for $0 \leq t \leq 1$. The two logarithmic likelihoods are $\beta X - \psi(1)$ and its negative; integrating them under their respective laws proves (69.2). For a fixed finite number of particles, X is bounded. $\square$

Remark 69.2 (The comparison law is part of the missing theorem). For a compactly supported truncation v of the mean defect, the proposition gives $\int v(\rho_0 - \rho_1) \leq 0$. The desired comparison with a constant c would follow from this particular argument if one could identify $\rho_1 = c$ on the relevant region and control removal of the truncation. Neither equality follows from convexity. In particular, removing a mean electrostatic potential from the energy does not remove the fluctuating electrostatic potential.

The next finite-volume example isolates that distinction using the same Green pair interaction. It is a comparison model, not the unweighted fixed- N disk gas, and its conclusion does not disprove the intensity conjecture for that gas.

Let Π_c be the Poisson process of constant intensity $c > 0$ on $\mathbb{D}$. Its total count is finite almost surely. Write

$$g(x, y) = -\log \left| \frac{x - y}{1 - \overline{x}y} \right|, \quad v_c(x) = c \int_{\mathbb{D}} g(x, y) dA(y) = \frac{\pi c}{2} (1 - |x|^2).$$

The last equality follows either by angular integration or by solving $-\Delta v_c = 2\pi c$ with zero boundary values. Define

$$H_c(\eta) = \sum_{\{x, y\} \subset \eta} g(x, y) - \sum_{x \in \eta} v_c(x), \quad \frac{dP_{\beta, c}}{d\Pi_c} = Z_{\beta, c}^{-1} e^{-\beta H_c}, \quad \beta \geq 0. \quad (69.3)$$

This interaction contains a constant compensating background in its one-particle term, and its reference activity is also c .

Proposition 69.3 (An exact free-energy decomposition with correlations). *The law in (69.3) is well-defined for every $\beta \geq 0$. For $\beta > 0$, let ρ and $\rho^{(2)}$ be its first and second factorial correlation densities, and put*

$$C^{(2)}(x, y) = \rho^{(2)}(x, y) - \rho(x)\rho(y), \quad f(x) = \rho(x) - c.$$

All terms below are finite, and

$$\begin{aligned} \text{Ent}(P_{\beta,c} \mid \Pi_c) &+ \frac{\beta}{2} \iint_{\mathbb{D}^2} g(x,y) f(x) f(y) dA(x) dA(y) \\ &\leq -\frac{\beta}{2} \iint_{\mathbb{D}^2} g(x,y) C^{(2)}(x,y) dA(x) dA(y). \end{aligned} \quad (69.4)$$

Moreover,

$$\text{Ent}(P_{\beta,c} \mid \Pi_c) \geq \int_{\mathbb{D}} \left\{ \rho \log \frac{\rho}{c} - \rho + c \right\} dA, \quad (69.5)$$

$$\iint_{\mathbb{D}^2} g(x,y) f(x) f(y) dA(x) dA(y) \geq 0. \quad (69.6)$$

Thus the variational principle bounds a mean-density defect by a correlation-energy contribution. It does not make the mean-density energy nonpositive.

Proof. Since $g \geq 0$ and $0 \leq v_c \leq \pi c/2$, the bound $H_c(\eta) \geq -(\pi c/2)\eta(\mathbb{D})$ implies $Z_{\beta,c} < \infty$ by the exponential moments of a Poisson random variable. The empty configuration has positive Poisson probability, so $Z_{\beta,c} > 0$. The resulting law has all exponential count moments. For every fixed integer k , all k th moments of H_c under a Poisson law whose intensity has been multiplied by a finite constant are finite. Indeed, after conditioning on its finite count, the inequality $(\sum_{j=1}^m a_j)^k \leq m^{k-1} \sum_j a_j^k$ for $a_j \geq 0$ reduces this assertion to integrability of powers of the logarithmic singularity of g and to Poisson count moments. On $\mathbb{D}$, $g(x,y) \leq \log(2/|x-y|)$, and every positive power of the latter function is integrable on $\mathbb{D}^2$. These observations also prove the energy and entropy integrability used below.

The Poisson insertion formula gives, for every nonnegative measurable f_0 ,

$$\mathbb{E} \sum_{x \in \Xi} f_0(x, \Xi \setminus \{x\}) = c \int_{\mathbb{D}} \mathbb{E} \left[f_0(x, \Xi) e^{\beta v_c(x) - \beta \sum_{y \in \Xi} g(x,y)} \right] dA(x).$$

To verify the formula, expand the Poisson expectation by total count, choose the distinguished point among the n coordinates, and cancel $n/n!$ against $1/(n-1)!$; the quotient of Gibbs weights is the exponential displayed above. In particular $0 < \rho(x) \leq ce^{\beta \pi c/2}$. Two insertions similarly give $0 \leq \rho^{(2)}(x,y) \leq c^2 e^{\beta \pi c}$. The event of no other points also gives a positive constant lower bound on ρ . Consequently all the displayed Green integrals are absolutely finite.

Jensen's inequality yields $\log Z_{\beta,c} \geq -\beta \mathbb{E}_{\Pi_c} H_c$. The exact entropy identity is

$$\text{Ent}(P_{\beta,c} \mid \Pi_c) = -\beta \mathbb{E}_{P_{\beta,c}} H_c - \log Z_{\beta,c}.$$

The Poisson factorial moment formulas and the definition of v_c give

$$\mathbb{E}_{\Pi_c} H_c = -\frac{c^2}{2} \iint g,$$

and a direct expansion gives

$$\mathbb{E}_{P_{\beta,c}} H_c - \mathbb{E}_{\Pi_c} H_c = \frac{1}{2} \iint g f f + \frac{1}{2} \iint g C^{(2)}.$$

Combining these equalities proves (69.4).

For any bounded real h , nonnegativity of relative entropy with respect to the exponential tilt of Π_c by $\sum h(x)$ gives

$$\text{Ent}(P_{\beta,c} \mid \Pi_c) \geq \int h \rho dA - c \int (e^h - 1) dA.$$

Take $h = \log(\rho/c)$, which is bounded by the intensity bounds just proved, to obtain (69.5). Finally, for $u(x) = \int_{\mathbb{D}} g(x, y) f(y) dA(y)$ one has $u \in H_0^1(\mathbb{D})$ and $-\Delta u = 2\pi f$. Testing this weak equation with u gives

$$\iint g f f = \int u f dA = \frac{1}{2\pi} \int_{\mathbb{D}} |\nabla u|^2 dA \geq 0.$$

These assertions follow first for smooth f , by the disk Green formula, and then for bounded f by approximation in L^2 and the energy estimate for the zero-boundary Poisson equation. $\square$

Proposition 69.4 (A constant background does not remove fluctuations). *For each fixed $x \in \mathbb{D}$, the intensity $\rho_{\beta, c}(x)$ of (69.3) has two right derivatives at $\beta = 0$, with*

$$\rho_{0, c}(x) = c, \quad \partial_{\beta} \rho_{\beta, c}(x)|_{0+} = 0, \quad \partial_{\beta}^2 \rho_{\beta, c}(x)|_{0+} = c^2 \int_{\mathbb{D}} g(x, y)^2 dA(y) > 0. \quad (69.7)$$

In particular, $\rho_{\beta, c}(x) > c$ for every sufficiently small positive β , with the allowed neighborhood depending on x . For any $\beta > 0$, the identity $\rho_{\beta, c} \equiv c$ on $\mathbb{D}$ is impossible.

Proof. Write $H = H_c$ and

$$S_x(\eta) = \sum_{y \in \eta} g(x, y), \quad T_x = S_x - v_c(x).$$

The insertion formula gives the exact identity

$$\frac{\rho_{\beta, c}(x)}{c} = \mathbb{E}_{P_{\beta, c}} e^{-\beta T_x} = \frac{\mathbb{E}_{\Pi_c} e^{-\beta(H+T_x)}}{\mathbb{E}_{\Pi_c} e^{-\beta H}}. \quad (69.8)$$

Here $H \geq -C\eta(\mathbb{D})$ and $H + T_x \geq -C\eta(\mathbb{D}) - v_c(x)$. Polynomial moments of H and S_x under any finite-intensity Poisson law are finite, by the logarithmic integrability argument in the preceding proof. Dominated convergence therefore permits two right differentiations of numerator and denominator, including continuity of those second derivatives at zero.

Under Π_c , the mean of T_x is zero and its variance is $c \int g(x, y)^2 dA(y)$. We claim also that $\text{Cov}_{\Pi_c}(T_x, H) = 0$. Indeed, the same insertion calculation for a linear statistic $L_a = \sum_y a(y)$ gives

$$\text{Cov}_{\Pi_c}(L_a, H) = c \int_{\mathbb{D}} a(y) \mathbb{E}_{\Pi_c} \{H(\eta + \delta_y) - H(\eta)\} dA(y).$$

The insertion difference is $S_y - v_c(y)$, whose Poisson mean is zero. Taking $a(y) = g(x, y)$ is justified by the integrability already proved; it establishes the claim. Thus $\mathbb{E}_{\Pi_c} H T_x = 0$. The first derivative of the quotient in (69.8) is zero. Its second derivative is

$$\mathbb{E}_{\Pi_c} (H + T_x)^2 - \mathbb{E}_{\Pi_c} H^2 = \mathbb{E}_{\Pi_c} T_x^2 = c \int_{\mathbb{D}} g(x, y)^2 dA(y),$$

which proves (69.7) and the asserted strict inequality for small positive β .

Finally suppose, for a fixed $\beta > 0$, that $\rho_{\beta, c} = c$ almost everywhere. Then for each fixed x , $\mathbb{E}_{P_{\beta, c}} T_x = \int g(x, y) c dA(y) - v_c(x) = 0$. The random variable T_x is not almost surely constant: the empty configuration has positive probability and gives $S_x = 0$, whereas exactly one particle in any nonempty open set away from x has positive probability and gives $S_x > 0$. The integrable variable $e^{-\beta T_x}$ is bounded above by $e^{\beta v_c(x)}$. Strict Jensen inequality in (69.8) consequently gives $\rho_{\beta, c}(x) > c$, a contradiction. $\square$

Remark 69.5 (The implication for the intensity conjecture). The comparison gas has a finite number of particles, a specified activity, and a zero-boundary Green interaction. It is not an actual subsequential limit of the original model. Its purpose is to establish precisely why a constant compensating background and convexity of Gibbs free energy do not by themselves identify the intensity: the potential variance and the connected pair correlation survive after subtracting the mean field. A successful screening proof for the disk-gas limit must control their contribution and the selected conditional count weights; replacing them by zero is a mean-field approximation, not an exact inference.

70 A coercive screening response at the determinantal point

This section proves the proposed screening sign in a precise class of one-body deformations of the Bergman process. The result is nonlinear, but its application to an actual change of inverse temperature would require an additional identification that is not established here. Put $\delta(z) = 1 - |z|^2$ and $d\lambda = \delta^{-2}dA$, and let $\mathbb{P}_2$ be the Bergman process, with kernel $K(z, w) = \pi^{-1}(1 - z\bar{w})^{-2}$ relative to dA .

70.1 An elementary strict contraction for the Berezin operator

Define

$$(Bf)(z) = \int_{\mathbb{D}} b(z, w)f(w) d\lambda(w), \quad b(z, w) = \frac{\delta(z)^2\delta(w)^2}{\pi|1 - z\bar{w}|^4}.$$

The kernel is symmetric, nonnegative, invariant under simultaneous disk automorphisms, and has row integral one. The last assertion follows from the reproducing identity for K , or by integrating at $z = 0$ and then applying invariance.

Proposition 70.1 (Coercive Bergman susceptibility). *The operator B extends to a positive self-adjoint operator on $L^2(\lambda)$ and*

$$0 \leq B \leq \frac{\pi}{4}I. \quad (70.1)$$

For every real bounded compactly supported f ,

$$\text{Var}_{\mathbb{P}_2} \Xi(f) = \frac{1}{\pi} \langle f, (I - B)f \rangle_{L^2(\lambda)}, \quad \frac{1 - \pi/4}{\pi} \|f\|_{L^2(\lambda)}^2 \leq \text{Var}_{\mathbb{P}_2} \Xi(f) \leq \frac{1}{\pi} \|f\|_{L^2(\lambda)}^2. \quad (70.2)$$

Proof. For $\xi \in \partial\mathbb{D}$, write $P_\xi(z) = \delta(z)/|\xi - z|^2$. For $0 < s < 1$, invariance and the Poisson-kernel transformation identity give

$$B(P_\xi^s)(z) = c_s P_\xi(z)^s, \quad c_s = \frac{1}{\pi} \int_{\mathbb{D}} P_1(w)^s dA(w).$$

Indeed, under a disk automorphism taking 0 to z , P_ξ becomes $P_\xi(z)P_\eta$ for some $\eta \in \partial\mathbb{D}$, while $b(0, w)d\lambda(w) = dA(w)/\pi$. Expanding $(1 - w)^{-s}$ and integrating in angle and radius, with nonnegative terms after angular integration, gives

$$c_s = \frac{1}{s+1} \sum_{k=0}^{\infty} \frac{(s)_k^2}{k!(s+2)_k} = \Gamma(1+s)\Gamma(2-s).$$

Here $(a)_k = a(a+1)\cdots(a+k-1)$, with $(a)_0 = 1$. To verify the sum without a summation formula, insert the beta integral for $(s)_k/(s+2)_k$. Tonelli's theorem and $\sum_k (s)_k t^k/k! = (1-t)^{-s}$ reduce the sum to $s \int_0^1 t^{s-1}(1-t)^{1-s} dt$. In particular, the strictly positive function $h = P_1^{1/2}$ satisfies $Bh = (\pi/4)h$.

The weighted Schur estimate is elementary. Cauchy–Schwarz gives

$$|Bf(z)|^2 \leq (Bh)(z) \int b(z, w) \frac{|f(w)|^2}{h(w)} d\lambda(w).$$

Integrating, using symmetry and $Bh = (\pi/4)h$ a second time, proves $\|Bf\|_2 \leq (\pi/4)\|f\|_2$. All exchanges for nonnegative integrands use Tonelli. Symmetry therefore gives a bounded self-adjoint extension. Positivity first follows for real or complex compactly supported f by expanding $|1 - z\bar{w}|^{-4}$: the quadratic form is a sum of nonnegative weighted squared moments. Uniform absolute convergence on the product of the compact supports justifies integration of the series. Positivity then extends by density.

The determinantal two-point identity gives

$$\text{Var } \Xi(f) = \int f(z)^2 K(z, z) dA(z) - \iint f(z) f(w) |K(z, w)|^2 dA(z) dA(w).$$

Since $K(z, z) = \pi^{-1}\delta(z)^{-2}$, this is precisely (70.2); the operator inequalities give the two bounds. $\square$

The constant function is not in $L^2(\lambda)$. Thus the strict L^2 contraction does not contradict $B1 = 1$, and it does not control a change in the constant hyperbolic intensity.

70.2 Regularized one-body tilts and their exact screening sign

For real $V \in L^\infty(\lambda) \cap L^2(\lambda)$, define the centered linear statistic $Y(V)$ as the $L^2(\mathbb{P}_2)$ limit of

$$\Xi(V\mathbf{1}_{B_r}) - \frac{1}{\pi} \int_{B_r} V d\lambda, \quad r \uparrow 1.$$

Proposition 70.1 proves existence, independence of the compact approximation in $L^2(\lambda)$, and

$$\frac{1 - \pi/4}{\pi} \|V\|_2^2 \leq \mathbb{E}_2 Y(V)^2 \leq \frac{1}{\pi} \|V\|_2^2. \quad (70.3)$$

The map extends linearly and continuously from real $L^2(\lambda)$ to centered $L^2(\mathbb{P}_2)$ random variables.

Lemma 70.2 (Exponential integrability and density defects). *For real $V \in L^\infty \cap L^2(\lambda)$ and every real t ,*

$$\mathbb{E}_2 e^{tY(V)} \leq \exp \left\{ \frac{t^2 e^{t\|V\|_\infty}}{2\pi} \|V\|_2^2 \right\}. \quad (70.4)$$

Consequently

$$\frac{d\mathbb{P}^V}{d\mathbb{P}_2} = \frac{e^{-2Y(V)}}{\mathbb{E}_2 e^{-2Y(V)}}$$

defines a probability measure equivalent to $\mathbb{P}_2$. It has locally finite intensity $\rho_V dA$, and

$$r_V := \delta^2 \rho_V - \frac{1}{\pi} \in L^2(\lambda), \quad \langle f, r_V \rangle = \mathbb{E}_{\mathbb{P}^V} Y(f) \quad (f \in L^2(\lambda)).$$

Proof. For bounded compactly supported f , the determinantal generating functional and $\log(1+x) \leq x$ for $x > -1$ give

$$\log \mathbb{E}_2 e^{t(\Xi(f) - \mathbb{E}_2 \Xi(f))} \leq \int_{\mathbb{D}} (e^{tf} - 1 - tf) K(z, z) dA(z) \leq \frac{t^2 e^{t\|f\|_\infty}}{2\pi} \|f\|_2^2.$$

For precision, restrict the kernel to a compact set containing the support of f . Its compression K_C is a positive trace-class contraction. The determinant can be written $\det(I + K_C^{1/2} M_{e^{tf}-1} K_C^{1/2})$. Its operator inside the determinant is bounded below by $\min(1, e^{-|t|\|f\|_\infty})I$, so the displayed logarithmic inequality applies eigenvalue by eigenvalue, also when f changes sign.

Apply this bound to $f = V\mathbf{1}_{B_r}$. An almost surely convergent subsequence of the L^2 -convergent centered statistics and Fatou's lemma prove (70.4). Bounds with an exponent larger than $|t|$ also give uniform integrability, so the exponential expectations converge along the entire approximation. The likelihood $D_V = d\mathbb{P}^V/d\mathbb{P}_2$ is strictly positive and belongs to every finite $L^p(\mathbb{P}_2)$.

For compactly supported bounded f , Cauchy–Schwarz and the variance upper bound give

$$|\mathbb{E}_2[(D_V - 1)Y(f)]| \leq \frac{\|D_V - 1\|_{L^2(\mathbb{P}_2)}}{\sqrt{\pi}} \|f\|_{L^2(\lambda)}.$$

The Riesz representation theorem supplies $r_V \in L^2(\lambda)$. Applying the identity to indicators of relatively compact Borel sets identifies the intensity as $(\pi^{-1} + r_V)d\lambda$, which is nonnegative because it is an intensity measure. Approximation in $L^2(\lambda)$ proves the identity for every f in that space. $\square$

Theorem 70.3 (Strict nonlinear screening monotonicity). *For real $V, W \in L^\infty \cap L^2(\lambda)$,*

$$\langle V - W, r_V - r_W \rangle = -2 \int_0^1 \text{Var}_{\mathbb{P}_{W+t(V-W)}} Y(V - W) dt \leq 0. \quad (70.5)$$

Equality holds if and only if $V = W$ in $L^2(\lambda)$. In particular, $\langle V, r_V \rangle < 0$ for every nonzero V in this class. Equivalently, the sum of the two finite relative entropies is

$$D(\mathbb{P}^V \|\mathbb{P}^W) + D(\mathbb{P}^W \|\mathbb{P}^V) = -2\langle V - W, r_V - r_W \rangle. \quad (70.6)$$

At $V = 0$ the weak directional derivative is

$$\left. \frac{d}{dt} \right|_{t=0} \langle f, r_{tV} \rangle = -\frac{2}{\pi} \langle f, (I - B)V \rangle, \quad f \in L^2(\lambda). \quad (70.7)$$

The operator $\frac{2}{\pi}(I - B)$ is bounded and invertible on $L^2(\lambda)$, with inverse norm at most $\pi/[2(1 - \pi/4)]$.

Proof. The exponential bounds justify differentiation under expectation on every compact t -interval, with all polynomial factors in the statistics absorbed by exponential moments. Differentiate $\mathbb{E}_{\mathbb{P}_{W+t(V-W)}} Y(V - W)$ to obtain minus twice its variance. Integrating from zero to one gives (70.5). If $V - W \neq 0$, its centered statistic has strictly positive variance under $\mathbb{P}_2$ by (70.3). Equivalence of all the tilted measures implies that the statistic remains nonconstant under every tilt, so the variance integrand is strictly positive. The logarithm of the likelihood ratio between the two tilts is $-2Y(V - W)$ plus a deterministic normalization constant. Integrating it under both laws with opposite signs cancels that constant and proves (70.6). The entropy integrals are finite by the proved exponential bounds.

Differentiation at zero and polarization of (70.2) prove (70.7). For arbitrary $f \in L^2$, no exponential moment of $Y(f)$ is needed: the exponential bounds for $Y(V)$ imply that the tilted likelihood is differentiable in $L^2(\mathbb{P}_2)$ at zero with derivative $-2Y(V)$, and $Y(f) \in L^2(\mathbb{P}_2)$ can be paired with that derivative by Cauchy–Schwarz. The claimed inverse bound follows from (70.1) and the spectral theorem. $\square$

Corollary 70.4 (Uniqueness for self-consistent one-body screening). *Let G be the positive $L^2(\lambda)$ Green operator of Proposition 59.1 and let $j \in L^2(\lambda)$ be fixed. The equation*

$$V = G(r_V + j), \quad V \in L^\infty \cap L^2(\lambda), \quad (70.8)$$

has at most one solution. For $j = 0$ its unique solution is $V = 0$.

Proof. If V, W are solutions, then

$$\langle V - W, r_V - r_W \rangle = \langle G(r_V - r_W), r_V - r_W \rangle \geq 0.$$

Strict monotonicity forces $V = W$. For $j = 0$, the zero function is a solution because $\mathbb{P}^0 = \mathbb{P}_2$ and $r_0 = 0$. $\square$

70.3 The precise limitation for the temperature conjecture

These results establish the desired screening sign for an actual, explicit family of interacting point processes, and a strict linear response gap on the whole hyperbolic disk. They do not assert that a disk-gas limit at $\beta \neq 2$ belongs to that family. A temperature change alters the pair interaction and cannot in general be represented by a one-body tilt. Moreover, a nonzero constant change of hyperbolic intensity is outside $L^2(\lambda)$. Applying Corollary 70.4 to the intensity conjecture would require both a justified reference process at the new temperature and a proved self-consistent one-body representation of its defect, with the stated boundary selection. Neither assertion follows from the DLR kernels.

There is one rigorous connection to the previously computed temperature responses. If $u_{N,\beta}$ denotes the normalized finite mean defect of (64.7), then, for $j = 0, 1, 2$,

$$\partial_\beta^j u_{N,\beta} \Big|_{\beta=2} \longrightarrow 0 \quad \text{locally in } C^2(\mathbb{D}). \quad (70.9)$$

Indeed $q_2 = 1$, $q'_2 = -1$, $q''_2 = 1$, and the proved local uniform zeroth, first, and second intensity responses are respectively 1, -1 , 1 times $[\pi\delta^2]^{-1}$. Substitute these in the radial integral formula for $u_{N,\beta}$. The same first- and second-radial derivative formulas used in Proposition 64.8 prove local C^2 convergence including the origin. This differentiates before taking $N \rightarrow \infty$ and supplies no remainder estimate on a nonzero temperature interval. In particular it does not imply the missing fixed-temperature screening sign.

70.4 A uniform response gap under bounded one-body tilts

The strict sign just obtained has a quantitative version on every set of uniformly bounded potentials. It also has an operator-theoretic proof that identifies the point processes in the reference family. For a real essentially bounded function V , put

$$\text{osc } V = \text{ess sup } V - \text{ess inf } V.$$

Proposition 70.5 (Weighted projections and a uniform variance gap). *Let $V \in L^\infty(\lambda) \cap L^2(\lambda)$ be real and let $\mathbb{P}^V$ be the regularized tilt of Lemma 70.2. On $L^2(\mathbb{D}, dA)$ let P be the Bergman projection and put $m = e^{-V}$. The process $\mathbb{P}^V$ is determinantal with orthogonal projection onto the closed subspace $m \text{Ran } P$. More precisely, if*

$$S = Pm^2P \Big|_{\text{Ran } P}, \quad P_V = mPS^{-1}Pm,$$

where the inverse acts on $\text{Ran } P$, then P_V is its correlation operator. For every real $f \in L^2(\lambda)$,

$$e^{-2\text{osc } V} \text{Var}_{\mathbb{P}_2} Y(f) \leq \text{Var}_{\mathbb{P}^V} Y(f) \leq e^{2\text{osc } V} \text{Var}_{\mathbb{P}_2} Y(f). \quad (70.10)$$

Consequently,

$$\text{Var}_{\mathbb{P}^V} Y(f) \geq e^{-2\text{osc } V} \frac{1 - \pi/4}{\pi} \|f\|_{L^2(\lambda)}^2. \quad (70.11)$$

Proof. Write $a = \text{ess inf } m > 0$ and $b = \text{ess sup } m < \infty$. Then $a^2 I \leq S \leq b^2 I$ on $\text{Ran } P$, so S is invertible. The operator $U = mPS^{-1/2}$, viewed as a map from $\text{Ran } P$ to $L^2(dA)$, is an isometry onto $m \text{Ran } P$. Hence $P_V = UU^*$ is its orthogonal projection. The range is closed because multiplication by m is boundedly invertible.

We verify that this projection describes the probabilistic tilt. First suppose that V is compactly supported. On $\text{Ran } P$ the operator $S - I = P(m^2 - 1)P$ is trace class: write it as the difference of the two positive compressions obtained from the positive and negative parts of $m^2 - 1$, each having trace $\int (m^2 - 1)_\pm K(z, z) dA(z) < \infty$. The determinantal multiplicative-functional identity gives

$$\mathbb{E}_2 e^{-2\Xi(V)} = \det_{\text{Ran } P} S.$$

For a nonnegative bounded compactly supported function h , the same identity applied with the additional factor e^{-h} yields

$$\frac{\mathbb{E}_2 e^{-2\Xi(V) - \Xi(h)}}{\mathbb{E}_2 e^{-2\Xi(V)}} = \det_{\text{Ran } P} (I + S^{-1/2} P m (e^{-h} - 1) m P S^{-1/2}).$$

The equality follows by conjugating the trace-class perturbation of S by $S^{-1/2}$ and using multiplicativity of Fredholm determinants for invertible trace-class perturbations of the identity. The cyclic determinant identity, with the localized Hilbert–Schmidt factors, turns this into

$$\det_{L^2(dA)} (I - \sqrt{1 - e^{-h}} P_V \sqrt{1 - e^{-h}}).$$

The deterministic centering in $Y(V)$ cancels between the numerator and denominator. These are exactly the Laplace functionals of the projection determinantal process with kernel P_V .

For general bounded $V \in L^2(\lambda)$, set $V_R = V \mathbf{1}_{B_R}$ and $m_R = e^{-V_R}$. All m_R and their inverses are bounded by common constants. Multiplication by m_R converges strongly to multiplication by m , by dominated convergence. Thus $S_R \rightarrow S$ strongly, and their inverses converge strongly as well: this follows from $S_R^{-1} - S^{-1} = S_R^{-1}(S - S_R)S^{-1}$ and their uniform norm bound. If $k_z = K(\cdot, z) \in \text{Ran } P$, the kernels of the weighted projections are given almost everywhere by

$$K_{V_R}(z, w) = m_R(z) m_R(w) \langle S_R^{-1} k_w, k_z \rangle,$$

with the inner-product convention chosen to agree with the displayed Bergman kernel. They converge almost everywhere to the analogous kernel for P_V . Moreover,

$$|K_{V_R}(z, w)| \leq C \sqrt{K(z, z) K(w, w)}$$

with C independent of R . On every relatively compact set the correlation determinants therefore converge by dominated convergence; Hadamard's inequality bounds their order- p integrals by the p th power of a fixed finite local trace. The Fredholm expansion for each compactly supported Laplace functional is consequently dominated by an exponential series and converges to that of P_V . On the other hand, $e^{-2Y(V_R)}/\mathbb{E}_2 e^{-2Y(V_R)}$ converges in $L^1(\mathbb{P}_2)$ to the likelihood of $\mathbb{P}^V$, by the exponential bounds in Lemma 70.2. The two limits of the Laplace functionals coincide. This proves the claimed determinantal representation. The kernel diagonal bound also proves local trace class for the limiting projection.

Put $Q = I - P$ and $Q_V = I - P_V$. For any $u \in L^2(dA)$, distance to the two closed ranges gives

$$a \|Qu\|_2 \leq \|Q_V m u\|_2 \leq b \|Qu\|_2. \quad (70.12)$$

Indeed the middle quantity is $\inf_{v \in \text{Ran } P} \|m(u - v)\|_2$, and multiplication by m changes every norm by a factor between a and b . For bounded compactly supported real f , the projection variance identity and $UU^* = P_V$ imply

$$\text{Var}_{\mathbb{P}^V} \Xi(f) = \|Q_V f P_V\|_{\text{HS}}^2 = \|Q_V f U\|_{\text{HS}}^2.$$

Since f commutes with m , apply (70.12) to each column of $f P S^{-1/2}$. This bounds the last squared norm between $a^2 \|Q f P S^{-1/2}\|_{\text{HS}}^2$ and $b^2 \|Q f P S^{-1/2}\|_{\text{HS}}^2$. The inequalities $b^{-2} I \leq S^{-1} \leq a^{-2} I$ then give

$$\frac{a^2}{b^2} \|Q f P\|_{\text{HS}}^2 \leq \text{Var}_{\mathbb{P}^V} \Xi(f) \leq \frac{b^2}{a^2} \|Q f P\|_{\text{HS}}^2.$$

Here $a^2/b^2 = e^{-2 \text{osc } V}$, and $\|Q f P\|_{\text{HS}}^2$ is the Bergman variance. Subtracting any deterministic center leaves the variances unchanged.

Finally, approximate an arbitrary real $f \in L^2(\lambda)$ by bounded compactly supported f_j in $L^2(\lambda)$. The variance upper comparison, together with $\mathbb{E}_{\mathbb{P}^V} Y(f_j) = \langle f_j, r_V \rangle$, makes $Y(f_j)$ Cauchy in $L^2(\mathbb{P}^V)$. Its limit agrees with $Y(f)$: the convergence in $L^2(\mathbb{P}_2)$ implies convergence in probability under $\mathbb{P}_2$, and absolute continuity implies convergence in probability under $\mathbb{P}^V$ to the same random variable. Passing to the limit proves (70.10) for every such f . The Bergman lower variance bound then proves (70.11). $\square$

Corollary 70.6 (Strong monotonicity on bounded potential sets). *If $V, W \in L^\infty \cap L^2(\lambda)$ are real and $\|V\|_\infty, \|W\|_\infty \leq M$, then*

$$\langle V - W, r_V - r_W \rangle \leq -\frac{2e^{-4M}(1 - \pi/4)}{\pi} \|V - W\|_{L^2(\lambda)}^2. \quad (70.13)$$

Proof. Every potential $W + t(V - W)$ on the joining segment has essential oscillation at most $2M$. Insert (70.11) into the exact integrated variance identity (70.5). $\square$

Example 70.7 (A boundary ambiguity outside the square-integrable class). The weighted-projection construction itself makes sense for every bounded real V , even when the centered exponential tilt is not defined by the preceding L^2 construction. In this larger class the intensity response need not determine V . Indeed, let $V = \Re h$ be bounded and harmonic on the disk, with h holomorphic. Then $a = e^{-h}$ and a^{-1} are bounded analytic functions, so $a \text{Ran } P = \text{Ran } P$. Writing $m = e^{-V}$,

$$m \text{Ran } P = e^{i\Im h} a \text{Ran } P = e^{i\Im h} \text{Ran } P.$$

Consequently the weighted projection has kernel

$$K_V(z, w) = e^{i\Im h(z)} K(z, w) e^{-i\Im h(w)}.$$

All its correlation determinants equal the Bergman determinants. Thus the resulting point-process law is exactly $\mathbb{P}_2$, and its intensity defect is zero, even for nonconstant V . For example, $V(z) = \Re z$ is bounded, has $V(0) = 0$, and is nonzero, while $-\delta^2 \Delta V = 0$. Hence a local screening PDE alone allows this nonzero potential with zero density defect. This V is outside $L^2(\lambda)$, since its angular squared mean is $r^2/2$ and $\int_0^1 r^3(1 - r^2)^{-2} dr = \infty$. The example does not contradict the preceding uniqueness theorem, which uses the L^2 Green operator and its boundary selection. Nor does it provide a radial counterexample to the intensity conjecture: a radial harmonic function regular at the origin is constant, and the normalization there fixes that constant.

71 Status of the uniqueness question

1. *Uniqueness proved for $\beta \geq 4$.* Theorem 20.1 gives full-sequence convergence to the empty configuration for $\beta > 4$, locally in total variation, with an explicit vanishing rate. The endpoint 4 follows from Theorem 36.1, without a rate. Every fixed correlation and compact count moment vanishes throughout $\beta \geq 4$ by Corollary 36.2; the explicit bounds of Corollary 21.1 apply above 4. This is a statement on the open disk: all N particles remain in the finite system and leave each fixed compact set as N increases. A second, independent proof, without a rate, follows from the Gaussian force regression and curvature inequality in Section 34.
2. *Almost sure infiniteness proved below 2.* Theorem 48.6 proves that every subsequential limit is almost surely infinite for $0 < \beta < 2$. The proof uses an exterior-only interpolation identity with a random polynomial, and yields quantitative hole and lower-count bounds. Theorems 48.2 and 48.4 give positive intensity and factorial lower bounds. The limiting intensity dominates $c_\beta/[\pi(1 - |z|^2)^2]$, with $c_\beta = 1$ for $\beta \leq 1$. There is also a positive almost sure logarithmic lower growth rate for boundary counts, and the sharp comparison upper bound $\limsup_{r \uparrow 1} (1 - r)M(r) \leq 2/\beta$. The stronger Theorem 49.2 proves infinite Blaschke mass almost surely, with quantitative Laplace decay and logarithmic growth of accumulated radial depth. Consequently the uncompensated Green potential diverges at every anchor, without an invariance assumption, and uncompensated grand-canonical equations cannot describe these actual limits. At $\beta = 2$ the full limit is the almost surely infinite Bergman process.
3. *The range still unresolved here.* Full-sequence uniqueness and hyperbolic invariance remain unproved for $0 < \beta < 4$, $\beta \neq 2$. Nonemptiness remains unresolved for $2 < \beta < 4$ only. Theorem 50.2 reduces that nonemptiness question to one uniform boundary moment estimate for a charged characteristic polynomial. By Corollary 50.3, any boundary exponent strictly below 2 suffices; such an estimate is still an unproved hypothesis. No claim is made that 4 is the exact threshold for emptiness. Gaussian mode convergence and intensity lower bounds do not by themselves identify the interior law.
4. *What the additional identities settle.* Fixed inserted charges shift the harmonic Gaussian means exactly as in Theorem 24.1; this does not identify the local conditional law. The full-density ratio theorem controls most particle numbers, and its radial-depth corollary concerns a noncompact observable. Neither assertion proves nonemptiness or uniqueness in the remaining range. The boundary-statistic estimates require a power strictly greater than one, whereas the centered exterior Green field contains the first power of the boundary distance. The proved local Möbius covariance is $O(N^{-1})$; the conditional symmetry criterion requires $o(N^{-1})$. These estimates cannot be substituted for one another.
5. *New constraints on the limiting law.* The canonical logarithmic derivative in Corollary 33.5 is a measurable function of the configuration and is uniquely determined by its integration-by-parts identity. This uniqueness is a statement about the score of a given law, not about the law itself. The force regression gives (34.4) and monotonicity of $(1 - r^2)^{\beta/2} \rho_1(r)$. Assuming a nonempty critical limit gives $(1 - r^2)^2 \rho_1(r) \rightarrow c$ with $0 < c \leq 1/\pi$. The residual score variance then vanishes under boundary rescaling; the resulting deterministic score is incompatible with pair repulsion. This closes the critical case by Theorems 36.1 and 37.1. Moreover, Theorem 38.1 excludes every finite positive total count for every limiting law with $\beta \geq 2$. Together with the new infiniteness

theorem below 2, this means that every limit at any $\beta > 0$ has total count either zero or infinity.

6. *What the polynomial representations reduce.* Theorem 39.7 shows that a compact portion of the actual process is determined to vanishing error by a terminal block of $O(\log N)$ Schur parameters. Their dependent law still needs to be analyzed. The independent reference has a Bergman limit for every parameter, but its Gram likelihood is asymptotically singular for $\beta \geq 4$; direct transfer of that limit is therefore impossible in that range. At $\beta = 4$, the explicit hypergeometric mixture in Theorem 40.2 reduces fixed orbital occupations to primary-sector masses. Critical emptiness implies decay of every fixed primary-sector occupation, although no direct norm asymptotic or quantitative critical rate is obtained.
7. *A remaining sufficient criterion.* Proposition 27.3 gives an explicit sufficient covariance estimate, (27.7). That estimate remains unproved. It is unnecessary for the proof above 4, and may be relevant to the remaining range. The deterministic example in Proposition 28.1 explains why exterior fields do not follow from vague convergence alone.
8. *What is now proved about DLR equations.* Theorem 41.3 proves canonical quasi-Gibbs densities for every subsequential limit at every positive β , without the inverse-field variance hypothesis. The densities are positive in every allowed count sector, real analytic after division by the Vandermonde factor, and give local diffeomorphism quasi-invariance. Every nonzero repulsion factor h_p is strictly positive even on the collision diagonals. Theorem 42.1 gives the stronger conditional Levi bound, which rules out a single harmonic exterior potential and forces genuine posterior randomness in the physical force marks. Theorem 43.2 retains this extra information as a meromorphic electric field and proves consistent canonical DLR kernels on the augmented space. Theorem 51.2 further proves that $(\Xi, \mathcal{A})$ is independent of the Gaussian modes, where $\mathcal{A} = \mathcal{C} + T_G$ on the nonempty branch. Exact Gaussian Bayes then gives Theorem 52.3: an explicit Green pair interaction and the one-body factor $(1 - |z|^2)^{-\beta/2}$, conditional on a retained exterior-field mark. The positive-kernel factorization and conditional Gaussian force convolution are stronger than the earlier covariance bounds. At $\beta = 2$, Theorem 46.3 gives the complete compensated DLR law; every finite conditional count has positive probability. This theorem concerns the Bergman law and does not prove uniqueness among all solutions of a specified infinite-volume equation. For other $0 < \beta < 4$, a Green interaction is now identified on the augmented space. Recovering its exterior field from the unmarked exterior and determining the relative conditional count weights remain unproved. Explicit finite-particle solutions show why the canonical equations alone do not give uniqueness; the new almost sure infiniteness theorem excludes those examples as actual gas limits. The stronger Theorem 54.2 constructs distinct infinite, rotationally invariant position laws satisfying the same canonical Green equations for every $0 < \beta < 4$. For $0 < \beta \leq 2$, their finite Blaschke mass makes them mutually singular with every actual gas limit. Thus uniqueness of the canonical specification fails, while uniqueness of the original limiting sequence remains a separate question. Canonical-tail extremality and position-changing decompositions are characterized in Theorem 55.2. Proposition 49.5 shows that all nontrivial actual candidates are nevertheless locally equivalent to area Poisson; this does not establish equality of their laws.
9. *What is now proved about hyperbolic symmetry.* The score relative to hyperbolic area transforms as a complex one-form, and the independent Gaussian force has the

corresponding invariant law. These statements do not establish invariance of the point process. Such invariance is known here at $\beta = 2$ and for the empty limits at $\beta \geq 4$. Theorem 45.1 proves that the uncompensated Green sum of every nonempty invariant locally finite process diverges everywhere, without a moment hypothesis. Consequently its positive-activity uncompensated DLR equation has no invariant solution. Propositions 42.3 and 45.5 impose necessary tests on a compensated description. The formula $c = (4/\beta - 1)/\pi$ remains conditional on the explicitly unproved canonical-DLR and field hypotheses in Section 45.3; it is not an established intensity formula for the general gas. The new affine-field criterion, Theorem 64.6, removes the need to construct an intrinsic centered potential from that conditional argument, but still assumes symmetry of the field law. The exact finite-particle mean defect and Palm dipole identities isolate the remaining expectations without assuming this symmetry. The Gaussian-free canonical kernels have an affine isometry covariance, which is distinct from invariance of the selected augmented law.

10. *Independent perturbative and spectral routes.* Theorem 56.4 identifies the limiting first inverse-temperature response of every fixed correlation order at $\beta = 2$ and proves its full hyperbolic covariance. In particular, $\partial_\beta \rho_{1,N}|_{\beta=2}$ converges locally uniformly to $-1/[\pi(1 - |z|^2)^2]$. Theorem 57.1 also bounds second derivatives of the correlation logarithms uniformly in N on compact sets of distinct tuples, at $\beta = 2$. Theorems 61.1 and 62.1 evaluate the second logarithmic response as zero and the second intensity response as $1/[\pi(1 - |z|^2)^2]$, locally uniformly throughout the disk. The proofs explicitly retain two finite-projection boundary terms, each converging to $1/4$, which cancel at every coherent Palm anchor. A fixed nonzero perturbation in β still requires control on a temperature interval. Theorem 63.2 identifies $2/\beta - 1/2$ as the radial-depth coefficient in the actual gas for $0 < \beta < 2$, in all positive moments on a density-one set of particle numbers, and proves signed-charge exponents throughout $t > -2$. The same leading depth coefficient holds at every fixed hyperbolic center. These logarithmic statements do not fix the order-one insertion prefactor or the mean-potential defect that determines the limiting intensity. The global gas laws at any two distinct temperatures become asymptotically singular, by Theorem 58.1. Thus a direct global change of temperature cannot have a uniformly integrable likelihood; this does not preclude local absolute continuity or local convergence. The sharp hyperbolic Green operator bound is $\|G\|_{L^2(\lambda)} = 2\pi$, but explicit invariant random boundary modes show why this spatial contraction does not prove uniqueness among invariant random fields. Under stated integrability and covariance hypotheses, two renormalized fields over the same configuration differ only by a tail-dependent constant; this does not identify the configuration law or the activity. Proposition 60.4 gives a different sufficient uniqueness criterion, whose monotonicity input has not been proved here. Proposition 60.1 also gives an exact necessary and sufficient oscillation test for multiple original gas subsequences, expressed in finite partition-integral ratios. Its required nonzero oscillation has not been proved. Fixed boundary regularizations are screened, but the boundary-uniform estimate needed to control adjacent particle numbers remains unproved.
11. *What the screening and energy method establishes.* The exact Euclidean-coordinate equation is $\text{div}(A\nabla U) = 2\pi(c - \tilde{\rho})$, with the anisotropic matrix A specified in Proposition 66.2. For a radial finite-energy defect, its finite trace $\ell = U(\infty)$ exists and testing with $U - \ell$ removes the cutoff boundary term; this yields the sufficient sign criterion in Proposition 66.4. The finite-energy hypothesis and the required sign have not been proved for the selected limits. Whole-plane finite-particle energy is infinite, and even a neutral finite-volume mean-charge energy is of order N at $\beta = 2$. Such

global estimates cannot replace a localized argument. The exact Ward identity gives a different sufficient criterion, Theorem 68.3, that requires no prior finite total energy but retains connected-force work and a cutoff boundary condition. The harmonic Gaussian modes do not remove this connected force. The variational decomposition in Proposition 69.3 shows explicitly where the correlation energy enters. For bounded square-integrable one-body Bergman tilts, strict screening monotonicity and positivity of the Green operator prove uniqueness of $V = G(r_V + j)$ for each prescribed source j . The weighted-projection formula makes this sign quantitatively coercive on uniformly bounded potential sets. Changing inverse temperature changes the pair interaction, so this theorem supplies no identification of the original general-temperature process. Neither the full intensity conjecture nor its uniqueness question is resolved by this revision.

12. *The temperature assumption.* At $\beta = 0$ the positions are independent uniform points of the disk and $M_N(r)$ is Binomial(N, r^2). Its mean is Nr^2 and variance at most N , so Chebyshev's inequality gives $M_N(r) \rightarrow \infty$ in probability for every $r > 0$. The fixed positive β hypothesis is therefore essential for unscaled local compactness.

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